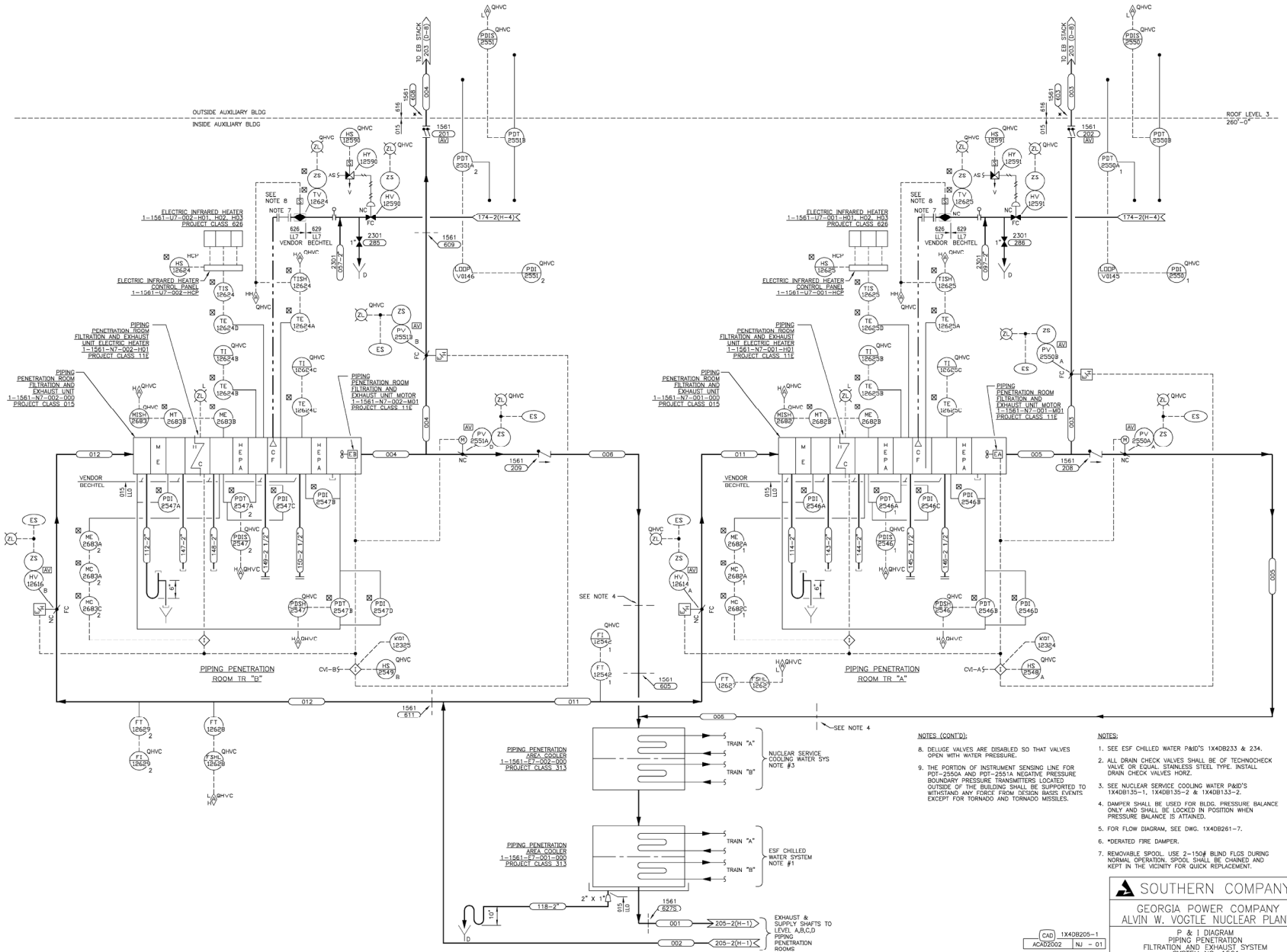

NOTES (CONT'D):

8. DELUGE VALVES ARE DISABLED SO THAT VALVES OPEN WITH WATER PRESSURE.
9. THE PORTION OF INSTRUMENT SENSING LINE FOR PDT-2550A AND PDT-2551A NEGATIVE PRESSURE BOUNDARY PRESSURE TRANSMITTERS LOCATED OUTSIDE OF THE BUILDING SHALL BE SUPPORTED TO WITHSTAND ANY FORCE FROM DESIGN BASIS EVENTS EXCEPT FOR TORNADO AND TORNADO MISSILES.

NOTES:

1. SEE ESF CHILLED WATER P&ID'S 1X4DB233 & 234.
2. ALL DRAIN CHECK VALVES SHALL BE OF TECHNOCHECK VALVE OR EQUAL. STAINLESS STEEL TYPE. INSTALL DRAIN CHECK VALVES HORZ.
3. SEE NUCLEAR SERVICE COOLING WATER P&ID'S 1X4DB135-1, 1X4DB135-2 & 1X4DB133-2.
4. DAMPER SHALL BE USED FOR BLDG. PRESSURE BALANCE ONLY AND SHALL BE LOCKED IN POSITION WHEN PRESSURE BALANCE IS ATTAINED.
5. FOR FLOW DIAGRAM, SEE DWG. 1X4DB261-7.
6. *DERATED FIRE DAMPER.
7. REMOVABLE SPOOL. USE 2-150# BLIND FLGS DURING NORMAL OPERATION. SPOOL SHALL BE CHAINED AND KEPT IN THE VICINITY FOR QUICK REPLACEMENT.

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NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
27.0	REVISED PER ABN-52213, VER. 1.0	8-30-16	NJ	WBJ	ASK	X

CAD 1X4DB205-1
ACAD2002 NJ - 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PIPING PENETRATION
FILTRATION AND EXHAUST SYSTEM
SYSTEM NO. 1561

SCALE: NONE

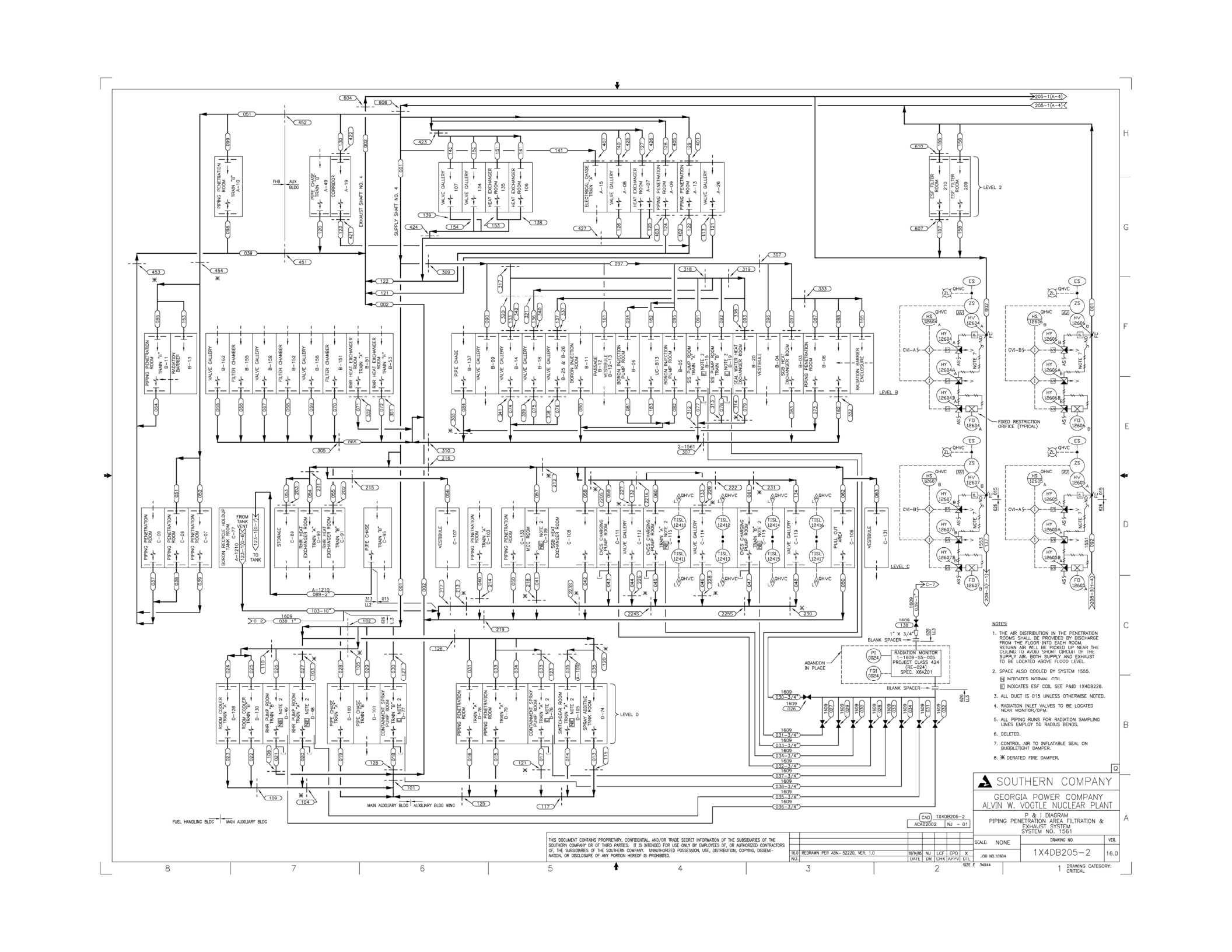
DRAWING NO. 1X4DB205-1

VER. 27.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL


NOTES:
1. THE AIR DISTRIBUTION IN THE PENETRATION ROOMS SHALL BE PROVIDED BY DISCHARGE FROM THE FLOOR INTO EACH ROOM. RETURN AIR WILL BE PICKED UP NEAR THE CEILING TO AVOID SHORT CIRCUIT OF THE SUPPLY AIR. BOTH SUPPLY AND EXHAUST TO BE LOCATED ABOVE FLOOD LEVEL.
2. SPACE ALSO COOLED BY SYSTEM 1555.
N INDICATES NORMAL COIL
E INDICATES ESF COIL SEE P&ID 1X4DB228.
3. ALL DUCT IS 015 UNLESS OTHERWISE NOTED.
4. RADIATION INLET VALVES TO BE LOCATED NEAR MONITOR/DPM.
5. ALL PIPING RUNS FOR RADIATION SAMPLING LINES EMPLOY 5D RADIUS BENDS.
6. DELETED.
7. CONTROL AIR TO INFLATABLE SEAL ON BUBBLETIGHT DAMPER.
8. ✳ DERATED FIRE DAMPER.
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
PIPING PENETRATION AREA FILTRATION & EXHAUST SYSTEM
SYSTEM NO. 1561
SCALE: NONE
DRAWING NO. 1X4DB205-2
VER. 16.0
JOB NO.10604
CAD 1X4DB205-2
ACAD2002 NU – 01
SIZE E 34X44
DRAWING CATEGORY: CRITICAL
16.0 REDRAWN PER ABN–52220, VER. 1.0 10/14/05 NU LCF EPD X
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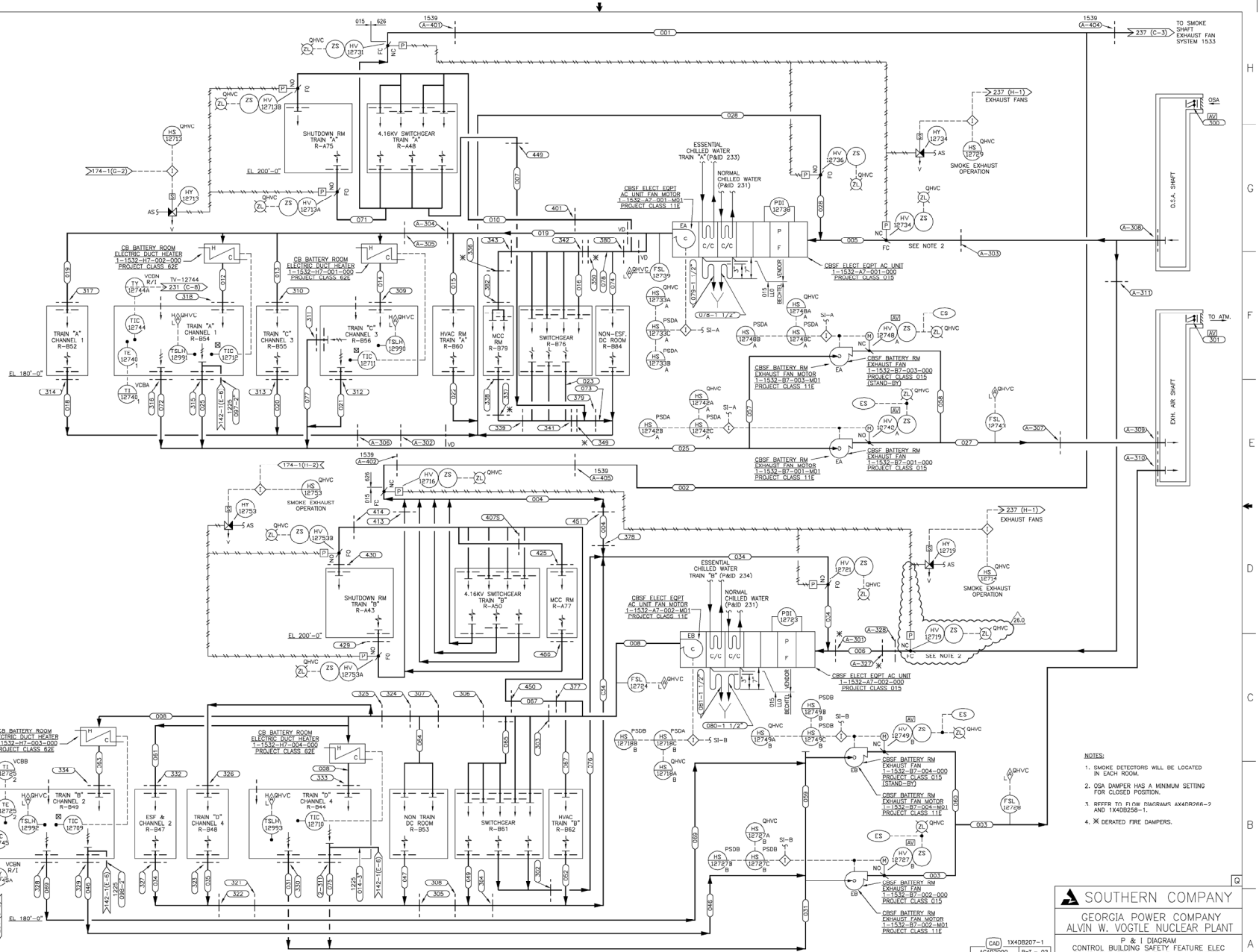
FUEL HANDLING BLDG
MAIN AUXILIARY BLDG
AUXILIARY BLDG WING
LEVEL 2
LEVEL B
LEVEL C
LEVEL D
FIXED RESTRICTION ORIFICE (TYPICAL)
BLANK SPACER
ABANDON IN PLACE
RADIATION MONITOR 1-1609-S5-005 PROJECT CLASS 424 (RE-024) SPEC. X6AZ01
PIPING PENETRATION ROOM TRAIN "B" A-10
PIPE CHASE TRAIN "B" A-49
CORRIDOR A-19
EXHAUST SHAFT NO. 4
SUPPLY SHAFT NO. 4
VALVE GALLERY 107
VALVE GALLERY 134
HEAT EXCHANGER ROOM 135
HEAT EXCHANGER ROOM 106
ELECTRICAL CHASE TRAIN "A" A-15
VALVE GALLERY A-08
HEAT EXCHANGER ROOM A-07
PIPING PENETRATION ROOM A-09
PIPING PENETRATION ROOM A-13
VALVE GALLERY A-26
ESF FILTER ROOM 210
ESF FILTER ROOM 209
PIPING PENETRATION ROOM TRAIN "B" B-11
RADIATION BARRIER B-13
VALVE GALLERY B-162
FILTER CHAMBER B-155
VALVE GALLERY B-159
FILTER CHAMBER B-152
VALVE GALLERY B-158
FILTER CHAMBER B-151
RHR HEAT EXCHANGER ROOM TRAIN "A" B-15
RHR HEAT EXCHANGER ROOM TRAIN "B" B-153
PIPE CHASE B-137
VALVE GALLERY B-09
VALVE GALLERY B-14
VALVE GALLERY B-16
VALVE GALLERY B-25 & B-26
BORON INJECTION ROOM B-11
PASSAGE VESTIBULE B-12
BORON INJECTION PUMP ROOM B-06
UC-B13
BORON INJECTION PUMP ROOM B-05
SIS PUMP ROOM TRAIN "A" B-1.5
SIS PUMP ROOM TRAIN "B" B-19
FEEDWATER HEATER ROOM B-20
VESTIBULE
SGB HEAT EXCHANGER ROOM B-04
PIPING PENETRATION ROOM B-03
PIPING PENETRATION ROOM B-08
BOUNTON BARRIER ENCLOSURE
PIPING PENETRATION ROOM C-06
PIPING PENETRATION ROOM C-08
PIPING PENETRATION ROOM C-07
BORON RECYCLE HOLDUP TANK A-1210 C-77
FROM TANK A-1210 C-77 ROOM C-7
STORAGE C-85
RHR HEAT EXCHANGER ROOM TRAIN "A" C-93
RHR HEAT EXCHANGER ROOM TRAIN "B" C-9
PIPE CHASE TRAIN "B" C-95
VESTIBULE C-107
PIPING PENETRATION ROOM TRAIN "A" C-103
PIPING PENETRATION ROOM C-133
MCC ROOM NOTE 2
SGB HEAT EXCHANGER ROOM C-109
CVCS CHARGING PUMP ROOM C-111
VALVE GALLERY C-112
CVCS CHARGING PUMP ROOM TRAIN "A" C-113 NOTE 2
VALVE GALLERY C-114
CVCS CHARGING PUMP ROOM TRAIN "B" C-115 NOTE 2
VALVE GALLERY C-119
PULL OUT AREA C-108
VESTIBULE C-131
RCWU COOLER TRAIN "A" D-128
RCWU COOLER TRAIN "B" D-130
RHR PUMP ROOM TRAIN "B" D-49 NOTE 2
RHR PUMP ROOM TRAIN "A" D-48 NOTE 2
PIPE CHASE TRAIN "A" D-190
PIPE CHASE TRAIN "B" D-101
CONTAINMENT SPRAY PUMP ROOM TRAIN "B" D-77 NOTE 2
PIPING PENETRATION ROOM TRAIN "A" D-76
PIPING PENETRATION ROOM TRAIN "A" D-79
CONTAINMENT SPRAY PUMP ROOM TRAIN "A" D-75 NOTE 2
SWITCHGEAR ROOM TRAIN "A" NOTE 2 D-105
SPRAY ADDITIVE TANK ROOM D-74
205-1(A-4)
208-5(F-3)
208-3(H-4)



NOTES:

1. SMOKE DETECTORS WILL BE LOCATED IN EACH ROOM.
2. OSA DAMPER HAS A MINIMUM SETTING FOR CLOSED POSITION.
3. REFER TO FLOW DIAGRAMS 1X4DB256-2 AND 1X4DB258-1.
4. ✳ DERATED FIRE DAMPERS.

CAD 1X4DB207-1
ACAD2000 BgT - 02

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NO.	REVISIONS	DATE	DN	CHK	APPV	DTL
26.0	REVISED PER ABN-V01208, VER.1.0	7-1-08	BgT	WMW	PML	X

SOUTHERN COMPANY

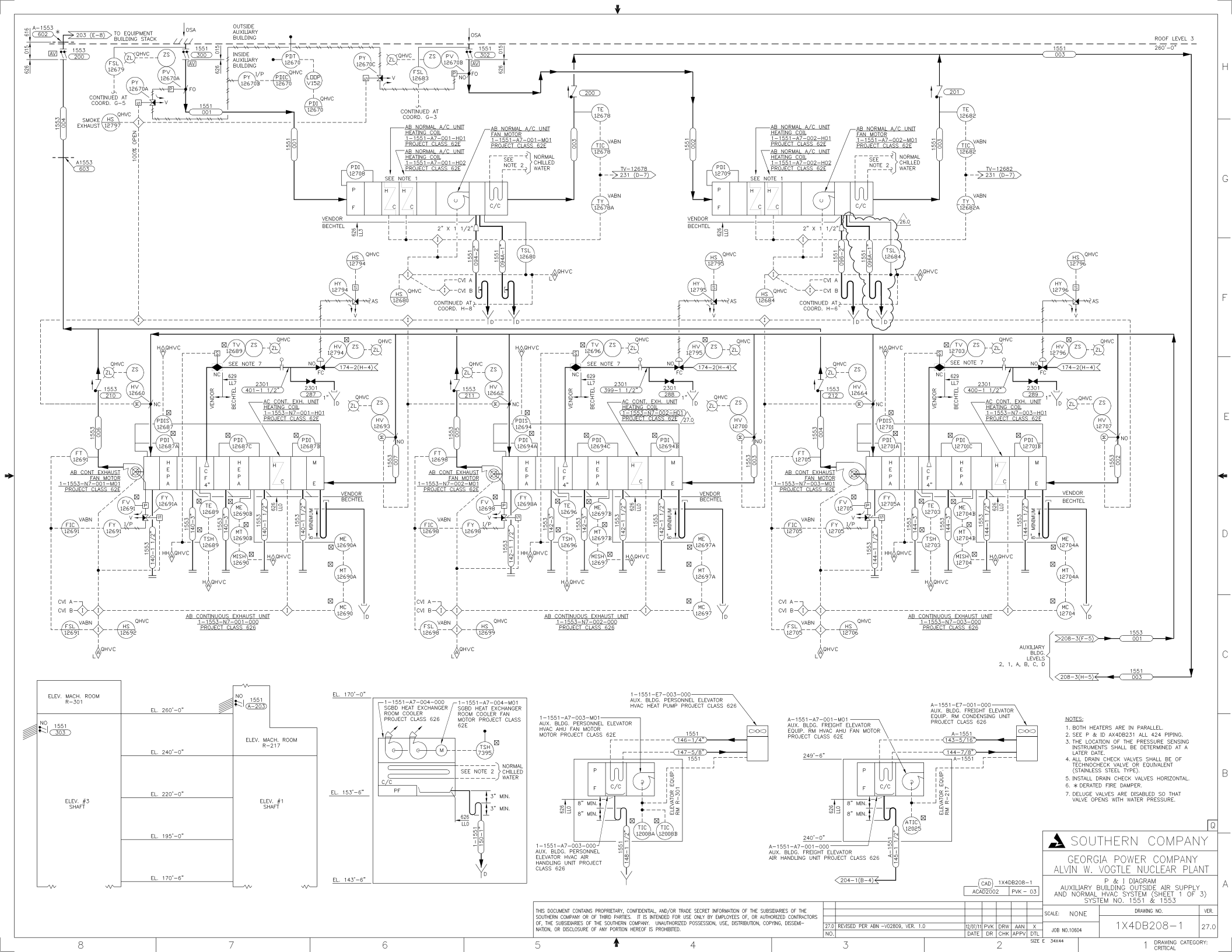
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING SAFETY FEATURE ELEC EQUIPMENT ROOM LEVEL B HVAC SYSTEM
SYSTEM NO. 1532

SCALE: NONE
SIZE E 34X44

DRAWING NO. 1X4DB207-1
VER. 26.0
JOB NO.10604

1 DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P. & I DIAGRAM
AUXILIARY BUILDING OUTSIDE AIR SUPPLY AND NORMAL HVAC SYSTEM (SHEET 1 OF 3)
SYSTEM NO. 1551 & 1553

CAD	1X4DB208-1
ACAD2002	PVK - 03

SCALE:	NONE	DRAWING NO.	VER.
JOB NO.	10604	1X4DB208-1	27.0

NO.	REVISED PER ABN -V02809, VER. 1.0	DATE	DR	CHK	APPV	DTL
27.0		12/07/11	PVK	DRW	AAN	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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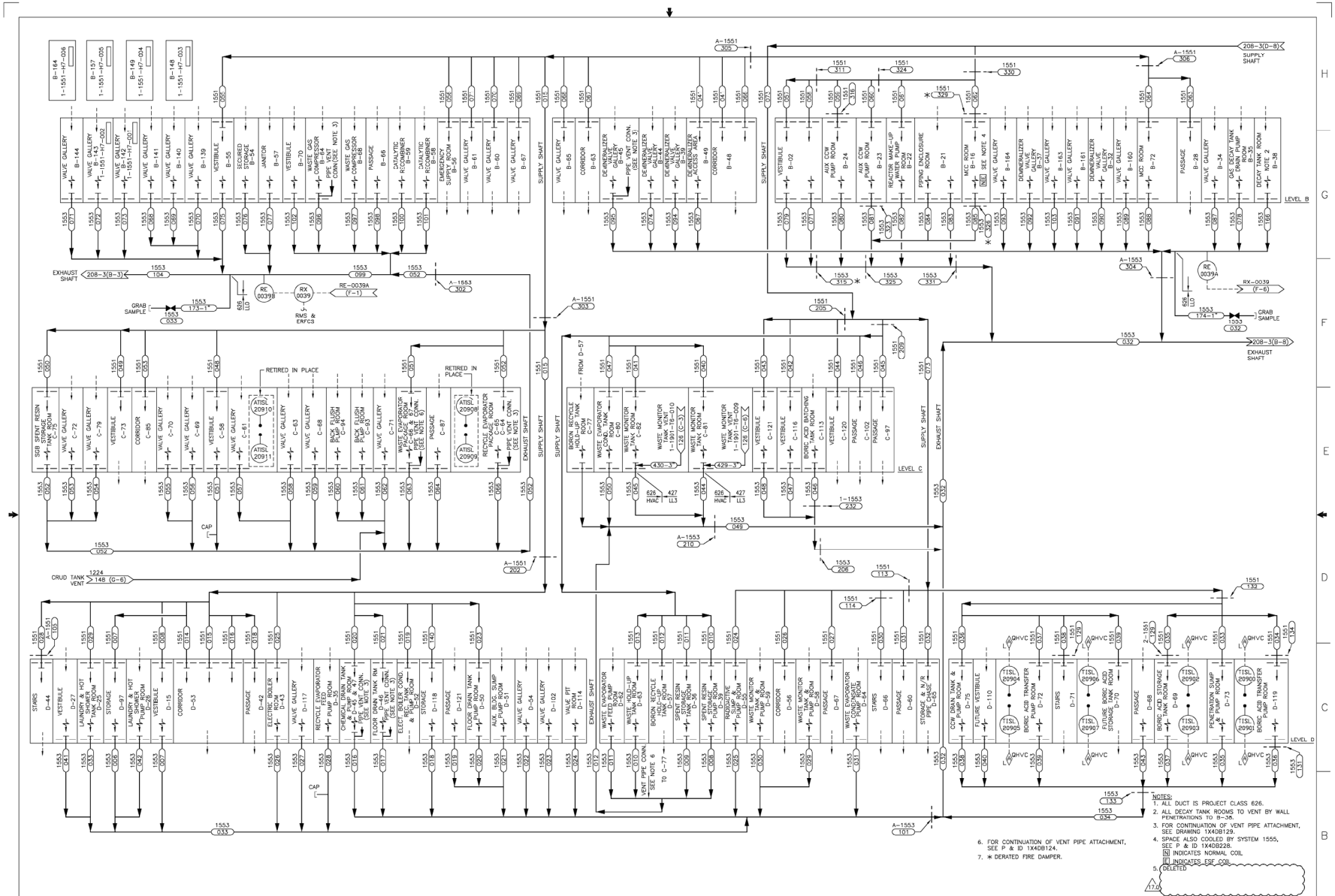
B-164 1-1551-H7-006
B-157 1-1551-H7-005
B-149 1-1551-H7-004
B-148 1-1551-H7-003
VALVE GALLERY B-144
VALVE GALLERY B-145
VALVE GALLERY B-142
VALVE GALLERY B-141
VALVE GALLERY B-140
VALVE GALLERY B-139
VESTIBULE B-55
SECURED STORAGE B-54
JANITOR B-57
VESTIBULE B-70
WASTE GAS COMPRESSOR B-64
PIPE VENT CONN. (SEE NOTE 3)
WASTE GAS COMPRESSOR B-58
PASSAGE B-66
CATALYTIC RECOMBINER B-65
CATALYTIC RECOMBINER B-59
EMERGENCY SUPPLY ROOM B-56
VALVE GALLERY B-61
VALVE GALLERY B-60
VALVE GALLERY B-67
SUPPLY SHAFT
VALVE GALLERY B-85
CORRIDOR B-63
DEMINERALIZER VALVE GALLERY B-62
PIPE VENT CONN. (SEE NOTE 3)
DEMINERALIZER VALVE GALLERY B-84
DEMINERALIZER VALVE GALLERY B-83
DEMINERALIZER ACCESS AREA B-49
CORRIDOR B-48
SUPPLY SHAFT
VESTIBULE B-02
AUX CCW PUMP ROOM B-24
AUX CCW PUMP ROOM B-23
REACTOR MAKE-UP WATER PUMP ROOM B-22
PIPING ENCLOSURE ROOM B-21
MCC ROOM B-16
SEE NOTE 4
VALVE GALLERY B-164
DEMINERALIZER VALVE GALLERY B-32
VALVE GALLERY B-163
VALVE GALLERY B-161
DEMINERALIZER VALVE GALLERY B-33
VALVE GALLERY B-160
MCC ROOM B-72
PASSAGE B-28
VALVE GALLERY B-34
GAS DECAY TANK DRAIN PUMP ROOM B-35
DECAY TANK ROOM NOTE 2 B-38
LEVEL B
208-3(B-3)
EXHAUST SHAFT
208-3(B-8)
SUPPLY SHAFT
GRAB SAMPLE
RE 0039B
RX 0039 (F-1)
RE-0039A (F-1)
RMS & EMFCS
RX-0039 (F-6)
SPENT RESIN STORAGE TANK C-75
VALVE GALLERY C-72
VALVE GALLERY C-79
VESTIBULE C-73
CORRIDOR C-85
VALVE GALLERY C-70
VALVE GALLERY C-69
VESTIBULE C-58
VALVE GALLERY C-61
RETIRED IN PLACE
ATISI 2091C
ATISI 2091
VALVE GALLERY C-63
VALVE GALLERY C-68
BACK FLUSH PACKAGE C-84
BACK FLUSH PUMP ROOM C-83
VALVE GALLERY C-82
WASTE EVAPORATOR PACKAGE ROOM C-71
PIPE VENT CONN. (SEE NOTE 6)
PASSAGE C-87
RETIRED IN PLACE
ATISI 2090C
ATISI 2090
RECYCLE EVAPORATOR PACKAGE ROOM C-64
PIPE VENT CONN. (SEE NOTE 3)
EXHAUST SHAFT
SUPPLY SHAFT
BORON RECYCLE HOLD-UP TANK ROOM C-57
FROM D-57
WASTE EVAPORATOR COND. TANK ROOM C-80
WASTE MONITOR TANK & PUMP ROOM C-62
WASTE MONITOR TANK VENT 1-1551-H7-010
WASTE MONITOR TANK C-61
WASTE MONITOR TANK VENT 1-1551-H7-009
VESTIBULE C-121
VESTIBULE C-116
BORIC ACID BATCHING TANK ROOM C-113
PASSAGE C-120
PASSAGE C-97
SUPPLY SHAFT
EXHAUST SHAFT
LEVEL C
CAP
CRUD TANK VENT
STAIRS D-44
VESTIBULE D-27
LAUNDRY & HOT SHOWER ROOM D-25
STORAGE D-97
LAUNDRY & HOT SHOWER PUMP ROOM D-26
VESTIBULE D-15
CORRIDOR D-53
PASSAGE D-42
ELECTRIC BOILER ROOM D-43
VALVE GALLERY D-117
RECYCLE EVAPORATOR PUMP ROOM D-14
CHEMICAL DRAIN TANK & PUMP ROOM D-13
PIPE VENT CONN. (SEE NOTE 3)
FLOOR DRAIN TANK RM D-12
PIPE VENT CONN. (SEE NOTE 3)
ELECT. EQUIP. ROOM & RECIRC. TANK ROOM D-52
STORAGE D-118
PASSAGE D-121
FLOOR DRAIN TANK PUMP ROOM D-10
FLOOR DRAIN TANK PUMP ROOM D-50
AUX. BLDG. SUMP PUMP ROOM D-51
VALVE GALLERY D-54
VALVE GALLERY D-102
VALVE PIT ROOM D-114
EXHAUST SHAFT
WASTE EVAPORATOR FEED PUMP ROOM D-81
WASTE HOLD-UP TANK ROOM D-82
BORON RECYCLE HOLD-UP TANK D-77
SPENT RESIN STORAGE TANK ROOM D-09
SPENT RESIN STORAGE ROOM D-39
RECIRC. PUMP ROOM D-08
RECIRC. PUMP ROOM D-38
WASTE MONITOR TANK & PUMP ROOM D-58
CORRIDOR D-56
WASTE MONITOR TANK & PUMP ROOM D-59
PASSAGE D-67
WASTE EVAPORATOR PUMP ROOM D-31
STAIRS D-46
PASSAGE D-60
STORAGE & N/R PIPE CHASE D-45
VENT PIPE CONN. SEE NOTE 6
TO C-77
CCW DRAIN TANK & PUMP ROOM D-36
FUTURE VESTIBULE D-110
BORIC ACID TRANSFER PUMP ROOM D-72
STAIRS D-71
FUTURE BORIC ACID STORAGE TANK ROOM D-70
PASSAGE D-88
BORIC ACID STORAGE TANK D-69
PENETRATION SUMP PUMP ROOM D-73
BORIC ACID TRANSFER PUMP ROOM D-119
LEVEL D
TISL 20904
TISL 20905
TISL 20906
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TISL 20903
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TISL 20902
QHVC
A-1551 305
A-1551 306
A-1551 303
A-1551 202
A-1553 210
A-1553 101
A-1553 302
A-1553 304
1224 148 (G-6)
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1553 174-1
1553 032
626 LLO
626 HVAC 427 LL3
430-3"
429-3"
128 (C-3)
128 (C-6)
1-1553 232
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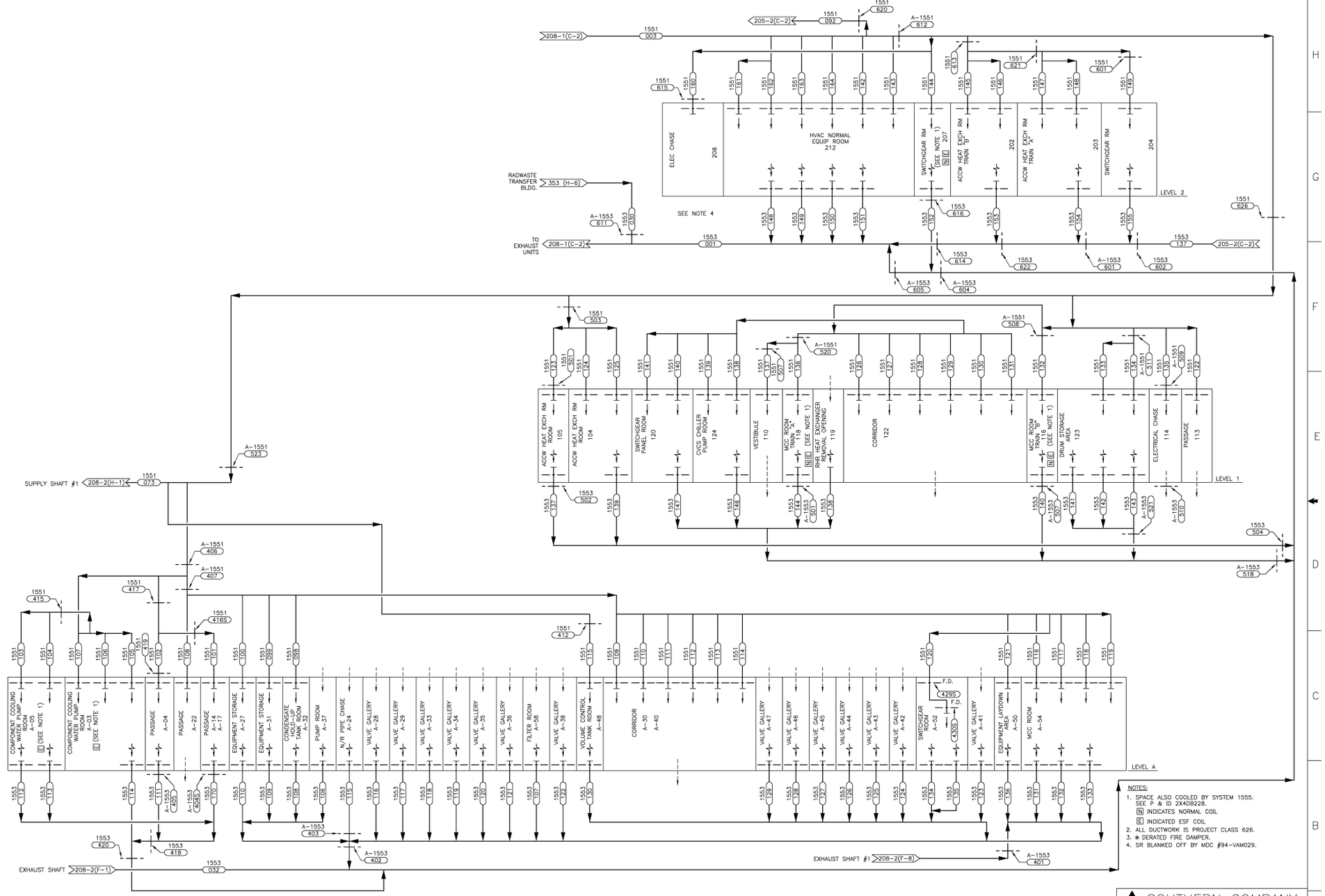
1551 006
1551 005
1551 004
1551 003
1551 002
1551 001
RETIRED IN PLACE
RETIRED IN PLACE
SEE NOTE 4
SEE NOTE 5
SEE NOTE 6
SEE NOTE 3



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING
VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

CAD 1X4DB208-3
ACAD2004 CYN-01

SCALE: NONE
DRAWING NO. 1X4DB208-3
VER. 12.0
JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
12.0	REDRAWN PER ABN-V70223, VER. 1.0.	1-4-07	CYN	LCF	SHS	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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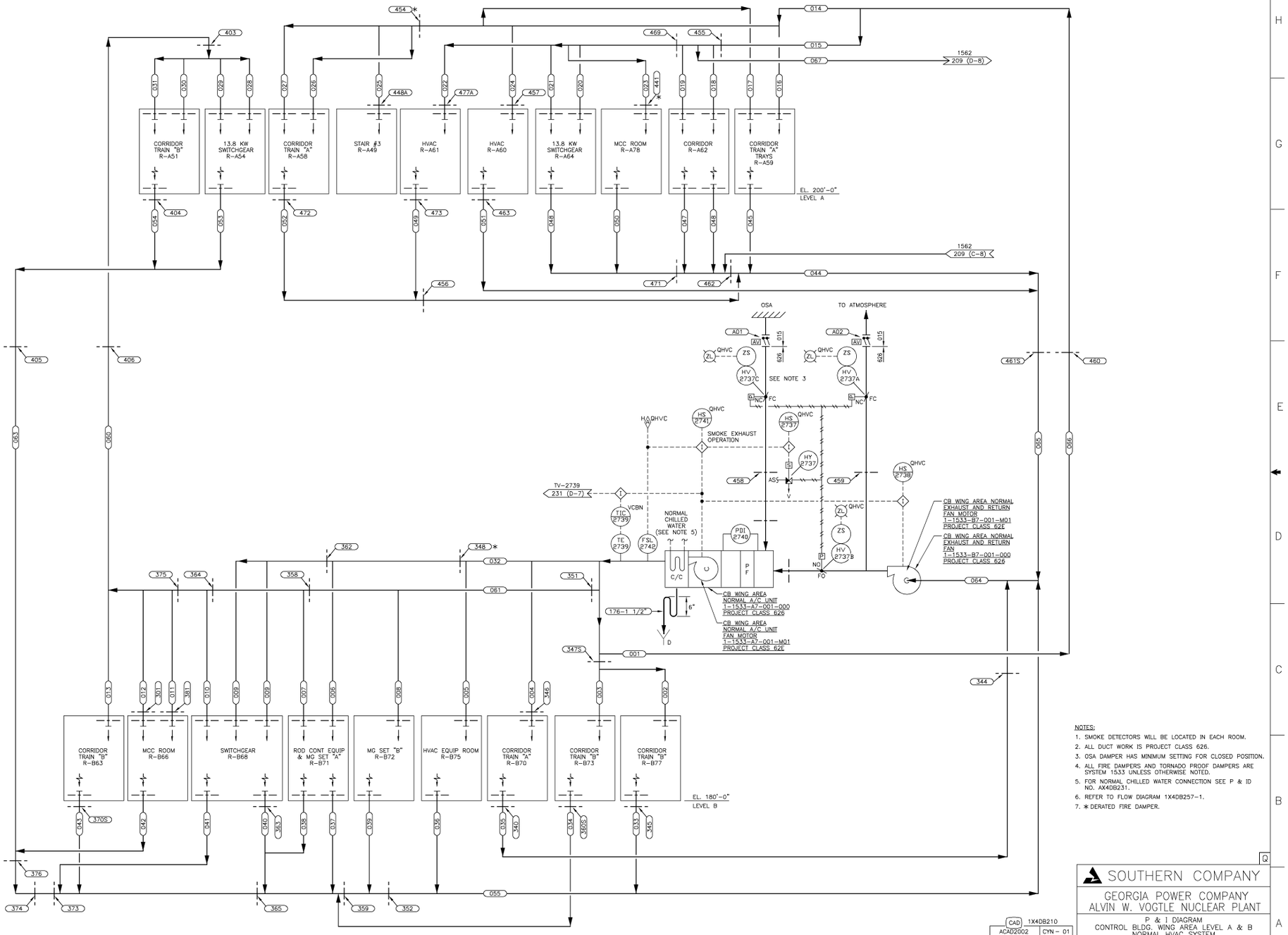

NOTES:

1. SMOKE DETECTORS WILL BE LOCATED IN EACH ROOM.
2. ALL DUCT WORK IS PROJECT CLASS 62G.
3. OSA DAMPER HAS MINIMUM SETTING FOR CLOSED POSITION.
4. ALL FIRE DAMPERS AND TORNADO PROOF DAMPERS ARE SYSTEM 1533 UNLESS OTHERWISE NOTED.
5. FOR NORMAL CHILLED WATER CONNECTION SEE P & ID NO. AX4DB231.
6. REFER TO FLOW DIAGRAM 1X4DB257-1.
7. * DERATED FIRE DAMPER.

CAD 1X4DB210
ACAD2002 CYN - 01

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NO.		DATE	DR	CHK	APPV	DTL
10.0	REDRAWN PER ABN-V70052, VER. 1.0	6-1-06	CYN	LCF	EPD	X

SIZE E 34X44

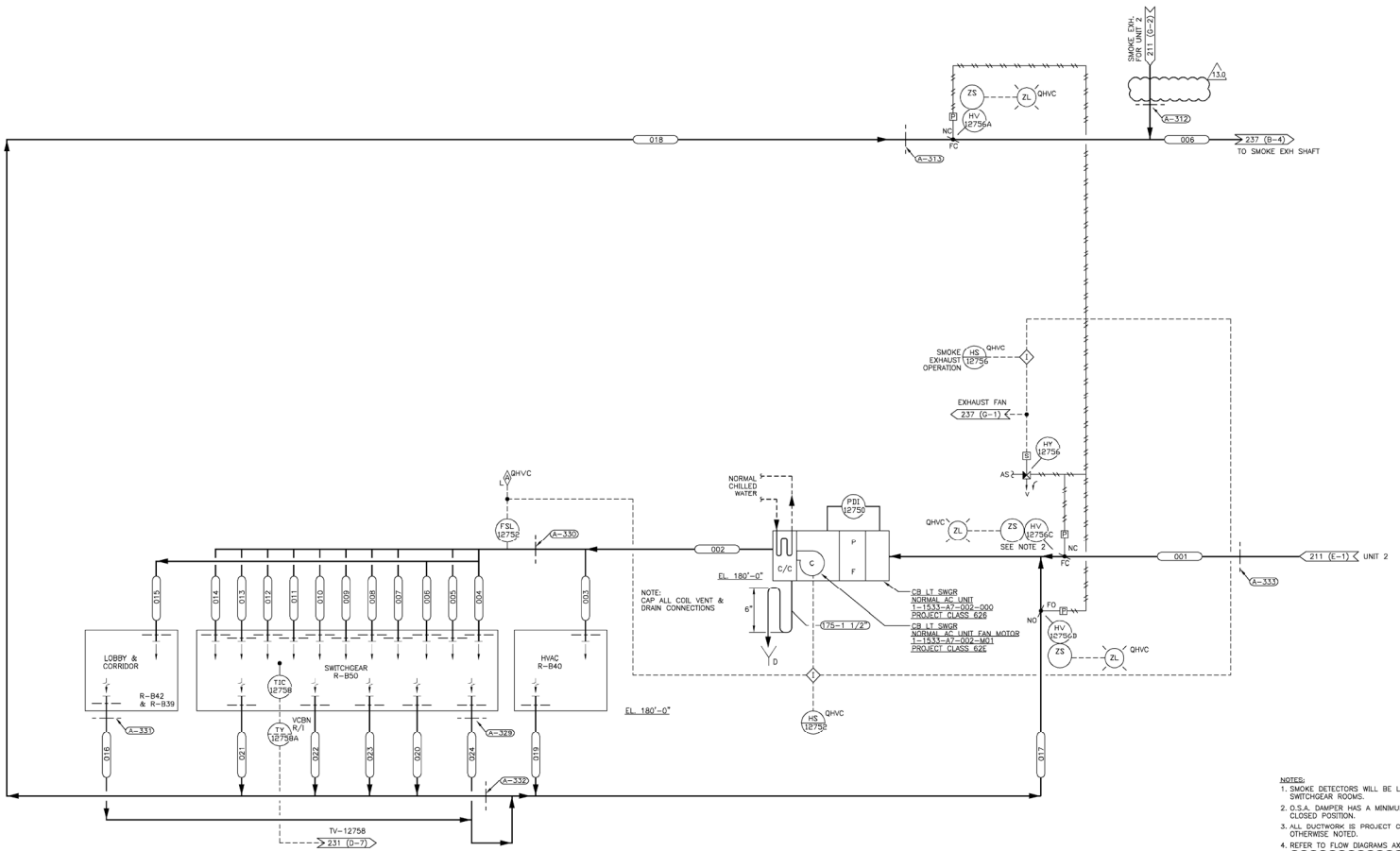
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BLDG. WING AREA LEVEL A & B
NORMAL HVAC SYSTEM
SYSTEM NO. 1533

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB210	10.0

DRAWING CATEGORY: CRITICAL



NOTES:
1. SMOKE DETECTORS WILL BE LOCATED IN THE NORMAL SWITCHGEAR ROOMS.
2. O.S.A. DAMPER HAS A MINIMUM SETTING FOR CLOSED POSITION.
3. ALL DUCTWORK IS PROJECT CLASS 62G UNLESS OTHERWISE NOTED.
4. REFER TO FLOW DIAGRAMS AX4DB266-2 & 1X4DB266-1.
5. DELETED

13.0

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING LEVEL B
NORMAL HVAC SYSTEM
SYTEM NO. 15.3.3

CAD 1X4DB211
ACAD2002 BgT - 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BgT	MWD	WLN	

SIZE E 34X44

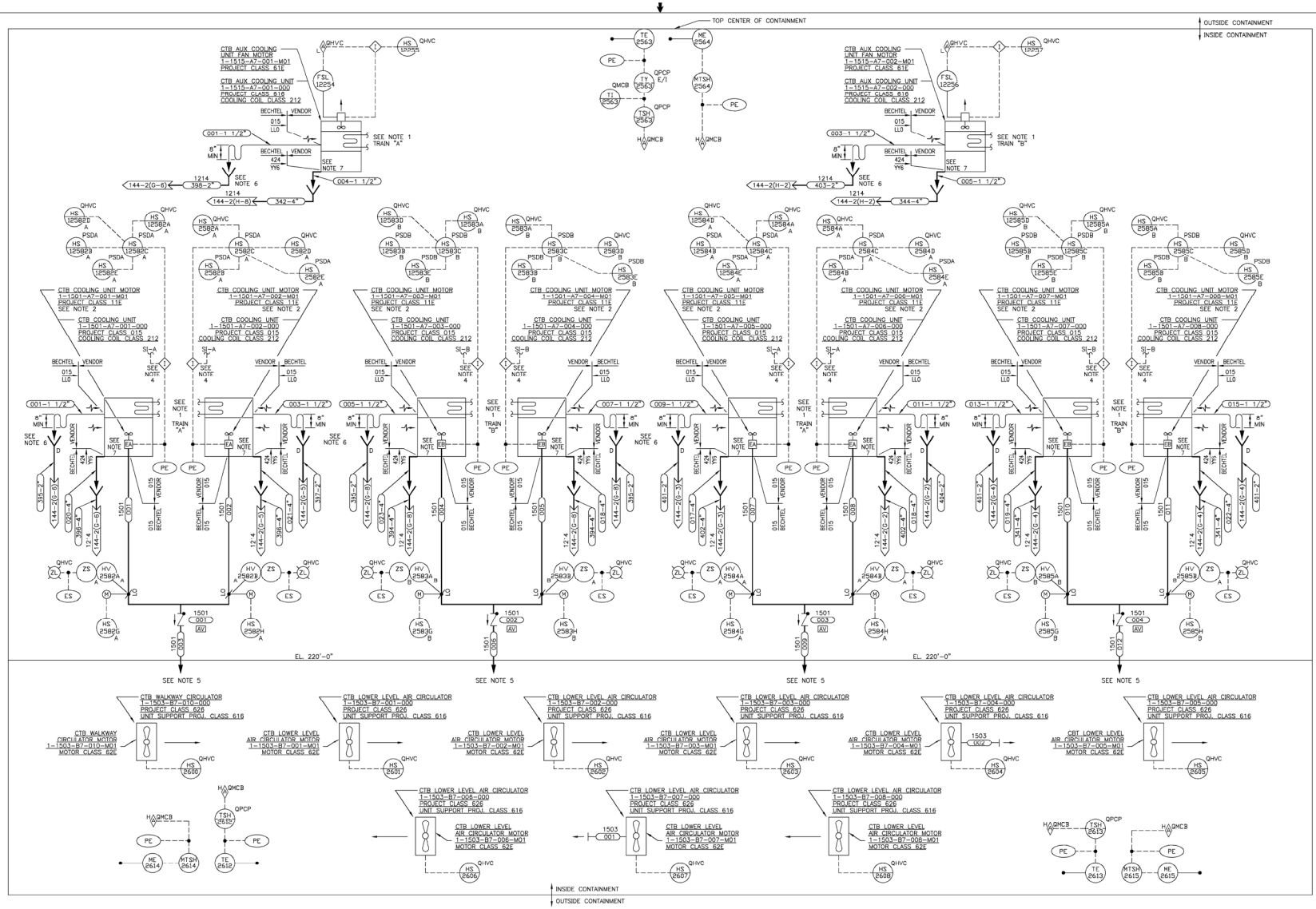
SCALE: NONE

DRAWING NO. 1X4DB211

VER. 13.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR NUCLEAR SERVICE COOLING WATER SEE P&ID 1X4DB135.
2. CTB COOLING UNIT FAN MOTORS ARE 2-SPEED.
3. FOR DRAIN LINES SEE P&ID 1X4DB144 2.
4. CTB COOLING UNIT FANS SHALL OPERATE AT LOW SPEED UPON RECEIPT OF THE SAFETY INJECTION (SI) SIGNAL.
5. REFER TO DWG. 1X4DB251-1 FOR AIR DISTRIBUTION.
6. SCREEN(S) INSTALLED TO PREVENT TRASH AND MISC. DEBRIS GETTING INTO AND BLOCKING DRAINS.
7. FLOOR DRAIN SCREEN(S) MAY BE INSTALLED IN THE AUX & MAIN COOLERS AS SHOWN IN MOD 92-V1M119. A RER SHOULD BE INITIATED TO ENSURE DWG IS REVISED TO REFLECT THE SCREEN(S) INSTALLATION.

CAD 1X4DB212
ACAD2002 NJ - 01

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NO.		DATE	DN	CHK	APPV	DTL
12.0	REDRAWN PER ABN-52221, VER. 1.0	11-2-05	NJ	LCF	EPD	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTAINMENT HEAT REMOVAL SYSTEM
SYSTEM NO. 1501 & 1515

SCALE: NONE

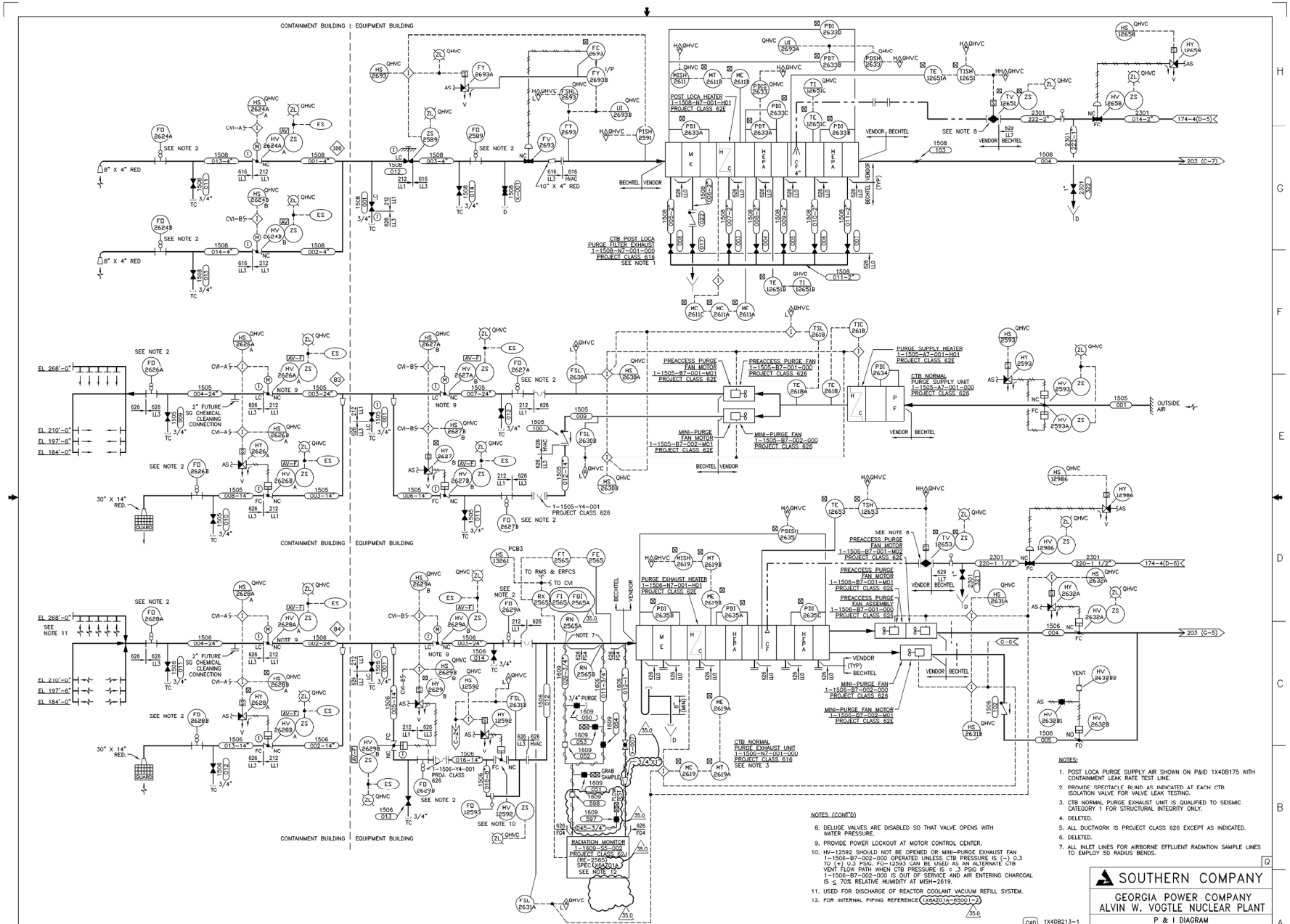
DRAWING NO. 1X4DB212

VER. 12.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. POST LOCA PURGE SUPPLY AIR SHOWN ON P&ID 1X4DB175 WITH CONTAINMENT LEAK RATE TEST LINE.
2. PROVIDE SPECTACLE BLIND AS INDICATED AT EACH CTB ISOLATION VALVE FOR VALVE LEAK TESTING.
3. CTB NORMAL PURGE EXHAUST UNIT IS QUALIFIED TO SEISMIC CATEGORY 1 FOR STRUCTURAL INTEGRITY ONLY.
4. DELETED.
5. ALL DUCTWORK IS PROJECT CLASS 626 EXCEPT AS INDICATED.
6. DELETED.
7. ALL INLET LINES FOR AIRBORNE EFFLUENT RADIATION SAMPLE LINES TO EMPLOY 5D RADIUS BENDS.

NOTES (CONT'D)

8. DELUGE VALVES ARE DISABLED SO THAT VALVE OPENS WITH WATER PRESSURE.
9. PROVIDE POWER LOCKOUT AT MOTOR CONTROL CENTER.
10. HV-12592 SHOULD NOT BE OPENED OR MINI-PURGE EXHAUST FAN 1-1506-B7-002-000 OPERATED UNLESS CTB PRESSURE IS (-) 0.3 TO (+) 0.3 PSIG. FD-12593 CAN BE USED AS AN ALTERNATE CTB VENT FLOW PATH WHEN CTB PRESSURE IS ≤ .3 PSIG IF 1-1506-B7-002-000 IS OUT OF SERVICE AND AIR ENTERING CHARCOAL IS ≤ 70% RELATIVE HUMIDITY AT MISH-2619.
11. USED FOR DISCHARGE OF REACTOR COOLANT VACUUM REFILL SYSTEM.
12. FOR INTERNAL PIPING REFERENCE X6AQ01A-R001-Z.

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NO.		DATE	DR	CHK	APPV	DTL
35.0	REVISED PER ABN 108013180TJ003, VER. 1.0.	04/11/13	TSL	GDM	JMR	X

CAD 1X4DB213-1 | Acad2007 | TSL – 04

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PURIFICATION & CLEAN-UP SYSTEM
SYSTEM NO. 1505, 1506, & 1508

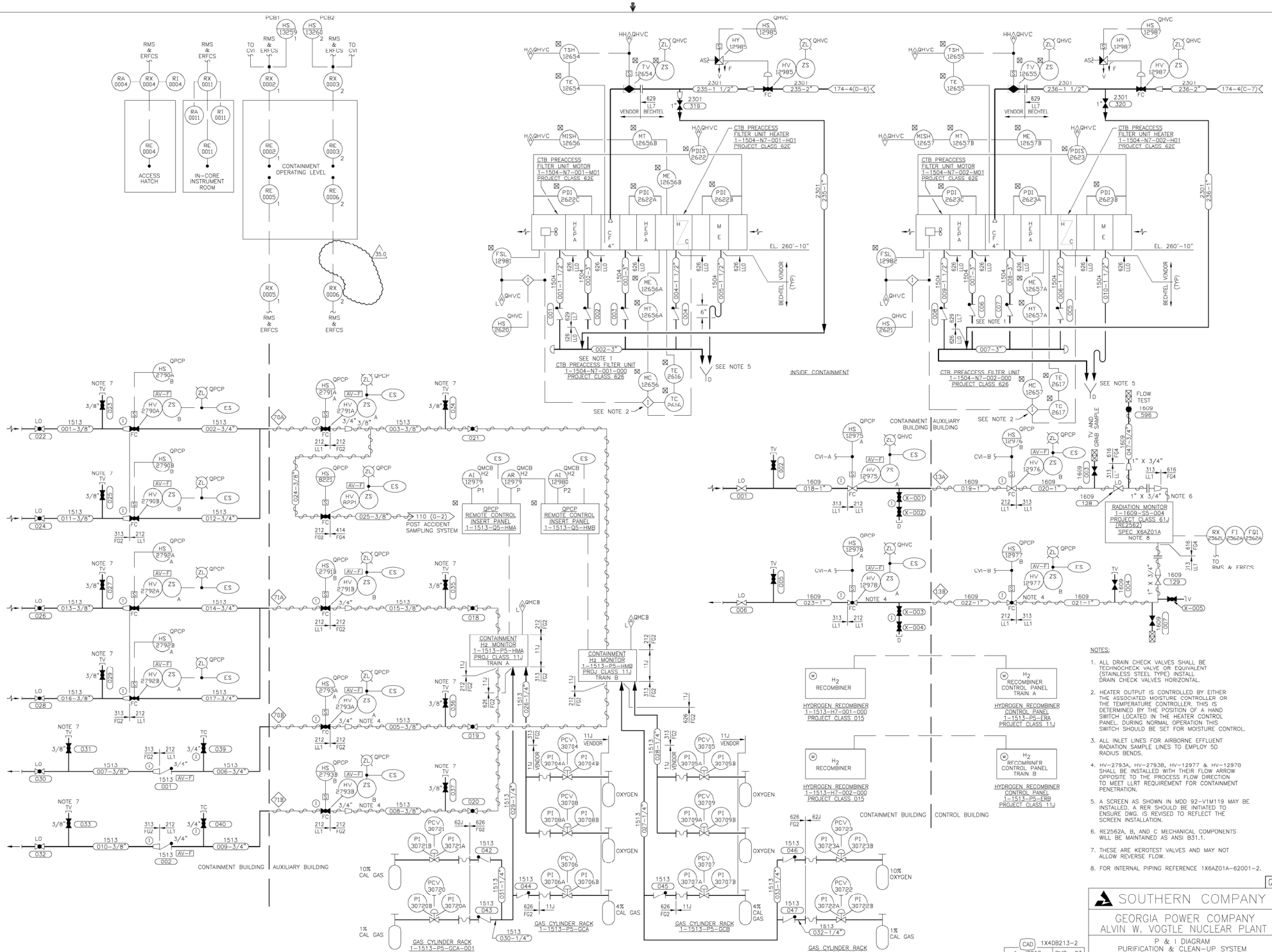
SCALE: NONE

DRAWING NO. 1X4DB213-1 VER. 35.0

JOB NO.10604

SIZE E 34044

DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL DRAIN CHECK VALVES SHALL BE TECHNOCHECK VALVE OR EQUIVALENT (STAINLESS STEEL TYPE) INSTALL DRAIN CHECK VALVES HORIZONTAL.
2. HEATER OUTPUT IS CONTROLLED BY EITHER THE ASSOCIATED MOISTURE CONTROLLER OR THE TEMPERATURE CONTROLLER. THIS IS DETERMINED BY THE POSITION OF A HAND SWITCH LOCATED IN THE HEATER CONTROL PANEL. DURING NORMAL OPERATION THIS SWITCH SHOULD BE SET FOR MOISTURE CONTROL.
3. ALL INLET LINES FOR AIRBORNE EFFLUENT RADIATION SAMPLE LINES TO EMPLOY 5D RADIUS BENDS.
4. HV-2793A, HV-2793B, HV-12977 & HV-12978 SHALL BE INSTALLED WITH THEIR FLOW ARROW OPPOSITE TO THE PROCESS FLOW DIRECTION TO MEET LLRT REQUIREMENT FOR CONTAINMENT PENETRATION.
5. A SCREEN AS SHOWN IN MDD 92-V1M119 MAY BE INSTALLED. A RER SHOULD BE INITIATED TO ENSURE DWG. IS REVISED TO REFLECT THE SCREEN INSTALLATION.
6. RE2562A, B, AND C MECHANICAL COMPONENTS WILL BE MAINTAINED AS ANSI B31.1.
7. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
8. FOR INTERNAL PIPING REFERENCE 1X6AZ01A-62001-2.

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NO.		DATE	DR	CHK	APPV	DTL
35.0	REVISED PER ABN SNC121508J002. VER. 1.0	11/20/15	SMD	SJO	WLN	

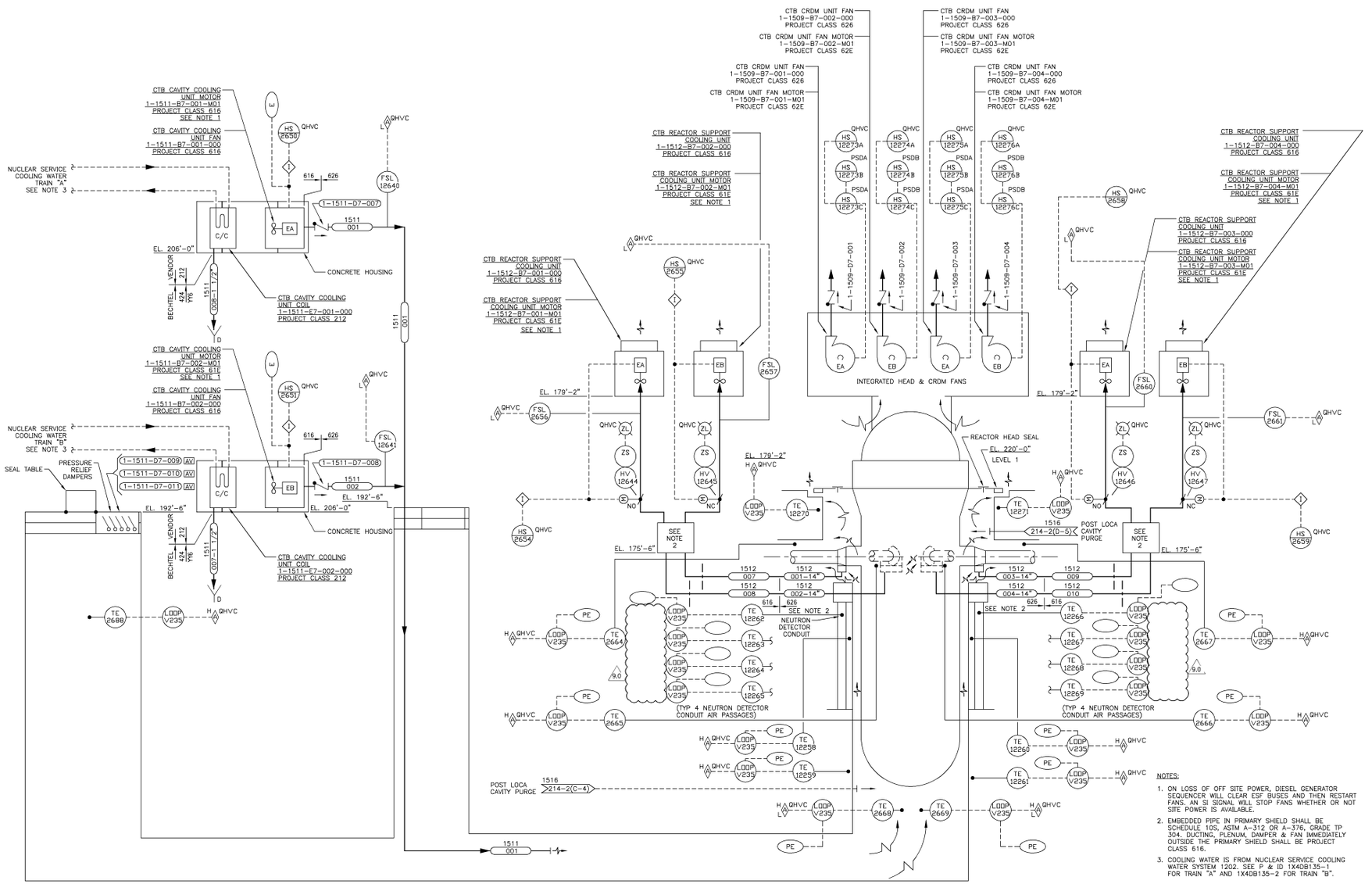
CAD 1X4DB213-2
Acad2012 SMD- 03

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PURIFICATION & CLEAN-UP SYSTEM
SYSTEM NO. 1504 & 1513

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB213-2
VER. 35.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. ON LOSS OF OFF SITE POWER, DIESEL GENERATOR SEQUENCER WILL CLEAR ESF BUSES AND THEN RESTART FANS. AN SI SIGNAL WILL STOP FANS WHETHER OR NOT SITE POWER IS AVAILABLE.
2. EMBEDDED PIPE IN PRIMARY SHIELD SHALL BE SCHEDULE 10S, ASTM A-312 OR A-376, GRADE TP 304. DUCTING, PLENUM, DAMPER & FAN IMMEDIATELY OUTSIDE THE PRIMARY SHIELD SHALL BE PROJECT CLASS 616.
3. COOLING WATER IS FROM NUCLEAR SERVICE COOLING WATER SYSTEM 1202. SEE P & ID 1X4DB135-1 FOR TRAIN "A" AND 1X4DB135-2 FOR TRAIN "B".

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
9.0	REVISED PER ABN-V00292, VER. 1.0	10-05-06	DFV	WMW	WLN	X

CAD 1X4DB214-1
ACAD2000 DFV - 02

SOUTHERN COMPANY

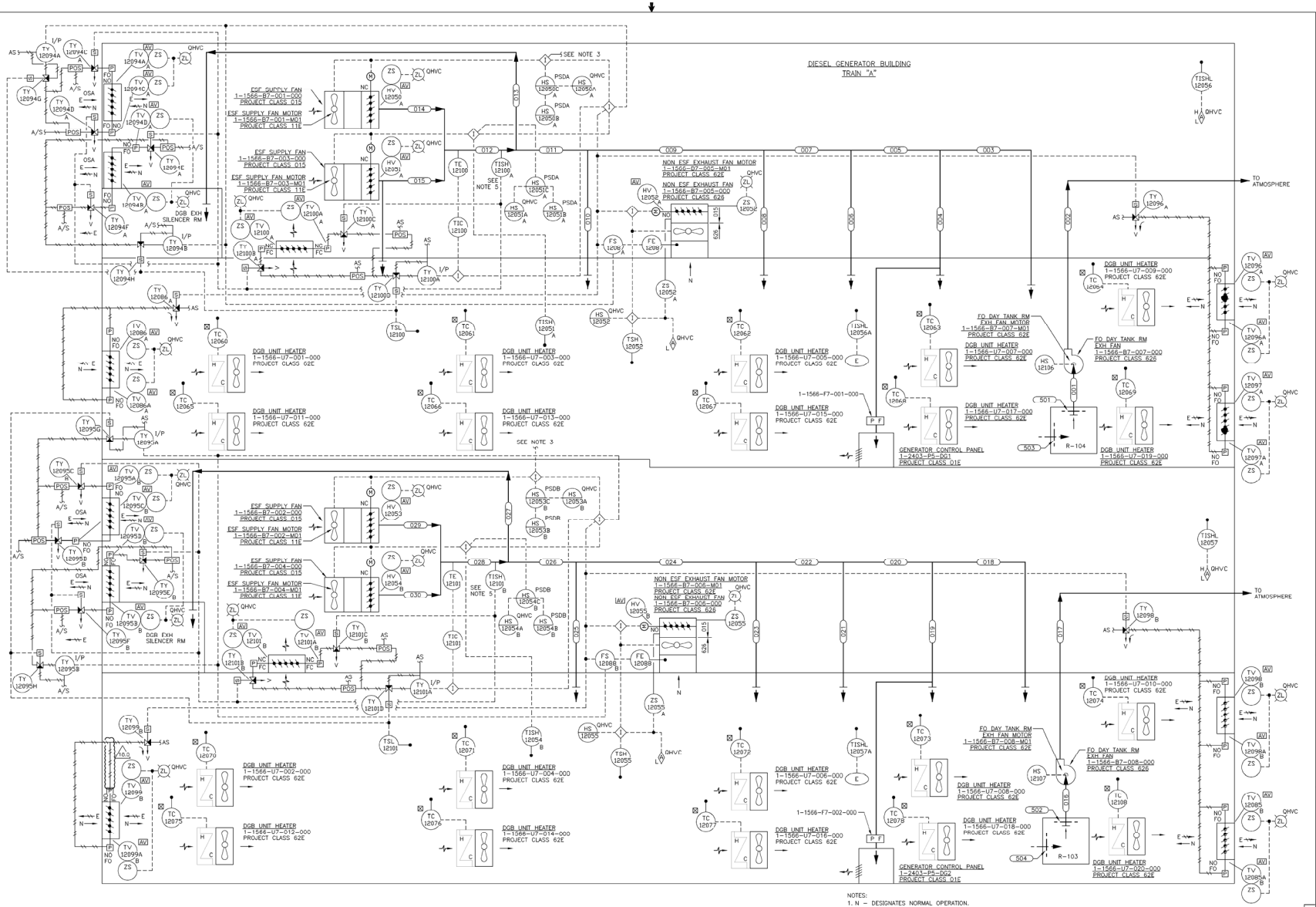
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTAINMENT, CRDM, CAVITY & REACTOR SUPPORT COOLING SYSTEM
SYSTEM NO. 1509, 1511 & 1512

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB214-1
VER. 9.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:
1. N – DESIGNATES NORMAL OPERATION.
2. E – DESIGNATES EMERGENCY OPERATION.
3. STARTS ON DIESEL GENERATOR RUNNING SIGNAL.
4. AV DENOTES ACTIVE VALVE OR DAMPER.
5. THIS INSTRUMENT IS NOT BEING USED AS AN INDICATOR.
6. OUTSIDE AIR DAMPERS TV–12094A, B, C & D, TV–12095A, B, C & D SHALL BE SET BETWEEN 35 TO 40% MAXIMUM CLOSE POSITION FROM THE FULL OPEN POSITION WHEN OPERATING IN THE MODULATION MODE. DAMPERS SHALL FAIL IN THE FAIL SAFE POSITION AS SHOWN.

CAD 1X4DB217
ACAD2000 BgT – 03

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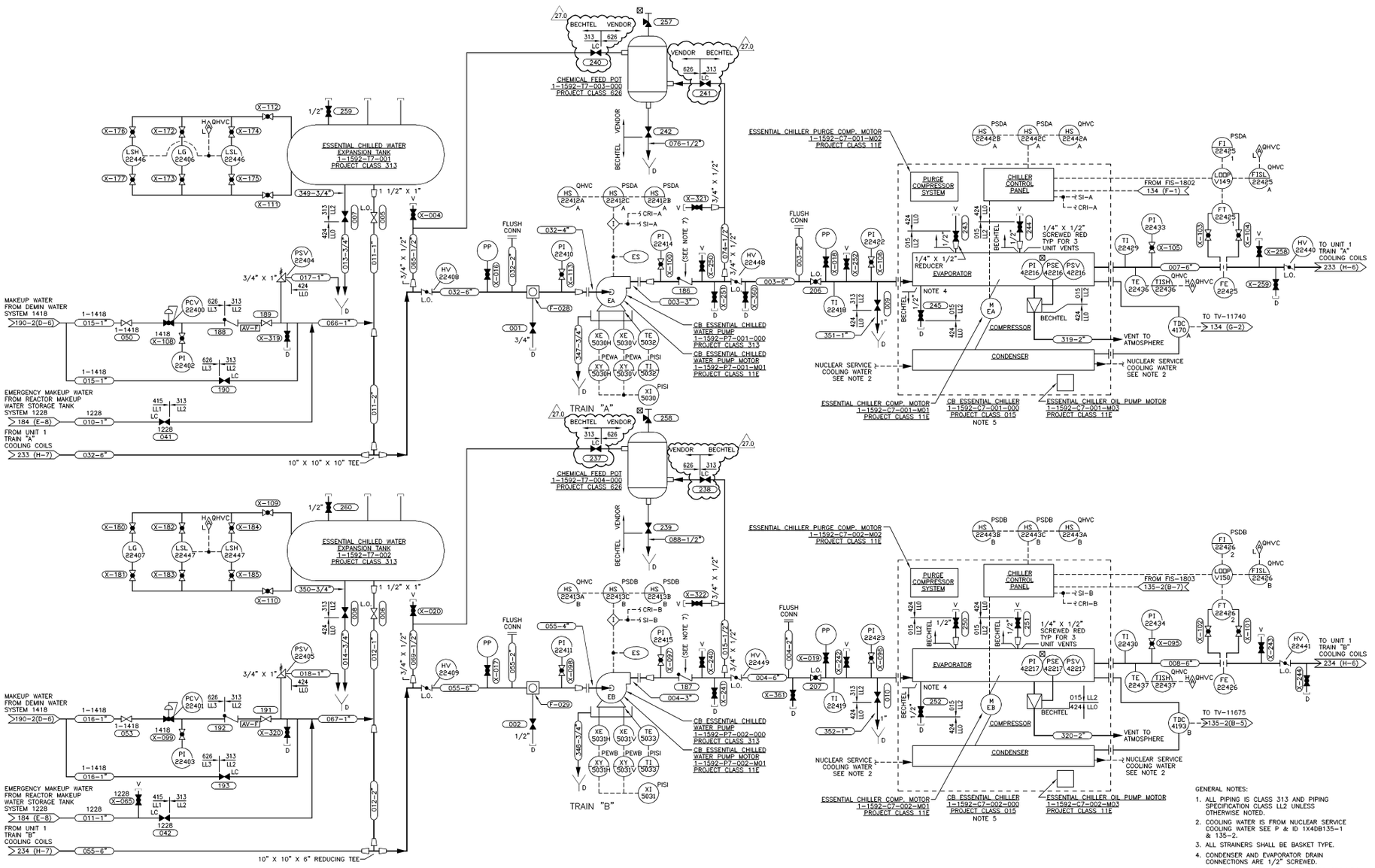
NO.		DATE	DH	CHK	APPV	DTL
18.0	REVISED PER ABN–V02004, VER.1.0	4–28–10	BgT	SRG	PML	

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
DIESEL GENERATOR BUILDING
HVAC SYSTEM NO. 1566

SCALE: NONE
DRAWING NO. 1X4DB217
VER. 18.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



GENERAL NOTES:

1. ALL PIPING IS CLASS 313 AND PIPING SPECIFICATION CLASS LL2 UNLESS OTHERWISE NOTED.
2. COOLING WATER IS FROM NUCLEAR SERVICE COOLING WATER SEE P & ID 1X4DB135-1 & 135-2.
3. ALL STRAINERS SHALL BE BASKET TYPE.
4. CONDENSER AND EVAPORATOR DRAIN CONNECTIONS ARE 1/2" SCREWED.
5. REFER TO VENDOR DRAWING AX4AJ04-104 FOR DETAILS OF SKID-MOUNTED PIPING, VALVES, AND INSTRUMENTATION.
6. REFER TO ED-85-008 FOR EQUIVALENT REPLACEMENT PART INFORMATION FOR 1592-P7-002-M01 BALL BEARING.
7. CHECK VALVES 1159U4186 & 187 INTERNAL PARTS REMOVED PER DCP 90-V1N0067-0-1.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY REALATED (ESSENTIAL) CHILLERS
UNIT 1 TRAINS A & B
SYSTEM NO. 1592

CAD 1X4DB221.DWG
ACAD2002 ELC - 03

SCALE:	NONE	DRAWING NO.	VER.
JOB NO.10604		1X4DB221	27.0

NO.		DATE	DR	CHK	APPV	DTL
27.0	REVISED PER ABN 1102110101M002, VER. 1.0	2-24-11	ELC	TSL	JMR	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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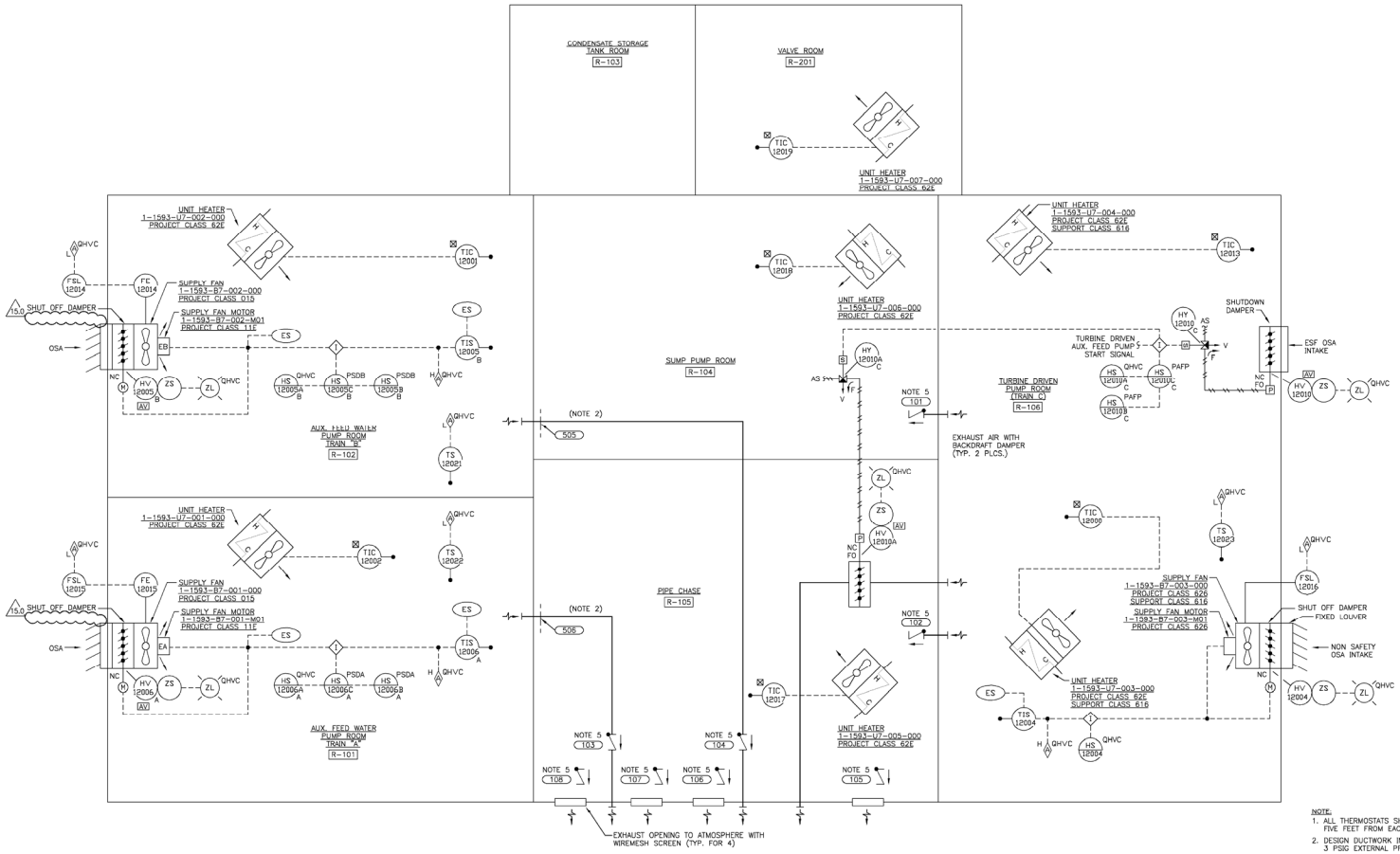

AUXILIARY FEEDWATER PUMP HOUSE

NOTE:
1. ALL THERMOSTATS SHALL BE LOCATED AT LEAST FIVE FEET FROM EACH PUMP.
2. DESIGN DUCTWORK IN PIPE CHASE TO WITHSTAND 3 PSIG EXTERNAL PRESSURE.
3. DELETED.
4. ALL DUCTWORK SHALL BE PROJECT CLASS 01S.
5. BACKDRAFT DAMPER #101, 102, 103, 104, 105, 106, 107 AND 108 HAVE BLADES REMOVED.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY FEEDWATER PUMP HOUSE
HEATING AND VENTILATING
SYSTEM NO. 15.93

CAD 1X4DB227
ACAD2002 BgT - 03

SCALE: NONE
DRAWING NO. 1X4DB227
VER. 15.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

NO.		DATE	DR	CHK	APPV	UTL
15.0	REVISED PER ABN-V00466, VER.1.0	1-19-07	BgT	MWD	WLN	

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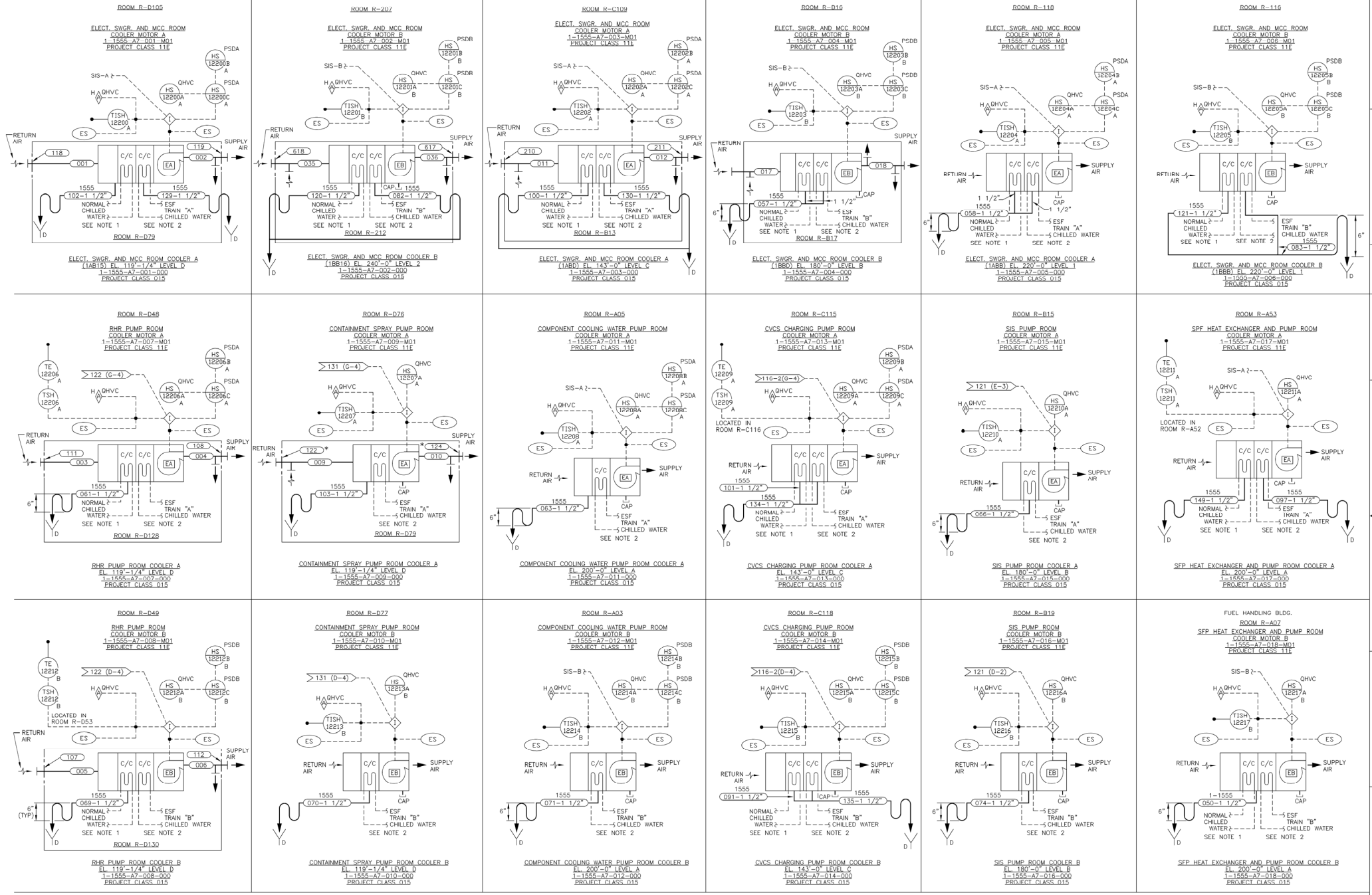

NOTES:
1. SEE P & ID'S AX4DB231 & 232.
2. SEE P & ID'S (1X4DB233) & 234. 13.0
3. ALL DUCT IS 015.
4. ALL DRAIN LINES TO BE PROJECT CLASS 626, PIPING CLASS LLO.
5. ✳ DERATED FIRE DAMPER.

CAD 1X4DB228
ACAD2007 | LPM- 07

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NO.		DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN-V02202, VER. 1.0	6-17-10	LPM	LCF	DEW	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ENGINEERED SAFETY FEATURE
ROOM COOLERS
SYSTEM 1555

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB228	13.0

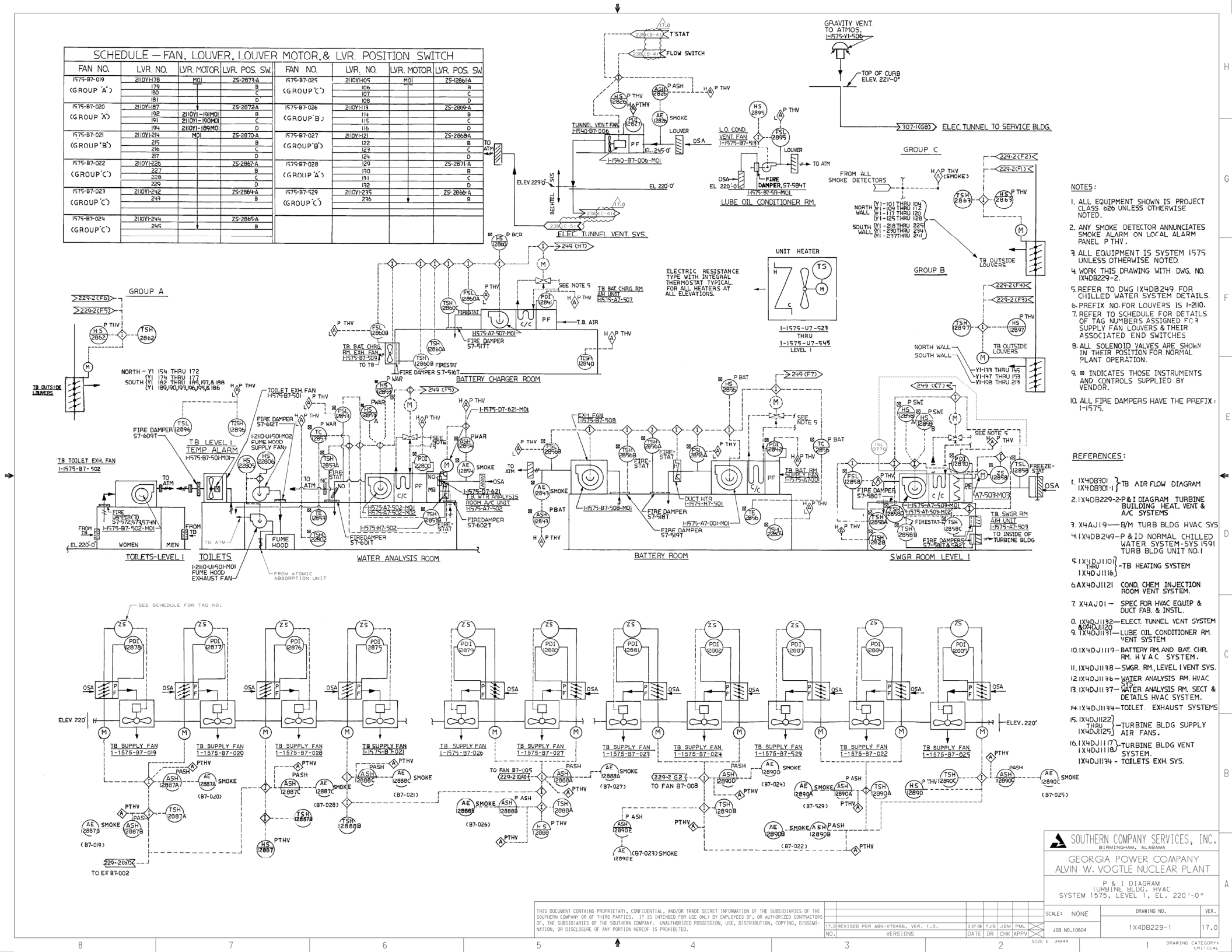
SIZE E 34X44

DRAWING CATEGORY: CRITICAL

SCHEDULE — FAN, LOUVER, LOUVER MOTOR, & LVR. POSITION SWITCH

FAN NO.	LVR. NO.	LVR. MOTOR	LVR. POS. SW.
1575-B7-019 (GROUP 'A')	2110YI-178	MOI	ZS-2873-A
	179		B
	180		C
	181		D
1575-B7-020 (GROUP 'A')	2110YI-187		ZS-2872-A
	192	2110YI-191MOI	B
	193	2110YI-190MOI	C
	194	2110YI-189MOI	D
1575-B7-021 (GROUP 'B')	2110YI-214	MOI	ZS-2870-A
	215		B
	216		C
	217		D
1575-B7-022 (GROUP 'C')	2110YI-226		ZS-2867-A
	227		B
	228		C
	229		D
1575-B7-023 (GROUP 'C')	2110YI-242		ZS-2864-A
	243		B
1575-B7-024 (GROUP 'C')	2110YI-244		ZS-2865-A
	245		B

FAN NO.	LVR. NO.	LVR. MOTOR	LVR. POS. SW.
1575-B7-025 (GROUP 'C')	2110YI-105	MOI	ZS-12861-A
	106		B
	107		C
	108		D
1575-B7-026 (GROUP 'B')	2110YI-113		ZS-2869-A
	114		B
	115		C
	116		D
1575-B7-027 (GROUP 'B')	2110YI-121		ZS-2868-A
	122		B
	123		C
	124		D
1575-B7-028 (GROUP 'A')	129		ZS-2871-A
	130		B
	131		C
	132		D
1575-B7-529 (GROUP 'C')	2110YI-235		ZS-2866-A
	236		B



NOTES:

1. ALL EQUIPMENT SHOWN IS PROJECT CLASS 026 UNLESS OTHERWISE NOTED.
2. ANY SMOKE DETECTOR ANNUNCIATES SMOKE ALARM ON LOCAL ALARM PANEL P THV.
3. ALL EQUIPMENT IS SYSTEM 1575 UNLESS OTHERWISE NOTED.
4. WORK THIS DRAWING WITH DWG. NO. 1X4DB229-2.
5. REFER TO DWG 1X4DB249 FOR CHILLED WATER SYSTEM DETAILS.
6. PREFIX NO. FOR LOUVERS IS 1-2110.
7. REFER TO SCHEDULE FOR DETAILS OF TAG NUMBERS ASSIGNED FOR SUPPLY FAN LOUVERS & THEIR ASSOCIATED END SWITCHES
8. ALL SOLENOID VALVES ARE SHOWN IN THEIR POSITION FOR NORMAL PLANT OPERATION.
9. ⊠ INDICATES THOSE INSTRUMENTS AND CONTROLS SUPPLIED BY VENDOR.
10. ALL FIRE DAMPERS HAVE THE PREFIX: I-1575.

REFERENCES:

1. 1X4DB301 / 1X4DB301-1 } -TB AIR FLOW DIAGRAM
2. 1X4DB229-2-P&I DIAGRAM TURBINE BUILDING HEAT, VENT & A/C SYSTEMS
3. X4AJ19 — B/M TURB BLDG HVAC SYS
4. 1X4DB249 — P & ID NORMAL CHILLED WATER SYSTEM-SYS 1591 TURB BLDG UNIT NO. 1
5. 1X4DJ1101 THRU 1X4DJ1116 } -TB HEATING SYSTEM
6. AX4DJ1121 COND. CHEM INJECTION ROOM VENT SYSTEM.
7. X4AJ01 — SPEC FOR HVAC EQUIP & DUCT FAB. & INSTL.
8. 1X4DJ1132 — ELECT. TUNNEL VENT SYSTEM & 1X4DJ1120
9. 1X4DJ1131 — LUBE OIL CONDITIONER RM. VENT SYSTEM
10. 1X4DJ1119 — BATTERY RM. AND BAT. CHR. RM. H V A C SYSTEM.
11. 1X4DJ1138 — SWGR. RM., LEVEL I VENT SYS.
12. 1X4DJ1136 — WATER ANALYSIS RM. HVAC SYS.
13. 1X4DJ1137 — WATER ANALYSIS RM. SECT & DETAILS HVAC SYSTEM.
14. 1X4DJ1134 — TOILET. EXHAUST SYSTEMS
15. 1X4DJ1122 THRU 1X4DJ1125 } -TURBINE BLDG SUPPLY AIR FANS.
16. 1X4DJ1117 / 1X4DJ1118 } -TURBINE BLDG VENT SYSTEM.
    1X4DJ1134 - TOILETS EXH. SYS.

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NO.	VERSIONS	DATE	DR	CHK	APPV
17.0	REVISED PER ABN-V70486, VER. 1.0.	3-27-08	TJS	JEM	PML

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

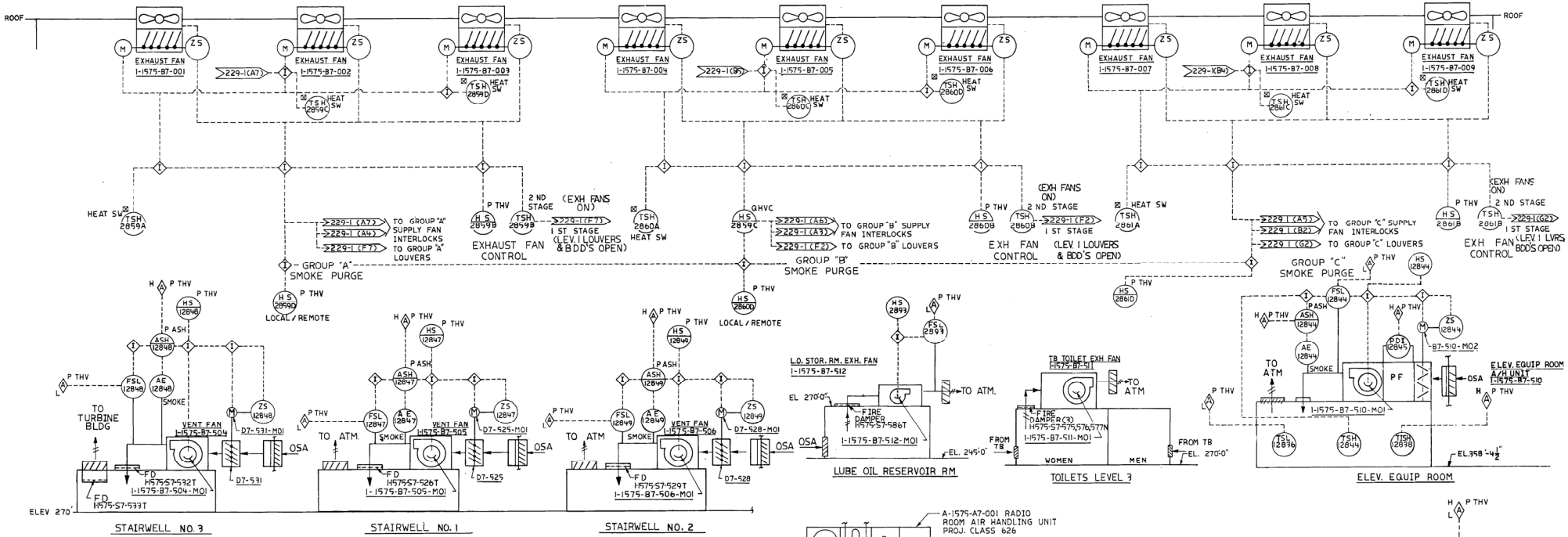
P & I DIAGRAM
TURBINE BLDG. HVAC
SYSTEM 1575, LEVEL 1, EL. 220'-0"

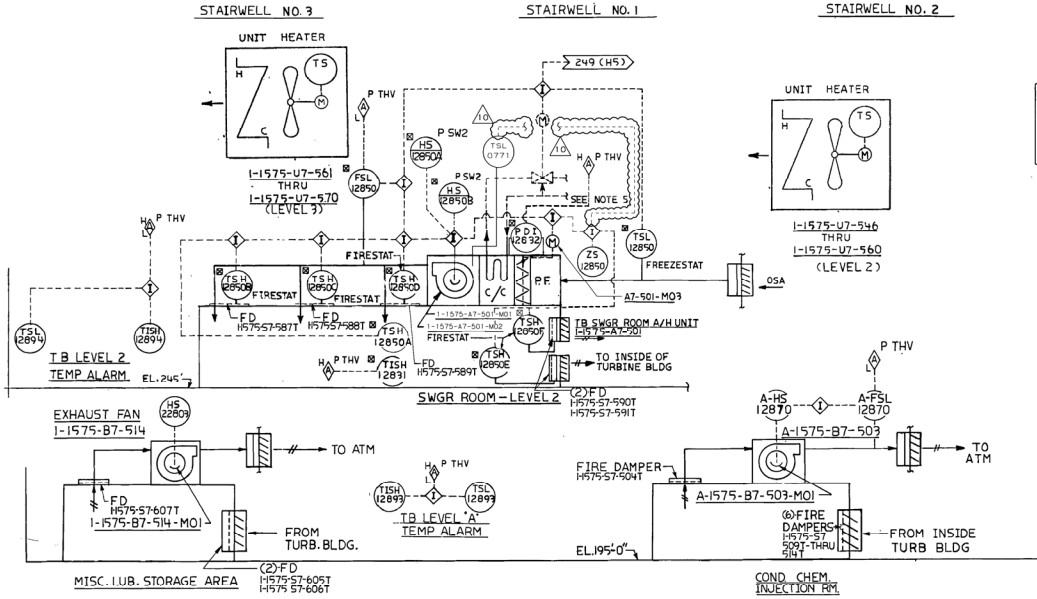
SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB229-1	17.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

SCHEDULE - FAN, DAMPER, DAMPER MOTOR, AND DAMPER POSITION SWITCH											
FAN NO.	DMPR. NO.	DMPR. MTR.	POS. SW.	FAN NO.	DMPR. NO.	DMPR. MTR.	POS. SW.	FAN NO.	DMPR. NO.	DMPR. MTR.	POS. SW.
1-1575-B7-001	D7-536	M01	ZS 2859 A	1-1575-B7-004	D7-548	M01	ZS 2860 A	1-1575-B7-007	D7-560	M01	ZS 2861 A
	D7-537	M01	ZS 2859 B		D7-549	M01	ZS 2860 B		D7-561	M01	ZS 2861 B
	D7-538	M01	ZS 2859 C		D7-550	M01	ZS 2860 C		D7-562	M01	ZS 2861 C
	D7-539	M01	ZS 2859 D		D7-551	M01	ZS 2860 D		D7-563	M01	ZS 2861 D
1-1575-B7-002	D7-540	M01	ZS 2859 E		D7-552	M01	ZS 2860 E		D7-564	M01	ZS 2861 E
	D7-541	M01	ZS 2859 F	1-1575-B7-005	D7-553	M01	ZS 2860 F	1-1575-B7-008	D7-565	M01	ZS 2861 F
	D7-542	M01	ZS 2859 G		D7-554	M01	ZS 2860 G		D7-566	M01	ZS 2861 G
1-1575-B7-003	D7-543	M01	ZS 2859 J		D7-555	M01	ZS 2860 J		D7-569	M01	ZS 2861 J
	D7-545	M01	ZS 2859 K	1-1575-B7-006	D7-556	M01	ZS 2860 K	1-1575-B7-009	D7-570	M01	ZS 2861 K
	D7-546	M01	ZS 2859 L		D7-559	M01	ZS 2860 L		D7-571	M01	ZS 2861 L





NOTES:

1. WORK THIS DWG WITH DWG 1X4DB229-1.
2. ☒ INDICATES THOSE INSTRUMENTS & CONTROLS TO BE FURNISHED BY VENDOR.
3. ALL SOLENOID VALVES ARE SHOWN IN THEIR POSITION FOR NORMAL PLANT OPERATION.
4. PREFIX FOR EQUIP. IS 1-1575 UNLESS OTHERWISE NOTED.
5. REFER TO SCHEDULE FOR TAG NO.'S ASSIGNED TO EXH. FAN DAMPERS AND THEIR ASSOCIATED POSITION SWITCHES.

REFERENCES

1X4DJ1133	TB FILTER UNIT
1X4DJ1135	TB SWGR ROOM LEV 2 VENT SYSTEM.
1X4DJ1126	ELEV. EQUIP. ROOM HEAT & VENT SYSTEM.
1X4DJ1134	TB TOILETS EXHAUST SYSTEM.
1X4DJ1130	TB LUBE OIL RESERVOIR VENT SYSTEM.
1X4DJ1127, 1X4DJ1128, 1X4DJ1129	TB STAIRWELLS VENT SYSTEMS.
1X4DB174-2	P&ID FIRE PROTECTION WATER SYSTEM.
AX4DJ1121, AX4DJ1121-1	COND. CHEM. INJ. ROOM VENT. SYSTEM

GEORGIA REGISTERED PROFESSIONAL WILLIAM T. GAMMILL No. 11173

William T. Gammill P.E.
GEORGIA NO. 11173

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM TURB BLDG. HVAC SYSTEM 1575, LEVELS A, 195'-0", 2, 245'-0", 3, 270'-0" & ROOF, 295'-0"

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB229-2	10

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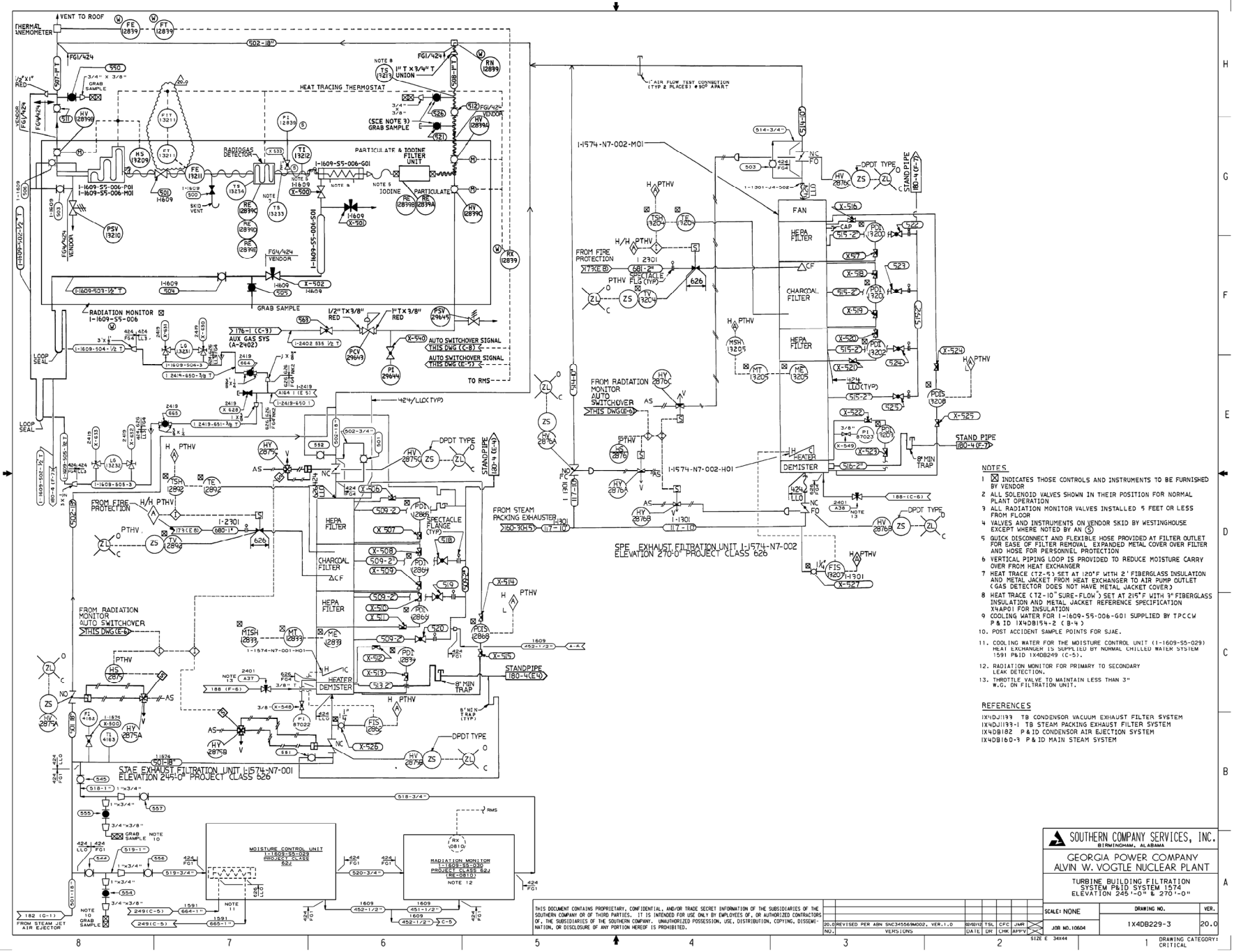
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-50839	06/04/99	KDL	GLB	MWD	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SIZE E 34X44

CAD NAME 1-40B2292



NOTES

1. ☒ INDICATES THOSE CONTROLS AND INSTRUMENTS TO BE FURNISHED BY VENDOR
2. ALL SOLENOID VALVES SHOWN IN THEIR POSITION FOR NORMAL PLANT OPERATION
3. ALL RADIATION MONITOR VALVES INSTALLED 5 FEET OR LESS FROM FLOOR
4. VALVES AND INSTRUMENTS ON VENDOR SKID BY WESTINGHOUSE EXCEPT WHERE NOTED BY AN Ⓢ
5. QUICK DISCONNECT AND FLEXIBLE HOSE PROVIDED AT FILTER OUTLET FOR EASE OF FILTER REMOVAL. EXPANDED METAL COVER OVER FILTER AND HOSE FOR PERSONNEL PROTECTION
6. VERTICAL PIPING LOOP IS PROVIDED TO REDUCE MOISTURE CARRY OVER FROM HEAT EXCHANGER
7. HEAT TRACE (TZ-5) SET AT 120°F WITH 2" FIBERGLASS INSULATION AND METAL JACKET FROM HEAT EXCHANGER TO AIR PUMP OUTLET (GAS DETECTOR DOES NOT HAVE METAL JACKET COVER)
8. HEAT TRACE (TZ-10 "SURE-FLOW") SET AT 215°F WITH 3" FIBERGLASS INSULATION AND METAL JACKET REFERENCE SPECIFICATION X4AP01 FOR INSULATION
9. COOLING WATER FOR I-1609-S5-006-G01 SUPPLIED BY TPCCW P&ID 1X4DB154-2 (B-4)
10. POST ACCIDENT SAMPLE POINTS FOR SJAE.
11. COOLING WATER FOR THE MOISTURE CONTROL UNIT (I-1609-S5-029) HEAT EXCHANGER IS SUPPLIED BY NORMAL CHILLED WATER SYSTEM 1591 P&ID 1X4DB249 (C-5).
12. RADIATION MONITOR FOR PRIMARY TO SECONDARY LEAK DETECTION.
13. THROTTLE VALVE TO MAINTAIN LESS THAN 3" W.G. ON FILTRATION UNIT.

REFERENCES

1X4DJ1139 TB CONDENSOR VACUUM EXHAUST FILTER SYSTEM
1X4DJ1193-1 TB STEAM PACKING EXHAUST FILTER SYSTEM
1X4DB182 P&ID CONDENSOR AIR EJECTION SYSTEM
1X4DB160-3 P&ID MAIN STEAM SYSTEM

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NO.	VERSIONS	DATE	DR	CHK	APP'V
20.0	REVISED PER ABN SNC345569M002, VER.1.0	02/02/12	TSL	CFC	JMR

SIZE E 34x44

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

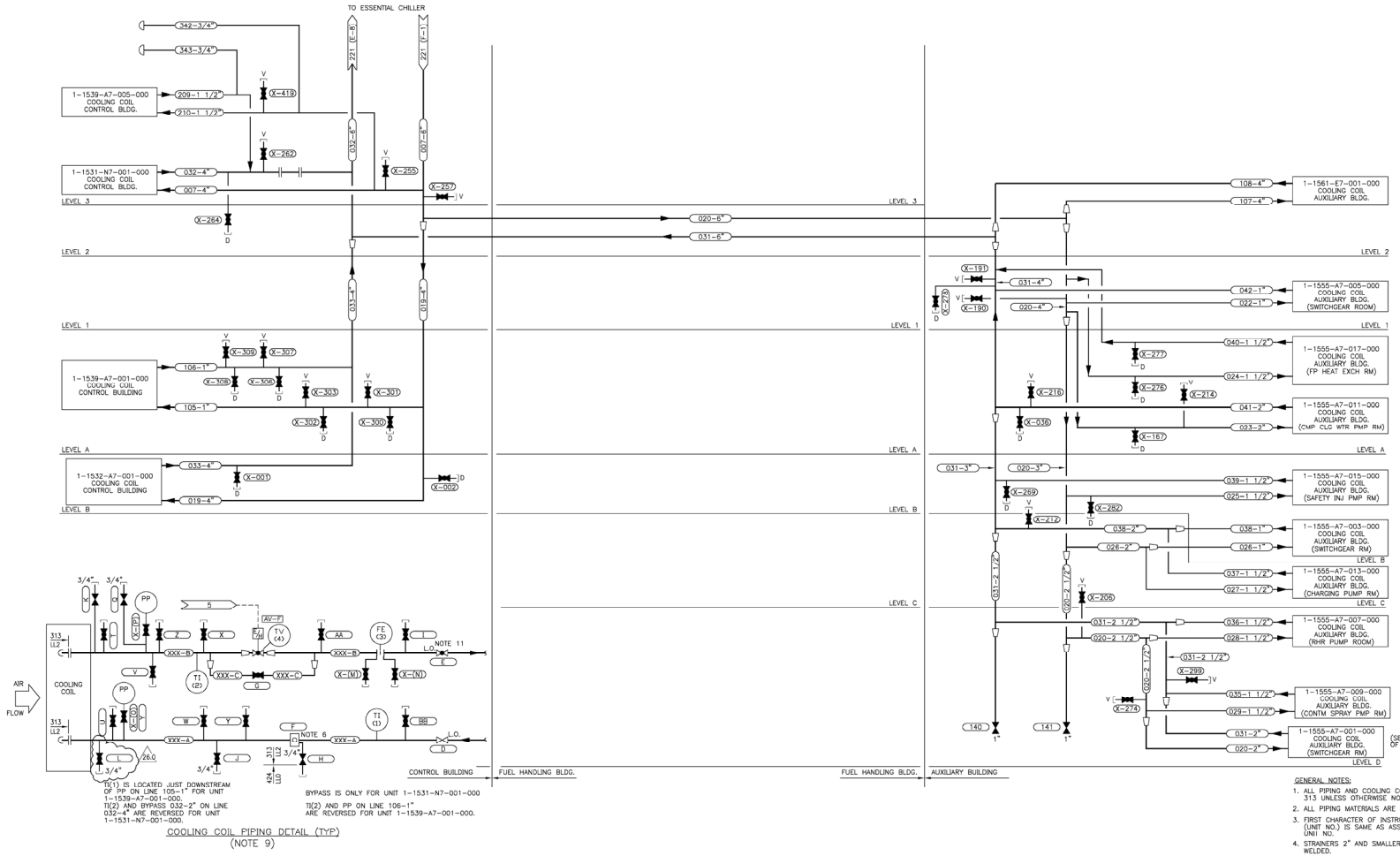
TURBINE BUILDING FILTRATION SYSTEM P&ID SYSTEM 1574
ELEVATION 245'-0" & 270'-0"

SCALE: NONE

DRAWING NO.	VER.
1X4DB229-3	20.0

JOB NO. 10604

DRAWING CATEGORY: CRITICAL



COOLING COIL PIPING DETAIL (TYP)
(NOTE 9)

CHILLED WATER COIL REFERENCE TABLE

EQUIPMENT NO.	INSTRUMENTATION NUMBER					CONTROL VALVE		PIPE & VALVE TAG SEQUENCE NUMBERS AND SIZE IN INCHES																								EQUIP AREA NO	INSTRUMENT ROOT VALVE TAG NUMBER							ROOT VALVES: V = VENT & D = DRAIN (SEE NOTE 10)									
								A		B		C		D		E		F		G		H	J	K	L								M	N	O	P	Q	R	S									
	TI (1)	TI (2)	FE (3)			TV (4)	P & ID (5)	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	NO.	NO.	NO.							NO.	NO.	NO.	NO.	NO.	NO.	NO.	I	T	U	V	W	X	Y	Z	AA	BB	
1-1531-N7-001-000	22132	22133	22134		—	12124	206-3 (H-3)	007	4	032	4	032	2	021	4	054	4	003	4	043	2	032	—	—	—							12B3	043	044	069	082	—	—	—		D-265	D-261	V-263	V-256		D-260				
1-1532-A7-001-000	22144	22145	22146					019	4	033	4			025	4	058	4	007	4	—	—	036	—	—	D-019							12DB	045	046	070	083	—	—	—				V-013			V-015	D-209			
1-1539-A7-001-000	21988	21989	21990	—	—	—	—	105	1	106	1	—	—	150	1	151	1	—	—	—	—	—	—	—	—							12AA	047	048	071	084	—	—	—				V-311	V-305		D-304	D-310			
1-1555-A7-001-000	22156	22157	22158	—	—	—	—	020	2	031	2	—	—	154	2	155	2	—	—	—	—	—	040	111	051							13GD	032	031	042	033	—	—	—					V-400		D-034				
1-1555-A7-003-000	22150	22151	22152	—	—	—	—	026	1	038	1	—	—	156	1	157	1	—	—	—	—	—	—	—	049							13GC	051	052	073	086	—	—	—		D-279					V-225	V-231			
1-1555-A7-005-000	22135	22136	22137	—	—	—	—	022	1	042	1	—	—	158	1	159	1	—	—	—	—	—	—	—	—							13E1	054	053	074	087	—	—	—								D-317			
1-1555-A7-007-000	22162	22163	22164	—	—	—	—	028	1 1/2	036	1 1/2	—	—	031	1 1/2	064	1 1/2	—	—	—	—	053	—	—	—							13ED	055	056	088	075	113	—	—											
1-1555-A7-009-000	22159	22160	22161	—	—	—	—	029	1 1/2	035	1 1/2	—	—	030	1 1/2	063	1 1/2	—	—	—	—	052	112	—	—							13GD	058	057	076	038	—	—	—								D-037			
1-1555-A7-011-000	22138	22139	22140	—	—	—	—	023	2	041	2	—	—	160	2	161	2	—	—	—	—	018	012	—	—							13HA	060	059	077	090	—	—	—	D-035										
1-1555-A7-013-000	21985	21986	21987	—	—	—	—	027	1 1/2	037	1 1/2	—	—	028	1 1/2	061	1 1/2	—	—	—	—	—	110	050	13HC							061	062	078	091	—	—	—		D-215	V-271				D-403				
1-1555-A7-015-000	22147	22148	22149	—	—	—	—	025	1 1/2	039	1 1/2	—	—	026	1 1/2	059	1 1/2	—	—	—	—	—	015	—	13H8							063	064	079	092	—	—	—											
1-1555-A7-017-000	22141	22142	22143	—	—	—	—	024	1 1/2	040	1 1/2	—	—	024	1 1/2	057	1 1/2	—	—	—	—	—	—	—	13BA							065	066	080	093	—	—	—											
1-1561-E7-001-000	22220	22221	22222	—	—	—	—	107	4	108	4	—	—	162	4	163	4	—	—	—	—	—	—	—	13E2							067	068	081	094	—	—	—								V-312	V-313		
1-1539-A7-005-000	22290	22291	22292	—	—	—	—	210	1 1/2	209	1 1/2	—	—	221	1 1/2	222	1 1/2	—	—	—	—	—	—	—	—							356	357	368	359	—	—	—											

GENERAL NOTES:

1. ALL PIPING AND COOLING COIL IS PROJECT CLASS 313 UNLESS OTHERWISE NOTED.
2. ALL PIPING MATERIALS ARE CLASS LL2.
3. FIRST CHARACTER OF INSTRUMENTATION TAG NO. (UNIT NO.) IS SAME AS ASSOCIATED EQUIPMENT'S UNIT NO.
4. STRAINERS 2" AND SMALLER SHALL BE SOCKET WELDED.
5. DELETED.
6. STRAINER SHALL BE BASKET TYPE.
7. DELETED.
8. VENT & DRAIN LINES SHALL BE 1" FOR PIPE HEADERS 1" & LARGER. FOR PIPE HEADERS 3/4" AND SMALLER, VENTS AND DRAINS SHALL BE THE SAME SIZE AS THE HEADER.
9. BREAKOUT FLANGES REQUIRED FOR COOLING COIL REMOVAL.
10. ALL ROOT VALVE NUMBERS SHALL HAVE THE PREFIX "X".
11. THE OUTLET ISOLATION VALVE FOR EACH COOLING COIL SHALL BE THROTTLED AS REQUIRED TO PROVIDE DESIGN FLOWRATES PER FLOW DIAGRAM 1X4DB271.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ESSENTIAL CHILLED WATER COOLING COILS
UNIT 1, TRAIN A
SYSTEM NO. 1592

CAD 1X4DB233
ACAD2002 CYN- 02

SCALE: NONE
DRAWING NO. 1X4DB233
VER. 26.0
JOB NO.10604
SIZE E 34X44

26.0 REVISED PER ABN-V70044, VER. 1.0 — 6-12-06 CYN LCF LWW X

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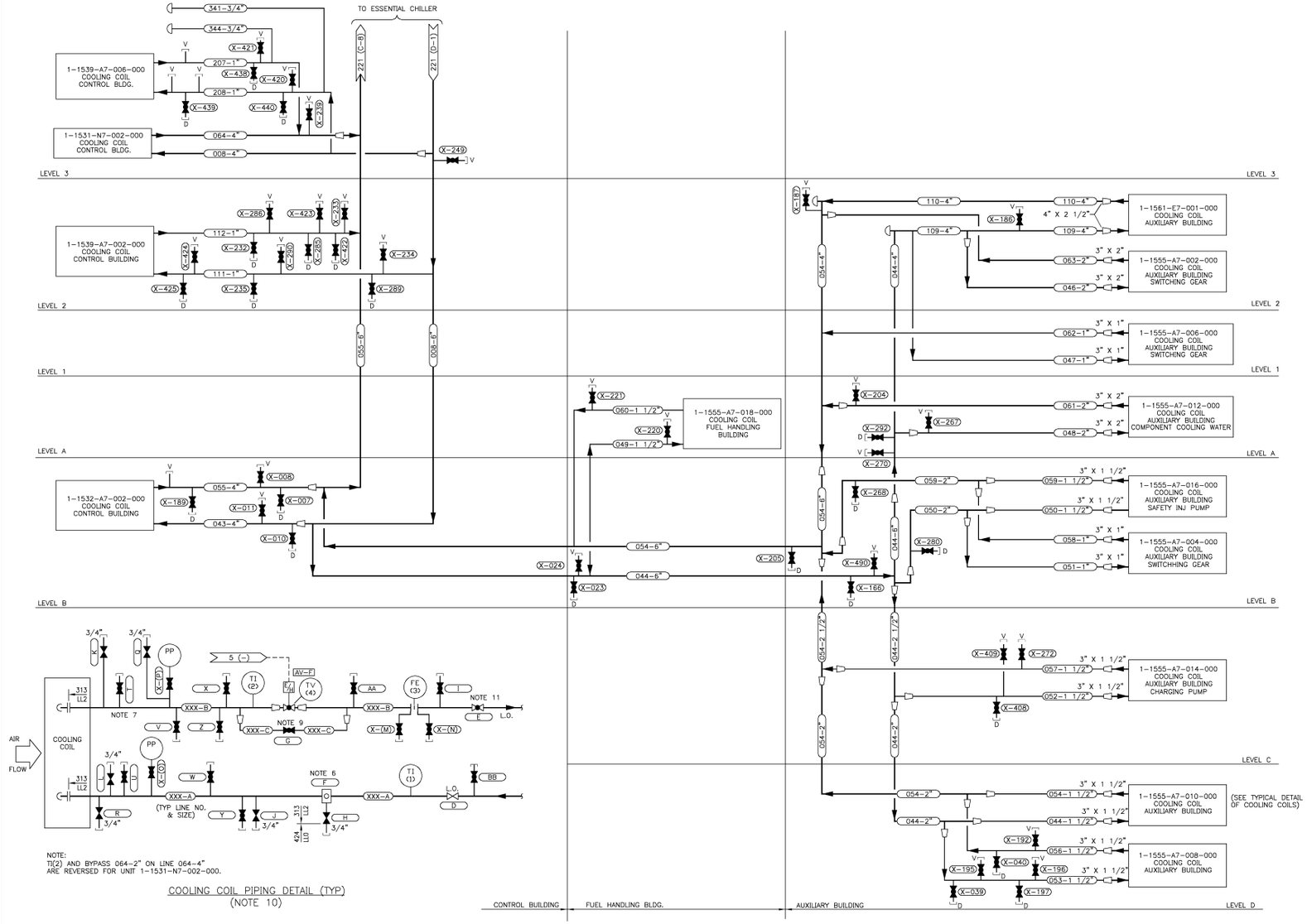
DRAWING CATEGORY: CRITICAL



NOTE:
TI(2) AND BYPASS 064-2" ON LINE 064-4" ARE REVERSED FOR UNIT 1-1531-N7-002-000.

COOLING COIL PIPING DETAIL (TYP)
(NOTE 10)

CHILLED WATER COIL REFERENCE TABLE

EQUIPMENT NO.	INSTRUMENTATION NUMBER					CONTROL VALVE		PIPE & VALVE TAG SEQUENCE NUMBERS AND SIZE IN INCHES																			EQUIP AREA NO	INSTRUMENT ROOT VALVE TAG NUMBER										ROOT VALVES: V = VENT ; D = DRAIN									
								A		B		C		D		E		F		G		H	J	K	L		M	N	O	P	Q	R	S													
	TI (1)	TI (2)	FE (3)			TV (4)	P & ID (5)	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	SIZE	NO.	NO.	NO.	NO.		NO.	NO.	NO.	NO.	NO.	NO.	NO.	I	T	U	V	W	X	Y	Z	AA	BB
1-1531-N7-002-000	22168	22169	22170	—		12125	206-3 (D-3)	008	4	064	4	064	2	065	4	099	4	014	4	088	2	077	—	—	—		12B3	114	115	318	153	—	—	—		D-236	D-248		V-247		D-246	V-237	D-238	
1-1532-A7-002-000	22183	22184	22185	—		—	—	043	4	055	4	—	—	070	4	104	4	019	4	—	—	082	—	—	—		12DB	116	117	141	154	—	116	—				V-210	V-207				D-208	
1-1539-A7-002-000	21982	21983	21984	—	—	—	—	111	1	112	1	—	—	167	1	168	1	—	—	—	—	—	—	—	—		12A2	118	119	142	155	—	170	—				V-287	V-288					
1-1555-A7-002-000	22171	22172	22173	—	—	—	—	046	2	063	2	—	—	172	2	173	2	—	—	—	—	—	—	—	—		13F2	120	121	143	156	—	—	—										
1-1555-A7-004-000	22189	22190	22191	—	—	—	—	051	1	058	1	—	—	174	1	175	1	—	—	—	—	—	—	—	—		13HB	122	123	144	157	—	—	—		V-284	V-281							
1-1555-A7-006-000	22174	22175	22176	—	—	—	—	047	1	062	1	—	—	176	1	177	1	—	—	—	—	—	—	—	—		13F1	125	124	145	158	—	—	—		D-314								D-315
1-1555-A7-008-000	22198	22199	22200	—	—	—	—	053	1 1/2	056	1 1/2	—	—	075	1 1/2	109	1 1/2	—	—	—	—	098	—	—	—		13ED	126	127	146	159	127	—	—										
1-1555-A7-010-000	22195	22196	22197	—	—	—	—	044	1 1/2	054	1 1/2	—	—	074	1 1/2	108	1 1/2	—	—	—	—	097	126	—	—		13GD	025	026	147	028	—	—	—		V-412	V-411					D-401		
1-1555-A7-012-000	22177	22178	22179	—	—	—	—	048	2	061	2	—	—	178	2	179	2	—	—	—	—	115	120	—	—		13HA	130	131	148	161	—	—	—									D-203	
1-1555-A7-014-000	22192	22193	22194	—	—	—	—	052	1 1/2	057	1 1/2	—	—	073	1 1/2	107	1 1/2	—	—	—	—	—	125	215	—		13HC	133	132	149	162	—	096	—				D-213			D-407			
1-1555-A7-016-000	22186	22187	22188	—	—	—	—	050	1 1/2	059	1 1/2	—	—	071	1 1/2	105	1 1/2	—	—	—	—	—	—	—	—		13HB	134	135	150	163	—	—	—										
1-1555-A7-018-000	22180	22181	22182	—	—	—	—	049	1 1/2	060	1 1/2	—	—	069	1 1/2	103	1 1/2	—	—	—	—	—	121	—	—		13DA	137	136	151	164	—	—	—				D-441						
1-1561-E7-001-000	22223	22224	22225	—	—	—	—	109	4	110	4	—	—	180	4	181	4	—	—	—	—	—	—	—	—		13E2	139	138	152	165	—	—	—										
1-1539-A7-006-000	22295	22296	22297	—	—	—	—	208	1	207	1	—	—	225	1	226	1	—	—	—	—	—	—	—	—		2B1	352	353	354	355	—	—	—										

NOTES:
1. ALL PIPING IS PROJECT CLASS 313 UNLESS OTHERWISE NOTED.
2. ALL PIPING MATERIALS ARE CLASS LL2.
3. FIRST CHARACTER OF INSTRUMENTATION TAG NO. (UNIT NO.) IS SAME AS ASSOCIATED EQUIPMENT'S UNIT NO.
4. STRAINERS 2" AND SMALLER SHALL BE SOCKET WELDED.
5. NOTE DELETED.
6. STRAINER SHALL BE BASKET TYPE.
7. NOTE DELETED.
8. VENT & DRAIN LINES SHALL BE 1" FOR PIPE HEADERS 1" AND LARGER. FOR PIPE HEADERS 3/4" AND SMALLER, VENTS AND DRAINS SHALL BE THE SAME SIZE AS THE HEADER.
9. BYPASS LINE SHALL BE PROVIDED ON UNIT 1-1531-N7-002-000 ONLY.
10. BREAKOUT FLANGES REQUIRED FOR COOLING COIL REMOVAL.
11. THE OUTLET ISOLATION VALVE FOR EACH COOLING COIL SHALL BE THROTTLED AS REQUIRED TO PROVIDE DESIGN FLOWRATES PER FLOW DIAGRAM 1X4DB272.

Q

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ESSENTIAL CHILLED WATER COOLING COILS
UNIT 1, TRAIN B
SYSTEM NO. 1592

CAD 1X4DB234
ACADZ002 CYN - 01

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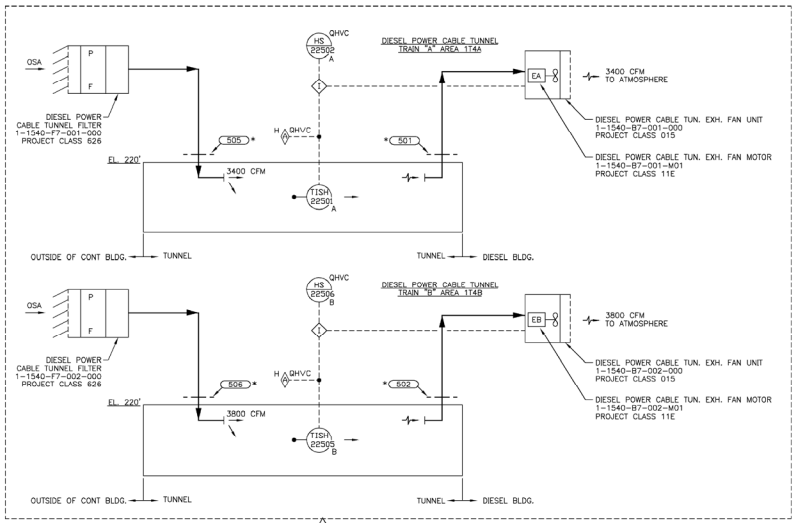
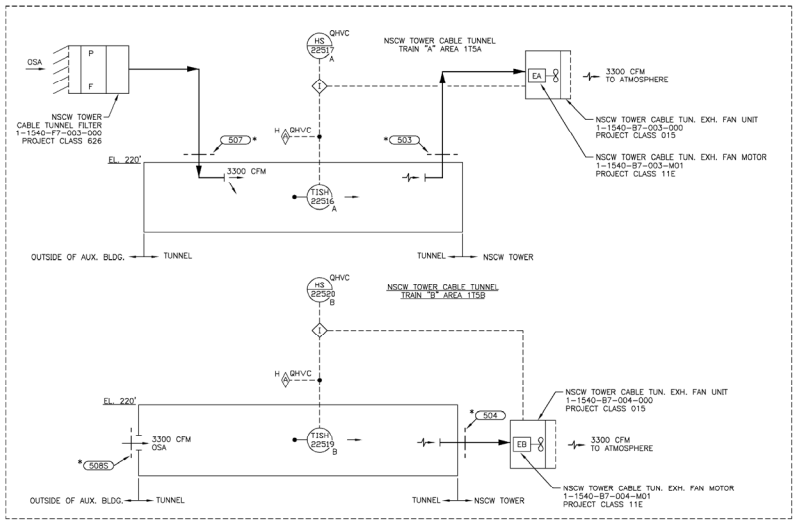
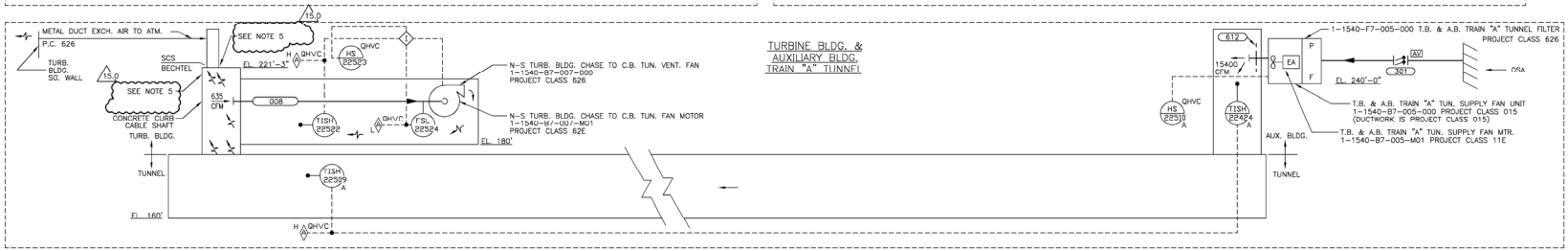
NO.		DATE	DR	CHK	APPV	DTL
25.0	REDRAWN PER ABN- 52221, VER. 1.0	11-2-05	CYN	LCF	EPD	X

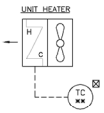
SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB234
VER. 25.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

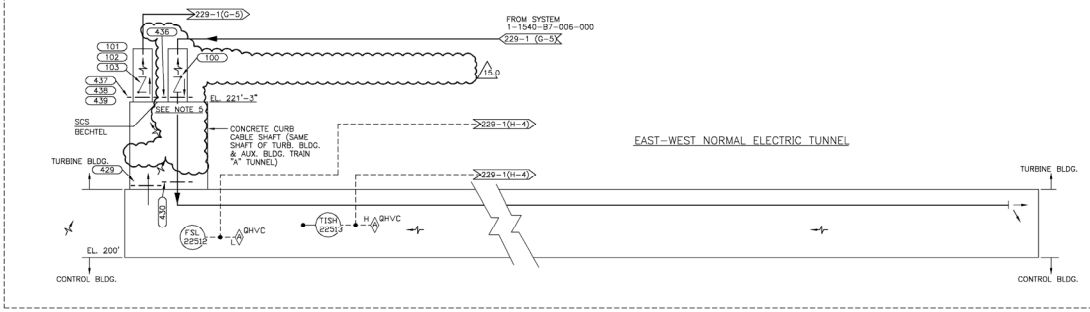






UNIT HEATERS	TEMP. CONTROL **	LOCATION
1-1540-U7-009	TC-22716	NSCW R-102
1-1540-U7-010	TC-22717	NSCW R-102
1-1540-U7-011	TC-22718	NSCW R-102
1-1540-U7-012	TC-22719	NSCW R-202
1-1540-U7-013	TC-22720	NSCW R-202
1-1540-U7-014	TC-22721	NSCW R-202
1-1540-U7-015	TC-22722	RWST R-101
1-1540-U7-016	TC-22723	RWWST R-101
1-1540-U7-017	TC-22724	TUNNEL 1T2B
1-1540-U7-019	TC-22726	TUNNEL 1T2A, 1T3A
1-1540-U7-020	TC-22727	TUNNEL 1T2A, 1T3A
1-1540-U7-022	TC-22729	TUNNEL 1T2A, 1T3A
1-1540-U7-023	TC-22730	TUNNEL 1T2B
1-1540-U7-026	TC-22733	CST R-103





NOTE:
1. DUCTWORK ASSOCIATED WITH CLASS 015 EQUIPMENT ARE CLASS 015; ALL OTHER DUCTWORK ARE CLASS 626.
2. * DERATED FIRE DAMPER.
3. "TC" (TEMPERATURE CONTROLLERS ARE UNIT MOUNTED THERMOSTATS).
4. ALL UNIT HEATERS ARE CLASS 62E.
5. FOR OPENING DETAILS SEE AX2D11A044. 15.0

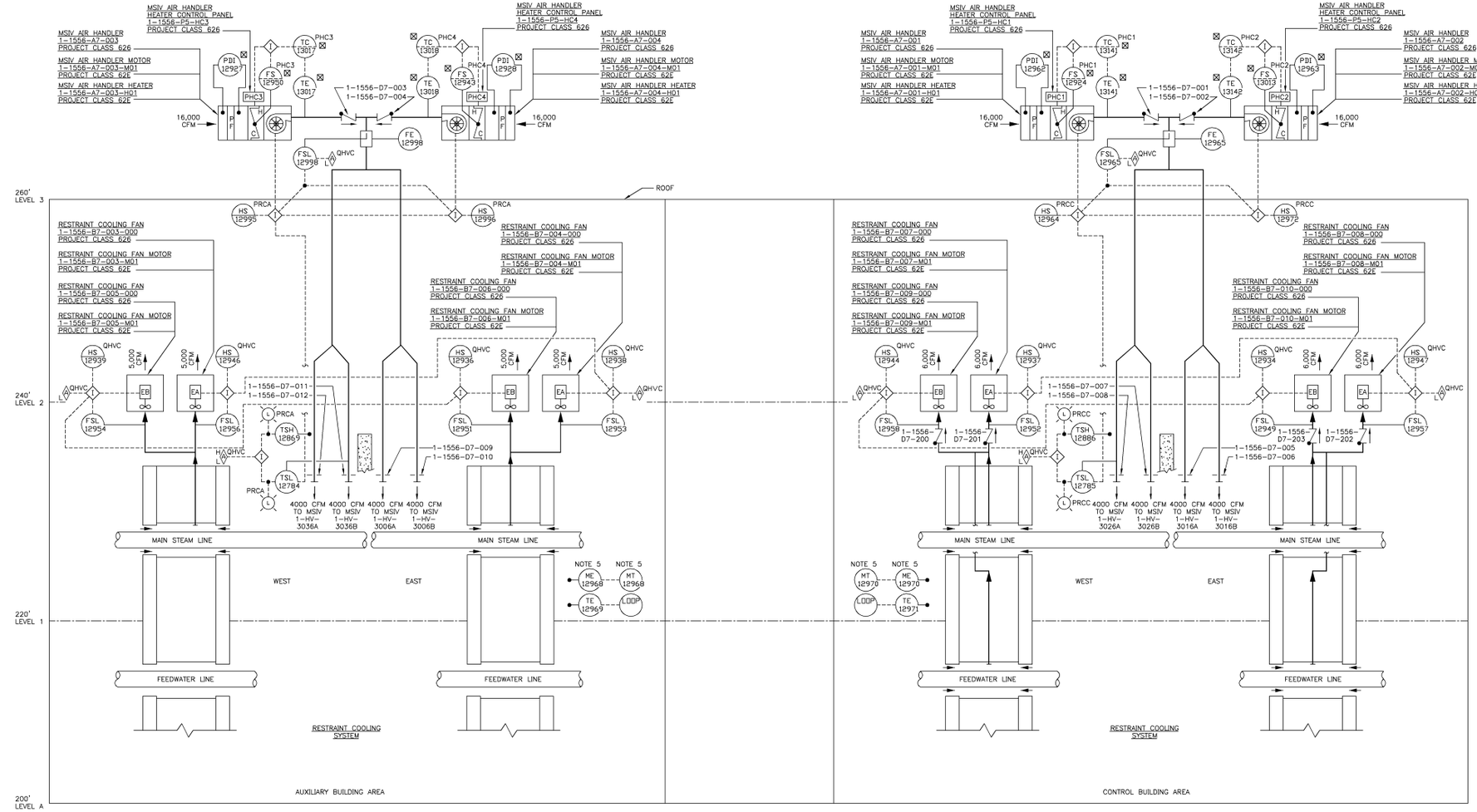
CAD 1X4DB238
Acad2000 TJS - 02

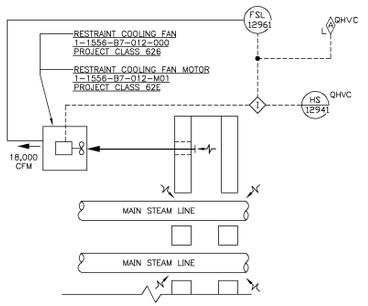
THIS DOCUMENT CONTAINS PROPRIETARY, CONFIDENTIAL, AND/OR TRADE SECRET INFORMATION OF THE SUBSIDIARIES OF THE SOUTHERN COMPANY OR OF THIRD PARTIES. IT IS INTENDED FOR USE ONLY BY EMPLOYEES OF, OR AUTHORIZED CONTRACTORS OF, THE SUBSIDIARIES OF THE SOUTHERN COMPANY. UNAUTHORIZED POSSESSION, USE, DISTRIBUTION, COPYING, DISSEMINATION, OR DISCLOSURE OF ANY PORTION HEREOF IS PROHIBITED.

NO.		DATE	DN	CHK	APPV	UTL
15.0	REVISED PER ABN-V01431, VER. 1.0.	12-09-08	TJS	PWG	PML	X

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
ELECTRICAL TUNNEL VENTILATION SYSTEM
SYSTEM NO. 1540

SCALE: NONE
DRAWING NO. 1X4DB238
VER. 15.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL





MAIN STEAM TUNNEL RESTRAINTS

NOTES:

1. ALL DUCT IS PROJECT CLASS 626 EXCEPT AS NOTED.
2. ⊠ FURNISHED WITH ASSOCIATED EQUIPMENT.
3. TEMPERATURE SENSORS TSH12886 (CONTROL BUILDING) AND TSH12869 (AUXILIARY BUILDING) ARE "HIGH TEMP" SENSORS LOCATED IN THE VICINITY OF THE MSIV'S. THESE SENSORS ARE INTENDED TO NOTIFY THE CONTROL ROOM THAT THE TEMPERATURE NEAR THE MSIV ACTUATORS IS APPROACHING THE ENVIRONMENTAL QUALIFICATION UPPER LIMIT FOR NORMAL OPERATION.
4. TEMPERATURE SENSORS TSL12785 (CONTROL BUILDING) AND TSL12784 (AUXILIARY BUILDING) ARE "LOW TEMP" SENSORS LOCATED IN THE AIR HANDLER DISCHARGE AIRSTREAM IN THE VICINITY OF THE MSIV'S. THESE SENSORS ARE INTENDED TO NOTIFY THE CONTROL ROOM THAT THE SUPPLY AIR TEMPERATURE IS APPROACHING THE VALVE ACTUATOR LOW OPERATING TEMPERATURE LIMIT.
5. 1MT-12968, 1ME-12968, 1MT-12970, & 1ME-12970 HAVE BEEN "ABANDONED IN PLACE".

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
9.0	REDRAWN PER ABN-V70252, VER. 1.0	2-12-07	CYN	LCF	EPD	X

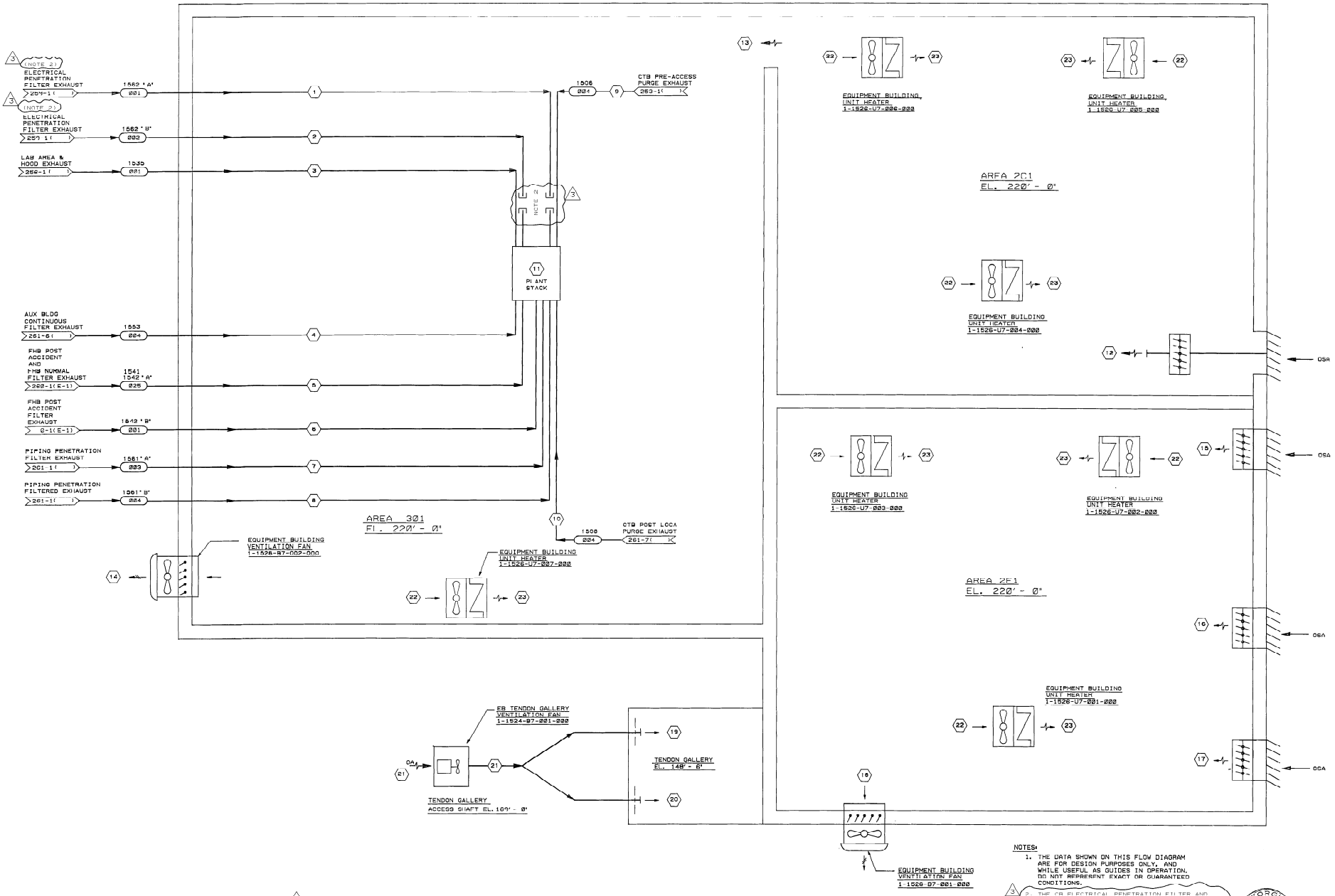
CAD 1X4DB245
ACAD2002 CYN – 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
PIPING PENETRATION
VENTILATION SYSTEM
SYSTEM NO. 1556

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB245
VER. 9.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



MODE		PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
NORMAL	POWER GENERATION	FLOW	CFM	—	—	11420	85400	30000	—	—	—	5000	—	144645	8900	8900	12,250	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	120	134	134	—	—	—	120	—	120	98	110	120	98	98	98	120	98	98	70	50	70
		PRESSURE	INCH OF WATER	—	—	0.75	0.75	0.75	—	—	—	0.75	—	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
	REFUELING	FLOW	CFM	—	—	11200	85400	30000	—	—	—	10000	—	154475	8900	8900	12,200	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	104	104	98	—	—	—	120	—	110	98	110	120	98	98	98	120	98	98	50	50	70
		PRESSURE	INCH OF WATER	—	—	0.75	0.75	0.75	—	—	—	0.75	—	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
POST LOCA	WITH OFFSITE POWER	FLOW	CFM	0	0	11200	85400	30000	—	2000	2000	—	500	142275	8900	8900	12,250	450	450	400	1300	2500	2500	5000	500	500
		TEMPERATURE	°F	—	—	104	104	104	—	120	100	—	120	120	98	110	120	98	98	98	120	98	98	50	50	70
		PRESSURE	INCH OF WATER	—	—	0.75	0.75	0.75	—	0.75	0.75	—	0.75	0.5	—	—	—	—	—	—	-0.5	0.25	0.25	—	—	—
	WITHOUT OFFSITE POWER	FLOW	CFM	0	0	—	—	5000	5000	2000	2000	—	500	76000	—	—	—	—	—	—	—	—	—	—	—	—
		TEMPERATURE	°F	—	—	—	—	104	104	100	100	—	120	110	—	—	—	—	—	—	—	—	—	—	—	—
		PRESSURE	INCH OF WATER	—	—	—	—	0.75	0.75	0.75	0.75	—	0.75	0.5	—	—	—	—	—	—	—	—	—	—	—	—

NOTES:
1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR DESIGN PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DO NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.
3. THE CR ELECTRICAL PENETRATION FILTER AND EXHAUST SYSTEM NO. 1562 IS RETIRED-IN-PLACE.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORPORATED PER DCP 95-VTN0003	10-11-96	CU	GLB	JGM	
1	ISSUED FOR CONSTRUCTION	SEE MASTER	M FOR	SIGNATURES		

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
EQUIPMENT BUILDING HVAC SYSTEM
SYSTEM NO. 1526 & 1524

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X4DB250-1
REV. 3
SIZE E 36x44
CAD NAME 14DB2501

SCALING RULES

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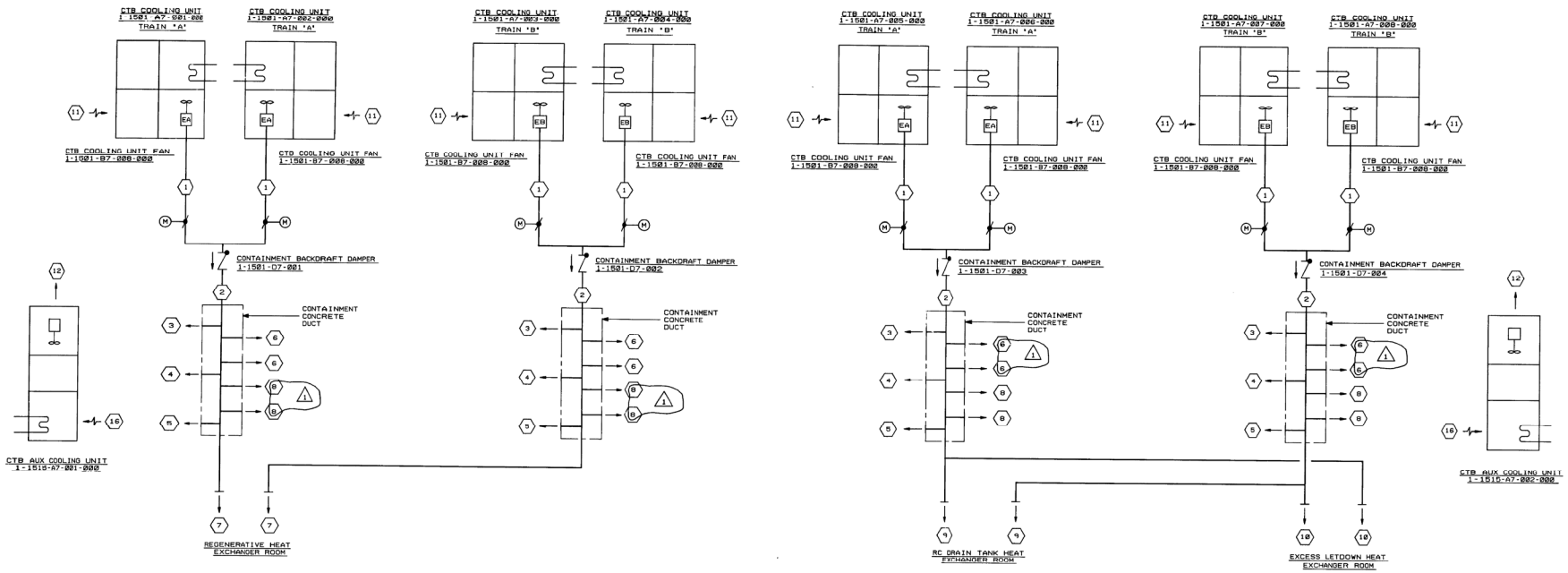
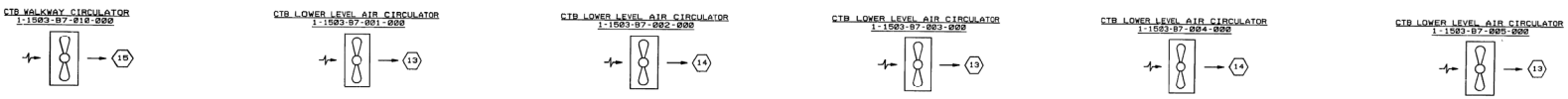
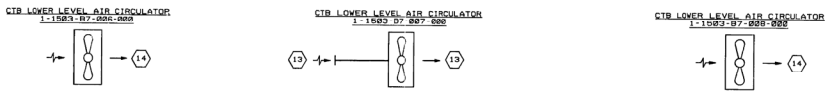






MODE	PARAMETER		①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪	⑫	⑬	⑭	⑮	⑯
POWER GENERATION & LOSS OF OFFSITE POWER	FLOW	CFM	N/A	110000	(9)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	N/A	183000	18360	14280	1450	183000
	TEMPERATURE	°F	98	98	98	98	98	98	98	98	98	98	120	98	120	120	120	120
	PRESSURE (1)	IN.OF WATER	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LOCA	FLOW	CFM	N/A	52000(8)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	N/A	—	—	—	—	—
	TEMPERATURE	°F	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	—	—	—	—	—
	PRESSURE	(J)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	—	—	—	—	—
MSLB	FLOW	CFM	N/A	52000(8)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	N/A	—	—	—	—	—
	TEMPERATURE	°F	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(4)	—	—	—	—	—
	PRESSURE		(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	—	—	—	—	—
RECIRCULATION	FLOW	CFM	N/A	52000(8)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	N/A	—	—	—	—	—
	TEMPERATURE	°F	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	(6)	—	—	—	—	—
	PRESSURE		(7)	(7)	(7)	(7)	(7)	(7)	(7)	(7)	(7)	(7)	(7)	—	—	—	—	—

NOTES:
1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR DESIGN PURPOSES ONLY AND WHILE USEFUL AS GUIDES IN OPERATION DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
2. AIR QUANTITIES SHOWN ARE ROUNDED TO THE NEAREST 10.
3. THE FLOW VALUES GIVEN ARE MINIMUM, WITH -18% MAXIMUM (-2%-18%), EXCEPT SYS. 1501-WHICH HAS NO MAXIMUM LIMIT.

(1) PRESSURES INDICATED ARE IN REFERENCE TO THE ATMOSPHERIC PRESSURE.
(2) CONTAINMENT TEMPERATURE MAY REACH 400°F FOR LOCA OPERATIONS.
(3) CONTAINMENT DESIGN PRESSURE IS 60 PSIA FOR LOCA CONDITION.
(4) CONTAINMENT TEMPERATURE MAY REACH 400°F ON MSLB ACCIDENT.
(5) CONTAINMENT PRESSURE MAY REACH 50 PSIA ON MSLB ACCIDENT.
(6) CONTAINMENT TEMPERATURE MAY REACH 250°F DURING RECIRCULATION MODE OF OPERATION.
(7) CONTAINMENT PRESSURE MAY REACH 30 PSIA DURING RECIRCULATION MODE OF OPERATION.
(8) S.T.P. EQUIVALENT FLOW IS 35000 CFM.
(9) REGISTERS SHALL BE FULL OPEN.

Daniel P. Stanley 4/29/76

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
1	ISSUED FOR CONSTRUCTION	3-30-76		CAO					

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTAINMENT BUILDING
CONTAINMENT HEAT REMOVAL SYSTEM
SYSTEM NO. 1501

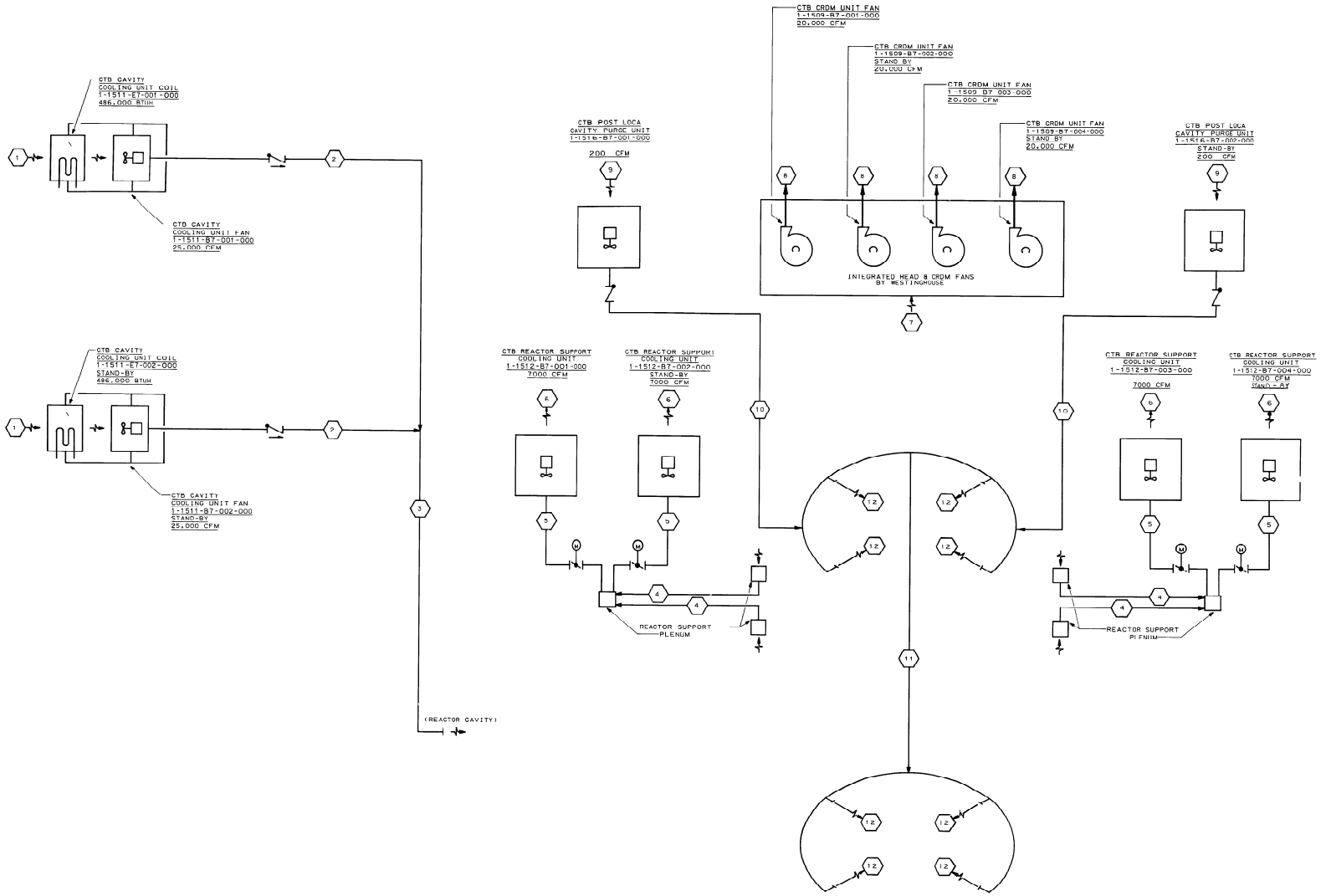
ENG. MGR. DATE

SCALE: NONE | DRAWING NO. 1X4DB251-1 | REV. 1

JOB NO. 9510

SEE MICROFILM FOR SIGNATURES

34X44 "E" SIZE



MODES	PARAMETER	1	2	3	4	5	6	7	8	9	10	11	12
NORMAL OPERATION	FLOW (CFM)	25,000	25,000	25,000	—	5300	5300	40,000	20,000	-	-	-	-
	TEMPERATURE (°F)	120	108	108	126	126	130	120	190	-	-	-	-
	PRESSURE (PSIG)	ATMOSPHERIC								-	-	-	-
POST LOCA OPERATION	FLOW (CFM)	-	-	-	-	-	-	-	-	200	200	100	25
	TEMPERATURE (°F)	-	-	-	-	-	-	-	-	340 MAXIMUM			
	PRESSURE (PSIG)	-	-	-	-	-	-	-	-	80 MAXIMUM			

NOTES:
1. THE DATA SHOWN ON THIS DIAGRAM ARE FOR DESIGN PURPOSES ONLY AND, WHILE USEFUL AS GUIDE IN OPERATION, DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF. E	P.E.	Q.A.E.
1	INCORP M-FCRB-1484 (MOD) (PFE)								
2	ISSUED FOR CONSTRUCTION								

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTAINMENT CRDM, CAVITY, & REACTOR SUPPORT COOLING SYSTEM
SYSTEM NO. 1509 & 1511

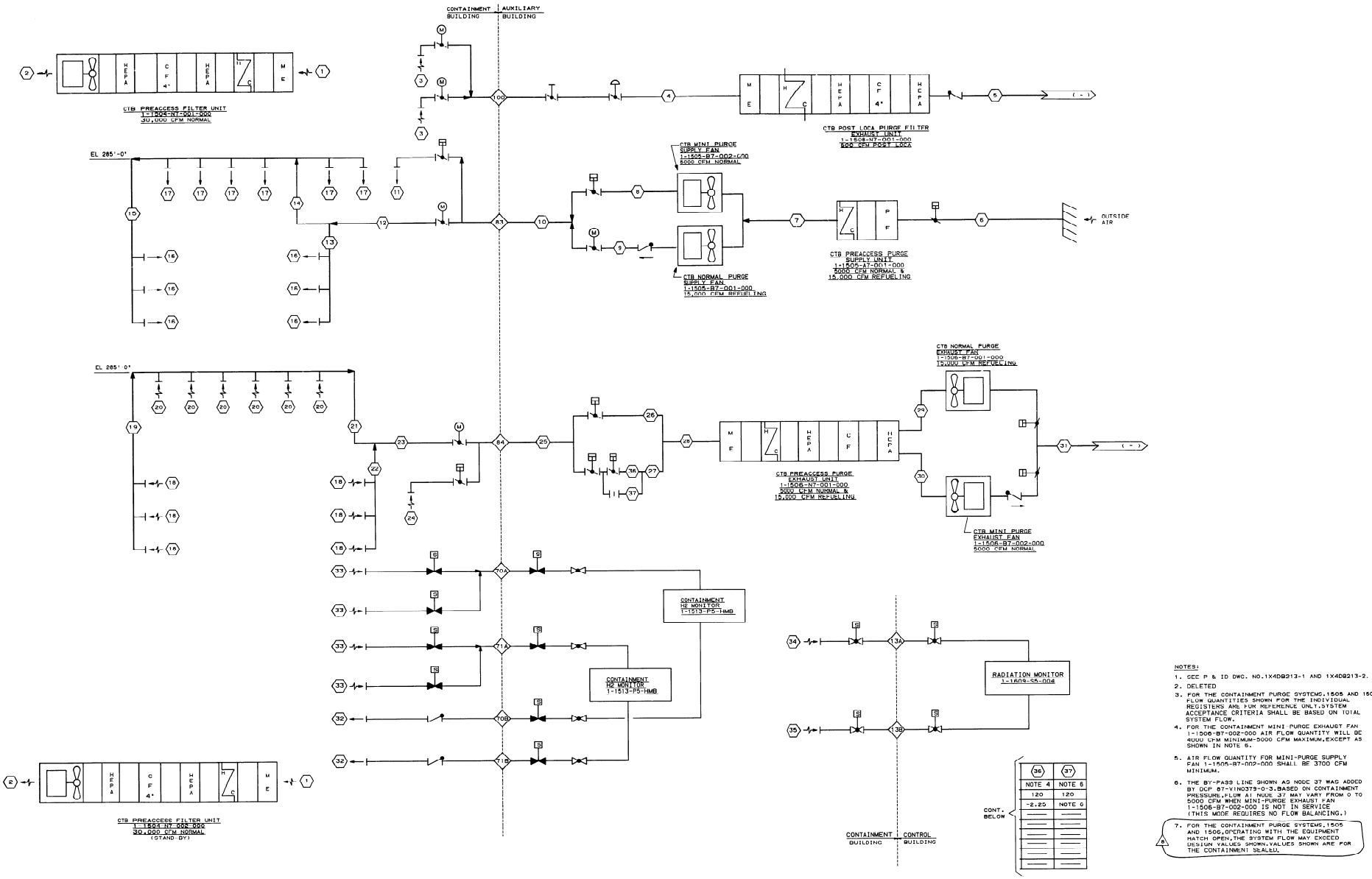
ENG. MGR. DATE

SCALE:

DRAWING NO. 1X4DB252 REV. 1

JOB NO. 9510

SIZE E 30444



36	37
NOTE 4	NOTE 6
120	120
-2.25	NOTE 6

NOTES:

1. SEE P & ID DWG. NO.1X4DB213-1 AND 1X4DB213-2.
2. DELETED
3. FOR THE CONTAINMENT PURGE SYSTEMS,1505 AND 1506 FLOW QUANTITIES SHOWN FOR THE INDIVIDUAL REGISTERS ARE FOR REFERENCE ONLY.SYSTEM ACCEPTANCE CRITERIA SHALL BE BASED ON TOTAL SYSTEM FLOW.
4. FOR THE CONTAINMENT MINI PURGE EXHAUST FAN 1-1508-B7-002-000 AIR FLOW QUANTITY WILL BE 4000 CFM MINIMUM-5000 CFM MAXIMUM,EXCEPT AS SHOWN IN NOTE 6.
5. AIR FLOW QUANTITY FOR MINI-PURGE SUPPLY FAN 1-1505-B7-002-000 SHALL BE 3700 CFM MINIMUM.
6. THE BY-PASS LINE SHOWN AS NODE 37 WAS ADDED BY DCP B7-V1NO379-0-3.BASED ON CONTAINMENT PRESSURE,FLOW AT NODE 37 MAY VARY FROM 0 TO 5000 CFM WHEN MINI-PURGE EXHAUST FAN 1-1508-B7-002-000 IS NOT IN SERVICE (THIS MODE REQUIRES NO FLOW BALANCING.)
7. FOR THE CONTAINMENT PURGE SYSTEMS,1505 AND 1506,OPERATING WITH THE EQUIPMENT HATCH OPEN,THE SYSTEM FLOW MAY EXCEED DESIGN VALUES SHOWN.VALUES SHOWN ARE FOR THE CONTAINMENT SEALED.

MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
NORMAL OPERATION	FLOW	CFM	30000	30000	—	—	—	3700	3700	3700	—	3700	3700	—	—	—	—	—	—	—	—	—	—	—	—	NOTE 4	NOTE 4	—	NOTE 4	NOTE 4	—	NOTE 4	NOTE 4	5 SCFH	5 SCFH	2-4	2-4
	TEMP	°F	120	127	—	—	—	98/17	98/60	101/63	—	101/63	101/63	—	—	—	—	—	—	—	—	—	—	—	—	120	120	—	120	120	—	120	125	120	120	120	120
	PRESSURE	IN W.G.	-0.15	-0.60	—	—	—	-0.25	-1.5	+2.9	—	+1.9	+0.9	—	—	—	—	—	—	—	—	—	—	—	—	-0.25	-0.75	—	-2.25	-2.25	—	-0.25	+0.45	—	—	—	—
POST LOCA OPERATION	FLOW	CFM	—	—	500	500	500	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	5 SCFH	5 SCFH	2-1	2-4
	TEMP	°F	—	—	150	150	150	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	300	300	300	300
	PRESSURE	IN W.G.	—	—	+55	+55	+47	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	50 PSIG	53 PSIG	50 PSIG	53 PSIG
REFUELING OPERATION	FLOW	CFM	—	—	—	—	—	15000	15000	—	15000	15000	—	15000	4500	10500	4500	1500	1000	1500	4500	1000	10500	4500	15000	—	15000	15000	—	—	15000	15000	—	15000	—	—	—	—
	TEMP	°F	—	—	—	—	—	98/17	98/60	—	102/64	102/64	—	102/64	102/64	102/64	102/64	102/64	102/64	120/60	120/60	120/60	120/60	120/60	120/60	—	120/60	120/60	—	120/60	120/60	—	130/70	—	—	—	—
	PRESSURE	IN W.G.	—	—	—	—	—	-0.45	-2.0	—	+3.0	+2.0	—	+1.0	+1.0	+1.0	+0.75	+0.60	+0.75	-0.25	-0.35	-0.25	-0.55	-0.55	-0.75	—	-1.75	2.75	—	2.75	-10.60	—	-0.7	—	—	—	—

CONT. ABOVE

NO.	REVISIONS	DATE	DR	CHK	CW	X	X	X	X	X
1	INCORP.ABN-00074	4-17-90	MMD							
2	REV.TO INC.ABN: R7-V1NO379A017	10-13-88	SEE MICROFILM FOR SIGNATURES							

NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	RPE	P.E.	QAE
1	INCORP.WFCRB 880H,WFCRB-881N & WFCRB-884N	1-23-87	SEE MICROFILM FOR SIGNATURES							
2	INCORPORATED W-FCRB-16,290	12-11-86	SEE MICROFILM FOR SIGNATURES							
3	INCORPORATED W-FCRB-16,134	12-8-86	SEE MICROFILM FOR SIGNATURES							

NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	RPE	P.E.	QAE
1	INCORP W-FCRB-16,427 & DCN = 2		SEE MICROFILM FOR SIGNATURES							
2	INCORPORATED DCN #11 (PPE)		SEE MICROFILM FOR SIGNATURES							
3	REDRAWN AND COMPLETED INFORMATION TABLE	8-19-93	SEE MICROFILM FOR SIGNATURES							
4	ISSUED FOR CONSTRUCTION	4-18-76	SEE MICROFILM FOR SIGNATURES							

ENG. MGR.

DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTAINMENT AIR PURIFICATION & CLEAN-UP SYSTEMS
SYS NO.1504,1505,1506,1508,1513,1609

SCALE: NONE

DRAWING NO. 1X4DB253-1

REV. 8

JOB NO.10604

SIZE E 34X44

12-02-86

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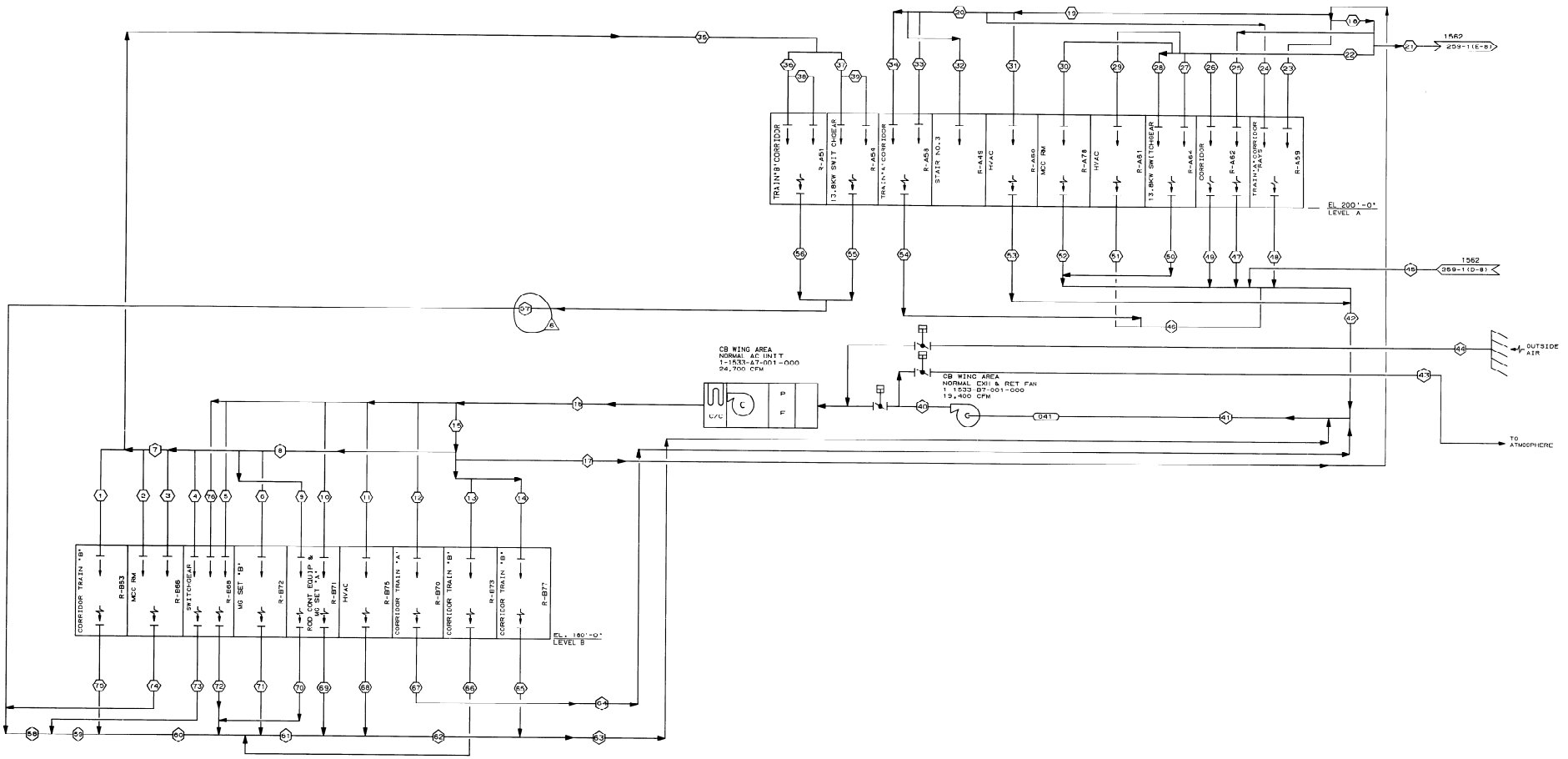


MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	26	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
SUMMER	FLOW	CFM	850	600	300	760	760	400	3375	5635	800	800	640	1300	900	1500	17665	21925	9630	7240	2055	1165	5305	1760	335	335	175	160	345	345	575	335	555	100	630	535	1925	675	1250	335	800	18555	18555	8660
	TEMP	°F	59	59	59	59	59	59	58	58	59	59	59	59	59	59	58	57	58	58	57	57	57	58	59	59	59	59	59	59	59	59	59	59	59	58	59	59	59	59	90	88	NA	
	PRESSURE	INCH OF WTR																																										
WINTER	FLOW	CFM	850	600	300	760	760	400	3375	5635	800	800	640	2150	245	1500	17665	21925	9630	7240	2055	1165	5305	1760	335	335	175	160	345	345	575	335	555	100	530	535	1925	675	1250	335	800	18555	18555	8660
	TEMP	°F	59	59	59	59	59	59	58	58	59	59	59	59	59	59	58	57	58	58	57	57	57	58	59	59	59	59	59	59	59	59	59	59	59	59	58	59	59	59	90	88	NA	
	PRESSURE	INCH OF WTR																																										

MODE	PARAMETER		43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84
SUMMER	FLOW	CFM	0	3370	4390	1775	50	670	130	590	575	400	555	1200	1300	675	1975	2875	3895	4335	7135	8495	8795	1100	300	200	1100	640	725	725	370	1500	1020	900	440	760								
	TEMP	°F		98	NA	NA	100	87	100	83	83	83	81	83	81	74	NA	NA	NA	NA	NA	NA	100	NA	71	88	89	108	103	103	103	100	100	94	72	59								
	PRESSURE	INCH OF WTR	——																																									
WINTER	FLOW	CFM	0	3370	4390	1775	50	670	130	590	575	400	555	1200	1300	675	1975	2875	3895	4335	7135	8495	8795	1100	300	200	1100	640	725	725	370	1500	1020	900	440	760								
	TEMP	°F	——	17	NA	NA	100	87	100	83	83	83	81	83	81	74	NA	NA	NA	NA	NA	NA	100	NA	71	88	89	108	103	103	103	100	100	94	72	59								
	PRESSURE	INCH OF WTR	——																																									

MODE	PARAMETER		21	43	44	45
EMERGENCY	FLOW	CFM	0	0	0	0
	PRESSURE	INCH OF WTR	——	——	——	——
SMOKE EXHAUST	FLOW	CFM	5305	18,555	21,955	4390
	PRESSURE	INCH OF WTR				
NORMAL	FLOW	CFM	5305	0	3370	4390
	PRESSURE	INCH OF WTR		——		

NOTES:

1. P & ID DWG 1X4DB210.
2. THE DATA SHOWN ON THIS FLOW DIAGRAM IS FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DOES NOT REPRESENT EXACT OR GUARANTEED CONDITIONS.

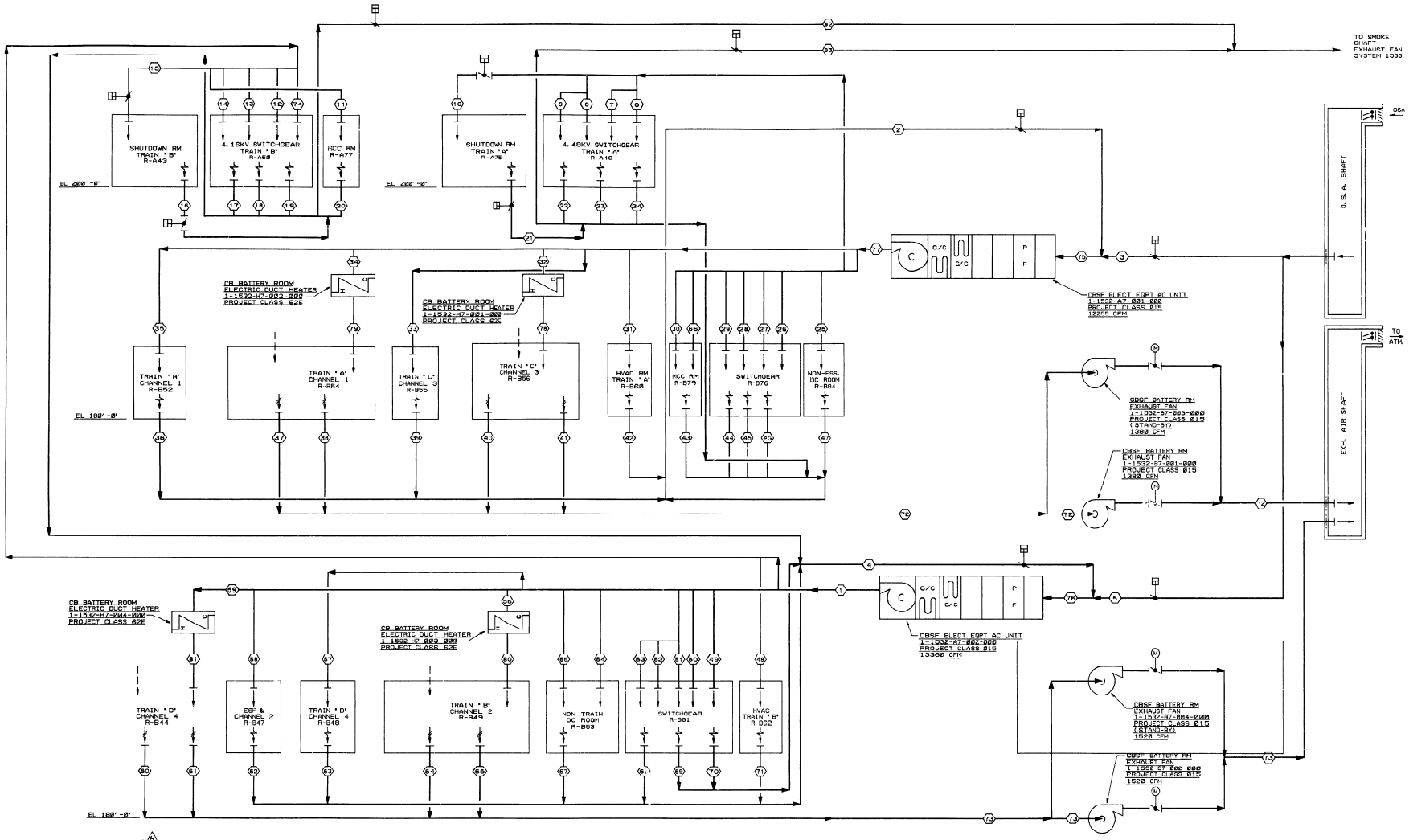
NO.	REVISIONS	DATE	DR.	CHK.	SUPV'EQL	EGS	CHF.E	P.E.	Q.A.E
4	INCORP. ABN 89-V1000A147	6-25-89	MMD	SLP	REP				
3	INC. ABN 89-V1000A304	8-18-89		SEE MICROFILM FOR SIGNATURES					
2	REVISIONS	DATE	DR.	CHK.	SW				
1	INCORPORATED M-FCRB-15,387	12-8-88		SEE MICROFILM FOR SIGNATURES					
NO.	REVISIONS	DATE	DR.	CHK.	SUPV'EQL	EGS	CHF.E	P.E.	Q.A.E
4	INCORPORATED DCN #4 (MOD)	11-3-86		SEE MICROFILM FOR SIGNATURES					
3	INCORPORATED DCN #2 & 3 (PPE)	10-16-86		SEE MICROFILM FOR SIGNATURES					
2	INCORPORATED DCN #1	10-10-86		SEE MICROFILM FOR SIGNATURES					
1	ISSUED FOR CONSTRUCTION	05-30-86		SEE MICROFILM FOR SIGNATURES					
NO.	REVISIONS	DATE	DR.	CHK.	SUPV	EGS	CHF.E	P.E.	Q.A.E

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
FLOW DIAGRAM
CONTROL BUILDING NORMAL HVAC SYSTEM
LEVEL A & B
SYSTEM NO. 1533

SCALE: NONE
DRAWING NO. 1X4DB257-1
REV. 6
JOB NO. 9510



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
SUMMER	FLOW	CFM	14655	12705	1320	11125	3530	480	160	480	480	500	725	350	350	350	140	110	460	470	470	480	500	550	525	525	175	1525	1525	1525	1525	1525	190	475	475	1100	625	2600	2730	430	430	1825	260	260	475
	TEMP	°F	67	93	98	94	98	59	59	59	59	59	59	59	59	59	59	75	100	100	100	98	98	98	98	98	98	59	59	59	59	59	59	59	59	59	59	59	98	75	75	100	75	75	88
	PRESSURE	INCH OF WTR																																											
WINTER	FLOW	CFM	14655	12705	1320	11125	3530	480	160	480	480	500	725	350	350	350	140	110	460	470	470	480	500	550	525	525	175	1525	1525	1525	1525	1525	190	475	475	1100	625	2600	2730	430	430	1825	260	260	475
	TEMP	°F	57	93	17	94	17	59	59	59	59	59	59	59	59	59	59	75	100	100	100	98	98	98	98	98	98	59	59	59	59	59	59	59	59	59	59	59	98	75	75	100	75	75	88
	PRESSURE	INCH OF WTR																																											

MODE	PARAMETER		43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84
SUMMER	FLOW	CFM	375	2850	2025	1750	175	470	1160	1160	1160	1160	1160	570	570	725	1575	1900	650	360	360	590	930	480	480	195	1140	2100	1900	2100	470	1300	1520	490	14825	14655	14825	475	625	725	650	0	0	
	TEMP	°F	98	98	98	98	98	59	59	59	59	59	59	59	59	59	59	59	59	75	75	98	100	75	75	59	100	98	98	98	85	77	77	59	93	94	57	59	59	59	59	——	——	
	PRESSURE	INCH OF WTR																																										
WINTER	FLOW	CFM	375	2850	2025	1750	175	470	1160	1160	1160	1160	1160	570	570	725	1575	1900	650	360	360	590	930	480	480	195	1140	2100	1900	2100	470	1300	1520	490	14825	14655	14825	475	625	725	650	0	0	
	TEMP	°F	98	98	98	98	98	59	59	59	59	59	59	59	59	59	59	59	59	75	75	98	100	75	75	59	100	98	98	98	85	77	77	59	93	94	57	59	59	59	59	——	——	
	PRESSURE	INCH OF WTR																																										

MODE	PARAMETER		82	83	2	3	4	5
SMOKE EXHAUST	FLOW	CFM	9980	8750	0	8750	2	9980

NOTES:

1. P & ID DWG 1X4DB207-1.
2. DELETED
3. AIR QUANTITIES FOR NODES 37,38,40,41,60, 61,64 AND 65 ARE MINIMUM VALUES ONLY.

GEORGIA NO. 14189 PE.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF.E	P.E.	Q.A.E.
△										
△										
△										
△										
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF.E	P.E.	Q.A.E.
△	INCORP. ABN NO. 87-V1000A052	5/16/87	MWD	DK	RCM					
△	INCORPORATED M-FCH8-16.388	12/9/87	DR	CHK	RPE	DM	ODPM	DPM	DPM	
△	INCORPORATED M-FCH8-16.391	2-5-88	SEE MICROFILM FOR SIGNATURES							
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF.E	P.E.	Q.A.E.
△	INCORPORATED DCN # 4 & 5	11-7-88	SEE MICROFILM FOR SIGNATURES							
△	INCORPORATED DCN # 3* (PFE)	12-14-88	SEE MICROFILM FOR SIGNATURES							
△	INCORPORATED DCN #1 & 2	10-13-89	SEE MICROFILM FOR SIGNATURES							
△	ISSUED FOR CONSTRUCTION	8-5-91	SEE MICROFILM FOR SIGNATURES							
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EQ.	EGS	CHF.E	P.E.	Q.A.E.

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING SAFETY FEATURE ELEC EQUIPMENT ROOM LEVEL A & B HVAC SYSTEM
SYSTEM NO. 1532

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X4DB258-1	6

34X44 E SIZE

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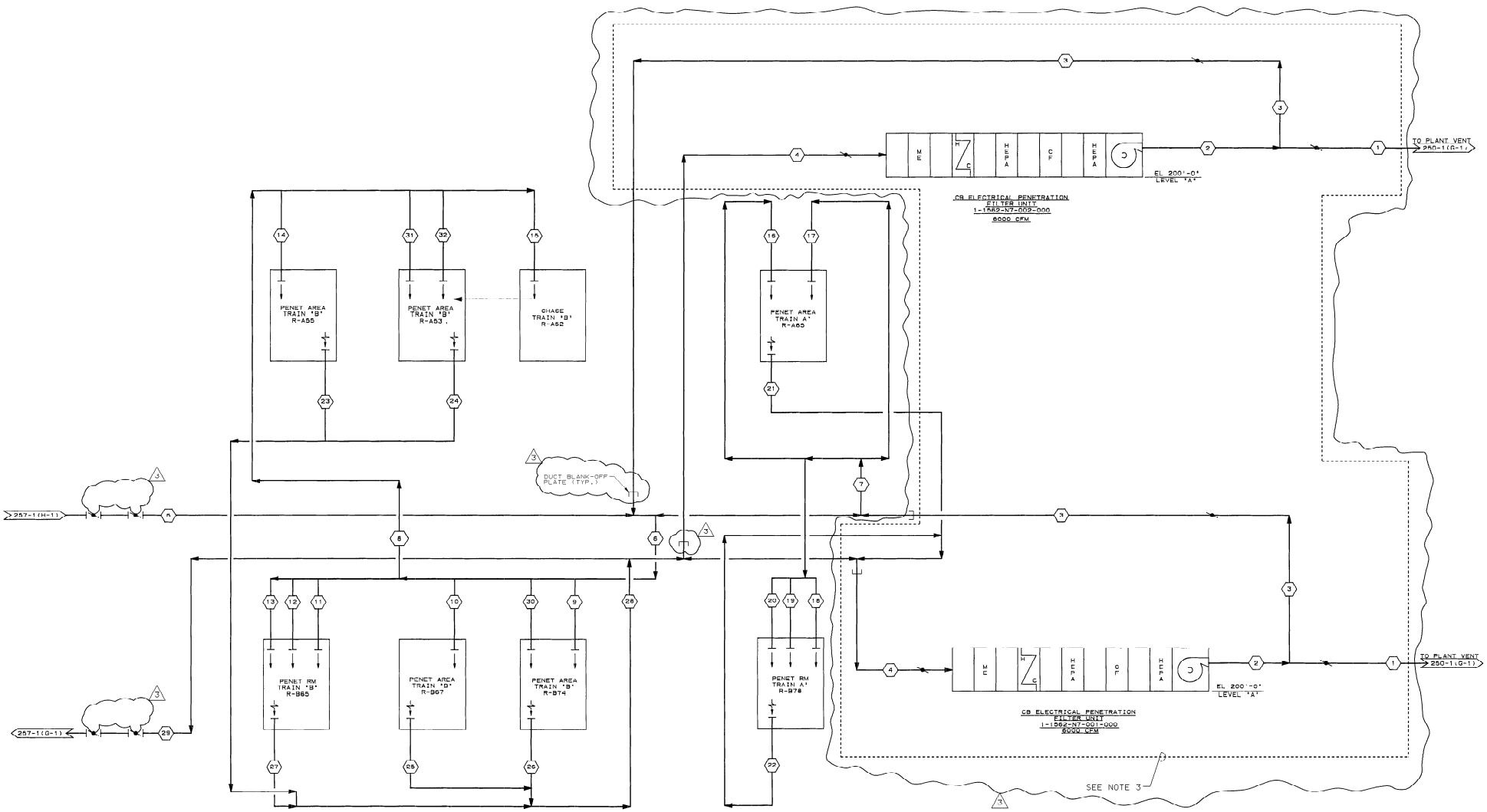

MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
NORMAL	FLOW	CFM	0	0	0	0	5305	2840	2465	1280	155	85	385	390	390	140	380	605	610	415	420	415	1100	1400	40	700	100	350	700	1890	4390	155	380	380
	TEMP	°F	—	—	—	—	57	57	57	57	59	59	59	59	59	59	59	59	59	59	59	59	63	76	86	69	84	93	82	NA	NA			
	PRESSURE	INCH OF WATER	—	—	—	—																												

NOTES:

1. DELETED
2. P & ID 1X4DB209.
3. THE CB ELECTRICAL & PIPE PENETRATION ROOM FILTER & EXHAUST SYSTEM IS RETIRED IN PLACE

GEORGIA REGISTERED PROFESSIONAL ENGINEER DANIEL C. STEARNS
12/7/94

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 95-VTN0003	10-11-95	CD	GLB	JGM	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
CONTROL BUILDING ELECTRICAL & PIPE PENETRATION ROOM FILTER & EXHAUST SYSTEM
SYSTEM NO. 1562

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB259-1	3

SIZE E 36x44

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TO EQUIPMENT BUILDING STACK 250-1(E-8)

OUTSIDE AUXILIARY BUILDING
INSIDE AUXILIARY BUILDING

ROOF LEVEL 3
260'-0"

AB NORMAL A/C UNIT
1-1551-A7-001-000
PROJECT CLASS 62B

AB NORMAL A/C UNIT
1-1551-A7-002-000
PROJECT CLASS 62B

AB CONTINUOUS EXHAUST UNIT
1-1553-N7-001-000
PROJECT CLASS 62B

AB CONTINUOUS EXHAUST UNIT
1-1553-N7-002-000
PROJECT CLASS 62B

AB CONTINUOUS EXHAUST UNIT
1-1553-N7-003-000
PROJECT CLASS 62B

376 (H-5)

>261-5(F-8)

261-5(H-8)

ELEV. EQUIP ROOM
R-217

A-1551-A7-001-000 AUX.BLDG.
FREIGHT ELEVATOR EQUIP. ROOM
HVAC AIR HANDLING UNIT
PROJECT CLASS 62B

ELEV. EQUIP ROOM
R 301

1-1551-A7-003-000 AUX. BLDG.
PERSONNEL ELEVATOR
HVAC AIR HANDLING UNIT
PROJECT CLASS 62B

ROOM 212
(SEE DRAWING 261-5 FOR SUPPLY AIR TO ROOM 212)

NOTE 1

MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
NORMAL	FLOW	CFM	38900	38900	38900	38900	77800	85200	1500	1650	1700	28400	28400	28400	28400	28400	28400	28400	28400	28400	73900	5400	650	1750	1750	700	700
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	80	50	60-85	50-74

NOTES:

1. ACTUAL FLOW FOR NODES 22,23,24 & 25 SHALL BE NO LESS THAN 90% OF VALUE SHOWN. NO MAXIMUM VALUE. Δ7
2. REFERENCE P & ID 1X4DB208-1.
3. THIS FLOW DIAGRAM IS TO BE USED IN CONJUNCTION WITH THE ALTERNATE RADWASTE BLDG. UNTIL SUCH TIME AS THE RADWASTE SOLIDIFICATION AND TRANSFER BUILDINGS ARE COMPLETED, AT WHICH TIME THE ORIGINAL DESIGN AIR FLOWS WILL BE REINSTATED.

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
AUXILIARY BUILDING OUTSIDE AIR SUPPLY AND NORMAL HVAC SYSTEM
SYSTEM NO. 1551 & 1553

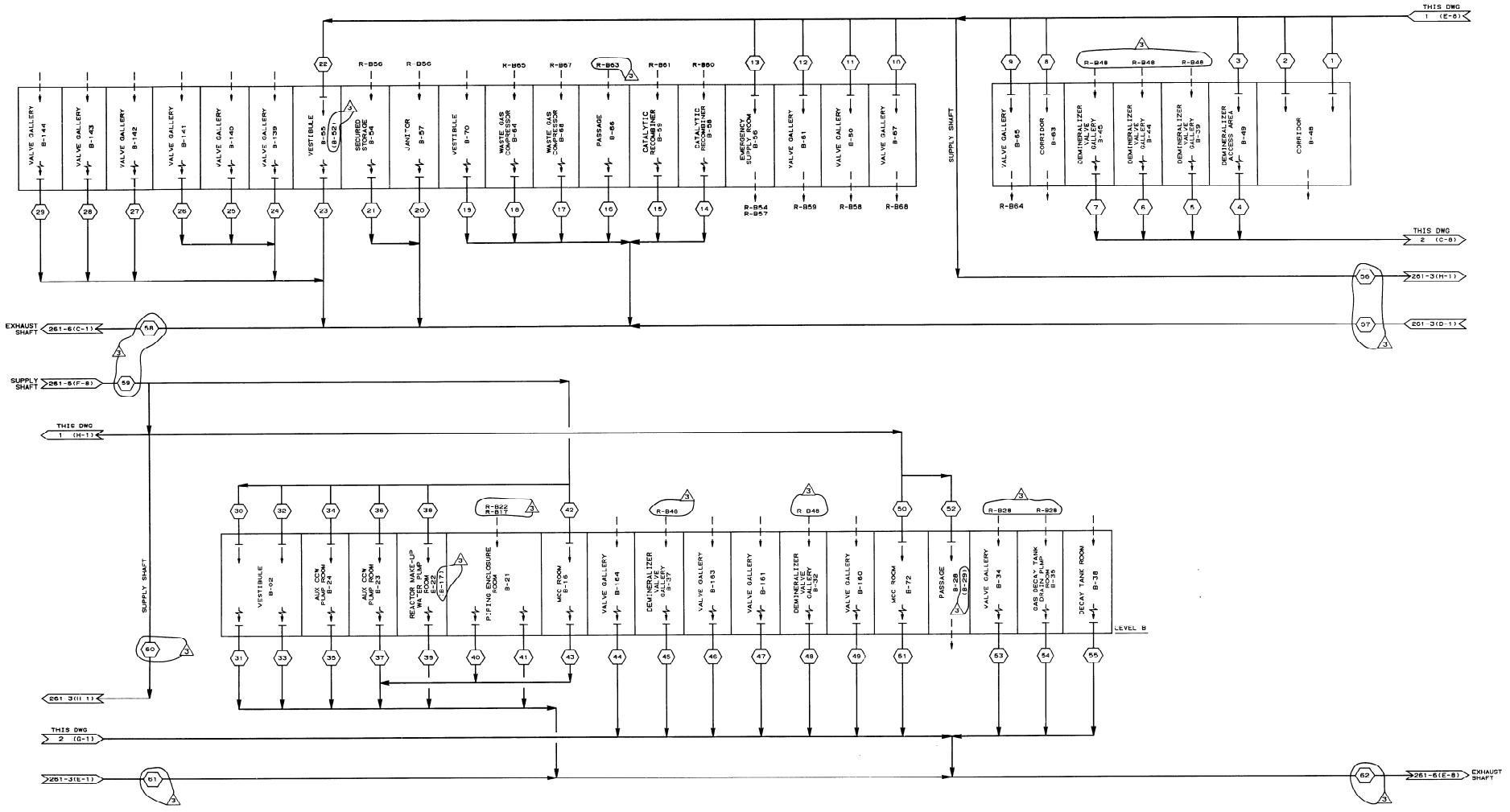
SCALE: NONE
DRAWING NO. 1X4DB261 1
REV. 7
JOB NO. 9510

"E" SIZE

NO.	REVISIONS	DATE	DR.	CHK.	DM	EGS	CHF.E	P.E.	Q.A.E.
Δ	INCORP. ASN 87-V1E0216A003	2-24-87	MER	[illegible]	DCD				
Δ	INCORP. ASN 87-V1G00A273	10-16-86	MWD	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Δ	INCORP. BY VERIFICATION	10-2-86	DR	CHK	MES	DM	[illegible]	[illegible]	[illegible]
Δ	REVISIONS								
Δ	INCORPORATED M-FCRB-16.044	8-19-86	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
Δ	INCORPORATE M-FCRB-16.404 & DCN #3 (PFE)	11-7-86	[illegible]	[illegible]	SEE MICROFILM FOR SIGNATURES				
Δ	INCORPORATE DCN'S 1 AND 2; TRANSFER TO PFE	4-4-86	[illegible]	[illegible]	SEE MICROFILM FOR SIGNATURES				
Δ	REVISED AND REDRAWN	1-11-86	[illegible]	[illegible]	SEE MICROFILM FOR SIGNATURES				
Δ	ISSUED FOR CONSTRUCTION	04-25-[illegible]	[illegible]	[illegible]	SEE MICROFILM FOR SIGNATURES				

ENG. MGR. DATE



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
NORMAL	FLOW	CFM	950	1325	265	375	115	145	100	1550	535	550	310	350	175	320	430	380	600	450	115	80	140	300	240	90	85	90	75	80	80	80	375	75	400	575	630	650	750	950	450	135
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
NORMAL	FLOW	CFM	55	150	85	30	150	85	95	110	110	230	210	300	115	160	330	9435	5575	8830	33095	14140	15775	20785
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

NOTES:

1. ROOM NUMBERS SHOWN IN PARENTHESIS ( ) INDICATE ADJOINING AREAS WHICH ARE NOT DIRECTLY SERVED BY THE VENTILATION SYSTEM BUT SHARE A COMMON AIR SPACE WITH ROOMS WHICH ARE SERVED BY THE VENTILATION SYSTEM.
2. REFERENCE P & ID 1X4DB208-2.
3. ROOMS B-26,B-29,B-46,B-63 AND B-70 HAVE DIRECTLY INTERCONNECTING AIR SPACES AND ARE CONSIDERED AS ONE AREA FOR DETERMINING AIR FLOW REQUIREMENTS.

CAD MICROSTATION NPSV-1

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EQL.	EGS	CNF.E	P.E.	Q.A.E
2	INCORP. ABN 89-V1000A279	10-4-89	WRD	SLP	REP.	X	X	X	X	X
1	REVISIONS	3-11-88	DR	CHK	DM	X	X	X	X	X
NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EQL.	EGS	CNF.E	P.E.	Q.A.E
3	INCORPORATED DCN #11 (PFE)	11-15-90	SEE MICROFILM FOR SIGNATURES							
2	REVISED AND REDRAWN	3-11-88	SEE MICROFILM FOR SIGNATURES							
1	ISSUED FOR CONSTRUCTION	08-30-79	SEE MICROFILM FOR SIGNATURES							
NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EQL.	EGS	CNF.E	P.E.	Q.A.E

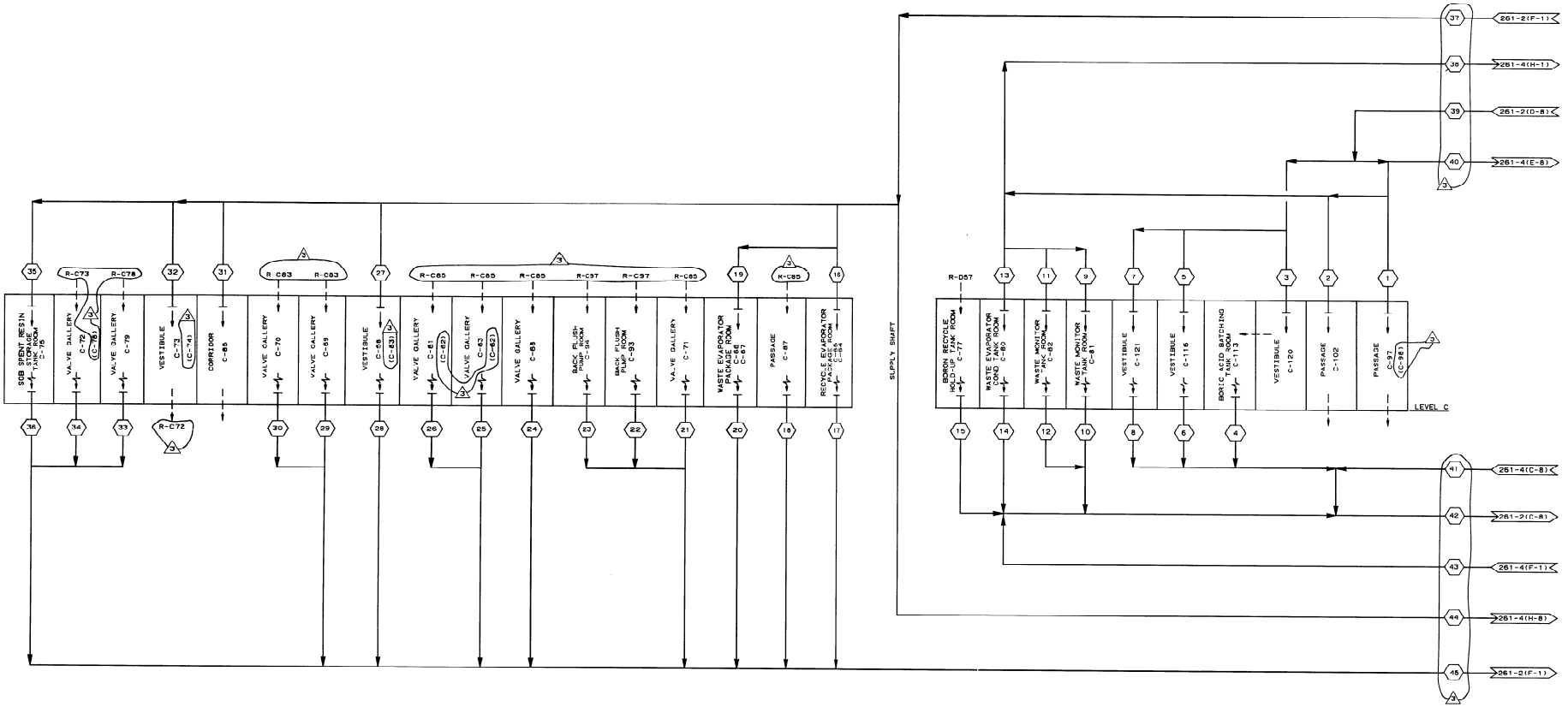
ENG. MGR. DATE

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
AUXILIARY BUILDING VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X4DB261-2	3



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
NORMAL	FLOW	CFM	2950	275	1350	720	150	110	2050	1850	115	190	115	190	200	400	750	1050	1050	550	1050	1175	100	140	55	85	800	700	350	260	160	95	650	235	40	175	385	150	9435	1395	14140	5540
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		41	42	43	44	45
NORMAL	FLOW	CFM	9875	15775	1690	5715	5575
	TEMP	°F	—	—	—	—	—

NOTES:

1. REFERENCE NOTE 1 ON DRAWING 1X4DB261-2.
2. REFERENCE P & ID 1X4DB208-2.
3. ROOMS C-58, C-83, C-85, C-97, C-98 AND C-102 HAVE DIRECTLY INTERCONNECTING AIR SPACES AND ARE CONSIDERED AS ONE AREA FOR DETERMINING AIR FLOW REQUIREMENTS.

CAD
MICROSTATION NPSV-1

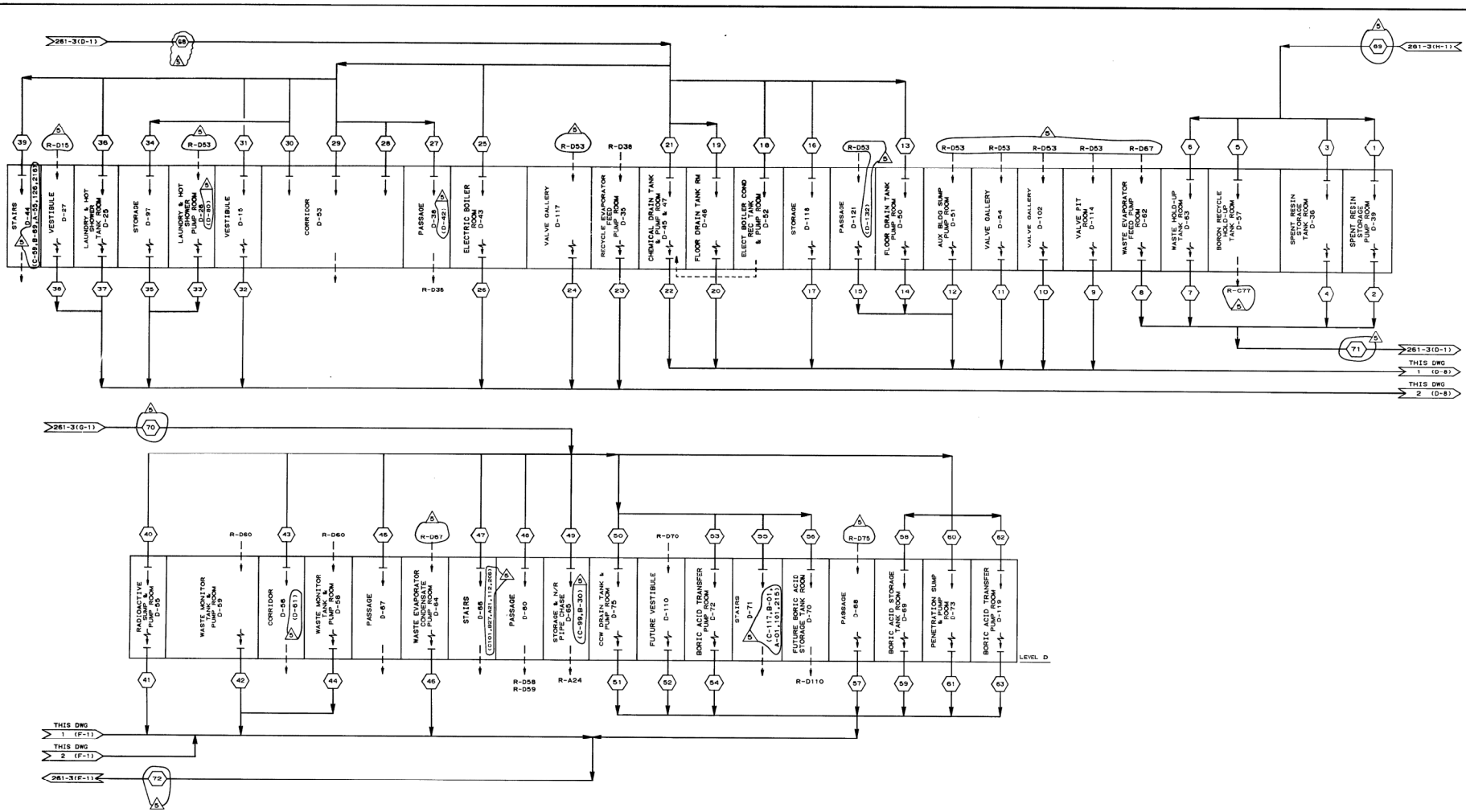
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
AUXILIARY BUILDING VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X4DB261-3	3

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EGL	EGS	CHF.E	P.E.	Q.A.E
	INCORP. ABN 89-Y1000A279	10-14-87	MWD	KJP	RCP	X	X	X	X	X
	REVISIONS	DATE	DR.	CHK.	SUP'V	EGL	EGS	CHF.E	P.E.	Q.A.E
2	INCORPORATED DCN #11 (RFE)	3-11-88	SEE MICROFILM FOR SIGNATURES							
1	REVISED AND REDRAWN	3-11-88	SEE MICROFILM FOR SIGNATURES							
0	ISSUED FOR CONSTRUCTION	06-30-78	SEE MICROFILM FOR SIGNATURES							
NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EGS	CHF.E	P.E.	Q.A.E	ENG. MGR. DATE



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
NORMAL	FLOW	CFM	480	550	190	275	625	100	165	700	100	425	250	550	135	200	300	115	100	750	450	400	175	875	450	150	300	250	420	975	700	500	600	600	525	140	100	155	160	100	300	135
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72								
NORMAL	FLOW	CFM	150	240	1050	250	530	525	225	475	950	740	750	425	190	250	240	425	550	240	750	150	200	190	250	100	100	100	100	5715	1395	5540	1690	9875								
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—								

NOTES:
1. REFERENCE NOTE 1 ON DRAWING 1X4DB261-2.
2. REFERENCE P & ID 1X4DB208-2.
3. DELETED
4. ROOMS D15,053,056,D61 AND D67 HAVE DIRECTLY INTERCONNECTING AIR SPACES AND ARE CONSIDERED AS ONE AREA FOR DETERMINING AIR FLOW REQUIREMENTS.

CAD
MICROSTATION NPSV- 2

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EQL	EGS	CHF.E	P.E.	Q.A.E
	INCORP. ABN 89-V1000A279	4-14-89	MMD	[illegible]	JP	X	K&P	X	X	X	X
	REV TO INC ABN 88-VCE112A001	5-1-89	SEE MICROFILM FOR SIGNATURES								
	REVISIONS	DATE	DR.	CHK.	RPE	DM	ODPM	DPM	DPM	X
	REVISIONS	DATE	DR.	CHK.	SUP'V	EQL	EGS	CHF.E	P.E.	Q.A.E
	INC NFORB-15,951 (MOD) & DCN 2	10-8-98	SEE MICROFILM FOR SIGNATURES							
	INCORPORATED DCN #13 (PFE)	10-15-96	SEE MICROFILM FOR SIGNATURES							
	REVISED AND RECRAWN	2-2-95	SEE MICROFILM FOR SIGNATURES							
	ISSUED FOR CONSTRUCTION	08-28-79	SEE MICROFILM FOR SIGNATURES							
	REVISIONS	DATE	DR.	CHK.	SUP'V	EGS	CHF.E	P.E.	Q.A.E	

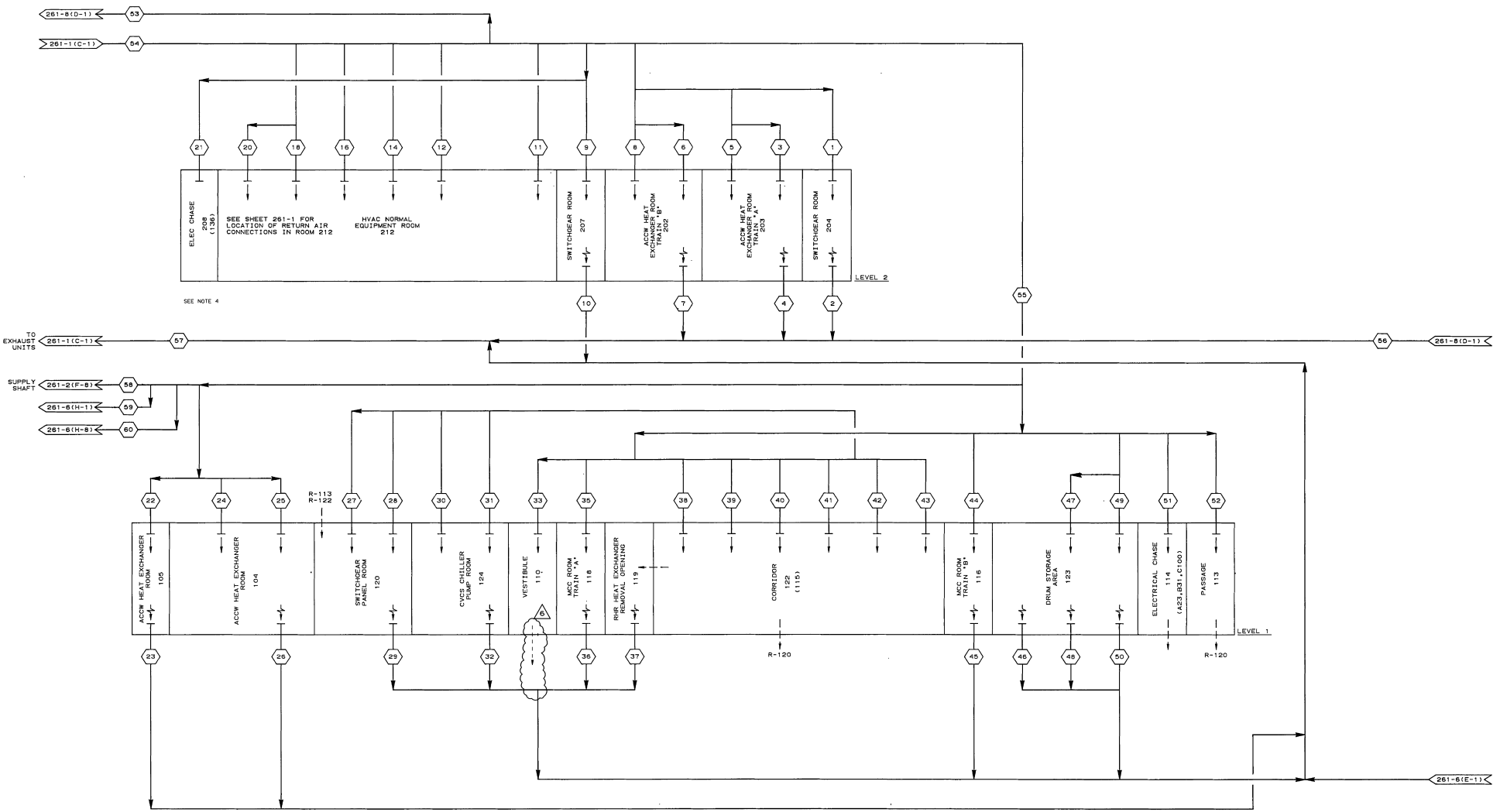
ENG. WOR. X
DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
AUXILIARY BUILDING VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

SCALE: NONE
DRAWING NO. 1X4DB261-4
REV. 5
JOB NO. 9510



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	35	36	37	38	39	40
NORMAL	FLOW	CFM	1450	1925	400	775	210	450	600	550	200	165	1500	1625	—	1400	—	1300	—	370	—	325		1250	900	575	550	1150	1000	925	3250	340	275	710	100	150	210	625	220	200	200
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

SEE NOTE 4

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
NORMAL	FLOW	CFM	200	200	100	100	150	250	320	250	265	200	260	220	14600	77800	53100	19800	73900	33095	190	12365	50435	42740
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

NOTES:

1. REFERENCE NOTE 1 ON DRAWING 1X4DB261-2.
2. REFERENCE P & ID 1X4DB208-3.
3. ROOMS 113,115 AND 122 HAVE DIRECTLY INTERCONNECTING AIR SPACES AND ARE CONSIDERED AS ONE AREA FOR DETERMINING AIR FLOW REQUIREMENTS.
4. SR BLANKED OFF BY MOC #94-VAM029

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
AUXILIARY BUILDING VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

SCALE: NONE

DRAWING NO. 1X4DB261-5

REV. 6

JOB NO. 10604

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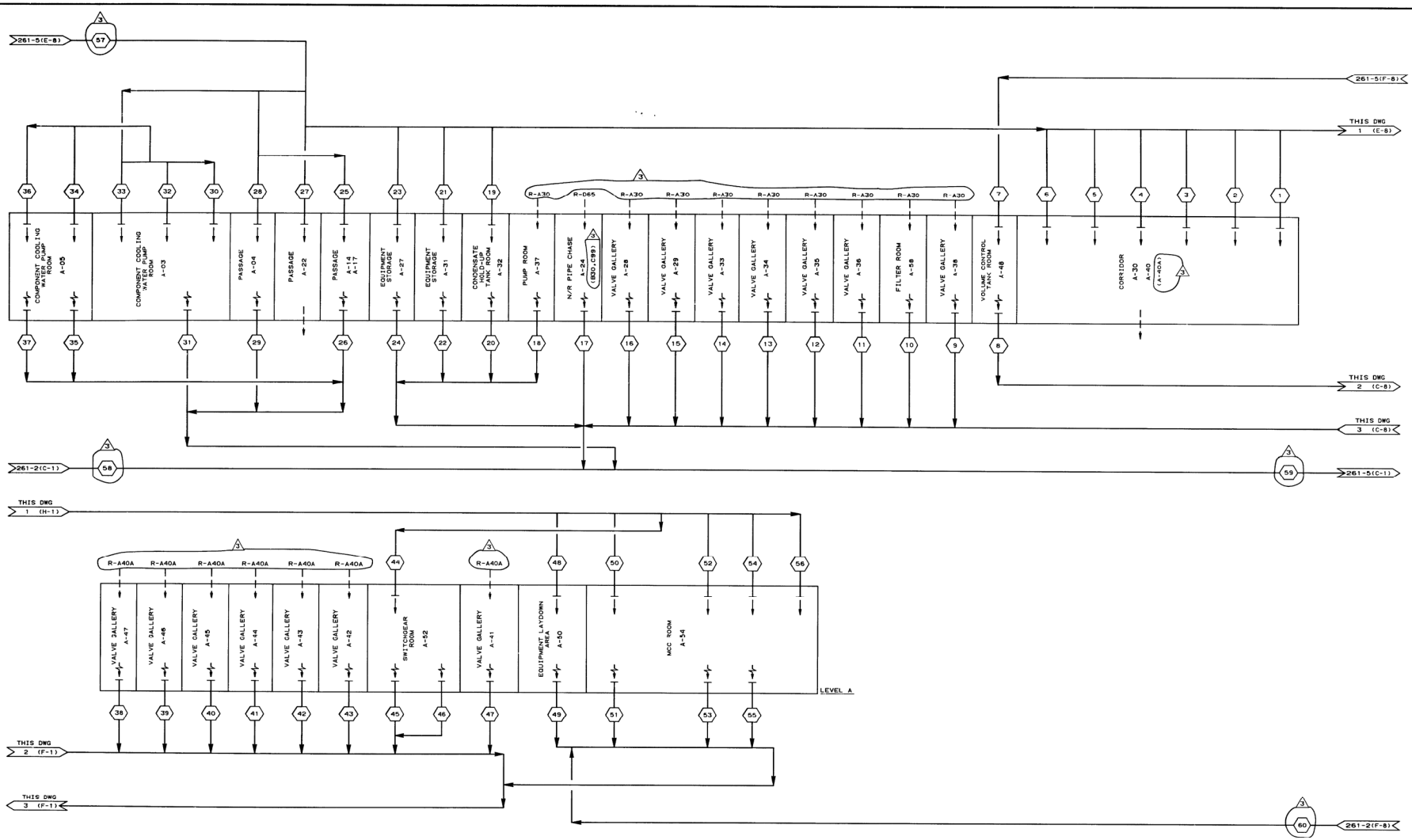
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORPORATED PER DCP 94-V1N0061	6/23/98	CD	MWD	EOG	
3	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SIZE E 34X44

CAD NAME 14DB261S



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
NORMAL	FLOW	CFM	750	675	750	1050	675	650	190	370	650	400	380	365	375	230	400	290	600	380	900	80	200	250	260	520	320	330	300	100	150	275	860	275	250	350	375	400	360	335	290	220
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
NORMAL	FLOW	CFM	225	250	225	900	500	575	220	550	625	500	650	480	680	530	725	600	12365	20785	42740	8830
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

NOTES:

1. REFERENCE NOTE 1 ON DWRAWING 1X4DB261-2.
2. REFERENCE P & ID 1X4DB208-3.
3. ROOMS A22,A30,A40 AND A40A HAVE DIRECTLY INTERCONNECTING AIR SPACES AND ARE CONSIDERED AS ONE AREA FOR DETERMINING AIR FLOW REQUIREMENTS.

CAD
MICROSTATION NPSV-1

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EGL	EGS	CHF.E	P.E.	Q.A.E
3	INCORP. ABN 89-V1000A279	10-16-89	MMD	[illegible]	[illegible]	X	X	X	X	X
2	REVISED AND REDRAWN	3-6-85	DR	CHK	DM	X	X	X	X	X
1	ISSUED FOR CONSTRUCTION	08-30-76								

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	EGS	CHF.E	P.E.	Q.A.E
3	INCORPORATED DCN #1: (PFE)	10-30-89		SEE MICROFILM FOR SIGNATURES					
2	REVISED AND REDRAWN	3-6-85		SEE MICROFILM FOR SIGNATURES					
1	ISSUED FOR CONSTRUCTION	08-30-76		SEE MICROFILM FOR SIGNATURES					

ENG. MGR.
DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

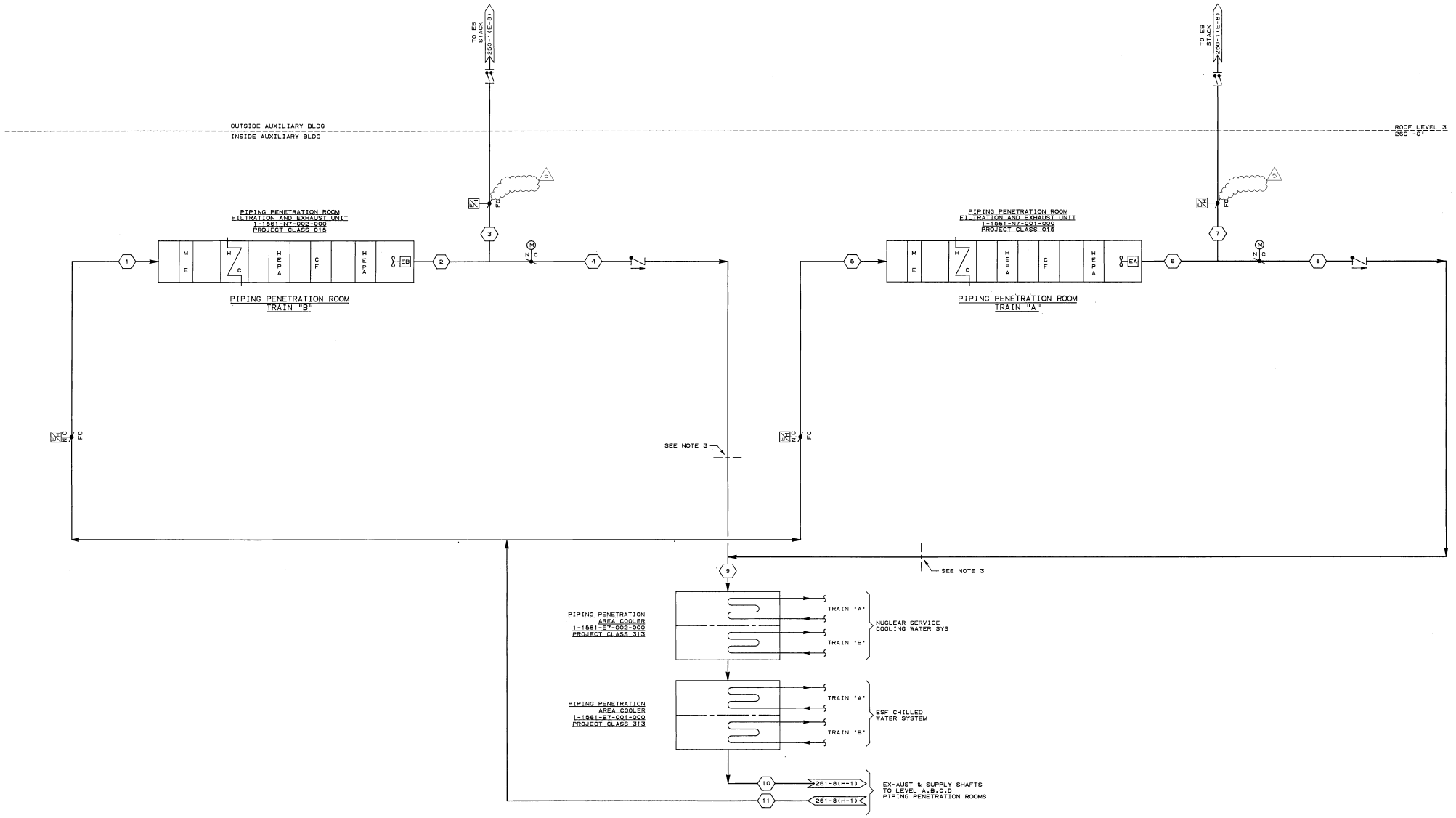
FLOW DIAGRAM
AUXILIARY BUILDING VENTILATION SYSTEM
SYSTEM NO. 1551 & 1553

SCALE: NONE

DRAWING NO. 1X4DB261-6

REV. 3

JOB NO. 9510



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11
EMERGENCY (MINIMUM)	FLOW	CFM	14460	14460	2700	11760	14460	14460	2700	11760	11760	11760	14460
	TEMP	°F			NOTE 1				NOTE 1				
					5				5				

NOTES:

1. THE STACK FLOW MINIMUM DESIGN VALUE HAS BEEN ESTABLISHED AT 2700 CFM. A VALUE OF 3000 CFM ± 10% MAY BE APPLIED TO FACILITATE TESTING.
2. REFERENCE P & ID 1X4DB205-1.
3. DAMPER SHALL BE USED FOR BLDG. PRESSURE BALANCE ONLY AND SHALL BE LOCKED IN POSITION WHEN PRESSURE BALANCE IS ATTAINED.
4. REFER TO NOTES 3 AND 4 SHOWN ON DRAWING 1X4DB261-9.

GEORGIA REGISTERED No. 14023 PROFESSIONAL DENNIS C. STANBERG ENGINEER
Dennis C. Stanberg 11/12/02

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1						
A	INCORP. PER ABN-51809	11/12/02	ELC	GLB	MWD	X

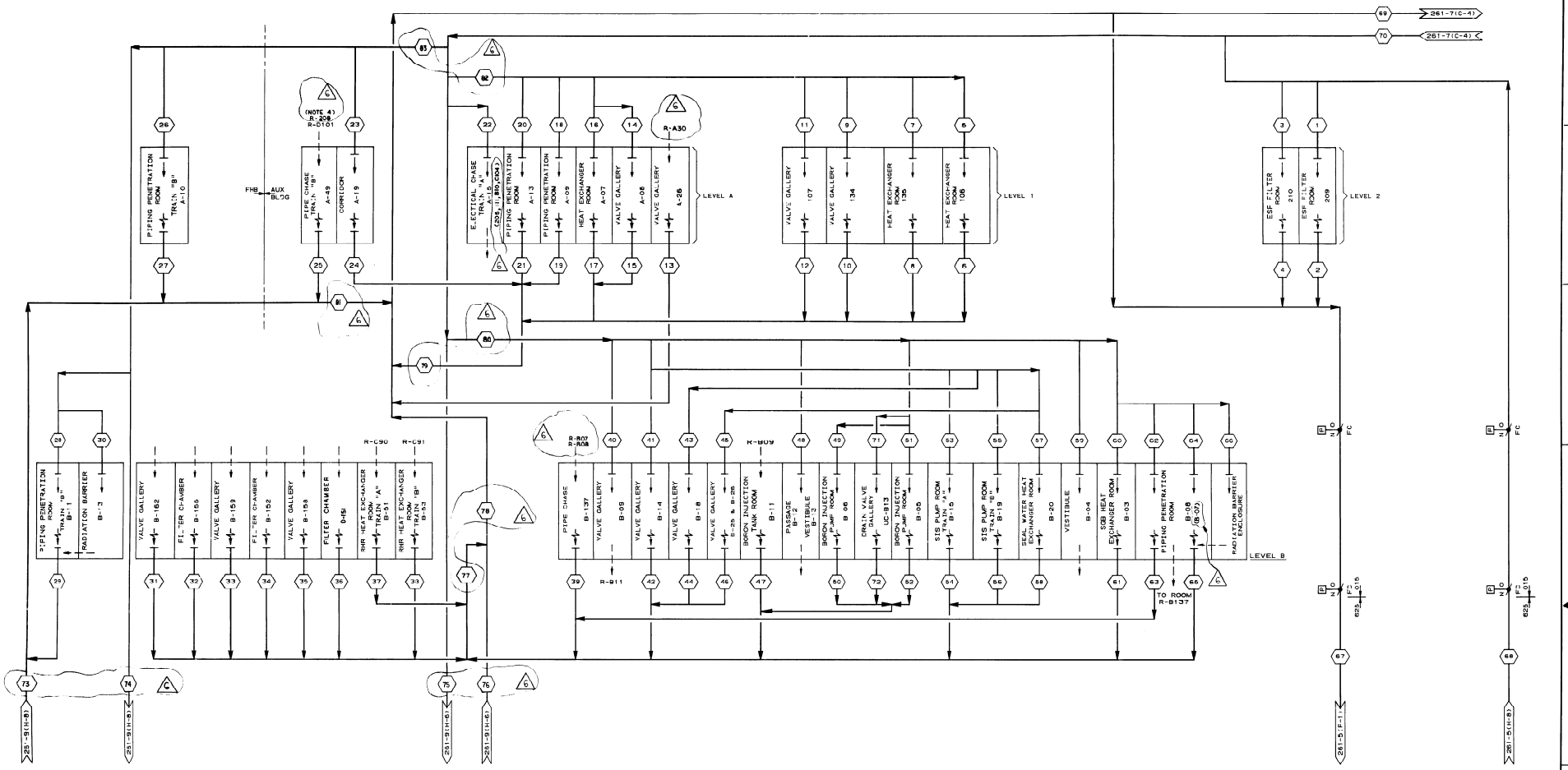
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
PIPING PENETRATION
FILTRATION AND EXHAUST SYSTEM
SYSTEM NO. 1561

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB261-7	5

SIZE E 34X44



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
NORMAL	FLOW	CFM	375	930	375	530	150	175	95	150	95	185	95	175	300	280	175	170	400	330	285	330	210	225	55	85	2050	550	1200	750	900	140	25	30	60	55	90	70	625	700	420	425
	TEMP	°F																																								
EMERGENCY (MINIMUM)	FLOW	CFM	300	376	290	396	119	126	74	105	75	135	76	126	215	158	126	134	287	260	203	260	148	178	42	60	1463	425	840	584	631	112	20	22	42	38	67	51	451	505	406	333
	TEMP	°F																																								

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
NORMAL	FLOW	CFM	75	90	60	90	100	80	180	400	90	35	145	95	145	200	150	210	125	625	250	500	400	500	750	425	350	275	19800	14800	0	0	150	50	785	490	6480	6435	5230	11665	1840	3815
	TEMP	°F																																								
EMERGENCY (MINIMUM)	FLOW	CFM	62	67	48	66	79	59	152	325	73	25	116	69	114	143	121	153	98	445	196	399	294	411	598	330	262	232	0	0	14400	11760	125	37	577	401	5190	4698	3966	8654	1316	3082
	TEMP	°F																																								

MODE	PARAMETER		81	82	83
NORMAL	FLOW	CFM	4935	1545	1985
	TEMP	°F			
EMERGENCY (MINIMUM)	FLOW	CFM	3511	1156	1564
	TEMP	°F			

NOTES:
1. ROOM NUMBERS SHOWN IN PARENTHESIS ( ) INDICATE ADJOINING AREAS WHICH ARE NOT DIRECTLY SERVED BY THE VENTILATION SYSTEM BUT SHARE A COMMON AIR SPACE WITH ROOMS WHICH ARE SERVED BY THE VENTILATION SYSTEM.
2. REFERENCE P & ID 1X4DB205-2.
3. AIR FLOW FROM ROOM 208 TO ROOM A49 OCCURS DURING NORMAL MODE ONLY.
4. REFER TO NOTES 3 AND 4 SHOWN ON DRAWING 0X4DB261-9.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
A	INCORP. ABN 89-V1000A279	10-07	C.I.B.	SJP	REP				
NO.	REVISIONS	DATE	DR.	CHKD.	DN				
A	INCORPORATED M-V-CHR-19,931	1/16		RC					
A	INCORPORATED DCN #3 M-FCRB-18,388 & M-FCRB-18,495	1/16	SEE MICROFILM FOR SIGNATURES						

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
A	INCORPORATED DCN #21 (PFE)	10/16	SEE MICROFILM FOR SIGNATURES						
A	INCORPORATED DCN 1	12-5-16	SEE MICROFILM FOR SIGNATURES						
A	REVISED AND REDRAWN	02-25-16	SEE MICROFILM FOR SIGNATURES						
A	ISSUED FOR CONSTRUCTION	06-30-16	SEE MICROFILM FOR SIGNATURES						

ENG. MGR. DATE 11/17/16

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
PIPING PENETRATION AREA FILTRATION AND EXHAUST SYSTEM
SYSTEM NO. 1061

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X4DB261-8	6

"C" SIZE

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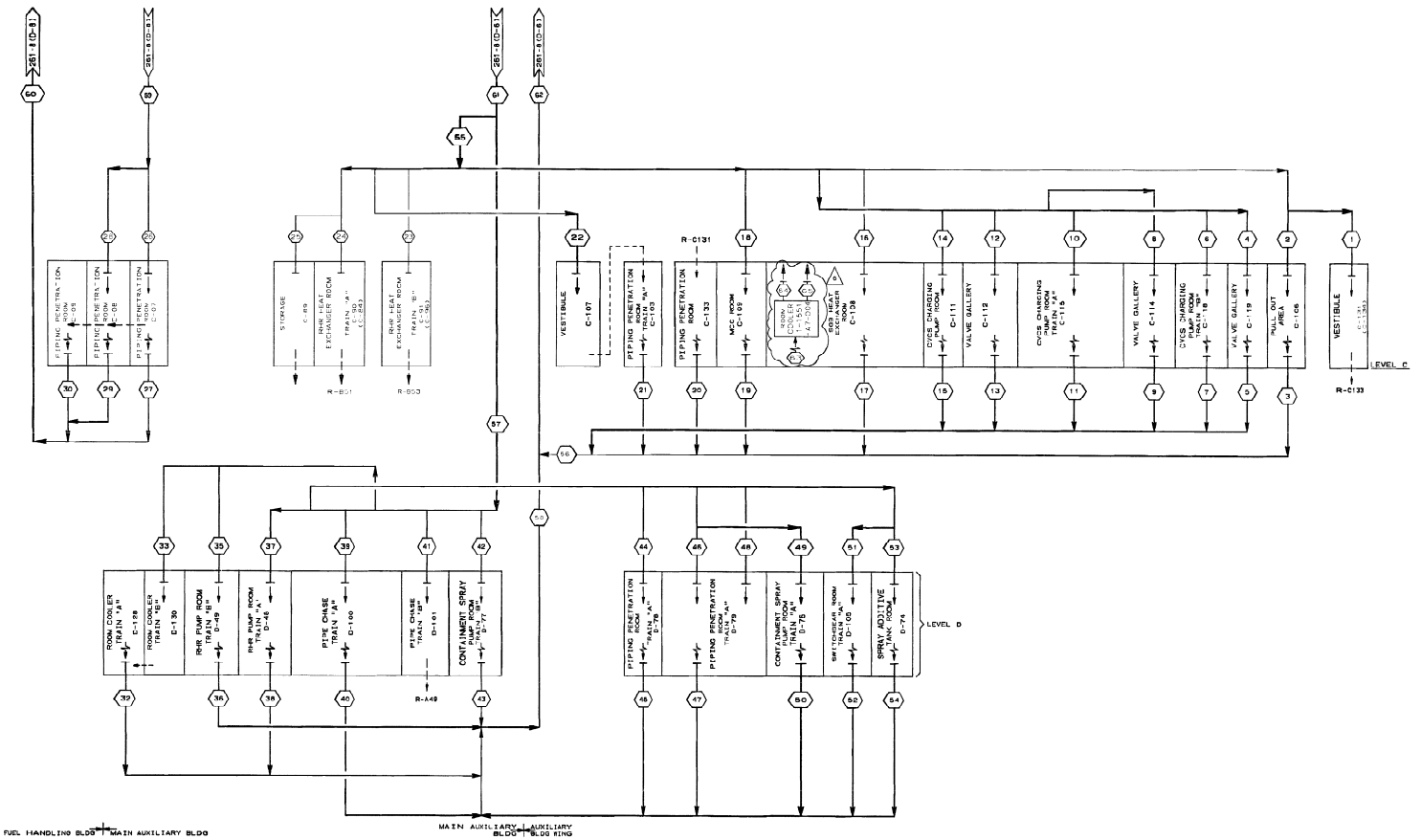


MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		32	33		35	36	37	38	39	40
NORMAL	FLOW	CFM	1150	350	275	45	90	140	300	75	130	75	105	60	110	205	350	1550	1550	35	275	1050	500	175	290	450	175	300	280	150	260	245		135	30		30	80	85	300	30	80
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		—	—		—	—	—	—	—	—
EMERGENCY (MINIMUM)	FLOW	CFM	923	286	204	35	65	112	187	62	95	60	121	46	77	210	268	1251	1131	27	198	772	363	139	232	360	139	246	208	155	191	178		90	25		26	44	68	219	27	61
	TEMP	°F	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		—	—		—	—	—	—	—	—

MODE	PARAMETER		41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65																
NORMAL	FLOW	CFM	875	65	200	80	220	60	300	85	75	115	80	85	175	210	4835	4755	4545	1680	490	785	6480	6435	7500	3750	3750																
	TEMP	°F																							104	70	70																
EMERGENCY (MINIMUM)	FLOW	CFM	699	52	147	62	162	45	221	46	57	85	65	40	134	151	3882	3472	1308	1275	401	577	5190	4698	N/A	N/A	N/A																
	TEMP	°F																							N/A	N/A	N/A																

NOTES
1. ROOM NUMBERS SHOWN IN PARENTHESIS ( ) INDICATE AREAS WHICH ARE NOT DIRECTLY SERVED BY THE VENTILATION SYSTEM BUT SHARE A COMMON AIR SPACE WITH ROOMS WHICH ARE SERVED BY THE VENTILATION SYSTEM.
2. REFERENCE P & ID 1X4DB205-2.
3. SUPPLY AND EXHAUST REGISTER FLOWS SHOWN ON THIS DRAWING AND 1X4DB261-8 REFLECT ADJUSTED TEST VALUES THE TEST VALUES HAVE BEEN CORRECTED BASED ON FLOW MEASUREMENTS BY DUCT TRAVERSE AND DOCUMENTED IN CALCULATION X4CI6C104. FUTURE FLOW MEASUREMENTS TO DETERMINE ACTUAL REGISTER FLOWS SHOULD CONSULT THIS CALCULATION FOR THE REQUIRED ADJUSTMENT FACTORS.
4. IF A SUPPLY FLOW IS INCREASED THE CORRESPONDING EXHAUST FLOW FOR THE SAME ROOM MUST BE INCREASED BY AN EQUAL OR GREATER AMOUNT TO MAINTAIN THE SAME OR GREATER DIFFERENTIAL BETWEEN SUPPLY AND EXHAUST FLOWS.

NO.	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM
5	INCORP. ARN-12793	8-3-92	GS	MJD					
NO.	ISSUED FOR CONSTRUCTION REVISIONS	DATE	DR	CHK	RPE	DM	DDPM	DPM	DRN

SEE MICROFILM FOR SIGNATURES

ENG. MGR. DATE

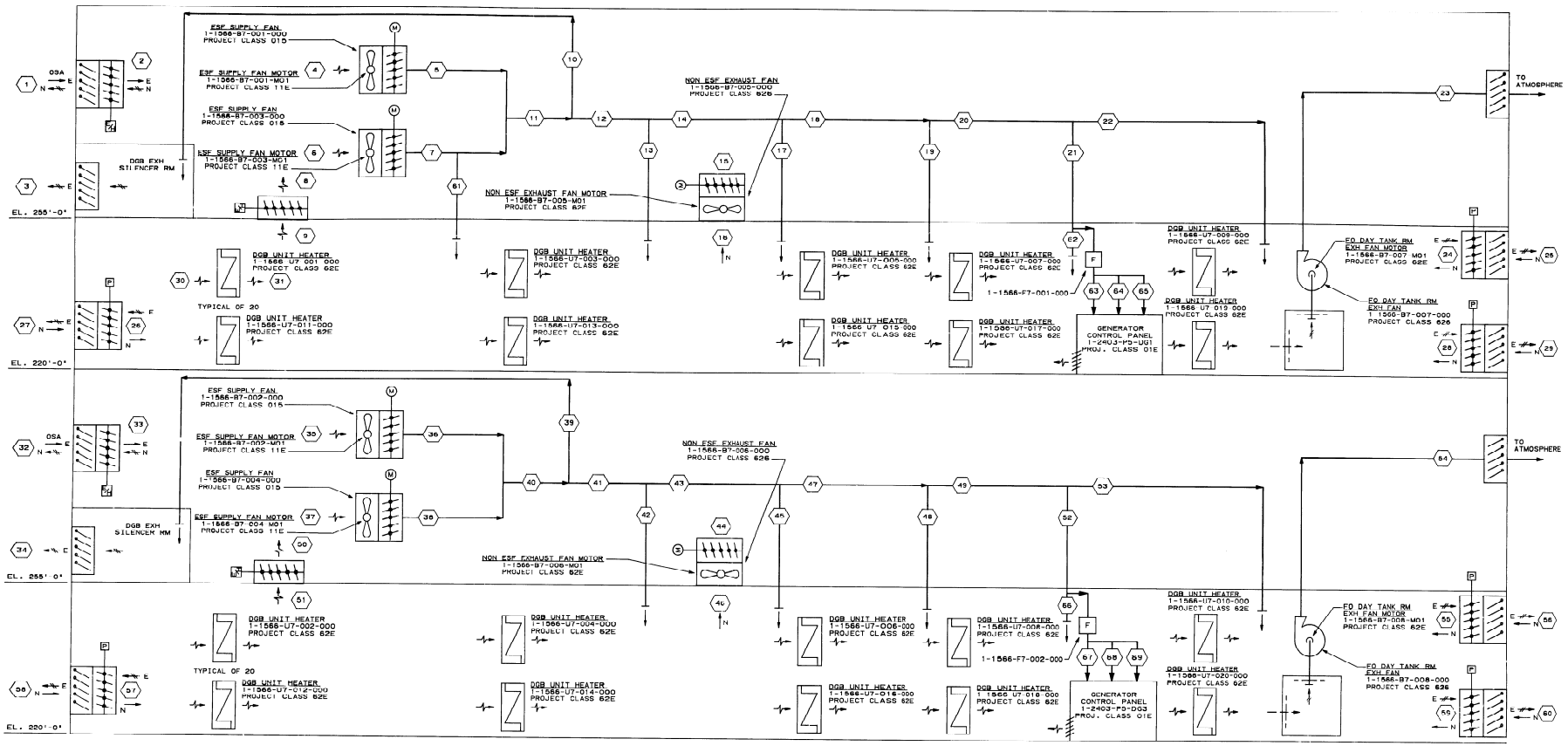
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
PIPING PENETRATION AREA FILTRATION AND EXHAUST SYSTEM
SYSTEM NO. 1561

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB261-9	6

SIZE E 34X44



MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
SUMMER NORMAL	FLOW	CFM	50,520	50,520	-	-	-	-	-	-	-	-	-	-	-	-	50,520	50,520	-	-	-	-	-	-	500	16,840	16,840	16,840	16,840	16,840	16,840	-
	TEMP	°F	120	120	-	-	-	-	-	-	-	-	-	-	-	-	120	120	-	-	-	-	-	-	120	98	98	98	98	98	98	
	PRESSURE	INCH OF WTR			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								-
SUMMER ESF 2 FAN	FLOW	CFM	150000	150000	7,500	75,000	76,000	75,000	75,000	-	-	7,500	100,000	97,500	30,000	67,500	-		30,000	37,500	22,000	15,500	1800	13900	500	47,300	47,300	47,400	47,400	47,300	47,300	-
	TEMP	°F	98	98	120	98	98	98	98	-	-	98	98	98	98	98	-		98	98	98	98	98	98	120	120	120	120	120	120	120	-
	PRESSURE	INCH OF WTR																														-
WINTER NORMAL	FLOW	CFM			-	-	-	-	-	-	-	-	-	-	-	-			-	-	-	-	-	-	500							1250
	TEMP	°F			-	-	-	-	-	-	-	-	-	-	-	-			-	-	-	-	-	-	50							50
	PRESSURE	INCH OF WTR			-																											
WINTER ESF 1 FAN	FLOW	CFM	80,000	80,000	5,700	97,500	97,500	97,500	97,500	32,000	32,000	5,700	88,000	82,300	30,000	52,300	-	-	17,800	34,800	20,000	14,800	1,400	13,400	500	19,800	19,800	19,700	19,700	19,800	19,800	1250
	TEMP	°F	17	17	120	28	28			50	50	28	28	28	28	28	-	-	28	28	28	28	28	28	50	50	50	50	50	50	50	
	PRESSURE	INCH OF WTR																														

MODE	PARAMETER		31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
SUMMER NORMAL	FLOW	CFM	-	50,520	50,520	-	-	-	-	-	-	-	-	-	-	50,520	-	50,520	-	-	-	-	-	-	-	500	16,840	16,840	16,840	16,840	16,840	16,840	—								
	TEMP	°F	-	120	120	-	-	-	-	-	-	-	-	-	-	120	-	120	-	-	-	-	-	-	-	120	98	98	98	98	98	98	—								
	PRESSURE	INCH OF WTR	-			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								—								
SUMMER ESF 2 FAN	FLOW	CFM	-	150000	150000	8,000	75,000	76,000	75,000	75,000	8,000	150000	142,000	34,000	108,000	-	34,000	-	74,000	50,000	24,000	-	-	2,000	22,000	500	47,200	47,300	47,100	47,100	47,200	47,200	45,000	525	350	350	475	800	350	420	430
	TEMP	°F	-	98	98	120	98	98	98	98	98	98	98	98	98	-	98		98	98	98	-	-	98	98	120	120	120	120	120	120	120	98	98	98	98	98	98	98	98	98
	PRESSURE	INCH OF WTR	-																			-	-										—								
WINTER NORMAL	FLOW	CFM	1250				-	-	-	-	-	-	-	-	-		-		-	-	-	-	-	-	-	500	16,840	16,840	16,840	16,840	16,640	16,840	—								
	TEMP	°F	74				-	-	-	-	-	-	-	-	-		-		-	-	-	-	-	-	-	50	17	17	17	17	17	17	—								
	PRESSURE	INCH OF WTR					-	-	-	-	-	-	-	-	-		-		-	-	-	-	-	-	-								—								
WINTER ESF 1 FAN	FLOW	CFM			80,000	80,000	4000	97,500	97,500	97,500	97,500	4,000	97,500	93,500	17,000	76,500	-	24,000	-	50,500	35,000	17,500	32,000	32,000	1,600	15,900	500	20,300	20,300	20,400	20,400	20,300	20,300	9,500	475	150	325	450	635	290	310	365
	TEMP	°F			17	17	120	28	28			28	28	28	28	28		28		28	28	28	50	50	28	28	50	50	50	50	50	50	50	28	28	28	28	28	28	28	28
	PRESSURE	INCH OF WTR																															—								

NOTES:
1. N - DESIGNATES NORMAL OPERATION.
2. E - DESIGNATES EMERGENCY OPERATION.
3. DELETED
4. FOR SUMMER MODE SINGLE FAN OPERATION (ESF), THE FLOW QUANTITIES SHALL BE SAME AS WINTER MODE ESF 1 FAN SHOWN AT NODES 31 & 35 & 36. SINGLE FAN OPERATION IS APPLICABLE TO FANS 1-1566-B7-001-000 AND 1-1566-B7-002-000 ONLY.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
5	INCORP. DCN 7, M-FCRB-16,727 & M-FCRB-19,881	11/88							

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGL	EGS	CHF.E	P.E.	Q.A.E.
4	INCORPORATED DCN NO. 6 (MOD)	11/88								
3	INCORPORATED DCN'S 2,3,4 AND 5(MOD)(PFE)	11-9-88								
2	INCORPORATED DCN 1 AND ADDED NOTE 3	1-9-89								
1	REVISED & REDRAWN	09-08-89								

ENG. MGR.	DATE
SEE MICROFILM FOR SIGNATURES	
SEE MICROFILM FOR SIGNATURES	
SEE MICROFILM FOR SIGNATURES	
SEE MICROFILM FOR SIGNATURES	

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
DIESEL GENERATOR BUILDING
HVAC SYSTEM 1566

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X4DB267	5

"E" SIZE

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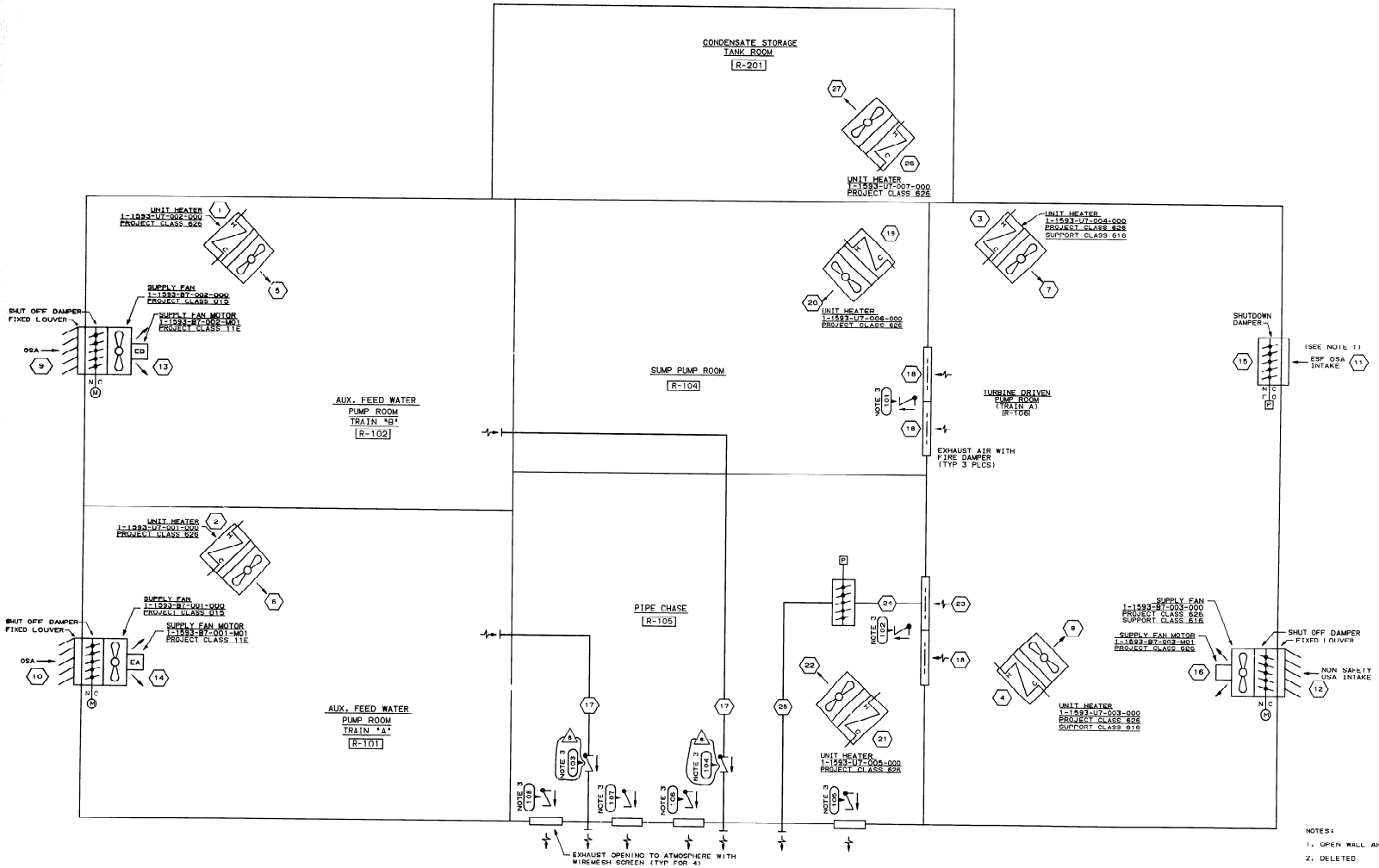


AUXILIARY FEEDWATER PUMP HOUSE

MODE	PARAMETER		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
NORMAL	FLOW	CFM	500	500	1250	1250	500	500	1250	1250	8400	8400	—	18000	8400	8400	—	18000	8400	6000	500	500	500	500	—	—	—	870	870
	TEMP	°F	VAR	VAR	VAR	VAR	VAR	VAR	VAR	VAR	17-98	17-98	—	17-98	17-98	17-98	—	17-98	40-104	40-104	VAR	VAR	VAR	VAR	—	—	—	VAR	VAR
	PRESSURE	INCH OF WTR																											
ACCIDENT	FLOW	CFM	—	—	—	—	—	—	—	—				—							—	—	—	—				—	—
	TEMP	°F	—	—	—	—	—	—	—	—	17-98	17-98	17-98	—	17-98	17-98	17-98	—	40-120	40-120	—	—	—	—				—	—
	PRESSURE	INCH OF WTR																							122	122	122		

NOTES:
1. OPEN WALL AREA TO BE 30 FT².
2. DELETED
3. BACKDRAFT DAMPER #101, 102, 103, 104, 105, 106, 107 AND 108 HAVE BLADES REMOVED.
4. REFER TO P&ID 1X4DB227
5. FOR ACCIDENT CONDITION, THE AIR FLOW SHALL VARY DEPENDING ON OUTSIDE AIR TEMPERATURE.

CAD MICROSTATION NPSV-1

NO.	REVISIONS	DATE	DR.	CHK.	SUP'V	ENGR	EGS	CHF.E	P.E.	Q.A.E
1	INCORPORATED ABN 88-V1N003A001	11-1-94								
2	INCORPORATED DCN 4 AND M-FCRB-15,560	11-21-94								
3	INCORPORATED DCN *218-FCRB 12881)((PFE)	10-18-95								
4	INCORPORATED DCN #1, & #2 UNDER BDG TRANSFICATION	11-1-96								
5	INCORPORATED DVR COMMENTS	03-31-97								
6	REVISED & REDRAWN RELOCATED TRAIN A & B EXHAUST DAMPER	05-26-98								
7	ISSUED FOR CONSTRUCTION	01-28-99								

	DES.	REP.
ELEC.	X	X
I&C	X	X
MECH.	X	X
CIVIL	X	X
CHK	DM	X
SUP'V	X	X

SEE MICROFILM FOR SIGNATURES

ENG. MGR. DATE

BECHTEL LOS ANGELES

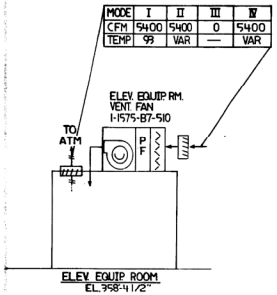
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

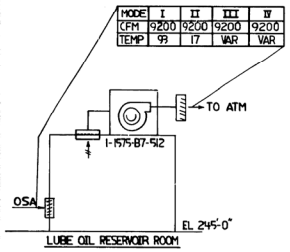
FLOW DIAGRAM
AUXILIARY FEEDWATER PUMP HOUSE
HEATING AND VENTILATING
SYSTEM NO. 1593

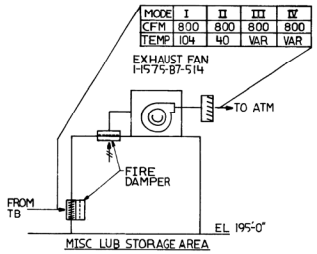
SCALE: NONE | DRAWING NO. 1X4DB268 | REV. 6 | JOB NO. 9510

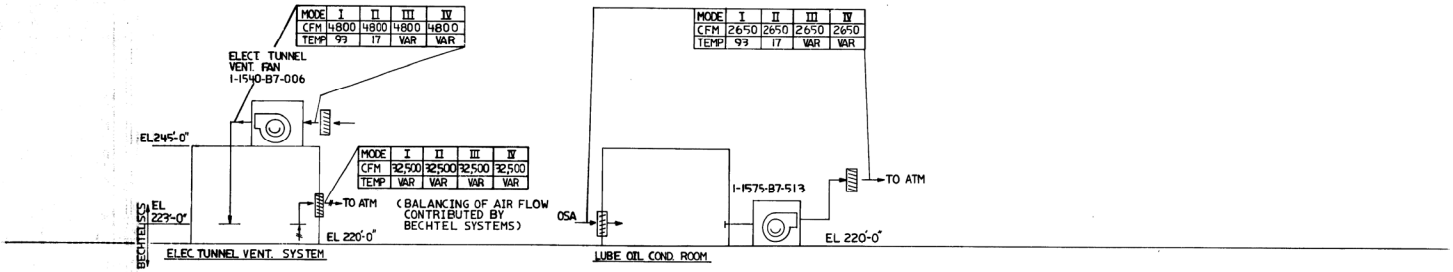
SIZE E 34X44

11-18-96









MODES OF OPERATION

I SUMMER
II WINTER
III SMOKE ISOLATION
IV SMOKE PURGE

NOTES:

1. WORK THIS DRAWING WITH 1X4DB301 AIR FLOW DIAGRAM
2. SEE DWG 1X4DD301 FOR ADDITIONAL NOTES AND REFERENCES

Southern Company Services, Inc.

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AIR FLOW DIAGRAM
TURBINE BUILDING HVAC SYSTEMS
SYSTEMS 1575

SCALE: NO SCALE
DRAWING NO. 1X4DB301-1
REV. 5
JOB NO. 9510

SIZE E 30244

REVISIONS

NO.	REVISIONS	DATE	DR.	CHK.	RPE	DM	ODPM	DPM	DPM	DRN
6	REV TO CLOSE MFCRS-109N LNO CHANGE REQUIRED	12/11	KKB	12/11	12/11	X	12/11	X	X	X
5	REV AIRFLOW AS PER M-FCRS-104N, 105N, 110N, AND MODIFIED M-FCRS-2896.	12-20-79	KKB	12-20-79	JWR	JWR	12-20-79			
5	REVISED SCS/BECHTEL INTERFACE ELEVATION	8-20-80	KKB	8-20-80	JWR	JWR	8-20-80			
5	SCSI APPROVAL RESPONSIBILITIES	10-23-80	DR	CHK	RPE	DM	ODPM	DPM	DPM	DRN
4	REVISED LUBE OIL COND RM. AIR FLOW DIAGRAM	10-23-80	TW	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80
4	ADDED MISC LUB STORAGE	10-23-80	DR	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80	10-23-80
4	ISSUE FOR CONSTRUCTION-	1-21-81	MR.S.	1-21-81	1-21-81	1-21-81	1-21-81	1-21-81	1-21-81	1-21-81
4	SCS RESPONSIBILITIES	12-20-79	DR.	CHK.	RPE	DM	ODPM	DPM	DPM	DRN
3		12-20-79		CHK.	SUPV.		EGS	CHF. E	P.E.	Q.A.E.

ENG. MGR. DATE

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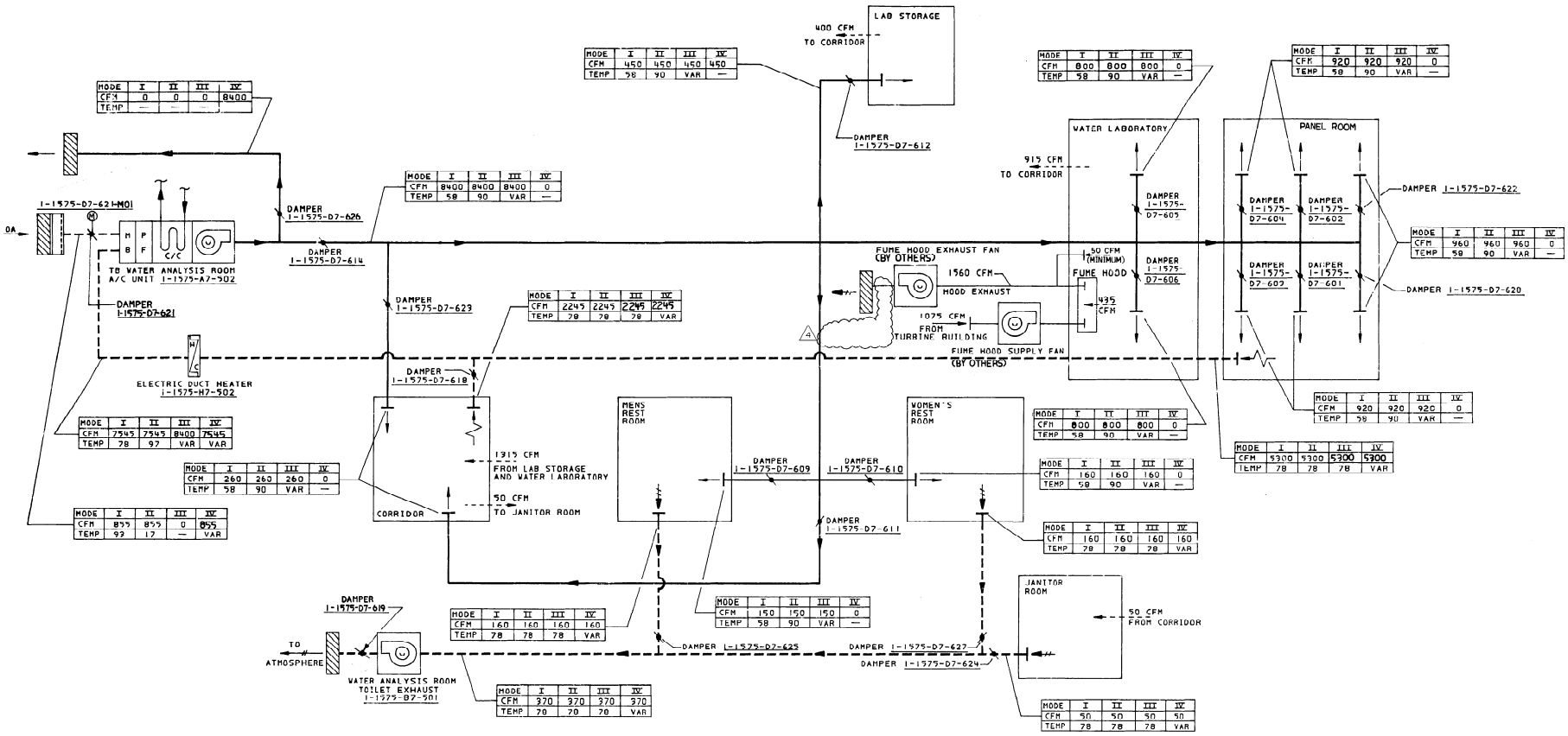

MODE	I	II	III	IV
CFM	0	0	0	8400
TEMP	—	—	—	—

MODE	I	II	III	IV
CFM	450	450	450	450
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	800	800	800	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	920	920	920	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	8400	8400	8400	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	960	960	960	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	2245	2245	2245	2245
TEMP	78	78	78	VAR

MODE	I	II	III	IV
CFM	7545	7545	8400	7545
TEMP	78	97	VAR	VAR

MODE	I	II	III	IV
CFM	800	800	800	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	920	920	920	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	260	260	260	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	5300	5300	5300	5300
TEMP	78	78	78	VAR

MODE	I	II	III	IV
CFM	855	855	0	855
TEMP	99	17	—	VAR

MODE	I	II	III	IV
CFM	160	160	160	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	160	160	160	160
TEMP	78	78	78	VAR

MODE	I	II	III	IV
CFM	160	160	160	160
TEMP	78	78	78	VAR

MODE	I	II	III	IV
CFM	150	150	150	0
TEMP	58	90	VAR	—

MODE	I	II	III	IV
CFM	970	970	970	970
TEMP	78	78	78	VAR

MODE	I	II	III	IV
CFM	50	50	50	50
TEMP	78	78	78	VAR

MODES OF OPERATION

I SUMMER
II WINTER
III SMOKE ISOLATION
IV SMOKE PURGE

NOTES:

1. TEMPERATURES SHOWN IN TABULATION ARE IN °F.
2. WORK THIS DWG WITH DWGS 1X4DJ1136 & 1X4DJ1197.
9. SEE DWG 1X4DB301 FOR ADDITIONAL NOTES AND REFERENCES.

REFERENCES:

1. 1X4DJ1136 - TURBINE BLDG-WATER ANALYSIS ROOM HVAC SYSTEM - PLAN AND ELEVATIONS LEVEL 1-EL 220'-0"
2. 1X4DJ1197 - TURBINE BLDG-WATER ANALYSIS ROOM HVAC SYSTEMS SECTIONS AND DETAILS LEVEL 1-EL 220'-0"
9. 1X4DB301 - AIR FLOW DIAGRAM-TURBINE BUILDING HVAC SYSTEMS-SYSTEMS 1575 & 1574

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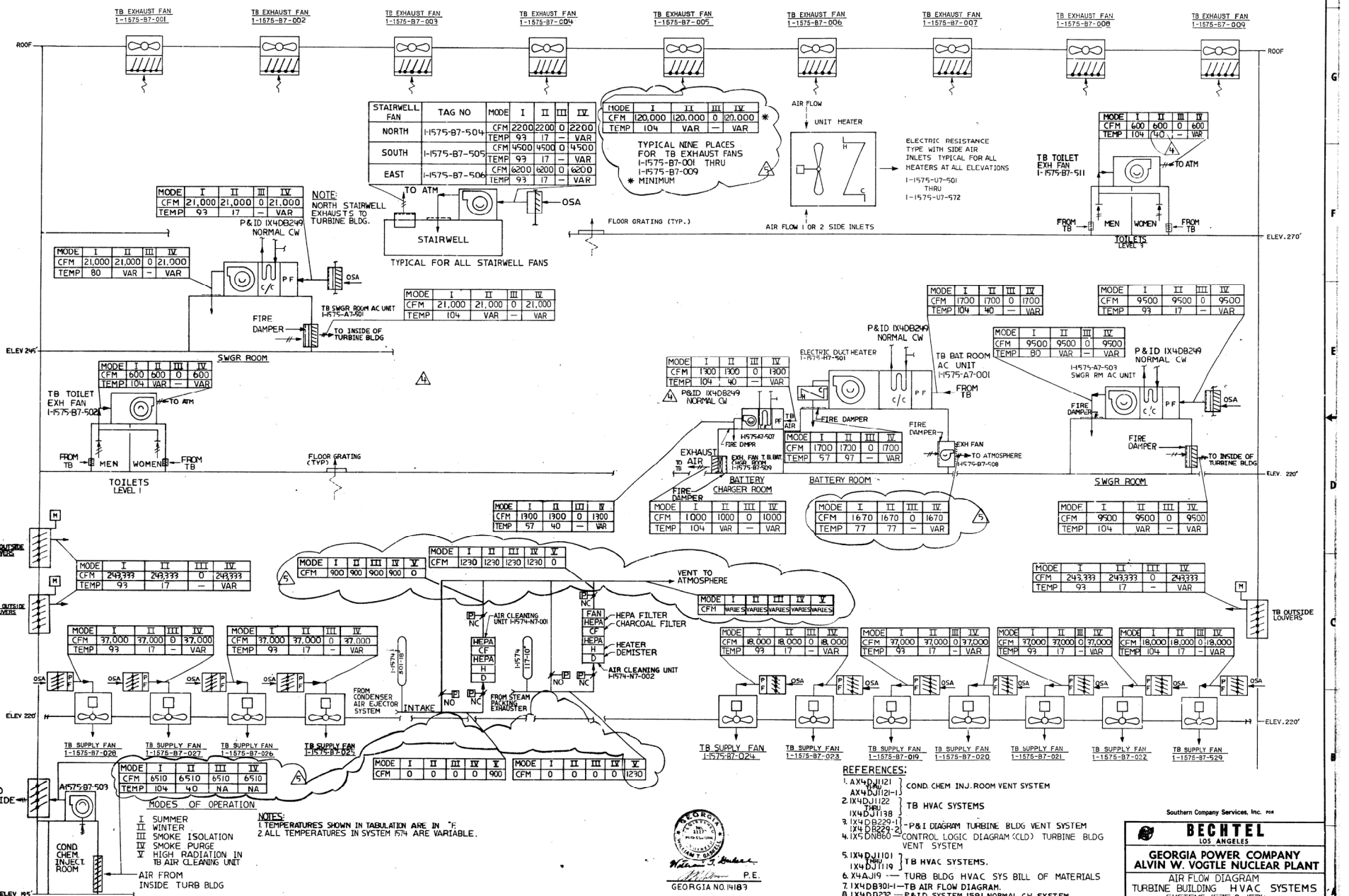
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 95-VAN0078	6/8/97	DS	EDG	JGM	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AIR FLOW DIAGRAM
TURBINE BLDG. HVAC SYSTEMS
WATER ANALYSIS ROOM SYSTEM 1575

SCALE: NONE | DRAWING NO. 1X4DB301-2 | REV. 4
JOB NO. 10604
SIZE E 36x44
CAD NAME 14DB3012



MODES OF OPERATION

I SUMMER
II WINTER
III SMOKE ISOLATION
IV SMOKE PURGE
V HIGH RADIATION IN TB AIR CLEANING UNIT

NOTES:
1. TEMPERATURES SHOWN IN TABULATION ARE IN °F.
2. ALL TEMPERATURES IN SYSTEM 1574 ARE VARIABLE.

REFERENCES:
1. AX4DJ1121 THRU AX4DJ1121-1 } COND. CHEM INJ. ROOM VENT SYSTEM
2. IX4DJ1122 THRU IX4DJ1138 } TB HVAC SYSTEMS
3. IX4DB229-1 THRU IX4DB229-2 } P&I DIAGRAM TURBINE BLDG VENT SYSTEM
4. IX5DN860 — CONTROL LOGIC DIAGRAM (CLD) TURBINE BLDG VENT SYSTEM
5. IX4DJ1101 THRU IX4DJ1119 } TB HVAC SYSTEMS.
6. X4AJ19 — TURB BLDG HVAC SYS BILL OF MATERIALS
7. IX4DB301-1 — TB AIR FLOW DIAGRAM.
8. IX4DB232 — P&ID SYSTEM 1591 NORMAL CW SYSTEM.

GEORGIA NO. 14183
P.E.

NO.	REVISIONS	DATE	DR.	CHK.	RPE	DM	ODPM	DPM	DPM	DRN
4	INCORP MFCRS-2897 & 2937 & 106N	9/27/74	S.K.							
3	DELETED WATER ANALYSIS R.M.	9/27/74	S.K.							
2	REVISED MODE I TEMP	9/27/74	S.K.							
1	REVISED SYSTEM 1574	9/20/74								
A	REVISED MODE 1,2 TEMPERATURES	6/19/75	BDK.							
B	RELOCATED 2 ELEC DUCT HTRS.	6/19/75	BDK.							
C	ADDED SMOKE PURGE LINE; ADDED 2 F.D.'S	6/19/75	BDK.							
D	REMOVED PRELIM. STICKER; ADDED CHART	2/3/76	MRS.							
E	CHNG. CFM NOS. MH-035	10/6/76	DWM							
F	REVISED TO FINAL DESIGN	3/3/77	MRS.							
G	ISSUED FOR CONSTRUCTION	7-20-78								

ENG. MGR. DATE

Southern Company Services, Inc. FOR
BECHTEL LOS ANGELES
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
AIR FLOW DIAGRAM
TURBINE BUILDING HVAC SYSTEMS
SYSTEMS 1575 & 1574

SCALE: NO SCALE
DRAWING NO. IX4DB301
REV. 5
JOB NO. 9510
SIZE E 36X24

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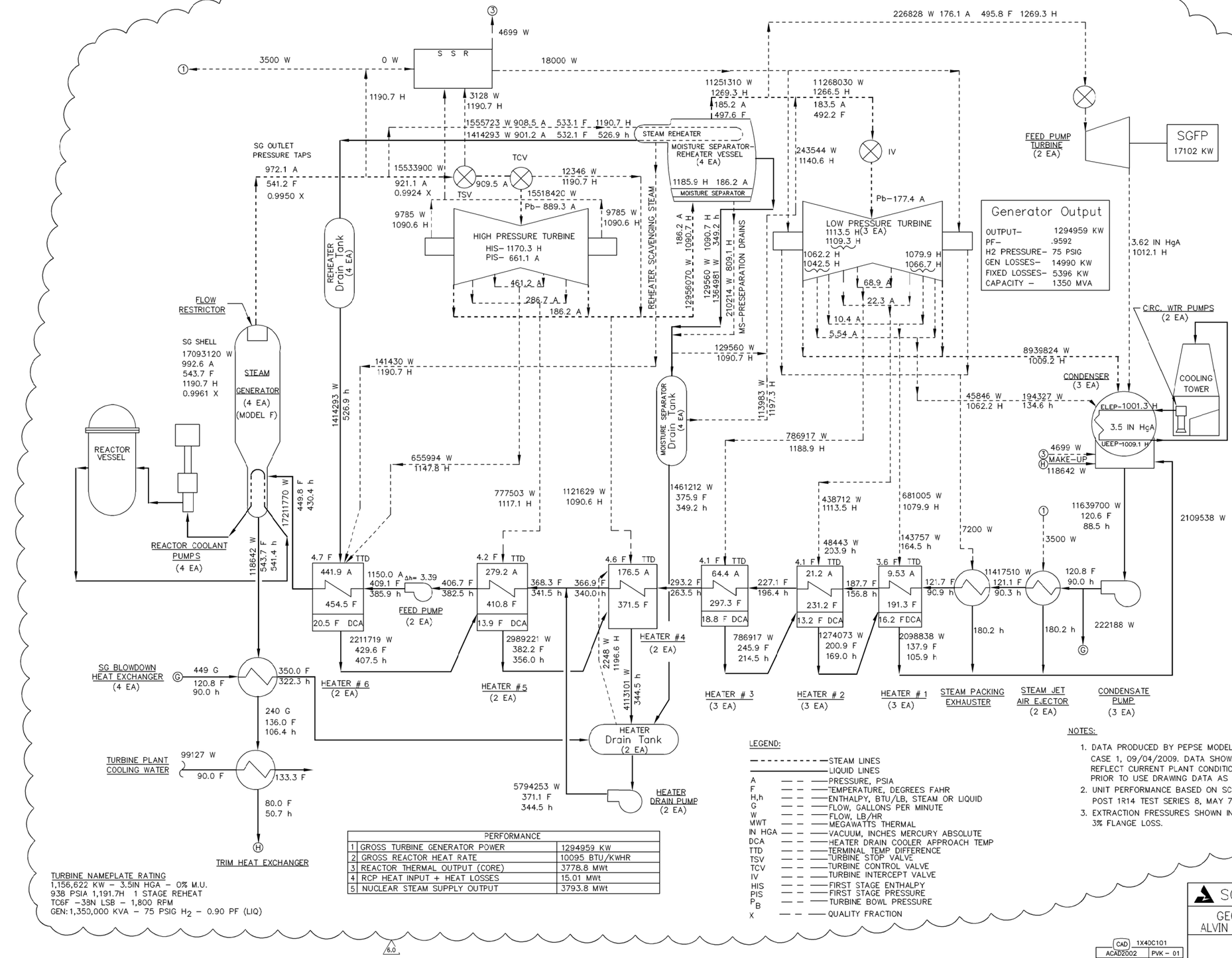

PERFORMANCE		
1	GROSS TURBINE GENERATOR POWER	1294959 KW
2	GROSS REACTOR HEAT RATE	10095 BTU/KWHR
3	REACTOR THERMAL OUTPUT (CORE)	3778.8 MWt
4	RCP HEAT INPUT + HEAT LOSSES	15.01 MWt
5	NUCLEAR STEAM SUPPLY OUTPUT	3793.8 MWt

TURBINE NAMEPLATE RATING
1,156,622 KW - 3.5IN HGA - 0% M.U.
938 PSIA 1,191.7H 1 STAGE REHEAT
TC6F -38N LSB - 1,800 RPM
GEN: 1,350,000 KVA - 75 PSIG H_2 - 0.90 PF (LIQ)

LEGEND:

	STEAM LINES
	LIQUID LINES
A	PRESSURE, PSIA
F	TEMPERATURE, DEGREES FAHR
H,h	ENTHALPY, BTU/LB, STEAM OR LIQUID
G	FLOW, GALLONS PER MINUTE
W	FLOW, LB/HR
MWT	MEGAWATTS THERMAL
IN HGA	VACUUM, INCHES MERCURY ABSOLUTE
DCA	HEATER DRAIN COOLER APPROACH TEMP
TTD	TERMINAL TEMP DIFFERENCE
TSV	TURBINE STOP VALVE
TCV	TURBINE CONTROL VALVE
IV	TURBINE INTERCEPT VALVE
HIS	FIRST STAGE ENTHALPY
PIS	FIRST STAGE PRESSURE
P_B	TURBINE BOWL PRESSURE
X	QUALITY FRACTION

NOTES:

1. DATA PRODUCED BY PEPSE MODEL V01AB5R2, CASE 1, 09/04/2009. DATA SHOWN MAY NOT REFLECT CURRENT PLANT CONDITIONS. USE CAUTION PRIOR TO USE DRAWING DATA AS A DESIGN INPUT.
2. UNIT PERFORMANCE BASED ON SCS/GPP POST 1R14 TEST SERIES 8, MAY 7-8, 2009.
3. EXTRACTION PRESSURES SHOWN INCLUDE 3% FLANGE LOSS.

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NO.		DATE	DR	CHK	APPV	DTL
5.0	REVISED PER ABN 1053871301M002, VER.2.0	5/22/13	PVK	RBH	JMR	

CAD 1X4DC101
ACAD2002 PVK - 01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
UNIT 1 HEAT BALANCE DIAGRAM
TURBINE VWO (MAX. CALCULATED)

SCALE: NONE
DRAWING NO. 1X4DC101
VER. 6.0
JOB NO.10604
SIZE E 34X44

Vogtle Electric Generating Plant

Unit No. 1

Document Number:	Sheet #
1X4DC102	1

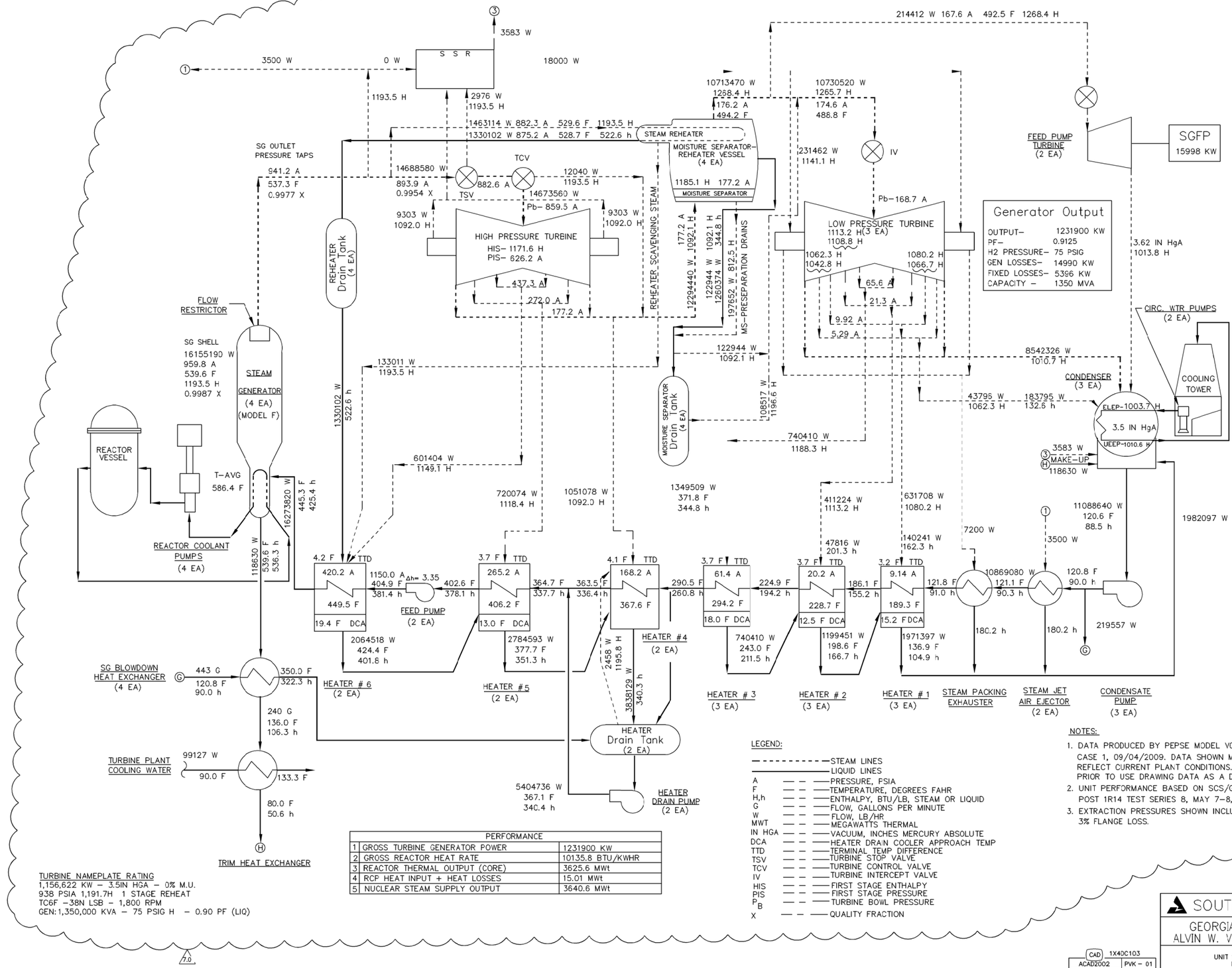
Document Title:

HEAT BALANCE DIAGRAM TURBINE GUARANTEE

VOIDED

VOIDED Per: ABN 1053871301M003, VER. 2.0

Ver. 6.0



PERFORMANCE		
1	GROSS TURBINE GENERATOR POWER	1231900 KW
2	GROSS REACTOR HEAT RATE	10135.8 BTU/KWHR
3	REACTOR THERMAL OUTPUT (CORE)	3625.6 MWt
4	RCP HEAT INPUT + HEAT LOSSES	15.01 MWt
5	NUCLEAR STEAM SUPPLY OUTPUT	3640.6 MWt

TURBINE NAMEPLATE RATING
1,156,622 KW - 3.5IN HGA - 0% M.U.
938 PSIA 1,191.7H 1 STAGE REHEAT
TC6F -38N LSB - 1,800 RPM
GEN: 1,350,000 KVA - 75 PSIG H - 0.90 PF (LIQ)

LEGEND:

----------	STEAM LINES
————	LIQUID LINES
A	PRESSURE, PSIA
F	TEMPERATURE, DEGREES FAHR
H,h	ENTHALPY, BTU/LB, STEAM OR LIQUID
G	FLOW, GALLONS PER MINUTE
W	FLOW, LB/HR
MWT	MEGAWATTS THERMAL
IN HGA	VACUUM, INCHES MERCURY ABSOLUTE
DCA	HEATER DRAIN COOLER APPROACH TEMP
TTD	TERMINAL TEMP DIFFERENCE
TSV	TURBINE STOP VALVE
TCV	TURBINE CONTROL VALVE
IV	TURBINE INTERCEPT VALVE
HIS	FIRST STAGE ENTHALPY
PIS	FIRST STAGE PRESSURE
PB	TURBINE BOWL PRESSURE
X	QUALITY FRACTION

NOTES:
1. DATA PRODUCED BY PEPSE MODEL V01AB5R2, CASE 1, 09/04/2009. DATA SHOWN MAY NOT REFLECT CURRENT PLANT CONDITIONS. USE CAUTION PRIOR TO USE DRAWING DATA AS A DESIGN INPUT.
2. UNIT PERFORMANCE BASED ON SCS/GPP POST 1R14 TEST SERIES 8, MAY 7-8, 2009.
3. EXTRACTION PRESSURES SHOWN INCLUDE 3% FLANGE LOSS.

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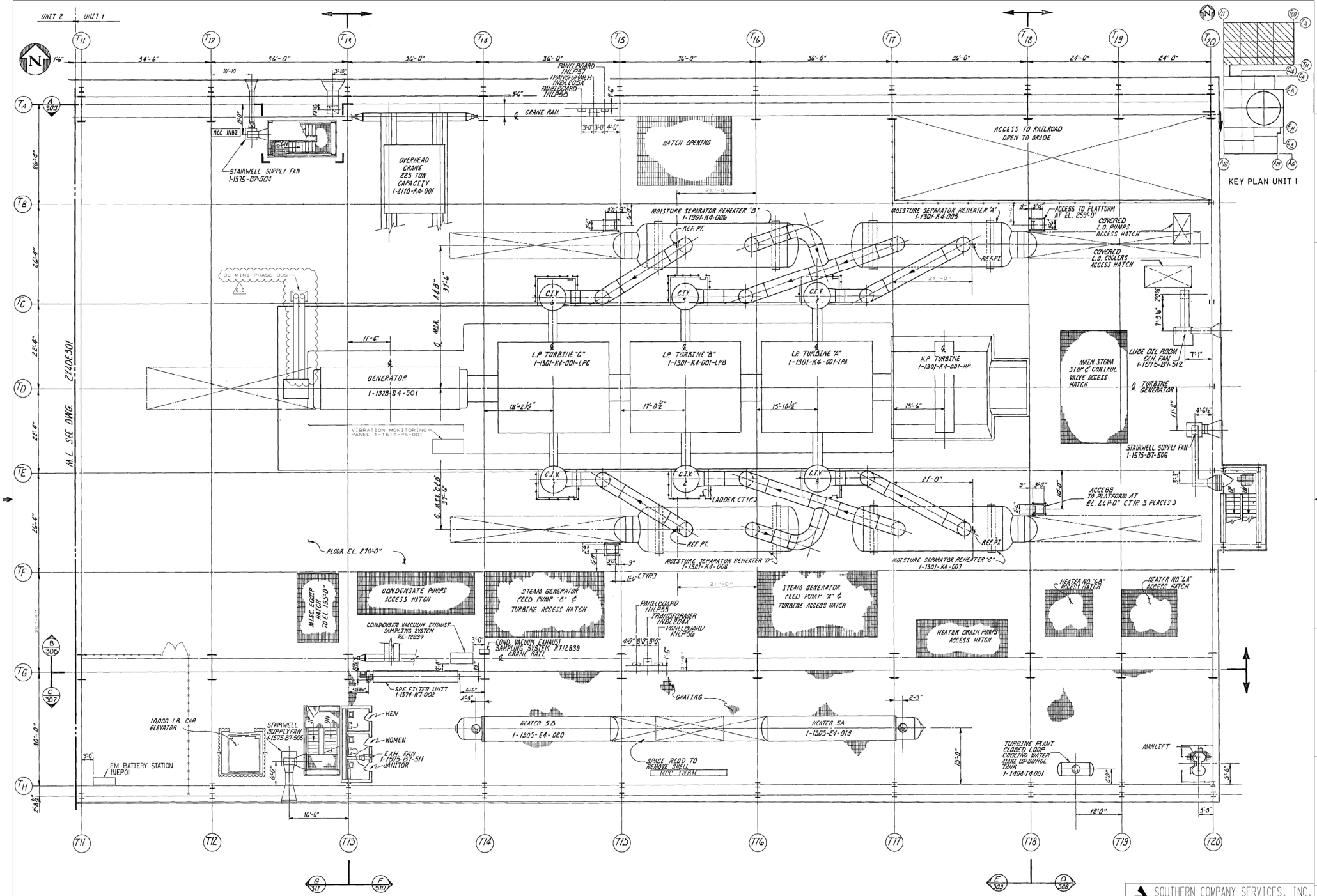
NO.		DATE	DR	CHK	APPV	DTL
7.0	REVISED PER ABN 1053871301M004, VER.2.0	5/22/13	PVK	RBH	JMR	

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

UNIT 1 HEAT BALANCE DIAGRAM
REACTOR WARRANTY

CAD 1X4DC103
ACAD2002 PVK - 01

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DC103
VER. 7.0
SIZE E 34X44



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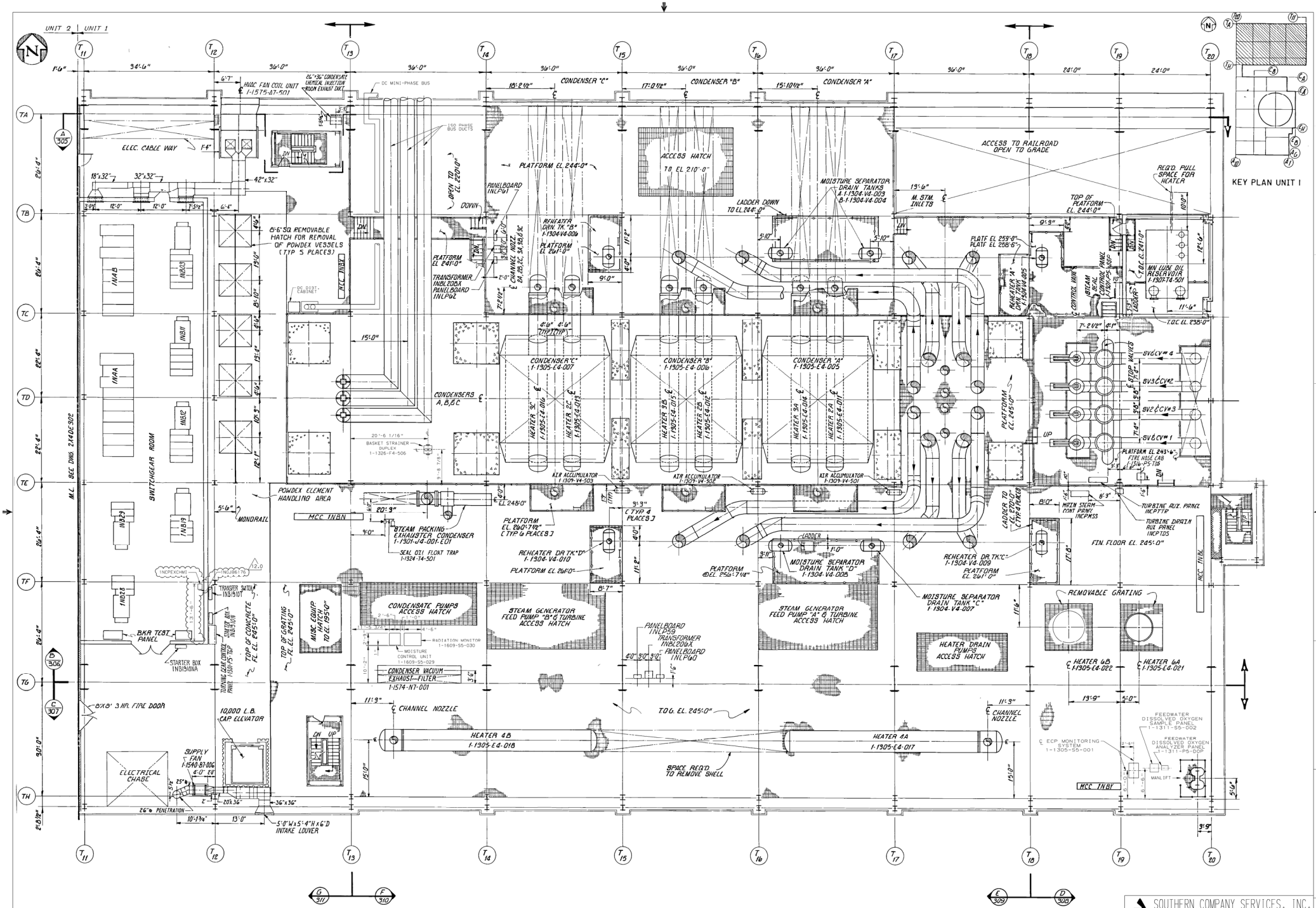
NO.	VERSIONS	DATE	DR	CHK	APPV
6.0	REVISED PER ABN 1049001501C009, VER. 1.0	6-5-08	JLO	RBH	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TURBINE BLDG.
EQUIPMENT LOCATION PLAN
LEVEL 3, ELEV. 270'-0" & ABOVE

SCALE: 1/8" = 1'-0"
JOB NO. 10604
DRAWING NO. 1X4DE301
VER. 6.0
SIZE E 34X44



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NO.	VERSIONS	DATE	DR	CHK	APPV
12.0	REVISED PER ABN 10490015O1E158, VER. 2.0	8-27-09	Vep	MWD	JMR

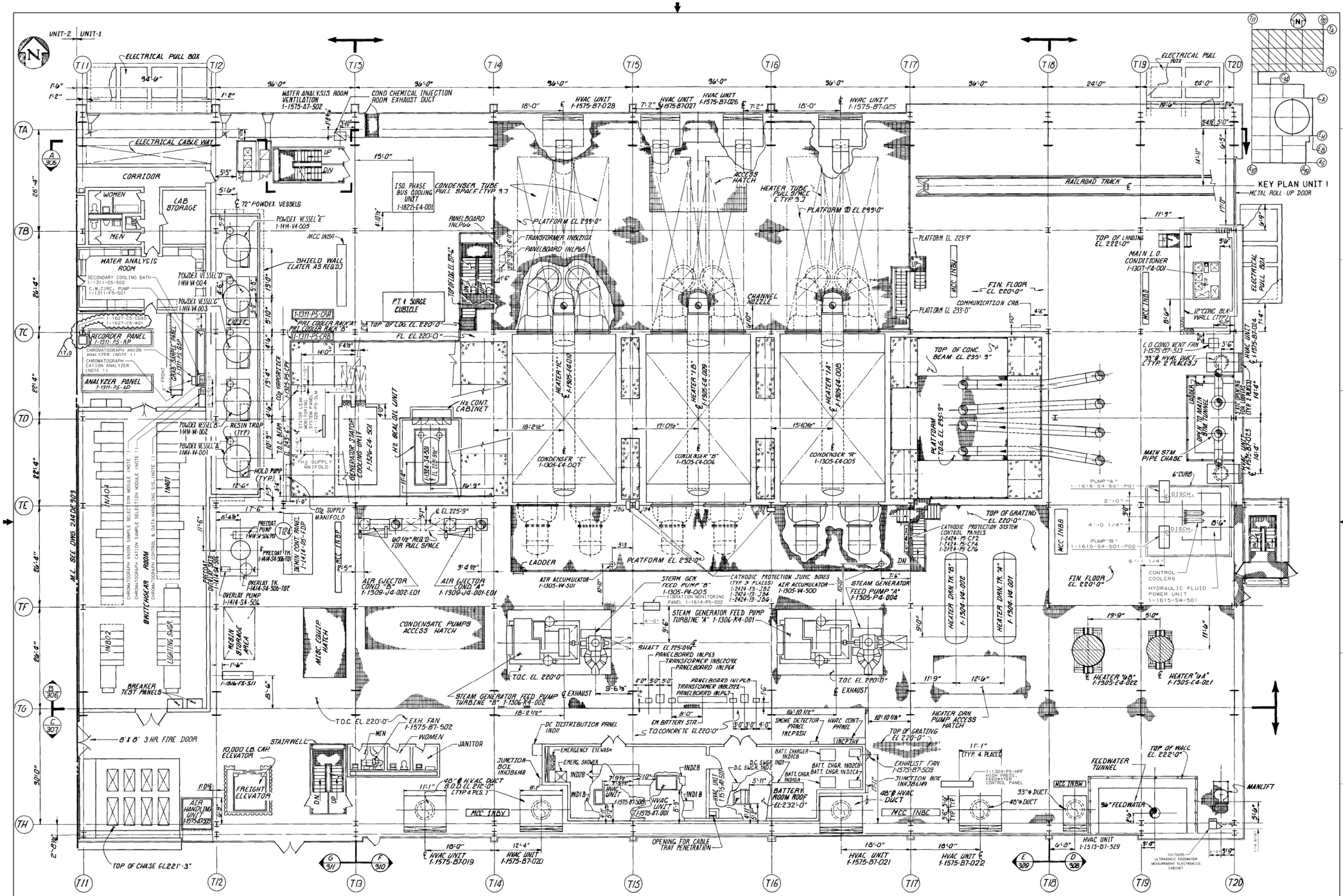
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TURBINE BLDG.
EQUIPMENT LOCATION PLAN
LEVEL 2, ELEV. 245'-0" TO 270'-0"

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DE302	12.0

SIZE E 34x44



NOTES:
1. TPSS CHROMATOGRAPH ANALYTICAL PROCESS ANALYZER (1-1311-P5-CPA) COMPONENT.

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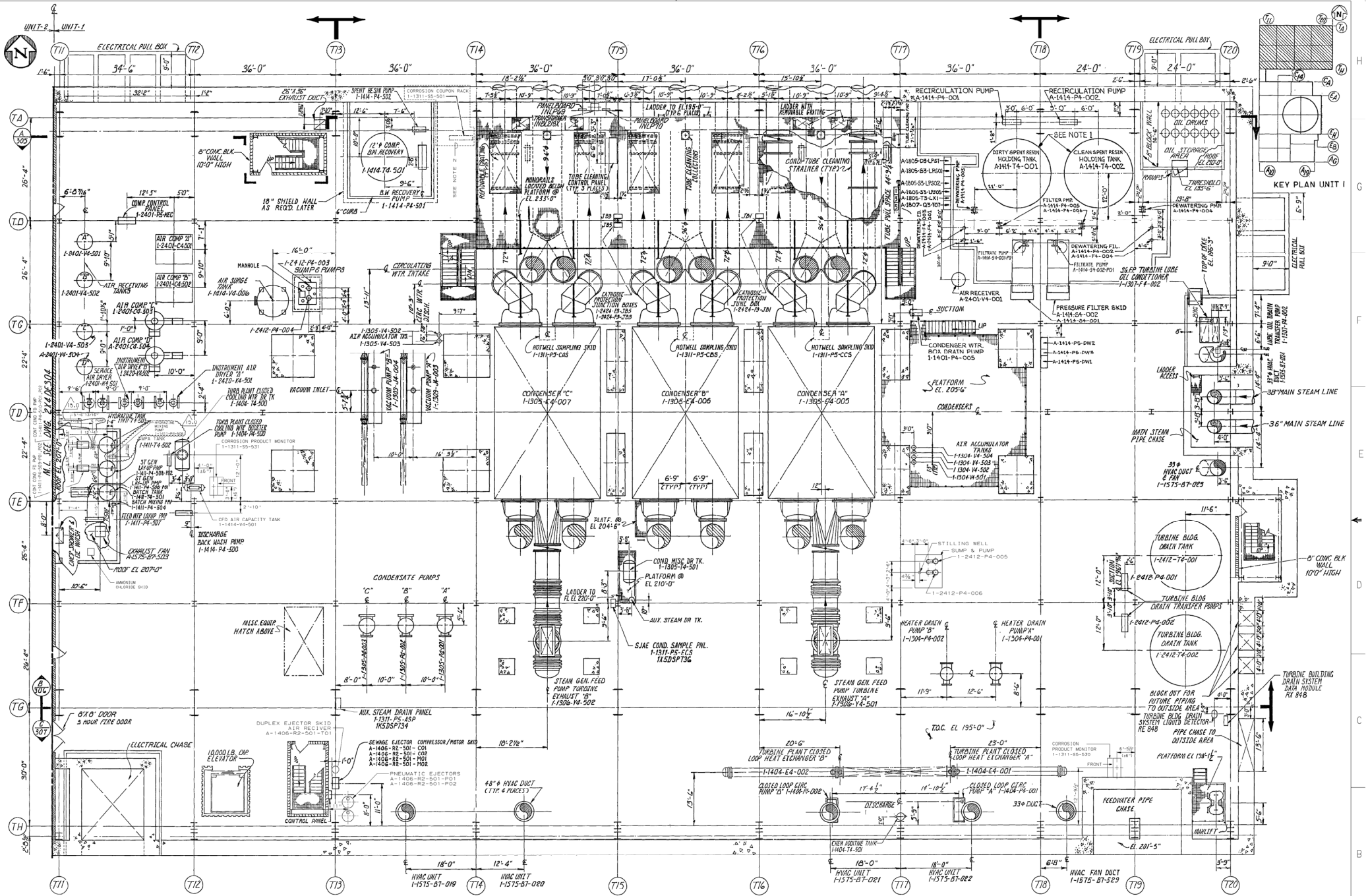
NO.	VERSIONS	DATE	DR	CHK	APPV
17.0	REVISED PER ABN 108184620IJ072, VER. 1.0	9-12-14	Vep	CYN	TSL

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TURBINE BLDG.
EQUIPMENT LOCATION PLAN
LEVEL 1, ELEV. 220'-0" TO 245'-0"

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DE303
VER. 17.0
SIZE E 34X44



NOTES:

1. COAT EXTERIOR SURFACES OF CLEAN & DIRTY SPENT RESIN TANKS AS FOLLOWS:
   A.) PRIMER - PER SYSTEM FD-3 IN SPEC. X1AJ07.
   B.) TOP COAT - PER SYSTEM FD-8 IN SPEC. X1AJ07.
2. "RETIRED IN PLACE"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TURBINE BLDG.
EQUIPMENT LOCATION PLAN
LEVEL A, ELEV. 195'-0" TO 220'-0"

SCALE: 1/8"=1'-0"

DRAWING NO. 1X4DE304

VER. 15.0

JOB NO. 10604

SIZE E 34X44

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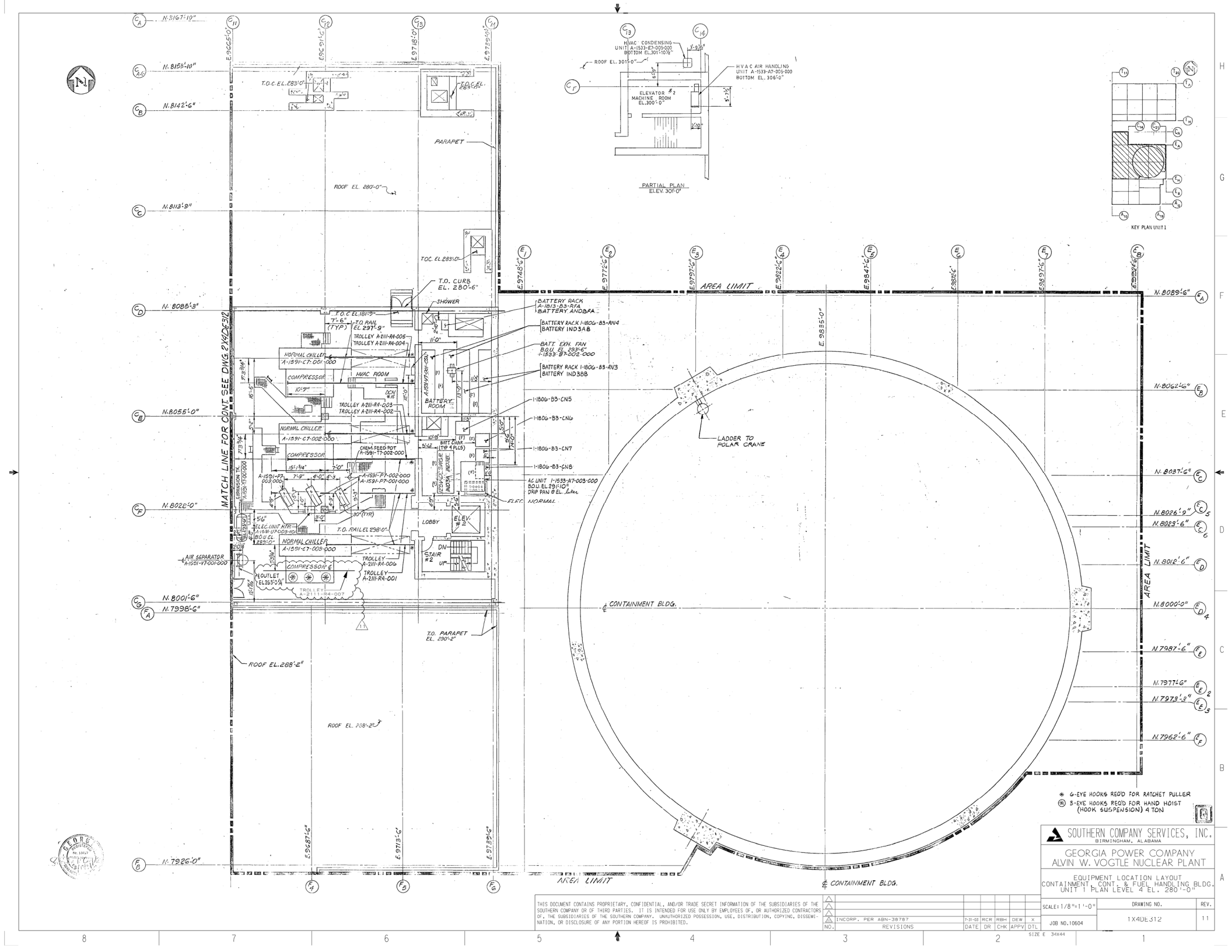
NO.	VERSIONS	DATE	DR	CHK	APPV
15.0	REVISED PER ABN 1039001101M052, VER 1.0	10-16-08	Vep	JWM	JMR



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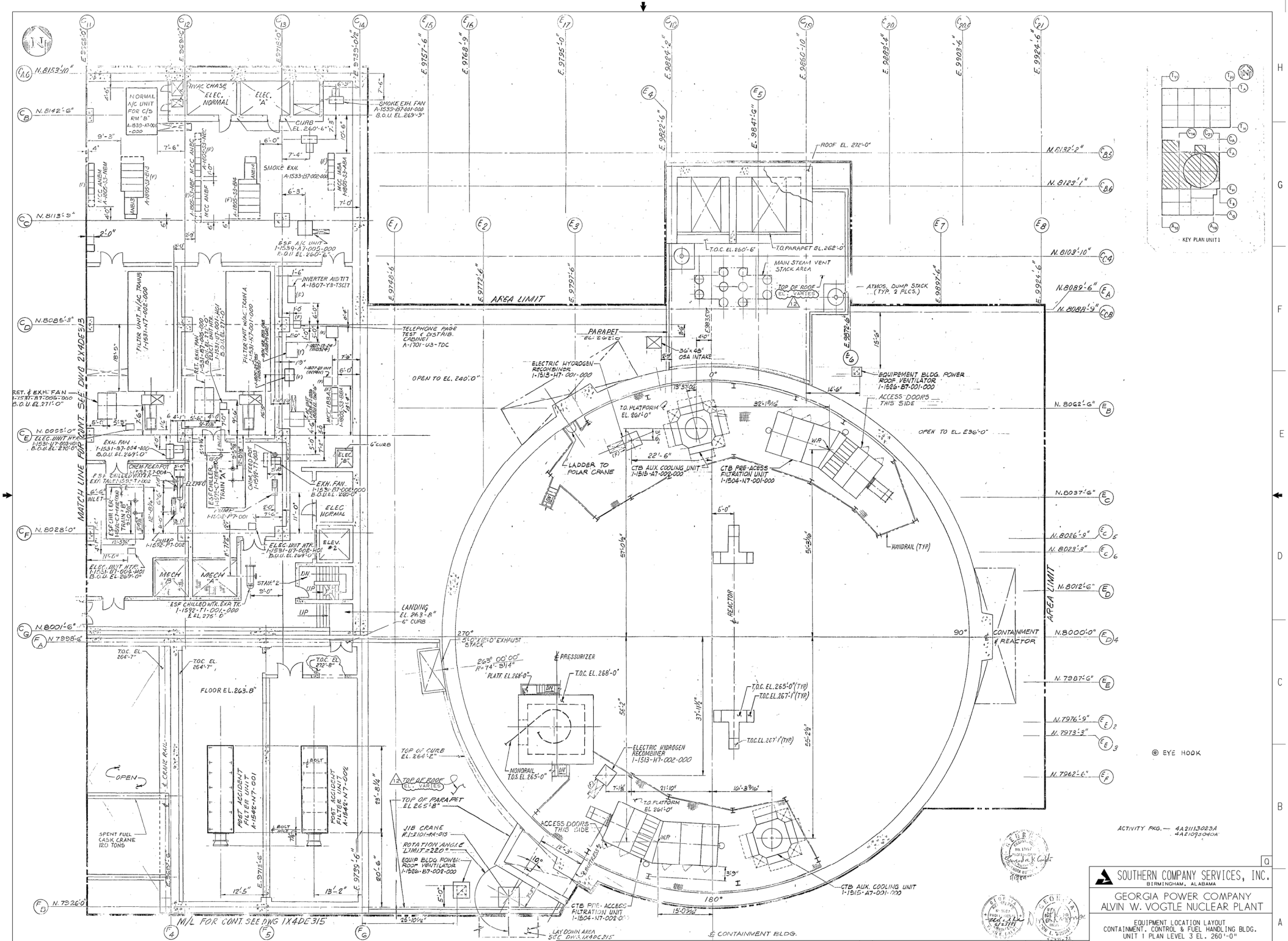
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	INCORP. PER ABN-38787	7-31-03	RCR	RBH	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

EQUIPMENT LOCATION LAYOUT
CONTAINMENT, CONT. & FUEL HANDLING BLDG.
UNIT 1 PLAN LEVEL 4' EL. 280'-0"

SCALE: 1/8"=1'-0" | DRAWING NO. 1X4DE312 | REV. 11 | JOB NO. 10604 | SIZE E 34X44



ACTIVITY PKG. — 4A2I113023A
4A2I0730404A

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 92-VAN0183	04/26/01	CD	EGG	ELC	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

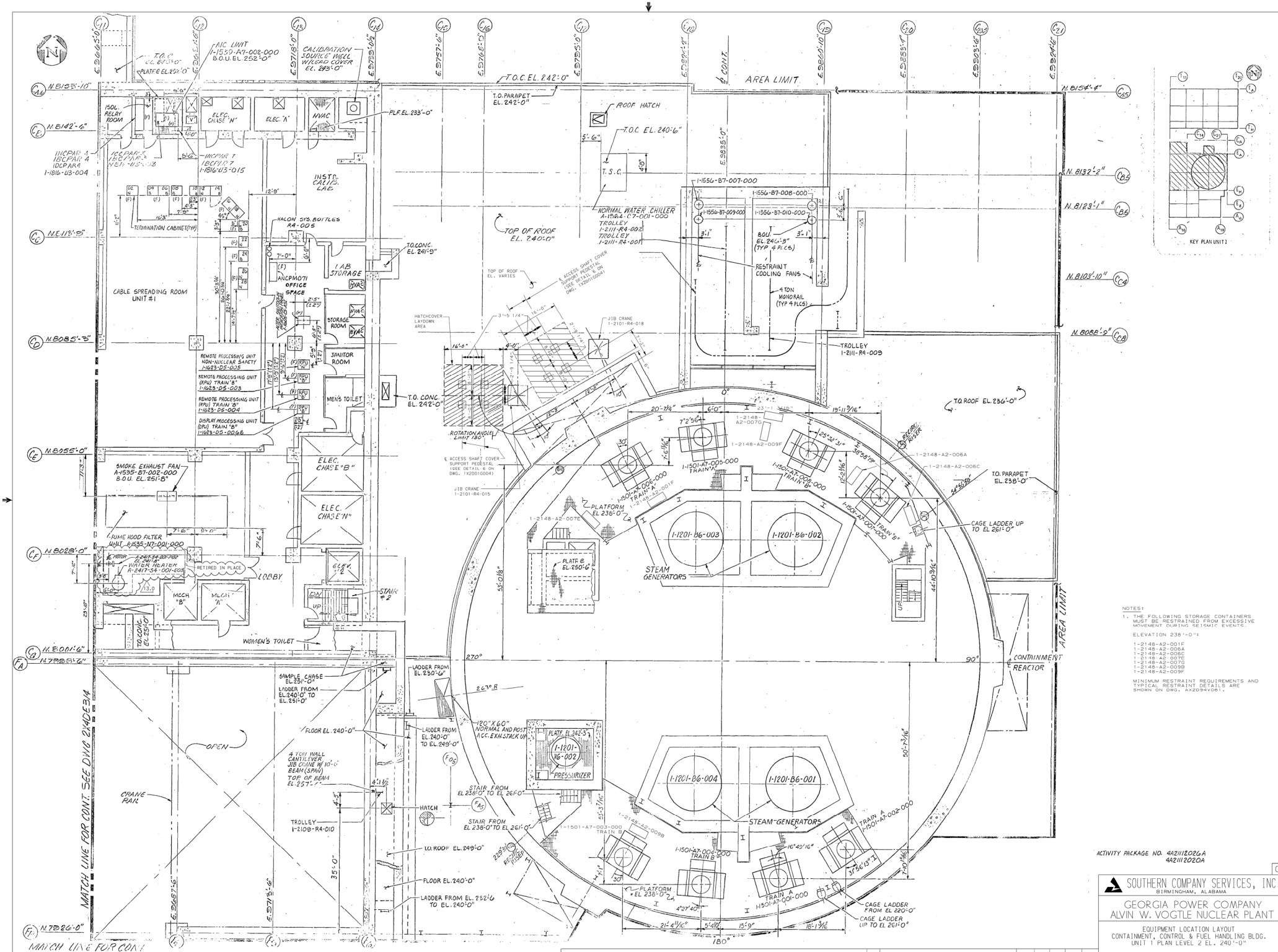
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

EQUIPMENT LOCATION LAYOUT
CONTAINMENT, CONTROL & FUEL HANDLING BLDG.
UNIT 1 PLAN LEVEL 3 EL. 260'-0"

SCALE: 1/8"=1'-0" | DRAWING NO. 1X4DE313 | REV. 12

JOB NO. 10604

SIZE E 34X44


T.O.C. EL. 271'-0"
PLATE EL. 252'-0"
AIC LIMIT
1-1539-A7-002-000
B.O.U. EL. 252'-0"
CALIBRATION SOURCE WELL W/LEAD COVER EL. 243'-0"
T.O.C. EL. 242'-0"
AREA LIMIT
T.O. PARAPET EL. 242'-0"
ROOF HATCH
T.O.C. EL. 240'-6"
PLF EL. 233'-0"
ISOL. RELAY ROOM
ELEC. CHASE "N"
ELEC. "A"
HVAC
INSTR. CALIB. LAB
HALON SYS. BOTTLES R4-005
TERMINATION CABINET(TYP)
I.A.B. STORAGE
ANCPM071 OFFICE SPACE
CABLE SPREADING ROOM UNIT #1
STORAGE ROOM
SANITOR ROOM
MEN'S TOILET
T.O. CONC. EL. 241'-9"
T.O. CONC. EL. 242'-0"
REMOTE PROCESSING UNIT NON-NUCLEAR SAFETY 1-1623-D5-005
REMOTE PROCESSING UNIT (RPU) TRAIN 'B' 1-1623-D5-003
REMOTE PROCESSING UNIT (RPU) TRAIN 'A' 1-1623-D5-004
DISPLAY PROCESSING UNIT (DPU) TRAIN "B" 1-1623-D5-006B
SMOKE EXHAUST FAN A-1535-B7-002-000 B.O.U. EL. 251'-8"
ELEC. CHASE "B"
ELEC. CHASE "N"
FUME HOOD FILTER UNIT A-1535-A7-001-000
WATER HEATER A-2417-S4-001-003
RETIRED IN PLACE
LOBBY
MECH "B"
MECH "A"
ELEV. #2
STAIR #2
T.O. CONC. EL. 251'-0"
WOMEN'S TOILET
SAMPLE CHASE EL. 251'-0"
LADDER FROM EL. 240'-0" TO EL. 251'-0"
FLOOR EL. 240'-0"
OPEN
4 TON WALL CANTILEVER JIB CRANE W/ 10'-6" BEAM (SPAN)
TOP OF BEAM EL. 257'-7"
CRANE RAIL
TROLLEY 1-2108-R4-010
HATCH
T.O. ROOF EL. 249'-0"
FLOOR EL. 240'-0"
LADDER FROM EL. 232'-6" TO EL. 240'-0"
LADDER FROM EL. 230'-6"
LADDER FROM EL. 240'-0" TO EL. 249'-0"
120"X60" NORMAL AND POST ACC. EXH. STACK UP
PLATE EL. 242'-3"
1-1201-B6-002
PRESSURIZER
STAIR FROM EL. 238'-0" TO EL. 261'-0"
STAIR FROM EL. 238'-0" TO EL. 261'-0"
MATCH LINE FOR CONT. SEE DWG 2H4DE314
MATCH LINE FOR CONT. SEE DWG 1H4DE315
TOP OF ROOF EL. 240'-0"
NORMAL WATER CHILLER A-1584-C7-001-000
TROLLEY 1-2111-R4-002
TROLLEY 1-2111-R4-001
1-1556-B7-007-000
1-1556-B7-008-000
1-1556-B7-009-000
1-1556-B7-010-000
B.O.U. EL. 240'-1 3/8" (TYP 4 PLCS)
RESTRAINT COOLING FANS
4 TON MONORAIL (TYP 4 PLCS)
TROLLEY 1-2111-R4-009
TOP OF ROOF EL. VARIES
ACCESS SHAFT COVER SUPPORT PEDESTAL (SEE DETAIL 6 ON DWG. 1X2001G004)
HATCHCOVER LAYDOWN AREA
JIB CRANE 1-2101-R4-018
JIB CRANE 1-2101-R4-015
ROTATION ANGLE LIMIT 180°
T.O. ROOF EL. 236'-0"
T.O. PARAPET EL. 238'-0"
1-2148-A2-006A
1-2148-A2-006C
1-2148-A2-009F
1-2148-A2-007E
1-2148-A2-001F
1-1501-A7-005-000 TRAIN 'A'
1-1501-A7-006-000 TRAIN 'B'
1-1501-A7-008-000 TRAIN 'B'
PLATFORM EL. 238'-0"
PLATE EL. 250'-6"
1-1201-B6-003
1-1201-B6-002
STEAM GENERATORS
CAGE LADDER UP TO EL. 261'-0"
CONTAINMENT REACTOR
1-1201-B6-004
1-1201-B6-001
STEAM GENERATORS
1-2148-A2-009B
1-1501-A7-003-000 TRAIN B
1-1501-A7-004-000 TRAIN B
1-1501-A7-002-000 TRAIN A
1-1501-A7-009-000 TRAIN A
PLATFORM EL. 238'-0"
CAGE LADDER FROM EL. 220'-0"
CAGE LADDER UP TO EL. 261'-0"
KEY PLAN UNIT 1
NOTES:
1. THE FOLLOWING STORAGE CONTAINERS MUST BE RESTRAINED FROM EXCESSIVE MOVEMENT DURING SEISMIC EVENTS.
ELEVATION 238'-0":
1-2148-A2-001F
1-2148-A2-006A
1-2148-A2-006C
1-2148-A2-007C
1-2148-A2-007E
1-2148-A2-009B
1-2148-A2-009F
MINIMUM RESTRAINT REQUIREMENTS AND TYPICAL RESTRAINT DETAILS ARE SHOWN ON DWG. A2D94V081.
ACTIVITY PACKAGE NO. 4A2111202GA
4A2111202OA
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
EQUIPMENT LOCATION LAYOUT
CONTAINMENT, CONTROL & FUEL HANDLING BLDG.
UNIT 1 PLAN LEVEL 2 EL. 240'-0"
SCALE: 1/8"=1'-0"
DRAWING NO. 1X4DE314
VER. 13.0
JOB NO. 10604
SIZE E 34X44
13.0 REVISED PER ABN-V01757, VER. 1.0 10-29-09 Vep WLM JMR
NO. VERSIONS DATE DR CHK APPV
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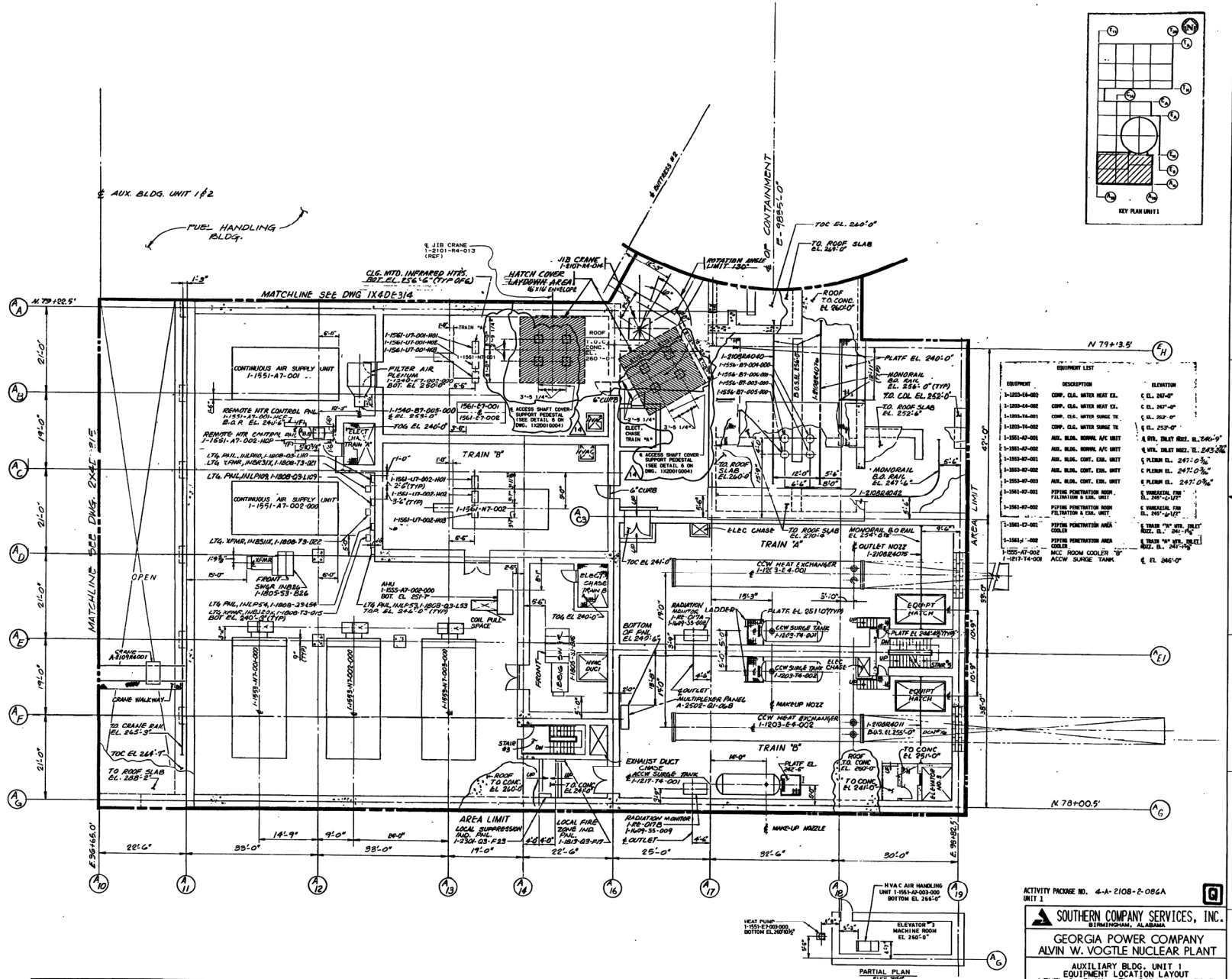

ACTIVITY PACKAGE NO. 4-A-2108-2-086A
UNIT 1

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AUXILIARY BLDG. UNIT 1
EQUIPMENT LOCATION LAYOUT
LEVEL 2. PLAN - EL. 240'-0" & ABOVE

SCALE: NONE | DRAWING NO. 1X4DE315 | REV. 14
JOB NO. 10604 | SIZE E 34X44

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
	INC. ABN-41233.	4/16/96	CD	24/96	CR	
	REVISIONS					
	ISSUED FOR CONSTRUCTION					
	REVISIONS					

ENG. MGR. DATE

SEE MICROFILM FOR SIGNATURES

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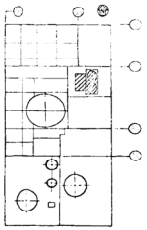

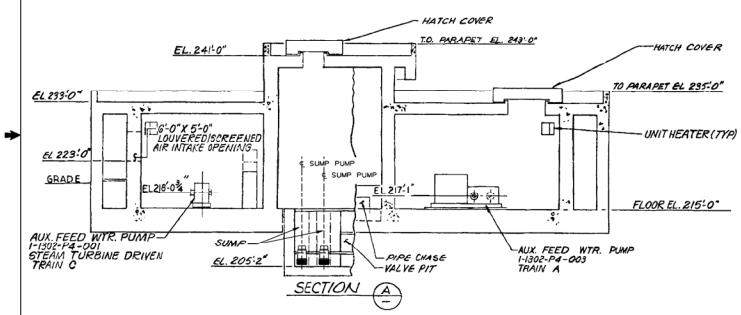

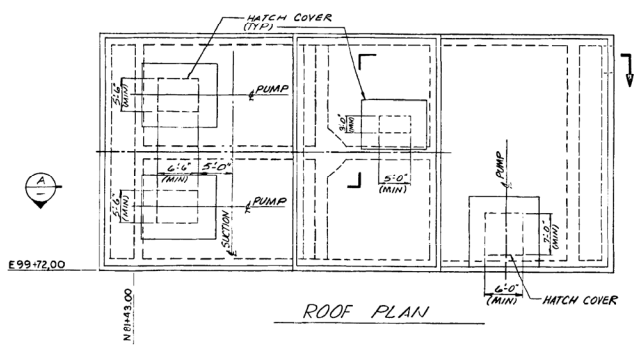

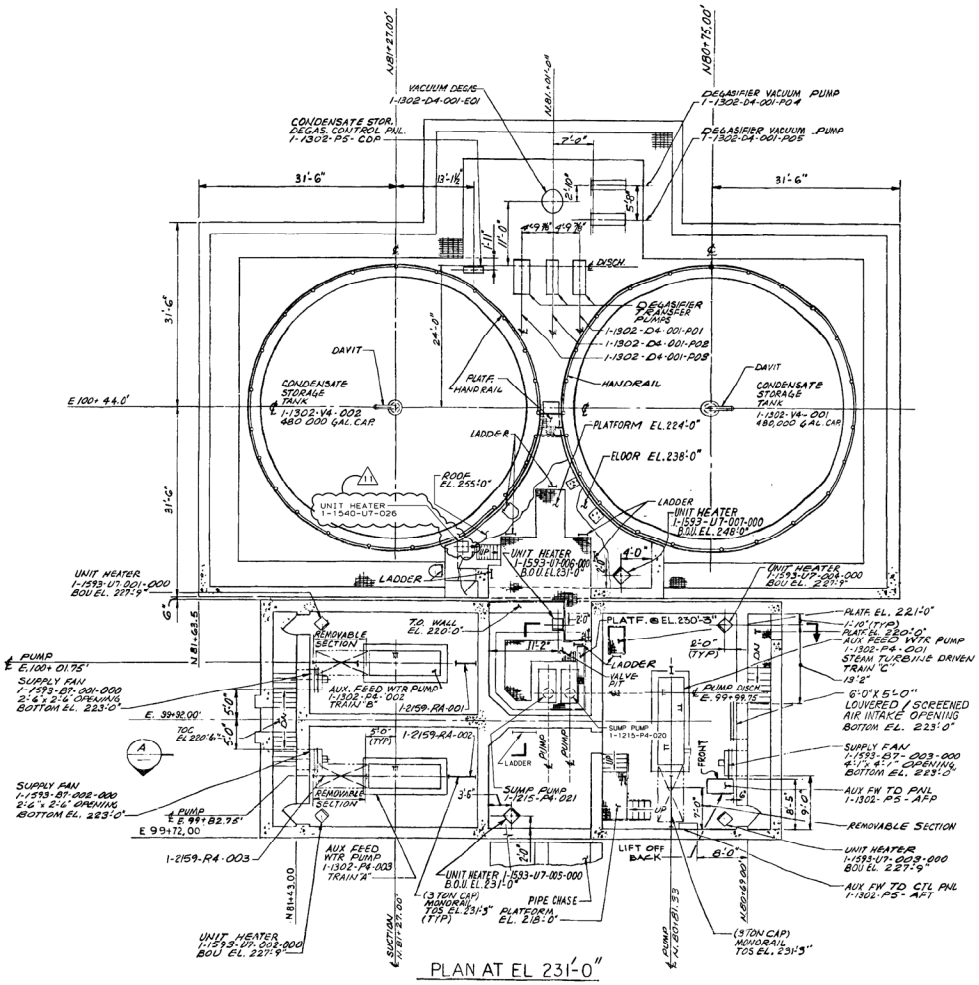

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	REVISED PER DCP 95-V1N0065	2-13-04	Vep	RBH	DEW	X

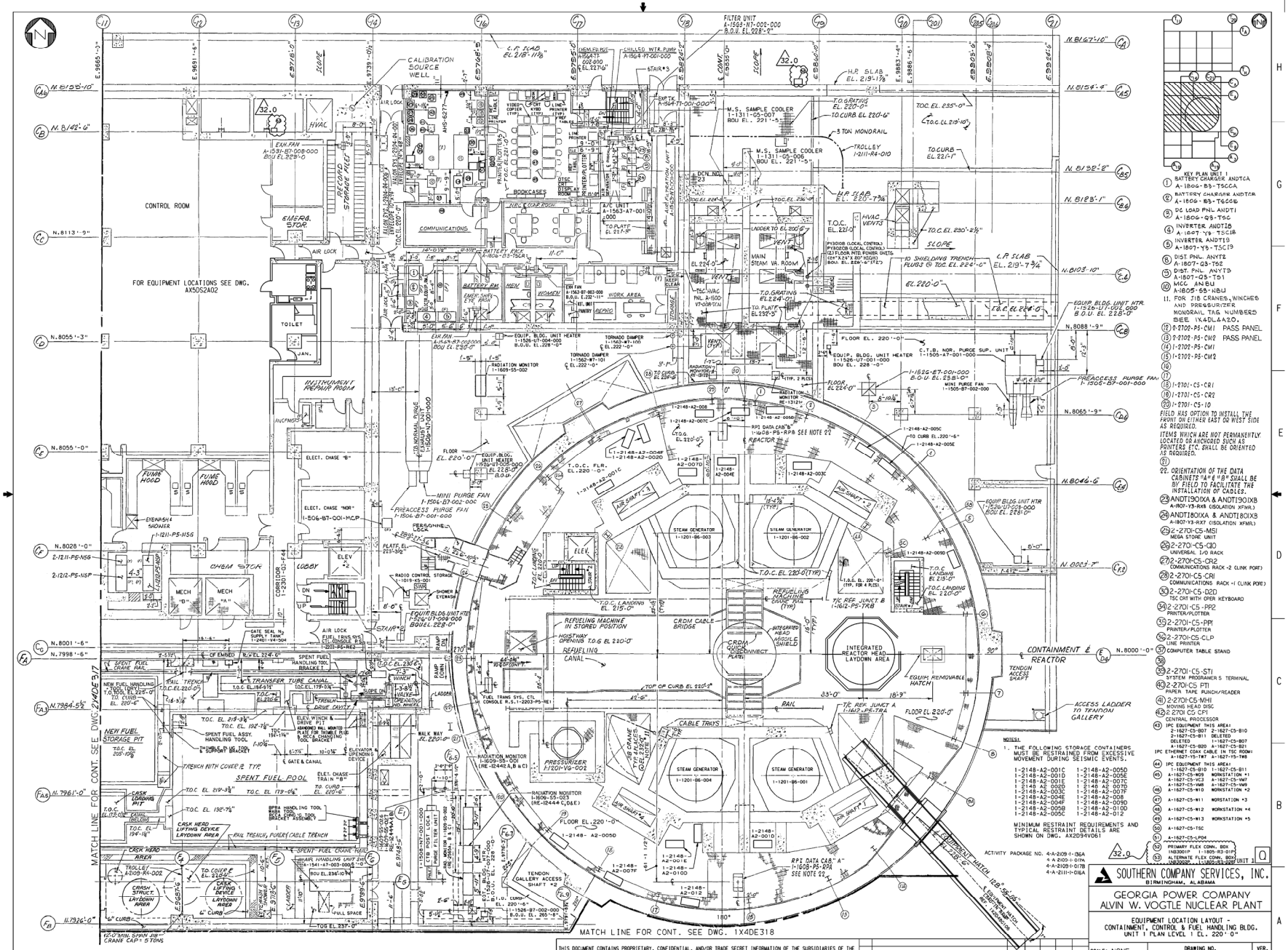
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

AUX. FEEDWATER PUMP HOUSE AND CONDENSATE STORAGE TANK EQUIPMENT LOCATION LAYOUT

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE316	11

SIZE E 34x44



KEY PLAN UNIT 1

1. BATTERY CHARGER ANDTCA A-1806-B5-TSCCA
2. BATTERY CHARGER ANDTBR A-1806-B5-T6CCB
3. DC LOAD PNL ANDTI A-1806-Q3-T5C
4. INVERTER ANDTIA A-1807-Y5-T5CIB
5. INVERTER ANDTIB A-1807-Y5-T5CI9
6. DIST PNL ANYTA A-1807-Q3-T52
7. DIST PNL ANYTB A-1807-Q5-T51
8. MCC ANBU A-1805-B5-NBU

11. FOR JIB CRANES, WINCHES AND PRESSURIZER MONORAIL TAG NUMBERS SEE 1X4DL4A2O.
12. P-2702-P5-CM1 PASS PANEL
13. P-2702-P5-CM2 PASS PANEL
14. I-2702-P5-CM1
15. I-2702-P5-CM2
16.
17.
18. I-2701-C5-CR1
19. I-2701-C5-CR2
20. I-2701-C5-10

FIELD HAS OPTION TO INSTALL THE FRONT ON EITHER EAST OR WEST SIDE AS REQUIRED.

ITEMS WHICH ARE NOT PERMANENTLY LOCATED OR ANCHORED SUCH AS PRINTERS ETC. SHALL BE ORIENTED AS REQUIRED.

21.

22. ORIENTATION OF THE DATA CABINETS "A" & "B" SHALL BE BY FIELD TO FACILITATE THE INSTALLATION OF CABLES.

23. ANDTI90IXA & ANDTI90IXB A-1807-Y3-RX8 (ISOLATION XFMR.)
24. ANDTI80IXA & ANDTI80IXB A-1807-Y3-RX7 (ISOLATION XFMR.)
25. I-2701-C5-MSI MEGA STORE UNIT
26. 2-2701-C5-CIO UNIVERSAL I/O RACK
27. 2-2701-C5-CR2 COMMUNICATIONS RACK -2 (LINK PORT)
28. 2-2701-C5-CRI COMMUNICATIONS RACK -1 (LINK PORT)
30. 2-2701-C5-D2O TSC CRT WITH OPER KEYBOARD
34. 2-2701-C5-PP2 PRINTER/PLOTTER
35. 2-2701-C5-PPI PRINTER/PLOTTER
36. 2-2701-C5-CLP LINE PRINTER
37. COMPUTER TABLE STAND
38.
39. 2-2701-C5-STI SYSTEM PROGRAMMER'S TERMINAL
40. 2-2701-C5-PTI PAPER TAPE PUNCH/READER
41. 2-2701-C5-MHI MOVING HEAD DISC
42. 2-2701-C5-CPI CENTRAL PROCESSOR
43. IPC EQUIPMENT THIS AREA1: 2-1627-C5-B07 2-1627-C5-B10 2-1627-C5-B11 DELETED IPC ETHERNET COAX CABLE IN TSC ROOM: A-1627-Y5-TWT A-1627-Y5-TW8
44. IPC EQUIPMENT THIS AREA1: 1-1627-C5-B10 1-1627-C5-B11
45. A-1627-C5-W09 WORKSTATION +1
46. A-1627-C5-VC3 A-1627-C5-VM7
47. A-1627-C5-VM5 A-1627-C5-VM9
48. A-1627-C5-W10 WORKSTATION +2
49. A-1627-C5-W11 WORKSTATION +3
50. A-1627-C5-W12 WORKSTATION +4
51. A-1627-C5-W13 WORKSTATION +5
52. A-1627-C5-TSC
53. A-1627-C5-LP04
54. PRIMARY FLEX CONN. BOX IND3001P 1-1805-R3-01P
55. ALTERNATE FLEX CONN. BOX IND3002P 1-1805-R4-02P

UNIT 1

NOTES:
1. THE FOLLOWING STORAGE CONTAINERS MUST BE RESTRAINED FROM EXCESSIVE MOVEMENT DURING SEISMIC EVENTS.

1-2148-A2-001C 1-2148-A2-005D
1-2148-A2-001D 1-2148-A2-005E
1-2148-A2-001E 1-2148-A2-007C
1-2148-A2-002D 1-2148-A2-007D
1-2148-A2-003C 1-2148-A2-007F
1-2148-A2-004E 1-2148-A2-008
1-2148-A2-004F 1-2148-A2-009D
1-2148-A2-005B 1-2148-A2-010D
1-2148-A2-005C 1-2148-A2-012

MINIMUM RESTRAINT REQUIREMENTS AND TYPICAL RESTRAINT DETAILS ARE SHOWN ON DWG. AX2D94V06!

ACTIVITY PACKAGE NO. 4-A-2109-I-06A
4-A-2105-I-017A
4-A-2105-I-017B
4-A-2111-I-016A

32.0

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

EQUIPMENT LOCATION LAYOUT - CONTAINMENT, CONTROL & FUEL HANDLING BLDG. UNIT 1 PLAN LEVEL 1 EL. 220'-0"

SCALE: NONE

DRAWING NO. 1X4DE317 VER. 32.0

JOB NO. 10604

SIZE E 34X44

NO.	VERSIONS	DATE	DR	CHK	APP'V
32.0	REVISED PER ABN SNC473105M005, VER. 2.0	01/28/16	EJ	MWD	WLN

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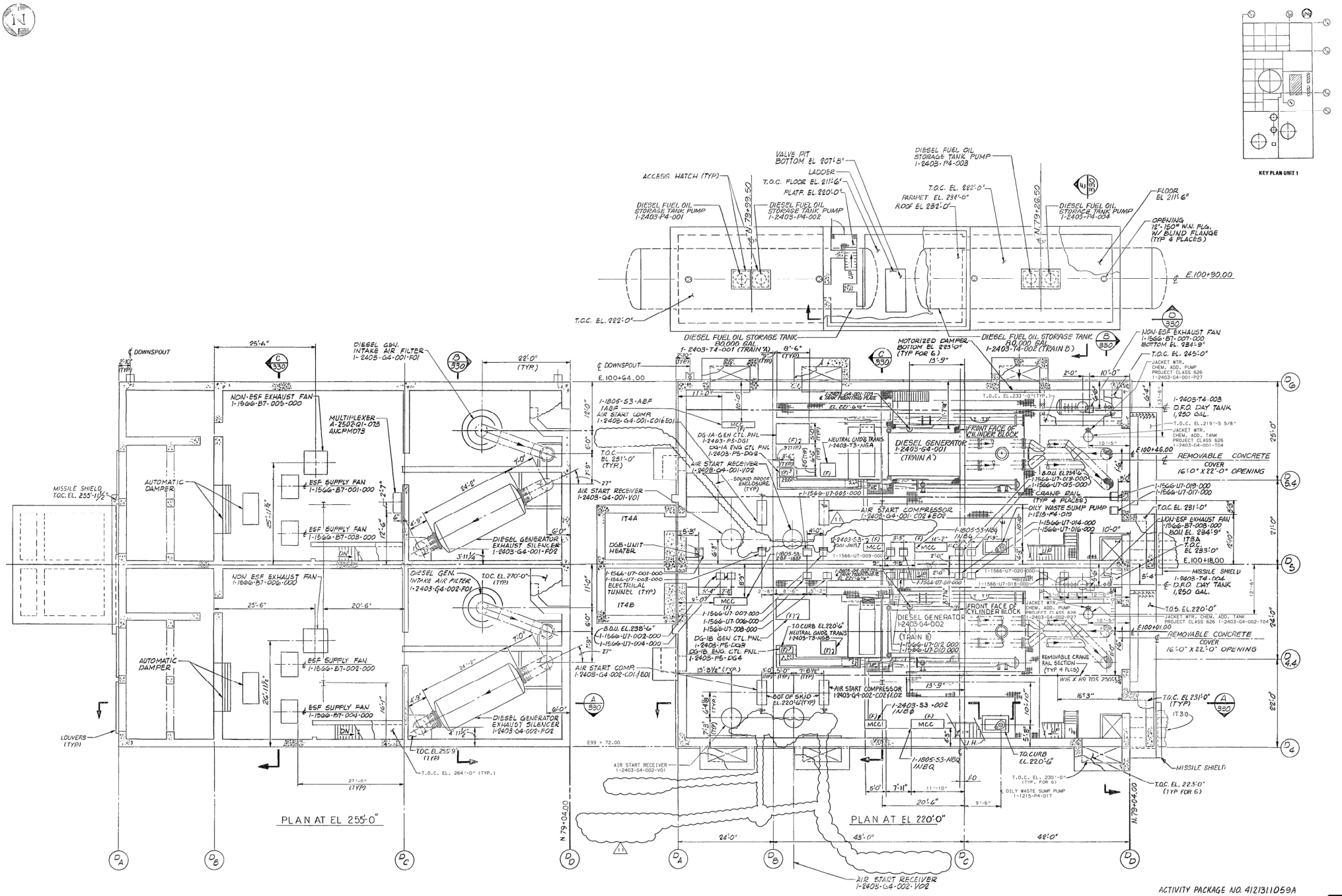

ACTIVITY PACKAGE NO. 4121311059A
4121072065A
4121071054A

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GEN. BLDG. UNIT 1
EQUIP. LOCATION LAYOUT

SCALE: 1/8"=1'-0" | DRAWING NO. 1X4DE327 | REV. 11
JOB NO. 10604

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 95-V1N0070	1-8-99	LVL	JGM	MWD	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
1						

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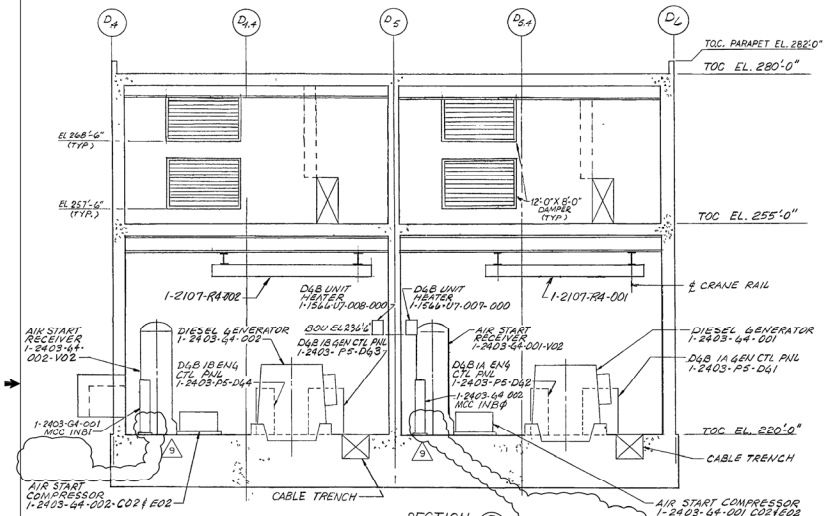
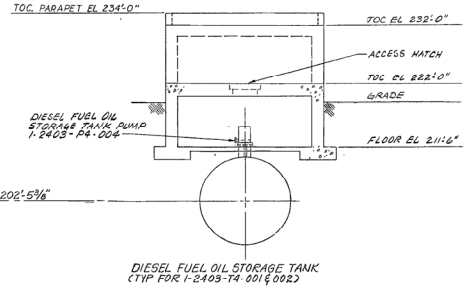
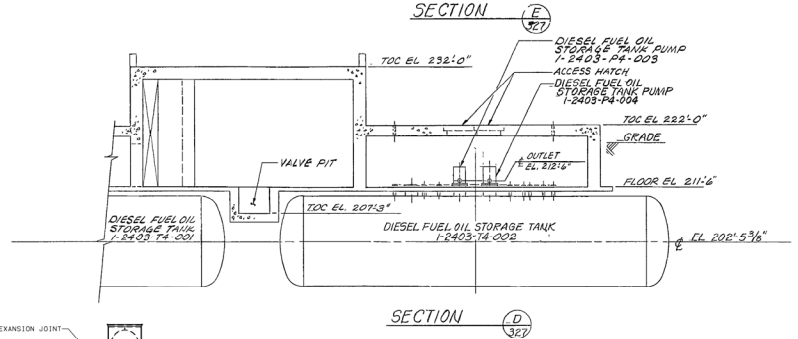
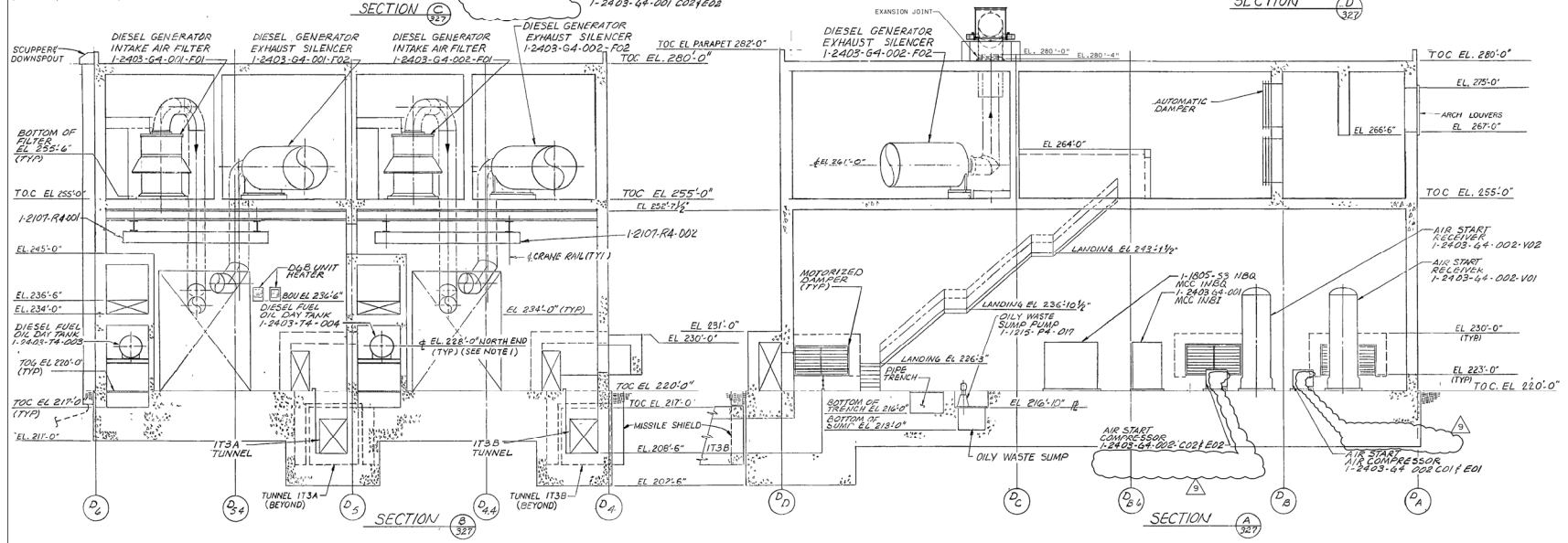
SIZE E 34X44
CAD NAME 1X4DE327









NOTE:
1. TANK SLOPES DOWN FROM NORTH END TO SOUTH END.

ACTIVITY PACKAGE NO. 4121071088A

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GEN. BLDG. UNIT 1
EQUIP. LOCATION LAYOUT SECTIONS

SCALE: 1/8" = 1'-0"

DRAWING NO. 1X4DE330

REV. 9

JOB NO. 10604

SIZE E 34X44

CAD NAME 1X4DE330

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 95-V1N0070	1-8-99	LVL	MWD	EOG	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

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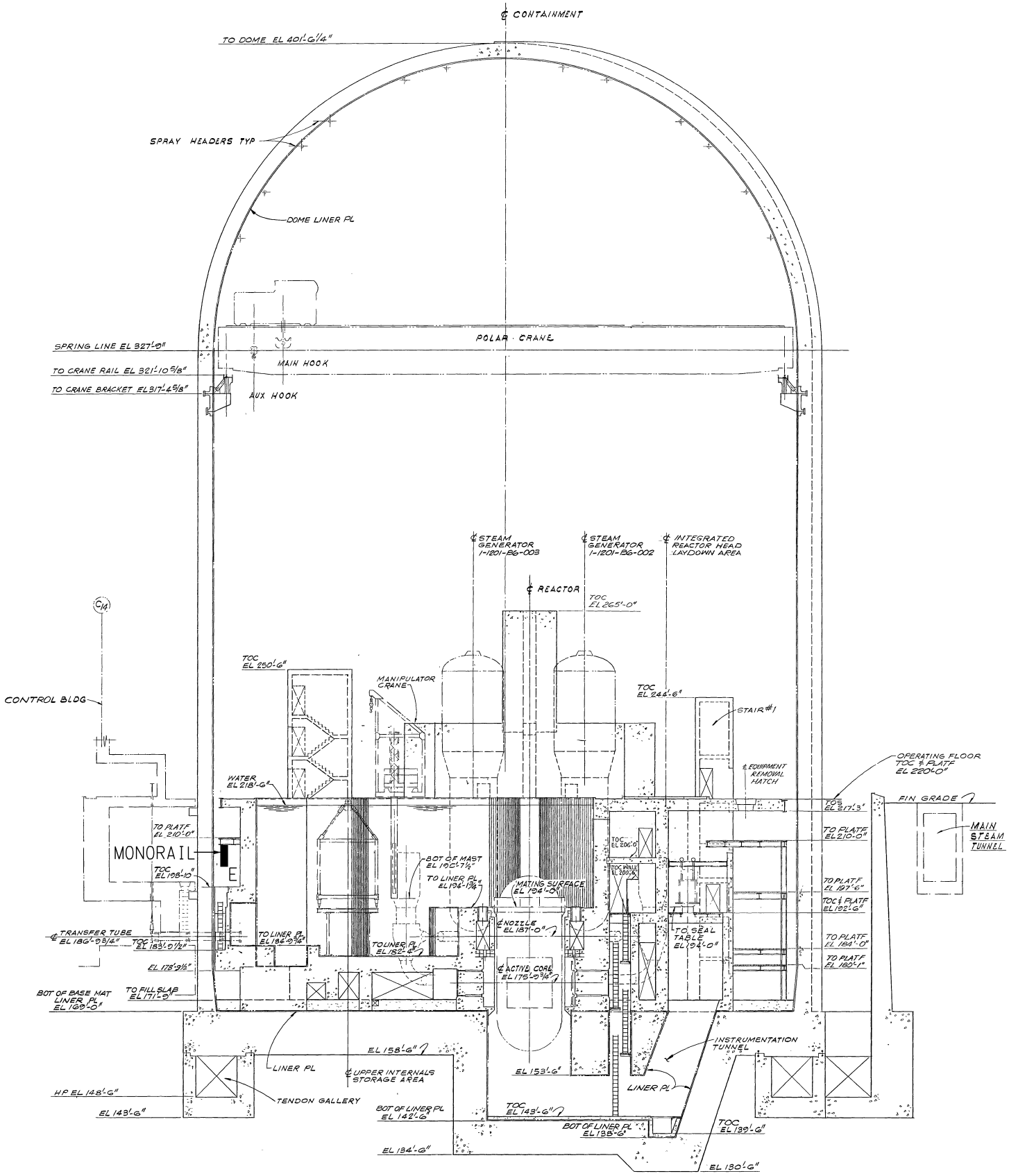

SECTION LOOKING NORTH

LEGEND:
- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▶ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ■ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- 1-1B CRANE AND LOAD PATH AREA

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	ISSUED PER REA 96-VAA054	7-29-98	LVL	RBH	CBH	WFP

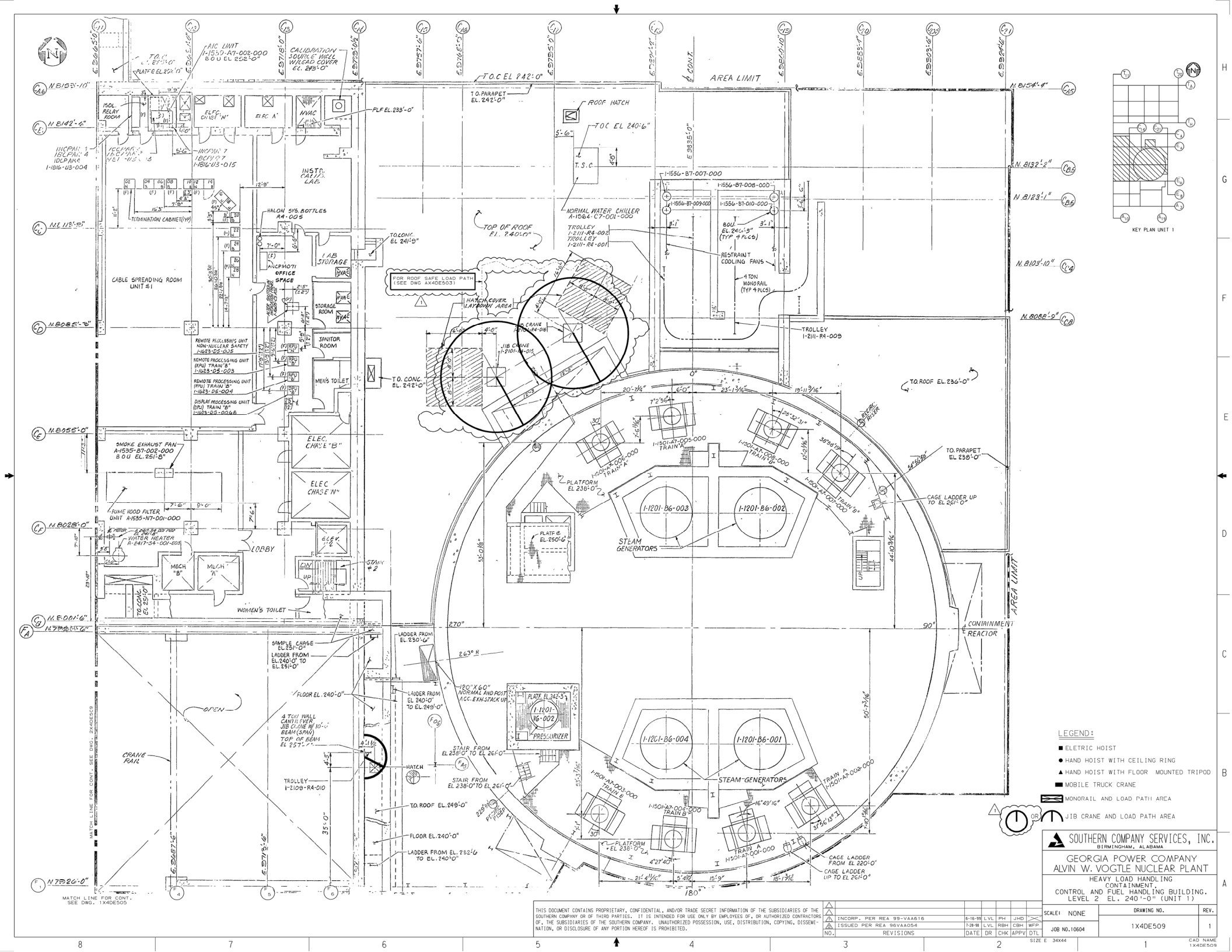
SIZE E 3MM4

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTAINMENT INTERNAL
SECTION LOOKING NORTH

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X4DE508
REV. 0
CAD NAME 1X4DE508



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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	INCORP. PER REA 99-VAA616	6-16-99	LVL	PH	JHD	
2	ISSUED PER REA 96VAA054	7-28-98	LVL	RBH	CBH	WFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

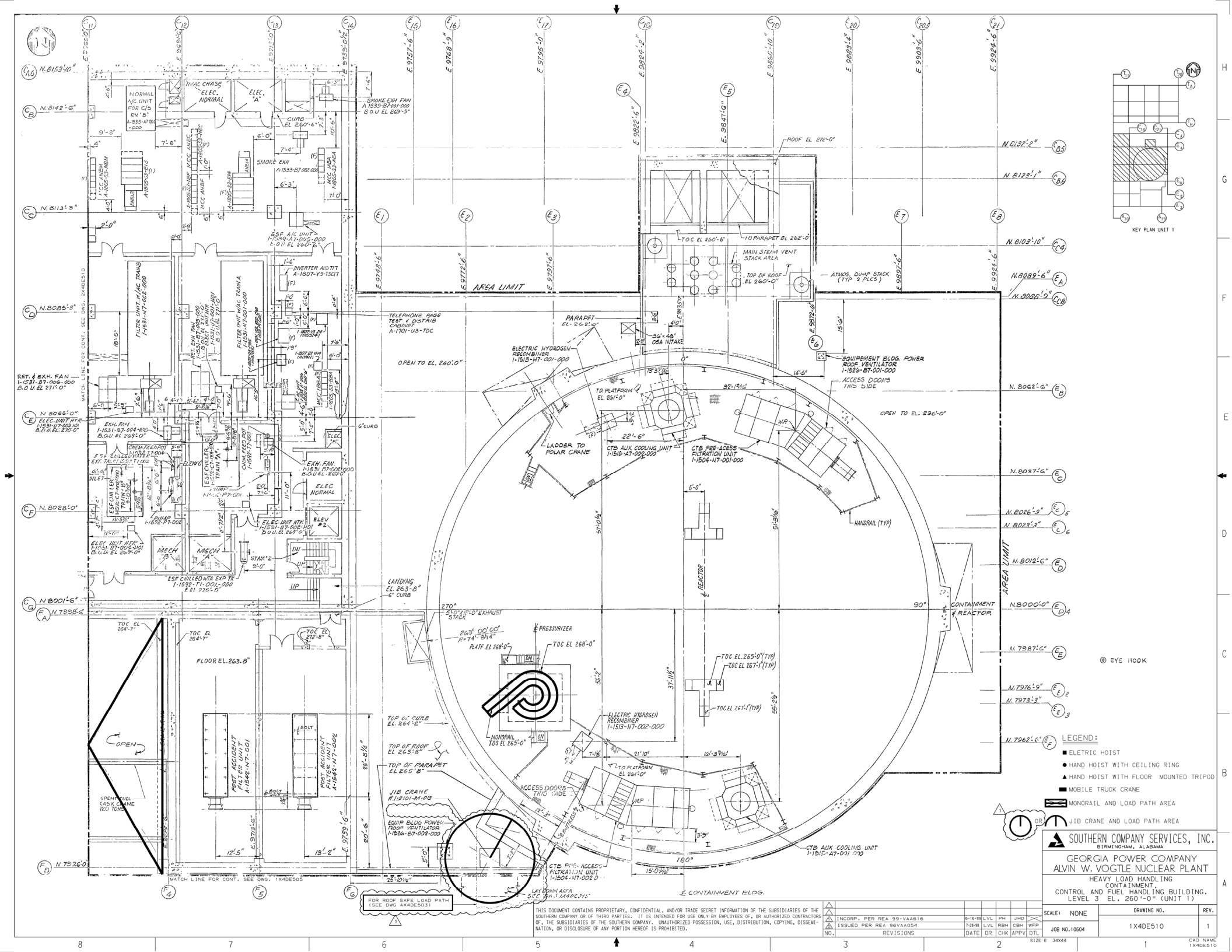
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING CONTAINMENT, CONTROL AND FUEL HANDLING BUILDING. LEVEL 2 EL. 240'-0" (UNIT 1)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE509	1

SIZE E 34X44

CAD NAME 1X4DE509



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING CONTAINMENT, CONTROL AND FUEL HANDLING BUILDING. LEVEL 3 EL. 260'-0" (UNIT 1)

SCALE: NONE | DRAWING NO. 1X4DE510 | REV. 1

JOB NO. 10604

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORP. PER REA 99-VAA616	6-16-99	LVL	PH	JHD	
1	ISSUED PER REA 96VAA054	7-28-98	LVL	RBH	CBH	WFP

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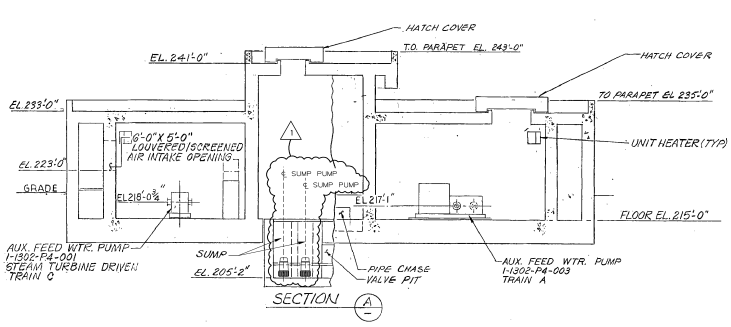
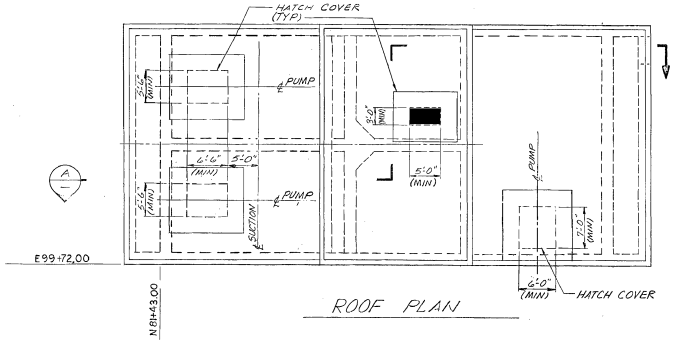
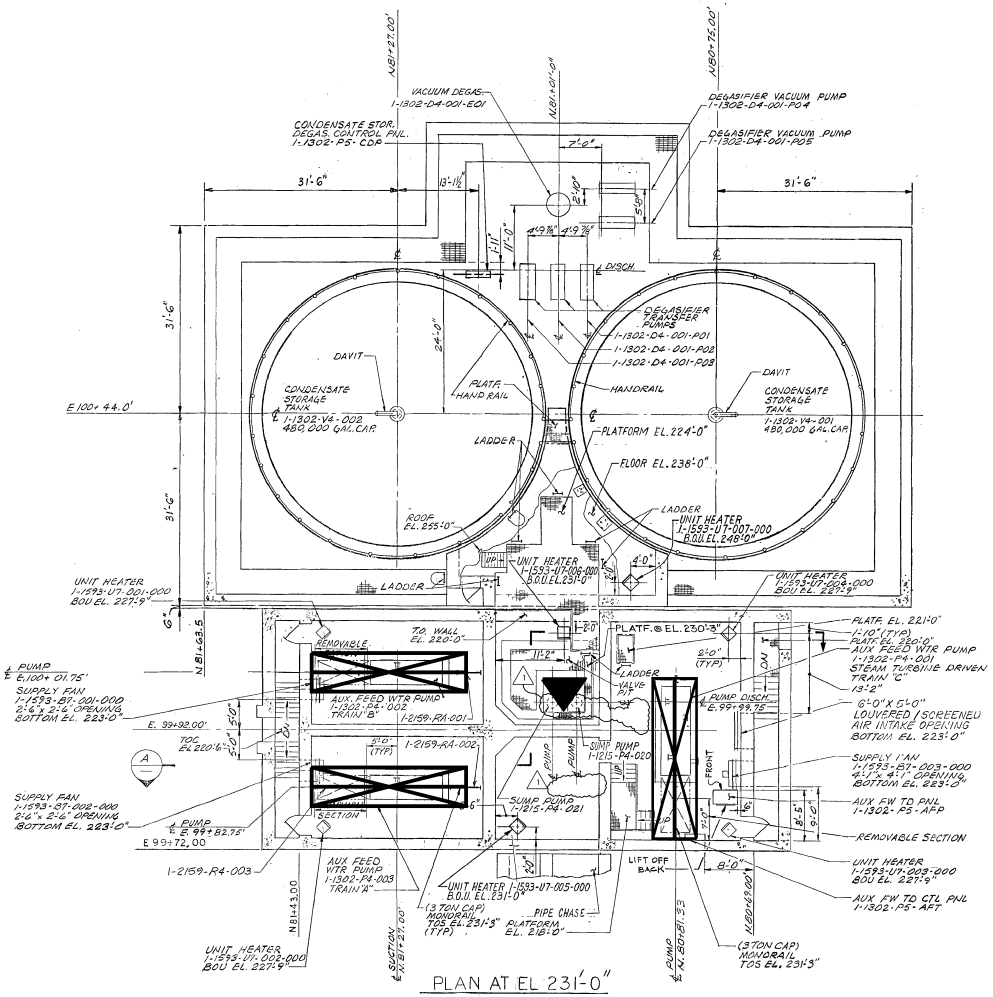
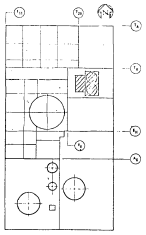
SIZE E 34X44

CAD NAME 1X4DE510









LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- ▭ MONORAIL AND LOAD PATH AREA
- ⌒ JIB CRANE AND LOAD PATH AREA

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3						
2						
1	INCORPORATED PER DCP 97-VAN0063	01/18/01	ELC	GLB	MWD	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

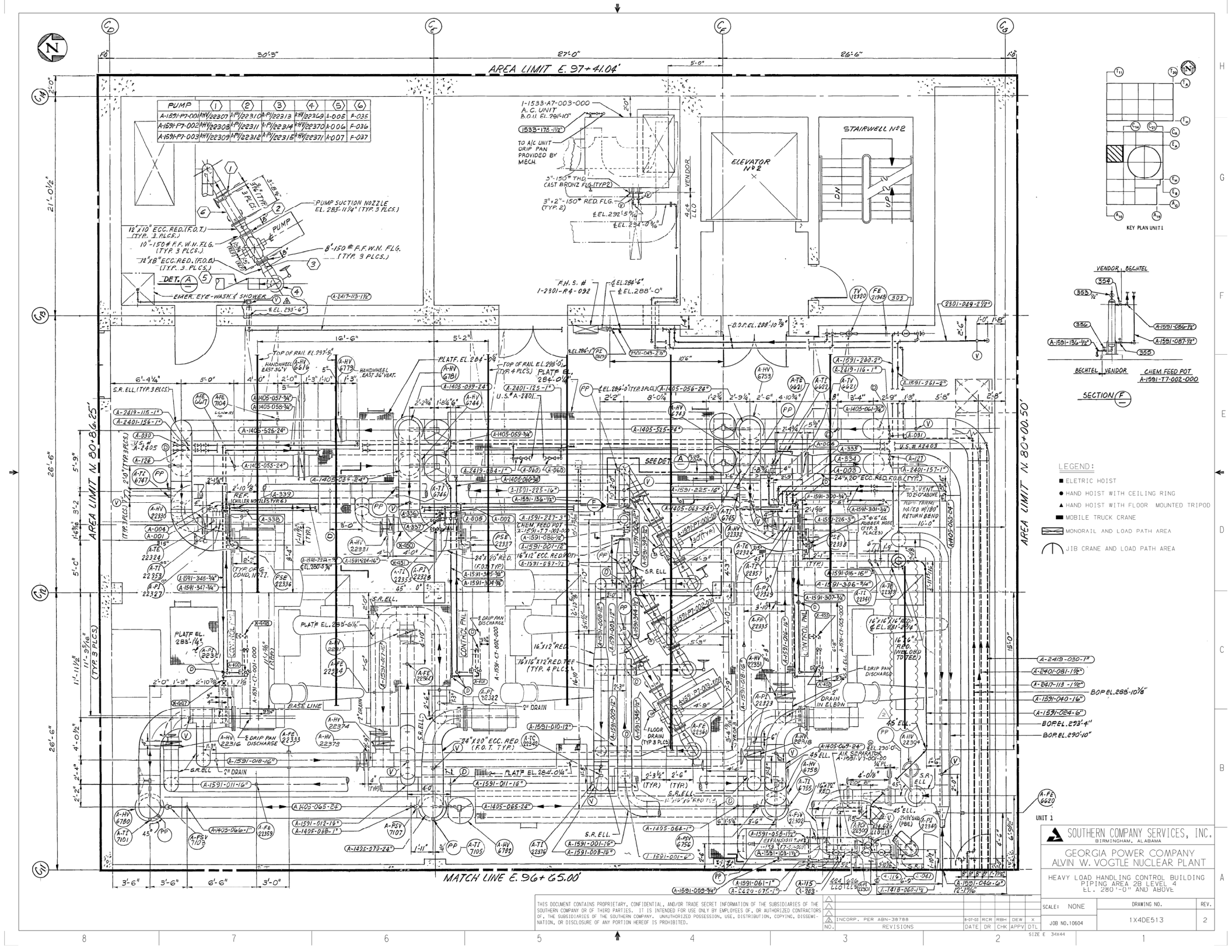
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
AUX. FEEDWATER
PUMP HOUSE AND CONDENSATE
STORAGE TANK

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE512	1

SIZE E 34x44



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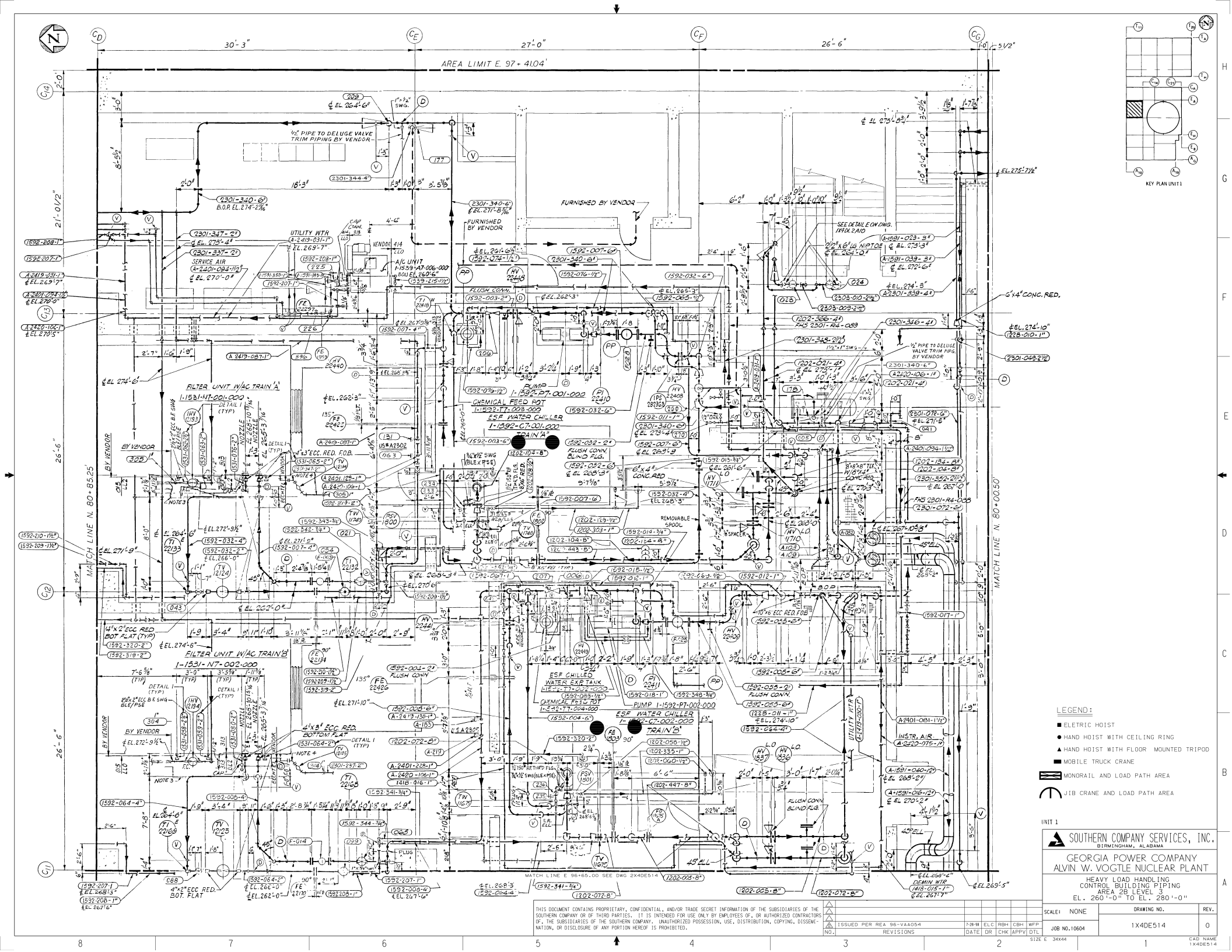
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-38788	8-07-03	RCR	RBH	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING CONTROL BUILDING
PIPING AREA 2B LEVEL 4
EL. 280'-0" AND ABOVE

SCALE: NONE | DRAWING NO. 1X4DE513 | REV. 2
JOB NO. 10604
SIZE E 34x44



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTROL BUILDING PIPING
AREA 2B LEVEL 3
EL. 260'-0" TO EL. 280'-0"

SCALE:	DRAWING NO.	REV.
NONE	1X4DE514	0
JOB NO. 10604		

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

SIZE E 34X44

CAD NAME 1X4DE514

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MAIN STEAM PIPING DOWN STREAM OF S WAY RESTRAINT 424/GK0

AN XXXX

FRODE MAT'L ASME SA-182 Gr F304L BAR STOCK (TYP 4 PLCS)

424 FGA

1"x 5/16" SHG

DETAIL 7

KEY PLAN UNIT1

MSIV/SV MONORAILS EL.254' (SEE NOTE #1)

4-TON MONORAIL (TYP FOR 3)

MATCH LINE E.98-62.33 SEE DWG. 1X4DE516

MATCH LINE N.88+08.00'

MATCH LINE 98+16.67

AREA LIMIT N 81+69.83'

FW VALVE MONORAIL EL. 236'

4 TON MONORAIL

M.S. SAMPLE COOLER 1-1311-G5-006

M.S. SAMPLE COOLER 1-1311-G5-007

STRESS FORGING (TYP.) 2 PLCS.

AIR FILTER UNIT A-1563-N7-001-000

SAMPLING NOZZLE 1" NOM. PIPE SIZE

WELDOLETS CAP (TYP. 4 PLCS.)

45° ELL (TYP.)

¼" THK. SPECTACLE BLIND

PLATE EL. 221'-8"

FOR STACK DRAIN LIT.# SEE TABLE BELOW (TYP 10 PLCS)

RADIATION MONITOR RE-13121

RADIATION MONITOR RE-13122

LEGEND:

- ELETRIC HOIST
- HAND HOIST WITH CEILING RING
- HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

NOTES:

1. THE MSIV/SV MONORAIL WITH TROLLEY 1-1211-R4-001 MAY BE USED FOR LIFTING THE ACTUATOR OF ATMOSPHERIC RELIEF VALVE 1PV-3020 OVER THE MAIN STEAM LINES UPSTREAM OF MSIV. DURING PLANT OPERATION, WITHOUT DOUBLE RIGGING. THE HOIST, THE SLING CONFIGURATION, AND THE CROSBY SHACKLE USED IN THE ACTUATOR LIFT SHALL HAVE A MINIMUM WORKING LOAD LIMIT OF TWO TONS.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN-51834	3-20-00	LRP	HAB	ASK	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
NORTH MAIN STEAM VALVE ROOM
PIPING AREA 2E LEVEL 1
EL. 220'-0" AND ABOVE

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE515	2

SIZE E 34x44

CAD NAME 1X4DE515

MATCH LINE E 99+26.25

AREA LIMIT N 81+69.83"

MATCH LINE E.98+62.33 SEE DWG. 1X4DE515

FW VALVE MONORAIL EL.236'

MSIV/SV MONORAIL EL.254'

ATMOSPHERIC RELIEF VALVE MONORAIL EL. 254'

KEY PLAN UNIT 1

LEGEND:
- ELETRIC HOIST
- HAND HOIST WITH CEILING RING
- HAND HOIST WITH FLOOR MOUNTED TRIPOD
- MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

NOTES:
1. THE MSIV/SV MONORAIL WITH TROLLEY 1-1211-R4-0001 MAY BE USED FOR LIFTING THE ACTUATOR OF ATOMSPHERIC RELIEF VALVE 1PV-3020 OVER THE MAIN STEAM LINES UPSTREAM OF MSIV, DURING PLANT OPERATIONS, WITHOUT DOUBLE RIGGING. THE HOIST, THE SLING CONFIGURATION, AND THE CROSBY SHACKLE USED IN THE ACTUATOR LIFT SHALL HAVE A MINIMUM WORKING LOAD LIMIT OF TWO TONS.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER AGN 51835	3-20-03	LRP	HAB	ASK	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

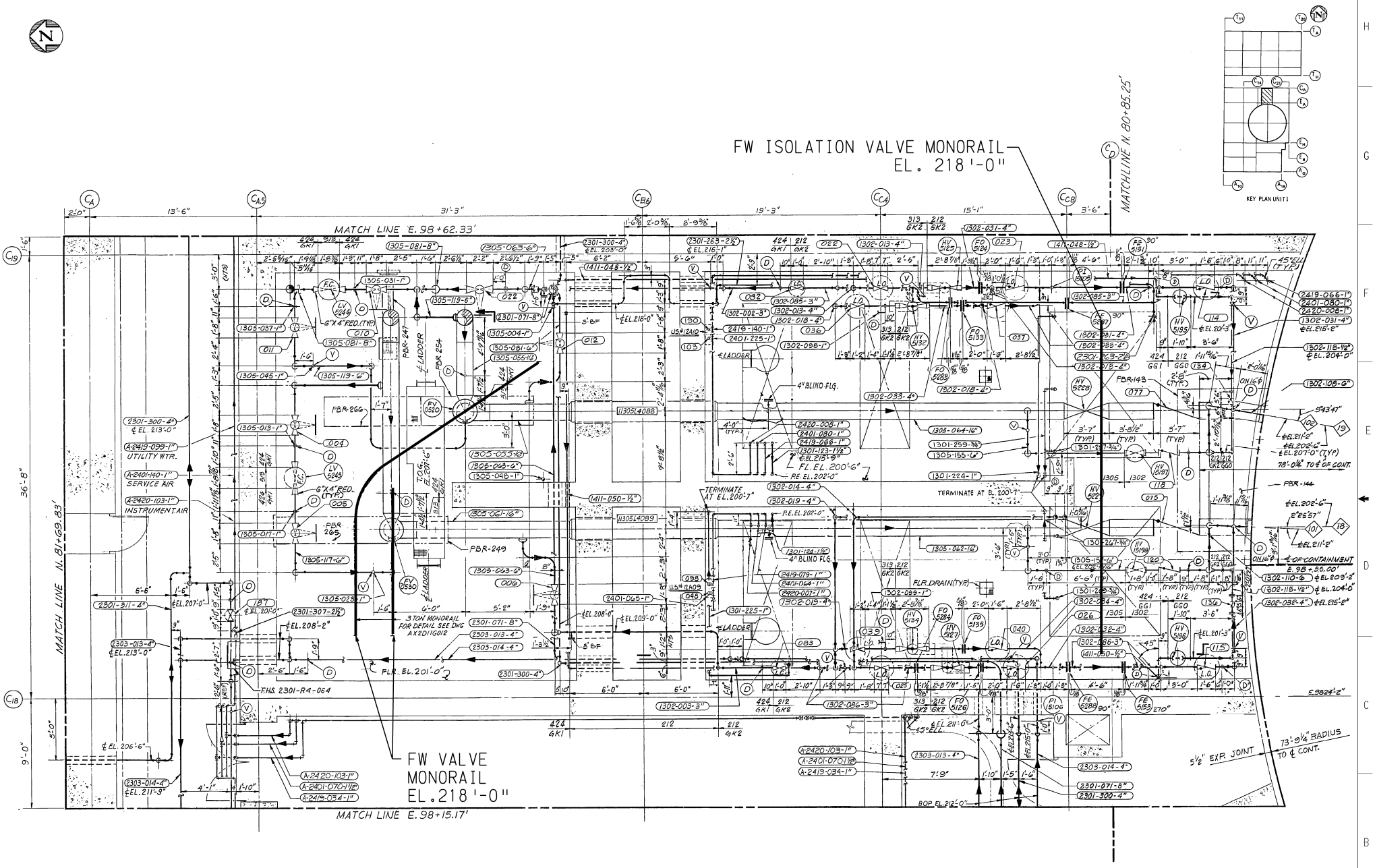
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
CONTROL BUILDING PIPING
AREA 2F LEVEL 1
EL. 220'-0" AND ABOVE

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	1X4DE516	1

SIZE E 34X44

CAD NAME 1X4DE516



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
NORTH FEEDWATER VALVE
ROOM PIPING AREA 2E LEVEL A
EL. 200'-0" TO EL.220'-0"

SCALE: NONE | DRAWING NO. 1X4DE517 | REV. 0
JOB NO.10604
SIZE E 34X44
CAD NAME 1X4DE517

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	ELC	RBH	CBH	WFP

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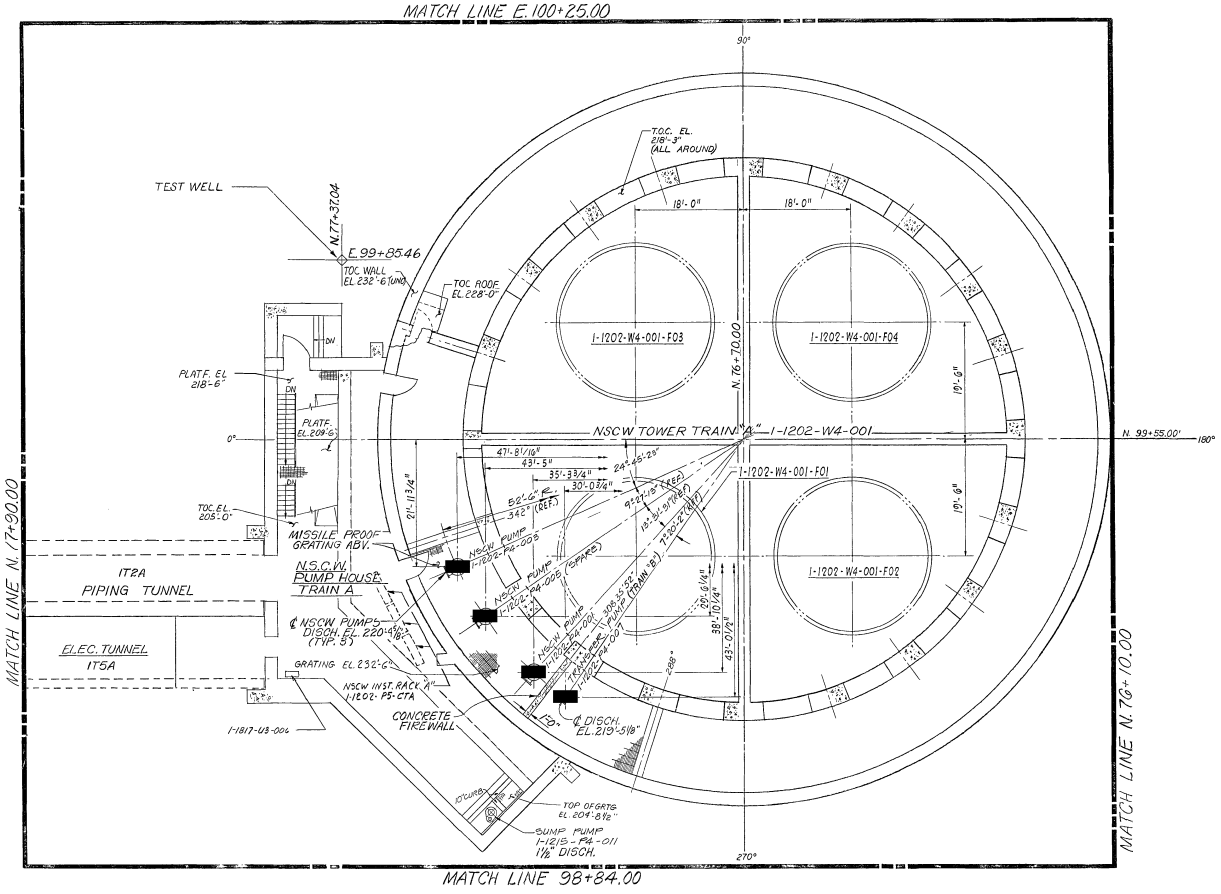

LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
TRAIN A
NUCLEAR SERVICE COOLING WATER TOWER
LEVEL 1, EL. 220'-0" (UNIT 1)

SCALE: NONE | DRAWING NO. 1X4DE518 | REV. 0
JOB NO. 10604
SIZE E 34X44
CAD NAME 1X4DE518

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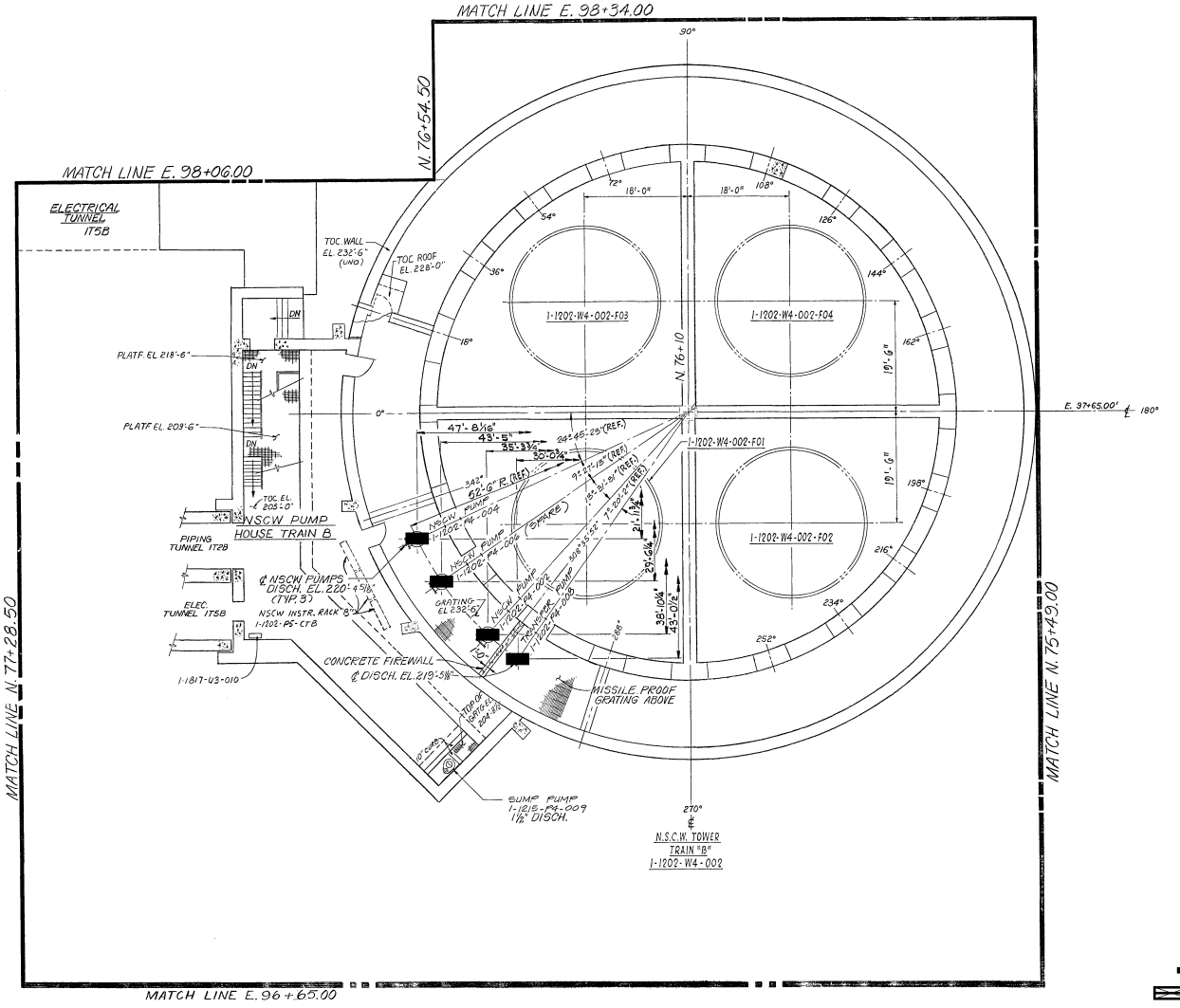
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	JLH	RBH	CBH	WFP



LEGEND:

- ■ ELETRIC HOIST
- ● HAND HOIST WITH CEILING RING
- ▲ HAND HOIST WITH FLOOR MOUNTED TRIPOD
- ▬ MOBILE TRUCK CRANE
- MONORAIL AND LOAD PATH AREA
- JIB CRANE AND LOAD PATH AREA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

HEAVY LOAD HANDLING
TRAIN B
NUCLEAR SERVICE COOLING WATER TOWER
LEVEL 1 EL. 220'-0" (UNIT 1)

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE519	0

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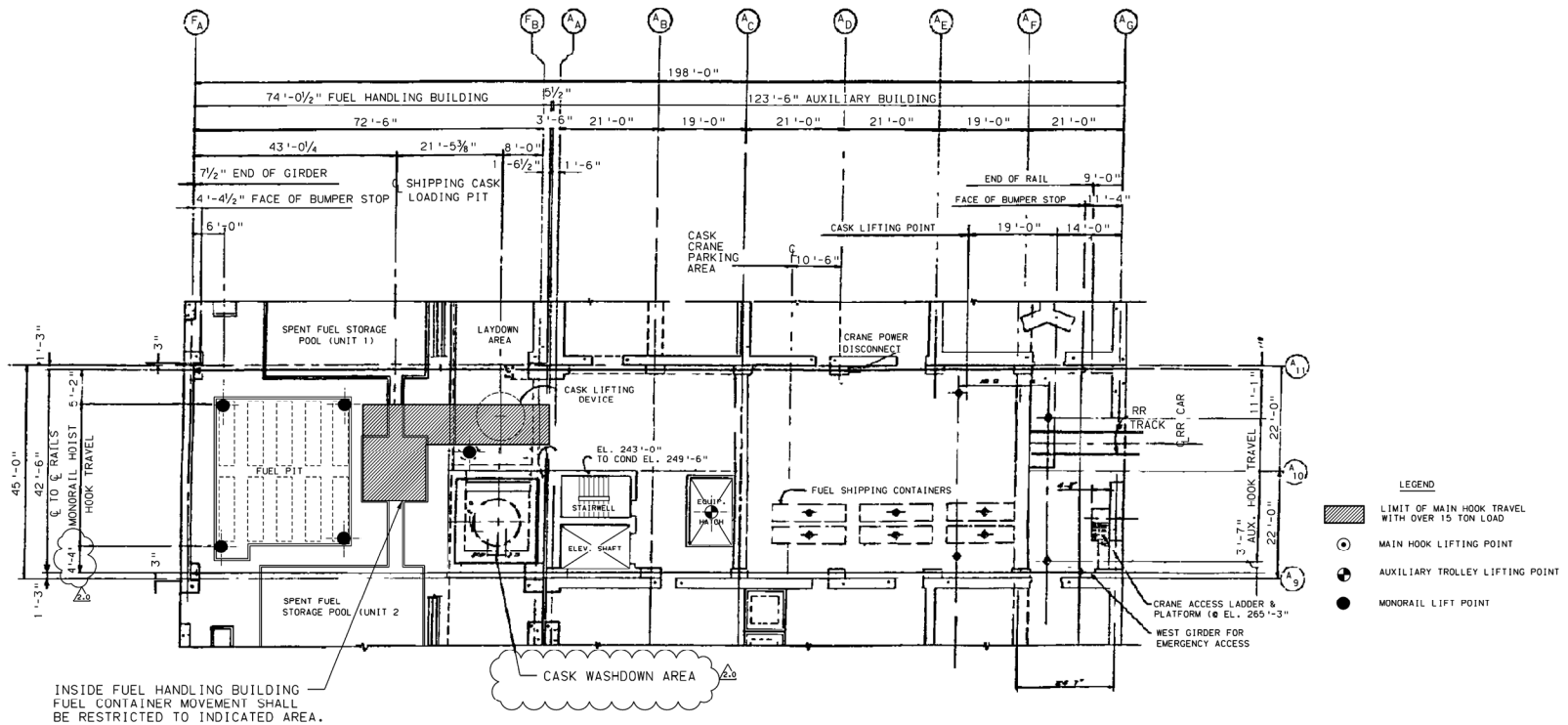
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-28-98	JLH	RBH	CBH	WFP

SIZE E 34X44

CAD NAME 1X4DE519



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NO.	VERSIONS	DATE	DR	CHK	APPV
2.0	REVISED PER ABN SNC142974M025, VER. 2.0	11/12/15	JLO	JMM	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SAFE LOAD PATH FOR MOVEMENT OF FUEL CONTAINERS UTILIZING SPENT FUEL CASK BRIDGE CRANE A-2109-R4-001

SCALE: NONE
DRAWING NO.: 1X4DE603
VER.: 2.0
JOB NO.10604
SIZE E 34x44

NOTES: (UNIT 1 ONLY)
1. LIFTS IN THE MEZZANINE AREA OVER THE SEAL INJECTION FILTER PITS, R-B151 OR R-B152, WHEN CONCRETE PLUGS ARE NOT INSTALLED SHALL BE LIMITED TO THE EQUIPMENT REQUIRED TO PERFORM CARTRIDGE FILTER ELEMENT REPLACEMENTS (I.E. WORKING SHIELD PLUG, CENTRAL PLUG, AND TRANSFER CASK).

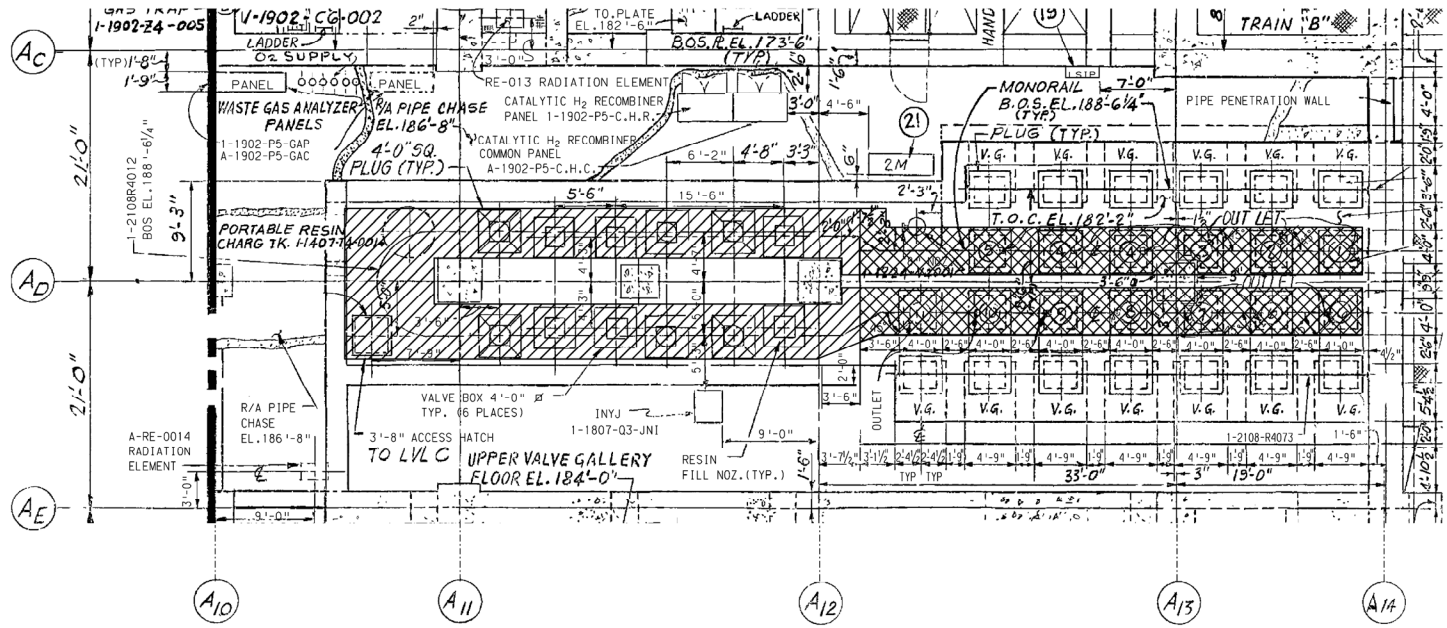
1.0

LEGEND

THE LOAD IN THIS AREA SHALL NOT BE LIFTED HIGHER THAN 3 FEET (AS MEASURED FROM SLAB TO BOTTOM OF LOAD).

THE LOAD HEIGHT IN THIS AREA IS NOT LIMITED.

1.0



1.0

NOTE: UNIT 1 SHOWN. UNIT 2 IS A MIRROR IMAGE.

NO.	VERSIONS	DATE	DR	CHK	APPV
1.0	REVISED PER ABN SNC517148C010, VER.1.0	09/15/14	JLO	CYN	TSL

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SAFE LOAD PATH FOR BACKFLUSHABLE FILTER MONORAIL
AUXILIARY BUILDING
LEVEL B MONORAIL 2108-R4-012

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X4DE605
VER. 1.0
SIZE D 22X34

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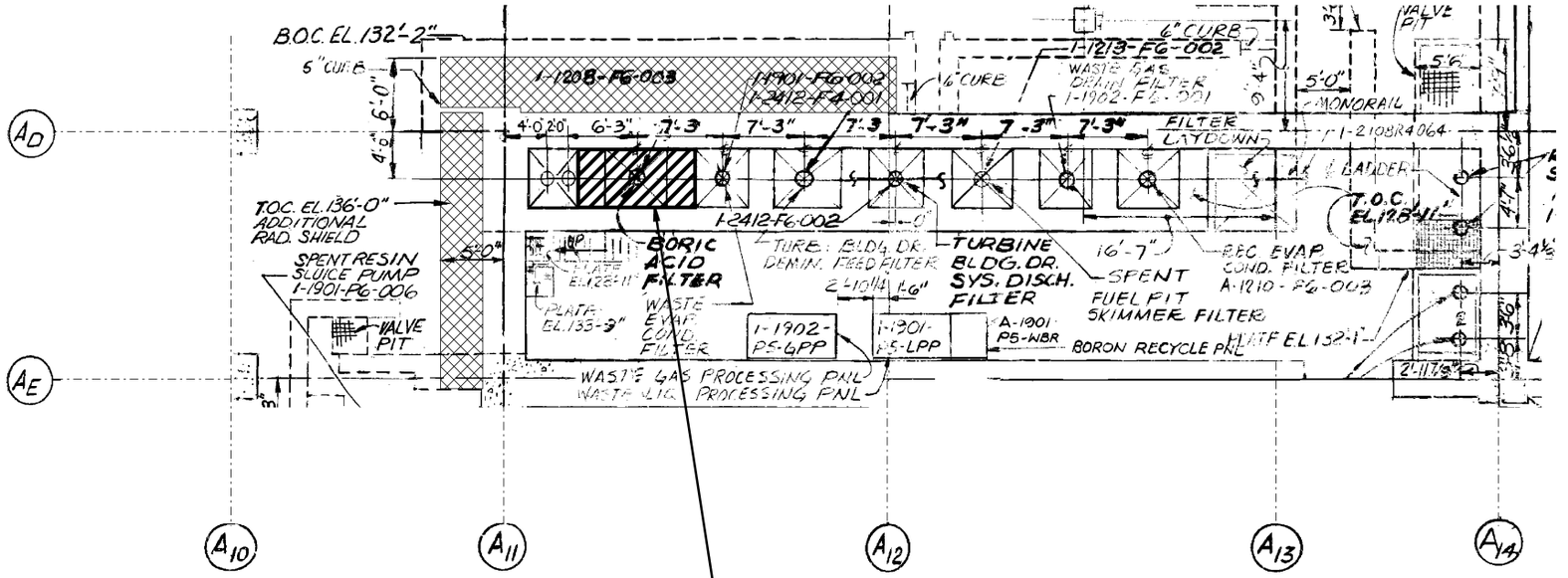


IN THE INDICATED AREA THE LOAD
SHALL NOT BE LIFTED HIGHER THAN 1 FOOT
(AS MEASURED FROM SLAB TO BOTTOM OF LOAD)

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	REVISED PER REA 96-VAA054	10-1-98	GS	RBH	CBH	
0	ISSUED PER REA 96-VAA054	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

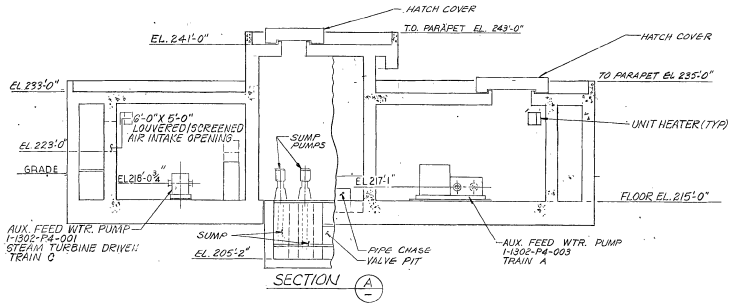
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

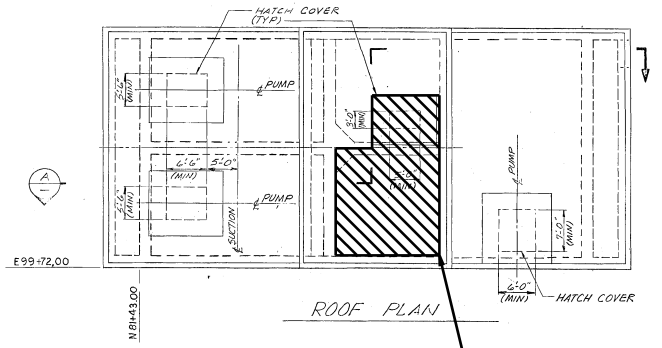
SAFE LOAD PATH FOR CARTRIDGE FILTER MONORAIL
AUXILIARY BUILDING
LEVEL D MONORAIL 2108-R4-064
(UNIT 1)

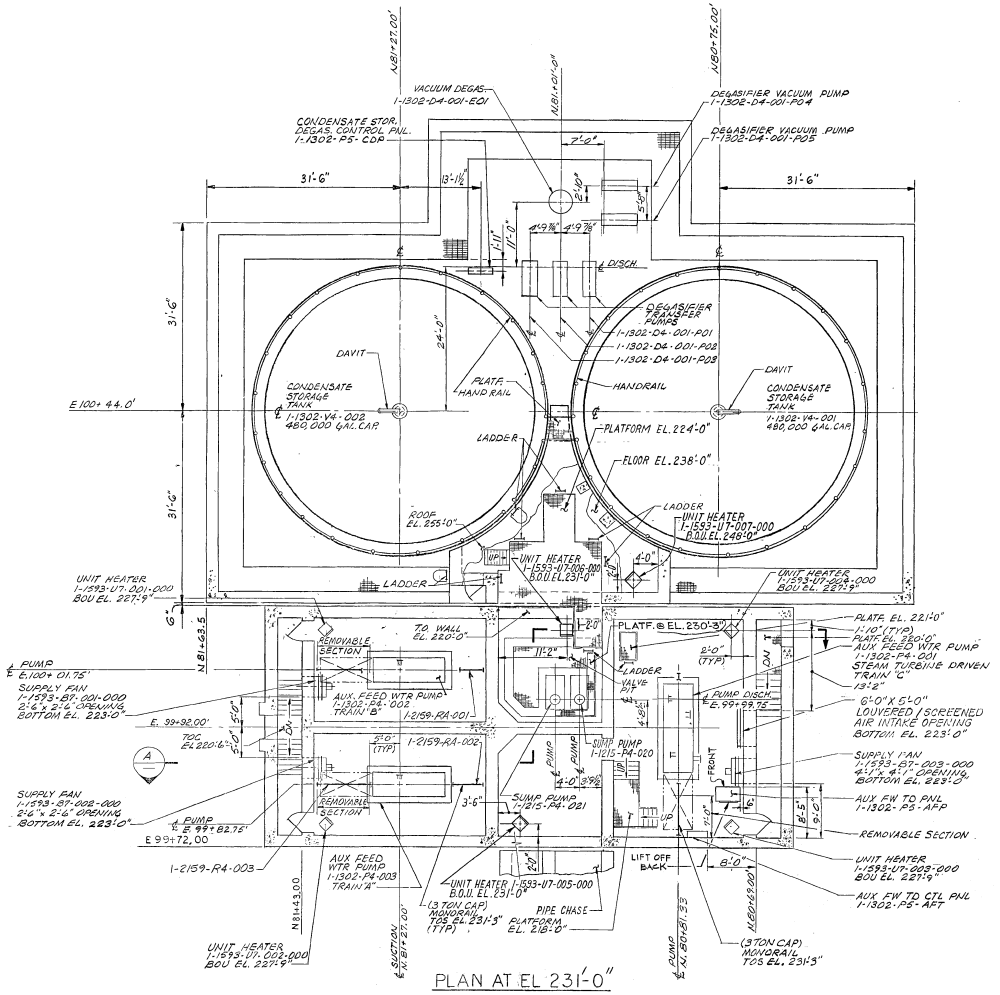
SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE606	1

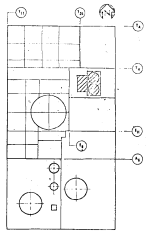
SIZE D 22X34

CAD NAME 1X4DE606









THE LOAD SHALL BE LIFTED ONLY IN THE INDICATED AREA AND SHALL NOT BE LIFTED HIGHER THAN 2 FT. (AS MEASURED FROM THE SLAB TO THE BOTTOM OF THE LOAD).

NOTE: UNIT 1 SHOWN. UNIT 2 IS A MIRROR IMAGE.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SAFE LOAD PATH
FOR AUXILIARY FEEDWATER PUMPHOUSE
SUMP PUMP HATCHES

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DE607	0

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2						
1	ISSUED PER REA 96-VAA054	7-26-96	SRM	RBH	CBH	WFP

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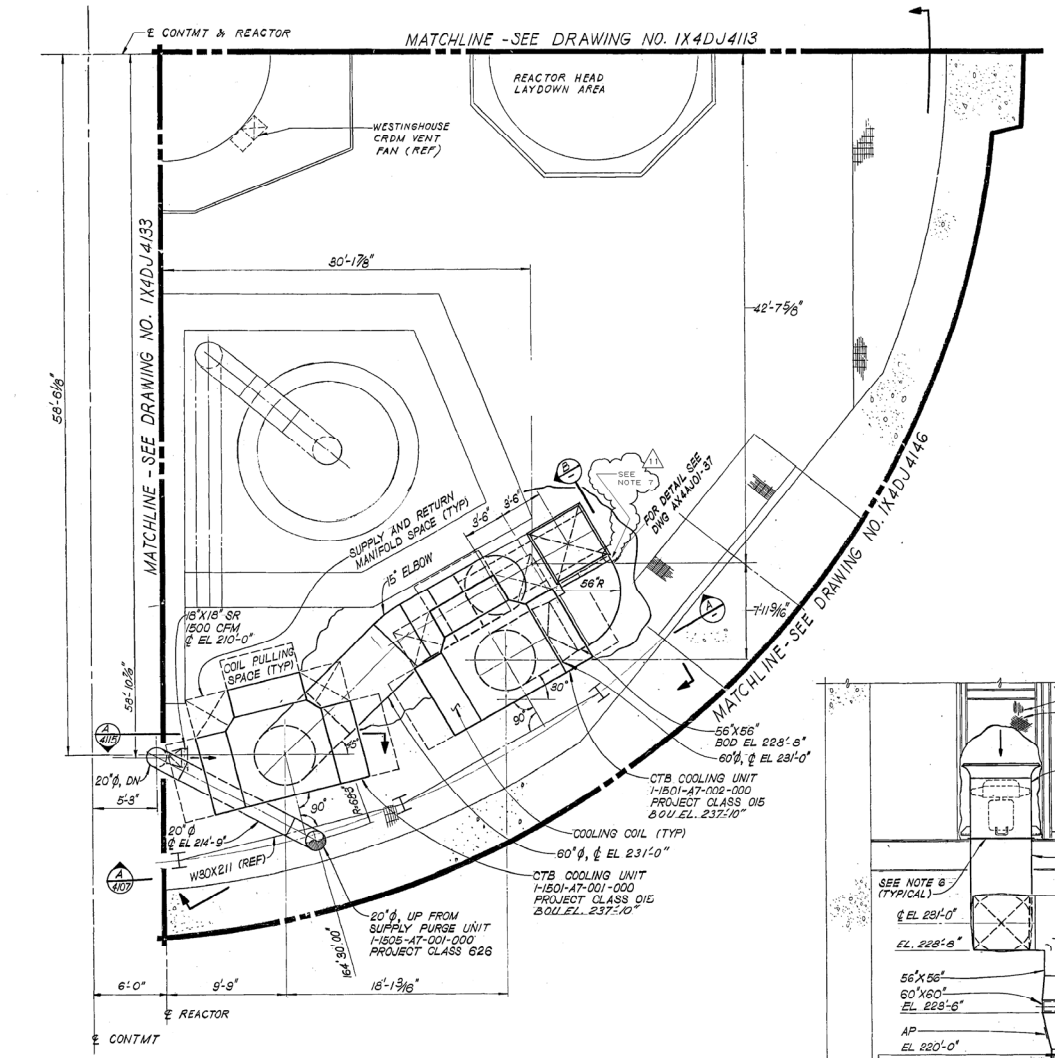
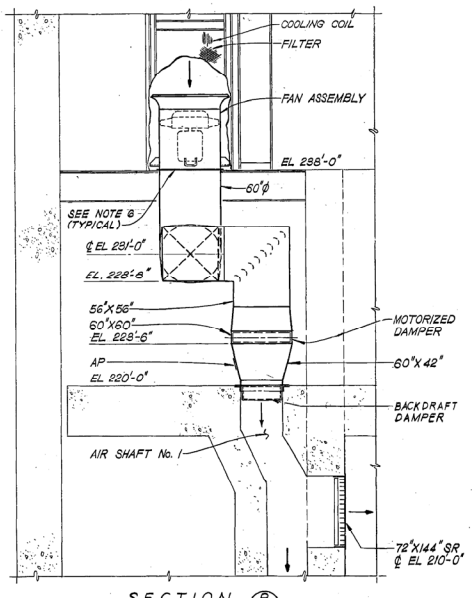
SIZE E 34X44

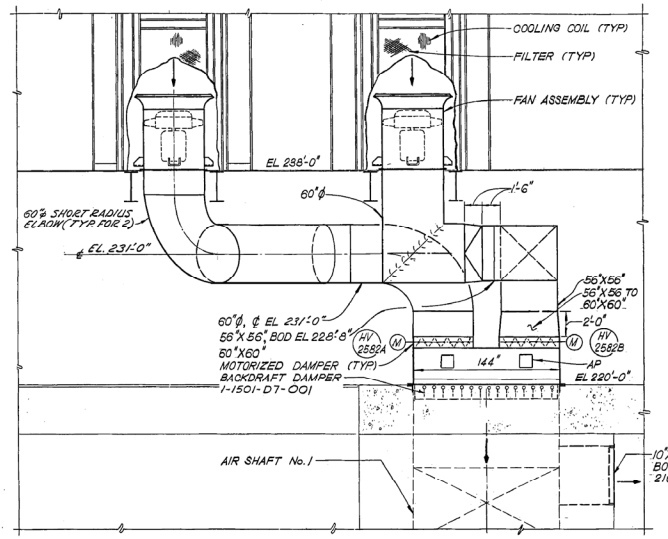
CAD NAME 1X4DE607



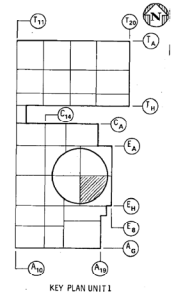
PLAN - EL. 210'-0" TO 238'-0"
SCALE: ¼"=1'-0"



SECTION (B)
SCALE: ¼"=1'-0"



SECTION (A)
SCALE: ¼"=1'-0"



NOTES:

1. FOR GENERAL NOTES AND LEGEND SEE DWG AX4DJ0001 AND AX4DJ0002.
2. FOR AREA ABOVE SEE DWG 1X4DJ4102.
3. FOR AREA BELOW SEE DWG 1X4DJ4105.
4. ALL DUCTWORK SHOWN IS PROJECT CLASS 01S UNLESS OTHERWISE NOTED.
5. FOR FLOW TRANSMITTERS LOCATIONS SEE CONTROL BLDG. DRAWING 1X4DJ2111
6. DUCTWORK SHALL BE BOLTED TO THE BOTTOM OF SYSTEM 1-1501-A7-001-000 & 002-000 UNITS. BOLT PATTERN TO CONFORM TO DWG AX2D67W001 REQUIREMENTS.
7. SEE MDC 97-V1M058 FOR TURNING VANE GAUGE AND STIFFENER DETAILS.

REFERENCE DRAWINGS

P&ID	1X4DB212, 213 & 251
CIVIL	1X2D48A012
STRUCTURAL	1X2D48F115, F118
ARCHITECTURAL	
PIPING	1X4DL4A02, A03
PLUMBING	
ELECTRICAL	1X3DK513, K514
DUCT SUPPORT	

4121018067A
ACTIVITY PACKAGE NO. 412101111GA

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG
AREA 4A LEVEL 1
PLAN EL. 210'-0" TO 238'-0"

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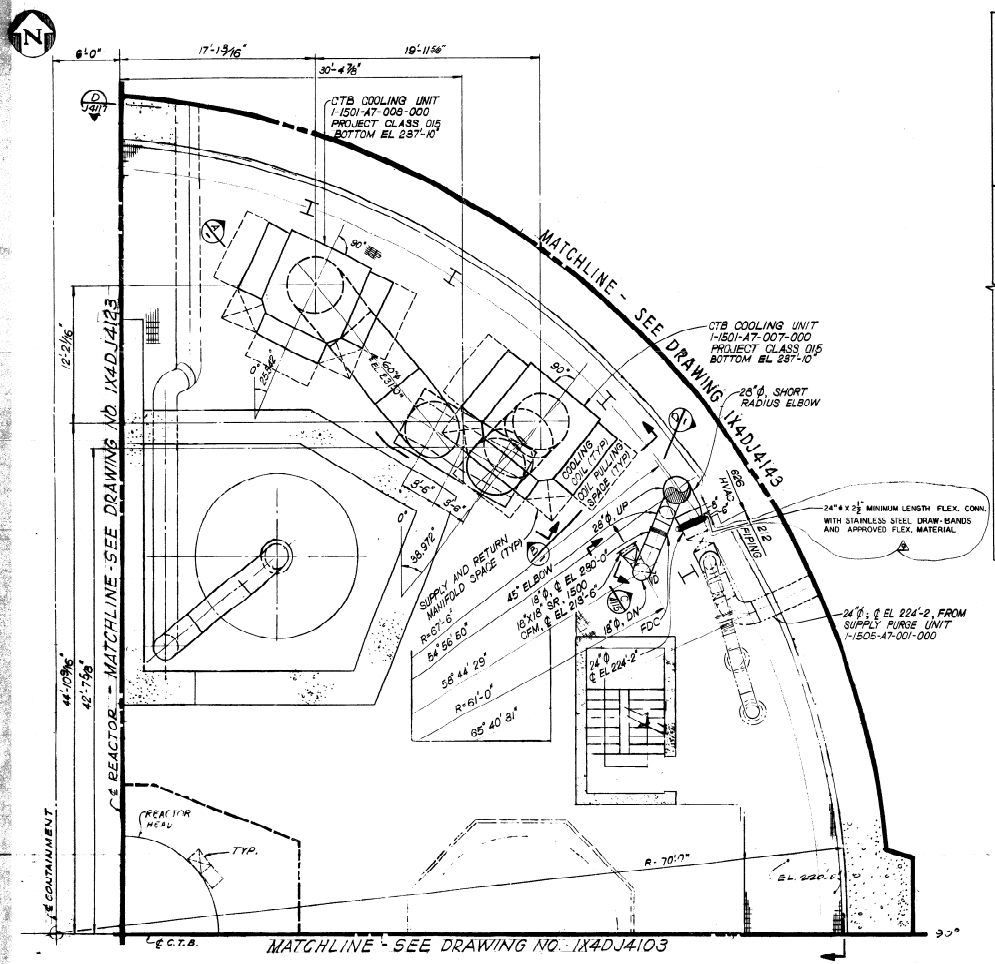
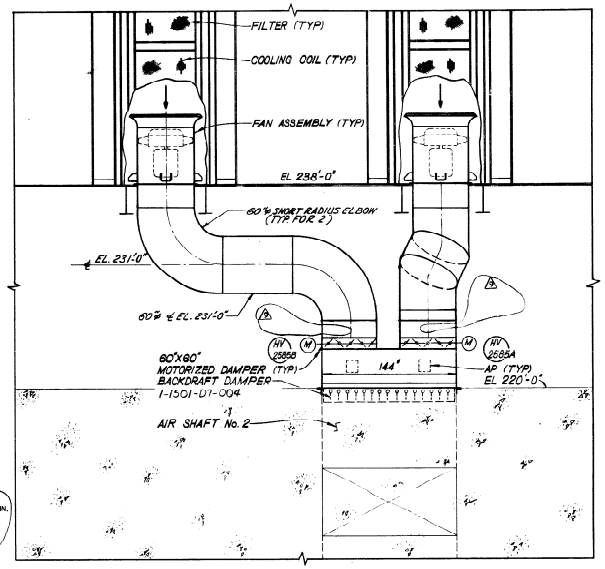
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	INCORP. PER ABN-42696	12-23-97	SRM	EJG	MWD	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

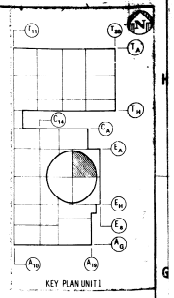
SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X4DJ4103
REV. 11
SIZE E 34x44
CAD NAME 14DJ4103

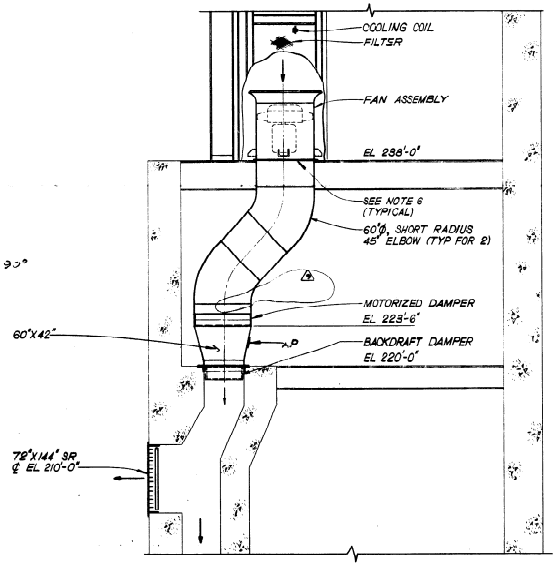


PLAN EL 210'-0" TO 238'-0"
SCALE: ¼"=1'-0"

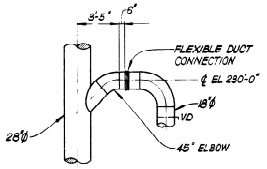


SECTION (A)
SCALE: ¼"=1'-0"





SECTION (B)
SCALE: ¼"=1'-0"



SECTION (D)
SCALE: ¼"=1'-0"

NOTES:

1. FOR GENERAL NOTES AND LEGEND SEE DWG AX4DJ0001 AND AX4DJ0002.
2. FOR AREA ABOVE SEE DWG 1X4DJ4112
3. FOR AREA BELOW SEE DWG 1X4DJ4115
4. ALL DUCTWORK SHOWN IS PROJECT CLASS 015 UNLESS OTHERWISE NOTED.
5. FOR FLOW TRANSMITTER LOCATION SEE CONTROL BLDG. DRAWING 1X4DJE2103.
6. DUCTWORK SHALL BE BOLTED TO THE BOTTOM OF SYSTEM 1-1501-A7-007-000 & 008-000 UNITS. BOLT PATTERN TO CONFORM TO DWG AX2D67W001 REQUIREMENTS.

REFFERENCE DRAWINGS

P&ID	1X4DB212, 213, 251 & 213-1
CIVIL	1X2D48A012
STRUCTURAL	1X2D48F114
ARCHITECTURAL	
PIPING	1X4DL802, 803
PLUMBING	1X4DH4103
ELECTRICAL	1X3DK503 & 513
DUCT SUPPORT	

4121013067A
ACTIVITY PACKAGE NO. 4121011116A

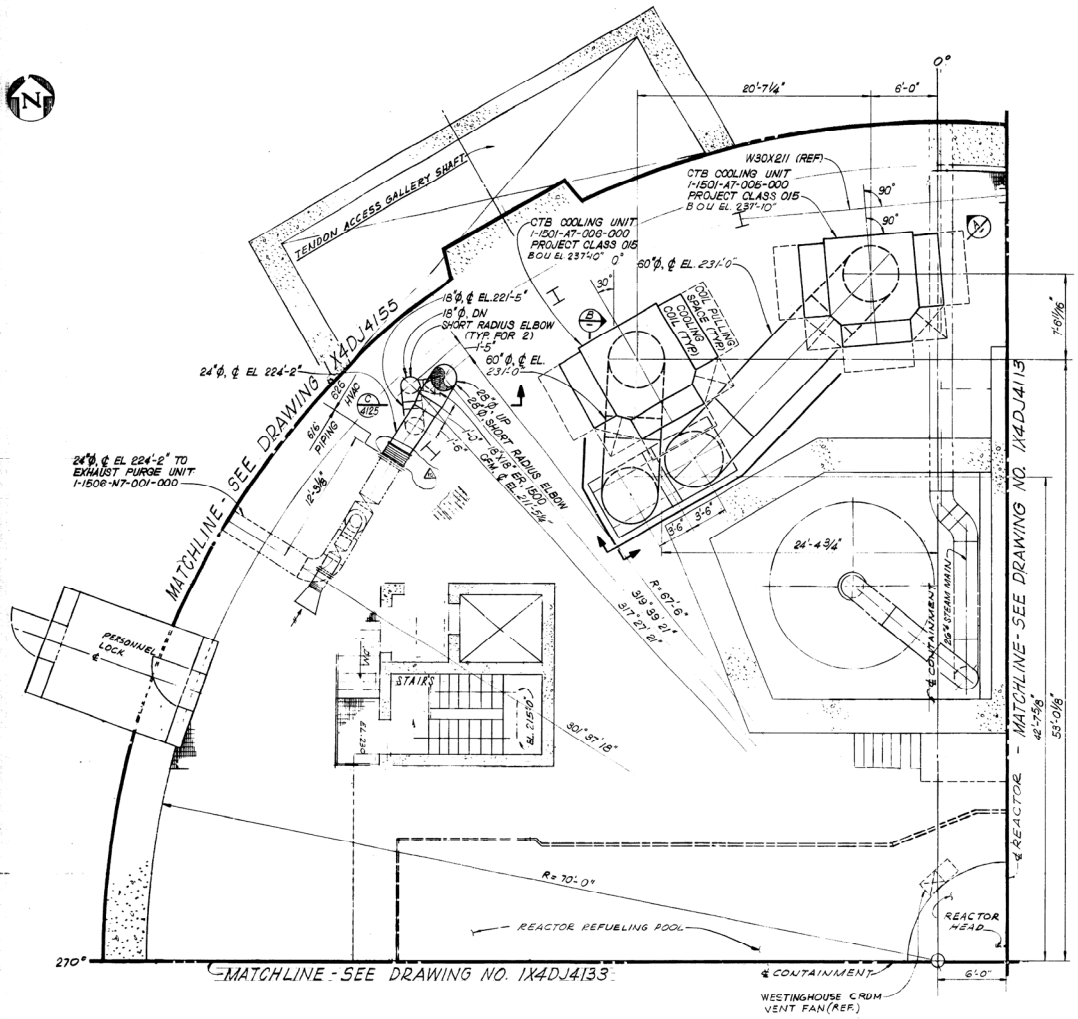
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

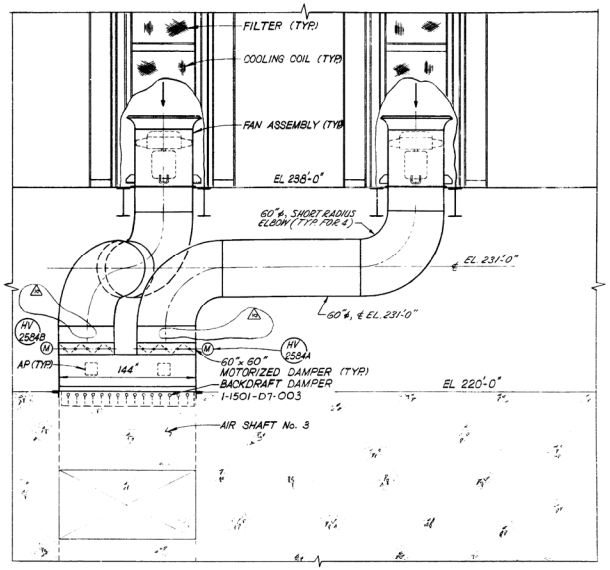
CONTAINMENT BLDG
AREA 4B LEVEL 1
PLAN EL. 210'-0" TO 238'-0"

SCALE: ¼"=1'-0" | DRAWING NO. 1X4DJ4113 | REV. 3 | JOB NO. 9510

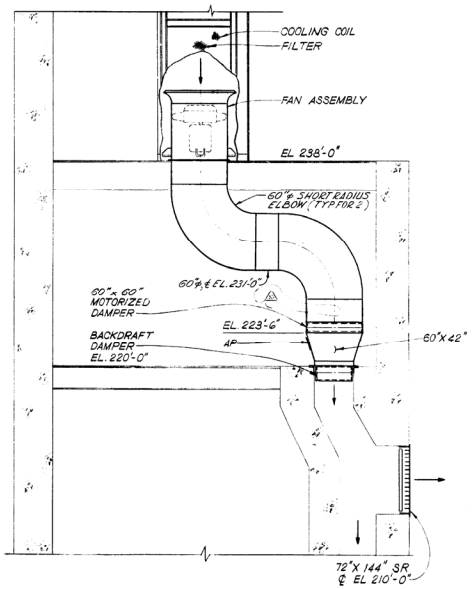
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
3	INCORP. DCN-R# 5 & 6	2-3-94	MT						
2	INCORP. DCN-R# 4 (C-FCRB-15,148)	2-3-94	MT						
1	INCORP. DCN-R# 3, REV AS SHOWN (TYP.)	7-12-94	MT						
0	ISSUED FOR CONSTRUCTION	2-3-94	MT						



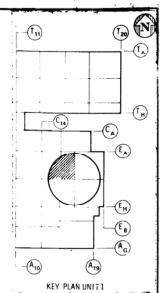
PLAN EL 210'-0" TO 238'-0"
SCALE: 1/4" = 1'-0"



SECTION A
SCALE 1/4"=1'-0"



SECTION B
SCALE 1/4"=1'-0"



NOTES:

1. FOR GENERAL NOTES AND LEGEND SEE DWG AX4DJ0001 AND AX4DJ0002.
2. FOR AREA ABOVE SEE DWG 1X4DJ4122.
3. FOR AREA BELOW SEE DWG 1X4DJ4125.
4. ALL DUCTWORK SHOWN IS PROJECT CLASS 015 UNLESS OTHERWISE NOTED
5. FOR FLOW TRANSMITTER LOCATION SEE CONTROL BLDG. DWG. 1X4DJ2111.
6. DUCTWORK SHALL BE BOLTED TO THE BOTTOM OF CTB COOLING UNITS 1-1501-A7-005-000 AND 1-1501-A7-006-000. PATTERN TO CONFORM TO DRAWING AX2DG7W001 REQUIREMENTS.

REFERENCE DRAWINGS

PAIR	1X4DB212, 213 & 251
CIVIL	1X2D483A14
STRUCTURAL	1X2D48F114
ARCHITECTURAL	
PIPING	1X4DL4C02, C03
PLUMBING	1X4DM4103
ELECTRICAL	1X8DK503 & 515
DUCT SUPPORTS	1X2D48W204 & 211

412101B057A
ACTIVITY PACKAGE NO. 4.12.10.1111.GA

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

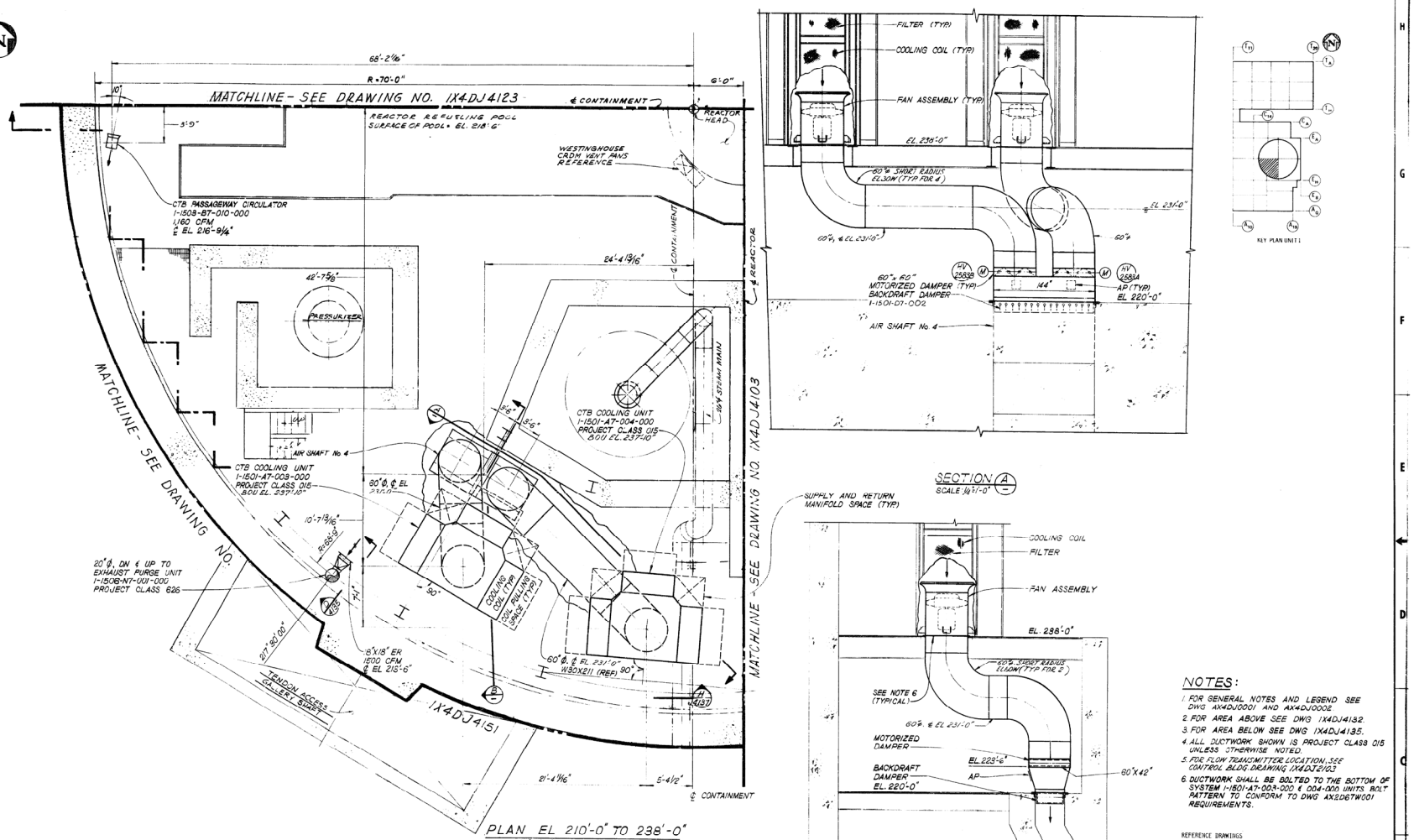
CONTAINMENT BLDG
AREA 4C LEVEL 1
PLAN EL. 210'-0" TO 238'-0"

SCALE: AS NOTED
DRAWING NO. 1X4DJ4123
REV. 10
JOB NO. 9510

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGL	EGS	CHF. E	P.E.	Q.A.E.
A	INCORP. M-FCRB-14955, M-FCRB-15302 & M-FCRB-15655.	11-14-89	C.T.B.	VL/L	REF.					
B	INCORP. DCN-R-# 9 (C-FCRB-18,625)	5-17-90	#	DM		AY		JR		
C	INCORP DCN-R-# 7 & 8	2-5-90	#	TY		LW		VL		
D	INCORP. DCN-R-# 6 (C-FCRB-18,141) (PFE)	9-13-90	#	M.S.F.		HT		VL		
E	INCORP. DCN-R-# 5 & 10 MOTORIZED DAMPER (ADDED NOTE 5)	10-16-90	#	M.S.F.		HT		VL		
F	INCORP. DCN-R-# 11 & 13 ADDED SECTION B. REVISED WP DWG. NO. REVISED 3D ROUTING	10-16-90	#	HT		VL		VL		
G	ADDED DUCTWORK PENETRATION. DUCTWORK REF. DWGS (ADDED	11-13-90	#	M.S.F.		HT		VL		
H	RELOCATED EQUIPMENT & REVISED DUCTWORK	12-5-90	#	M.S.F.		HT		VL		
I	DIMENSIONS REVISED & REGISTERS ADDED	2-27-90	R.E.	DC		VL		VL		
J	ISSUED FOR CONSTRUCTION	2-5-91	R.E.	DC		VL		VL		

ENG. MGR. DATE

SIZE E 36 x 44



NOTES:

1. FOR GENERAL NOTES AND LEGEND SEE DWG AX4DJ0001 AND AX4DJ0002.
2. FOR AREA ABOVE SEE DWG IX4DJ4132.
3. FOR AREA BELOW SEE DWG IX4DJ4135.
4. ALL DUCTWORK SHOWN IS PROJECT CLASS 015 UNLESS OTHERWISE NOTED.
5. FOR FLOW TRANSMITTER LOCATION, SEE CONTROL BLDG. DRAWING IX4DJ2103.
6. DUCTWORK SHALL BE BOLTED TO THE BOTTOM OF SYSTEM 1-1501-A7-003-000 & 004-000 UNITS. BOLT PATTERN TO CONFORM TO DWG AX2D6TW001 REQUIREMENTS.

REFERENCE DRAWINGS

P&ID	IX4DB212, 213 & 251
CIVIL	IX2D48A015
STRUCTURAL	IX2D48F118, F113
ARCHITECTURAL	
PIPING	IX4DL4002, D03
PLUMBING	
ELECTRICAL	IX3DK518, K514
DUCT SUPPORT	

4121011B067A
ACTIVITY PACKAGE NO. 4121011116A

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG
AREA 4D LEVEL 1
PLAN EL. 210'-0" TO 238'-0"

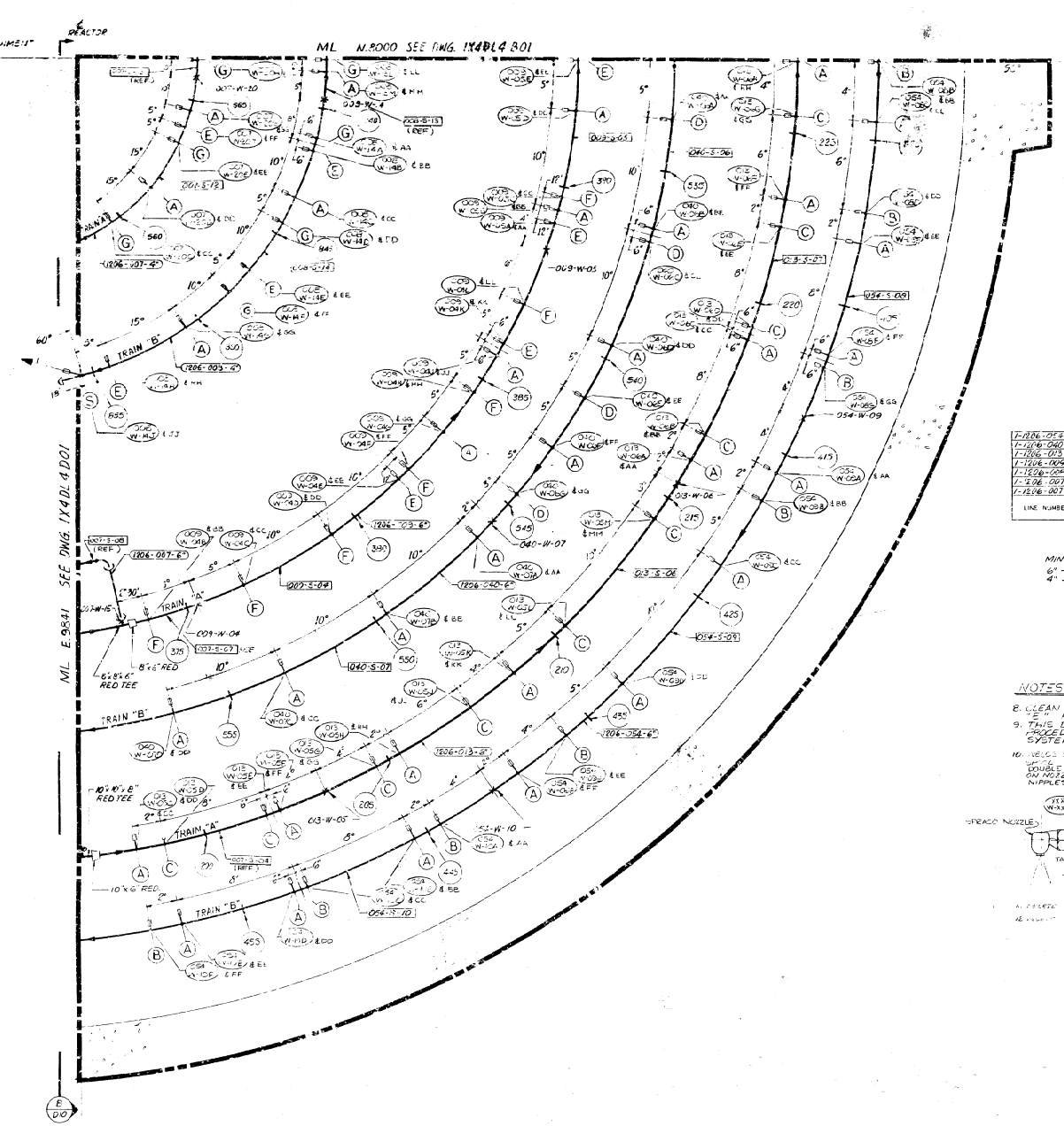
SCALE: ¼"=1'-0" | DRAWING NO. IX4DJ4133 | REV.

JOB NO. 9516

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF. E	P.E.	Q.A.E.
5	INCORP DCN-R# 5 & 6	2-5-86	#	JV	KAC	MM			
4	INCORP DCN-R # 4, REV AS SHOWN	7-11-84	#	BWC	KAC	DLO			
3	INCORP DCN 3 (ISSUED BY PRE)	11-20-85	#	DMC	KAC	SJ			
2	REVISED PER DCN-R#1 & 2, ADDED SYMBOLS, CHANGED SIZE OF DUCTWORK, CHANGED FAN & DAMPER	6-25-85	#	HT	KAC	SJ			
1	ISSUED FOR CONSTRUCTION	2-5-85	M.S.A.	GAA	KAC	DG			
0	REVISED DUCT LOCATION, DUCTWORK & ADDED INSTRUMENTS	8-16-84	M.S.A.	HT	KAC	DG			
1	RELOCATED EQUIPMENT & REVISED DUCTWORK	1-22-84	#	HT	KAC	DG			
2	DUCT DIMENSION & AZIMUTH REVISED	9-27-83	R.T.C.	DC	KAC	DG			
3	ISSUED FOR CONSTRUCTION	2-28-83	M.S.A.	GAA	KAC	DG			
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF. E	P.E.	Q.A.E.

ENG. MGR. DATE

SIZE E 36 x 44


CONTAINMENT
REACTOR
ML N.8000 SEE DWG. 1X4DL4B01
ML E.9641 SEE DWG. 1X4DL4D01
TRAIN "A"
TRAIN "B"
10"x10"x6" RED TEE
8"x6" RED
10"x6" RED
-009-W-05
-009-W-04
-009-W-06
-009-W-07
-009-W-08
-009-W-09
-009-W-10
-009-W-11
-009-W-12
-009-W-13
-009-W-14
-009-W-15
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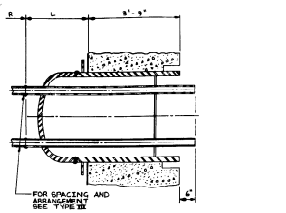
-009-W-37
-009-W-38
-009-W-39
-009-W-40
-009-W-41
-009-W-42
-009-W-43
-009-W-44
-009-W-45

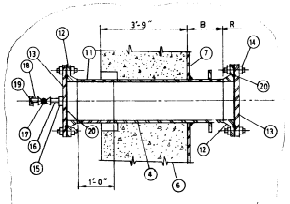
SEE DWG. 1X4DLA13

PENETRATION NO.	LINE NO. SYSTEM	SEQ. NO.	NOM. SIZE	PIPE SPEC.	SYSTEM DESCRIPTION	TYPE	INSERVICE INSPECTION (SEE NOTE#1)	NORM. OPERATING PRESS. (PSI.G)	NORM. OPERATING TEMP. (°F)	MAX. DESIGN PRESS. (PSI.G)	MAX. DESIGN TEMP. (°F)	PIPE O.D. (INCH)	PIPE I.D. (INCH)	WALL THK.	MATERIAL (A.S.M.E. SPEC. NO.) (SEE NOTE#15) ①⑤ PIPE REFERENCE	② FLUED HEAD SEE NOTE#15	③ INTERMEDIATE SLEEVE	④ PENETRATION SLEEVE	PENETRATION ASSEMBLY DIMENSIONS(INCHES) (SEE NOTE#15) L	B	C	D	t	℄ ELEVATION	℄ AZIMITH	P. & I.D. DWG. NO.	MECH. DWG. NO.	RADIUS "R"	PENETRATION NO.
70 (J)	1513	002	3/4"	LL1	CONTAINMENT HYDROGEN MONITOR-PE-HMB (OUT)	III	VISUAL	80	340	80	340	1.050	0.824	SCH.40S	SA-312, TP304L	SA-182, GR F304L	SA-312, TP304L	SA-333, GR.6	12.750	6.250	6.000	12.750	0.375	210'-1"	233°30'46"	1X4DB213-2	1X4DL4D03	68'-11½"	70A
(J)	↓	006	↓	↓	(IN)	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	B
(J)	1206		1½"	SEE NOTE#7	CONTAINMENT PRESS. INSTR. IPT-0936 (NOTE 7)			0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L										1X4DB131			C
(J)	1513	014	3/4"	LL1	CONTAINMENT HYDROGEN MONITOR-PE-HMA (OUT)			80	340	80	340	1.050	0.824	SCH.40S	SA-312, TP304L									189'-6"	153°14'33"	1X4DB213-2	1X4DL4D06		71A
(B)	↓	009	↓	↓	(IN)	↓	↓	80	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	B
(C)	1206		½"	SEE NOTE#7	CONTAINMENT PRESS. INSTR. IPT-0935 (NOTE 7)			0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L										1X4DB131			C
72 (A)	1204	106	3/4"	HG0	LOOP1 ACCUMULATOR VG-002 SAMPLE			100	120	100	300	1.050	0.824	SCH.40S	SA-312, TP304L									181'-4½"	177°34'03"	1X4DB120	1X4DL4D07		72A
(B)	↓	108	↓	↓	LOOP3 VG-004	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	B
(C)	—	—	—	SEE NOTE#11	SPARE NECK ASS'Y.			—	—	—	—		0.612	SCH 160	SEE NOTE #18													C
73 (A)	1204	107		HG0	LOOP2 ACCUMULATOR VG-003 SAMPLE			100	120	100	300		0.824	SCH.40S	SA-312, TP304L										180°	1X4DB120			73A
(B)	↓	109	↓	↓	LOOP4 VG-005	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	B
(C)	↓	↓	↓	SEE NOTE#11	SPARE NECK ASS'Y	↓	↓	—	—	—	—	↓	0.612	SCH 160	SEE NOTE #18	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	C
74	—	—	—	—	NOT USED	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	74
75	2830,32 34,36	64 66	2"-2" 2"-4"	—	BULK GASES SUPPLY SYSTEM TEMPORARY	V	N/A	—	—	—	—	—	—	—	—	—	—	SA-333, GR.6	—	8.250	—	10.750	0.365	201'-11½"	246°6'28"	AX4DB012 022,032,042	1X4DL4D04	—	75
76	—	—	—	—	SPARE PENETRATION SLEEVE	V	—	—	—	—	—	—	—	—	—	—	—		—	8.250	6.000			189'-6"	182°25'57"	—	1X4DL4D06	—	76
77	1901	022	3"	LL1	RCDT-EE-001 HT EX. D SCH TO WPS WASTE INCLUDE TANK T-002	II	VISUAL	125	130	150	250	3.500	3.068	SCH. 40S	SA-312, TP304L	SA-182, GR F304L	SA-312, TP304L		15.750	6.250	6.000			201'-11½"	245°40'31"	1X4DB 127	1X4DL4 D04	68'-8½"	77
78	1214	028	2"		CONTAINMENT SUMP PUMPS DISCH. TO WPS. FOL. T-002 & WHT T-002			57	AMB	60	AMB	2.375	2.067											202'-6"	153°41'33"	1X4DB 143	1X4DL4A04		78
79	1901	029	3/4"	LL2	RCDT T-001 H2 SUPPLY & WASTE GAS REMOVAL			02	120	150	250	1.050	0.742	SCH.80	SA-106, GR.B	SA-350, LF2	NONE		12.000					201'-11½"	243°14'34"	1X4DB 127	1X4DL4D04	68'-0¼"	79
80	2401	050	4"	LL1	SERVICE AIR SUPPLY			120	115	132	115	4.500	4.026	SCH.40S	SA-312, TP304	SA-182, GR F304L			16.000			18.000	0.500	189'-6"	181°17'51"	1X4DB186-1	1X4DL4D06	68'-8¼"	80
81	2420, 043		2"		INSTRUMENT AIR							2.375	2.067	SCH40S					12.000			10.750	0.365	184'-5/56	184°5'56	1X4DB186-2		68'-0¼"	81
82	—	—	—	—	SPARE PENETRATION SLEEVE	V	N/A	—	—	—	—	—	—	—	—	—	—		—	8.250	—	16.000	0.500	216'-1"	245°40'31"	—	1X4DL4D03	—	82
83	1505	003	24"	LL1	CTB NORMAL PROCESS PURGE SUPPLY	NOTE 6	VISUAL	01	98	80	340	24.000	23.250	STD.WT.	SA-312, TP304	—	NONE		10.000	—	—	24.000		224'-2"	65°40'31"	1X4DB2134	1X4DJ4113 1X4DL4B02	69'-2¼"	83
84	1506	002			EXHAUST			01	120							—				—	—				30°37'18"		1X4DJ4123 1X4PLA602		84
85 (A)	—	—	¾"	SEE NOTE#11	SPARE NECK ASS'Y.	III	VISUAL	—	—	—	—	1.050	0.612	SCH 160	SEE NOTE #18	SA-182, GR F304L	SA-312, TP304L		12.750	6.250	6.000	12.750	0.375	182 3½	250°30'25"	—	1X4DL4D07	68'-11½"	85A
(B)	—	—	—	—			VISUAL	—	—	—	—	—	—	—	—	—	—									—			B
(C)	1206		½"	SEE NOTE 1	CONTAINMENT PRESSURE INSTR. IPT-0936 (NOTE 7)		VISUAL	0	100	60	320	0.840	0.252	SCH XXS	SA-312, TP304L										1X4DB131			C
86 (A)	2702	002	½"	LL1	POST ACCIDENT SAMPLING SYSTEM	III		50	300	65	400	0.840	0.622	SCH 80S	SA-312, TP304L	SA 182, GR304L	SA-312, TP304L		23.67	8.250				235°56'48"	1X4DB110		68'-0 9/16	86A
(B)	—		½"	SEE NOTE #11	SPARE			—	—	—	—	—	—	SCH40S											—			B
(C)	2702	008	1"	FG0	POST ACCIDENT SAMPLING SYSTEM			2235	617	2485	650	1.315	0.815	SCH 160	SA-312, TP304L	SA-182, GR304L	SA-312, TP304L								1X4DB110			C
87	1222	015	8"	LL2	CONTAINMENT LEAK RATE TEST	II	VISUAL	60	100	60	100	8.625	7.981	SCH40	SA-106, GRB	SA-350, LF2	SA-106, GRB	SA-333, GR 6	16.000	6.250	6.000	18.000	0.500	189'-6"	167°50'19"	1X4DB132	1X4DL4A06	68'-8¼"	87
88	1201		½"	FG2	RESERVED FOR RVLIS	VII	VISUAL	2235	120	2485	320	0.840	0.252	XXS	SA-312, TP304L	SA-182, F304L	NONE		15.25	8.250	6.000	10.750	0.365	189'-1½"	226°8'52"	1X4DB113	1X4DL4D06	68'-0"	88
89	—	—	26"	—	FUEL TRANSFER TUBE (SEE NOTE#2&#8)	—	—	—	—	—	—	—	—	—	—	—	—	SA-516, GR 70	—	—	—	—	—	186'-9⅞"	—	1X4DB130	1X4DL4D06	SEE NOTE#2	89
90	—	—	—	—	DEDICATED PENETRATION	VII	VISUAL	—	—	—	—	—	—	—	—	—	—	SA-333, GR 6	—	8.250	6.000	10.750	0.365	185'-6"	155°4'36"	—	1X4DL4A06	68'-9¾"	90
91	1202	216	8"	LL8	N.C.R.W. SUPPLY TO STMT. COOLER	II	VISUAL	145	95	200	280	8.625	7.981	SCH.40	SA-312, TP304L	SA-182, GRF304L	SA-312, TP304L		19.750	6.250	6.000	18.000	0.750	183'-1½"	245°40'31"	1X4DB135-2	1X4DL4D06	68'-6½"	91
92		209																							240°48'37"				92
93		188																						181'-4½"	148°28'39"	1X4DJ135-1	1X4DL4A07		93
94		196																						189'-6"			1X4DL4A06		94
95		220			RETURN FM				105															189'-1½"	246°6'28"	1X4DJ135-2	1X4DL4D06		95
96		212																							243°14'34"				96
97		001																						189'-6"	150°48'36"	1X4DJ135-1	1X4DL4A06		97
98	↓	192	↓	↓		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	181'-4½"	↓	↓	1X4DL4A07	↓	98
99	—	—	—	—	NOT USED	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	99
100	1508	001	4"	LL1	CTB POST LOCA PURGE EXHAUST	NOTE 6	VISUAL	03	150	80	340	4.500	4.026	SCH.40S	SA-312, TP304	--	NONE	SA-333, GR.6	0.250	—	—	4.500	0.237	264'-3"	177°34'-01"	1X4DB213-1	1X4DJ4132 1X4DL4D16	69'-3"	100
101	1302	110	6"	GA2	AUX. FEEDWATER	I	NOT REQ'D MET/RC	1150	445	1800	660	6.625	5.501	SCH 120	SA-106, GRB	SA-350, LF2	NONE		20.55	6.250	6.000	18.000	0.750	211'-2"	21°25'51"	1X4DB113-3	1X4DL4C03	68'-3 9/16	101
102		108	6"																					211'-2"	97°33'41"		1X4DL4B03		102
103		109	6"																					209'10"	177°34'01"		1X4DL4A03		103
104		107	6"																					207'-10	170°16'12"		1X4DL4A03		104



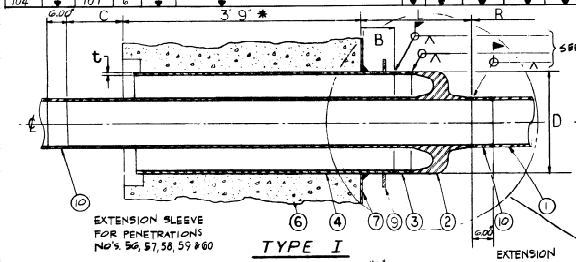
FOR NOTES AND COMPLETE DETAILED INFORMATION SEE VENDOR DWG. NO. 7-000685.

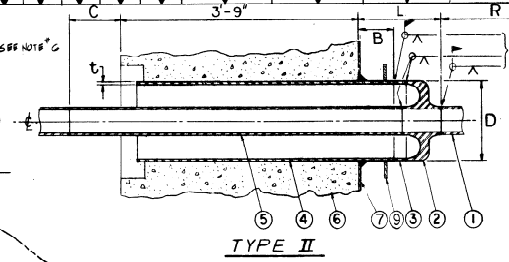
TYPE VII

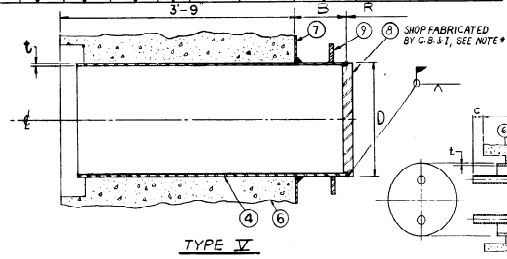


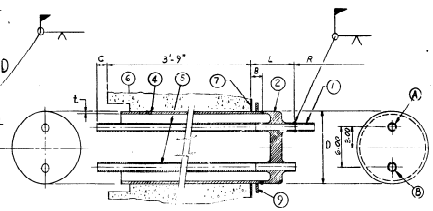
TYPE VII
(1, 5S,& 90)

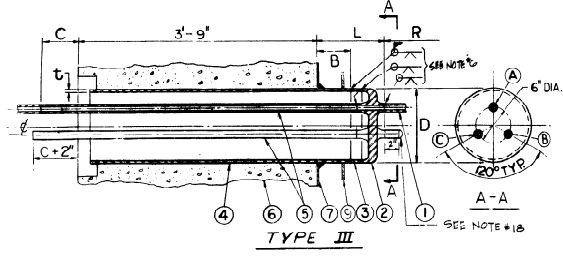
(CONTINUED FROM BELOW)
(11) EXTENSION SLEEVE FOR PENETRATION NO'S 5,55 & 90.
(12) WELDING NECK FLANGE("O"RING JOINTS).
(13) BLIND FLANGE-10"150 LBS. WITH"O"RING JOINTS.
(14) NUTS & BOLTS.
(15) HALF COUPLING.
(16) ¾"S X 5" LONG NIPPLE (P.B.E.).
(17) ¾" GLOBE VALVE.
(18) ¾"S X 3"LONG NIPPLE (T.O.E.).
(19) ¾"THREADED CAP
(20) RING JOINT GASKET.

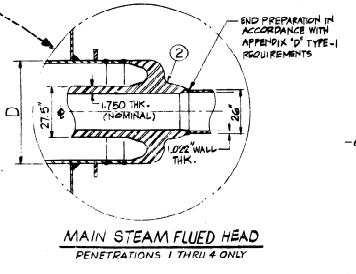


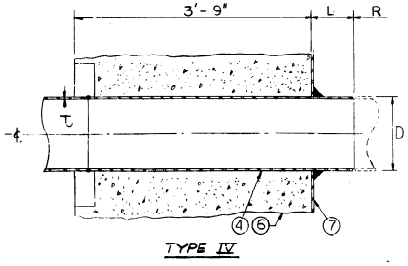












LEGEND
(1) CONNECTING PIPE OR COMPONENT
(2) FLUED HEAD FITTING
(3) INTERMEDIATE SLEEVE (NOTE #3 & #13)
(4) PENETRATION SLEEVE (NOTE #2)
(5) NECK EXTENSION (NOTE #11)
(6) CONTAINMENT WALL
(7) LINER PLATE (INSIDE CONTAINMENT WALL)
(8) PERMANENT END CLOSURE
(9) REINFORCING RING PLATE
(10) EXTENSION SLEEVE FOR PENETRATION NO'S 56 THRU 60 NOTE # 16)
(CONTINUED ABOVE)

THIS DRAWING IS REQUIRED FOR "W" PROOF OF DESIGN

FOR NOTES & REFERENCES SEE DWG. NO. 1X4DLA813

ACT. PKG. 2-1-2101-C-008A UNIT 1

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG. UNIT 1
(CONTAINMENT WALL PIPING PENETRATION DESIGN LIST)

SCALE: NONE | DRAWING NO. 1X4DLA414 | REV. 9
JOB NO. 9510

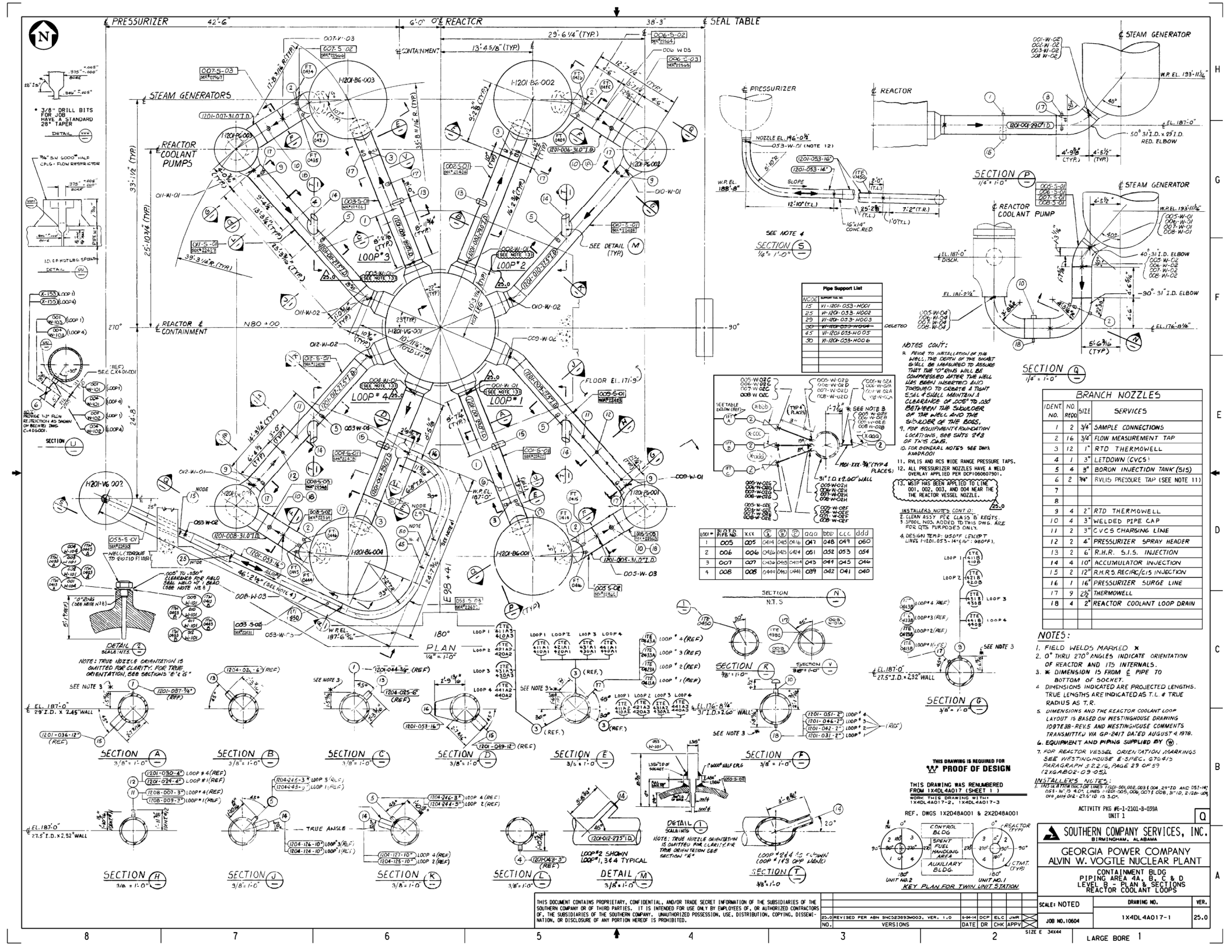
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGL	EGS	CHF.E	P.E	Q.A.E
6	INCORP. DCN 4 (M-FCRD-56-BF.) 5 & 6, M-FCRD-63449.	10-31-79	W.L.B.							
5	INTERCHANGED PENET. #76 & 88 PER LTR. LOG #PFE-5073	5-15-80	DAG							
4	REVISED PRESS & TEMP. PER LATEST LINE LIST REF #22	10-31-79	DIL							
3	REVISED PER LATEST P&I.D., VENDOR & LINE LISTS	1-28-80	SMP							
2	INCORP. DCN # 3	1-28-80	SMP							
1	INCORPORATED DCN # 5	6-11-80	AMH							
0	INCORP A&N 00465 (MODIFIED)	11-7-80	W.K.G.							
9	REVISED PER LATEST P.I.D., VENDOR, STRUC. DWGS; ADDED (12) & (13)	1-7-83	AMK							

NO.	REVISIONS	DATE	DR.	CHK.	RE	DM	DH	HDM
8	ADDED (14) THRU (20)	1-5-84	GAC					

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CH.E	P.E	Q.A.E
7	REVISED PER LATEST P&I.D.	1-7-85							

ENG. APPR. DATE

SIZE E 34 x 44



• 3/8" DRILL BITS FOR JOB HAVE A STANDARD 28° TAPER

DETAIL UUU

3/4" SW 6000# HALF CPLG - FLOW RESTRICTOR

I.D. OF HOT LEG SPOOL

DETAIL VV

Pipe Support List

NODE	SUPPORT TAG NO.
15	V1-1201-053-H001
25	W-1201-053-H002
29	W1-1201-053-H003
30	W1-1201-053-H004
45	W1-1201-053-H005
30	V1-1201-053-H006

LOOP #	3/4 O.D. PIPE NO.	X&X	Ⓐ	Ⓑ	Ⓒ	aaa	bbb	ccc	ddd
1	005	005	0414	0415	0416	047	048	049	050
2	006	006	0420	0421	0422	051	052	053	054
3	007	007	0426	0427	0428	043	044	045	046
4	008	008	0444	0445	0446	039	042	041	040

BRANCH NOZZLES

IDENT. NO.	NO. REQD	SIZE	SERVICES
1	2	3/4"	SAMPLE CONNECTIONS
2	16	3/4"	FLOW MEASUREMENT TAP
3	12	1"	RTD THERMOWELL
4	1	3"	LETDOWN (CVCS)
5	4	3"	BORON INJECTION TANK (SIS)
6	2	3/4"	RVLIS PRESSURE TAP (SEE NOTE 11)
7			
8			
9	4	2"	RTD THERMOWELL
10	4	3"	WELDED PIPE CAP
11	2	3"	CVCS CHARGING LINE
12	2	4"	PRESSURIZER SPRAY HEADER
13	2	6"	R.H.R. S.I.S. INJECTION
14	4	10"	ACCUMULATOR INJECTION
15	2	12"	R.H.R.S. RECIRC/C.I.S INJECTION
16	1	16"	PRESSURIZER SURGE LINE
17	9	2½"	THERMOWELL
18	4	2"	REACTOR COOLANT LOOP DRAIN

NOTES CONT:

8. PRIOR TO INSTALLATION OF THE WELL, THE DEPTH OF THE SOCKET SHALL BE MEASURED TO ASSURE THAT THE "O" RINGS WILL BE COMPRESSED AFTER THE WELL HAS BEEN INSERTED AND TORQUED TO CREATE A TIGHT SEAL & SHALL MAINTAIN A CLEARANCE OF .005" TO .030" BETWEEN THE SHOULDER OF THE WELL AND THE SHOULDER OF THE BORE.
9. FOR EQUIPMENT FOUNDATION LOCATIONS, SEE SHTS 2&3 OF THIS DWG.
10. FOR GENERAL NOTES SEE DWG. 1X4DR001
11. RVLIS AND RCS WIDE RANGE PRESSURE TAPS.
12. ALL PRESSURIZER NOZZLES HAVE A WELD OVERLAY APPLIED PER DCP1060807901.
13. WSIP HAS BEEN APPLIED TO LINE 001, 002, 003, AND 004 NEAR THE THE REACTOR VESSEL NOZZLE.

INSTALLERS NOTE'S CONT D:

2. CLEAN ASSY PER CLASS 'B' REQTS.
3. SPOOL NOS. ADDED TO THIS DWG. ARE FOR QTS PURPOSES ONLY.
4. DESIGN TEMP: 650°F EXCEPT LINES 1-1201-053-14"-EJW-1: 980°F).

NOTES:

1. FIELD WELDS MARKED ✕
2. 0° THRU 270° ANGLES INDICATE ORIENTATION OF REACTOR AND ITS INTERNALS.
3. ✱ DIMENSION IS FROM ℄ PIPE TO BOTTOM OF SOCKET.
4. DIMENSIONS INDICATED ARE PROJECTED LENGTHS. TRUE LENGTHS ARE INDICATED AS T.L. & TRUE RADIUS AS T.R.
5. DIMENSIONS AND THE REACTOR COOLANT LOOP LAYOUT IS BASED ON WESTINGHOUSE DRAWING 1097F3B REV.5 AND WESTINGHOUSE COMMENTS TRANSMITTED VIA GP-2417 DATED AUGUST 4, 1978.
6. EQUIPMENT AND PIPING SUPPLIED BY (W).
7. FOR REACTOR VESSEL ORIENTATION MARKINGS SEE WESTINGHOUSE E-SPEC. 675415 PARAGRAPH 3.2.2.1.4, PAGE 29 OF 53 (2XGABOR-09-05).

INSTALLERS NOTE'S:

1. INSULATION THICK OR LINES 1-1201-001, 002, 003 & 004 24"ID AND 053-14" 053-16" IS 4.0". LINES 1-1201-005, 006, 007 & 008, 31"ID, 2-1201-009, 010, 011 & 012 - 27.5"ID IS 3.0".

THIS DRAWING IS REQUIRED FOR W PROOF OF DESIGN

THIS DRAWING WAS RENUMBERED FROM 1X4DL4A017 (SHEET 1)

WORK THIS DRAWING WITH: 1X4DL4A017-2, 1X4DL4A017-3

REF. DWGS 1X2D48A001 & 2X2D48A001

ACTIVITY PKG #6-1-2101-B-039A UNIT 1

Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG
PIPING AREA 4A, B, C & D
LEVEL B - PLAN & SECTIONS
REACTOR COOLANT LOOPS

SCALE: NOTED

DRAWING NO. 1X4DL4A017-1

VER. 25.0

JOB NO. 10604

SIZE E 34X44

LARGE BORE 1

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NO.	VERSIONS	DATE	DR	CHK	APPV
25.0	REVISED PER ABN SNC523693M003, VER. 1.0	6-04-14	DCP	ELC	JMR

PLAN DIM. FROM REF. DWG. 1X4DL4A17

N

REACTOR COOLANT PUMP CASING ELEV. & LOC.

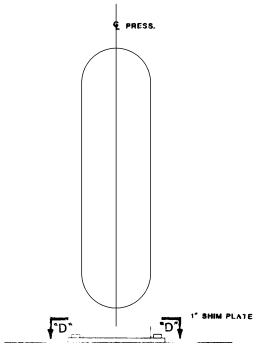
STA	PLAN	LOOP 1	LOOP 2	LOOP 3	LOOP 4
PLAN	N/A	195-7.4498 (N/A CL 195-7.5378 ATAZ 195-7.5378)	195-7.4008 195-7.4008	195-7.4708 195-7.4708	195-7.4148 195-7.5748
A	N/A	189-7.4560	189-7.5516	189-7.6810	189-7.3081
A1	N/A	189-6.9700	189-7.041-	189-7.173[illegible]	189-6.8002
B	N/A	189-7.4690	189-7.5144	189-7.6810	189-7.3112
B1	N/A	189-6.9590	189-7.0032	189-7.1740	189-6.8042
C	N/A	189-7.4490	189-7.5540	189-7.6340	189-7.3012
C1	N/A	189-6.9390	189-7.0426	189-7.1290	189-6.7902
HORZ LOC NORTH SOUTH	25.9074	-25.9010	+ 25.9190	+ 25.9074	-25.8963
HORZ LOC EAST WEST	29.5016	+ 29.4636	+ 29.4611	-29.5016	-29.4963
AZ	48-42-41	48°-42'-04"	48°-39'-55"	48°42'--1"	48°-39'-26"
FIELD BOOK		SCI-40	SCI-38	SCI-38	SCI-41

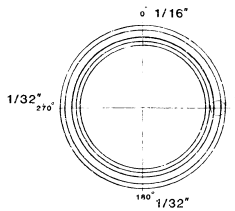
+ NORTH / EAST FROM ℄ R.V.
− SOUTH / WEST FROM ℄ R. V.

STEAM GENERATOR ELEV., LOC. PLUM.

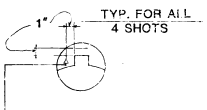
STA	PLAN	LOOP 1	LOOP 2	LOOP 3	LOOP 4
PLAN	N/A	195-5.0520"	195-4.9442	195-5.0887	195-5.1277
W	N/A	195-4.8892	195-4.9786	195-4.8650	195-4.8850
X	N/A	193-5.0092	195-4.9156	195-4.9258	195-4.7340
Y	N/A	195-4.6432	195-4.8276	195-4.8238	195-4.5390
Z	N/A	195-4.4862	195-4.9106	195-4.8138	195-4.5450
HORZ LOC NORTH SOUTH	33.125	-33.0512	+ 33.0743	-33.0817	-33.0987
HORZ LOC EAST WEST	13.3854	+13.5252	+ 13.4044	-13.4814	-13.3595
PLUM NORTH SOUTH	0	-13/16"	+ 1/8"	0	-1/2"
PLUM EAST WEST	0	0	- 7/16"	-7/16"	-1/2"
AZ	22-00-11	157-44-41	22-03-42"	337-10-37	201-58-49
FIELD BOOK		SCI-40	SCI-38	SCI-38, SCI-39	SCI-41

+ NORTH / EAST FROM ℄ R.V.
− SOUTH / WEST FROM ℄ R. V.



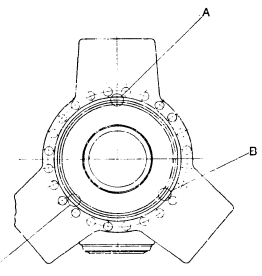


@ CORE SUPPORT LEDGE
SECTION "A"-"A"

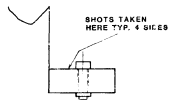


ELEVATIONS

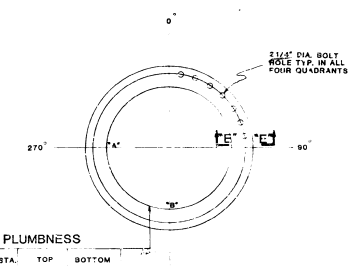
0	192 - 11.062
90	192 - 11.059
180	192 - 11.053
270	192 - 11.050
AVG. PLAN	192 - 11.0126



SECTION "B"-"B"



SECTION "E"-"E"



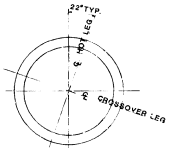
PLUMBNESS

STA.	TOP	BOTTOM
"A"	4"	4 1/8
"B"	4"	4 1/8"
ERROR	WEST 1/8"	SOUTH 1/8"

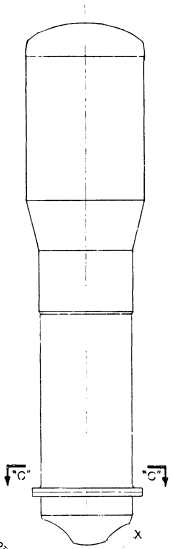
PLAN	195 - 11.4998"
0°	195 - 11.5920"
90°	195 - 11.6556"
180°	195 - 11.6016"
270°	195 - 11.6392"

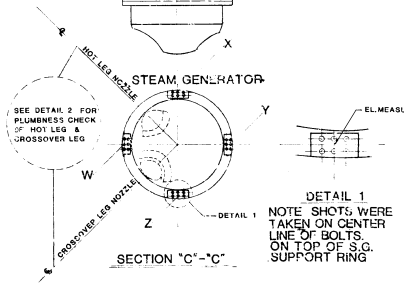
SECTION "D"-"D"

PLUMBNESS CHECK



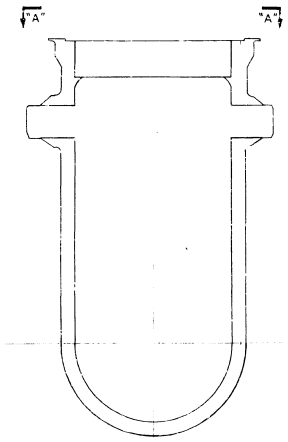
DETAIL 2
TYP. 4 PLACES



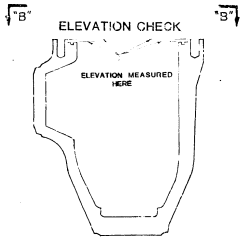


DETAIL 1
NOTE: SHOTS WERE
TAKEN ON CENTER
LINE OF BOLTS
ON TOP OF S.G.
SUPPORT RING

SECTION "C"-"C"



REACTOR VESSEL



REACTOR COOLING PUMP CASING
TYP. 3 PLACES

NOTES: 1) DETAIL 2 IS TYPICAL FOR ALL FOUR (4) STEAM GENERATOR
2) DETAIL 3 IS TYPICAL FOR ALL FOUR (4) REACTOR COOLANT PUMP CASING

22 THIS DRAWING WAS RENUMBERED FROM 1X4DL4A017 (SHEET 2)
WORK THIS DRAWING WITH: 1X4DL4A017-A, 1X4DL4A017-3

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BLDG
PIPING AREA 4A, B, C & D
LEVEL B - PLAN & SECTIONS
REACTOR COOLANT LOOPS

SCALE: NTS
DRAWING NO. 1X4DL4A017-2
REV. 22
JOB NO. 10604

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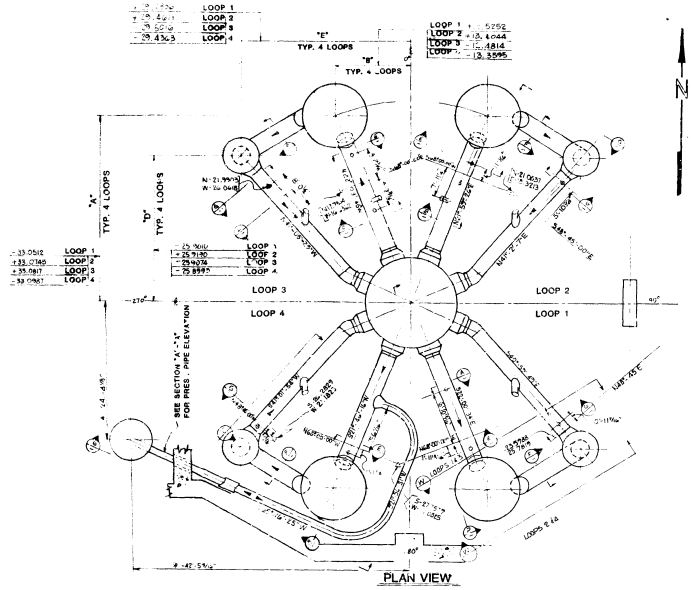
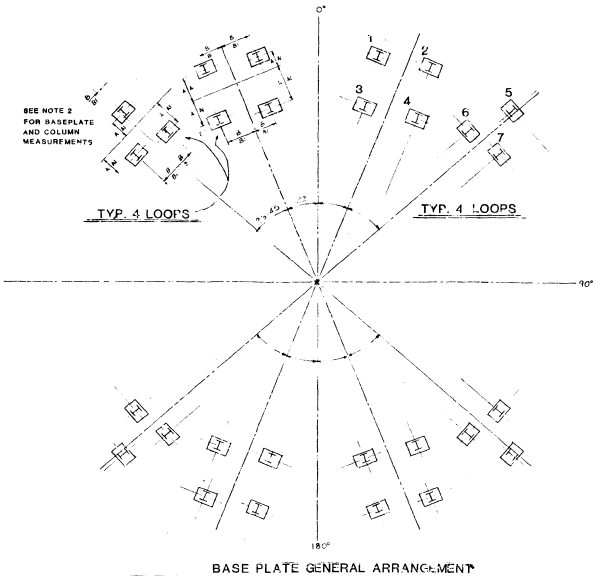
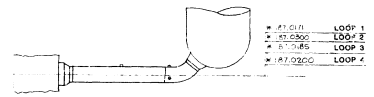
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP PER ABN 38665	4-25-03	DFV	LRP	ASK	-

SIZE E 34x44
CAD NAME 1X4DL4A0172

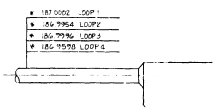




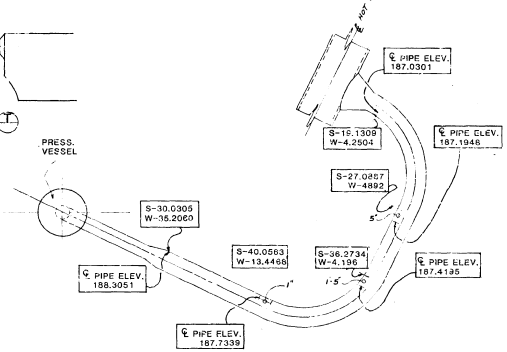
BASE PLATE GENERAL ARRANGEMENT
FOR BASE PLATE DETAILS SEE DWGS 1185F12 & 1187F36

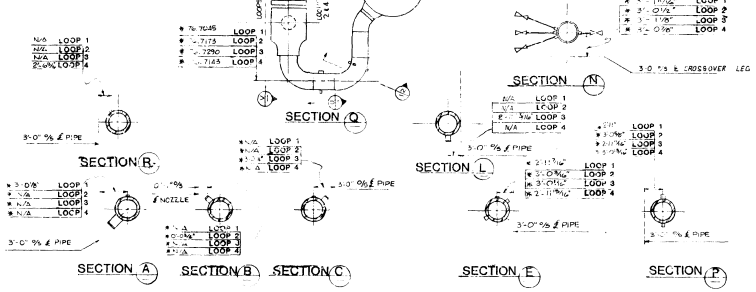


SECTION P

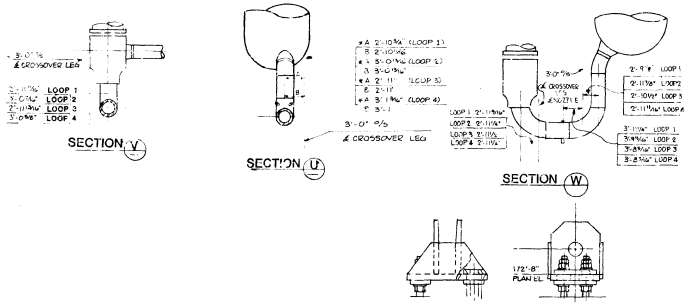


SECTION T



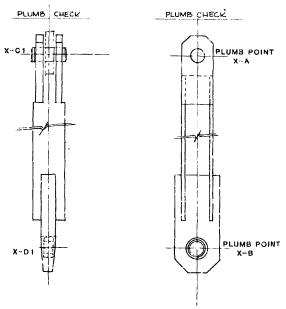


SECTION A SECTION B SECTION C SECTION L SECTION N SECTION Q SECTION E SECTION F SECTION G SECTION R

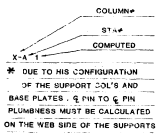


SECTION V SECTION U SECTION W

R.C. PUMP & S.G. COLUMN BASE
7 BASE REQD PER LOOP



VERTICAL COLUMNS
S.G. - 4REQD PER LOOP
R.C. PUMP - 3 REQD PER LOOP



SECTION AA

BASE PLATES LOCATION & EL.

LOOP #	PLATE #	ACTUAL "A"	PLAN "A"	ACTUAL "B"	PLAN "B"	ACTUAL EL.	PLAN EL.	FIELD BOOK
1	1	3'-8 13/16"	3'-8"	5'-1	5'-0 15/16"	172.8490	172.8667	SCI- 40
1	2	5'-0 5/16"	5'-0 13/16"	5'-0 7/16"	5'-0"	172.8533		"
1	3	5'-1 7/16"	5'-0 13/16"	3'-0"	3'-0"	172.7331		"
1	4	3'-0 7/16"	3'-0"	3'-0 7/16"	3'-0"	172.6943		"
1	5	4'-2 9/16"	4'-2 9/16"	0'-0 5/8"	0'-0 1/16"	.172.6733		"
1	6	1'-10"	3'-7 9/16"	2'-0 9/16"	2'-0 7/8"	172.673"		"
1	7	2'-0 11/16"	2'-3 3/4"	3'-1 7/16"	3'-2"	172.6578		"
2	1	1'-0 1/16"	3'-8"	5'-0 5/8"	5'-0 13/16"	172.8699		SCI - 38
2	2	5'-1 1/8"	5'-0 13/16"	4'-8 5/16"	5'-0"	172.6743		"
2	3	5'-0 7/16"	5'-0 13/16"	3'-0"	3'-0"	172.6975		"
2	4	3'-0 13/16"	3'-0"	5'-1 1/2"	5'-0 13/16"	172.6953		"
2	5	4'-2 5/8"	4'-2 9/16"	0'-0 1/2"	0'-0 11/16"	172.675.		"
2	6	3'-0 1/8"	3'-7 9/16"	1'-2 11/16"	2'-0 7/8"	172.6775		"
2	7	2'-3 1/4"	2'-3 3/8"	3'-7 1/4	3'-2"	172.6722		"
3	1	3'-8"	3'-8"	5-1 7/16	5'-0 13/16"	172.8717		SCI - 39
3	2	5'-1 1/16	5'-0 13/16"	3'-8 7/16"	3'-0"	172.8938		"
3	3	5'-0 5/16"	5'-0 13/16"	3'-0 3/4"	3'-0"	172.8808		"
3	4	3'-0 13/16"	3'-0"	3'-0 1/2"	5'-0 13/16"	172.7078		"
3	5	4'-2 15/16"	4'-2 9/16"	0'-0 11/16"	0'-0 11/16"	172.8529		"
3	6	3'-0 9/16"	3'-7 9/16"	2'-10 1/16"	3'-0 7/8"	172.8830		"
3	7	2'-3 3/8"	2'-3 3/8"	3'-0 9/16"	3'-2"	172.8685		"
4	1	3'-0 1/16"	3'-8"	5'-0 1/2"	5'-0 13/16"	172.8968		SCI - 41
4	2	5'-0 7/16"	5'-0 13/16"	3'-8 13/16"	3'-0"	172.8613		"
4	3	5'-0 5/16"	5'-0 13/16"	3'-0 3/4"	3'-0"	172.8953		"
4	4	3'-0 11/16"	3'-0"	5'-0 15/16"	5'-0 13/16"	172.6906		"
4	5	4'-2 1/2"	4'-2 9/16"	0'-0 7/16"	0'-0 11/16"	172.8410		"
4	6	3'-0 7/8"	3'-7 9/16"	2'-0 9/16"	2'-0 7/8"	172.8623		"
4	7	2'-3 1/2"	2'-3 3/8"	3'-7 7/8"	3'-2"	172.8565		"

SUPPORT COL'S - LOCATION & PLUMBNESS*

LOOP #	COL. #	ACTUAL A1	PLAN A1	ACTUAL B1	PLAN B1	DIST. FROM ℄ LOOP EAST	DIST. FROM ℄ LOOP WEST	DIST. TOP EAST	DIST. TOP WEST	FIELD BOOK
1	1	3'-8 15/16"	3'-8"	5'-11 7/16"	5'-0 13/16"	0.2"	—	0.2/8	—	SCI-40
1	2	5'-0 1/16"	5'-0 13/16	3'-0 5/8"	3'-0"	0.2/8	—	0.2/8	—	"
1	3	5-1 7/16	5-0 13/16	3-8 3/4	3-0"	0.2/8	—	0.2/8	—	"
1	4	3'-0 1/2"	3'-0"	5'-1 1/16"	5'-0 13/16"	0.3/8"	—	0.2/8	—	"
1	5	4'-2 1/4"	4'-2 9/16"	0'-0 5/8"	0'-0 11/16"	0.2/4"	—	0.2/8	—	"
1	6	3-8 13/16	3'-7"	2'-0 1/2"	2'-0 7/8"	—	0.2"	—	0.2"	"
1	7	2'-3 7/8"	2'-3 3/8"	3'-0 7/8"	3'-2"	0.2/8	—	0.2/8	—	"
2	1	3'-8 15/16"	3'-8"	5'-0 11/16"	5'-0 13/16"	0.2"	—	0.2/8	—	SCI-38
2	2	5'-1"	5'-0 13/16"	3'-8 5/16"	3'-0"	0.3"	—	0.2/8	—	"
2	3	5'-1 1/16"	5'-0 13/16"	3'-8 7/8"	3'-0"	0.3/8"	—	0.2/8	—	"
2	4	3-8 7/16	3'-0"	5'-1 1/16"	5'0 13/16"	0.2/8"	—	0.2/8	—	"
2	5	4'-2 15/16"	4'-2 9/16"	0'-0 7/16"	0'-0 11/16"	0.2/8"	—	0.2/8	—	"
2	6	3'-8 3/8"	3'-7 9/16"	2'-0 5/8"	2'-0 7/8"	—	0.2/8"	—	0.2/8	"
2	7	2-3 5/8"	2'-3 3/8"	3'-0 7/8"	3'-2"	0.2/8"	—	0.2/8	—	"
3	1	3'-2"	3'-8"	5'-0 3/4"	5'-0 13/16"	0.1	—	0.2/8	—	SCI-39
3	2	5'-0 7/8"	5'-0 13/16"	3'-0 7/16"	3'-0"	0.3	—	0.2/8	—	"
3	3	5'-0 15/16"	5'-0 13/16"	3'-0 1/2"	3'-0"	0.3/8	—	0.2/8	—	"
3	4	3'-0"	3'-0"	5'-0 15/16"	5'-0 13/16"	0.3/8"	—	0.2/8	—	"
3	5	4'-2 1/2"	4'-2 9/16"	0'-0 1/16"	0'-0 11/16"	0.2/8"	—	0.2/8	—	"
3	6	3'-0 5/16"	3'-7 9/16"	2'-10 1/16"	2'-0 7/8"	—	0.2/8"	0.07	—	"
3	7	2'-3 7/16"	2'-3 3/8"	3'-0 5/8"	3'-2"	0.2/8"	—	—	0.2/8	"
4	1	3'-8 5/16"	3'-8"	5'-0 5/8"	5'-0 13/16"	0.2/8	—	—	0.2/8	SCI-41
4	2	4'-11 15/16"	5'-0 13/16"	3'-9 7/16"	3'-0"	0.1/8	—	—	0.2/8	"
4	3	5'-1 3/8"	5'-0 13/16"	3'-8 13/16"	3'-0"	0.3/8"	—	—	0.2/8	"
4	4	3'-0 3/16"	3'-0"	5'-0 15/16"	5'-0 13/16"	0.2/8"	—	—	0.2/8	"
4	5	4'-2 1/2"	4'-2 9/16"	0'-0 11/16"	0'-0 11/16"	0.2/8"	—	—	0.2/8	"
4	6	3'-8 15/16"	3'-7 9/16"	2'-10 1/16"	2'-0 7/8"	—	0.2/8"	—	0.2/8	"
4	7	2'-4 3/16"	2'-3 3/8"	3'-0 7/8"	3'-2"	—	0.2/8"	—	0.2/8	"

NOTES:

1. DENOTES ASBUILT DIMENSION.
2. A- MEASURED TO ℄ BASE PLATE A1 - MEASURED TO ℄ COL
3. ALL C L PIPE O/S'S ARE FROM ACTUAL C L PIPE, NOT THEORETICAL
4. FROM VESSEL OR TOWARD VESSEL, MEANS THE VESSEL THAT THE COLUMN'S ARE HOLDING UP.
5. N/A : NOT APPLICABLE
6. O/S : OFF SET

22 THIS DRAWING WAS RENUMBERED FROM 1X4DL4A017 (SHEET 3)
WORK THIS DRAWING WITH: 1X4DL4A17-1, 1X4DL4A17-2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

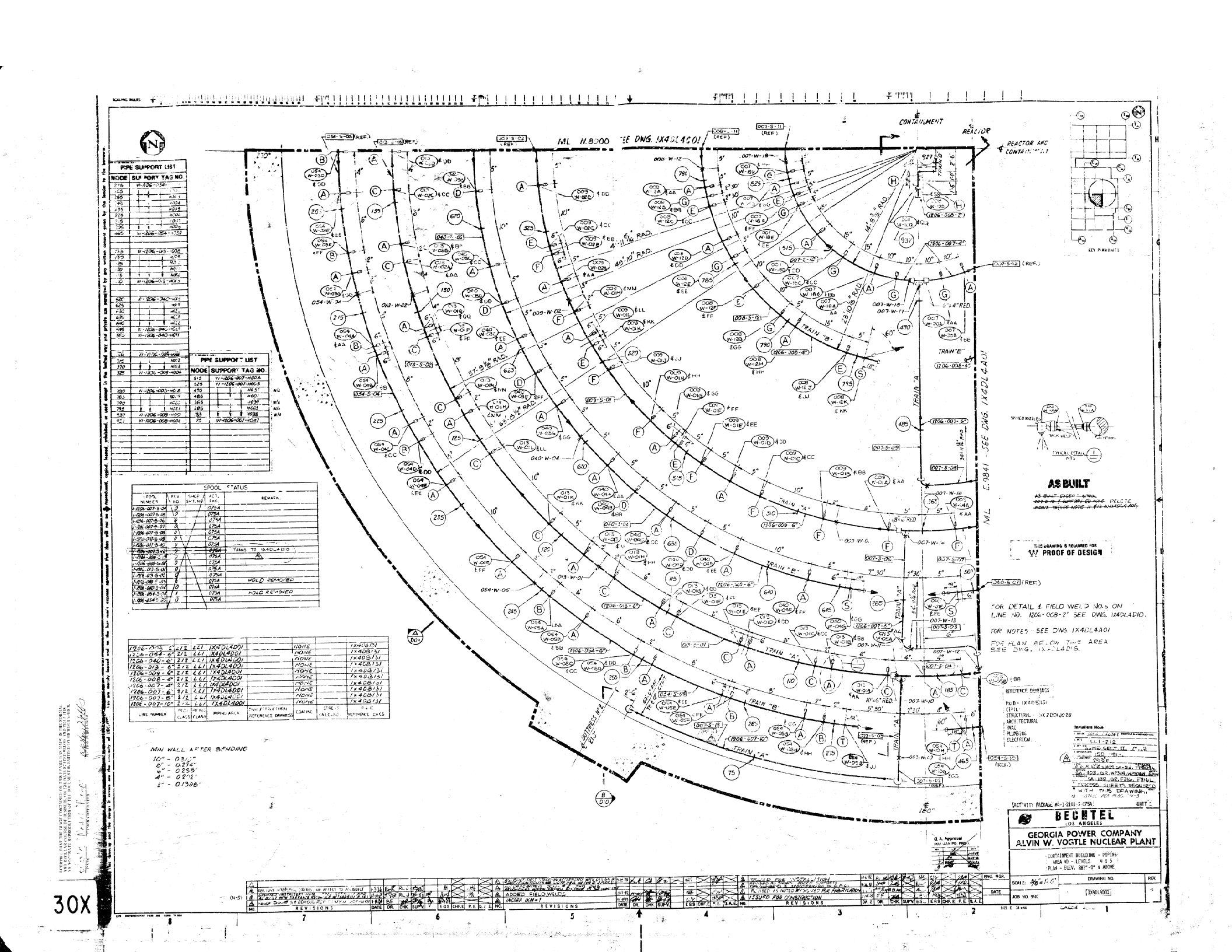
CONTAINMENT BLDG
PIPING AREA 4A, B, C & D
LEVEL B - PLAN & SECTIONS
REACTOR COOLANT LOOPS

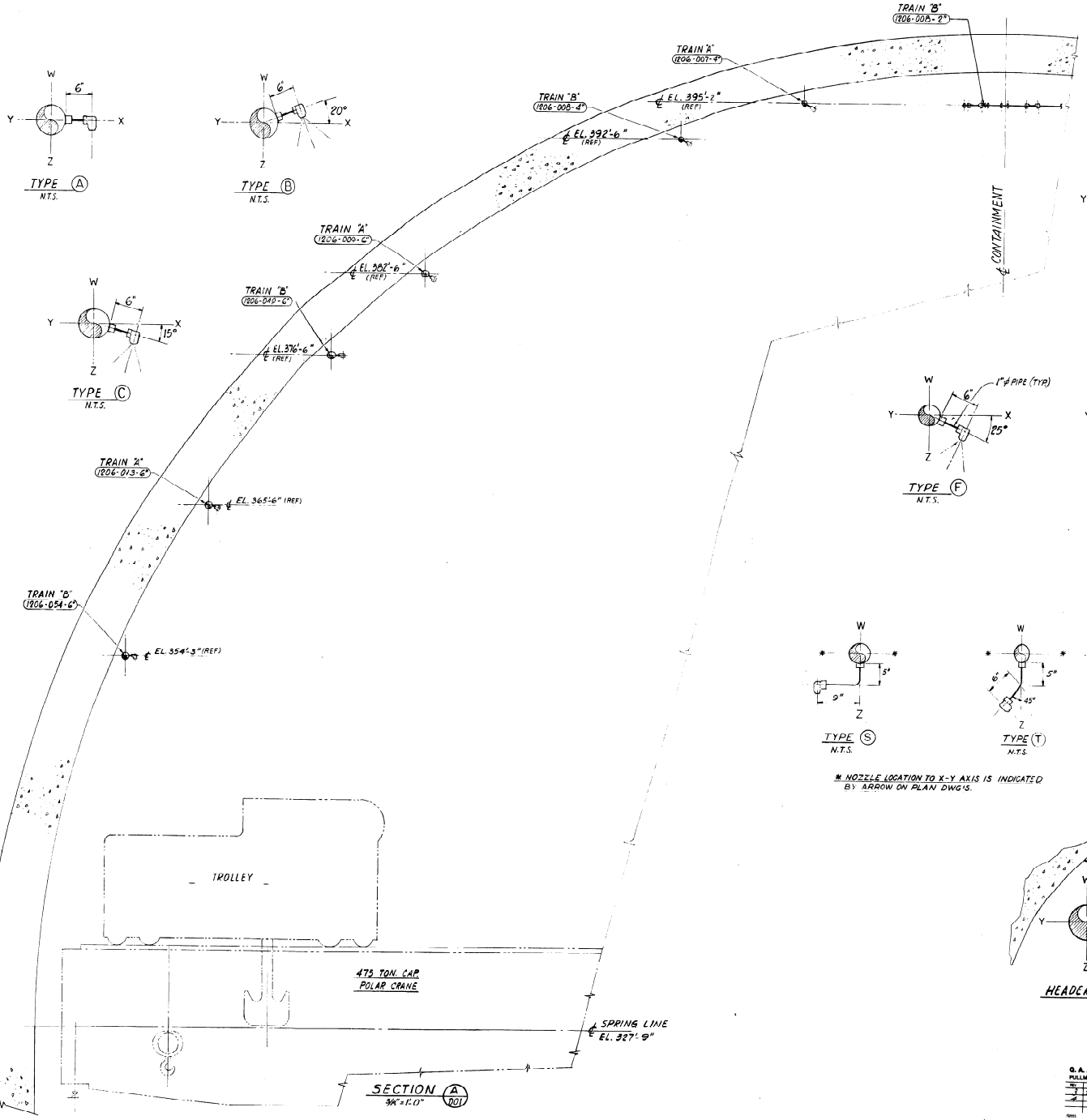
SCALE: NOTED
DRAWING NO. 1X4DL4A017-3
REV. 22
JOB NO. 10604

THIS DOCUMENT CONTAINS PROPRIETARY, CONFIDENTIAL, AND/OR TRADE SECRET INFORMATION OF THE SUBSIDIARIES OF THE SOUTHERN COMPANY OR OF THIRD PARTIES. IT IS INTENDED FOR USE ONLY BY EMPLOYEES OF, OR AUTHORIZED CONTRACTORS OF, THE SUBSIDIARIES OF THE SOUTHERN COMPANY. UNAUTHORIZED POSSESSION, USE, DISTRIBUTION, COPYING, DISSEMINATION, OR DISCLOSURE OF ANY PORTION HEREOF IS PROHIBITED.

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP PER ABN 38665	4-25-03	DFV	LRP	ASK	-

SIZE E 34x44
LARGE BORE 1
CAD NAME 1X4DL4A0173


PIPE SUPPORT LIST
NODE
SUPPORT TAG NO.
PIPE SUPPORT LIST
NODE
SUPPORT TAG NO.
SPOOL STATUS
SPOOL NUMBER
REV. NO.
SHOP SHT. NO.
ACT. PKG.
REMARK
LINE NUMBER
REV. NO.
CLASS
PIPING AREA
CWP / STRUCTURAL REFERENCE DRAWING
COATING
STRESS CALC. NO.
P & ID REFERENCE DWGS.
MIN WALL AFTER BENDING
10" - 0.310"
8" - 0.274"
6" - 0.255"
4" - 0.212"
2" - 0.1308"
CONTAINMENT
REACTOR
REACTOR AND CONTAINMENT
ML N.8000
SEE DWG. 1X4DL4A01
ML E.9841 - SEE DWG. 1X4DL4A01
TRAIN "A"
TRAIN "B"
AS BUILT
THIS DRAWING IS REQUIRED FOR PROOF OF DESIGN
FOR DETAIL & FIELD WELD NOS ON LINE NO. 1206-008-2" SEE DWG. 1X4DL4D10.
FOR NOTES - SEE DWG. 1X4DL4A01
FOR PLAN BELOW THIS AREA SEE DWG. 1X4DL4D10.
REFERENCE DRAWINGS
P&ID - 1X4DB131
CIVIL
STRUCTURAL - 1X2D0NJC28
ARCHITECTURAL
HVAC
PLUMBING
ELECTRICAL
Installers Note
ASME SEC. III, CL. 2
ASME SEC. II, PART A
SA-106 GR. B
SA-234 GR. WPB
SA-420 GR. WPL6, WPBW, WPLW
SA-182 GR. F316, F316L
EXCESS SHEETS REQUIRED WITH THIS DRAWING
BECHTEL
LOS ANGELES
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
CONTAINMENT BUILDING - PIPING
AREA 40 - LEVELS 4 & 5
PLAN - ELEV. 287'-0" & ABOVE
DRAWING NO. 1X4DL4001
REVISIONS
30X



THIS DRAWING IS REQUIRED FOR W PROOF OF DESIGN (A)

FOR NOTES SEE DWG. NO. 1X4DL4A01

ACTIVITY PACKAGE #4-1-2101-3-075A UNIT 1

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT BUILDING - PIPING
AREA 4A, B, C & D - LEVEL 5
SECTION & DETAILS

SCALE: AS NOTED | DRAWING NO. 1X4DL4009 | REV. 4
JOB NO. 9510

SIZE E 36 X 44 LARGE BORE

Q. A. Approval PULLMAN PO. PROD.

NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
(A)	NOZZLE PER (1X4DL4C01, D01 DCN-1)	11-15-91	APPM						
(A)	ISSUED FOR INSTALLATION & REV. TYPE (H)	6-21-91	SMP						
(A)	TRANSFERRED TO G.P.C.	1-31-91	AMR						
(A)	REVISED AS NOTED & ISSUED FOR FABRICATION	11-13-90	FF						
(A)	ISSUED FOR CONSTRUCTION	11-6-90	SMP						
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	EGS	CHF.E	P.E.	Q.A.E.
(A)	INCORP. DCN-R #3	1-11-90	MB	SMP	SMP	FGS	HF		

ENG. MGR. DATE

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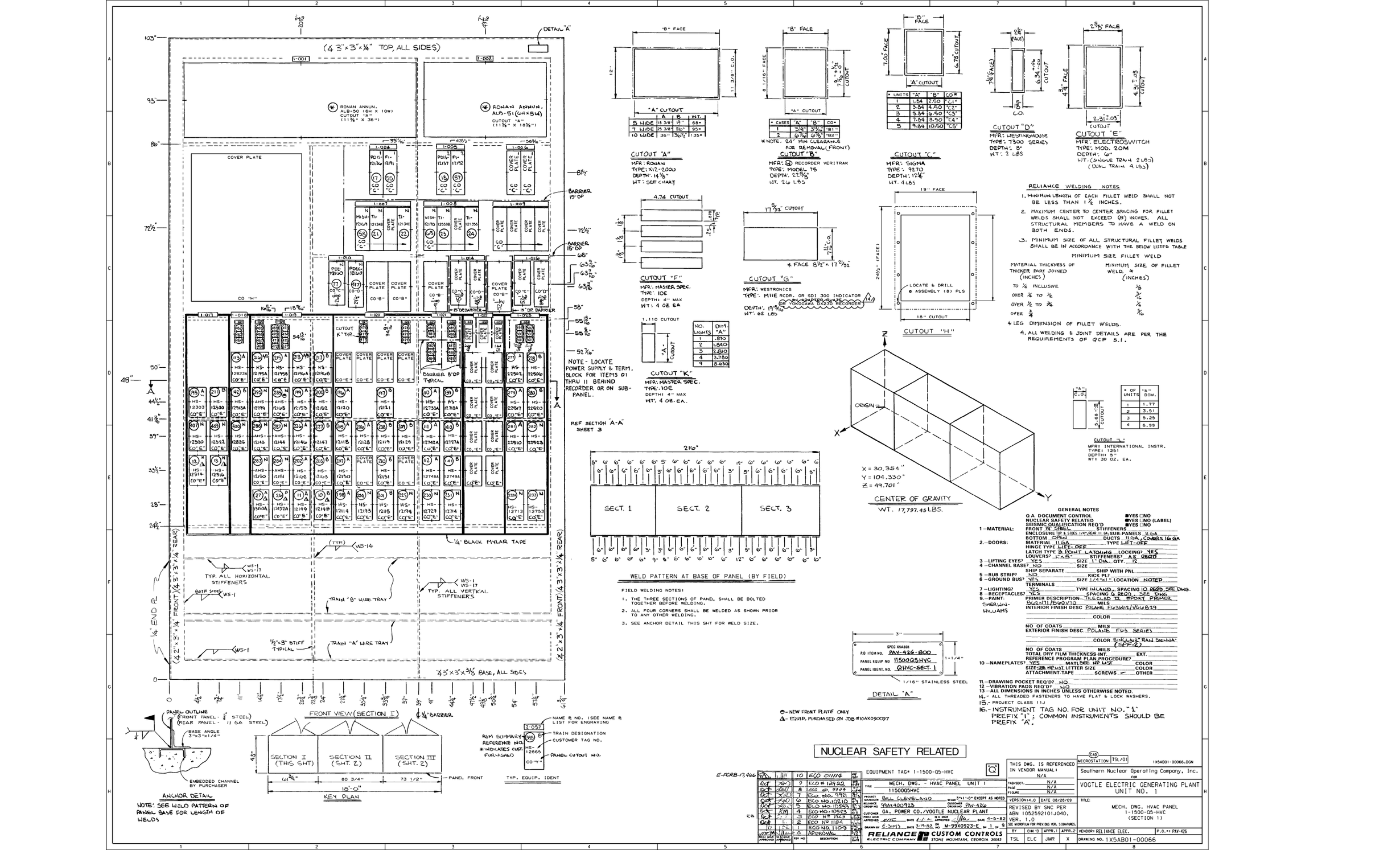


* UNITS	"A"	"B"	CO#
1	1.54	2.50	"C1"
2	3.54	4.50	"C2"
3	5.54	6.50	"C3"
4	7.54	8.50	"C4"
5	9.54	10.50	"C5"

5 WIDE	18 3/8"	19"	68*
7 WIDE	25 3/8"	26"	95*
10 WIDE	36"	36½"	135*

* CASES	"A"	"B"	CO*
1	3¼"	3½"	"B1"
2	6⅝"	6¾"	"B2"

NO. LIGHTS	DIM "A"
1	.810
2	1.840
3	2.810
4	3.780
5	8.650

* OF UNITS	"A" DIM.
1	1.77
2	3.51
3	5.25
4	6.99

E-FCRB-17,466

REV					
10	ECO 011114				
9	ECO # 12922				
8	ECO NO. 9327				
7	ECO NO. 9751				
6	ECO NO. 10210				
5	ECO NO. 10593				
4	ECO NO. 10923				
3	ECO N° 11673				
2	ECO N° 11841				
1	ECO NO. 11099				
0	APPROVAL				

EQUIPMENT TAG* 1-1500-05-HVC

TITLE: MECH. DWG. - HVAC PANEL UNIT 1
1150005HVC

PROJECT MANAGER: BILL CLEVELAND
RELIANCE ORDER NO: 91AX090925
CUSTOMER ORDER NO: PAV-426
SCALE 3"=1'-0" EXCEPT AS NOTED
CUSTOMER: GA. POWER CO./VOGTLE NUCLEAR PLANT
DRAWN BY: E-SIMS DATE 3-19-92 DR. NO. M-99X0923-E SH. 1 OF 9
Q.A. HIGH APPROVED: DATE 4-5-82
RELIANCE CUSTOM CONTROLS ELECTRIC COMPANY STONE MOUNTAIN, GEORGIA 30083

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A
VERSION 14.0 DATE 08/28/09
REVISED BY SNC PER ABN 1052592101J040, VER. 1.0
SEE MICROFILM FOR PREVIOS VER. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
TSL	ELC	JMR	X

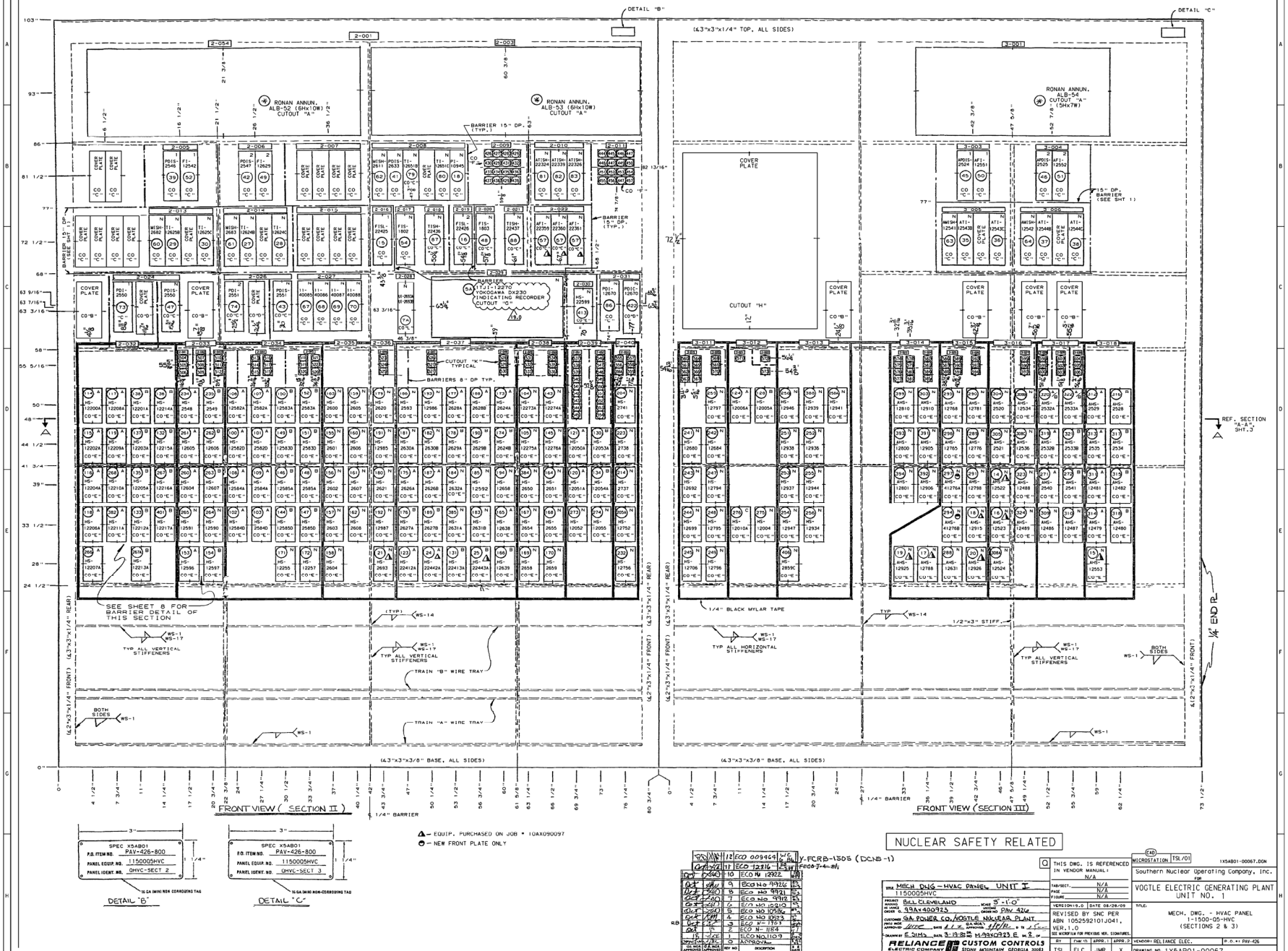
CAD MICROSTATION TSL/01 1X5AB01-00066.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

TITLE: MECH. DWG. HVAC PANEL 1-1500-05-HVC (SECTION 1)

VENDOR: RELIANCE ELEC. P.O.#: PAV-426

DRAWING NO. 1X5AB01-00066



SPEC X5AB01
P.O. ITEM NO. PAV-426-800
PANEL EQUIP. NO. 1150005HVC
PANEL IDENT. NO. OHVC-SECT 2
16 GA (MIN) NON CORRODING TAG
DETAIL "B"

SPEC X5AB01
P.O. ITEM NO. PAV-426-800
PANEL EQUIP. NO. 1150005HVC
PANEL IDENT. NO. OHVC-SECT 3
16 GA (MIN) NON-CORRODING TAG
DETAIL "C"

▲ — EQUIP. PURCHASED ON JOB # 10AX090097
● — NEW FRONT PLATE ONLY

REV	BY	NO	DESCRIPTION
12	ECO 009464		
11	ECO-12816-1		
10	ECO N 12922		
9	ECO NO 9926		
8	ECO NO 9921		
7	ECO NO 9912		
6	ECO NO 10210		
5	ECO NO 10256		
4	ECO NO 10523		
3	ECO N-1793		
2	ECO N-1184		
1	ECO NO.1109		
0	APPROVAL		

Y-FCRB-1305 (DCNB-1)
FECO-J-4-91;

NUCLEAR SAFETY RELATED

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A

TITLE: MECH DWG - HVAC PANEL UNIT I
1150005HVC
PROJECT MANAGER: BILL CLEVELAND
SCALE: 3"=1'-0"
DRAWING NO. 93AX900923
CUSTOMER: GA POWER CO. VOGTLE NUCLEAR PLANT
PAV 426
DRAWN BY: E SIMS DATE: 3-19-93
APPROVED: DATE:
RELIANCE ELECTRIC COMPANY
CUSTOM CONTROLS STONE MOUNTAIN GEORGIA 30083

TAB/SECT. N/A
PAGE N/A
FIGURE N/A
VERSION 19.0 DATE 08/28/09
REVISED BY SNC PER ABN 10529210101J041, VER.1.0
SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

BY	CHK. (1)	APPR. 1	APPR. 2
TSL	ELC	JMR	X

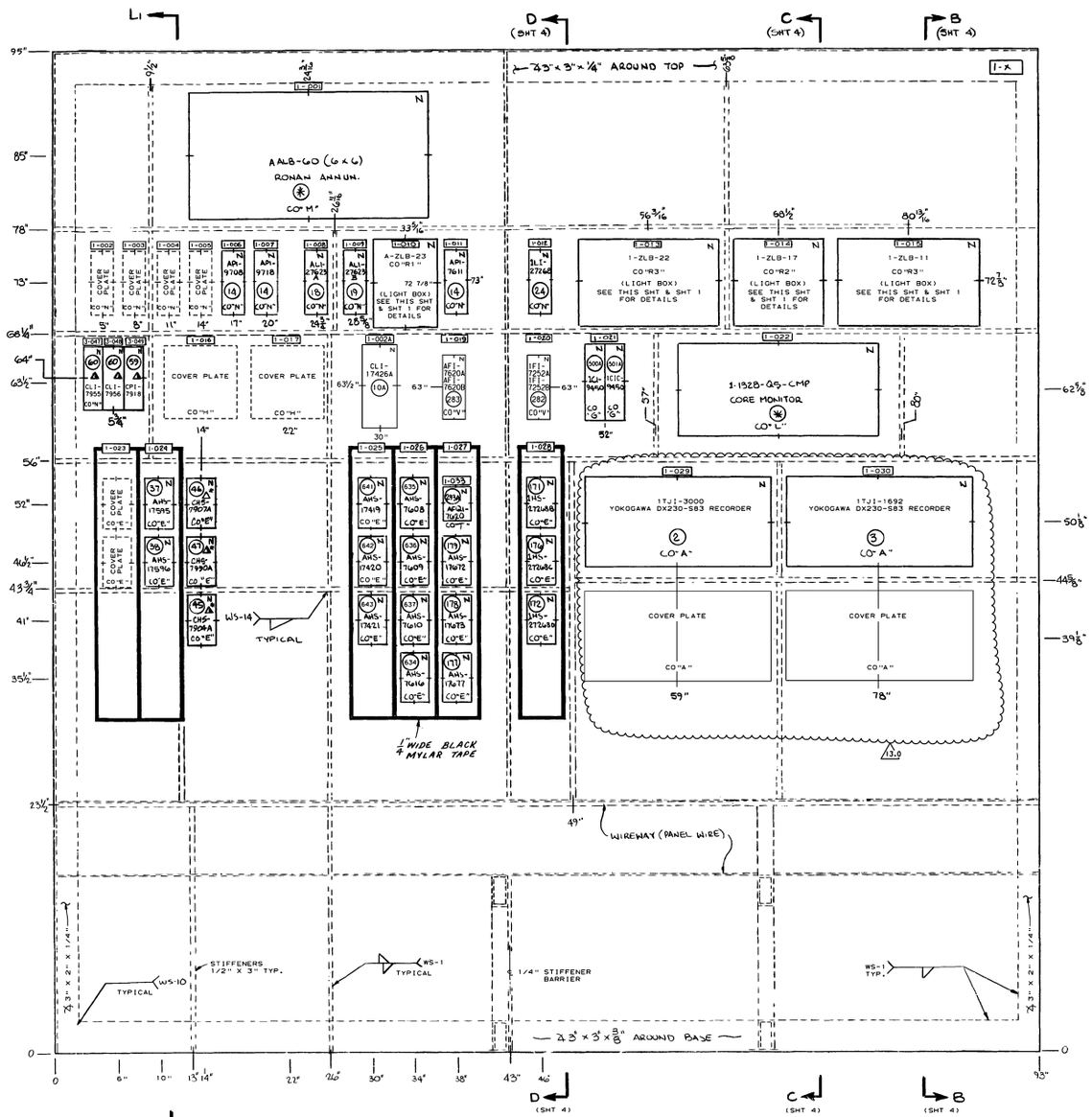
MICROSTATION TSL101 1X5AB01-00067.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
MECH. DWG. - HVAC PANEL
1-1500-05-HVC
(SECTIONS 2 & 3)

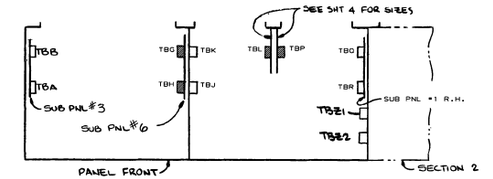
VENDOR: RELIANCE ELEC. P.O. # PAV-426

DRAWING NO. 1X5AB01-00067



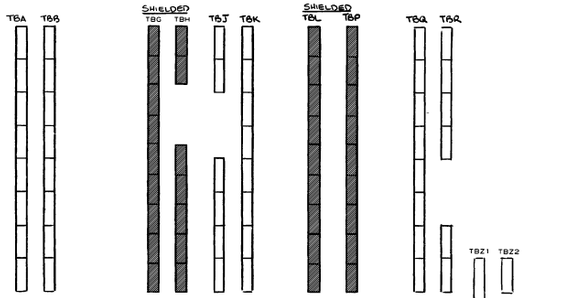
FRONT ELEVATION "SECTION 1"

Δ - EQUIP. PURCH. ON JOB # 10AX090097



"TOP VIEW" TERMINAL BLOCK ARRANGEMENT

(SEE SHT 11 FOR SUB PANEL DETAILS)



SHADED AREAS INDICATES GE CR151-12 TERM. BLOCKS

NON-SHADED AREAS INDICATES STATES TERM. BLOCKS (12 POINTS)

"ELEVATION" TERMINAL BLOCK ARRANGEMENT

1-010 A-ZLB-23

599	598	598	598
601	598	598	598
603	598	598	598
605	598	598	598
595	598	598	598
598	598	598	598

1-013 1-ZLB-22

563	564	565	566	567	568	585	586	587	588
569	570	571	572	573	574	589	590	591	588
575	576	577	576A	577A	578	592	593	594	588
579	580	581	582	583	584	595	596	597	588
588	588	588	588	588	588	588	588	588	588
588	588	588	588	588	588	588	588	588	588

1-014 1-ZLB-17

539	541	542	543	545	540
546	547	548	549	550	540
551	544	552	553	554	540
555	556	557	558	540	540
559	560	561	562	540	540
658	540	540	540	540	540

1-015 1-ZLB-11

300	301	302	303	303A	304	305	306	307	308
309	310	311	312	313	314	315	316	317	318
319	320	321	322	303A	323	324	325	326	327
328	329	330	331	332	333	334	335	336	337
338	339	340	341	342	343	344	345	346	347
348	349	350	351	303A	303A	303A	303A	303A	299

MASTER SPEC. LIGHTS ARRANGEMENT

NOTE: NUMBER INSIDE SQUARE INDICATES RELIANCE'S ITEM NUMBER

EQUIPMENT TAG NO:
1-1604-05-PCP

* REFER TO SUMMARY OF MATERIALS BM-10AX090097-A, BECHTEL LOG Nº 1X5AB01-1108

NUCLEAR SAFETY RELATED

Q

REV	DES	NO	DESCRIPTION	DATE
13			ECO Nº 11530	
12			ECO 013361	
11			ECO 12932	
10			ECO 12979	
9			ECO NO. 9941	
8			ECO NO. 9937	
7			ECO NO 9910	
6			ECO NO. 9928	
5			ECO NO. 10212	
4			ECO NO. 10249	
3			ECO NO. 10375	
2			ECO NO. 10918	
1			ECO NO. 10918	
0			APPROVAL	

TITLE: MISCELLANEOUS SYSTEMS/EQUIPMENT PANEL (QPCP) SECTION 1, UNIT 1

PROJ MANAGER: B. ALLEN
SCALE: 3"=1'0"
LEAD/JOB NO: 99AX490924
CUSTOMER ORDER NO: PAV-426
CUSTOMER: GEORGIA POWER CO.
DESIGN MGR APPROVED: [signature] DATE 2/10/99
Q.A. MGR APPROVED: [signature] DATE 2/10/99
DRAWN BY: E. SIMS DATE 2-10-99
DR. NO: M-99X0924-F SH 2 OF 11

RELIANCE ELECTRIC COMPANY CUSTOM CONTROLS STONE MOUNTAIN, GEORGIA 30083

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT.: N/A
PAGE: N/A
FIGURE: N/A

VERSION 13.0 DATE 08/18/09
REVISED BY SNC PER ABN 1052592101J002, VER. 1.0
SEE MICROFILM FOR PREVIOS VER. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
TSL	ELC	JMR	X

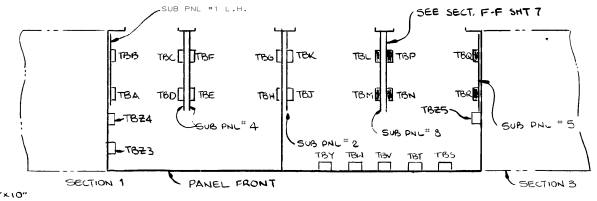
CAD MICROSTATION TSL01 1X5AB01-00557.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

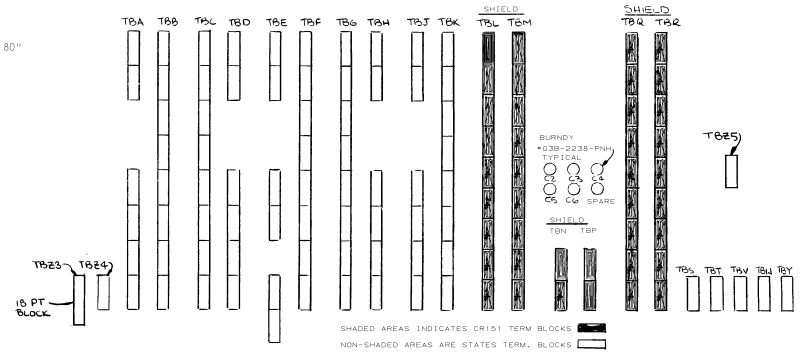
TITLE: MISCELLANEOUS SYSTEMS EQUIPMENT PANEL (QPCP) SECTION 1

VENDOR: RELIANCE ELEC. P.O.#: PAV-426

DRAWING NO. 1X5AB01-00557



"TOP VIEW" TERMINAL BLOCK ARRANGEMENT
(SEE SHT 11 FOR SUB PANEL DETAILS)



"ELEVATION" TERM. BLOCK ARRANGEMENT

2-004 1-ZLB-12

353	354	355	356	357	358	359	360	361	362
363	364	365	366	367	357	368	369	370	371
372	373	374	375	376	377	378	379	380	381
382	383	384	385	386	387	388	389	390	391
392	393	394	395	396	397	398	399	400	357
401	402	357	403	404	405	406	357	357	352

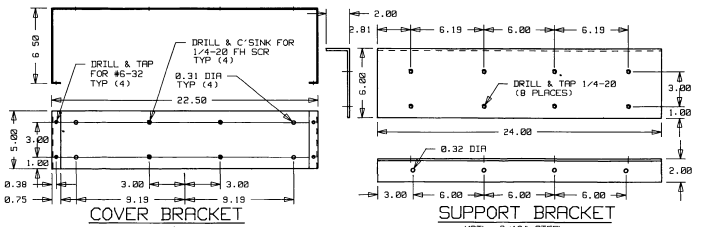
2-005 1-ZLB-13

409	410	411	412	413	414	415	416	417	418
419	420	421	422	423	424	425	426	427	428
429	430	431	432	433	434	435	436	437	438
439	440	441	442	443	444	445	446	447	448
449	450	451	452	453	454	455	456	457	458
459	460	461	462	408	408	408	408	408	407

2-007 1-ZLB-14

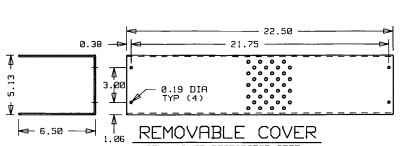
465	463	463	463	466	467	468	463	463	463
469	470	471	472	473	474	463	463	475	463
476	477	478	479	480	481	482	483	484	485
486	487	488	489	490	491	492	493	494	495
496	497	498	499	500	501	502	503	504	505
506	507	508	509	510	511	512	513	514	464

MASTER SPEC. LIGHTS ARRANGEMENT
NOTE: NUMBER INSIDE SQUARE INDICATES RELIANCE'S ITEM NUMBER

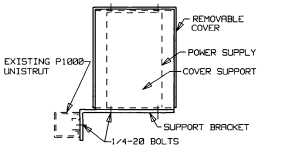


COVER BRACKET
MATL: 16 GA STEEL

SUPPORT BRACKET
MATL: 3/16" STEEL



REMOVABLE COVER
MATL: 16 GA PERFORATED STEEL



POWER SUPPLY MTG ASSY

NUCLEAR SAFETY RELATED

REV NO	DESCRIPTION	DATE
10	ECO 12747	
9	ECO NO. 9941	
8	ECO NO. 9932	
7	ECO NO. 9910	
6	ECO NO. 9928	
5	ECO NO. 10212	
4	ECO NO. 10511	
3	ECO NO. 10248	
2	ECO NO. 10175	
1	ECO NO. 10132	
0	APPROVAL	

11 ECO ISSUED

DCNS-1 (E-TCBB-15,503)

TITLE: MISCELLANEOUS SYSTEMS/EQUIPMENT PANEL (QPCP) SECTION 2, UNIT 1

PROJECT MANAGER: R. CLEVELAND
RELIANCE ORDER NO: 99AX400924
SCALE: 3"-1'0"
CUSTOMER ORDER NO: PAV-426

CUSTOMER: GEORGIA POWER CO.
PROJ. MGR APPROVED
Q.A. MGR APPROVED
DATE
DRAWN BY: E. SIMS DATE 2-17-92
DR. M-99X0924-E SH 5 OF 11

RELIANCE ELECTRIC CUSTOM CONTROLS
STONE MOUNTAIN GEORGIA 30083

SIZE E

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT.: N/A
PAGE: N/A
FIGURE: N/A

VERSION 12.0 DATE 5-14-07

REVISED BY SNC PER ABN-V00606, VER. 1.0

SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.

CAD MICROSTATION DFV/02 1X5AB01-00560.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

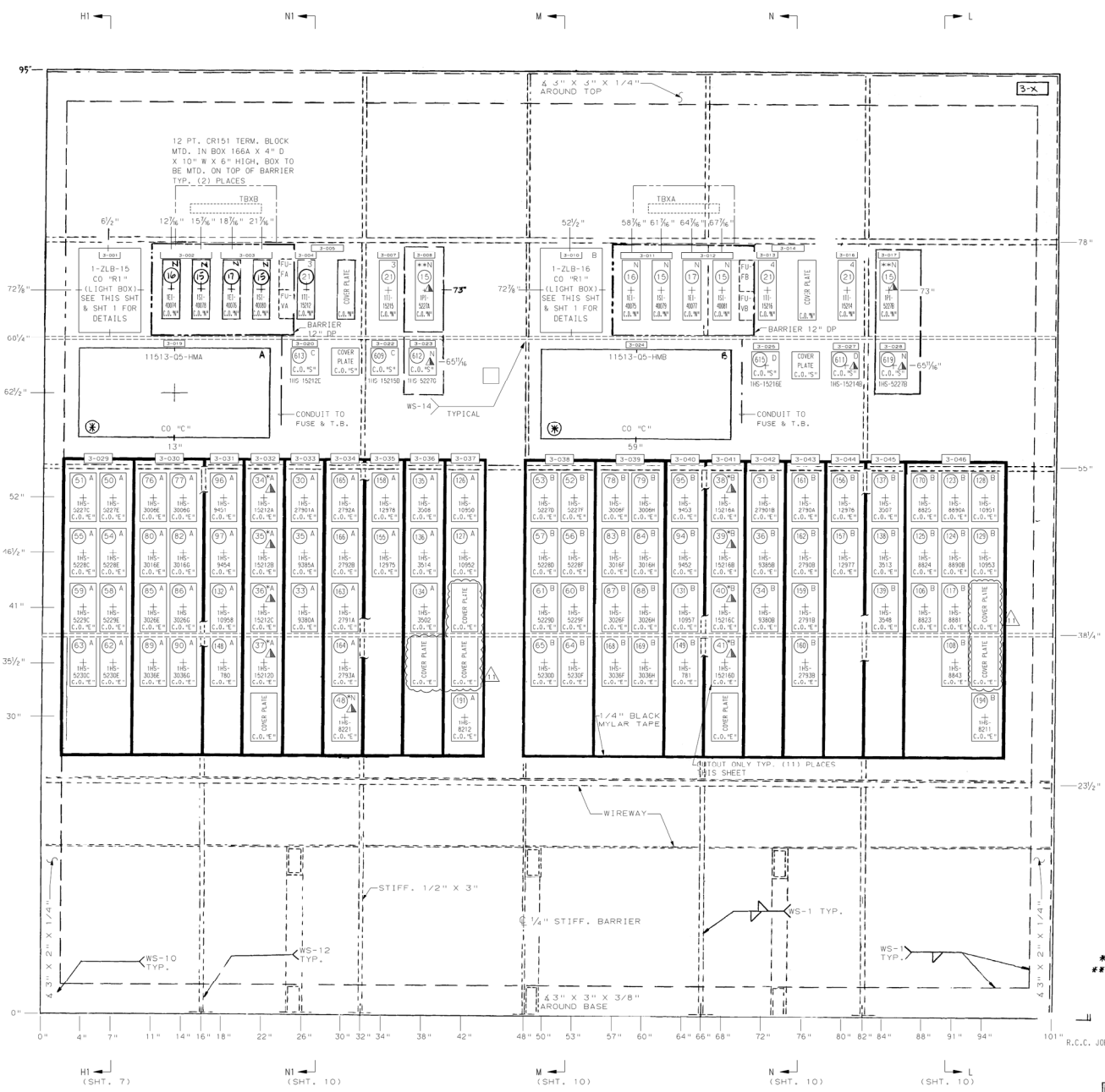
TITLE: MISCELLANEOUS SYSTEMS EQUIPMENT PANEL (QPCP) SECTION 2

BY	CHK'D	APPR.1	APPR.2
DFV	WMW	WLN	x

VENDOR: RELIANCE ELECTRIC P.O.# PAV-426

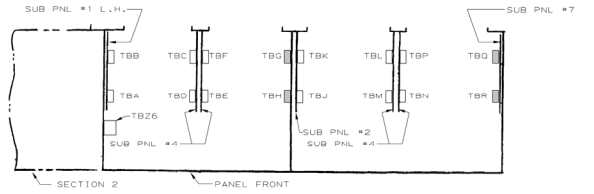
DRAWING NO. 1X5AB01-00560

SIZE E



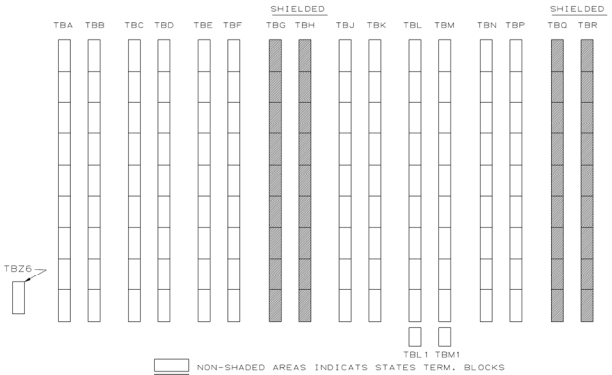
FRONT ELEVATION "SECTION 3"

▲1 - EQUIPMENT PURCHASED ON JOB # 10AX090097



"TOP VIEW" TERM. BLOCK ARRANGEMENT

(SEE SHT 11 FOR SUB PANEL DETAILS)



"ELEVATION" TERMINAL BLOCK ARRANGEMENT

3-001 1-ZLB-15

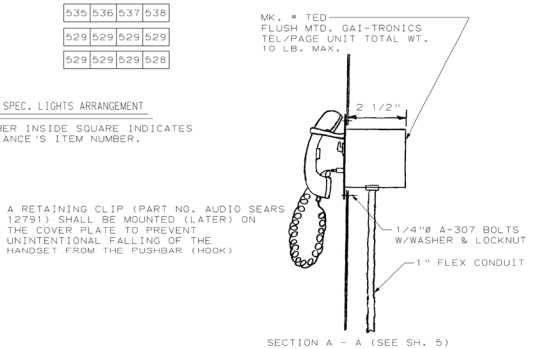
517	518	519	520
521	516	516	516
522	523	516	516
524	525	526	527
516	516	516	516
516	516	516	515

3-010 1-ZLB-16

530	531	532	533
534	529	529	529
529	529	529	529
535	536	537	538
529	529	529	529
529	529	529	528

MASTER SPEC. LIGHTS ARRANGEMENT

NOTE: NUMBER INSIDE SQUARE INDICATES RELIANCE'S ITEM NUMBER.



SECTION A - A (SEE SH. 5)
N.T.S.

* REFER TO SUMMARY OF MATERIALS BM-10X0097-A , BECHTEL LOG Nº 1X5AB01-1108

** REFER TO SUMMARY OF MATERIALS BM-10X0199-A BECHTEL LOG Nº AX5AB01-1136

REV		NO	DESCRIPTION	DATE
		15	ECO Nº 11530	
		14	ECO 011564	
		13	ECO 013961	(DCNS-1) FCRB-1718
		12	ECO 12742	
		11	ECO 12932	
		10	ECO 12919	
		9	ECO NO. 9941	
		8	ECO NO. 9934	
		7	ECO NO 9918	
		6	ECO NO. 9928	
		5	ECO NO. 10812	
		4	ECO NO. 10569	
		3	ECO NO. 10249	
		2	ECO NO. 1378	
		1	ECO NO. 1333	
		0	APPROVAL	

TITLE: MISCELLANEOUS SYSTEMS/EQUIPMENT PANEL (QPCP) SECTION 3, UNIT 1

PROJECT MANAGER: B. CLEVELAND — SCALE: 3"=1'-0"

RELIANCE ORDER NO.: 93AX400924 — CUSTOMER ORDER NO.: PAV-426

CUSTOMER: BECHTEL

PROJ. MGR APPROVED: — Q.A. MGR APPROVED: — DATE:

DRAWN BY: E.SIMS DATE: 2-5-82 — DR NO: M-99X0924-F SH 8 OF 11

RELIANCE ELECTRIC COMPANY — CUSTOM CONTROLS — STONE MOUNTAIN, GEORGIA 30083

NUCLEAR SAFETY RELATED

CAD MICROSTATION LDT/01 — 1X5AB01-00563.DGN — Q

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A

TAB/SECT.: N/A
PAGE: N/A
FIGURE: N/A

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

REVISION 11 | DATE 2/17/04

REVISED BY SNC PER DCP 01-V1N0021

SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

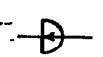
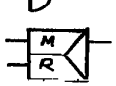
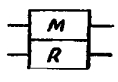
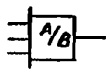
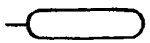
TITLE: MISCELLANEOUS SYSTEMS/EQUIPMENT PANEL (QPCP) SECTION 3

BY	CHK'D	APPR.1	APPR.2
LDT	WND	DEW	X

VENDOR: RELIANCE ELECTRIC — P.O.#/LINK

DRAWING NO. 1X5AB01-00563

SIZE E

FUNCTION	SYMBOL	MEANING
AND		OUTPUT EXISTS ONLY WHEN ALL INPUTS ARE PRESENT.
OR		OUTPUT EXISTS ONLY WHEN ANY INPUT IS PRESENT.
NOT		OUTPUT EXISTS ONLY WHEN THE INPUT IS NOT PRESENT.
TIME DELAY (ON DELAY)		OUTPUT EXISTS AFTER A TIME DELAY WHEN THE INPUT IS CONTINUOUSLY PRESENT. OUTPUT CEASES WHEN THE INPUT IS NOT PRESENT.
TIMED DELAY (OFF DELAY)		OUTPUT EXISTS WHEN THE INPUT IS PRESENT AND CONTINUES TO EXIST FOR A TIME AFTER THE INPUT CEASES
OFF RETURN MEMORY		MEMORY OUTPUT EXISTS WHEN THE MEMORY INPUT IS PRESENT AND CONTINUES TO EXIST UNTIL THE RESET INPUT IS PRESENT, EXCEPT UPON INTERRUPTION OF POWER, MEMORY OUTPUT DISAPPEARS.
RETENTIVE MEMORY		MEMORY OUTPUT EXISTS WHEN THE MEMORY INPUT IS PRESENT AND CONTINUES TO EXIST (ALSO UPON INTERRUPTION OF POWER) UNTIL THE RESET INPUT IS PRESENT. RESET OUTPUT EXIST ONLY WHEN THE MEMORY OUTPUT IS NOT PRESENT.
COINCIDENCE MATRIX		OUTPUT EXISTS ONLY WHEN AT LEAST A OUT OF B INPUTS ARE PRESENT.
SPECIAL		A FUNCTION WHICH PRODUCES AN OUTPUT UNDER SPECIAL CONDITIONS STATED IN OR ADJACENT TO THE SYMBOL.
LIGHT		R-RED - OPERATING, FLOWING, OR INCREASING G-GREEN - NOT OPERATING, NOT FLOWING, OR DECREASING A-AMBER- ELECTRICAL OR MECHANICAL PROTECTION TRIP W-WHITE- MONITOR B-BLUE - MISCELLANEOUS
ANNUNCIATOR		INPUT TO ANNUNCIATOR
ANNUNCIATOR		COMMON INPUT TO ANNUNCIATOR
COMPUTER		INPUT TO COMPUTER
DEMARCATION LINES		DASHED LINES INDICATE OUTLINED AREAS OF LOGIC THAT ARE SUPPLIED BY A STATED VENDOR, OR AS NOTED
CROSS REFERENCE		CROSS REFERENCE TO ANOTHER LOGIC DRAWING

GENERAL NOTES

1. PROCESS EQUIPMENT WILL CHANGE STATE WHEN A CHANGE IS INITIATED, AND WILL REMAIN IN THAT STATE UNTIL A CHANGE TO ANOTHER STATE IS INITIATED.
2. PROCESS EQUIPMENT WILL REMAIN IN, OR RETURN TO, THE ORIGINAL STATE AFTER A LOSS AND RESTORATION OF POWER, UNLESS OTHERWISE NOTED.
3. INHERENT EQUIPMENT INTERLOCKS, SUCH AS CIRCUIT BREAKER TRIP FREE AND REVERSING STARTER CROSS INTERLOCKS, ARE NOT SHOWN. EQUIPMENT CANNOT BE OPERATED WHEN PROTECTIVE INPUTS EXIST.
4. THE BLOCKING OF EQUIPMENT OPERATING CIRCUITS WHEN OVERLOAD OR GROUNDS OCCURS IS GENERALLY NOT SHOWN. SWITCHGEAR CIRCUITS WHICH USE LOCKOUT RELAYS FOR PROTECTIVE TRIPS AND ALL MCC CIRCUITS REQUIRE OPERATION OF A SWITCH AT THE SWITCHGEAR OR MCC TO RESET A PROTECTIVE TRIP. (2)
5. IF A CIRCUIT BREAKER HAS BEEN TRIPPED ON BUS VOLTAGE FAILURE, IT WILL NOT RECLOSE ON BUS VOLTAGE RESTORATION
6. THE TEST CONTROL SWITCHES AT THE SWITCHGEAR WHICH FUNCTION ONLY WHEN BREAKER IS IN THE TEST POSITION ARE NOT SHOWN. WHEN A CIRCUIT BREAKER IS IN THE TEST POSITION, REMOTE CONTROL IS CUT OFF.
7. FINAL INSTRUMENT SET POINTS ARE SHOWN ELSEWHERE. SET POINTS SHOWN ON SYSTEM CONTROL LOGIC DIAGRAMS ARE APPROXIMATE.
8. THE LOGIC TO SHOW THAT VALVE AND DAMPER POSITION LIGHTS ARE BOTH ON WHEN THE EQUIPMENT IS IN AN INTERMEDIATE POSITION IS NOT SHOWN.
9. MOTOR OPERATED VALVES GO TO FULLY OPEN OR FULLY CLOSED POSITION WITH MOMENTARY COMMAND UNLESS JOG PUSHBUTTON IS NOTED.
10. LIMIT AND TORQUE SWITCHES TO STOP VALVE AND DAMPER MOTOR ACTUATORS AT THE END OF TRAVEL ARE NOT SHOWN IN THE LOGIC. THE VALVE TYPE AND REQUIRED ACTIONS WILL BE NOTED ON THE DIAGRAM WHEN AVAILABLE.
11. THE HAND SWITCH FUNCTION ABBREVATION WILL BE USED ON SYSTEM CONTROL LOGIC DIAGRAMS AS FOLLOWS:
    (M)- MAINTAINED CONTACT SWITCH
    (P)- MOMENTARY CONTACT SWITCH

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL	ENG MGR
2	INC. ABN-20808.	6/2/95	CD	LJH	DMH		
1	REVISIONS	DATE	DR	CHK	APPV	DTL	DATE
0	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES				
NO.	REVISIONS	DATE	DR				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM LEGEND

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN002-1	2

SIZE B 11x17

SPECIAL NOTES

1. SELECTED ENGINEERED SAFETY FEATURE EQUIPMENTS REQUIRED FOR (1) SHUTDOWN OF THE PLANT WILL HAVE TWO POINTS OF CONTROL, ONLY IF ONE OF THE POINTS IS IN THE CONTROL ROOM. A TRANSFER SWITCH ON THE SHUTDOWN PANEL SHALL DISABLE THE CONTROL ROOM POINT OF CONTROL WHEN THE TRANSFER SWITCH IS ON LOCAL POSITION. THIS CONDITION WILL BE ANNUNCIATED IN THE CONTROL ROOM.

2. THE THERMAL OVERLOAD PROTECTION DEVICE OF SAFETY RELATED MOTOR OPERATED VALVE SHALL BE BYPASSED AND TEMPORARILY PLACED IN FORCE ONLY WHEN THE VALVE MOTOR IS UNDERGOING PERIODIC OR MAINTENANCE TESTING.

GENERAL NOTES (CONT.)

12. THE FOLLOWING SAFETY SIGNALS ABBREVATIONS WILL BE USED ON SYSTEM CONTROL LOGIC DIAGRAMS.

- CI-A-A – CONTAINMENT ISOLATION-PHASE A-TRAIN A
- CI-A-B – CONTAINMENT ISOLATION-PHASE A-TRAIN B
- CI-B-A – CONTAINMENT ISOLATION-PHASE B-TRAIN A
- CI-B-B – CONTAINMENT ISOLATION-PHASE B-TRAIN B
- SI-A – SAFETY INJECTION SIGNAL-TRAIN A
- SI-B – SAFETY INJECTION SIGNAL-TRAIN B
- CVI-A – CONTAINMENT VENTILATION ISOLATION-TRAIN A
- CVI-B – CONTAINMENT VENTILATION ISOLATION-TRAIN B
- CRI-A – CONTROL ROOM ISOLATION-TRAIN A
- CRI-B – CONTROL ROOM ISOLATION-TRAIN B
- SLI-A – STEAM LINE ISOLATION-TRAIN A
- SLI-B – STEAM LINE ISOLATION-TRAIN B
- FI-A – FEEDWATER ISOLATION-TRAIN A
- FI-B – FEEDWATER ISOLATION-TRAIN B
- FHBI-A – FUEL HANDLING BUILDING ISOLATION TRAIN A
- FHBI-B – FUEL HANDLING BUILDING ISOLATION TRAIN B

13. THE FOLLOWING ABBREVATIONS WILL BE USED FOR CONTROL BOARDS, PANELS AND STATIONS. OTHERS REFER TO CONTROL PANEL LIST.

(1) 2X EQUIPT INDEX

- QMCB – MAIN CONTROL BOARD;
- QHVC – HVAC PANEL
- QPCP – MISC. SYSTEMS EQUIPMENT PANEL
- QSPS – SYSTEMS STATUS MONITOR PANEL
- PAFP – AUX FD WTR SHUTDOWN PNL; PAFT – AUX FD WTR TURB PNL.
- PRP – TURBINE PLANT SAMPLING
- PSDA – SHUTDOWN PANEL TRAIN A; PSDB – SHUTDOWN PANEL TRAIN B
- PNSS – NUCLEAR SERVICE WATER SAMPLE PANEL
- PDG1 – DIESEL GEN. PANEL NO. 1; PDG2 – DIESEL GEN. PANEL NO. 2
- PNSP – NUCLEAR SAMPLING SYSTEM PANEL
- PWHP – WASTE HOLD-UP PANEL
- PAFT – AUX FDWTR TURB PNL
- YTCN – TECHNICAL SUPPORT PNL
- L – LOCAL STATION
- ERF – EMERGENCY RESPONSE FACILITY

GENERAL NOTES (CONT)

14. THE BLOCKING OF AIR CIRCUIT BREAKER MANUAL STOP ACTIONS FROM THE CONTROL ROOM DURING ESF SEQUENCER OPERATION IS NOT SHOWN ON INDIVIDUAL LOGIC DIAGRAMS. REFER TO ELEMENTARY DIAGRAMS. (1)

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	R. E.	Q.A.E
3										
2										
1	REVISED SPECIAL NOTE 1 & GENERAL NOTE 13, 14	7-11-84	ETL	ALV						
0	ISSUED FOR CONSTRUCTION	1-4-79	NBD	FAL	MKK					

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM

LEGEND

SCALE: NONE | DRAWING NO. 1X5DN002-2 | REV. 1

JOB NO. 9510

BM 9510

SIZE B 11 x 17

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DATE: 7-11-84 CAMERA OPERATOR — SECTION SUPERVISOR M. Pavlakoff

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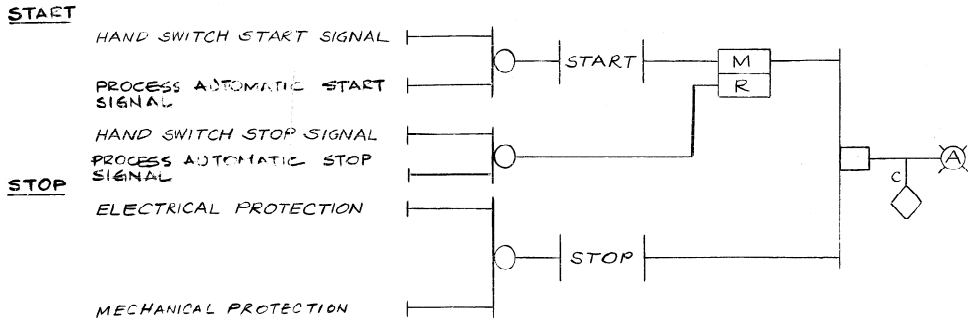


TYPICAL AUTOMATIC PROTECTION ALARM AND INDICATION

NO.	REVISIONS	DATE	DR	CHK.	SUP'V.		EGS	CHF. E.	P. E.	Q.A. E.	ENG. MGR.
△											
△											DATE
△											
0	ISSUED FOR CONSTRUCTION	5/1/79	HL	FAL	MKV						

BM 9510

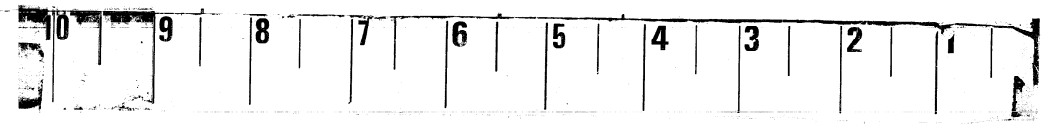
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
TYPICAL

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN002-3	0

SIZE B 11 x 17



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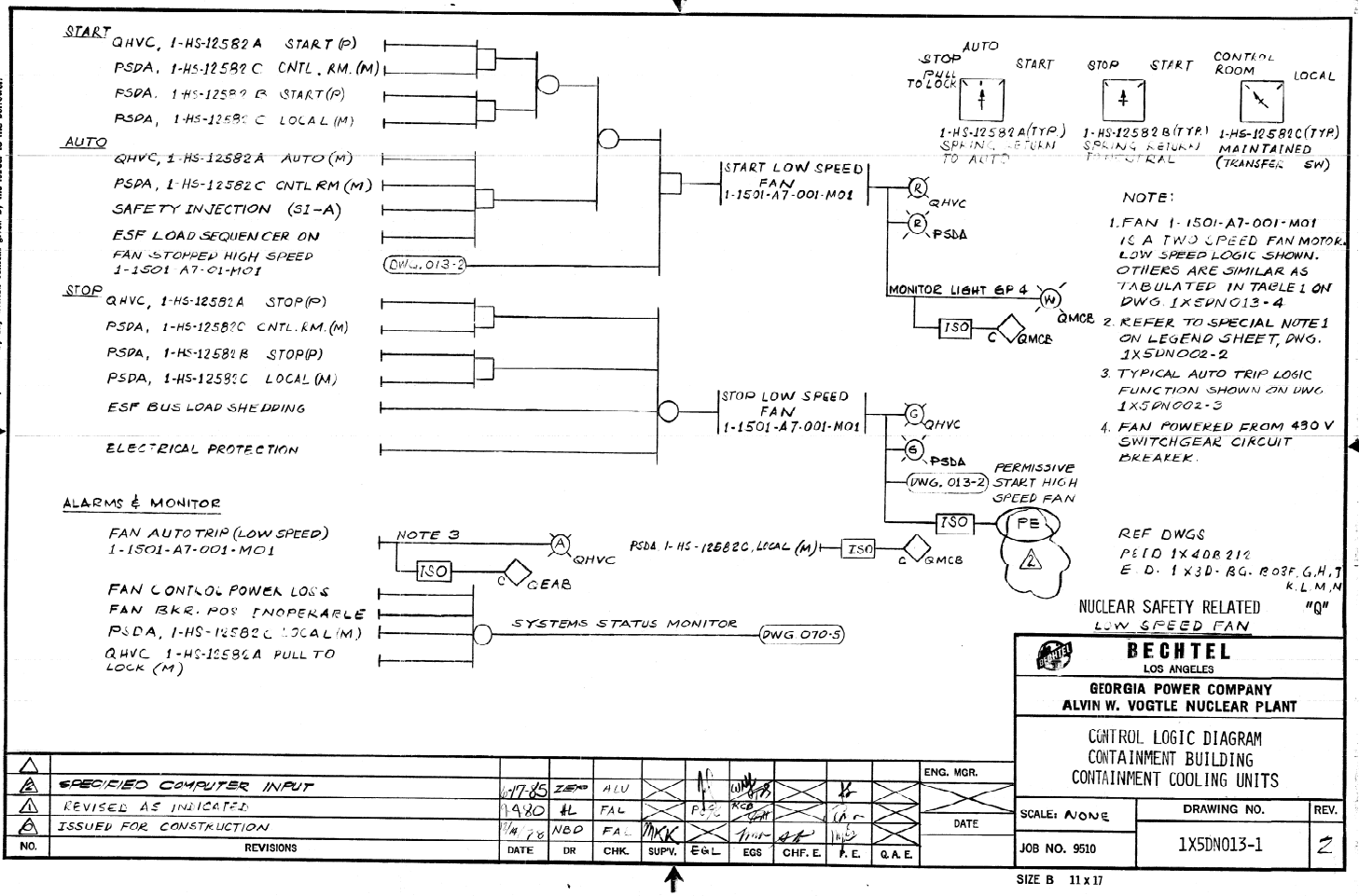
6-1-79 Cynthia Kim — D.K. Vaughn
DATE: CAMERA OPERATOR — SECTION SUPERVISOR

LAD-2591 6/78



NOTE:

1. FAN 1-1501-A7-001-M01 IS A TWO SPEED FAN MOTOR. LOW SPEED LOGIC SHOWN. OTHERS ARE SIMILAR AS TABULATED IN TABLE 1 ON DWG. 1X5DN013-4
2. REFER TO SPECIAL NOTE 1 ON LEGEND SHEET, DWG. 1X5DN002-2
3. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG. 1X5DN002-3
4. FAN POWERED FROM 480 V SWITCHGEAR CIRCUIT BREAKER.

REF DWGS
P&ID 1X40B212
E D. 1X3D-BG. B03F, G, H, J
K, L, M, N

NUCLEAR SAFETY RELATED "Q"
LOW SPEED FAN

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NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
4	SPECIFIED COMPUTER INPUT	6-17-85	ZERO	ALV						ENG. MGR.
3	REVISED AS INDICATED	11-4-80	NBD	FAL						DATE
2	ISSUED FOR CONSTRUCTION	11-4-78	NBD	FAL						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT BUILDING
CONTAINMENT COOLING UNITS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN013-1	2

SIZE B 11 x 17

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DATE: 6-17-85 CAMERA OPERATOR SECTION SUPERVISOR

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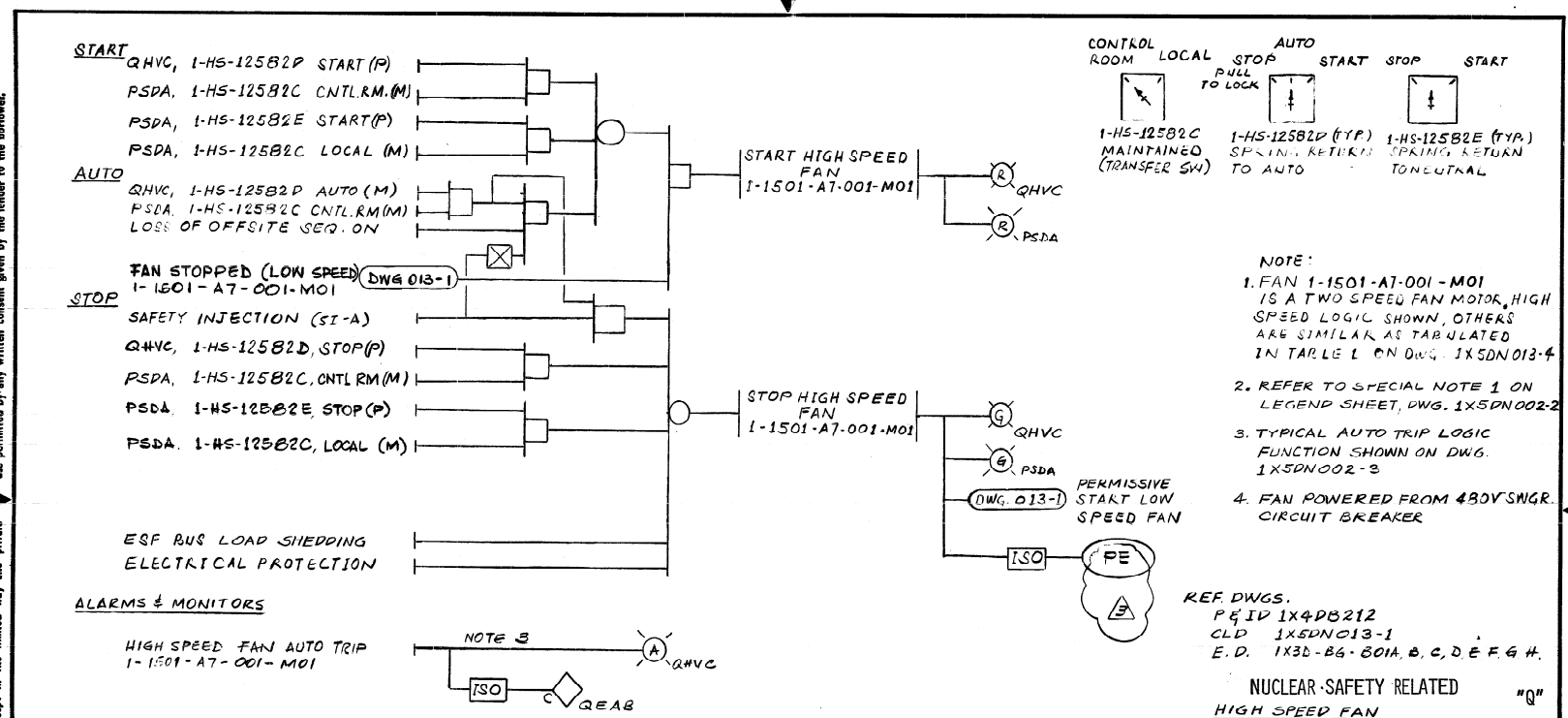
DATE: 6-17-85 CAMERA OPERATOR SECTION SUPERVISOR



NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	R. E.	Q. A. E.
4	SPECIFIED COMPUTER INPUT	6-17-85	ZER	AIV						
3	ADDED INPUT INTLK. WITH SI	12-7-81	AV	FAL						
2	REVISED AS INDICATED	9-9-80	HL	FAL						
1	ISSUED FOR CONSTRUCTION	12/14/78	NBD	FAL						

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT BUILDING
CONTAINMENT COOLING UNITS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN013-2	3

P&ID 212 REV.

SIZE B 11 x 17

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6-17-85
DATE: CAMERA OPERATOR SECTION SUPERVISOR

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6-17-85
DATE: CAMERA OPERATOR SECTION SUPERVISOR

14.5K1

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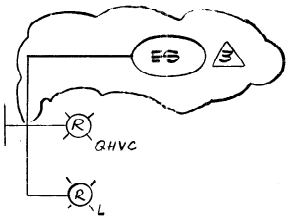
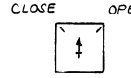
OPEN

L, 1-HS-2582G, OPEN (P) — OPEN DAMPER 1-HV-2582A — (R) QHVC, (R) L

CLOSE

L, 1-HS-2582G, CLOSE (P) — CLOSE DAMPER 1-HV-2582A — (G) QHVC, (G) L





1-HS-2582G SPRING RETURN TO NEUTRAL

NOTE:
1. DAMPER 1-HV-2582A SHOWN, OTHER DAMPERS LISTED IN TABLE 2 SIMILAR.
2. TABLE 1 LOGIC SHOWN IN DWG 1X5DN013-1 & 1X5DN013-2.
3. HAND SWITCHES LISTED IN TABLE 2 ARE LOCATED OUTSIDE THE CONTAINMENT.

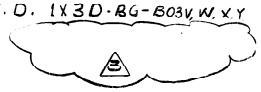
TABLE 1

TRAIN	EQUIP TAG NO.	LOW SPEED CNTL SW		TRANSF. SW	HIGH SPEED CNTL SW				SAFETY SIG.	PANEL
A	1-1501-A7-001-M01	1-HS-12582A	1-HS-12582B	1-HS-12582C	1-HS-12582D	1-HS-12582E			SI-A	PSDA
A	-002	-2582A	-2582B	-2582C	-2582D	-2582E				PSDA
B	-003	-12583A	-12583B	-12583C	-12583D	-12583E			SI-B	PSDB
B	-004	-2583A	-2583B	-2583C	-2583D	-2583E				PSDB
A	-005	-12584A	-12584B	-12584C	-12584D	-12584E			SI-A	PSDA
A	-006	-2584A	-2584B	-2584C	-2584D	-2584E				PSDA
B	-007	-12585A	-12585B	-12585C	-12585D	-12585E			SI-B	PSDB
B	-008	-2585A	-2585B	-2585C	-2585D	-2585E				PSDB

TABLE 2

DAMPER	HAND SW	TRAIN
1-HV-2582A	1-HS-2582G	A
2B	2H	
3A	3G	B
3B	3H	
4A	4G	A
4B	4H	
5A	5G	B
5B	5H	

REF. DWGS:
P&ID 1X4DB212
CLD 1X5DN013-1
1X5DN013-2
E.D. 1X3D-BG-B03V,W,X,Y



NUCLEAR SAFETY RELATED
EQUIPMENT LIST & FAN DAMPERS
"Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	R. E.	Q.A.E.	ENG. MGR.
5	SHOWN COMPUTER INPUT & DELETED REF. DWGS.	6-17-85	ZED	HLV							DATE
4	REVISED LOCATION OF INDICATING LIGHT & REF. DWGS.	12-8-81	AV	FAL		PCL	DCP				
3	REVISED AS INDICATED	3/1/80	HL	FAL		DJC	MKK				
2	ISSUED FOR CONSTRUCTION	12/15/78	NBD	FAL	MKK			AW			

BECHTEL
LOS ANGELES
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT BUILDING
CONTAINMENT COOLING UNITS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN013-4	5

SIZE B 11 x 17

P&ID 212 REV 1 BM 9510

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6-17-85
DATE: CAMERA OPERATOR SECTION SUPERVISOR

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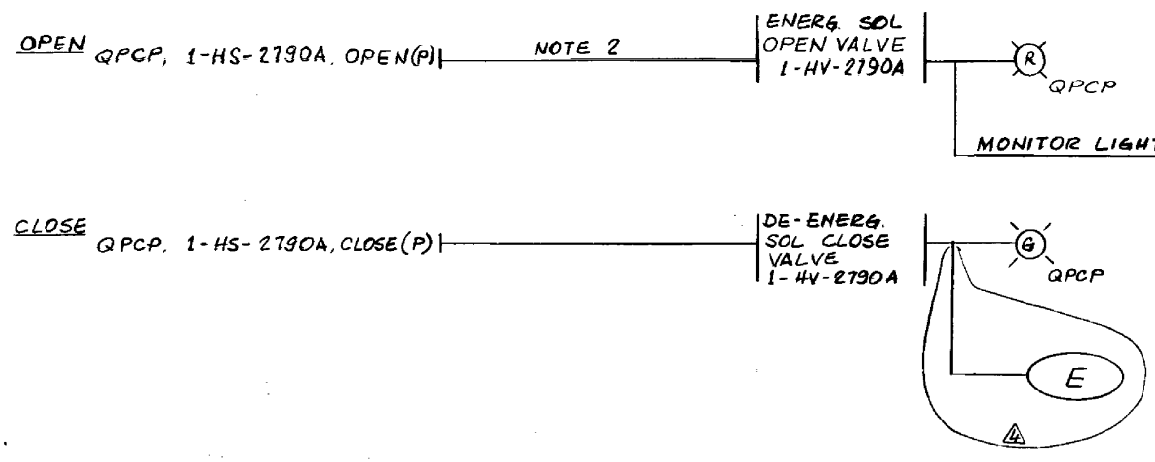

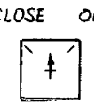

NOTE:

1. VALVE 1-HV-2790A SHOWN, OTHER VALVE LISTED IN TABLE 1 SIMILAR.
2. HAND SWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH
3. (4) VALVE 1-HV-8221 IS A NON 1E VALVE AND DOES NOTE REQUIRE MONITOR LIGHT ON QMCB.

REF. DWGS:
P&ID 1X4DB213-2
E.D 1X3D-BG-B05B,E

TABLE 1

SERVICE	TRAIN	SOL VALVE	SWITCH
HYDROGEN SAMPLE INLET - IRC	B	1-HV-2790A	1-HS-2190A
INLET - IRC	B	-2790B	-2790B
INLET - ORC	A	-2791A	-2791A
INLET - ORC	B	-2791B	-2791B
INLET - IRC	A	-2792A	-2792A
INLET - IRC	A	-2792B	-2792B
OUTLET - ORC	A	-2793A	-2793A
OUTLET - ORC	B	-2793B	-2793B
INLET - ORC	NOTE 3 (4)	-8221	-8221

NUCLEAR SAFETY RELATED "Q"
CNMT HYDROGEN SAMPLING

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT AIR PURIFICATION AND CLEAN-UP SYSTEM

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P.E.	Q.A.E.
3	ADDED SAMPLE VALVE FOR PASS HV-8221	9/1/82	HV	FAL		PCL			KT	
2	REVISED REF DWGS.	12-8-81	AT	FAL		PCL			IAT	
1	DEGRADED VALVE 1-HV-8221 PER DCN 3 ADDED ERF INPUT	3-27-84	ALV	FAL		PCL				
0	ISSUED FOR CONSTRUCTION	12/19/78	NBL	FAL	MKK			MP		

ENG. MGR. / DATE

SCALE: NONE | DRAWING NO. 1X5DN015-1 | REV. 4

JOB NO. 9510

SIZE B 11 x 17

BM 9510
P&ID 213-2 REV.1

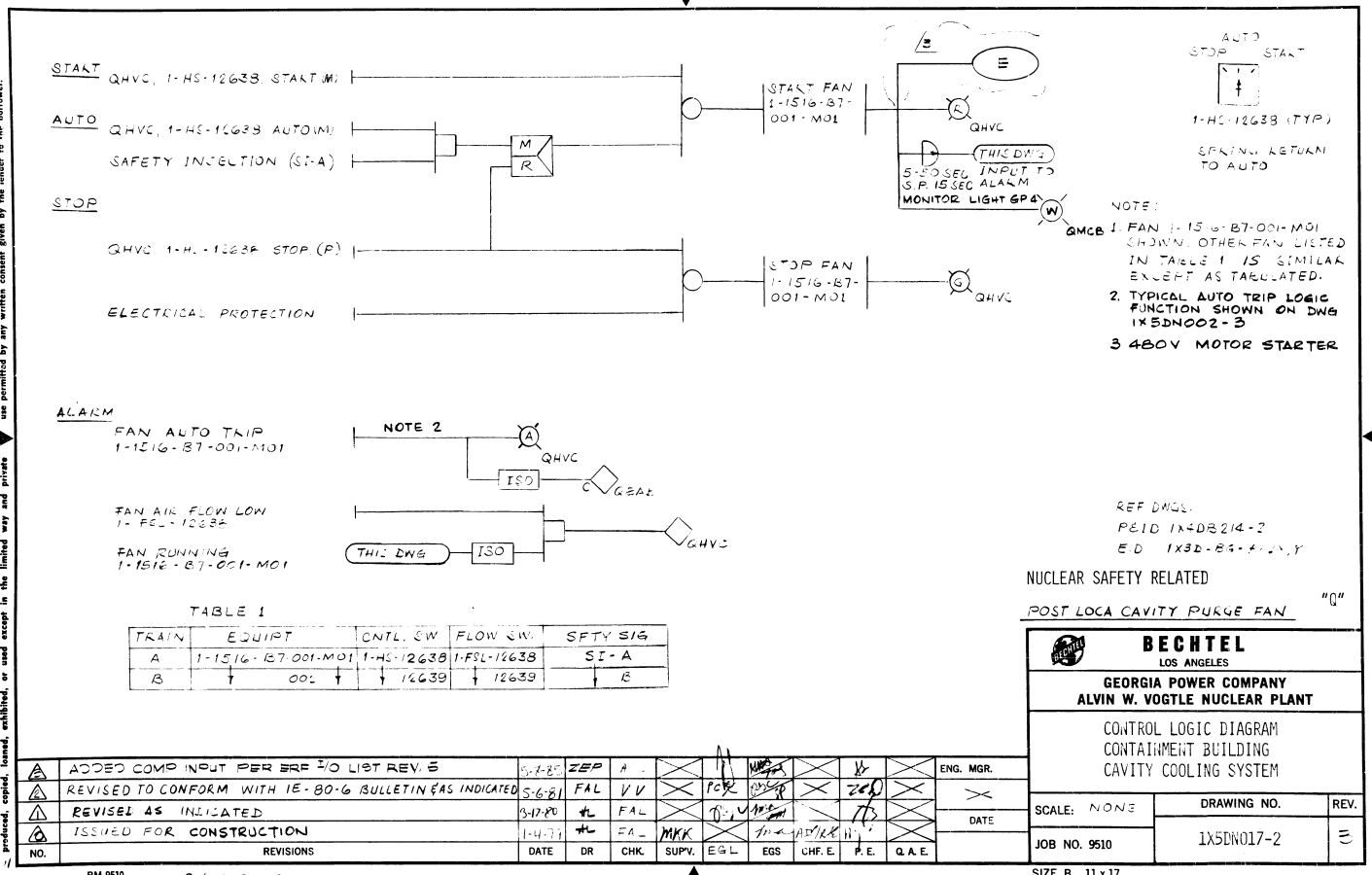

TABLE 1

TRAIN	EQUIPT	CNTL. SW.	FLOW SW.	SFTY SIG
A	1-1516-B7-001-M01	1-HS-12638	1-FSL-12638	SI-A
B	001	12639	12639	B

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.
5	ADDED COMP INPUT PER SRE I/O LIST REV. 5	5-1-85	ZEP	A						
4	REVISED TO CONFORM WITH IE-80-6 BULLETIN AS INDICATED	5-6-81	FAL	VV		FCK				
3	REVISED AS INDICATED	3-17-80	HL	FAL						
2	ISSUED FOR CONSTRUCTION	1-4-77	HL	FAL	MKK					

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT BUILDING
CAVITY COOLING SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN017-2	5

SIZE B 11 x 17

BM 9510 P&ID REV 3

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5/9/85
DATE
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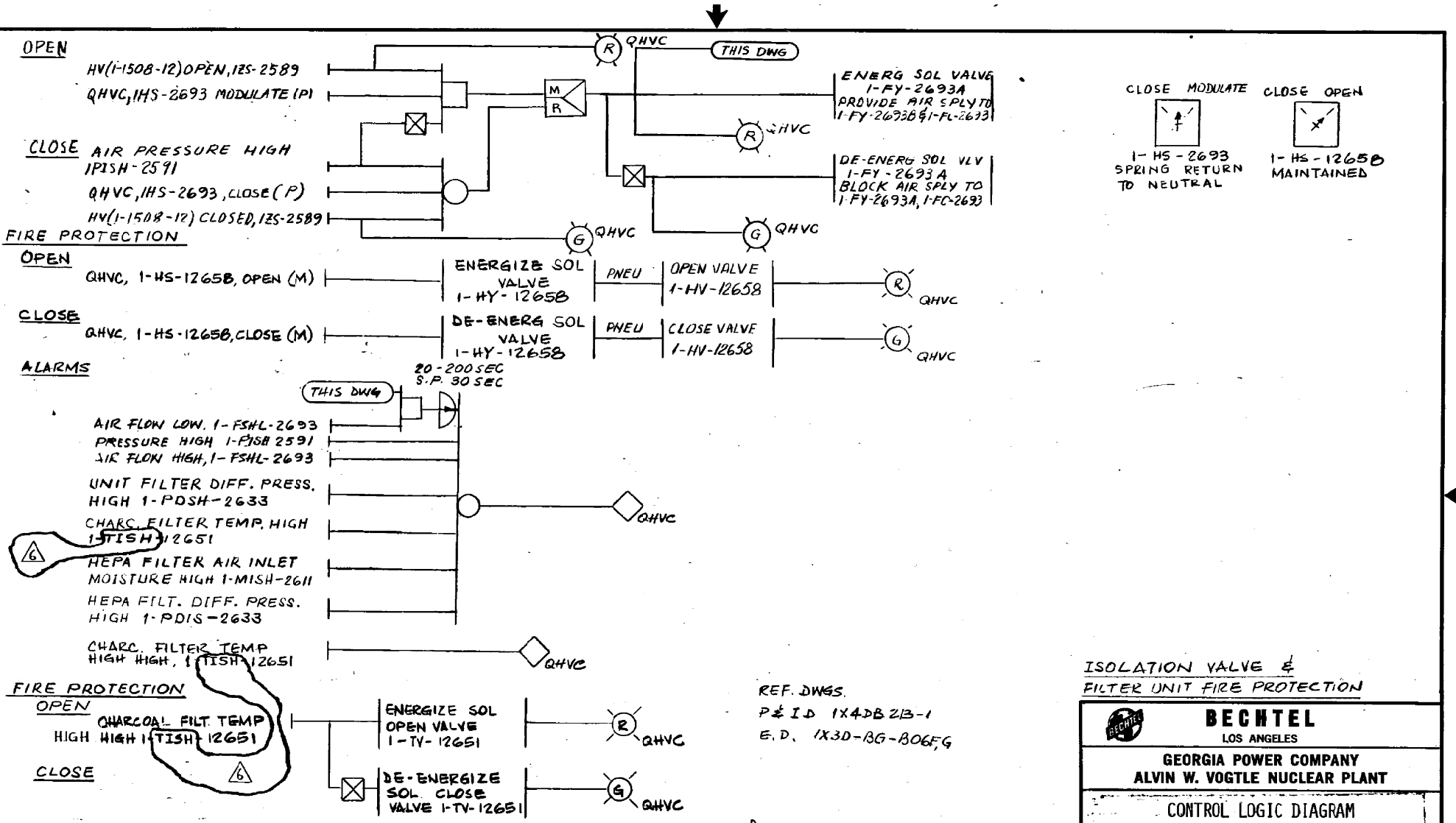
DATE
CAMERA OPERATOR
SECTION SUPERVISOR



ISOLATION VALVE & FILTER UNIT FIRE PROTECTION

NO.	REVISIONS	DATE	DR	CHK.	ELEC	EGL	EGS	CHF. E.	P.E.	Q.A.E.	ENG. MGR.
6	CORRECTED TAG NO. 1-TISH-12651	3/6/85	AS	AIV							DATE
5	CORRECTED SW POS. AND GATE	1-7-85	ALV								
4	REVISED PER P&ID DCN 11	9-13-84	ETL	ALV							
3	ISSUED FOR CONSTRUCTION	1-4-74		FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTAINMENT POST LOCA PURGE
EXHAUST DUCT ISOLATION VALVES

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN019-1	6

P&ID REV

SCANNED DATE: 11/19/91

SIZE B 11 x 17

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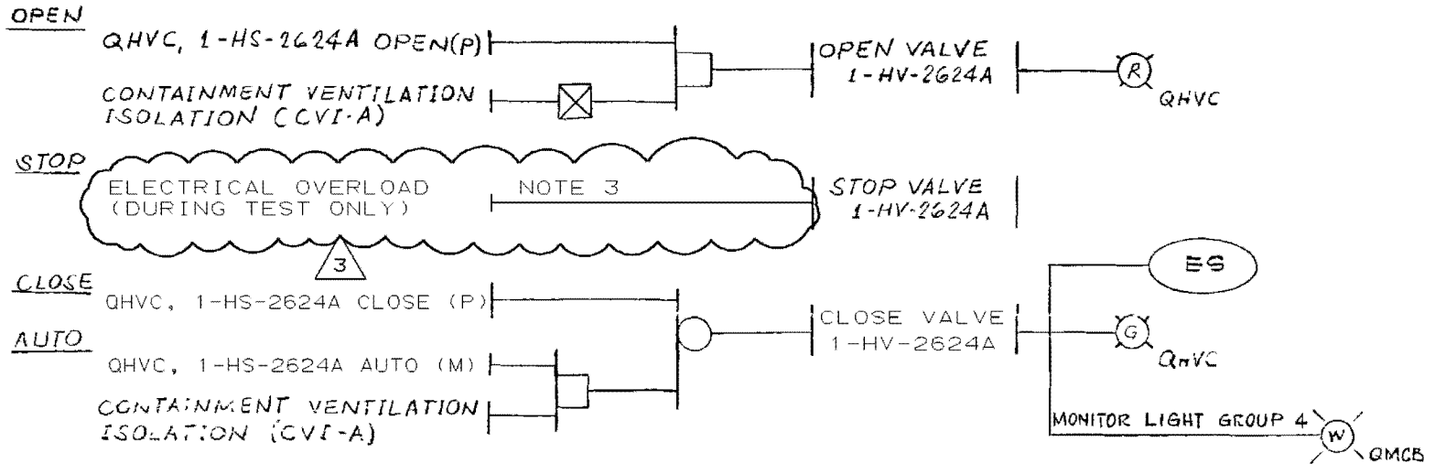
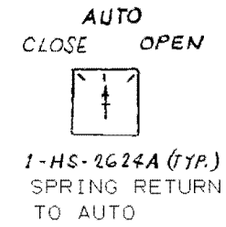
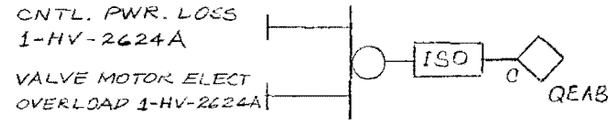




NOTE

1. VALVE 1-HV-2624A SHOWN. 1-HV-2624B SIMILAR.
2. VALVES LISTED IN TABLE 1 ARE BUTTERFLY VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
3. REFER TO SPECIAL NOTE 2 ON DWG 1X5DN002-2

VALVE NO.	TRAIN	SAFETY SIGNAL	SWITCH
1-HV-2624A	A	CVI-A	1-HS-2624A
⏚ -2624B	B	CVI-B	⏚ -2624B



REF. DWGS.
P & ID 1X4DB213-1
E.D. 1X3D-BG-B04A,4B

NUCLEAR SAFETY RELATED "Q"

INSIDE CONTAINMENT ISOL VALVES

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM CONTAINMENT POST LOCA PURGE EXHAUST DUCT ISOLATION VALVES

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
△						
△						
3	INCORPORATED PER DCP 95-V1N0035	12/5/97	CD	GLB	WFP	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE | DRAWING NO. 1X5DN019-2 | REV 3

JOB NO. 10604

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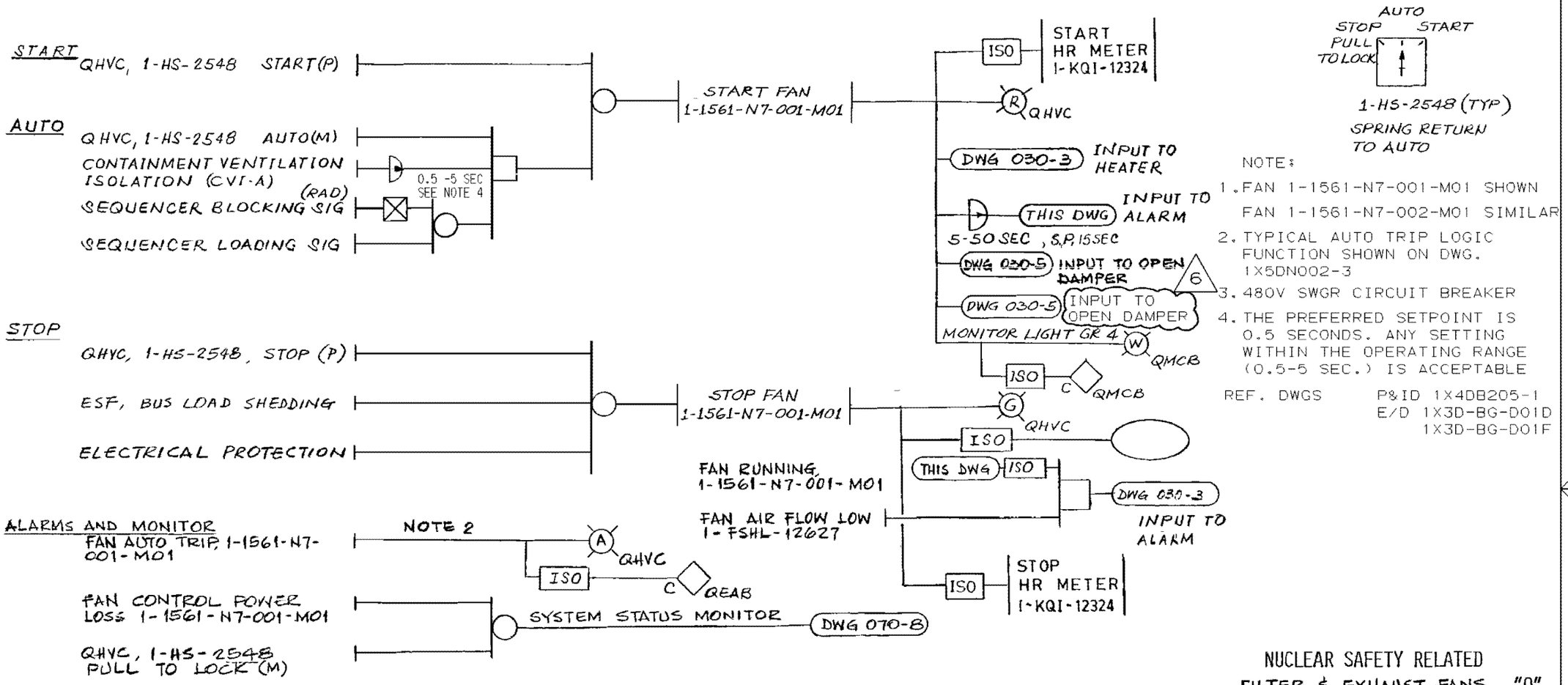
CAD NAME 15DN0192 SIZE B 11x17



NOTE:

1. FAN 1-1561-N7-001-M01 SHOWN
   FAN 1-1561-N7-002-M01 SIMILAR
2. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG. 1X5DN002-3
3. 480V SWGR CIRCUIT BREAKER
4. THE PREFERRED SETPOINT IS 0.5 SECONDS. ANY SETTING WITHIN THE OPERATING RANGE (0.5-5 SEC.) IS ACCEPTABLE

REF. DWGS P&ID 1X4DB205-1
E/D 1X3D-BG-D01D
1X3D-BG-D01F

NUCLEAR SAFETY RELATED
FILTER & EXHAUST FANS "Q"

TRAIN	FAN No	SWITCH	SFTY SIG	FLOW SW	HR METER
A	1-1561-N7-001-M01	1-HS-2548	CVI-A	1-FSHL-12627	1-KQI-12324
B	002	9	B	B	1 KQI 12325

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
6	INCORPORATED PER DCP 95-VAN0056	12-31-97	CD	WMW	EOG	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM PIPING PENETRATION ROOM FILTER AND EXHAUST UNITS

SCALE: NONE

DRAWING NO. 1X5DN030-1

REV 6

JOB NO. 10604

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CAD NAME 15DN0301

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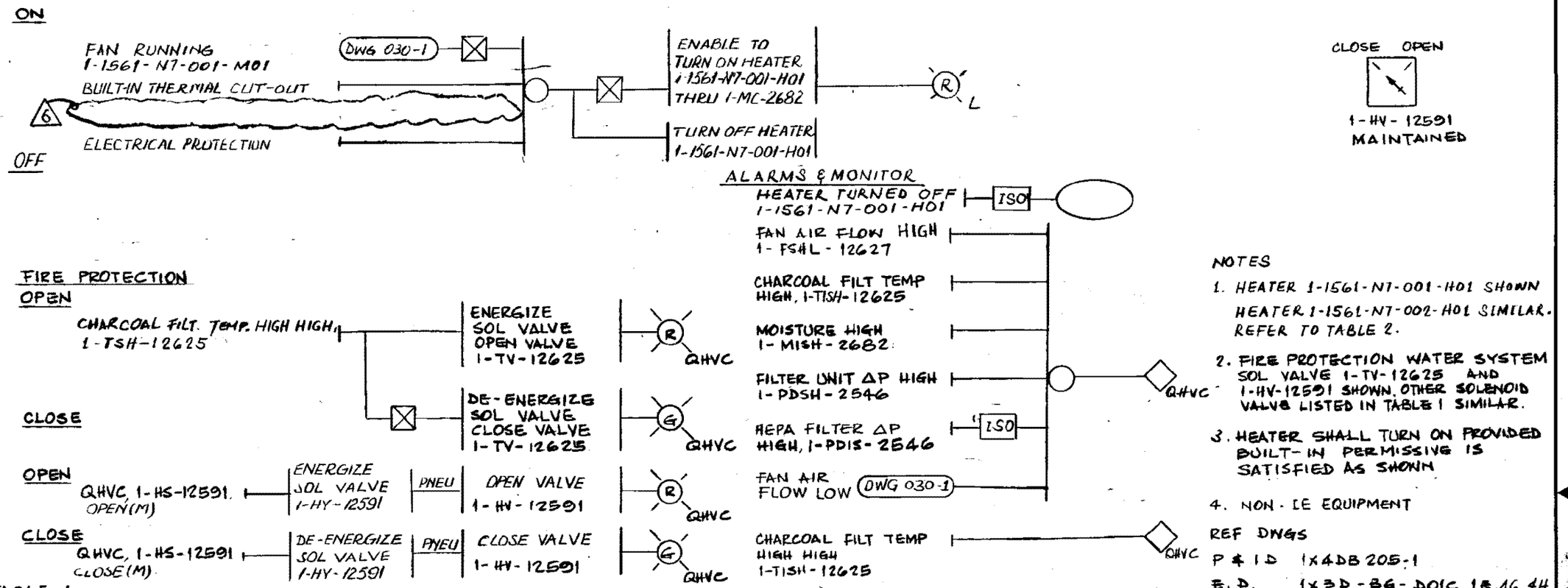

NOTES

1. HEATER 1-1561-N7-001-H01 SHOWN HEATER 1-1561-N7-002-H01 SIMILAR. REFER TO TABLE 2.
2. FIRE PROTECTION WATER SYSTEM SOL VALVE 1-TV-12625 AND 1-HV-12591 SHOWN. OTHER SOLENOID VALVE LISTED IN TABLE 1 SIMILAR.
3. HEATER SHALL TURN ON PROVIDED BUILT-IN PERMISSIVE IS SATISFIED AS SHOWN
4. NON-1E EQUIPMENT

REF DWGS

P & I D 1X4DB 205-1

E. D. 1X5D-BG-D01C, 1E, 4G, 4H

TABLE 1

FILT UNIT NO.	SOL VLV	TEMP SW	SOL VLV (NOTE 4)	HAND SW (NOTE 4)	MOIST SW	FLOW SW	UNIT ΔP SW	HEPA ΔP SW
1-1561-N7-001-000	1-TV-12625	1-TISH-12625	1-HV-12591	1-HS-12591	1-MISH-2682	1-FSHL-12627	1-PDSH-2546	1-PDIS-2546
002	4	4	0	0	3	8	7	7

NUCLEAR SAFETY-RELATED

ELECTRIC HEATING COIL "Q"

TABLE 2

TRAIN	HEATER NO	FAN RUNNING	MOIST. CNTL
A	1-1561-N7-001-H01	1-1561-N7-001-M01	1-MC-2682
B	-002-H01	002-M01	3

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	ENG. MGR.
6	DELETED FLOW SW. INPUT PER VIP AX4AJ01-400-6	2/6/87	ALV	ACG							DATE
5	ADDED HTR LIGHT PER DCN #22 ON P&ID	5-2-85	ZEP	ALV							
4	CORRECTED TAG NO AND PRESENTATION OF HTR ACTUATION	10 16 84	ALV	WJW							
6	ISSUED FOR CONSTRUCTION	3/7/79	HC	FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
PIPING PENETRATION ROOM FILTER AND EXHAUST UNITS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN030-3	6

SIZE B 11 x 17

BM 9510 P&ID 205-1 REV.

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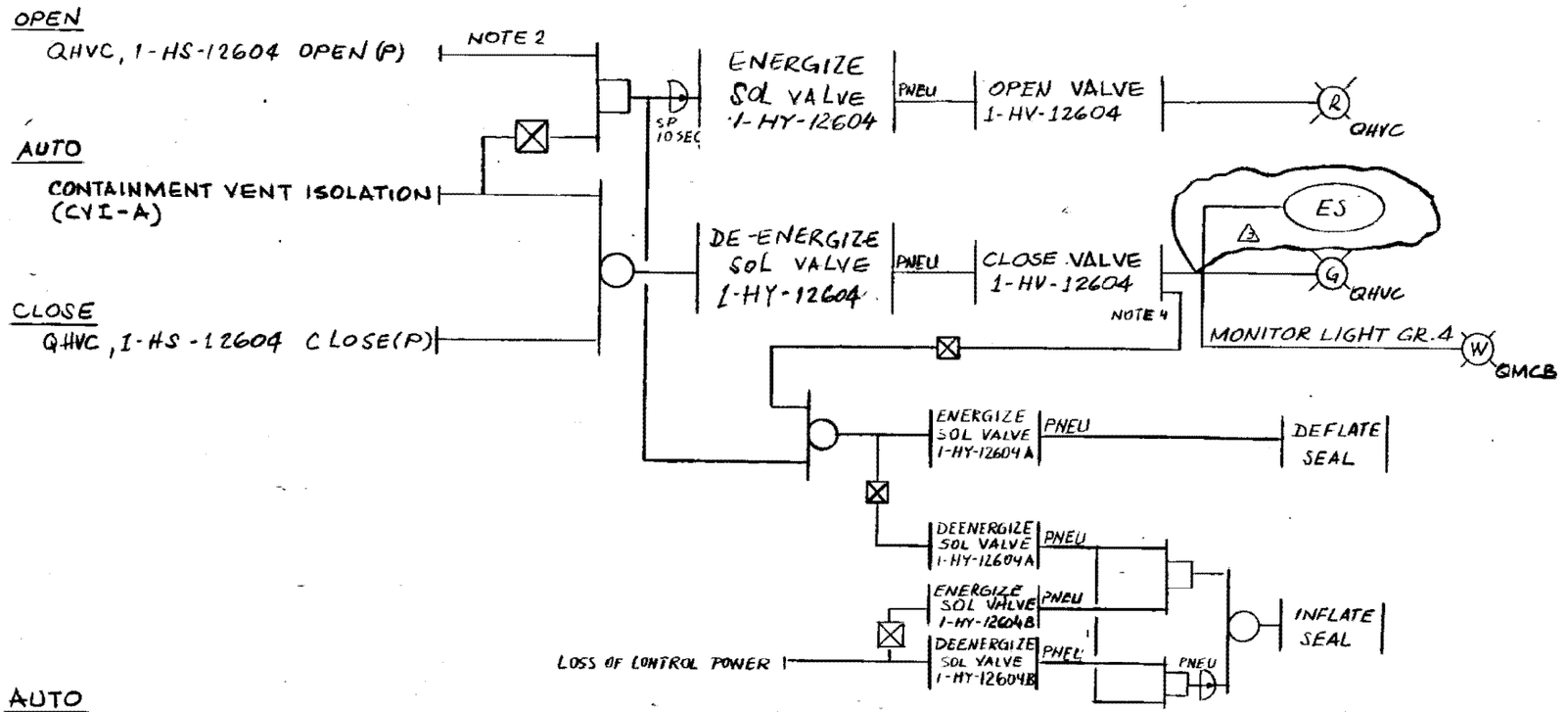
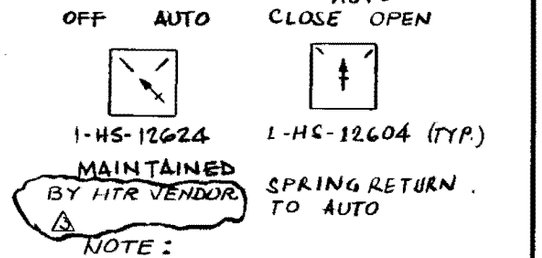




NOTE:

1. VALVE 1-HV-12604 SHOWN. OTHER VALVES LISTED IN TABLE 1 SIMILAR.
2. TO OPEN VALVE, CONTROL SWITCH SHALL BE HELD ON OPEN POSITION UNTIL DAMPER IS FULLY OPEN TO ALLOW THE CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH.
3. ELECTRIC INFRARED HEATER 1-1561-U7-001 SHOWN 1-1561-U7-002 IS SIMILAR.
4. POSITION SWITCH CONTACT 33ac VALVE NOT FULLY CLOSED.

REF. DWGS.
P&ID 1X4DB 205-2
E.D. 1X3D-BG-D03X,Y, D04A,B

NUCLEAR SAFETY RELATED
ISOLATION VALVE & INFRARED HEATER
"Q"

TABLE 1

INFRARED HEATER	HAND SW	TEMP SW
1-1561-07-001	1-HS-12624	1-TIS-12624
1-1561-07-002	↓ 5	↓ 5

TABLE 2

TRN	VALVE	SOL VALVE	CNTL SW	SAFETY SIG	SEAL CNTL	DMPR. PROT
A	1-HV-12604	1-HY-12604	1-HS-12604	CVI-A	1-HY-12604A	1-HY-12604B
↓	5	5	5	↓	5	5
B	6	6	6	CVI-B	6	6
↓	7	7	7	↓	7	7

SCANNED DEC 1991

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	E G L	EGS	CHF. E.	P. E.	Q. A. E.	ENG. MGR. / DATE
3	ADDED COMPUTER INPUT PER I/O LIST REV 4	10-16-84	ALU								
2	ADDED CONTROL FOR QAD BUBLE TIGHT DAMPER PER DCN 6	5-11-84	ALU								
1	REVISED AS INDICATED	3-17-80	HL	FAL							
0	ISSUED FOR CONSTRUCTION	3-15-79	HL	FAL	MKK.						

BECHTEL
LOS ANGELES

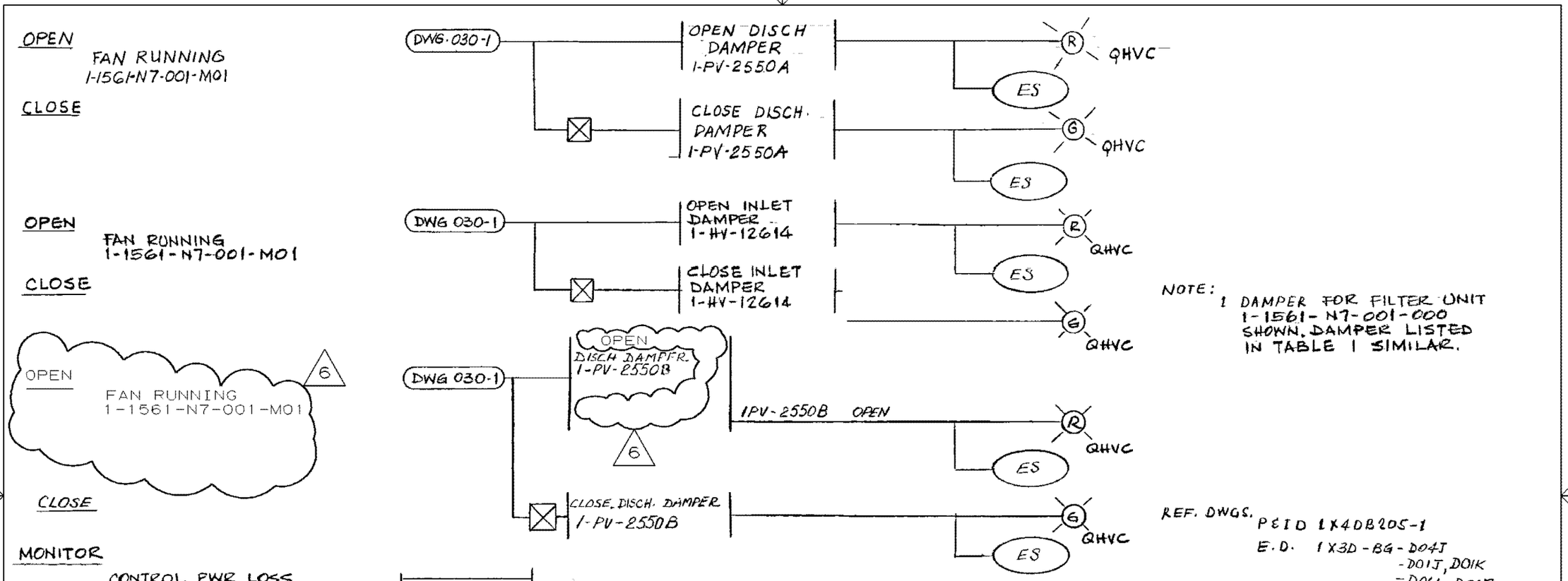
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
PIPING PENETRATION ROOM FILTER AND EXHAUST UNITS

SCALE: NONE	DRAWING NO. 1X5DN030-4	REV. 3
JOB NO. 9510		

SIZE B 11 x 17

BM 9510 P&ID REV 5



NOTE: 1 DAMPER FOR FILTER UNIT 1-1561-N7-001-000 SHOWN. DAMPER LISTED IN TABLE 1 SIMILAR.

REF. DWGS.
P&ID 1X4DB205-1
E.D. 1X3D-BG-D04J
-D01J, D01K
-D01L, D01M

NUCLEAR SAFETY RELATED
FAN INLET & OUTLET DAMPERS

"Q"

TABLE 1

TRAIN	FAN RUNNING	DISCH DAMPER	DISCH. DAMPER	INLET DAMPER	FILTER UNIT	ΔP SWITCH
A	1-1561-N7-001-M01	1-PV-2550A	1-PV-2550B	1-HV-12614	1-1561-N7-001-000	1-PDIS-2550
B	002	1A	1B	6	002	1-PDIS-2551

6

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM

PIPING PENETRATION ROOM
FILTER AND EXHAUST UNITS

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
6	INCORPORATED PER DCP 95-VAN0056	12-31-97	CD	WMW	EOG	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE | DRAWING NO. 1X5DN030-5 | REV 6
JOB NO. 10604

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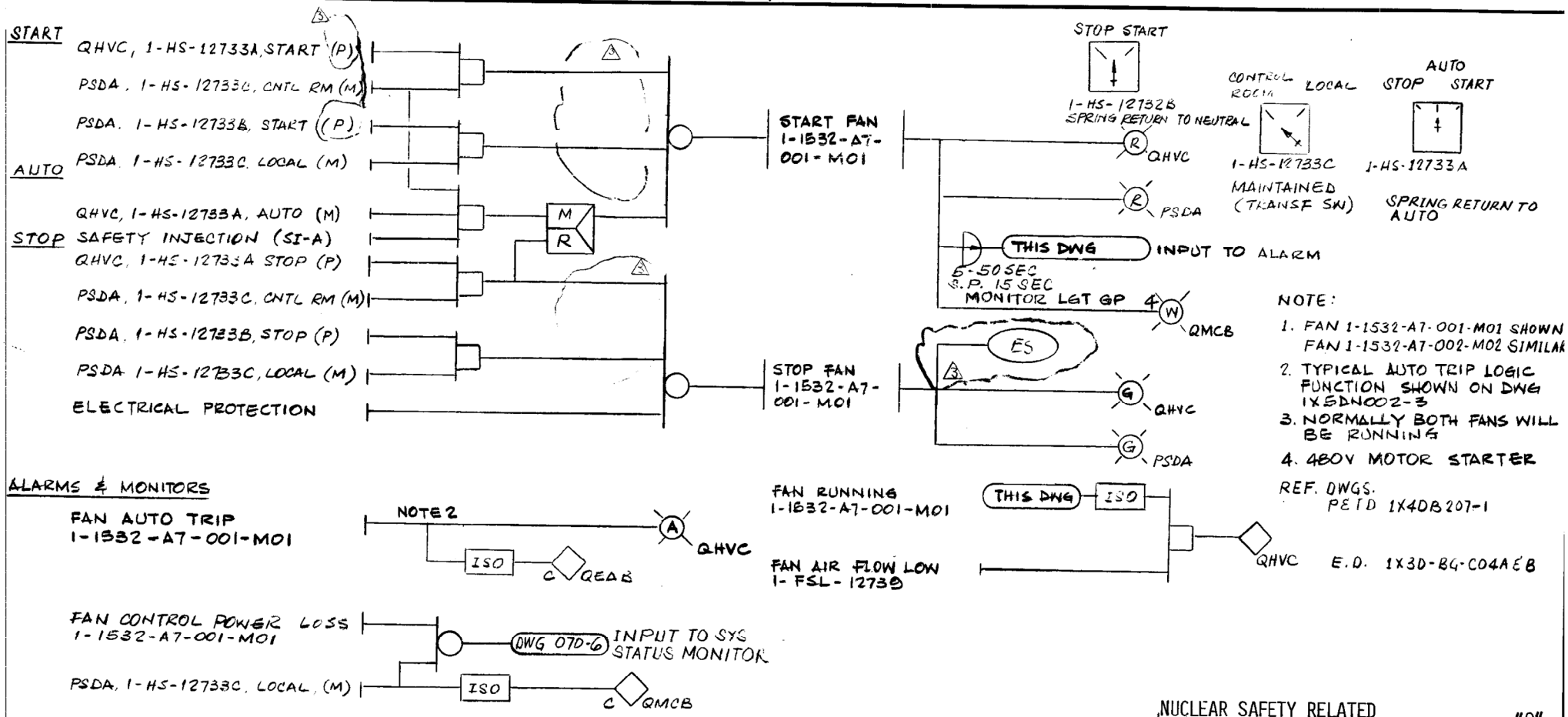
CAD NAME 15DN0305 SIZE B 11x17



TRAIN	FAN	HAND SWITCH			FLOW SWITCH	SFTY SIG
A	1-1532-A7-001-M01	1-HS-12733A	1-HS-12733B	1-HS-12733C	1-FSL-12739	SI-A
B	002-	18A	18B	18C	24	SI-B

NUCLEAR SAFETY RELATED
ELECTRICAL EQUIPMENT SUPPLY FAN "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
3	ADDED COMPUTER INPUT AND CLARIFY SW ACTION	1-7-85	ALV							
2	REVISED TO CONFORM WITH IE 80-06 BULLETIN	3-6-81	FAL							
1	REVISED AS INDICATED (COLD SHUTDOWN REQMT)	6-18-80	HL	FAL						
0	ISSUED FOR CONSTRUCTION	12/15/78	HL	FAL	MKK					

ENG. MGR.
DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTROL BUILDING
SAFETY FEATURES ELECTRICAL EQUIPMENT ROOM HVAC

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN044-1	3

SID REV 9

SIZE B 11 x 17

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MODULATE

SAFETY INJECTION (SI-A)

ENABLE TO MODULATE VALVE 1-TV-12740 THRU 1-TIC-12740

CLOSE VALVE 1-TV-12740 △3

TABLE 1

VALVE	SFTY SIG.	TEMP CNTL
1-TV-12740	SI - A	1-TIC-12740
↓ 725	↓ B	↓ 725

NOTES:

1. VALVE 1-TV-12740 SHOWN OTHER VALVES LISTED IN TABLE 1 SIMILAR AS TABULATED

REF DWGS

P & ID 1X4DB207-1

V/P 1X4AJ20-356, 357 △3

NUCLEAR SAFETY RELATED

ESF CHILLED WATER VALVE "Q"

SCANNED DATE SEP 14 [illegible]

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	ENG. MGR.
△3	ADDED VALVE CLOSING SIGN AND REF DWGS	6-25-84	ALV	WLL							DATE
△2	REVISED REF. DWGS.	12-8-81	AV	FAL		PER					
△1	REVISED AS INDICATED.	3/28/80	HL	FAL							
△0	ISSUED FOR CONSTRUCTION	5-31-79	HL	FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTROL BUILDING
SAFETY FEATURES ELECTRICAL
EQUIPMENT ROOM HVAC

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN044-2	3

SIZE B 11 x 17

1-12-94

29

P&ID 207-1 REV. 3

JCH 9-14-91

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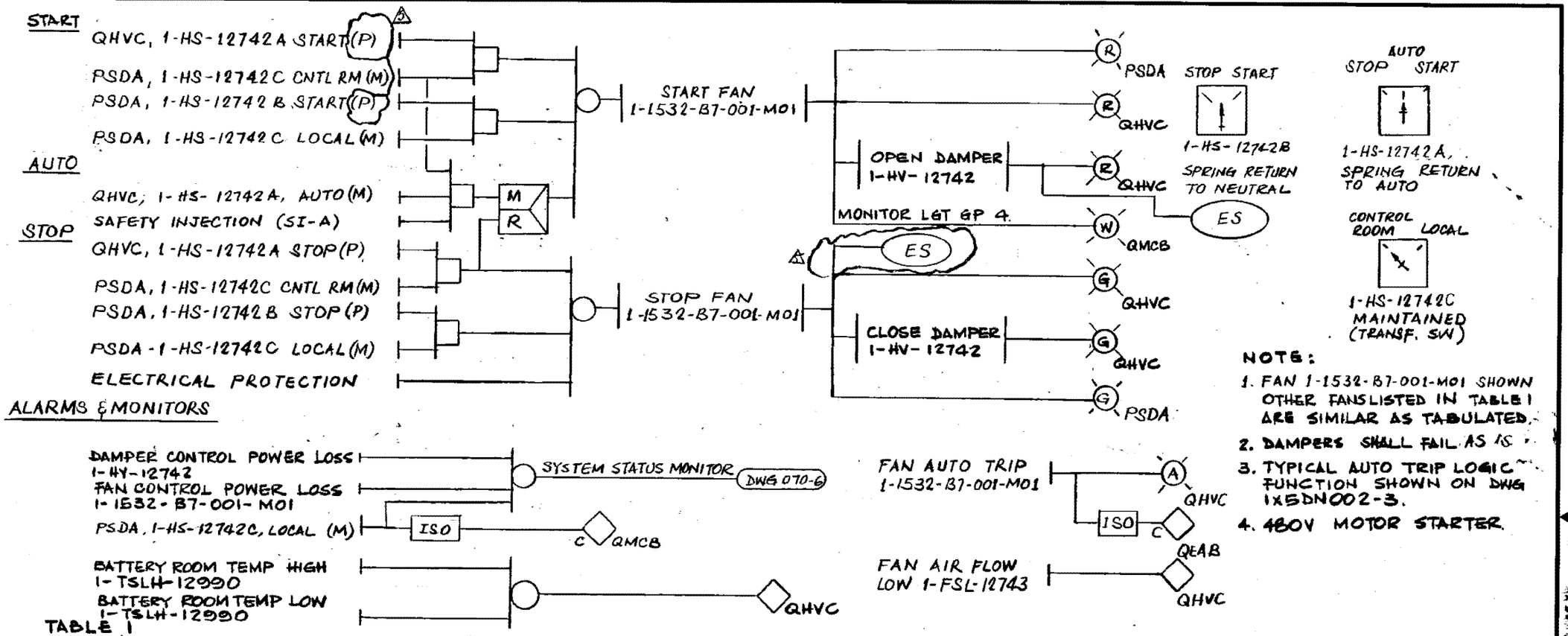

NOTES:

1. FAN 1-1532-B7-001-M01 SHOWN OTHER FANS LISTED IN TABLE I ARE SIMILAR AS TABULATED.
2. DAMPERS SHALL FAIL AS IS.
3. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG 1X5DN002-3.
4. 480V MOTOR STARTER.

TABLE I

TRAIN	EQUIP. TAG NOS.	HAND SWITCH	FLOW SW.	DAMPER	ROOM TEMP	HAND SWITCH		SFTY SIG
A	1-1532-B7-001-M01	1-HS-12742A	1-FSL-12743	1-HV-12742	1-TSLH-12990	1-HS-12742B	1-HS-12742C	SI-A
B	002-	12727A	12728	12727	2	12727B	12727C	SI-B
A	003	12748A	12743	12748	1	12748B	12748C	SI-A
B	004	12749A	12728	12749	3	12749B	12749C	SI-B

REF. DWGS.

P&I.D. 1X4DB 207-1
E.D. 1X3D-BG-C04N, P, Q, R, S, T, U, V

NUCLEAR SAFETY RELATED "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.	ENG. MGR.
3	REVISED REF DWGS & TAG NO	12-8-81	AV	FAL							DATE
2	ADDED COMPUTER INPUT AND CLARIFY SW ACTION	1-7-85	ALV								
1	CORRECTED NOTE 2 PER P&ID, REV 7, ADDED COMP INPUT	6-25-84	ALV								
0	ISSUED FOR CONSTRUCTION	12/15/78	HL	FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CONTROL BUILDING
SAFETY FEATURES BATTERY
ROOM EXHAUST FAN

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN045-1	5

SIZE B 11 x 17

P&ID 207-1 REV 9

JLH 9-14-91

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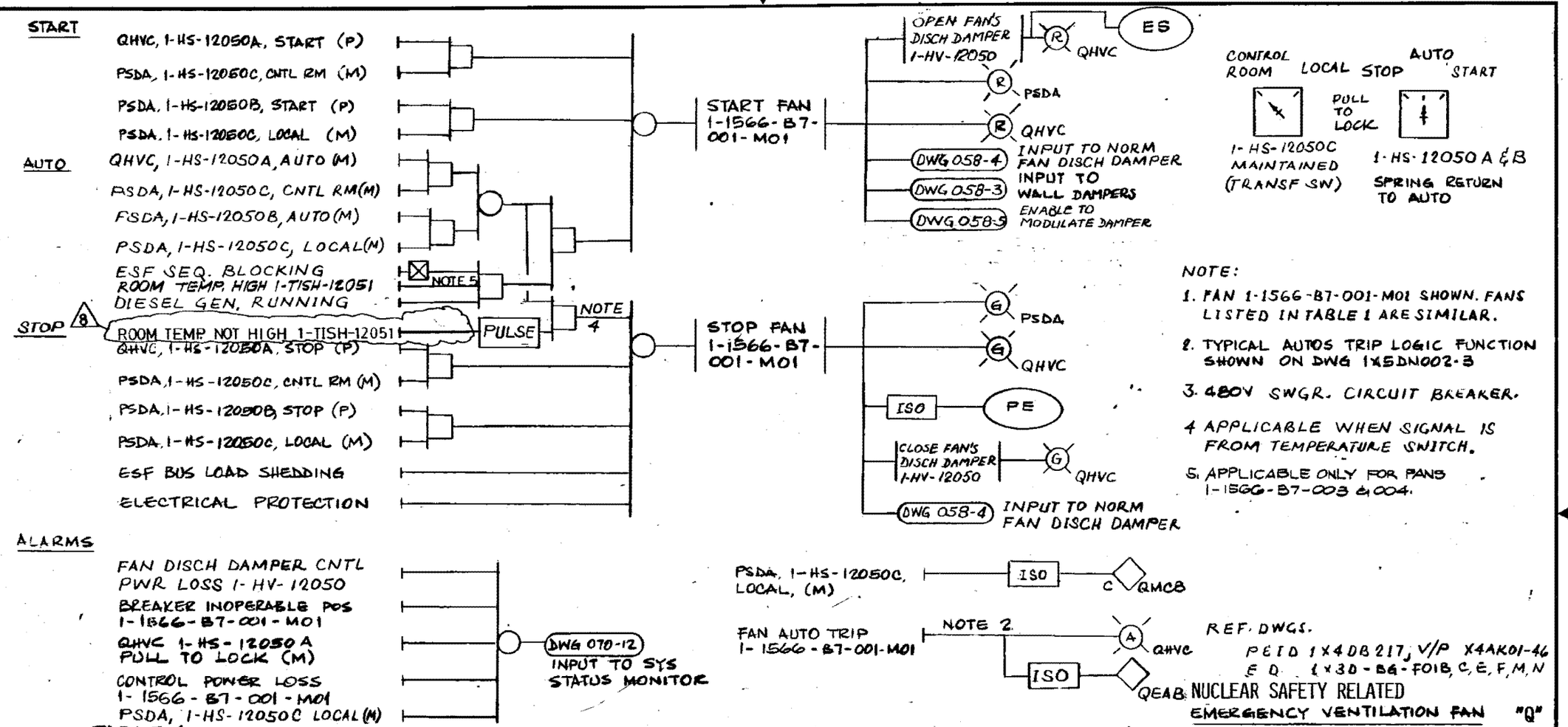

TABLE 1

TRAIN	FAN	HAND SW	HAND SW	TEMP. SW	ESF SEQ		DISCH DAMPER
A	1-1566-B7-001-M01	1-HS-12050A	1-HS-12050B&C	———	TRAIN A		1-HV-12050
↓	003	1A	1	1-TISH-12051	A		51
B	002	3A	3	———	B		53
↓	004	4A	4	1-TISH-12054	B		54

NO	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM		ENG. MGR.
B	INCORP. ABN-01296	12 4 9	W.V.G.	RLW	CLQ	JJ					
A	ISSUED FOR CONSTRUCTION	4-10-19	HL	FAL	MKK						DATE
NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
DIESEL GENERATOR BUILDING
HVAC SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN058-1	8

BM 9510
P&ID REV.

SCANNED DATE: 11/19/91

SIZE B 11 x 17

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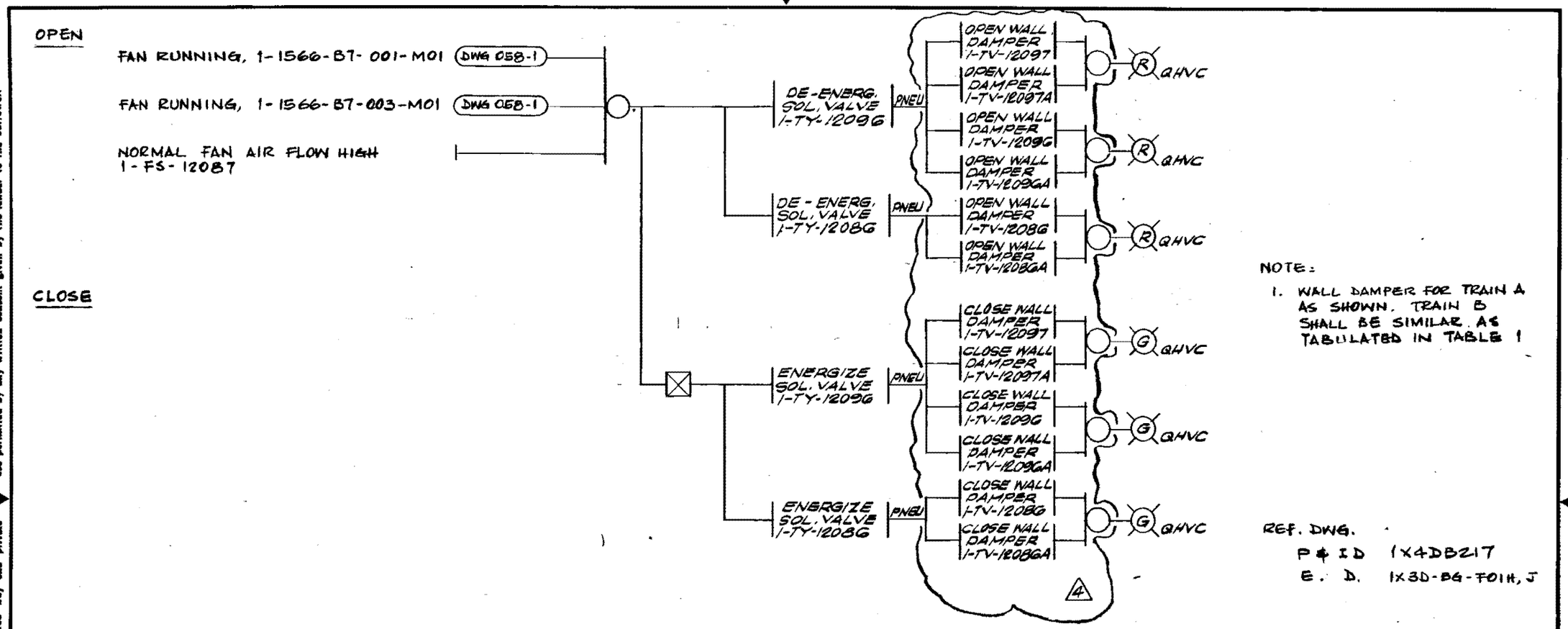

NOTE:

1. WALL DAMPER FOR TRAIN A AS SHOWN. TRAIN B SHALL BE SIMILAR AS TABULATED IN TABLE 1

REF. DWG.
P & I D 1X4DB217
E. D. 1X3D-B6-F01H, J

TABLE 1

SERVICE	TRAIN	SOL VALVE	WALL DAMPER	FAN RUNNING SIGNAL	FLOW SW
WALL DAMPER, 1-TV-12097, 1-TV 12096	A	1-TY-12096	1-TV-12096, 1-TV-12096A, 1-TV-12097, 1-TV-12097A	1-1566-B7-001-M01, 1-1566-B7-003-M01 (4)	1-FS-12087
1-TV-12086		12086	1-TV-12086, 1-TV-12086A		
1-TV-12098, 1-TV-12085	B	12098	1-TV-12098, 1-TV-12098A, 1-TV-12085, 1-TV-12085A	1-1566-B7-002-M01, 1-1566-B7-004-M01	1-FS-12088
1-TV-12099		12099	1-TV-12099, 1-TV-12099A		

4

NUCLEAR SAFETY RELATED

WALL DAMPERS "0"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
3	CHANGED DAMPER TO PNEUMATIC FROM MOTOR OPERATED	7-14-82	AV	FAL		PCK			JJT	
2	REVISED TAG NO	12-9-81	AV	FAL		PCK			JJT	
1	ADDED DOUBLE DMPR. PER DCN #10 ON P&ID 1X4DB217 & AS SHOWN	2-12-85	ZEP	ALV						
0	ISSUED FOR CONSTRUCTION	4-27-79	HL	FAL	MKK				TS	

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
DIESEL GENERATOR BUILDING
HVAC SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN058-3	4

BM 9510 P&ID 217

SCANNED DATE: 11/19/91

SIZE B 11 x 17

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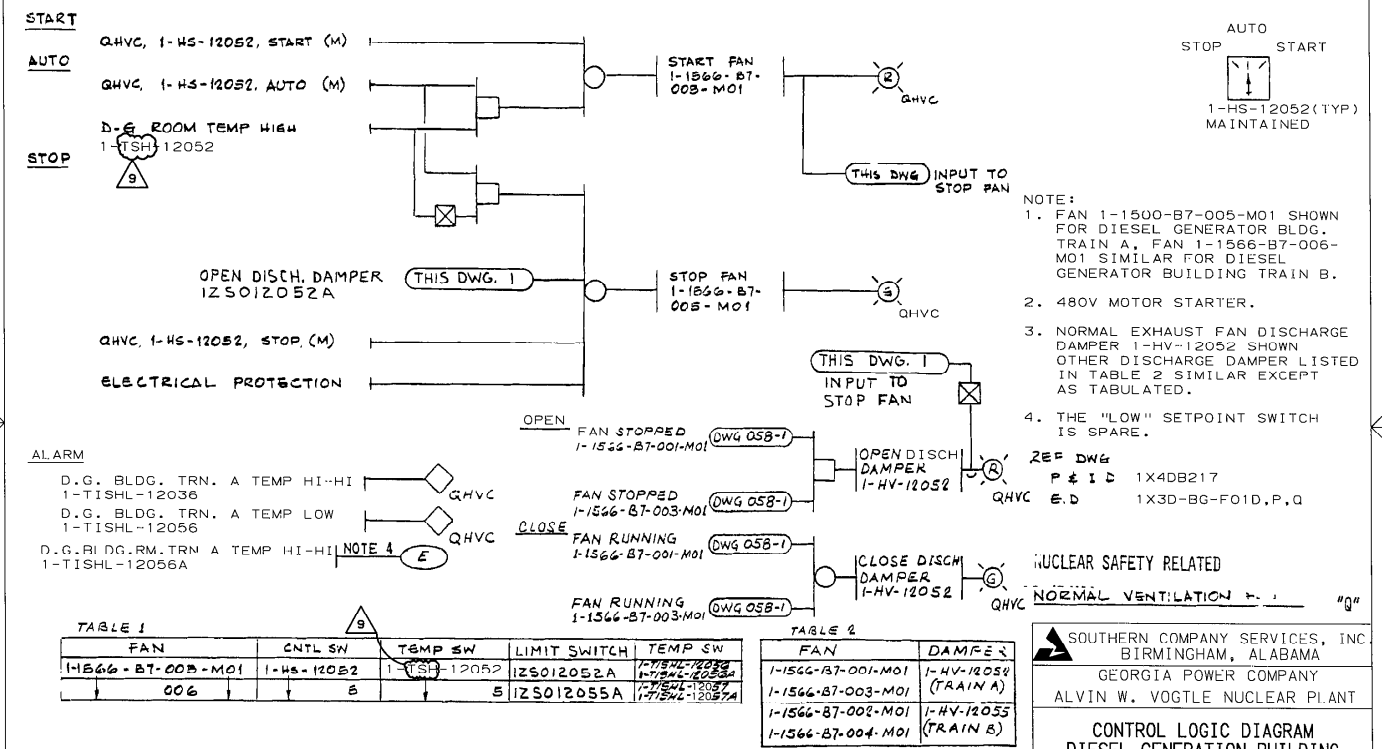


TABLE 1

FAN	CNTL SW	TEMP SW	LIMIT SWITCH	TEMP SW
1-1566-B7-003-M01	1-HS-12052	1-TSH-12052	1ZS012052A	1-TISHL-12036 1-TISHL-12036A
006	5	5	1ZS012055A	1-TISHL-12057 1-TISHL-12057A

TABLE 2

FAN	DAMPER
1-1566-B7-001-M01	1-HV-12054 (TRAIN A)
1-1566-B7-003-M01	
1-1566-B7-002-M01	1-HV-12055 (TRAIN B)
1-1566-B7-004-M01	

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
9	INC. ABN-22216.	9/18/95	CD	LJE	EDT	
0	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES			

ENG MGR

DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

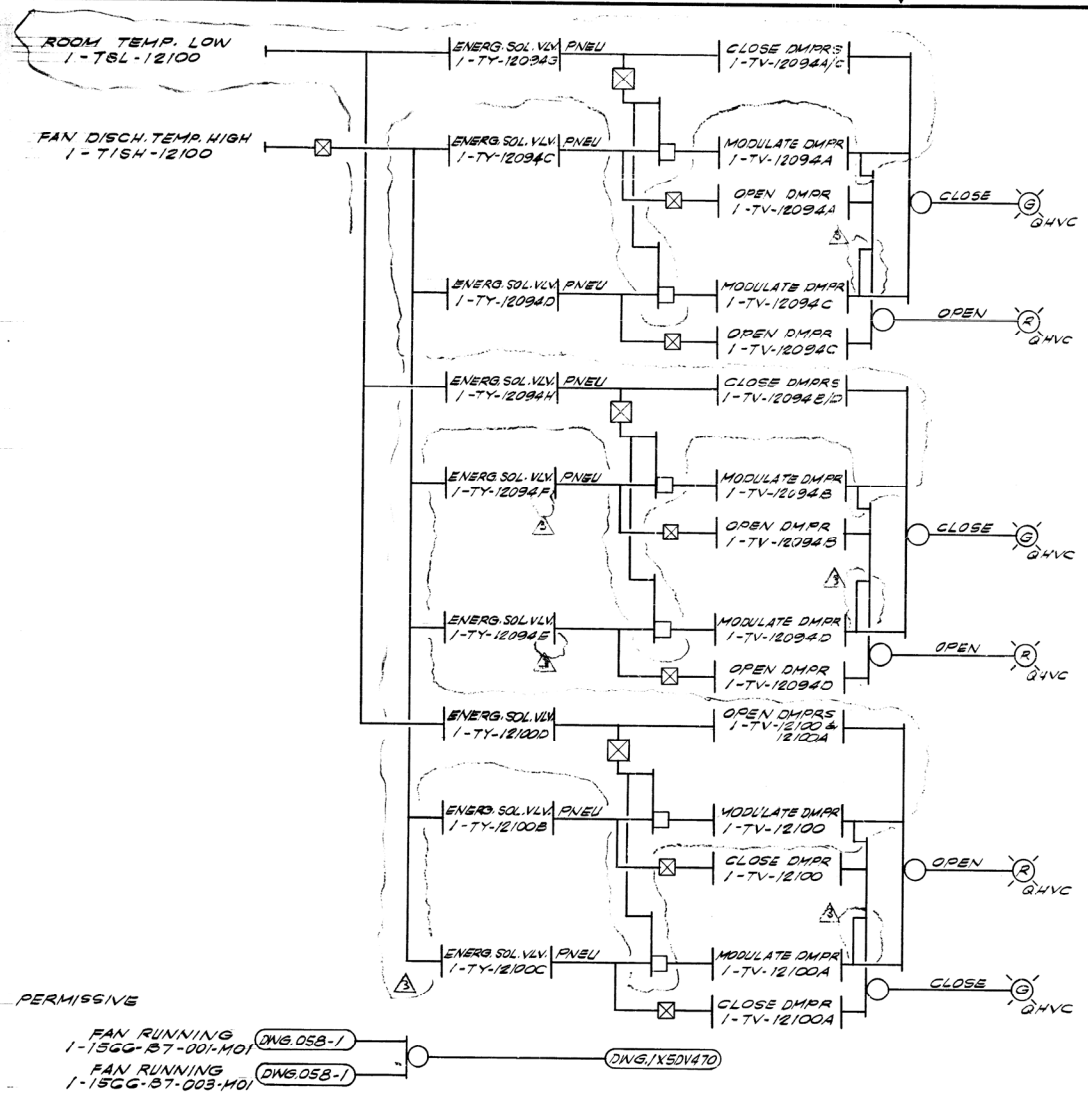
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM DIESEL GENERATION BUILDING HVAC SYSTEM

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN058-4	9

SIZE B 11x17

k:\vog\archive\abn\15dn0584.dgn Sep. 7, 1995 15:30:34 CD



NOTE :

1. DAMPERS 1-TV-12094 & 1-TV-12100, SHOWN. OTHER DAMPERS LISTED IN TABLE 1 SIMILAR.
2. TEMP. SWITCHES ITSL-12100, ITSL-12101 AND SOL. VALVES ITY-12094G, H & ITY-12100D ARE POWERED FROM NON 1-E SOURCE.

3

REF. DWGS. :
P & I.D. 1X4DB217
E.D. 1X30-BG-F01K, L

TABLE 1

TRAIN	FAN	TEMP. CONTROLLER	LOOP DWG	WALL DAMPER	RECIRC. DAMPER	WALL DAMPER TEMP. SOL. VLV.		RECIRC. DAMPER TEMP. SOL. VLV.	TEMP. SW.
A	1-15GG-B7-001-M01 1-15GG-B7-003-M01	1-TIC-12100	1X5DV470	1-TV-12094A, C 1-TV-12094B, D	1-TV-12100 1-TV-12100A	1-TY-12094C -12094F -12094G	1-TY-12094D -12094E -12094H	1-TY-12100B -12100C -12100D	1-TISH-12100 1-TSL-12100
B	1-15GG-B7-002-M01 1-15GG-B7-004-M01	1-TIC-12101	1X5DV471	1-TV-12095A, C 1-TV-12095B, D	1-TV-12101 1-TV-12101A	-12095C -12095F -12095G	-12095D -12095E -12095H	-12101B -12101C -12101D	1-TISH-12101 1-TSL-12101

OUTSIDE AIR AND RECIRCULATION DAMPERS

NUCLEAR SAFETY RELATED "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.
3	REDRAWN PER CCP B102ZGM, DCN #1	8-26-86	JEP	ALU			SGG			
2	ISSUED FOR CONSTRUCTION	7-14-86	(SEE MICROFILM FOR SIGNATURES)							

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
DIESEL GENERATOR BUILDING
HVAC SYSTEM

SCALE: NONE | DRAWING NO. 1X5DN058-5 | REV. 3

JOB NO. 9510

SIZE C 17X22

2/78 DIETERICH-POST 2112B .003 FILM FORM CM 9510

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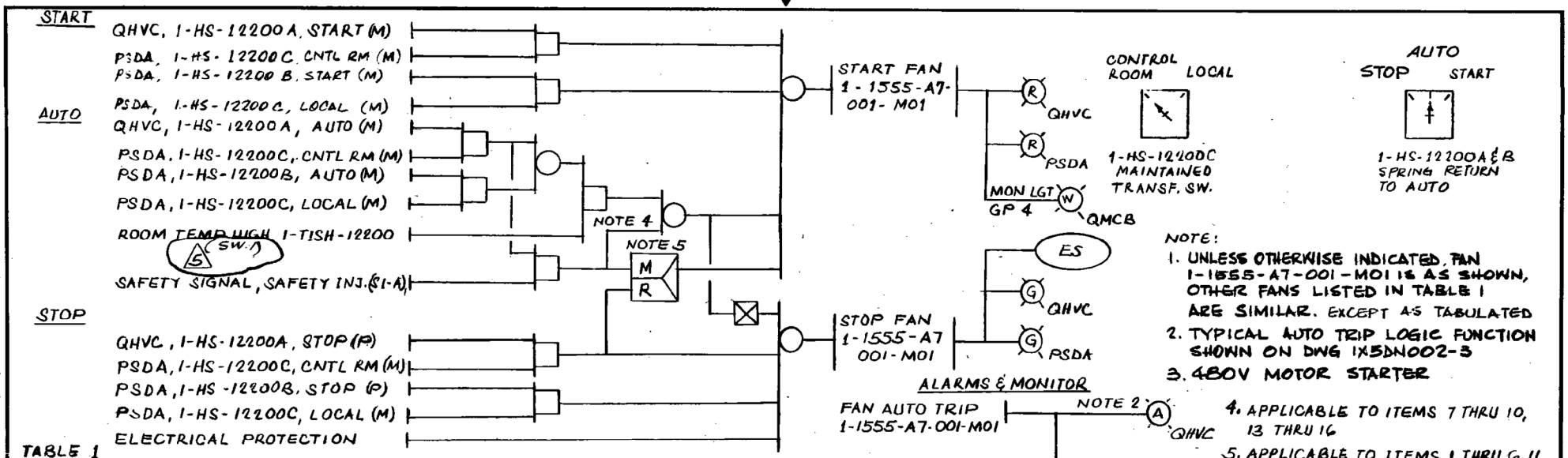

NOTE:

1. UNLESS OTHERWISE INDICATED, FAN 1-1555-A7-001-M01 IS AS SHOWN, OTHER FANS LISTED IN TABLE 1 ARE SIMILAR. EXCEPT AS TABULATED
2. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG 1X5DN002-3
3. 480V MOTOR STARTER
4. APPLICABLE TO ITEMS 7 THRU 10, 13 THRU 16
5. APPLICABLE TO ITEMS 1 THRU 6, 11, 12, 17 & 18.
6. SET POINT FOR ALARM AND COMP. INPUT IS HIGHER THAN THE SET POINT FOR FAN ACTUATION. (5)

REF DWGS

P & I D 1X4DB228

E. D 1X3D-BG-D05A, B, C, D, E, F, G, H, J, K, N, P, Q, R, S, T, U, V

TABLE 1

ITEM	LOCATION	TEMP SW	TRAIN	FAN 1-1555-A7	HAND SWITCHES	SAFETY SIGNAL
1	ELECT SWGR & MCC RM	1-TISH-12200	A	001-M01	1-HS-12200A, B&C	SI-A
2		-12201	B	002-	-12201A, B&C	B
3		-12202	A	003-	-12202A, B&C	A
4		-12203	B	004-	-12203A, B&C	B
5		-12204	A	005-	-12204A, B&C	A
6		-12205	B	006-	-12205A, B&C	B
7	RHR PUMP ROOM	1-TSH-12206	A	007-	-12206A, B&C	1-1205-P6-001
8		-12212	B	008-	-12212A, B&C	-002
9	CNMT SPRAY PUMP ROOM	1-TISH-12207	A	009-	-12207A	1-1206-P6-001
10		-12213	B	010-	-12213A	-002
11	CCW PUMP ROOM	-12208	A	011-	-12208A, B&C	SI-A
12		-12214	B	012-	-12214A, B&C	B
13	CHARG PUMP ROOM	1-TSH-12209	A	013-	-12209A, B&C	1-1208-P6-002
14		1-TISH-12215	B	014-	-12215A, B&C	-003
15	SI PUMP ROOM	-12210	A	015-	-12210A	1-1204-P6-003
16		-12216	B	016-	-12216A	-004
17	SFP HE & PUMP ROOM	1-TSH-12211	A	017-	-12211A	SI-A
18		1-TISH-12217	B	018-	-12217A	B

NUCLEAR SAFETY RELATED

COOLER FANS

"Q"

NO.	REVISIONS	DATE	DR	CHK.		ENG. MGR.	EGS	CHF. E.	P&E.	Q.A.E.
3	ADDED COMPUTER INPUT PER 110 LIST REV. 4	10-16-84	ALV	JJ						
5	REVISED AS SHOWN & ADDED NOTE 6	2-22-85	ZEP	ALV						
4	CORRECTED TAG NOS PER DCN 9	1-17-85	ALV	W.H.						
0	ISSUED FOR CONSTRUCTION	1-12-79	HL	FAL	MKK		EGL			

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
SAFETY FEATURE ROOM COOLERS

SCALE: NONE

DRAWING NO. 1X5DN065-1

REV. 3

JOB NO. 9510

SIZE B 11 x 17

BM 9510

P&ID REV. 6

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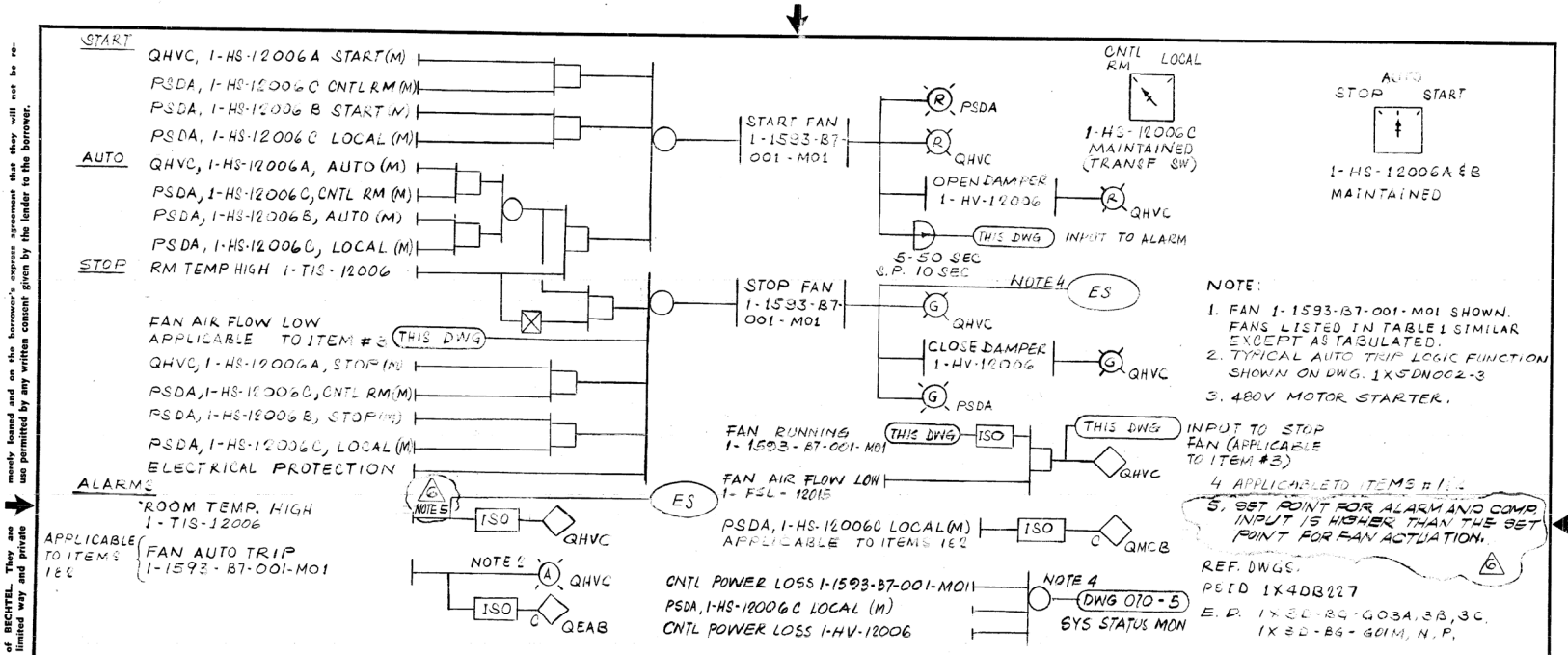

NOTE:
1. FAN 1-1593-B7-001-M01 SHOWN. FANS LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATED.
2. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG. 1X5DN062-3
3. 480V MOTOR STARTER.
4 APPLICABLE TO ITEMS # 1&2
5. SET POINT FOR ALARM AND COMP. INPUT IS HIGHER THAN THE SET POINT FOR FAN ACTUATION.

REF. DWGS:
P&ID 1X4DB227
E.D. 1X5D-BG-G03A,3B,3C.
1X5D-BG-G01M, N, P.

TABLE 1

ITEM	TRAIN	FAN NO.	HAND SW.	TEMP SW.	DAMPER	FLOW SW	HAND SW
1	A	1-1593-B7-001-M01	1-HS-12006A	1-TIS-12006	1-HV-12006	1-FSL-12015	1-HS-12006 B&C
2	B	002-M01	12005A	12005	12005	12014	1-HS-12005 B&C
3	—	003-M01	12004	12004	12004	12016	—

NUCLEAR SAFETY RELATED
VENTILATION FANS "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P.E.	Q.A.E.	ENG. MGR.
E	ADDED NOTE 5	2-8-85	ZEN	ALV							DATE
D	REVISED AS SHOWN	10-16-84	ALV								
C	ADDED COMPUTER INPUTS PER I/O LIST REV. 4	9-10-84	ALV								
B	ISSUED FOR CONSTRUCTION	5-3-79	NBD	FAL	MKK						

BM 9510

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER PUMPHOUSE
HEATING AND VENTILATION

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1XBDN068-1	6

SIZE B 11 x 17

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2-8-85 DATE — CAMERA OPERATOR — SECTION SUPERVISOR

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DATE: 2-10-84 CAMERA OPERATOR SECTION SUPERVISOR

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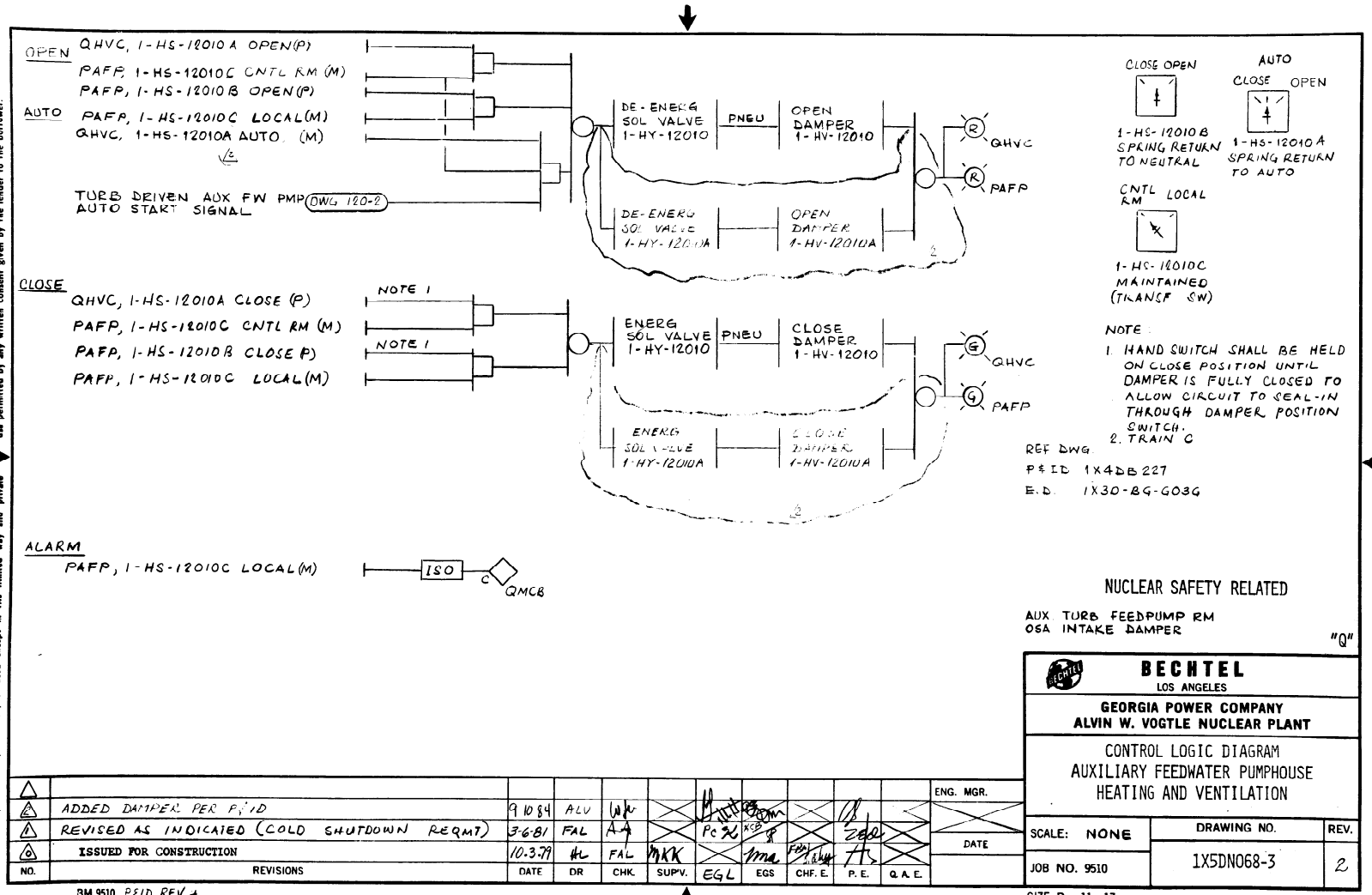

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	ENG. MGR.
3											
2	ADDED DAMPER PER P&ID	9 10 84	ALV								
1	REVISED AS INDICATED (COLD SHUTDOWN REQMT)	3-6-81	FAL			PC 21	KCB 8				DATE
0	ISSUED FOR CONSTRUCTION	10.3.79	HL	FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER PUMPHOUSE
HEATING AND VENTILATION

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN068-3	2

BM 9510 P&ID REV 4

SIZE B 11 x 17

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS, ON THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT SUBMITTED TO MICROGRAPHICS.
DATE 9-10-84 CAMERA OPERATOR SECTION SUPERVISOR

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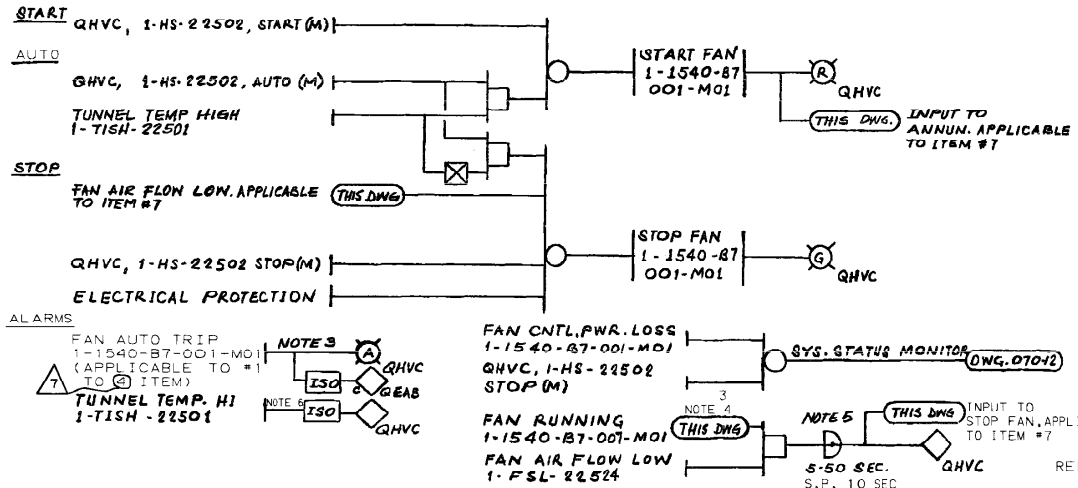

NOTE

1. FAN 1-1540-B7-001-M01 SHOWN OTHER FANS LISTED IN TABLE 1 ARE SIMILAR AS TABULATED.
2. 480V MOTOR STARTER
3. TYPICAL LOGIC FUNCTION SHOWN ON DRAWING 1X5DN002-3.
4. FOR ITEM 6 SEE FAN RUNNING INPUT ON DWG. 1X5DN863-4
5. FOR ITEM 6 TIME DELAY RELAY IS .1-10 MIN, S.P. 4 MIN
6. SET POINT FOR ALARM, IS HIGHER THAN THE SET POINT FOR FAN ACTUATION.

TABLE 1

ITEM	FUNCTION	TRAIN		FAN	CNTL. SW.	TEMP. SW.	FLOW SW
1	DIESEL GEN. PWR. CABLE TUN.	A	—	1-1540-B7-001-M01	1-HS-22502	1-TISH-22501	
2	DIESEL GEN. PWR. CABLE TUN.	B	—	-002	-22506	-22505	
3	NSCW TOWER PWR CABLE TUN.	A	—	-003	-22517	-22516	
4	NSCW TOWER PWR. CABLE TUN.	B	—	-004	-22520	-22519	
5							
6	TUNEL VENT FAN	—	—	1 1540 B7 006 M01	—	1-TISH -22513	1-FSL-22512
7	TURB. BLDG. CHASE TO CNTL. BLDG	—	—	1-1540-B7-007-M01	1-HS-22523	-22522	1-FSL-22524

REF. DWG.

P & ID 1X4DB238
E.D. 1X3D-BG-K01C, 1D, 1G

NUCLEAR SAFETY RELATED "Q"

ELECTRICAL TUNNEL VENT FAN

NO.	REVISIONS	DATE	DR	CHK	ELECT	EGL	EGS	CHF E	PE	QAE
7	INCORP. ABN-12461.	6-22-79	GTD	MWD	LDM	WPP				
6	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM	
5	ISSUED FOR CONSTRUCTION	3/7/79		SEE MICROFILM FOR SIGNATURES						

ENG MGR

DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM TUNNEL VENTILATION SYSTEM

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN069-1	7

SIZE B 11x17

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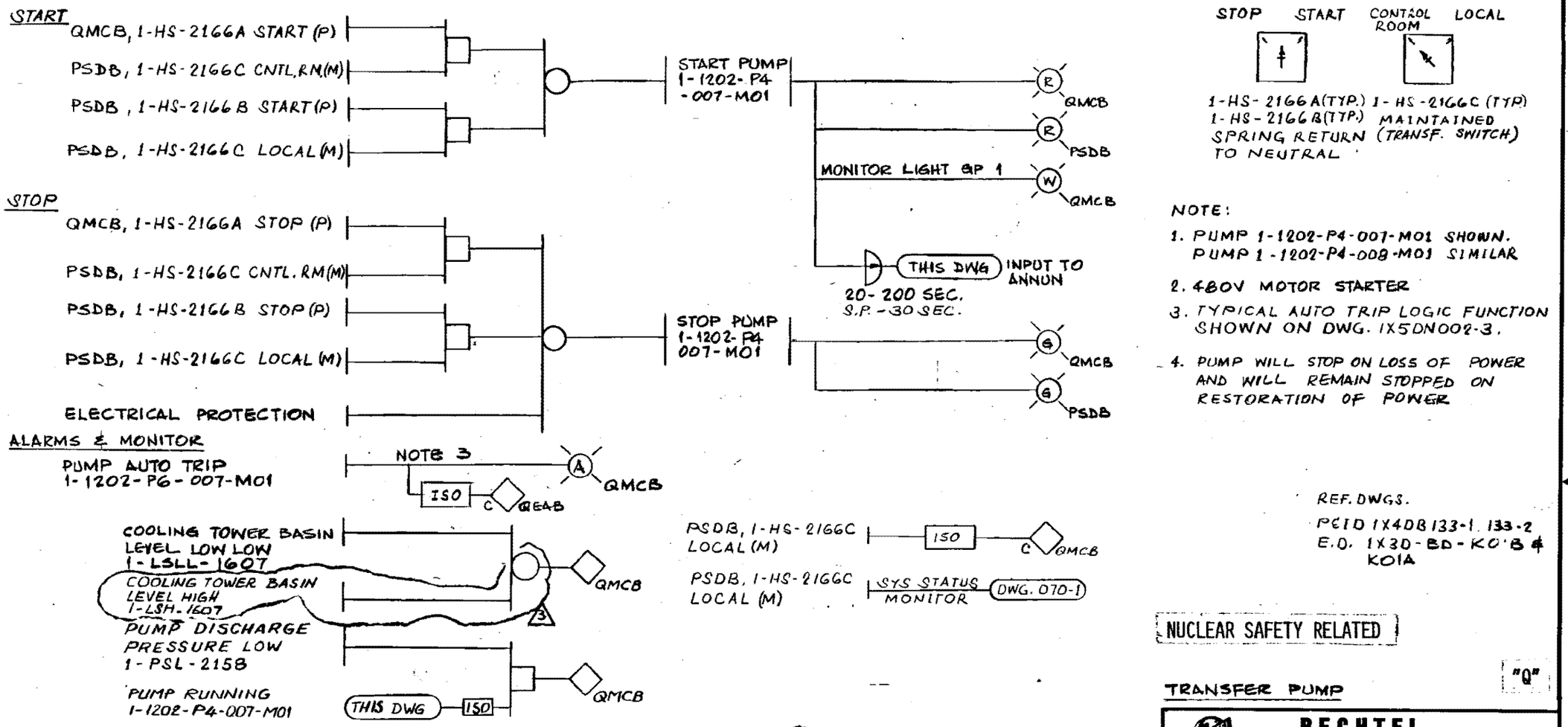


NOTE:

1. PUMP 1-1202-P4-007-M01 SHOWN. PUMP 1-1202-P4-008-M01 SIMILAR
2. 480V MOTOR STARTER
3. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG. 1X5DN002-3.
4. PUMP WILL STOP ON LOSS OF POWER AND WILL REMAIN STOPPED ON RESTORATION OF POWER

REF. DWGS.
P&ID 1X4DB133-1, 133-2
E.D. 1X3D-BD-KO1B & KO1A

NUCLEAR SAFETY RELATED

"Q"

TRANSFER PUMP

TRAIN	PUMP	SWITCHES			PRESS SW	LVL SW	LVL SW
B	1-1202-P4-007-M01	1-HS-2166A	1-HS-2166B	1-HS-2166C	1-PSL-2158	1-LSLL-1607	1-LSH-1607
A	008	2167A	2167B	2167C	2159	1606	-1606

3

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
5	ADDED HIGH ALARM TO REFLECT THE ACTUAL DESIGN	3/21/85	ZEP	ALV						
4	ADDED TIME DELAY SET POINT & REVISED AS INDICATED	3-6-81	FAL	AAT		PCL				
1	REVISED AS INDICATED	2-15-79	HL	FAL						
0	ISSUED FOR CONSTRUCTION	2-15-79	HL	FAL	MKK					

ENG. MGR.

DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE COOLING TOWER
TRANSFER PUMPS

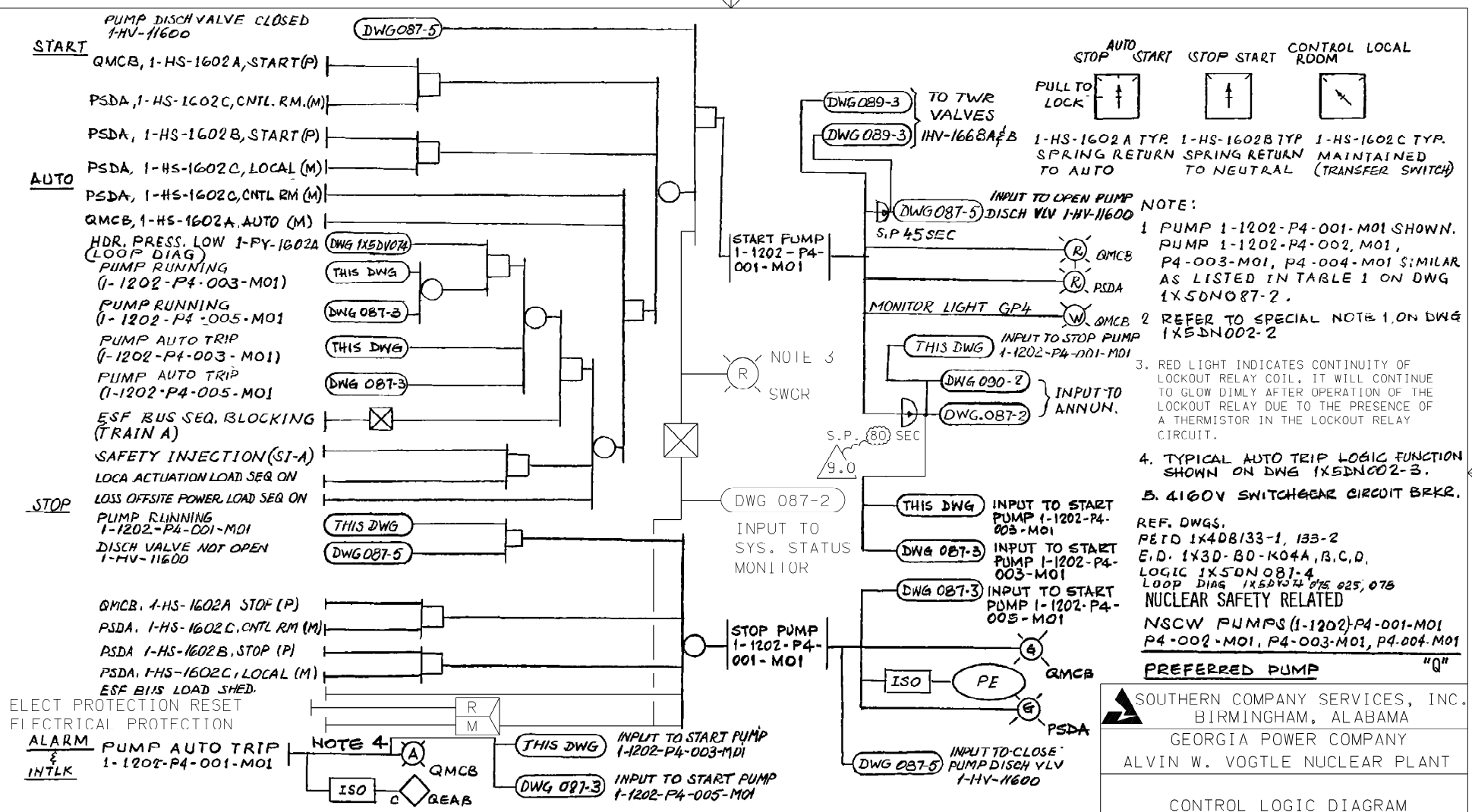
SCALE: NONE

DRAWING NO. 1X5DN086-1

REV. 3

JOB NO. 9510

SIZE B 11x17



NOTE:

1. PUMP 1-1202-P4-001-M01 SHOWN. PUMP 1-1202-P4-002, M01, P4-003-M01, P4-004-M01 SIMILAR AS LISTED IN TABLE 1 ON DWG 1X5DN087-2.
2. REFER TO SPECIAL NOTE 1, ON DWG 1X5DN002-2
3. RED LIGHT INDICATES CONTINUITY OF LOCKOUT RELAY COIL. IT WILL CONTINUE TO GLOW DIMLY AFTER OPERATION OF THE LOCKOUT RELAY DUE TO THE PRESENCE OF A THERMISTOR IN THE LOCKOUT RELAY CIRCUIT.
4. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG 1X5DN002-3.
5. 4160V SWITCHGEAR CIRCUIT BRKR.

REF. DWGS.
P&ID 1X4DB133-1, 133-2
E.I.D. 1X3D-B0-K04A, B, C, D,
LOGIC 1X5DN087-4
LOOP DIAG 1X5DV074 075, 025, 078

NUCLEAR SAFETY RELATED

NSCW PUMPS (1-1202-P4-001-M01 P4-002-M01, P4-003-M01, P4-004-M01

PREFERRED PUMP "Q"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING WATER PUMPS

NO.	VERSIONS	DATE	DR	CHK	APPV
9.0	REVISED PER ABN-V00180, VER. 1.0	1/29/07	JLO	ELC	JMR

SCALE: NONE | DRAWING NO. 1X5DN087-1 | VER. 9.0

JOB NO. 10604

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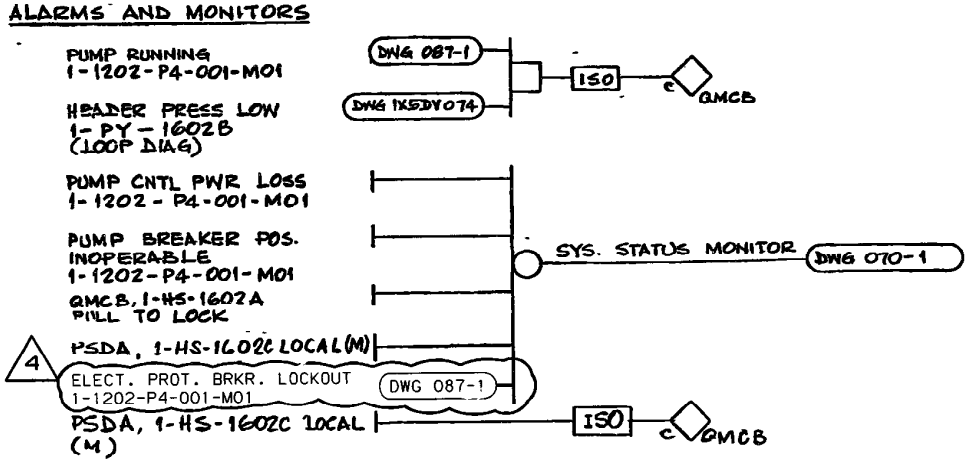
ALARMS AND MONITORS



NOTE:

1. TRAIN A ALARMS & MONITORS SHOWN, OTHER ALARMS & MONITORS, SIMILAR AS LISTED IN TABLE 1 & 2
2. OTHER LOGIC FUNCTION SHOWN ON DWG 1X5DN087-1, 1X5DN087-3.

TABLE 1

LOOP DWG 1X5DV-	TRAIN	PUMP NO	SAF. SIG.	SWITCHES			PRESS SW LOOP DIAG	PUMP RUNNING, ELEC OLVD	
-074	A	1-1202-P4-001-M01	SI-A	1-HS-1602A	1-HS-1602B	1-HS-1602C	1-PY-1602A&B	1-1202-P4-003-M01	1-1202-P4-005-M01
-075	B	-002	B	03A	03B	03C	03	004	006
-025	A	003	A	34A	34B	34C	56	001	005
-078	B	004	B	35A	35B	35C	57	002	006

TABLE 2

LOOP DWG 1X5DV	TRAIN	PUMP NO.	SAF. SIG	SWITCHES			PRESS SW LOOP DIAG	PUMP RUNNING, ELEC OLVD	
-076	A	1-1202-P4-005-M01	SI-A	1-HS-1608A	1-HS-1608B	1-HS-1608C	1-PY-1608A&B	1-1202-P4-001-M01	1-1202-P4-003-M01
-077	B	006	B	09A	09B	09C	09	002	004

REF DWGS:
PEID 1X4DB133-1 , 133-2
E.D 1X30-BD-K04A,B,C,D,E&F
LOOP DIAGRAM 1X5DV074,075,025,078,076,077

NUCLEAR SAFETY RELATED "Q"

ALARMS, MONITORS & EQUIP. LIST.

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. ABN-21049	11/29/95	MWD	WLM	EOL	
3	REVISIONS	DATE	DR			
0	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES			

ENG MGR

DATE

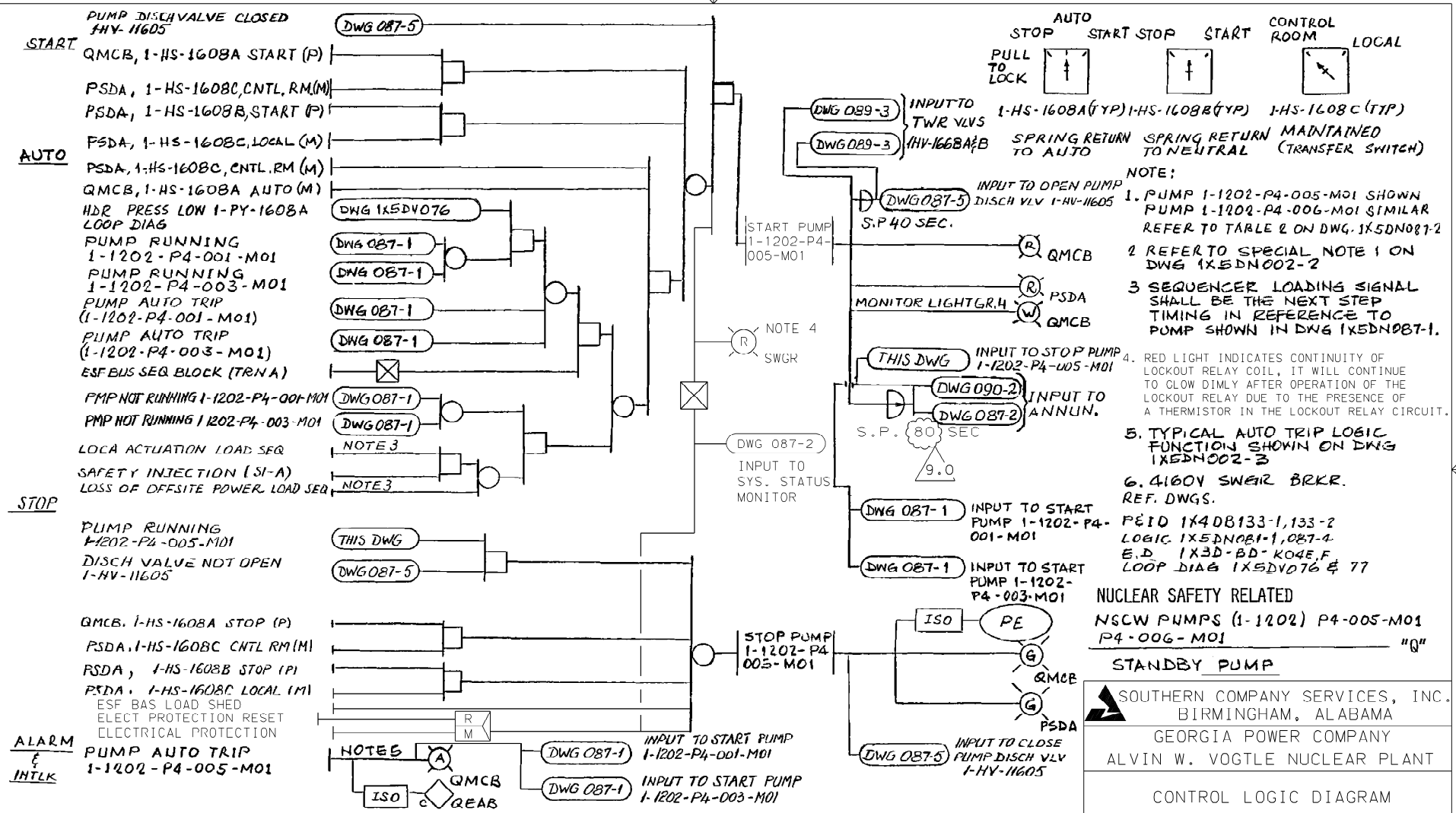
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM NUCLEAR SERVICE COOLING WATER PUMPS

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN087-2	4

SIZE D 11-13



NO.	VERSIONS	DATE	DR	CHK	APPV
9.0	REVISED PER ABN-V00480, VER. 1.0	1/29/07	JLO	ELC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING WATER PUMPS

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X5DN087-3	9.0

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SIZE B 11x17

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7-25-84 D. COLLINS CAMERA OPERATOR
DATE: SECTION SUPERVISOR

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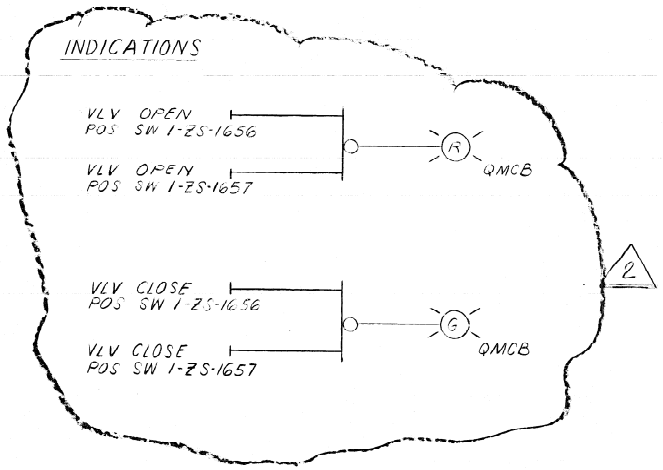

NOTE:
1. POSITION INDICATION SHOWN FOR MANUALLY OPERATED VALVE. OTHER POSITION INDICATION LISTED IN TABLE 1 SIMILAR.

REF. DWGS:
P&ID 1X4DB 133-1, 133-2
E.D. 1X5D-BD-K01E, K01F

TABLE 1

FUNCTION	VLV POS SW
COOLING TOWER BASIN MAKE UP TRAIN A	1-ZS-1656
	7
COOLING TOWER BASIN MAKE UP TRAIN B	8
	9

MANUAL MAKE-UP VALVE MONITOR

NO.	REVISIONS	DATE	DR	CHK.	SUP'V.	EGL	EGS	CHF. E.	P.E.	Q.A.E.
3										
2	CORRECTED THE LIGHT POS INDICATION	7-24-84	ETL	7/24						
1	ADDED REF DWGS.	5-14-80	HL	FAL						
0	ISSUED FOR CONSTRUCTION	2-10-78	HL	FAL	MKK					

ENG. MGR. DATE

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GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING WATER PUMPS

SCALE: NONE | DRAWING NO. 1X5DN087-4 | REV. 2
JOB NO. 9510

SIZE B 11 x 17

DM 9510

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7-25-84 D. COLLINS CAMERA OPERATOR
DATE: SECTION SUPERVISOR

OPEN

NSCW PUMP RUNNING
1-1202-P4-001-M01 — DWG 087-1 — OPEN VALVE 1-HV-11600

- DWG 90-6 — INPUT TO CLOSE BLOWDOWN VLV
- (R) — QMCB (NOTE 4)

STOP

ELECTRICAL OVERLOAD (ONLY DURING PERIODIC TEST) — NOTE 3 — STOP VALVE 1-HV-11600

3

CLOSE

NSCW PUMP STOPPED
1-1202-P4-001-M01 — DWG 087-1 — CLOSE VALVE 1-HV-11600

- (G) — QMCB (NOTE4)
- DWG087-1 — PERMISSIVE INPUT TO NSCW TO START
- DWG087-1 — INPUT TO STOP NSCW PUMP

MONITOR & ALARM

VALVE 1-HV-11600
CONTROL POWER LOSS — SYSTEMS STATUS MONITOR — DWG 070-2

NOTE:

1. VALVE 1-HV-11600 SHOWN, OTHER VALVES LISTED IN TABLE 1 SIMILAR.
2. VALVES LISTED IN TABLE 1 ARE BUTTERFLY VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
3. SPECIAL NOTE 2 ON DWG 1X5DN002-2,
4. INPUT FROM STEM MOUNTED POSITION SWITCH

REF. DWGS:
P&ID 1X4DB133-1, 133 2
E.D 1X3D-BD-K04Y,,-K04Z

TABLE 1

TRAIN	VALVE	PUMP	DWG
A	1-HV-11600	1-1202-P4-001-M01	087-1
B	11607	2	
A	11606	3	
B	11613	4	
A	11605	5	087-3
B	11612	6	

VALVE CNTL POWER LOSS 1-HV-11600

VALVE MOTOR ELECT OVERLOAD 1-HV-11600

ISO — C — QEAB

NUCLEAR SAFETY RELATED

PUMP DISCHARGE VALVES "Q"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM NUCLEAR SERVICE COOLING WATER PUMPS

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 95-V1N0035	12-5-97	CD	GLB	WFP	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE

JOB NO. 10604

DRAWING NO. 1X5DN087-5

REV 3

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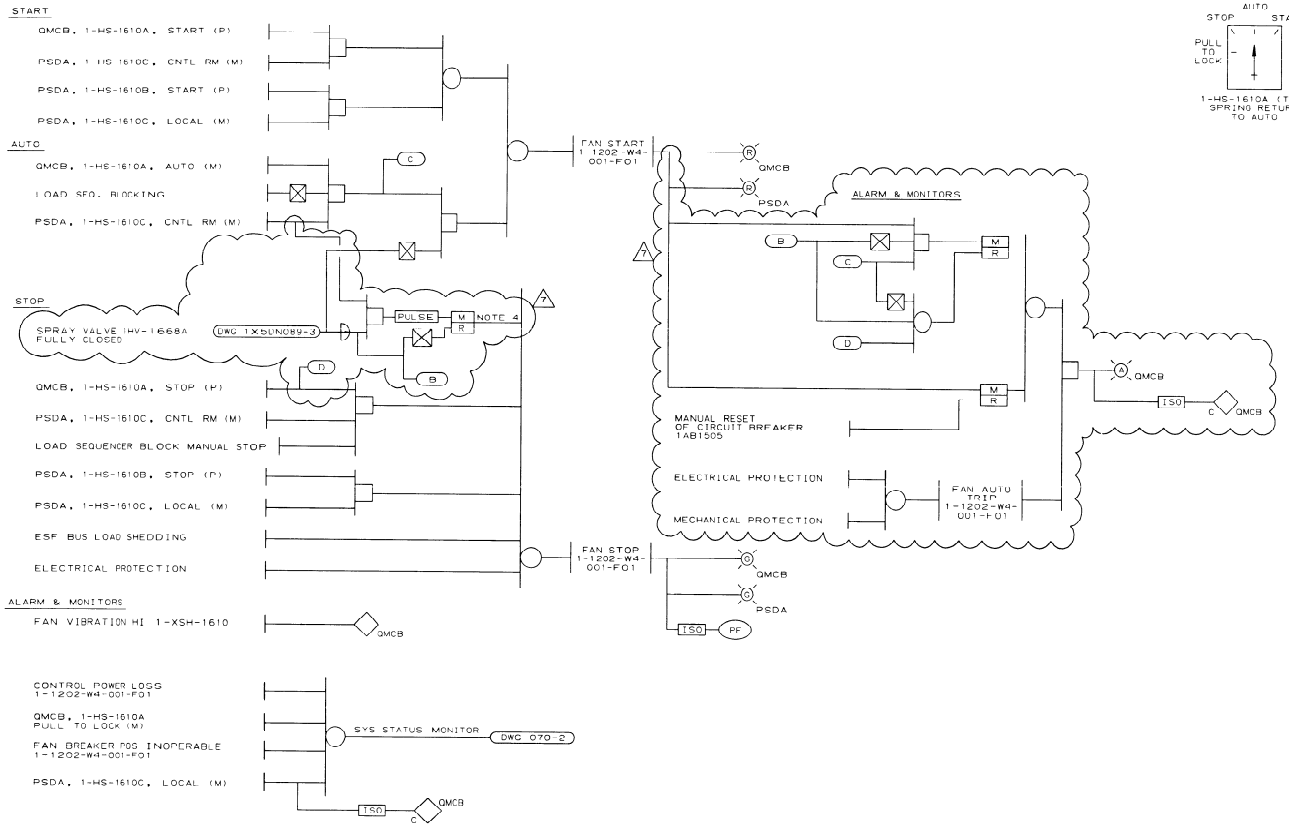
CAD NAME 15DN0875

SIZE B 11x17



NOTES:

1. FAN 1-1202-W4-001-F01 SHOWN, FAN LISTED IN TABLE 1 SIMILAR.
2. 480V SWGR CIRCUIT BREAKER.
3. REFER TO SPECIAL NOTE 1 ON DWG 1X5DN002-2.
4. ONE PULSE SIGNAL IS GENERATED TO AUTO STOP THE FAN. FOR ANOTHER PULSE SIGNAL TO BE GENERATED TO AUTO STOP THE FAN THE SPRAY VALVE 1HV-1668A MUST BE IN THE NOT CLOSED POSITION.

REFERENCES:

P & I D 1X4DB133-1, 133-2

E.D. 1X3D-BD-K03A, 1X3D-BD-K03E, 1X3D-BD-K05U, 1X3D-BD-K05W

LOCATION 1X4DE328, 329

TABLE 1

TRAIN	CLG TOWER FANS	SWITCHES			VIBRATION SWITCHES	CIRCUIT BREAKER	SPRAY VALVE
A	1-1202-W4-001-F01	1-HS-1610A	1-HS-1610B	1-HS-1610C	1-XSH-1610	1AB1505	1HV-1668A
B	1-1202-W4-002-F01	1-HS-1611A	1-HS-1611B	1-HS-1611C	1-XSH-1611	1BB1505	1HV-1669A

NUCLEAR SAFETY RELATED
NSCW TOWER FANS

Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING TOWER SYSTEM

NO.	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM
2	INCORP. ARN-11637.	11-21-91	GWM	JDM	MTP				
1	REVISIONS	7/8/91	SFF	MICROFILM	FOR	SIGNATURES			
0	ISSUED FOR CONSTRUCTION	7/8/91	SFF	MICROFILM	FOR	SIGNATURES			

NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHFE	PE	QAE

ENG. MGR.
DATE
SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X5DN089-1
REV. 7
SIZE D 22X34

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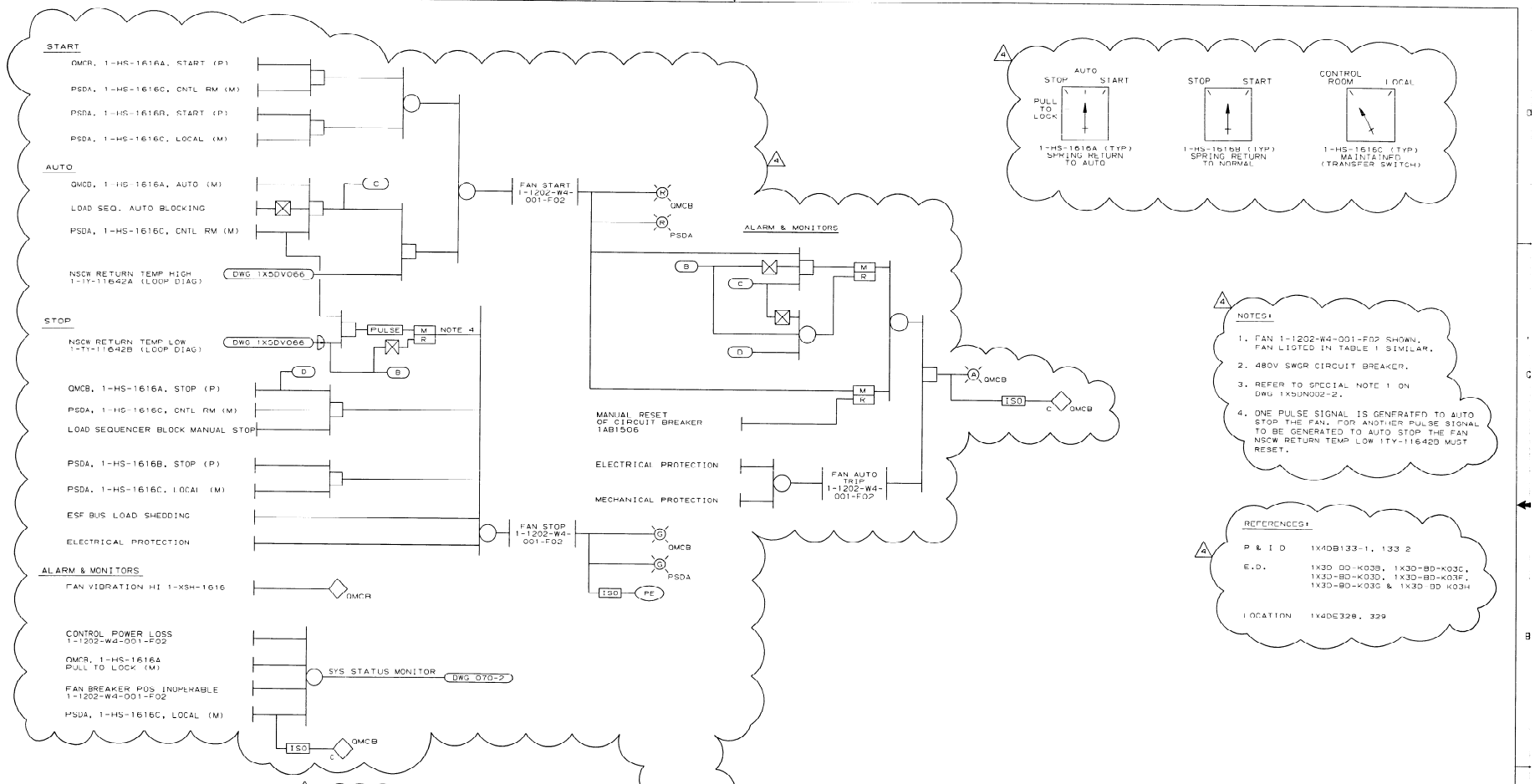

NOTES:

1. FAN 1-1202-W4-001-F02 SHOWN, FAN LISTED IN TABLE 1 SIMILAR.
2. 480V SWGR CIRCUIT BREAKER.
3. REFER TO SPECIAL NOTE 1 ON DWG 1X5DN002-2.
4. ONE PULSE SIGNAL IS GENERATED TO AUTO STOP THE FAN. FOR ANOTHER PULSE SIGNAL TO BE GENERATED TO AUTO STOP THE FAN NSCW RETURN TEMP LOW (TY-11642B MUST RESET.

REFERENCES:

P & I D	1X4DB133-1, 133 2
E.D.	1X3D-BD-K03B, 1X3D-BD-K03C, 1X3D-BD-K03D, 1X3D-BD-K03F, 1X3D-BD-K03G & 1X3D-BD-K03H
LOCATION	1X4DE328, 329

TABLE 1

TRAIN	CLG TOWER FANS	SWITCHES			VIBRATION SWITCHES	TEMP SW	LOOP DIAGRAM	CIRCUIT BREAKER
A	1-1202-W4-001-F02	1-HS-1616A	1-HS-1616B	1-HS-1616C	1-XSH-1616	1-TY-11642A/B	1X5DV066	1AB1506
A	1-1202-W4-001-F03	1-HS-1622A	1-HS-1622B	1-HS-1622C	1-XSH-1622	1-TY-11643A/B	1X5DV067	1AB1513
A	1-1202-W4-001-F04	1-HS-1628A	1-HS-1628B	1-HS-1628C	1-XSH-1628	1-TY-11644A/B	1X5DV068	1AB1506
B	1-1202-W4-002-F02	1-HS-1617A	1-HS-1617B	1-HS-1617C	1-XSH-1617	1-TY-11647A/B	1X5DV070	1BB1606
B	1-1202-W4-002-F03	1-HS-1623A	1-HS-1623B	1-HS-1623C	1-XSH-1623	1-TY-11648A/B	1X5DV071	1BB1613
B	1-1202-W4-002-F04	1-HS-1629A	1-HS-1629B	1-HS-1629C	1-XSH-1629	1-TY-11649A/B	1X5DV072	1BB1608

NUCLEAR SAFETY RELATED
NSCW TOWER FANS

Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING TOWER SYSTEM

NO.	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM	ENG. MGR.
4	INCORP. ADN-11636.	10-22-96	GLO							
3	REVISIONS	2/15/95	SEE MICROFILM FOR SIGNATURES							
2	ISSUED FOR CONSTRUCTION	DATE	DR	CHK	SUPV	CGL	EGS	CHFE	PE	QAE

SCALE: NONE | DRAWING NO. 1X5DN089-2 | REV. 4
JOB NO. 10604
SIZE D 22X34

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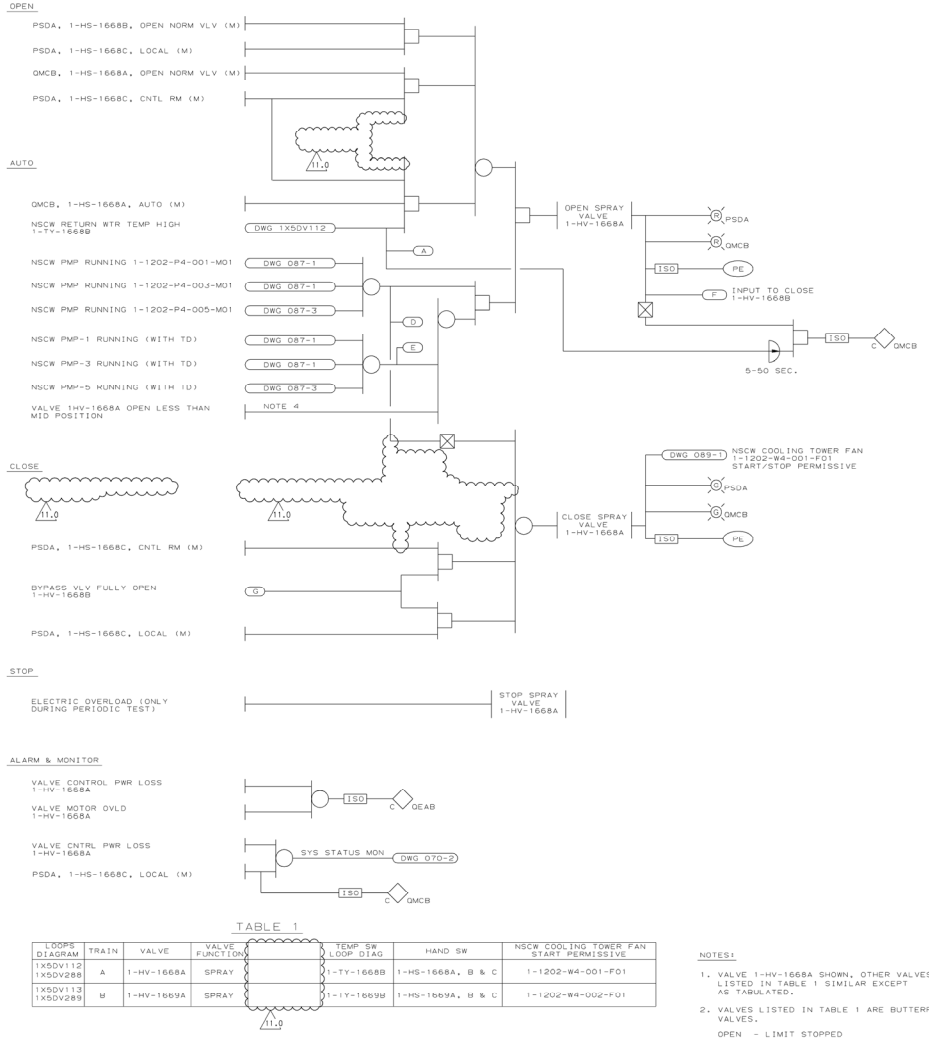

TABLE 1

LOOPS DIAGRAM	TRAIN	VALVE	VALVE FUNCTION	TEMP SW LOOP DIAG	HAND SW	NSCW COOLING TOWER FAN START PERMISSIVE
1X5DV112 1X5DV288	A	1-HV-1668A	SPRAY	1-TY-1668B	1-HS-1668A, B & C	1-1202-W4-001-F01
1X5DV113 1X5DV289	B	1-HV-1669A	SPRAY	1-TY-1669B	1-HS-1669A, B & C	1-1202-W4-002-F01

11.0

NOTES:

1. VALVE 1-HV-1668A SHOWN, OTHER VALVES LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATED.
2. VALVES LISTED IN TABLE 1 ARE BUTTERFLY VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
3. REFER TO SPECIAL NOTE NO. 2 ON DWG 1X5DN002-2.
4. VALVE MID POSITION ANGLE IS:
   10° FOR 1HV-1668A
   7.5° FOR 1HV-1669A

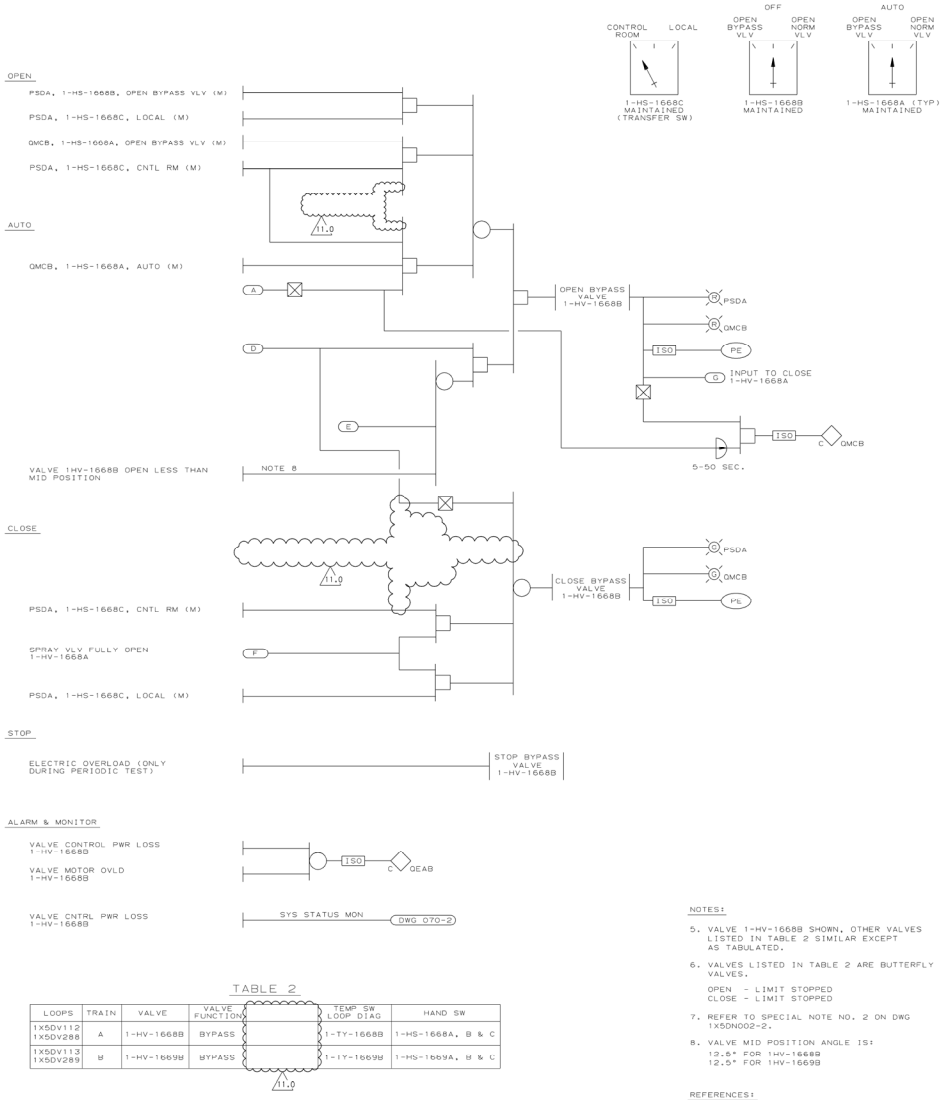

TABLE 2

LOOPS	TRAIN	VALVE	VALVE FUNCTION	TEMP SW LOOP DIAG	HAND SW
1X5DV112 1X5DV288	A	1-HV-1668B	BYPASS	1-TY-1668B	1-HS-1668A, B & C
1X5DV113 1X5DV289	B	1-HV-1669B	BYPASS	1-TY-1669B	1-HS-1669A, B & C

11.0

NOTES:

5. VALVE 1-HV-1668B SHOWN, OTHER VALVES LISTED IN TABLE 2 SIMILAR EXCEPT AS TABULATED.
6. VALVES LISTED IN TABLE 2 ARE BUTTERFLY VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
7. REFER TO SPECIAL NOTE NO. 2 ON DWG 1X5DN002-2.
8. VALVE MID POSITION ANGLE IS:
   12.5° FOR 1HV-1668B
   12.5° FOR 1HV-1669B

REFERENCES:

P & I D 1X4DB133-1, 133-8

E.D. 1X3D-BD-K05U, 1X3D-BD-K05V, 1X3D-BD-K05W, 1X3D-BD-K05X, 1X3D-BD-K03A & 1X3D-BD-K03E

LOOP DIAG. 1X5DV112, 113

LOCATION 1X4DE328, 329

COOLING TOWER SPRAY HEADER VALVES
NUCLEAR SAFETY RELATED Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE
COOLING TOWER SYSTEM

SCALE: NONE | DRAWING NO. 1X5DN089-3 | VER. 11.0 | JOB NO. 10604

NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN-V01148, VER. 1.0	7-7-08	vep	ELC	JMR

SIZE E 34x44

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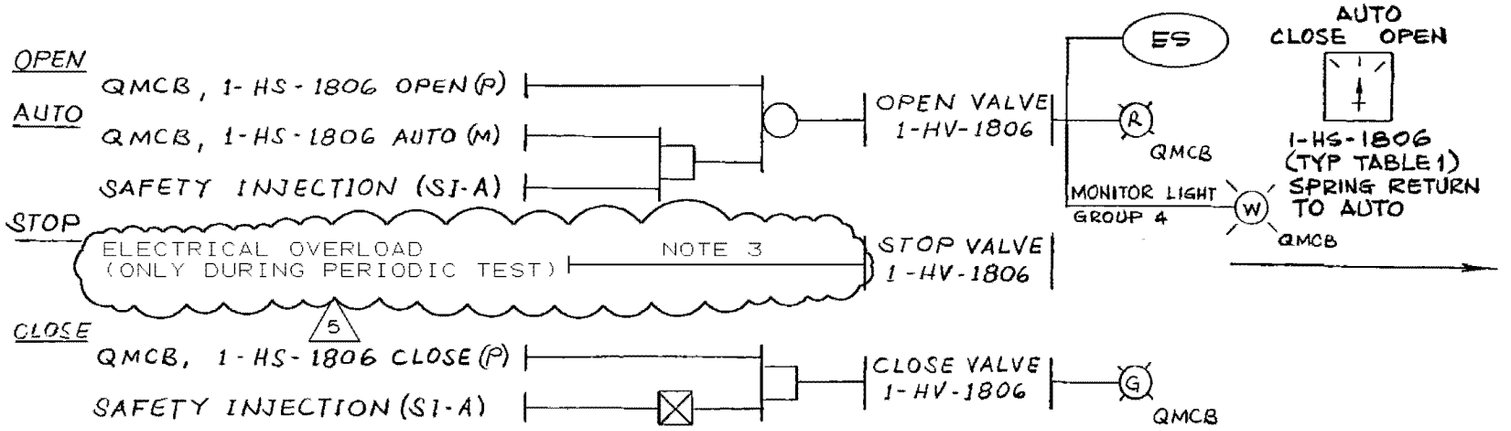


TABLE 1

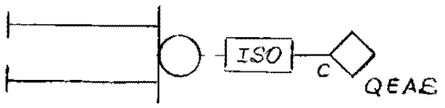
TRAIN	EQUIPMENT TAG NOS.		SAFETY SIG.
A	1-HS-1806	1-HV-1806	SI-A
B	-1807	-1807	SI-B
A	-1808	-1808	SI-A
B	-1809	-1809	SI-B
A	-1822	-1822	SI-A
B	-1823	-1823	SI-B
A	-1830	-1830	SI-A
B	-1831	-1831	SI-B

MONITOR (TYPICAL FOR TABLE 1)

VALVE CONTROL POWER LOSS 1-HV-1806

SYSTEMS STATUS MONITOR DWG 070-1

VALVE CNTL PWR LOSS 1-HV-1806
VALVE MOTOR OVLD 1-HV-1806



NOTE:

1. VALVES LISTED IN TABLE 1 ARE BUTTERFLY TYPE VALVES
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED

NUCLEAR SAFETY RELATED

CONTAINMENT COOLER ISOLATION VALVE Q

3. REFER TO SPECIAL NOTE 2 ON DWG 1X5DN002-2.

REF. DWGS
P&ID 1X4DB135-1,-2
E.D. 1X3D-BD-K04J,K,N,P,S,T,U,V,W & X

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
△						
△						
5	INCORPORATED PER DCP 95-V1N0035	12/5/97	CD	GLB	WFP	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM NUCLEAR SERVICE COOLING WATER SYSTEM AUXILIARIES AND ALARMS

SCALE: NONE

DRAWING NO. 1X5DN090-1

REV 5

JOB NO. 10604

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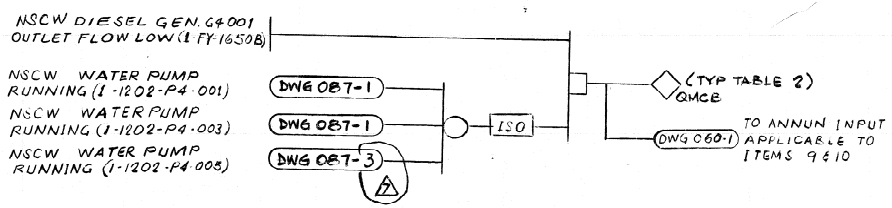
CAD NAME 15DN0901 SIZE B 11x17

ALARMS



REF DWGS.
P & I D. 1X4DB133-2, 134, 135-1&2
E.D. 1X3D-BD-K04A #
1X3D-BD-K04F

TABLE 2

ITEM NO.	ANNUN WINDOW LOCATION	SERVICE	FLOW SW	LOOP DIAG	NSCWS PUMP 1-1202- P4-001	P4-003	P4-005
1	QMCB	NSCW DIESEL GEN G4-001	1-FY-1650B	1X5DV082	2	4	6
2		2	1651B	33	1	3	5
3		CNTMT AIR CLR A7-001/002	1818B	15	2	4	6
4		-003/004	1819B	16	1	3	5
5		-005/006	1820B	17	2	4	6
6		-007/008	1821B	18	1	3	5
7		RX CAVT CLG COIL -001	2132B	19	2	4	6
8		002	2133B	20	1	3	5
9	QHVC	ESF CHLR COND C7-001	1-FIS-1804	185	2	4	6
10		2	1803	186	1	3	5
11	QMCB	RHR PMP & MTR COOLER	1-FS-1746	—	2	4	6
12			1747				

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
5	CORRECTED TAG #'S	12-13-84	HCV	[illegible]						
4	CORRECTED TAG NOS PER DCN E3 ON P&ID 134	9-11-84	EJL	HCV	X			X		X
3	INCORPORATED DCN #1	1/21/87	ALV	[illegible]						
2	ISSUED FOR CONSTRUCTION	12/3/84	HL	FAL	MKK					

ENG. MGR. / DATE

BM 9510 P&ID 133-2 REV. 134 REV. P&ID 135-1 REV. 135-2 REV.

ALARMS

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE COOLING WATER
SYSTEM AUXILIARIES AND ALARMS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN090-2	7

SIZE B 11 x 17

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DATE: 1-26-88 CAMERA OPERATOR SECTION SUPERVISOR

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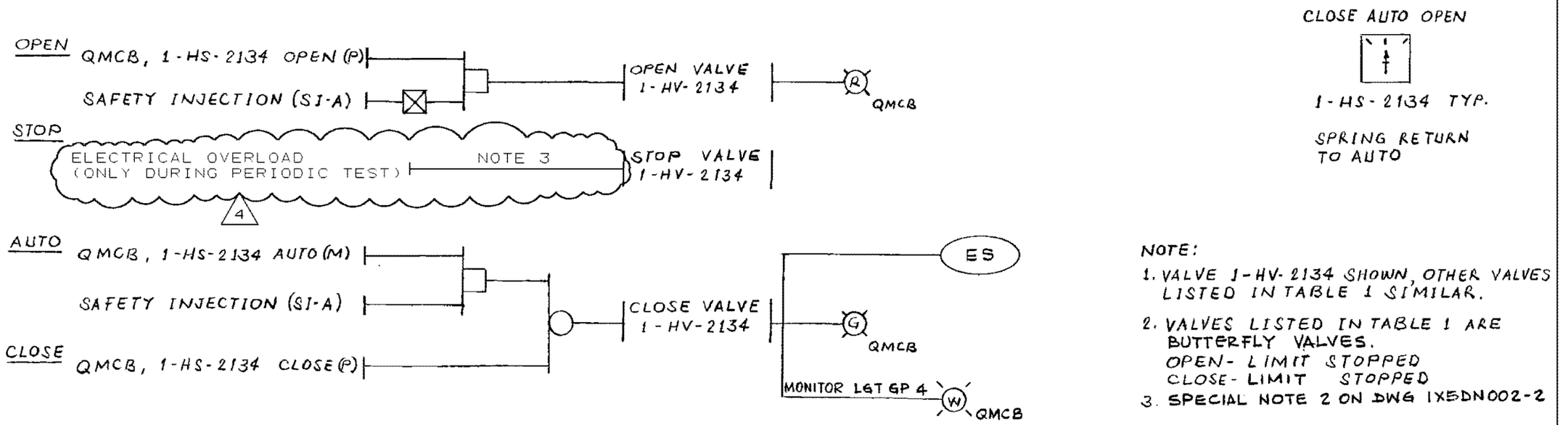
DATE: 1-26-88 CAMERA OPERATOR SECTION SUPERVISOR



CLOSE AUTO OPEN

1-HS-2134 TYP.

SPRING RETURN TO AUTO

NOTE:

1. VALVE 1-HV-2134 SHOWN, OTHER VALVES LISTED IN TABLE 1 SIMILAR.
2. VALVES LISTED IN TABLE 1 ARE BUTTERFLY VALVES.
   OPEN- LIMIT STOPPED
   CLOSE- LIMIT STOPPED
3. SPECIAL NOTE 2 ON DWG 1X5DN002-2

MONITOR & ALARM

VALVE 1-HV-2134 CONTROL POWER LOSS — SYSTEMS STATUS MONITOR — DWG 070-1

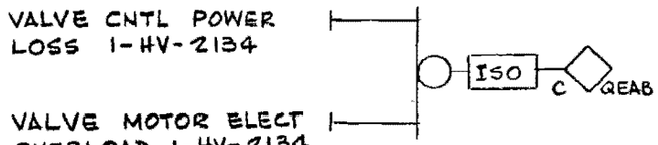
REF. DWGS.

P&ID 1X4DB135-1,2

E.D. 1X3D-BD-K04Q, K04R, K04L & K04M

TABLE 1

TRAIN	VALVE	SWITCH	SAFETY SIG
A	1-HV-2134	1-HS-2134	SI-A
B	-2135	-2135	SI-B
A	-2138	-2138	SI-A
B	-2139	-2139	SI-B



NUCLEAR SAFETY RELATED

REACTOR CAVITY AND CRDM COOLING COIL ISOLATION VALVES Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM NUCLEAR SERVICE COOLING WATER SYSTEM AUXILIARIES AND ALARMS

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORPORATED PER DCP 95-V1N0035	12/5/97	CD	GLB	WFP	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE

JOB NO. 10604

DRAWING NO. 1X5DN090-3

REV 4

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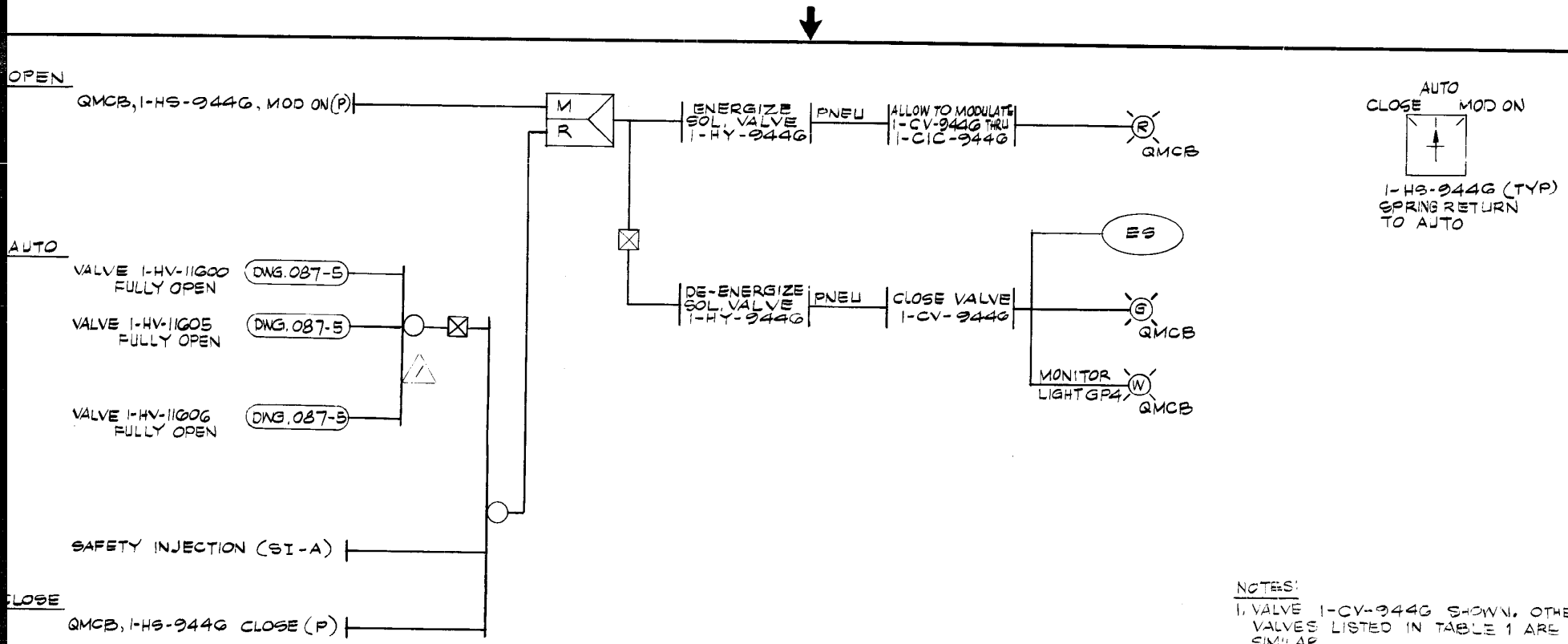
CAD NAME 15DN0903

SIZE B 11x17



AUTO
CLOSE MOD ON
1-HS-9446 (TYP)
SPRING RETURN
TO AUTO

NOTES:
1. VALVE 1-CV-9446 SHOWN. OTHER VALVES LISTED IN TABLE 1 ARE SIMILAR.

REF. DWGS.:
P&ID 1X4DB133-1,2
E.D. 1X3D-BD-K04J,K

TABLE 1

TRAIN	SWITCH	VALVE	SOL. VALVE NOS.	SFTY SIG.	CONTR.	NSCW DISCH. VALVES
A	1-HS-9446	1-CV-9446	1-HY-9446	SI-A	1-CIC-9446	1-HV-11600, 11605, 11606
B	↓ -9447	↓ -9447	↓ -9447	SI-B	↓ -9447	1-HV-11607, 11612, 11613

2

NSCW TOWER BLOWDOWN VALVE "Q"

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
NUCLEAR SERVICE COOLING WATER
SYSTEM AUXILIARIES AND ALARMS

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.	ENG. MGR.
3	DELETED NOTE PER DCN #1	1/21/87	ALV								DATE
2	CHANGED RESET LOGIC, ADDED NOTES.	1-7-86	ZEP	ALV							
1	ISSUED FOR CONSTRUCTION	3/29/85	ZEP								

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN090-6	2

BM 9510

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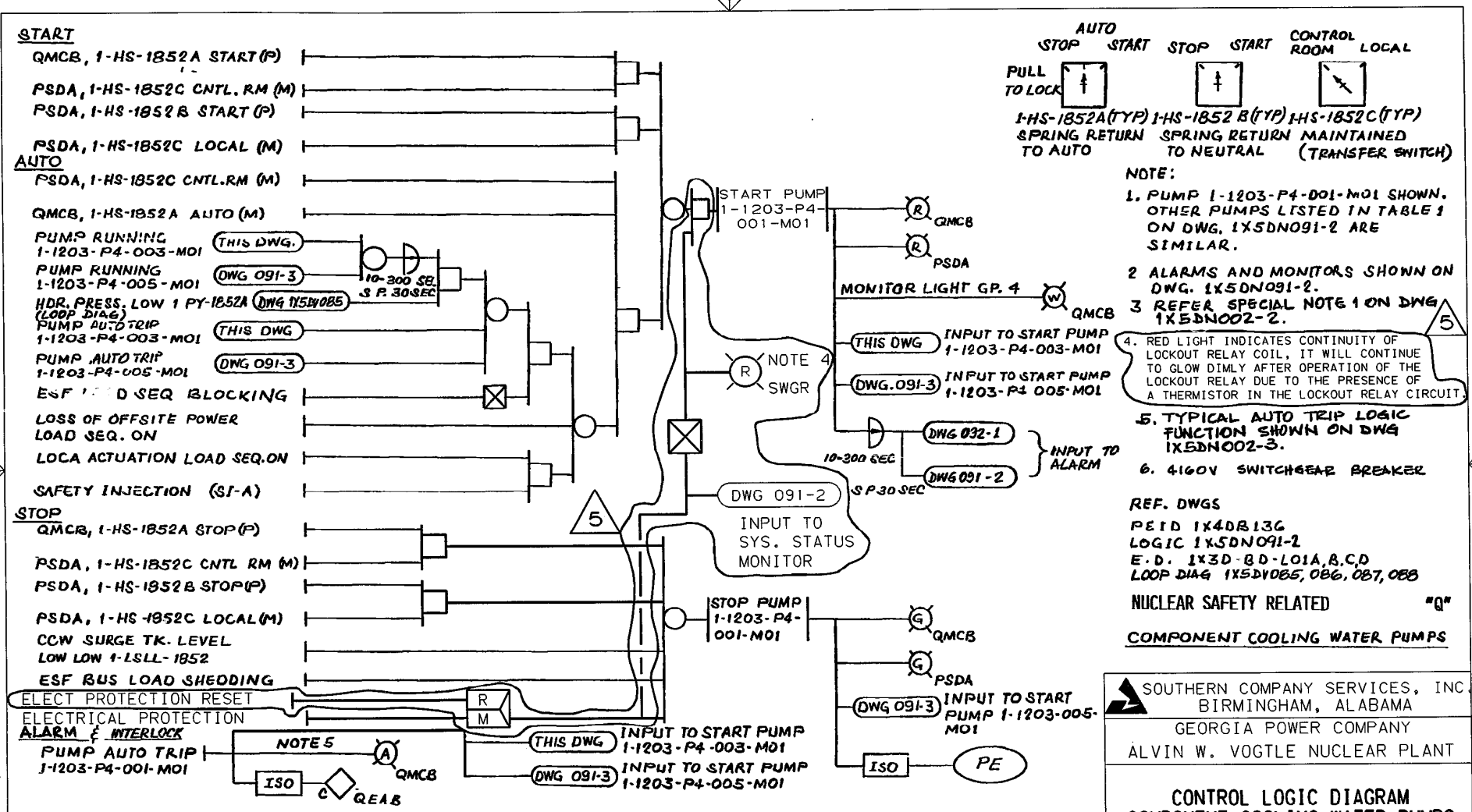

NO	REVISIONS	DATE	DR	CHK	APPV	DTL
5	INCORP. ABN-21051	11/30/95	MWD	WLM	EDZ	
0	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES			

ENG MGR

DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM COMPONENT COOLING WATER PUMPS

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN091-1	5

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ALARMS

PUMP RUNNING 1-1203-P4-001-M01 (DWG 091-1)
HEADER PRESS LOW 1-PY-1852B (LOOP DIAG) (DWG 1X5DN085)
→ ISO → C → QMCB

CCW SURGE TANK LEVEL LOW LOW (1-LSLL-1852)
→ ISO → C → QMCB

PSDA, 1-HS-1852C LOCAL, (M)
→ ISO → C → QMCB

MONITORS

ELECT. PROT. BRKR. LOCKOUT 1-1203-P4-001-M01 (DWG 091-1) — 3
PSDA, 1-HS-1852C, LOCAL (M)
CONTROL POWER LOSS 1-1203-P4-001-M01
QMCB, 1-HS-1852A PULL TO LOCK (M)
PUMP BREAKER INOPERABLE POSITION 1-1202-P4-001-M01
→ SYSTEM STATUS MONITOR (DWG 070-3)

NOTE:
1. TYPICAL ALARMS AND MONITORS SHOWN, OTHER DEVICES LISTED IN TABLE 1 AND 2 SIMILAR.
2. TABLE 1 LOGIC SHOWN ON DWG. 1X5DN091-1 AND TABLE 2 LOGIC SHOWN ON DWG. 1X5DN091-3.

TABLE 1

TRAIN	PUMP NO. (1-1203-)	SAFETY ACT. SIG.	SWITCHES			PRESS SW LOOP DIAG	LEVEL SW	PUMP START/OVLD. (1-1203-)	
A	P4-001-M01	SI-A	1-HS-1852A	1-HS-1852B	1-HS-1852C	1-PY-1852A/B	1-LSLL-1852	P4-003-M01	P4-005-M01
B	P4-002-M01	SI-B	1853A	1853B	1853C	1853A/B	-1853	P4-004-M01	P4-006-M01
A	P4-003-M01	SI-A	1854A	1854B	1854C	1854A/B	1854	P4-001-M01	P4-005-M01
B	P4-004-M01	SI-B	1855A	1855B	1855C	1855A/B	1855	P4-002-M01	P4-006-M01

TABLE 2

TRAIN	PUMP NO. (1-1203-)	SAFETY ACT. SIG	SWITCHES			PRESS SW LOOP DIAG	LEVEL SW	PUMP START/OVLD. (1-1203-)	
A	P4-005-M01	SI-A	1-HS-1856A	1-HS-1856B	1-HS-1856C	1-PY-1856A/B	1-LSLL-1856	P4-001-M01	P4-003-M01
B	P4-006-M01	SI-B	1857A	1857B	1857C	1857A/B	1857	P4-002-M01	P4-004-M01

REF. DWGS
P&ID 1X4DB136
LOGIC 1X5DN091-1, 091-3
E.D 1X3D-BD-LO1A,B,C,D,E,F
LOOP DIAG 1X5DV085, 086, 087, 088, 089, 090

NUCLEAR SAFETY RELATED

ALARMS, MONITOR AND EQPT. LIST "Q"

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. ABN-21052	11/29/95	MWD	WLM	EBL	
2	REVISIONS					
1	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES			
0	REVISIONS	DATE	DR			

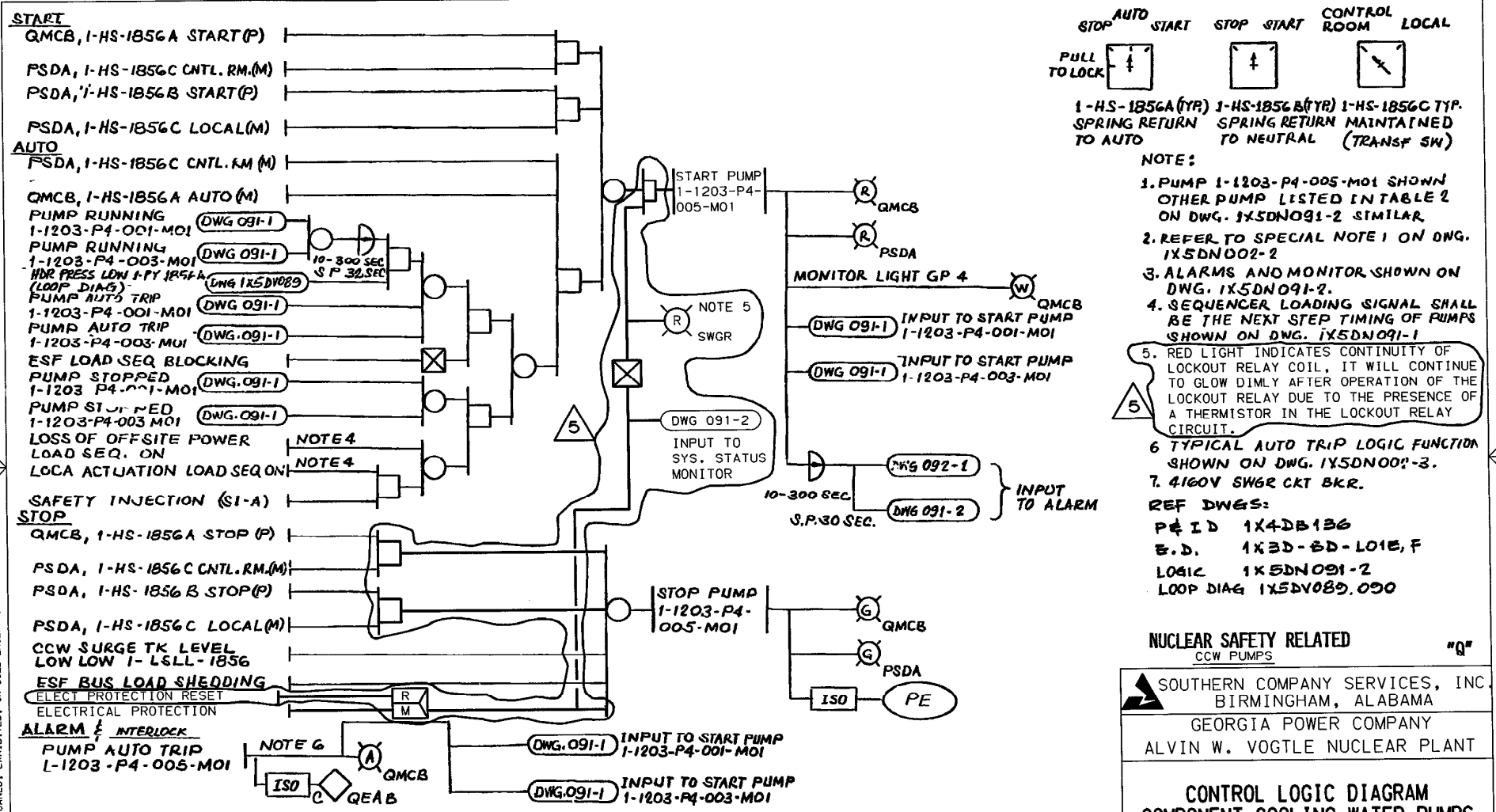
ENG MGR
DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM COMPONENT COOLING WATER PUMPS

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X5DN091-2
REV 3



STOP AUTO START STOP START CONTROL ROOM LOCAL

PULL TO LOCK

1-HS-1856A (TYP) SPRING RETURN TO AUTO

1-HS-1856B (TYP) SPRING RETURN TO NEUTRAL

1-HS-1856C TYP. MAINTAINED (TRANSF SW)

NOTE:

1. PUMP 1-1203-P4-005-M01 SHOWN OTHER PUMP LISTED IN TABLE 2 ON DWG. 1X5DN091-2 SIMILAR.
2. REFER TO SPECIAL NOTE 1 ON DWG. 1X5DN002-2
3. ALARMS AND MONITOR SHOWN ON DWG. 1X5DN091-2.
4. SEQUENCER LOADING SIGNAL SHALL BE THE NEXT STEP TIMING OF PUMPS SHOWN ON DWG. 1X5DN091-1
5. RED LIGHT INDICATES CONTINUITY OF LOCKOUT RELAY COIL, IT WILL CONTINUE TO GLOW DIMLY AFTER OPERATION OF THE LOCKOUT RELAY DUE TO THE PRESENCE OF A THERMISTOR IN THE LOCKOUT RELAY CIRCUIT.
6. TYPICAL AUTO TRIP LOGIC FUNCTION SHOWN ON DWG. 1X5DN002-3.
7. 4160V SWGR CKT BKR.

REF DWGS:

P & I D 1X4DB136

E.D. 1X3D-BD-L01E, F

LOGIC 1X5DN091-2

LOOP DIAG 1X5DV089, 090

NUCLEAR SAFETY RELATED "Q"

CCW PUMPS

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM COMPONENT COOLING WATER PUMPS

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
5	INCORP. ABN-21053	11/30/95	MWD	WLM	EOZ	
4	REVISIONS	DATE	DR			
0	ISSUED FOR CONSTRUCTION		SEE MICROFILM FOR SIGNATURES			

ENG MGR

DATE

SCALE: NONE

JOB NO. 10604

DRAWING NO. 1X5DN091-3

REV 5

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OPEN — QMCB, 1-HS-1850 OPEN (M)

AUTO — QMCB, 1-HS-1850 AUTO (M)

CCW SURGE TANK LEVEL LOW 1-LSL-1850

→ OPEN VALVE 1-LV-1850 → (R) QMCB

STOP — TORQUE PROTECTION

ELECTRICAL OVERLOAD

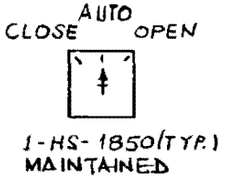
→ STOP VALVE 1-LV-1850

CLOSE — QMCB, 1-HS-1850 CLOSE (M)

AUTO — QMCB, 1-HS-1850 AUTO (M)

CCW SURGE TANK LEVEL HIGH 1-LSH-1850

→ CLOSE VALVE 1-LV-1850 → (G) QMCB



1-HS-1850 (TYP.)
MAINTAINED

TABLE 1

VALVE	SWITCH	LEVEL SW	
1-LV-1850	1-HS-1850	1-LSH-1850	1-LSL-1850
↓ 1851	↓ 1851	↓ 1851	↓ 1851

NOTE:

1. VALVE 1-LV-1850 SHOWN, 1-LV-1851 SIMILAR.
2. VALVE LISTED IN TABLE 1 ARE GATE VALVES
   OPEN - LIMIT STOPPED
   CLOSE - TORQUE STOPPED

ALARMS

CCW SURGE TANK LEVEL LOW 1-LSL-1850 → ◇ QMCB

RHR HEAT EXH TRN A CCW FLOW LOW 1-FSHL-1928 (4) → ISO → QMCB

CCW PMPS DISCH HDR FLOW LOW 1-FY-1876 → 1X5DV026 → ISO → QMCB

RHR PUMP GLAND HEAT EXCH. CCW FLOW LOW 1-FSL-1936

PUMP RUNNING 1-1203-P4-001-M01 → DWG. 091-1

PUMP RUNNING 1-1203-P4-003-M01 → DWG 091-1

PUMP RUNNING 1-1203-P4-005-M01 → DWG. 091-3

→ ISO → QMCB

FLOW SW	PUMP	FLOW SW
1-FSHL-1928	1-1203-P4-001-M01	
1-FSL-1936	-003	1-FY-1876 (LOOP DIAG 026)
	-005	
1-FSHL-1929	-002	
1-FSL-1937	-004	1-FY-1877 (LOOP DIAG 027)
	-006	

(4)

REF. DWG.

P&ID 1X4DB136 & 137

E.D. 1X3D-BD-L016, L01K, L01A, L01E

LOOP DIAG 1X5DN026, 027

NORMAL MAKE-UP VALVE

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	R.E.	Q.A.E.	
3	ADDED ALARM INPUT PER P&ID 137 DCN 11	8-3-92	HT	FAL		RCG					ENG. MGR.
2	ADDED ALARM INPUT AND REVISED AS INDICATED	7-9-91	FAL	V.V.		PCG					
1	INCORPORATED DCN #1	2-26-91	RBR	ALJ							DATE
0	ISSUED FOR CONSTRUCTION	1-12-17	HL	FAL	MKK						

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
COMPONENT COOLING SYSTEM
AUXILIARIES AND ALARMS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN092-1	4

P&ID 136 REV. 2
137 REV. 7 DCN 11

SIZE B 11 x 17

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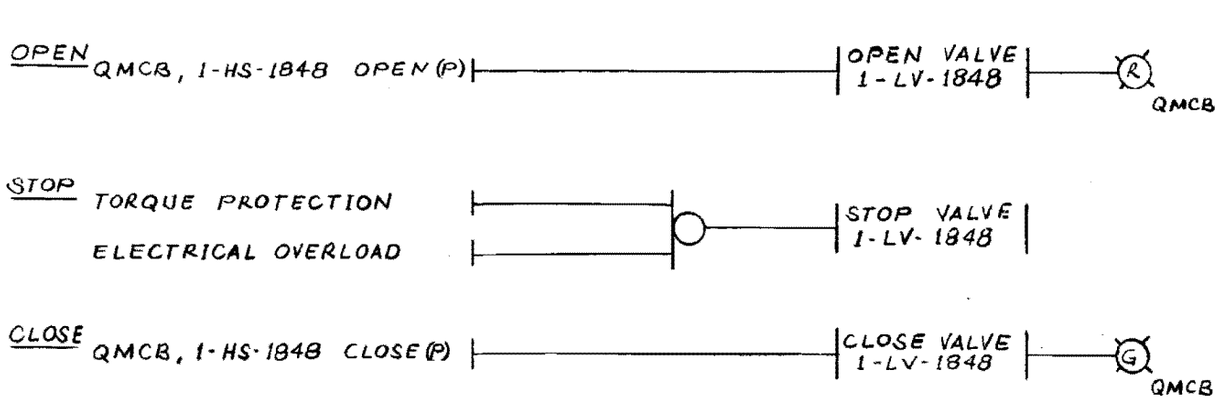
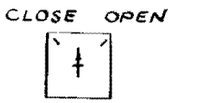




1-HS-1848 (TYP.)

SPRING RETURN TO NEUTRAL

NOTE:

1. VALVE 1-LV-1848 SHOWN. 1-LV-1849 SIMILAR.
2. VALVES LISTED IN TABLE 1 ARE GATE VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - TORQUE STOPPED

TABLE 1

VALVE	SWITCH
1-LV-1848	1-HS-1848
↓ 1849	↓ 1849

REF. DWGS
P&ID 1X4DB136
E.D. 1X3D-BD-L01H, L01M

SCAN ID DATE

BACK-UP MAKE-UP VALVE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
COMPONENT COOLING SYSTEM
AUXILIARIES AND ALARMS

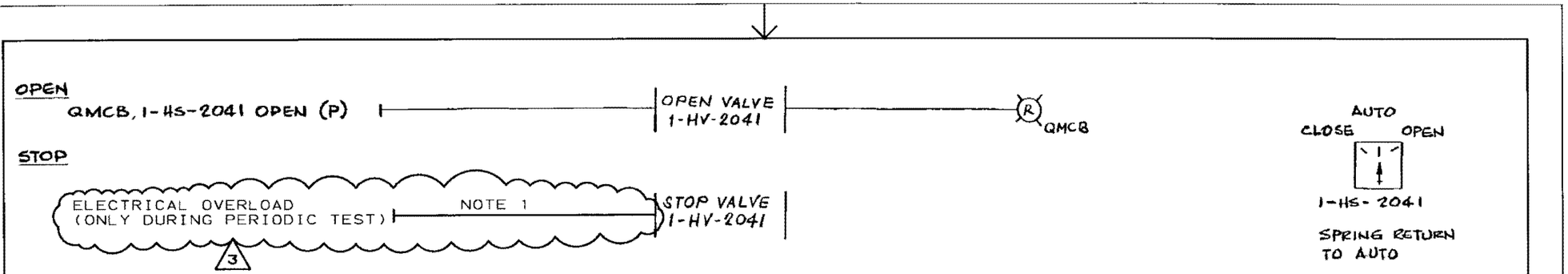
NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.
2										
1										
1	ADDED REF DWGS.	3-18-80	HL	FAL						
0	ISSUED FOR CONSTRUCTION	1-12-79	HL	FAL	MKK					

ENG. MGR.
DATE

SCALE: NONE
JOB NO. 9510
DRAWING NO. 1X5DN092-2
REV. 1

BM 9510

SIZE B 11 x 17



NOTE:

1. REFER TO SPECIAL NOTE 2 ON DWG 1X5DN002-2
2. VALVE IS A GATE VALVE
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
3. VALVE 1-HV-2041 POWERED FROM TRAIN B

REF DWGS
P&ID 1X4DB138-2
E.D. 1X3D-BD-L03P
LOOP DIAG 1X5DV060

NUCLEAR SAFETY RELATED "Q"

REACTOR COOLANT PUMPS THERMAL BARRIER ACCWS OUTLET HEADER VALVE

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER DCP 95-V1N0022.	9/23/96	CD	[illegible]	SM	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM AUXILIARY COMPONENT COOLING WATER AUXILIARIES

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN094-3	3

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CAD NAME ~~15N94003~~ SIZE B 11x17

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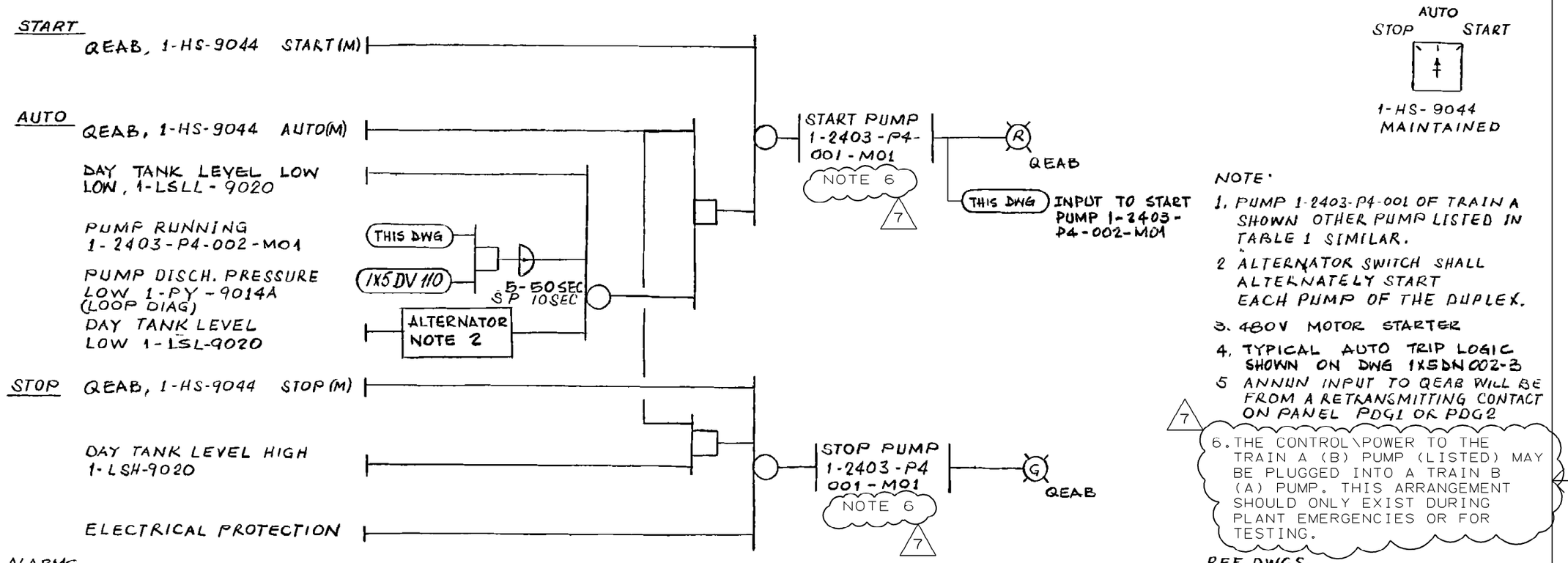
15DN0943



ALARMS

PUMP AUTO TRIP 1-2403-P4-001-M01 — NOTE 4 — A — QEAB; ISO — C — QEAB

CONTROL POWER LOSS 1-2403-P4-001-M01
QEAB, 1-HS-9044 STOP(M)
SYS STATUS MONITOR — DWG 070-12

STORAGE TANK LEVEL HIGH, 1-LIS-9022
STORAGE TANK LEVEL LOW, 1-LIS-9022
NOTE 5 — QEAB; PDG2

DAY TANK LEVEL HIGH, 1-LSHL-9018
DAY TANK LEVEL LOW 1-LSHL-9018
NOTE 5 — QEAB; PDG2

NOTE:

1. PUMP 1-2403-P4-001 OF TRAIN A SHOWN OTHER PUMP LISTED IN TABLE 1 SIMILAR.
2. ALTERNATOR SWITCH SHALL ALTERNATELY START EACH PUMP OF THE DUPLEX.
3. 480V MOTOR STARTER
4. TYPICAL AUTO TRIP LOGIC SHOWN ON DWG 1X5DN002-3
5. ANNUN INPUT TO QEAB WILL BE FROM A RETRANSMITTING CONTACT ON PANEL PDG1 OR PDG2
7
6. THE CONTROL\POWER TO THE TRAIN A (B) PUMP (LISTED) MAY BE PLUGGED INTO A TRAIN B (A) PUMP. THIS ARRANGEMENT SHOULD ONLY EXIST DURING PLANT EMERGENCIES OR FOR TESTING.

REF. DWGS.
P&ID 1X4DB170-1,170-2
E D 1X3D-BH-G01X, G01Z, G02B & G02D
LOOP DIAG. 1X5DV110,111,106 & 107

NUCLEAR SAFETY RELATED "Q"

FUEL OILSTORAGE TK PUMP

TABLE 1

TRN	PANEL	PUMP	SWITCH	DAY TK. LEVEL SW.	STOR. TK. LEVEL SW.	PRESS. SW. (LOOP DIAG)
A	PDG2	1-2403-P4-001-M01	1-HS-9044	1-LSHL-9018	1-LIS-9022	1-PY-9014A
		-002	-9046	1-LSH, 1-LSL, 1-LSLL 9020		9014B
B	PDG4	003	9045	1-LSHL-9019	1-LIS-9023	9015A
		004	9047	1-LSH, 1-LSL, 1-LSLL 9021		9015B

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
7	INCORP. PER DCP 99-V1N0058	12/22/03	JGM	TSL	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM DIESEL GENERATOR FUEL OIL SYSTEM

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN107-1	7

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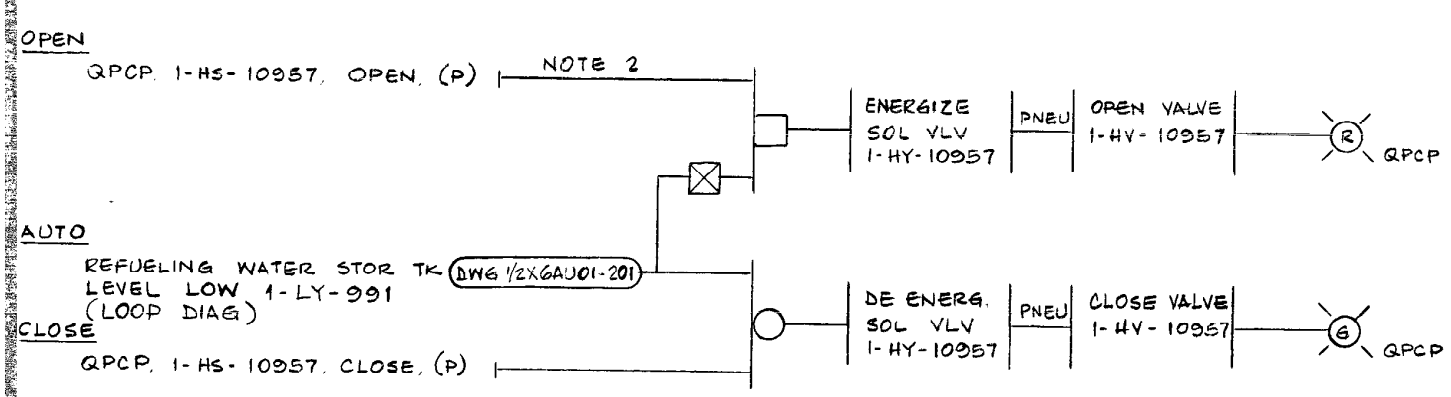
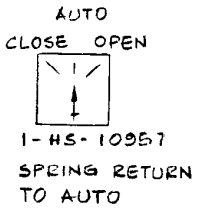
SIZE B 11x17





NOTE:

1. VALVE 1-HV-10957 SHOWN VALVES LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATE
2. HAND SWITCH SHALL BE HELD IN OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH

REF. DWGS.

P&ID 1X4DB121

E.D. 1X3D-BD-D058 (4)

LOOP DIAG 1/2XGAU01-201

TABLE 1.

TRAIN	VALVE	HAND SWITCH	LEVEL SWITCH (LOOP DIAG)	SOLENOID VALVE
B	1-HV-10957	1-HS-10957	1-LY-991	1-HY-10957
A	↓ 10958	↓ 10958	↓ 990	↓ 10958

(2)

NUCLEAR SAFETY RELATED

ISOLATION VALVES "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P.E.	Q.A.E.	ENG. MGR.
2	ADDED SOLENOID VALVES TO TABLE 1	2-8-85	AS	ALV							
1	ADDED REF. DWG	7-9-81	FAL	V.V							DATE
0	ISSUED FOR CONSTRUCTION	12-6-79	HL	FAL							

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
REFUELING WATER SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN114-4	2

SIZE B 11 x 17

BM 9510

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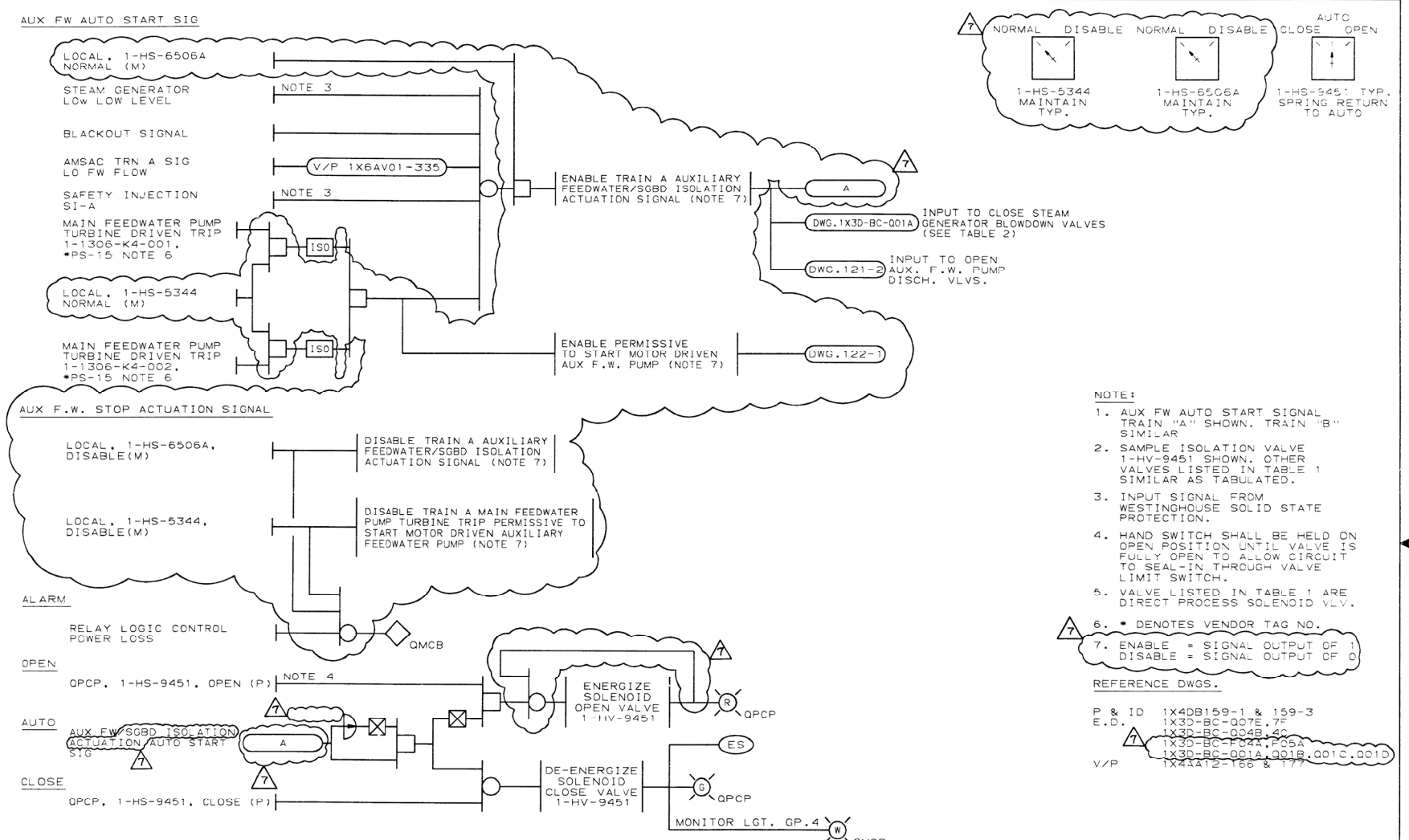

NOTE:

1. AUX FW AUTO START SIGNAL TRAIN "A" SHOWN. TRAIN "B" SIMILAR
2. SAMPLE ISOLATION VALVE 1-HV-9451 SHOWN. OTHER VALVES LISTED IN TABLE 1 SIMILAR AS TABULATED.
3. INPUT SIGNAL FROM WESTINGHOUSE SOLID STATE PROTECTION.
4. HAND SWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL-IN THROUGH VALVE LIMIT SWITCH.
5. VALVE LISTED IN TABLE 1 ARE DIRECT PROCESS SOLENOID VLV.
6. * DENOTES VENDOR TAG NO.
7. ENABLE = SIGNAL OUTPUT OF 1
   DISABLE = SIGNAL OUTPUT OF 0

REFERENCE DWGS.

P & ID 1X4DB159-1 & 159-3
E.D. 1X3D-BC-Q07E,7F
1X3D-BC-Q04B,4C
1X3D-BC-F04A,F05A
1X3D-BC-Q01A,Q01B,Q01C,Q01D
V/P 1X4AA12-166 & 177

TABLE 1

TRAIN	VALVE	HAND SW.
A	1-HV-9451	1-HS-9451
B	9452	9452
B	9453	9453
A	9454	9454

TABLE 2

SOL VLV TRAIN A	SOL VLV TRAIN B	SG. BLOW DOWN VALVES	E/D
1-HY-7603E	1-HY-7603J	1-HV-7603A	1X3D-BC-Q01A
F	K	B	-Q01B
G	L	C	-Q01C
H	M	D	-Q01D

TABLE 3

TRAIN	SAFETY RELATED HAND SWITCH	NON-SAFETY RELATED HAND SWITCH
A	1-HS-6506A	1-HS-5344
B	1-HS-6506B	1-HS-5345

AUX FW AUTO SIG & STM GEN SAMPLE ISO VLV

NUCLEAR SAFETY RELATED Q

NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHFE	PE	QAE
4	INCORP. ABN-11574.	10-22-93	CIB	MJD						
3	REVISIONS	DATE	DR	CHK	RE	DM	DM	DM	PDM	
2	ISSUED FOR CONSTRUCTION	3/9/79	SEE MICROFILM FOR SIGNATURES							
1	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHFE	PE	QAE

ENG. MGR. / DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
STEAM GENERATOR BLOWDOWN
AND SAMPLE SYSTEM

SCALE: NONE | DRAWING NO. 1X5DN117-1 | REV. 7
JOB NO. 10604

SIZE C 17X22

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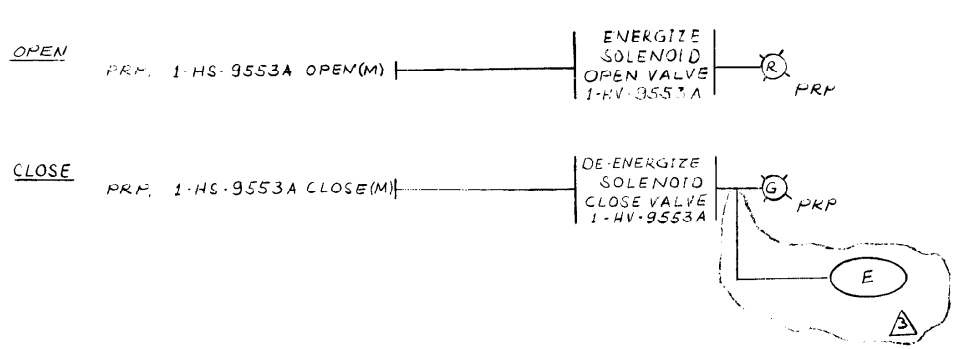

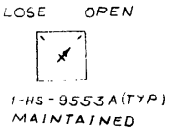

NOTE:

1. PROCESS SOLENOID VALVE 1-HV-9553A SHOWN. OTHER SOLENOID VALVES LISTED IN TABLE 1 ARE SIMILAR.

TABLE 1

VALVE	CNTL. SW.
1-HV-9553A	1-HS-9553A
9553B	9553B
9554A	9554A
9554B	9554B
9555A	9555A
9555B	9555B
9556A	9556A
9556B	9556B

REF. DWGS
P&ID 1X4DB159-1 & 159-3
E.D. 1X3D-BC-2035, T

SAMPLING VALVES

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	
4	SHOWN COMP INPUT PER TIO LIST REVS (NO PHYSICAL IMPACT)	3/27/85	ZEP	AIV							ENG. MGR.
3	REVISED REF. DWGS	12-8-81	AL	FAL							
2	CHANGED REF DWGS	1-10-80	HL	FAL							DATE
1	ISSUED FOR CONSTRUCTION	3-9-79	NRD	FAL							

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LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
STEAM GENERATOR BLOWDOWN
AND SAMPLE SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN117-2	3

SIZE B 11x17

BM 9510 P&ID 1X4-1 REV -3

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3/29/85 Ben Mallory
DATE CAMERA OPERATOR
SECTION SUPERVISOR

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OPEN — QPCP, 1-HS-5194, OPEN, (P) — NOTE 2 — ENERGIZE SOL VALVE 1-HY-5194 — PNEU — OPEN VALVE 1-HV-5194 — (R) QPCP

CLOSE — QPCP, 1-HS-5194, CLOSE (P) — DE-ENERGIZE SOL VALVE 1-HY-5194 — PNEU — CLOSE VALVE 1-HV-5194 — (G) QPCP, (E), (3)

CLOSE OPEN

1-HS-5194 (TYP)

SPRING RETURN TO NEUTRAL

NOTE:

1. VALVE 1-HV-5194 SHOWN, OTHER VALVES LISTED IN TABLE 1 ARE SIMILAR.
2. HAND SWITCH SHALL BE HELD IN OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE LIMIT SWITCH.

TABLE 1

SERVICE	HAND SWITCH	SOL VALVE	VALVE
CHEMICAL FEED	1-HS-5194	1-HY-5194	1-HV-5194
	5	5	5
	6	6	6
	7	7	7
WET LAYUP	1-HS-5278	1-HY-5278	1-HV-5278
	79	79	79
	80	80	80
	81	81	81

REF:

P & I D 1X4DB159-1, 159-3 & 168-3 (2)

E. D. 1X3D-BC-C04N, QC04A

STM GEN CHEM FEED & WET LAYUP VALVE

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
3	SHOWN COMP INPUT PER I/D LIST REV 5 (NO PHISICAL IMPACT)	3/27/85	ZEP	ALV		PCL				
2	ADDED REF DWG	12 8 81	AJF	FAL						
1	ADDED REF. DWGS.	1-10-80	HL	FAL						
0	ISSUED FOR CONSTRUCTION	4 17 79	HL	FAL	MKK					

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
STEAM GENERATOR BLOWDOWN
AND SAMPLE SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN117-3	3

SIZE B 11x17

BM 9510 P&ID 159-1 REV. -3 P&ID 168-3 REV.

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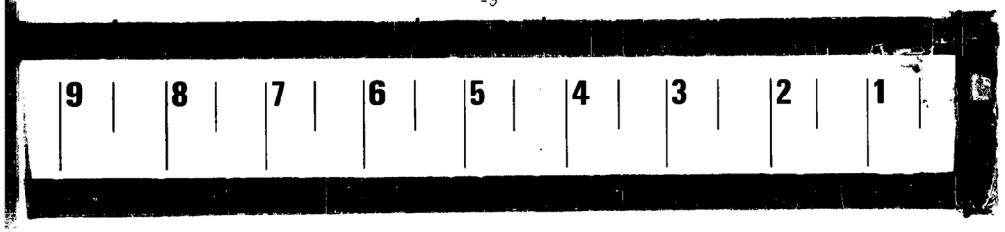
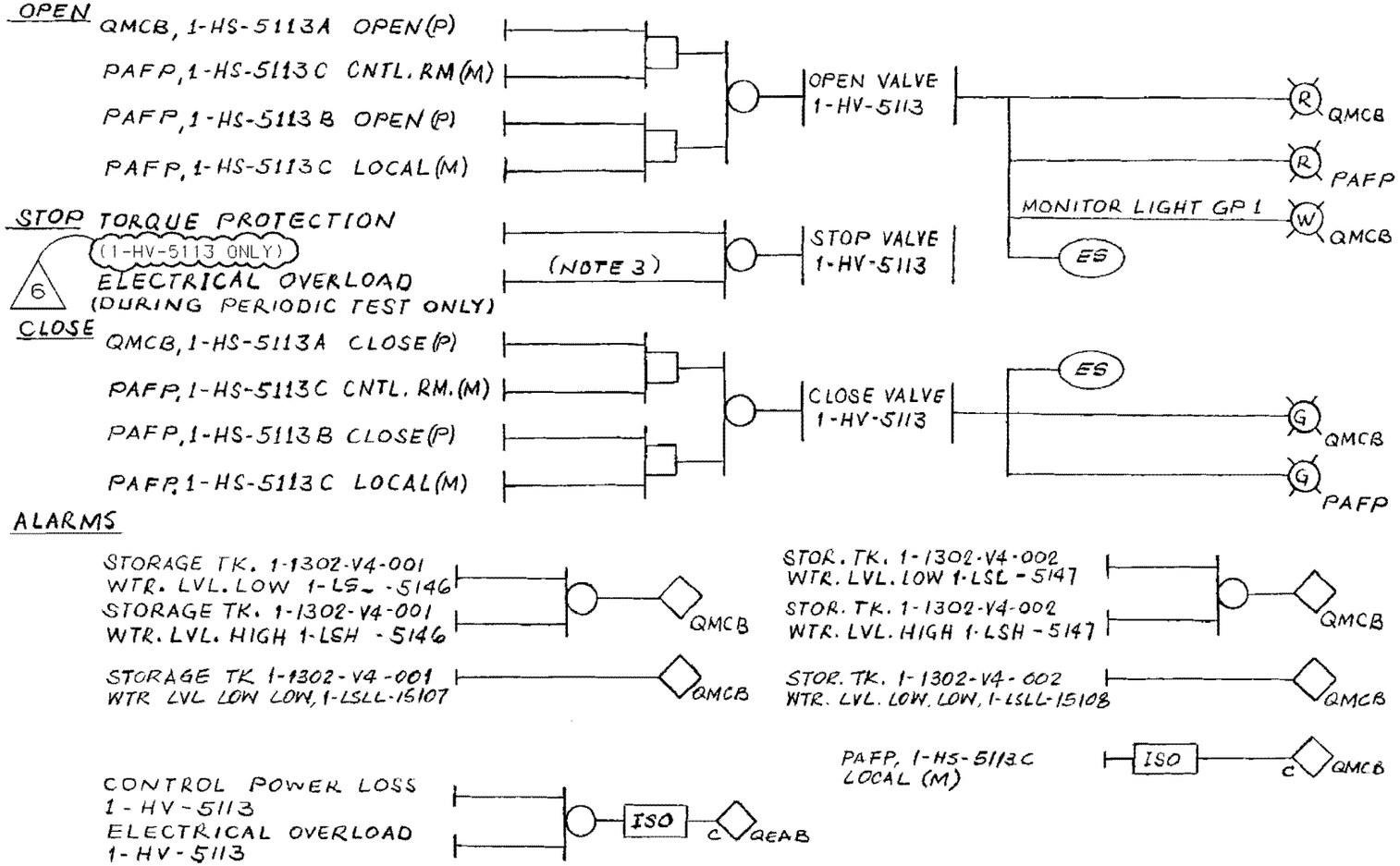
3/27/85 DATE — CAMERA OPERATOR — SECTION SUPERVISOR





CONTROL ROOM

CLOSE OPEN — LOCAL

1-HS-5113A
1-HS-5113B
SPRING RETURN TO NEUTRAL

1-HS-5113C
MAINTAINED (TRANSFER SWITCH)

NOTE:

1. VALVE 1-HV-5113 SHOWN. OTHER VALVE LISTED IN TABLE 2 ARE SIMILAR.
2. REFER TO SPECIAL NOTE 1 ON DWG. 1X5DN002-2.
3. REFER TO SPECIAL NOTE 2 ON DWG. 1X5DN002-2.
4. VALVES LISTED IN TABLE 2 ARE BUTTERFLY VALVES
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED
5. VALVE 1-HV-5113 HAS A D.C. MOTOR OTHERS ARE A.C. MOTORS.

REF. DWGS.
P&ID 1X5DN161-1 & 2
E.D. 1X3D-BC-F02C, D, B

TABLE 2

SERVICE	TRAIN	VALVE	SWITCHES			PANEL
AUX. FW PUMP TURB. DRIVEN	C	1-HV-5113	1-HS-5113A	1-HS-5113B	1-HS 5113C	PAFP
AUX. FW PUMP MOTOR DRIVEN	B	5118	5118A	5118B	5118C	PSDB
AUX. FW PUMP MOTOR DRIVEN	A	5119	5119A	5119B	5119C	PSDA

NUCLEAR SAFETY RELATED
PUMP SUCTION VALVE
Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM AUXILIARY FEEDWATER SYSTEM

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
6	INCORPORATED PER DCP 95-V1N0035	12-5-97	CD	GLB	WFP	X
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X5DN120-1
REV 6

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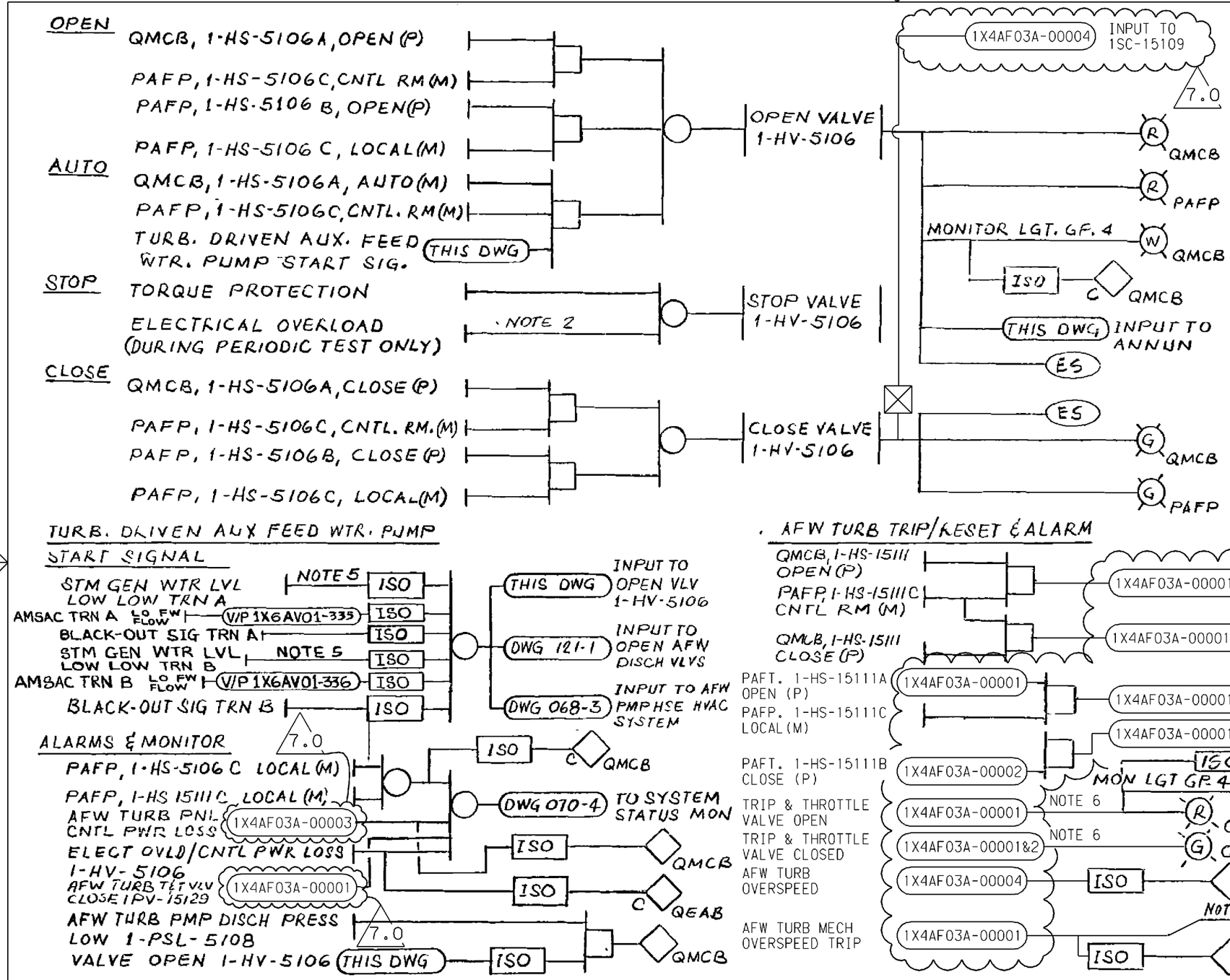
CAD NAME 15DN1201 SIZE B 11x17



NO.	VERSION	DATE	DR	CHK	APPV	DTL
7.0	REVISED PER ABN 1019003501J023, VER. 3.0	8-2-05	LPM	ELC	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER SYSTEM

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X5DN120-2	7.0

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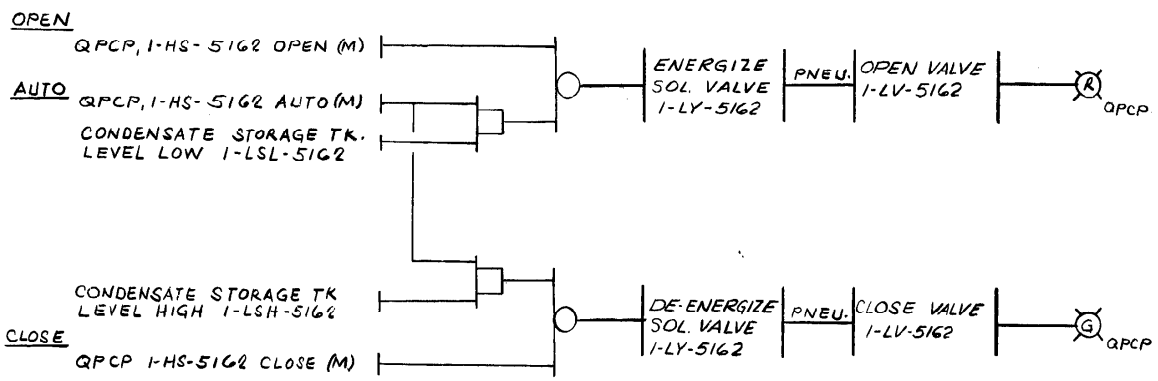
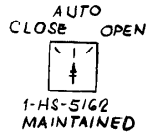
SIZE B 11x17

14.5X





NOTE:
1. VALVE 1-LV-5162 SHOWN
1-LV-5158 SIMILAR

REF. DWGS:
P&ID 1X4DB161-1
ED 1X3D-BC-F02E, F

COND. STOR. TANK	VALVE	SOL. VALVE	LEVEL SW.	LEVEL SW.	HAND SW.
1-1302-V4-002	1-LV-5162	1-LY-5162	1-LSH-5162	1-LSL-5162	1-HS-5162
↓ -001	↓ -5158	↓ -5158	↓ -5158	↓ -5158	↓ 5158

CONDENSATE TANK MAKE-UP VALVE

NO.	REVISIONS	DATE	DR	CHK.	SUP'V.	EGL	EGS	CHF. E.	P. E.	Q. A. E.	ENG. MGR.
△											
△											
1	ADDED REF DWGS.	1-10-80	AL	FAL							DATE
0	ISSUED FOR CONSTRUCTION	3-15-79	NBD	FAL	MKK.						

BM 9510

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER SYSTEM

SCALE:	DRAWING NO.	REV.
JOB NO. 9510	1X5DN120-3	1

SIZE B 11 x 17

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1-10-80 Denise Turner Frances Luckie
DATE: CAMERA OPERATOR SECTION SUPERVISOR

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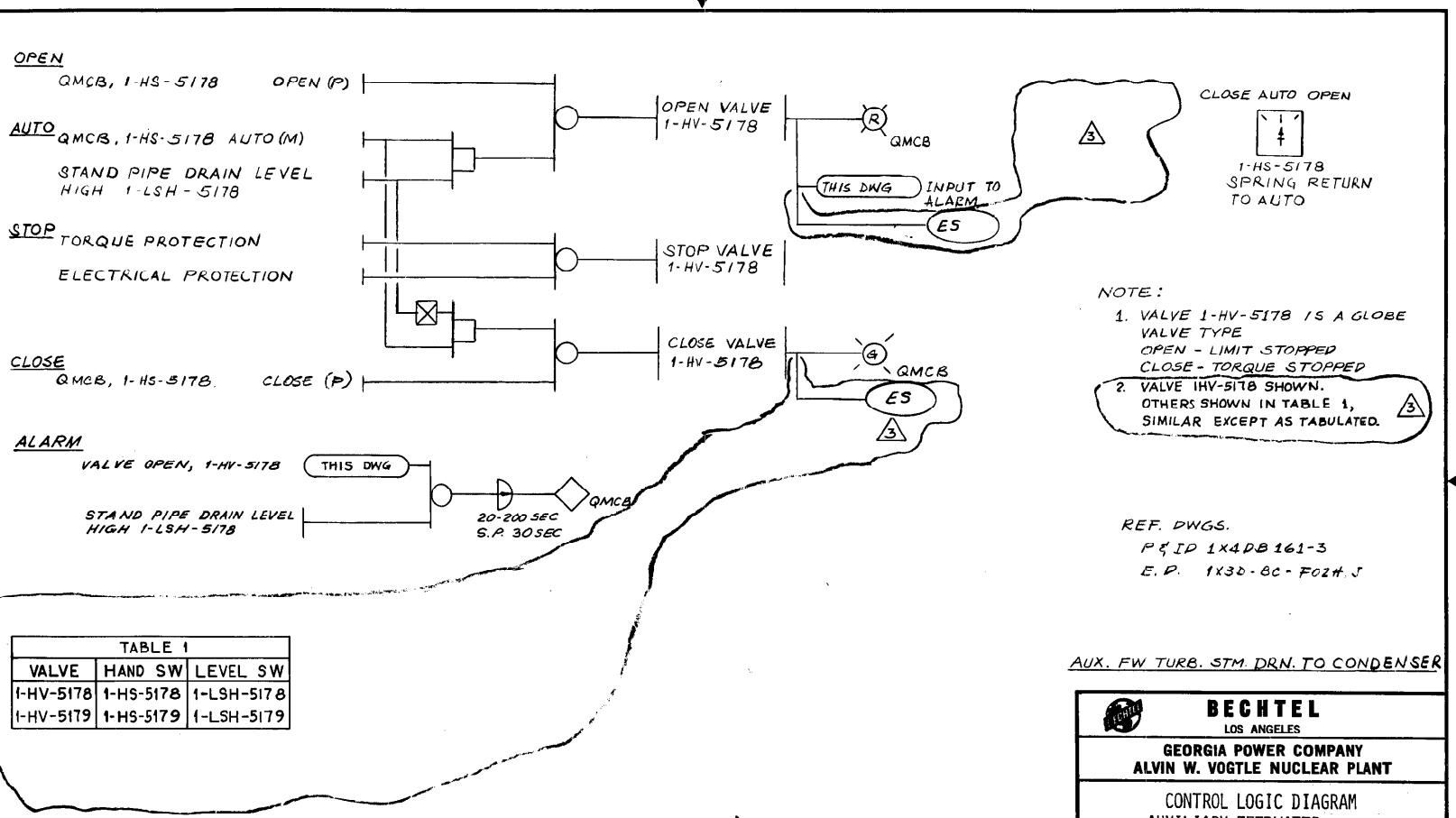

TABLE 1		
VALVE	HAND SW	LEVEL SW
1-HV-5178	1-HS-5178	1-LSH-5178
1-HV-5179	1-HS-5179	1-LSH-5179

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
5	REVISED AS SHOWN PER P&ID 1X4DB161-3, DCN NO. 25	11-14-84	CA							
4	CHANGED VALVE HV-5181 FROM MOV TO SOLENOID VALVE.	7-9-81	FAL	V.V.		PCL				
3	REVISED 1-HV-5178 CLOSE INPUT & ANNUN INPUT	1-10-80	HL	FAL		GDC				
2	ISSUED FOR CONSTRUCTION	4-17-79	NBD	FAL	MKK				MRI	

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN120-5	3

SIZE B 11 x 17

BM 9510

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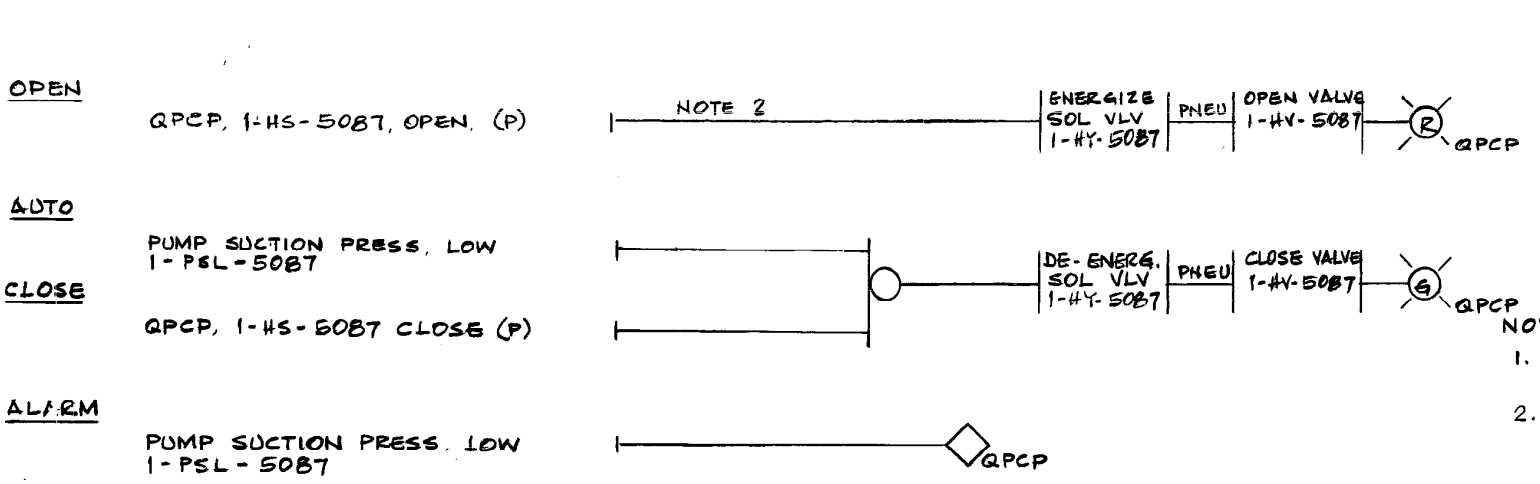
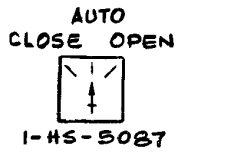
11-14-84 DATE: CAMERA OPERATOR SECTION SUPERVISOR





NOTE:

1. VALVE 1-HV-5087 SHOWN, 1-HV-5088 IS SIMILAR.
2. TO OPEN VALVE, CONTROL SWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW THE CIRCUIT TO SEAL-IN THROUGH VALVE POSITION SWITCH. AND TO RESET PRESSURE SWITCH AFTER LINE IS DEPRESSURIED.

REF DWG

P&ID 1X4DB161-1
E.D. 1X3D-BC-F02M

TABLE 1

VALVE	SWITCH	SOL VLV	PRESS SW
1-HV-5087	1-HS-5087	1-HY-5087	1-PSL-5087
8	8	8	8

VACUUM DEGASIFIER ISOL. VALVE

NO.	REVISIONS	DATE	DR	CHK.	SUP'V.	EGL	EGS	CHF. E.	P. E.	Q.A.E.	ENG. MGR.
4	CORRECTED REF. DWG.	2-15-85	ZBP	ALV							
3	REVISED OPEN INPUT, NOTE 2 AND ADDED REF DWGS	1-10-80	HL	FAL							DATE
2	ISSUED FOR CONSTRUCTION	4-6-79	HL	FAL	MKK			JBM	MRT		

BECHTEL
LOS ANGELES

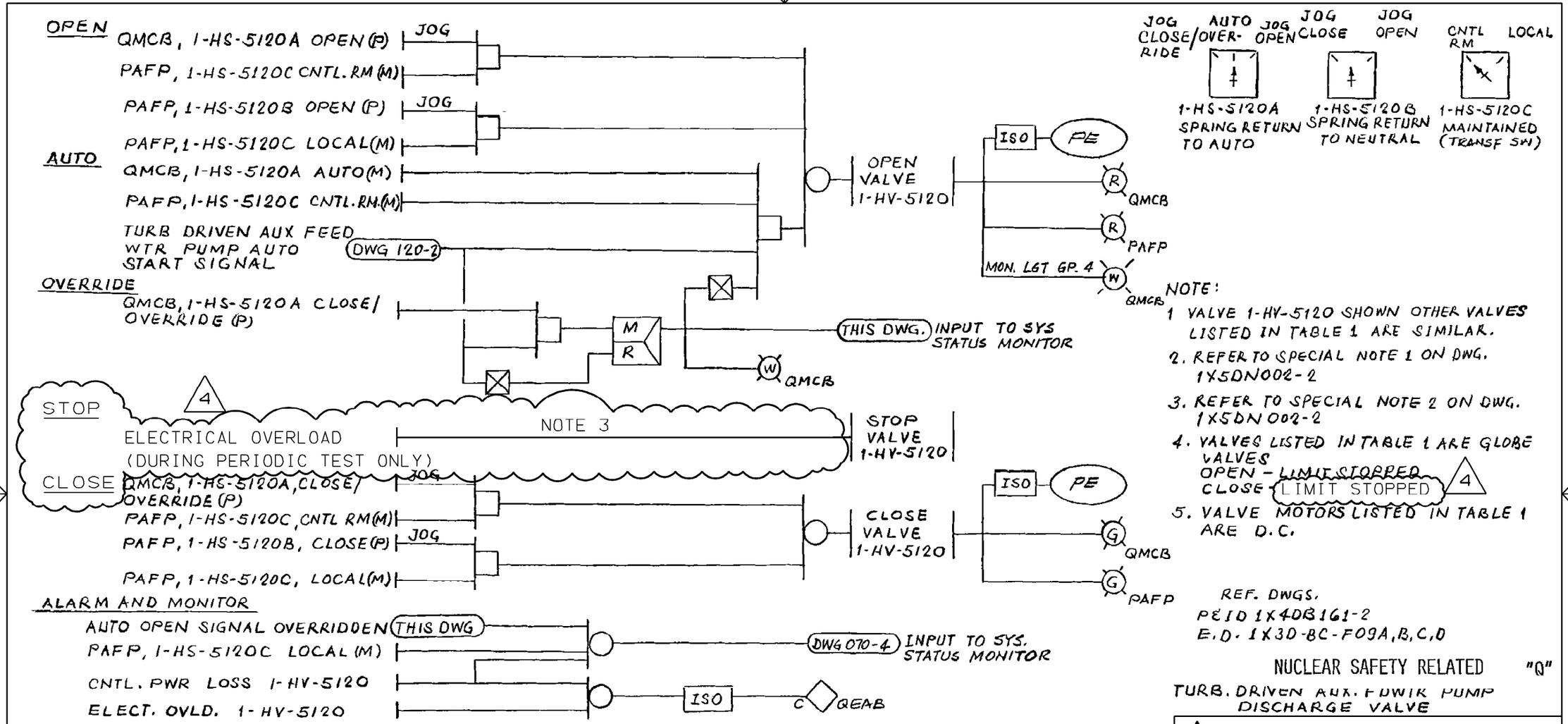
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN120-6	2

BM 9510

SIZE B 11 x 17



NOTE:

1. VALVE 1-HV-5120 SHOWN OTHER VALVES LISTED IN TABLE 1 ARE SIMILAR.
2. REFER TO SPECIAL NOTE 1 ON DWG. 1X5DN002-2
3. REFER TO SPECIAL NOTE 2 ON DWG. 1X5DN002-2
4. VALVES LISTED IN TABLE 1 ARE GLOBE VALVES
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED (4)
5. VALVE MOTORS LISTED IN TABLE 1 ARE D.C.

REF. DWGS.
P&ID 1X4DB161-2
E.D. 1X3D-BC-F09A,B,C,D

NUCLEAR SAFETY RELATED "Q"

TURB. DRIVEN AUX. FDWTR PUMP DISCHARGE VALVE

TABLE 1

TRAIN	PUMP NO	VALVE	SWITCHES		
		1-HV-5120	1-HS-5120A	1-HS-5120B	1-HS-5120C
C	1-1302-P4-001-000	5122	5122A	5122B	5122C
		5125	5125A	5125B	5125C
		5127	5127A	5127B	5127C

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM

AUXILIARY FEEDWATER VALVES TO STEAM GENERATOR

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER DCP 98-V1N0025	04/16/02	PM	TSL	RBH	

SCALE: NONE | DRAWING NO. 1X5DN121-1 | REV 4

JOB NO. 10604

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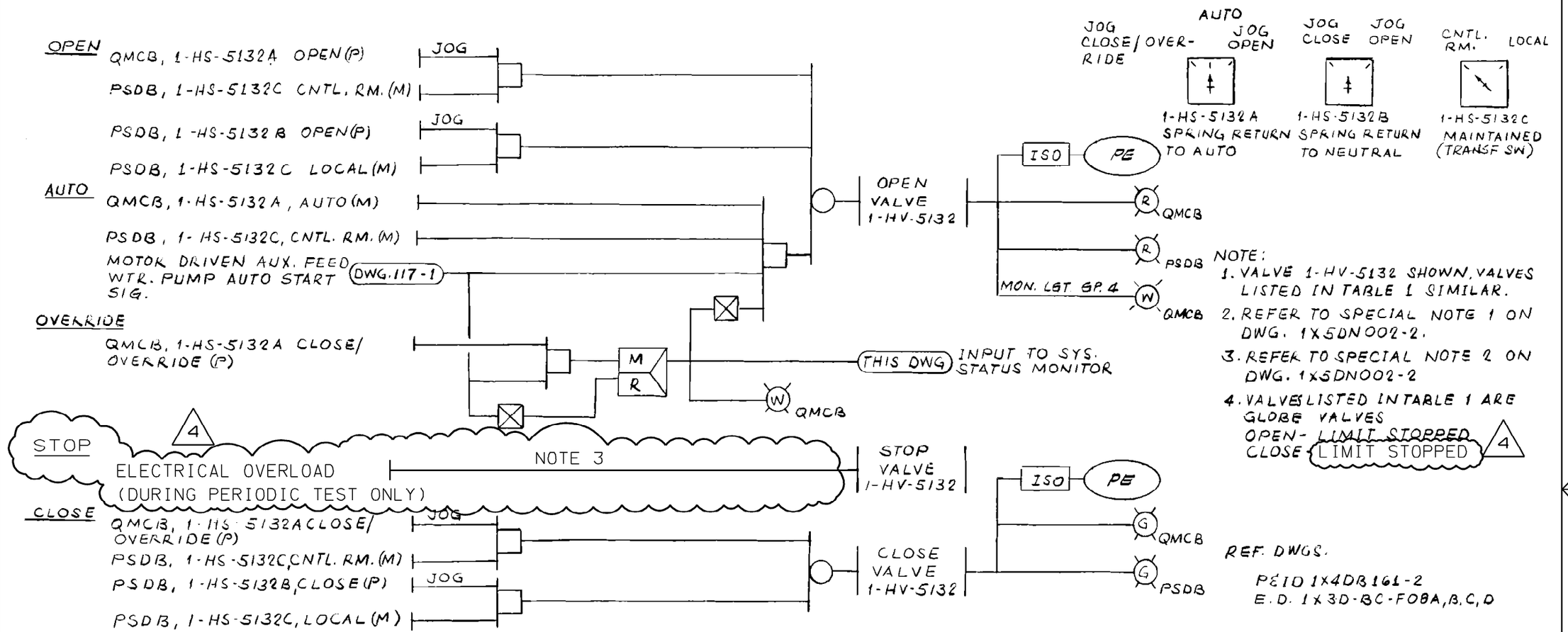

NOTE:
1. VALVE 1-HV-5132 SHOWN, VALVES LISTED IN TABLE 1 SIMILAR.
2. REFER TO SPECIAL NOTE 1 ON DWG. 1X5DN002-2.
3. REFER TO SPECIAL NOTE 2 ON DWG. 1X5DN002-2
4. VALVES LISTED IN TABLE 1 ARE GLOBE VALVES
   OPEN - LIMIT STOPPED
   CLOSE - LIMIT STOPPED

REF. DWGS.
P&ID 1X4DB161-2
E.D. 1X3D-BC-F08A,B,C,D

NUCLEAR SAFETY RELATED "Q"
MOTOR DRIVEN AUX. FDWTR. PUMPS DISCHARGE VALVE

TABLE 1

TRAIN	PUMP NO.	VALVE	SWITCHES		
			1-HS-5132A	1-HS-5132B	1-HS-5132C
B	1-1302-P4-002-M01	1-HV-5132	1-HS-5132A	1-HS-5132B	1-HS-5132C
		5134	5134A	5134B	5134C
A	1-1302-P4-003-M01	5137	5137A	5137B	5137C
		5139	5139A	5139B	5139C

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER DCP 98-V1N0025	04/16/02	PM	TSL	RBH	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AUXILIARY FEEDWATER VALVES TO STEAM GENERATOR

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN121-2	4

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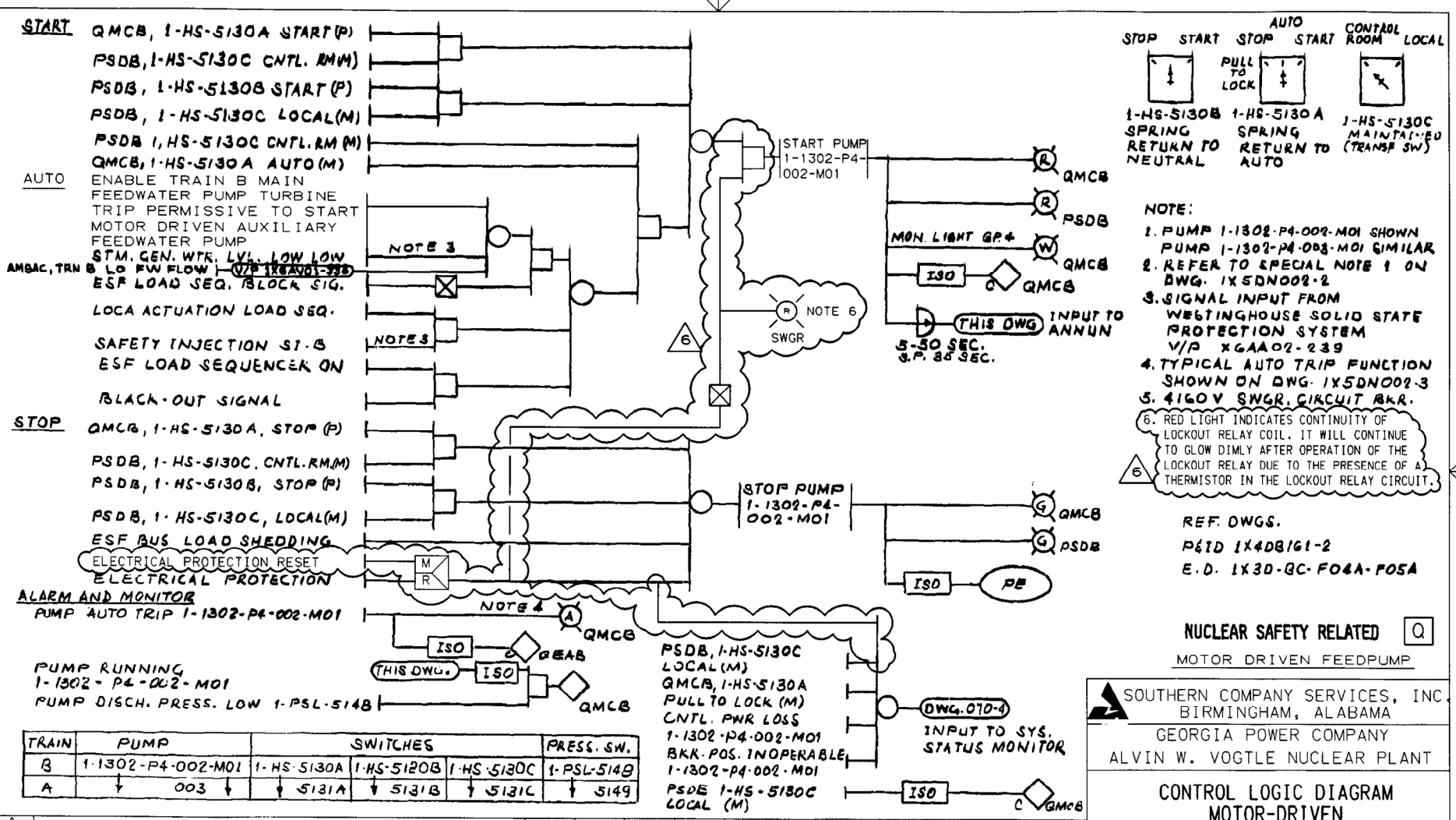
SIZE B 11x17



NOTE:

1. PUMP 1-1302-P4-002-M01 SHOWN PUMP 1-1302-P4-003-M01 SIMILAR
2. REFER TO SPECIAL NOTE 1 ON DWG. 1X5DN002-2
3. SIGNAL INPUT FROM WESTINGHOUSE SOLID STATE PROTECTION SYSTEM V/P XGAA02-239
4. TYPICAL AUTO TRIP FUNCTION SHOWN ON DWG. 1X5DN002-3
5. 4160 V SWGR. CIRCUIT BKR.
6. RED LIGHT INDICATES CONTINUITY OF LOCKOUT RELAY COIL. IT WILL CONTINUE TO GLOW DIMLY AFTER OPERATION OF THE LOCKOUT RELAY DUE TO THE PRESENCE OF A THERMISTOR IN THE LOCKOUT RELAY CIRCUIT.

REF. DWGS.

P&ID 1X4DB161-2

E.D. 1X3D-BC-F04A-F05A

NUCLEAR SAFETY RELATED Q

MOTOR DRIVEN FEEDPUMP

TRAIN	PUMP	SWITCHES			PRESS. SW.
B	1-1302-P4-002-M01	1-HS-5130A	1-HS-5120B	1-HS-5130C	1-PSL-5148
A	↓ 003	↓ 5131A	↓ 5131B	↓ 5131C	↓ 5149

NO.	REVISIONS	DATE	DR	CHK	SUPV	EGL	EGS	CHF E	PE	QAE
6	INCORP. ABN-20848	5/17/95	LVL	JMM	LFG					
5	REVISIONS	DATE	DR	CHK	APPV	DTL				
0	ISSUED FOR CONSTRUCTION	2/28/79		SEE MICROFILM FOR SIGNATURES						

ENG MGR

DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM MOTOR-DRIVEN AUXILIARY FEEDWATER PUMPS

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN122-1	6

SIZE B 11x17

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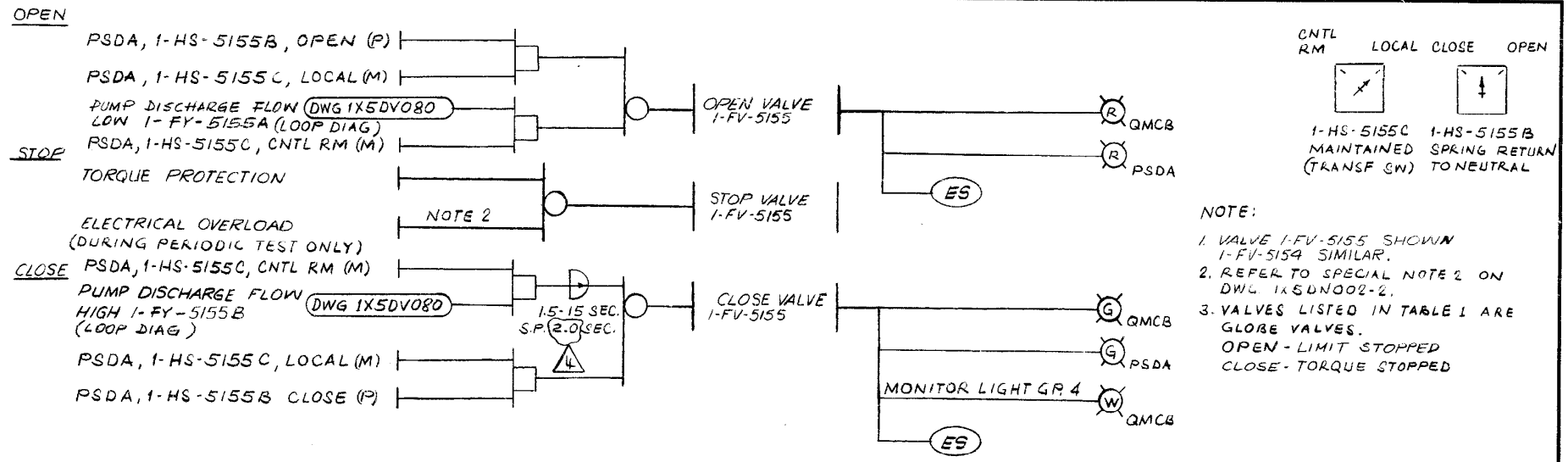

NOTE:

1. VALVE 1-FV-5155 SHOWN 1-FV-5154 SIMILAR.
2. REFER TO SPECIAL NOTE 2 ON DWG. 1X5DN002-2.
3. VALVES LISTED IN TABLE 1 ARE GLOBE VALVES.
   OPEN - LIMIT STOPPED
   CLOSE - TORQUE STOPPED

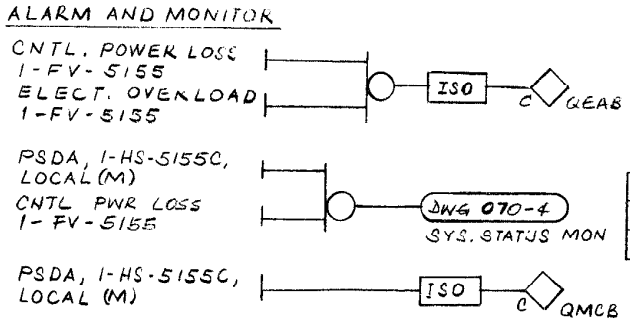

TABLE 1

TRAIN	HAND SWITCH	VALVE	FLOW SW (LOOP DIAG)	
A	1-HS-5155 B&C	1-FV-5155	1-FY-5155A	1-FY-5155B
B	5154 B&C	5154	5154A	5154B

REF. DWGS.
P&ID 1X4DB161-2
E.D. 1X3D-BC-F04B, F05B
LOOP DIAG 1X3D1073 & 080

NUCLEAR SAFETY RELATED "Q"

PUMP MINI FLOW VALVE

NO.	REVISIONS	DATE	DR.	CHK.	RPE	DM	ODPM	DPM	DPM	
3	ADDED COMPUTER INPUT PER I/O LIST REV. 5	2-15-85	ZEP	ACV						ENG. MGR.
4	INCORP. ABN 86-VIE007A001	10/22/87	TJS	SJP	JBP		RPZ			
2	ISSUED FOR CONSTRUCTION	2-28-79	NRD	FAL	MKK					DATE
NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM

MOTOR-DRIVEN AUXILIARY FEED-WATER PUMPS

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN122-2	4

SIZE B 11 x 17

BM 9510

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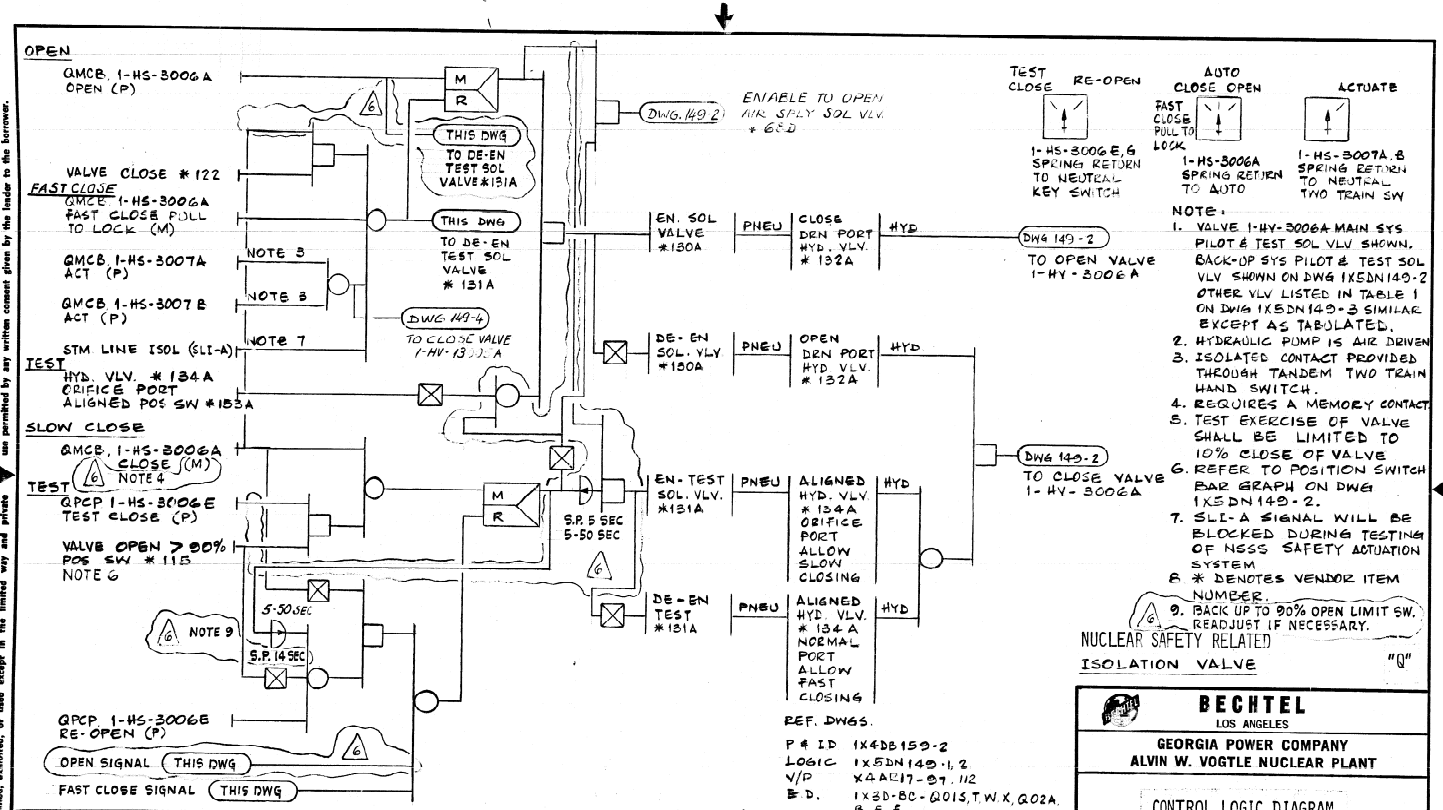

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
6	INCORPORATED DCN #1	10-31-86	RBR	ALV						
5	CLARIFIED "FAST CLOSE" FUNCTION (EDITORIAL ONLY)	6-11-86	ALV							
4	REVISED AS SHOWN	1-17-85	ALV							
6	ISSUED FOR CONSTRUCTION	5-31-79		FAL	PK					

ENG. MGR. | DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN STEAM SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN149-1	6

BM 9510

SIZE B 11 x 17

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DATE:
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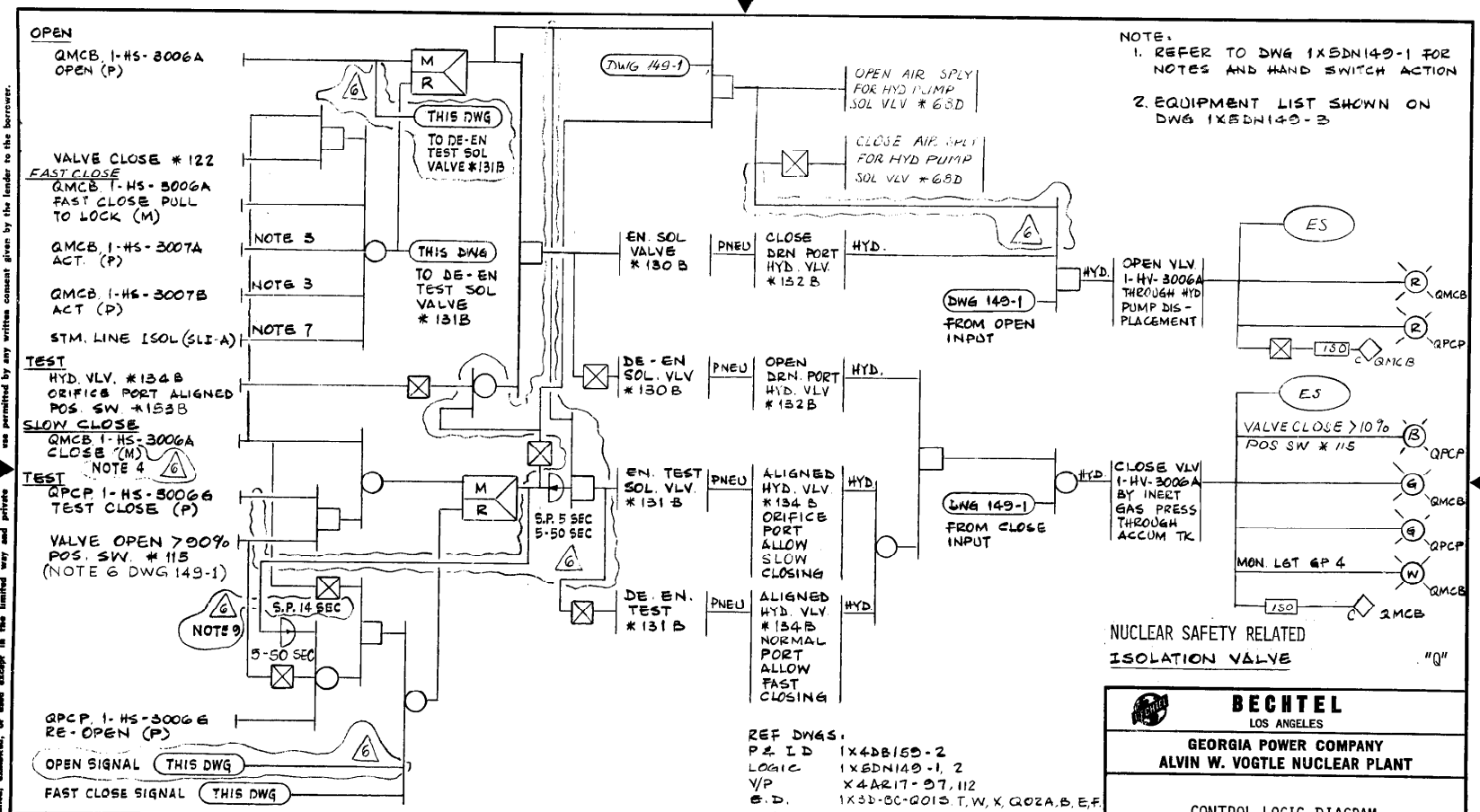

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q.A.E.
6	INCORPORATED DCN #1	10-30-86	RBR	AJJ						
5	CLARIFIED "FAST CLOSE" FUNCTION (EDITORIAL ONLY)	6-11-86	ALV							
4	REVISED AS SHOWN	1-17-85	ZEP							
0	ISSUED FOR CONSTRUCTION	5-31-79		FAL	PK					

ENG. MGR. / DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN STEAM SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN149-2	6

SIZE B 11 x 17

BM 9510

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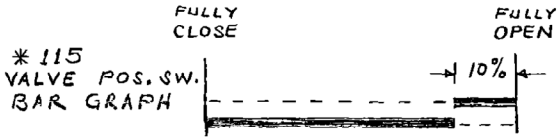
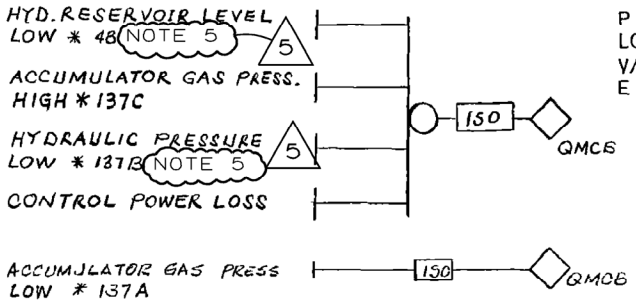
ALARMS AND MONITORS





REF DWGS

P & ID	1X4DB159-2
LOGIC	1X5DN149-1,2
V/P	X4AR17-97, 101
E.D.	1X3D-BC-Q01S,T,W,X Q02A,B,E,F

TABLE 1

TRAIN	SFTY SIG.	VALVE	HAND SW MAIN AND BACK-UP SYSTEM	HAND SW. MANUAL ISOL. ACTUATION	TEST SW MAIN SYS.	TEST SW. BACK-UP	SOL VALVE MAIN SYS.	SOL. VALVE BACK-UP SYS.	HYD. VALVE MAIN SYS.	HYD VALVE BACK-UP SYS	SOL.VLV TEST EXERC. MAIN SYS.	SOL VLV. TEST EXERC. BACK UP SYS.	HYD ORIFICE VALVE MAIN SYS.	HYD. ORIFICE VALVE BACK-UP SYS	HYD. ORIFICE VLV POS. SW MAIN SYS	HYD ORIFICE VLV. POS. SW BACK UP SYS.	TEST EXERC POS SW.	VALVE POS SW	HYD RSVR LEVEL SWITCH	HYD PRESS SWITCH	ACCUM GAS PRESS SWITCH	AIR SUPPLY SHUT OFF SOL VLV
A	SLI-A	1-HV-3006A	1-HS-3006A	1-HS 3007A 1-HS-3007B	1-HS-3006E	1-HS-3006G	*130A	*130B	*132A	*132B	*131A	*131B	*134A	*134B	*153A	*153B	*115	*122	* 48	*137B	*137A	*68D
B	-B	-3006B	-3006B		-3006F	-3006H																
A	-A	-3016A	-3016A		-3016E	-3016G																
B	-B	-3016B	-3016B		-3016F	-3016H																
A	-A	-3026A	-3026A		-3026E	-3026G																
B	-B	-3026B	-3026B		-3026F	-3026H																
A	-A	-3036A	-3036A		-3036E	-3036G																
B	-B	-3036B	-3036B		-3036F	-3036H																

VALVE	OPEN POSITION SWITCH		TEST POSITION SWITCH		CLOSE POSITION SWITCH	
	VENDOR	BECHTEL	VENDOR	BECHTEL	VENDOR	BECHTEL
1HV-3006A	* 111	1ZSO-3006A	* 115	1ZSOT-3006A	* 122	1ZSC-3006A
1HV-3006B		1ZSO-3006B		1ZSOT-3006B		1ZSC-3006B
1HV-3016A		1ZSO-3016A		1ZSOT-3016A		1ZSC-3016A
1HV-3016B		1ZSO-3016B		1ZSOT-3016B		1ZSC-3016B
1HV-3026A		1ZSO-3026A		1ZSOT-3026A		1ZSC-3026A
1HV-3026B		1ZSO-3026B		1ZSOT-3026B		1ZSC-3026B
1HV-3036A		1ZSO-3036A		1ZSOT-3036A		1ZSC-3036A
1HV-3036B		1ZSO-3036B		1ZSOT-3036B		1ZSC-3036B

NOTE.

1. LOGIC SHOWN ON DWG. 1X5DN149-1 & 2
2. * DENOTES VENDOR ITEM NUMBER.
3. VALVE BAR GRAPH POSITION SWITCH SOLID LINE DENOTES CONTACT CLOSED.
4. REFER TO DWG 1X5DN149-1 FOR OTHER NOTES
5. THE HYD RESV LEVEL LOW SWITCH AND HYD PRESSURE LOW SWITCH ARE NOT SAFETY RELATED. SEE CALC. X5CFEMA1816. (5)

NUCLEAR SAFETY RELATED

EQUIPMENT LIST AND ALARM

Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN STEAM SYSTEM

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
5	INCORP. PER ABN-50579	7-16-98	ELC	EOG	MWD	
0	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SCALE: NONE | JOB NO. 10604 | DRAWING NO. 1X5DN149-3 | REV 5

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CAD NAME 15DN1493 SIZE B 11x17

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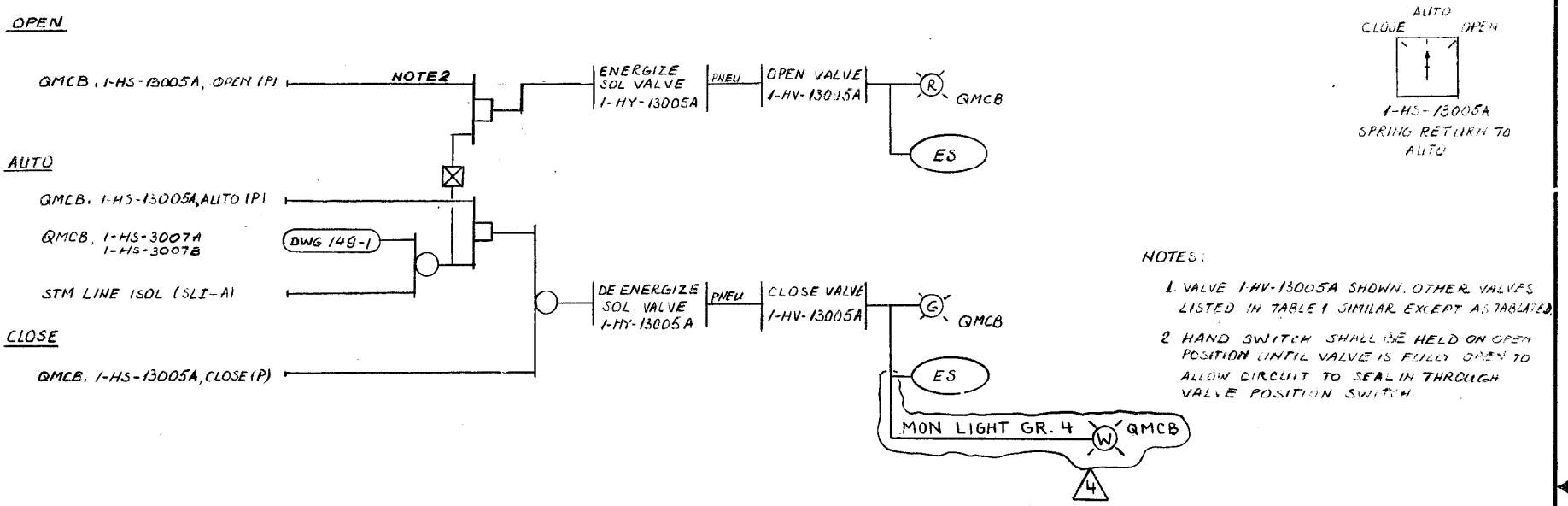


NOTES:

1. VALVE 1-HV-13005A SHOWN. OTHER VALVES LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATED.
2. HAND SWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH.

REF DWGS
P&ID 1X4DB15[illegible]-2
E.D. 1X3D-BC-Q02H, J, 4

TABLE 1

HAND VALVE	HAND SWITCH	SAFETY SIGNAL	SOLENOID VALVE
1-HV-13005A	1-HS-13005A	SLI-A	1-HY-13005A
↓ B	↓ B	-B	↓ B
1-HV-13006A	1-HS-13006A	-A	1-HY-13006A
↓ B	↓ B	-B	↓ B
1-HV-13007A	1-HS-13007A	-A	1-HY-13007A
↓ B	↓ B	-B	↓ B
1-HV-13008A	1-HS-13008A	-A	1-HY-13008A
↓ B	↓ B	-B	↓ B

MSIV BYPASS VALVES

NUCLEAR SAFETY RELATED "Q"

NO.	REVISIONS	DATE	DR	CHK.	RPE	DM	ODPM	DPM	DPM
4	REV. TO INC. DCN 1	10-24-84	JLH	[illegible]	JBP		[illegible]		
3	CLARIFY HS INPUT	SEE MICROFILM FOR SIGNATURES							
0	ISSUED FOR CONSTRUCTION	5-25-84	ALV	[illegible]		[illegible]	[illegible]	[illegible]	[illegible]
NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.

ENG. MGR. DATE

BECHTEL LOS ANGELES

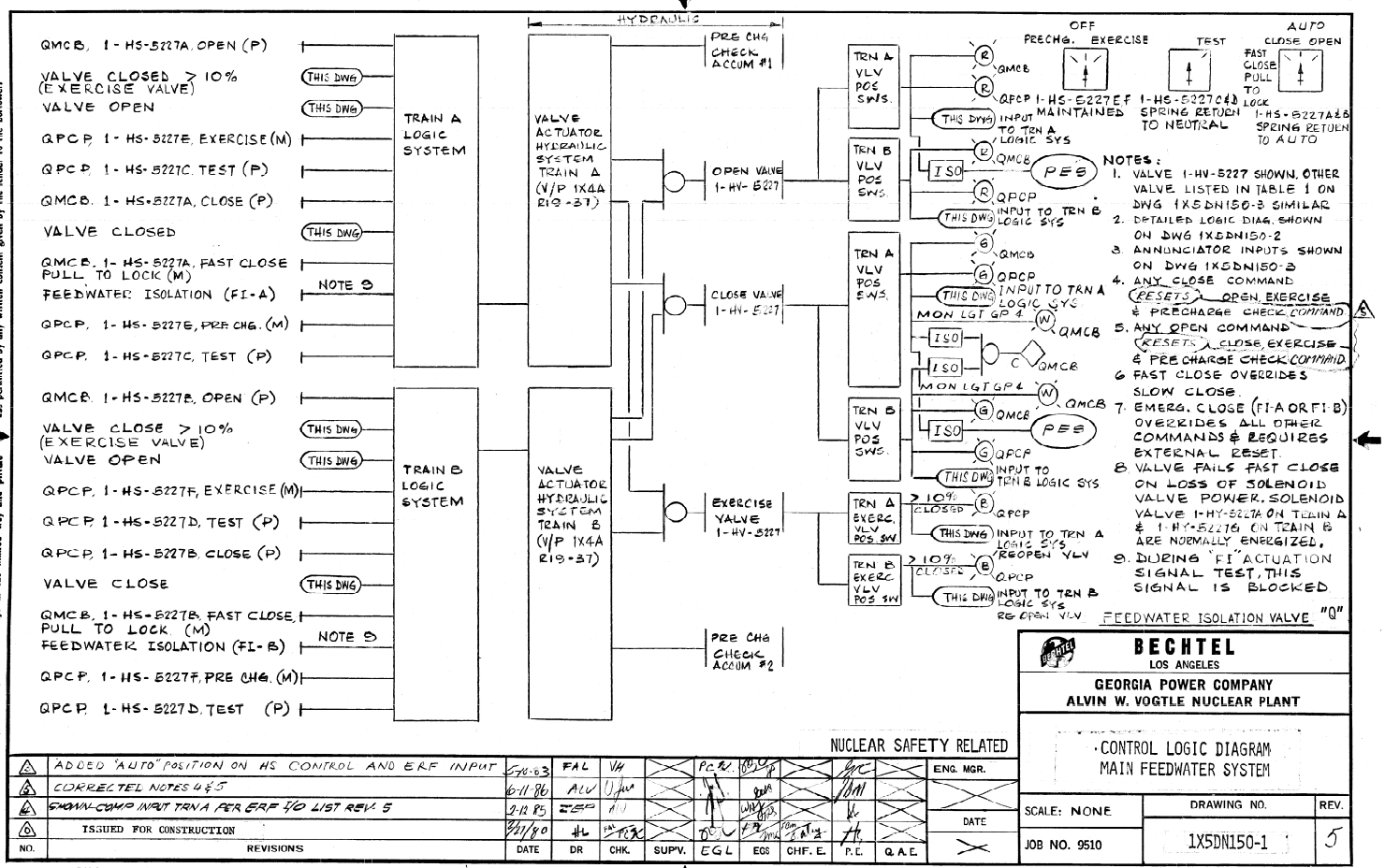
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN STEAM SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN149-4	4

BM 9510

SIZE B 11 x 17



NUCLEAR SAFETY RELATED

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EQS	CHF. E.	P.E.	Q.A.E.
3	ADDED "AUTO" POSITION ON HS CONTROL AND ERF INPUT	5/6-03	FAL	VH		PCR				
2	CORRECTED NOTES 4 & 5	6-11-86	ALV							
1	SHOWN COMP INPUT TRN A PER ERF I/O LIST REV. 5	2-12-85	REP							
0	ISSUED FOR CONSTRUCTION	8/21/80	HL							

ENG. MGR. | DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN FEEDWATER SYSTEM

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN150-1	5

BM 9510 P&ID REV.

SIZE B 11 x 17

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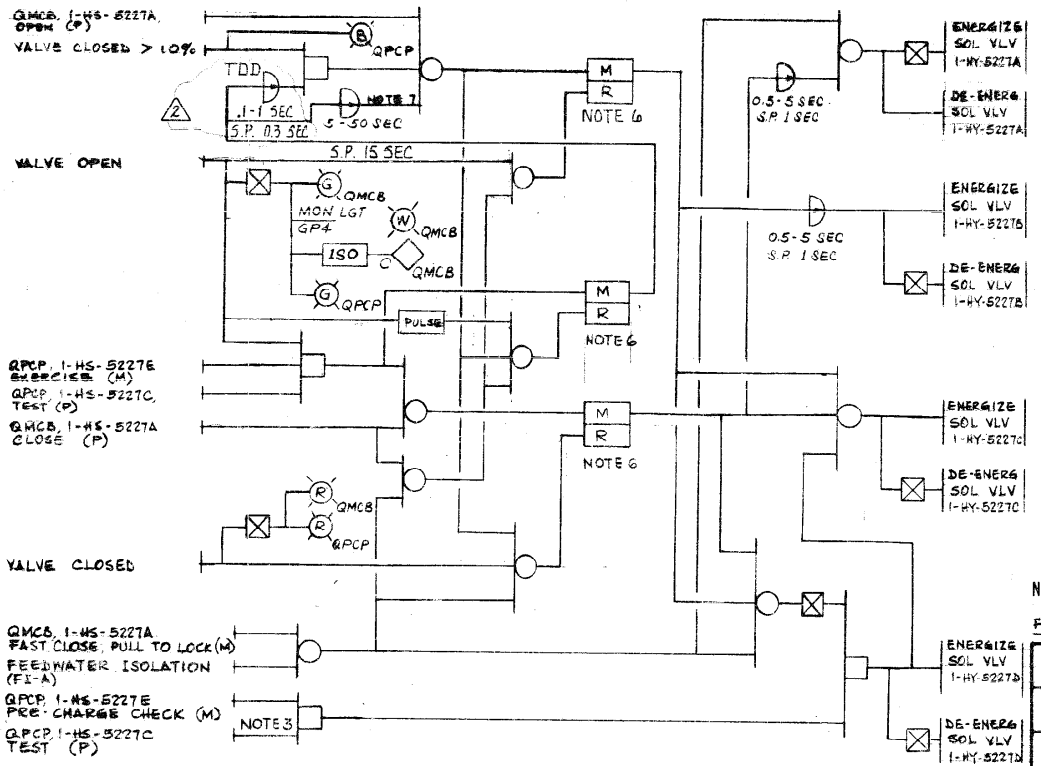
6-12-86 DATE — CAMERA OPERATOR — SECTION SUPERVISOR



NOTES:

1. VALVE 1-HV-8227 LOGIC TRAIN A SHOWN, TRAIN B SIMILAR, OTHER VALVES LISTED IN TABLE 1 ON DWG 1X5DN150-3.
2. HAND SWITCH POSITION SHOWN ON DWG 1X5DN180-1.
3. HOLD DURING PRECHARGE ACCUMULATOR PRESSURE CHECK AND THEN RELEASE.
4. SOLENOID VALVES ARE 125VDC.
5. SEQUENCE OF OPERATION SHOWN IN TABLE 2 ON DWG 1X5DN150-3
6. RESET HAS PRIORITY OVER MEMORY
7. TIME DELAY SHALL BE SET MORE THAN THE TIME THE VALVE REACHES 10% CLOSE FROM FULLY OPEN POSITION, THIS TIME DELAY SIGNAL IS A BACK-UP TO TEST EXERCISE.

REF. DWGS.

P&ID 1X4DB168-3

E.D. 1X30-BC-0060, 60, 7C, 7D, 8C, 8D, 9C, 9D

V/P 1X4AR19-37, 39 & 57

NUCLEAR SAFETY RELATED "Q"

FEEDWATER ISOLATION VALVE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN FEEDWATER SYSTEM

NO.	REVISIONS	DATE	DR.	CHK	RPE	DM	ODPM	DPM	DPM
2	INCORP. ABN 87-VIE207A001	2/20	M.W.D.	CJP	SLW				
1	ADDED TIME DELAY SET POINT & REVISED AS INDICATED	7-9-8	FAL	V.V.					
2	[illegible] FOR CONSTRUCTION	3/31/80	sh.	FAL PCK					
NO.	REVISIONS	DATE	DR.	CHK.	SUPV.	E.G.L.	EGS	CHF. E.	P.E.	Q.A.E.

ENG. MGR. DATE

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN150-2	2

SIZE B 11 x 17

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TABLE 1 (EQUIPMENT LIST)

VALVE	HAND SWITCHES			SOLENOID VALVES				SFTY ACT	TRAIN	HYDRAULIC PRESS SW	AIR RESERV. PRESS SW	HYD. TK LVL SW	HYD. PRESS. RELIEF VLV.
1-HV-5227	1-HS-5227A	1-HS-5227C	1-HS-5227E	1-HY-5227A	1-HY-5227B	1-HY-5227C	1-HY-5227D	FI-A	A	1-PS-5227A	1-PS-5227B	1-LS-5227	1PSV-5227C
	27B	27D	27F	27G	27H	27J	27K	B	B		27C		1PSV-5227D
28	28A	28C	28E	28A	28B	28C	28D	A	A	28A	28B	28	1PSV-5228C
	28B	28D	28F	28G	28H	28J	28K	B	B		28C		1PSV-5228D
29	29A	29C	29E	29A	29B	29C	29D	A	A	29A	29B	29	1PSV-5229C
	29B	29D	29F	29G	29H	29J	29K	B	B		29C		1PSV-5229D
30	30A	30C	30E	30A	30B	30C	30D	A	A	30A	30B	30	1PSV-5230C
	30B	30D	30F	30G	30H	30J	30K	B	B		30C		1PSV-5230D

4

NOTES.
1. TABLE 2 SHOWS VALVE 1-HV-5227 SOLENOID VALVE'S SEQUENCE OF OPERATION, OTHER VALVES SIMILAR
2. E - ENERGIZE
   D - DE-ENERGIZE
   TD - TIME DELAY
3. LOGIC DIAGRAM SHOWN ON DWG 1X5DN150-1 & 2
4. ANNUNCIATOR INPUTS FOR VALVE 1-HV-5227 SHOWN OTHER VALVES ALARM INPUT SIMILAR AS LISTED IN TABLE 1.

TABLE 2 (SEQUENCE OF OPERATION)

CONTROL ACTION	STEP	SOLENOID VALVE STATUS (TRAIN A)				SOLENOID VALVE STATUS (TRAIN B)			
		1-HY-5227A	1-HY-5227B	1-HY-5227C	1-HY-5227D	1-HY-5227G	1-HY-5227H	1-HY-5227J	1-HY-5227K
SLOW OPEN	FULL CLOSE	E	D	D	D	E	D	D	D
	OPENING		TD, E	E			TD, E	E	
	FULL OPEN		D	D			D	D	
SLOW CLOSE	FULL OPEN								
	CLOSING	TD, D		E		TD, D		E	
	FULL CLOSE	E		D		E		D	
FAST OR EMERG CLOSE	FULL OPEN								
	CLOSING	D				D			
	FULL CLOSE								
EXERCISE 10% VALVE STROKE	FULL OPEN	E				E			
	CLOSING	TD, D		E		TD, D		E	
	RE-OPEN	E	TD, E			E	TD, E		
	FULL OPEN		D	D			D	D	
ACCUM INERT GAS PRE CHG CHECK	OPEN OR CLOSE								
	ACCUM HYDR FLUID DRN			E	E			E	E
	RECHG ACCUM HYDR FLUID			D	D			D	D

ALARMS

VALVE HYDRAULIC PRESS HIGH 1-PS-5227A → QMCB

TRAIN A AIR RESERVOIR PRESS LOW 1-PS-5227B
TRAIN B AIR RESERVOIR PRESS LOW 1-PS-5227C
→ QMCB

HYDRAULIC TANK LEVEL LOW, 1-LS-5227 → QMCB

REF DWGS:
P&ID 1X4DB168-3
E.D. 1X3D-BC-C06C, 6D, 7C, 7D, 8C, 8D, 9C, 9D

NUCLEAR SAFETY RELATED "Q"

ALARMS & EQUIPMENT LIST

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER DCP 00-V1N0045	12-12-03	RCR	RBH	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM MAIN FEEDWATER SYSTEM

SCALE: NONE	DRAWING NO.	REV
JOB NO. 10604	1X5DN150-3	4

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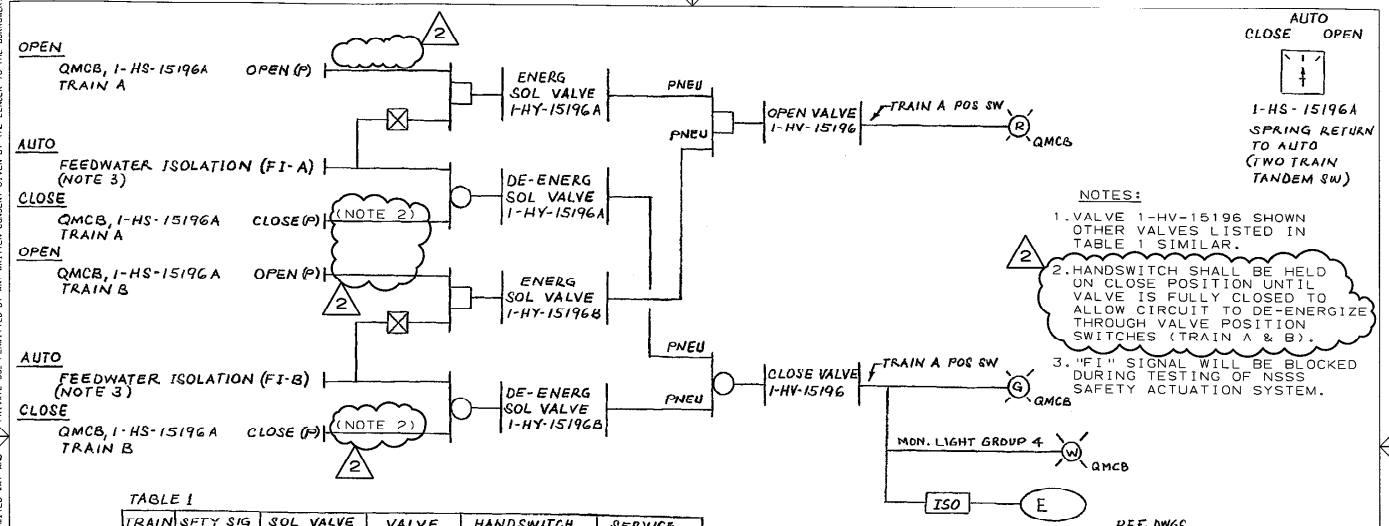
SIZE B 11x17



NOTES:

1. VALVE 1-HV-15196 SHOWN OTHER VALVES LISTED IN TABLE 1 SIMILAR.
2. HANDSWITCH SHALL BE HELD ON CLOSE POSITION UNTIL VALVE IS FULLY CLOSED TO ALLOW CIRCUIT TO DE-ENERGIZE THROUGH VALVE POSITION SWITCHES (TRAIN A & B).
3. "FI" SIGNAL WILL BE BLOCKED DURING TESTING OF NSSS SAFETY ACTUATION SYSTEM.

TABLE 1

TRAIN	SFTY SIG	SOL VALVE	VALVE	HANDSWITCH	SERVICE
A	FI-A	1-HY-15196A	1-HV-15196	1-HS-15196A	STM GEN. 001
B	B	15196B			
A	A	15197A	15197	15197A	002
B	B	15197B			
A	A	15198A	15198	15198A	003
B	B	15198B			
A	A	15199A	15199	15199A	004
B	B	15199B			

REF. DWGS
P&ID 1X4DB168-3
E D 1X3D-BC-C04P

MAIN FDWTR. ISOLATION BYPASS VALVE

NUCLEAR SAFETY RELATED "Q"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
MAIN FEEDWATER SYSTEM

2	INCORP. ABN-87-V1E0311A001	4/11/96	GS	RB	EDH							ENG MGR
	REVISIONS	DATE	DR	CHK	APPV	DTL						
1	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES										DATE
0	REVISIONS											

SCALE: NONE | DRAWING NO. 1X5DN150-4 | REV 2
JOB NO. 10604
SIZE B 11x17

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OPEN

QMCB, 1-HS-15214 OPEN (P) NOTE 1

AUTO

AB, CVCS. LETDOWN HX RM R-A07
HIGH TEMP. 1-TY-15214D (1X5DV202)

AB, CVCS VLV GALLERY RM R-A08
HIGH TEMP 1-TY-15214F (1X5DV203)

AB, PIPE PENET RM R-A09
HIGH TEMP 1-TY-15214G (1X5DV204)

CLOSE

QMCB, 1-HS-15214 CLOSE (P)

ENERGIZE SOL VALVE 1-HY-15214 | PNEU | OPEN VALVE 1-HV-15214 | (R) QMCB

DE-ENERG SOL VALVE 1-HY-15214 | PNEU | CLOSE VALVE 1-HV-15214 | (G) QMCB

AUTO

CLOSE OPEN

1-HS-15214
SPRING RETURN TO AUTO

NOTES.

1. HANDSWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH.
2. TRAIN B POWER SUPPLY.

ALARM

AB, CVCS LETDOWN HX RM R-A07
HIGH TEMP 1-TY-15214J (1X5DV202)

AB, CVCS VLV GALLERY RM R-A08
HIGH TEMP 1-TY-15214K (1X5DV203)

AB, PIPE PENET RM R-A09
HIGH TEMP 1-TY-15214L (1X5DV204)

(C) QPCP

REF DWGS.
P & ID 1X4DB114
E. D. 1X3D-BD-C015.

NUCLEAR SAFETY RELATED "Q"

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EE&L	EGS	CHF. E.	P. E.	Q.A.E.	
3	INPUT PARAMETER CHANGED FROM FY-15214A TO PY-15214A PER P&ID	6-13-83	AV	FAL		PC					ENG. MGR.
2	INCORPORATED DCN #1 (YFCRB-4353)	9-17-80	ALV	NCB		PC	JGG		WHK		
1	CHANGED SWITCH ACTION PER PROCESS REQUIREMENTS	4-2-81	ALV	FAL		PC					DATE
0	ISSUED FOR CONSTRUCTION	1-29-80	FAL	XCB		PC			ZER		

BM 9510 P&ID REV.

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CVCS LETDOWN HX INLET
PIPE BREAK ROOM PROTECTION

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN151-1	5

SIZE B 11 x 17

ILN 9-14-91

4-11-84

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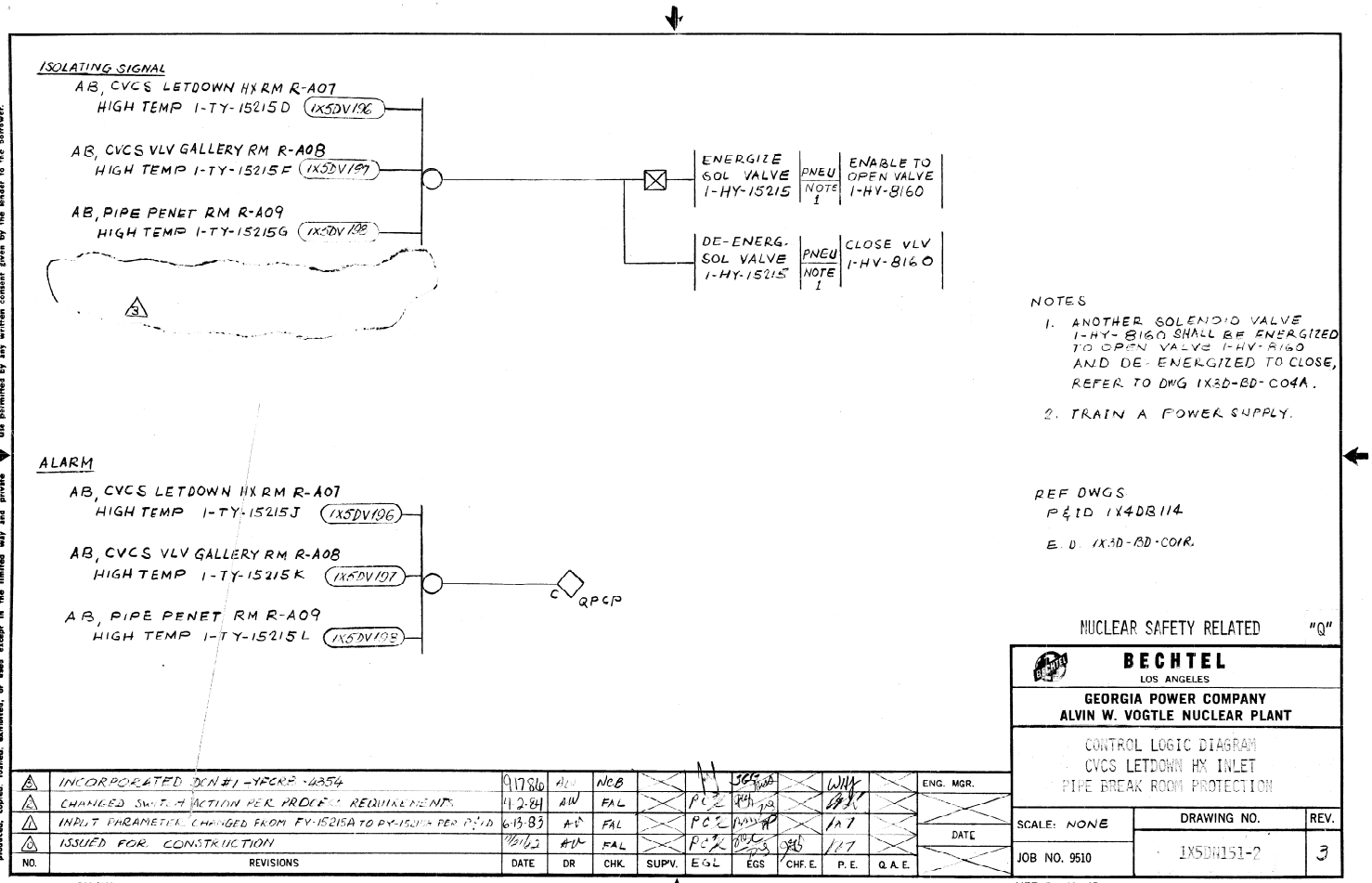

NOTES

1. ANOTHER SOLENOID VALVE 1-HY-8160 SHALL BE ENERGIZED TO OPEN VALVE 1-HV-8160 AND DE-ENERGIZED TO CLOSE, REFER TO DWG 1X3D-BD-C04A.
2. TRAIN A POWER SUPPLY.

REF DWGS

P&ID 1X4DB114

E.D. 1X3D-BD-C01R.

NUCLEAR SAFETY RELATED "Q"

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
CVCS LETDOWN HX INLET
PIPE BREAK ROOM PROTECTION

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DW151-2	3

SIZE B 11 x 17

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.
4	INCORPORATED DCN #1 -YFCRF-4854	9/17/86	ALL	NCB						
3	CHANGED SWITCH ACTION PER PROCESS REQUIREMENTS	4-2-84	AW	FAL		PCE				
2	INPUT PARAMETER CHANGED FROM FY-15215A TO PY-15215 PER P&ID	6-13-83	AW	FAL		PCE				
1	ISSUED FOR CONSTRUCTION	11/16/82	AW	FAL		PCE				

ENG. MGR. / DATE

BM 9510 P&ID REV

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS, ON THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT SUBMITTED TO MICROGRAPHICS.

DATE: 5-18-86 CAMERA OPERATOR SECTION SUPERVISOR

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OPEN

QPCP, 1-HS-15212A OPEN (P) — NOTE 2

AUTO

- AB, PIPE PENET RM R-B08 HIGH TEMP 1-TY-15212E — 1X5DV209
- AB, PULL OUT AREA RM R-C106 HIGH TEMP 1-TY-15212F — 1X5DV210
- AB, VESTIBULE AREA RM R-C107 HIGH TEMP 1-TY-15212G — 1X5DV211
- AB, SGB HX RM R-C108 HIGH TEMP 1-TY-15212H — 1X5DV212
- SG1 BLOWDOWN LINE HIGH FLOW 1-FY-15212A — 1X5DV228

ENERGIZE SOL VALVE 1-HY-15212A — PNEU — OPEN VALVE 1-HV-15212A — (R) QPCP

DE-ENERG SOL VALVE 1-HY-15212A — PNEU — CLOSE VALVE 1-HV-15212A — (G) QPCP

CLOSE

QPCP, 1-HS-15212A CLOSE (P)

ALARM

- AB, PIPE PENET RM R-B08 HIGH TEMP 1-TY-15212J — 1X5DV209
- AB, PULL OUT AREA RM R-C106 HIGH TEMP 1-TY-15212K — 1X5DV210
- AB, VESTIBULE AREA RM R-C107 HIGH TEMP 1-TY-15212L — 1X5DV211
- AB, SGB HX RM R-C108 HIGH TEMP 1-TY-15212M — 1X5DV212

△2 — (C) QPCP

AUTO
CLOSE OPEN

1-HS-15212A SPRING RETURN TO AUTO

NOTES:

1. VALVE 1-HV-15212A SHOWN. OTHER VALVES LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATED
2. HANDSWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH.

REF. DWGS.

P & ID 1X4DB179-1, 159-1, 159-3
LOOP DIAG. 1X5DV228, 230, 232, 234
E.D. 1X3D-BC-Q01J

SCANNED DATE: SEP 16 1991

NUCLEAR SAFETY RELATED "Q"

TABLE 1

SERVICE	TRAIN	SOL VALVE	VALVE	HAND SW.	TEMP SW	FLOW SW.
SG 1 BLOWDOWN LINE	A	1-HY-15212A	1-HV-15212A	1-HS-15212A	1-TY-15212 E, F, G & H	1-FY-15212A
2	A	-15212B	-15212B	-15212B		-15212B
3	A	-15212C	-15212C	-15212C		-15212C
4	A	-15212D	-15212D	-15212D		-15212D

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.	ENG. MGR. / DATE
△3	DELETED MONITORING LIGHTS. VALVES NOT CONSIDERED CNMT. ISOLATION	10-18-82	AS	FAL		PCS					
△2	DELETED PRESSURE PARAMETER AND ADDED ISOLATION VALVE	6-25-82	AS	FAL		PCS					
△1	REVISED PER REF P&ID REV. 5 MARKED-UP	2-8-82	FAL			PCS					
△0	ISSUED FOR CONSTRUCTION	12-21-80	FAL	KSB		PCS					

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AB STM GEN BLOWDOWN
PIPE BREAK ROOM PROTECTION

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 9510	1X5DN152-1	3

BM 9510 P&ID 179-1 REV. 5, 159-1 REV 7, 159-3 REV. 1

SIZE B 11 x 17

11-14-94

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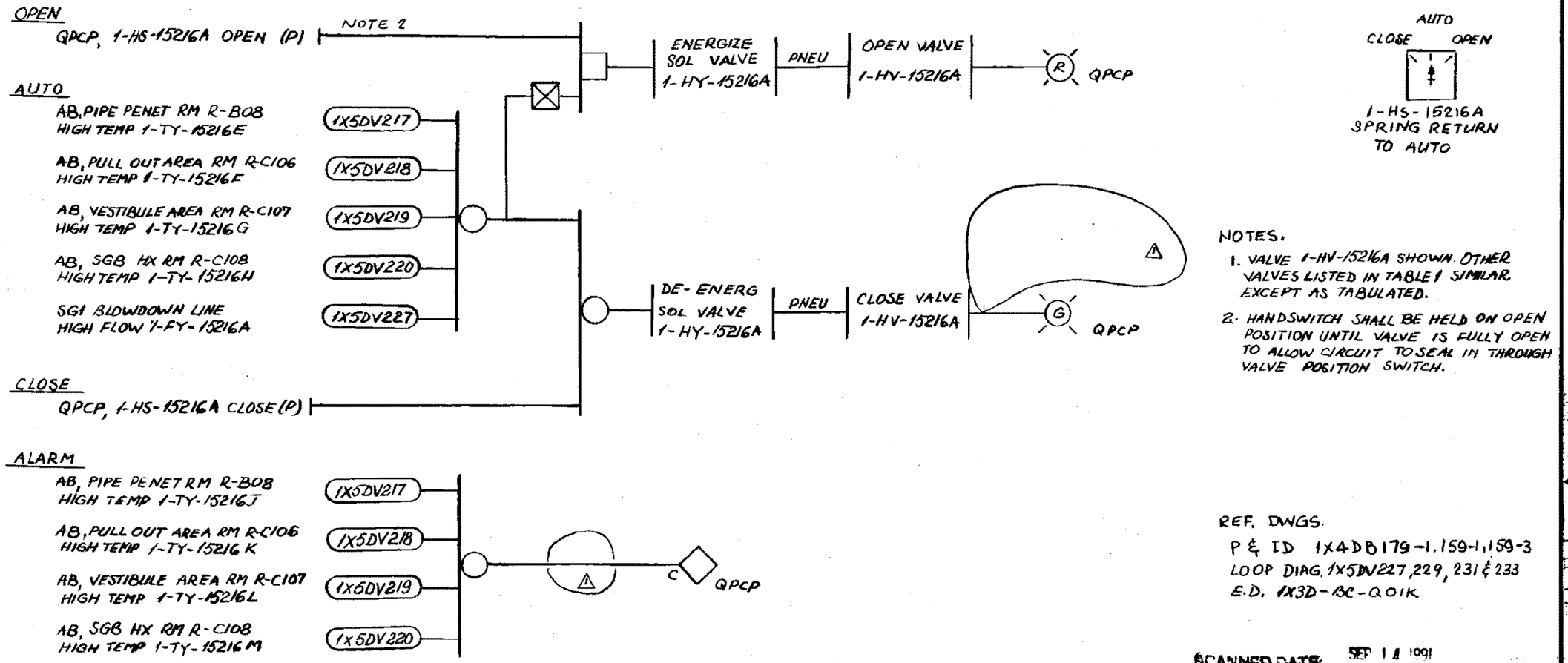

ALARM

NOTES.

1. VALVE 1-HV-15216A SHOWN. OTHER VALVES LISTED IN TABLE 1 SIMILAR EXCEPT AS TABULATED.
2. HANDSWITCH SHALL BE HELD ON OPEN POSITION UNTIL VALVE IS FULLY OPEN TO ALLOW CIRCUIT TO SEAL IN THROUGH VALVE POSITION SWITCH.

REF. DWGS.
P & ID 1X4DB179-1, 159-1, 159-3
LOOP DIAG. 1X5DV227, 229, 231 & 233
E.D. 1X3D-BC-Q01K

SCANNED DATE SEP 1 1 1991

NUCLEAR SAFETY RELATED "Q"

TABLE 1

SERVICE	TRAIN	SOL VALVE	VALVE	HAND SW.	TEMP SW.	FLOW SW.
SG 1 BLOWDOWN LINE	B	1-HY-15216A	1-HV-15216A	1-HS-15216A	1-TY-15216E, F, G & H	1-FY-15216A
2	B	-15216B	-15216B	-15216B		-15216B
3	B	-15216C	-15216C	-15216C		-15216C
4	B	-15216D	-15216D	-15216D		-15216D

NO.	REVISIONS	DATE	DR	CHK.	SUPV.	EGL	EGS	CHF. E.	P. E.	Q. A. E.
△										
△										
1	DELETED MONITORING LIGHTS. VALVES NOT CONSIDERED CNMT. ISOLATION	10-18-82	RV	FAL	X	PCK	PCK	X	MN	X
0	ISSUED FOR CONSTRUCTION	6-25-82	AV	FAL	X	PCK	PCK	JB	IAT	X

ENG. MGR. DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTROL LOGIC DIAGRAM
AB STM GEN BLOWDOWN
PIPE BREAK ROOM PROTECTION

SCALE: NONE | DRAWING NO. 1X5DM152-2 | REV. 4

JOB NO. 9510

BM 9510

SIZE B 11 x 17

1-11-94

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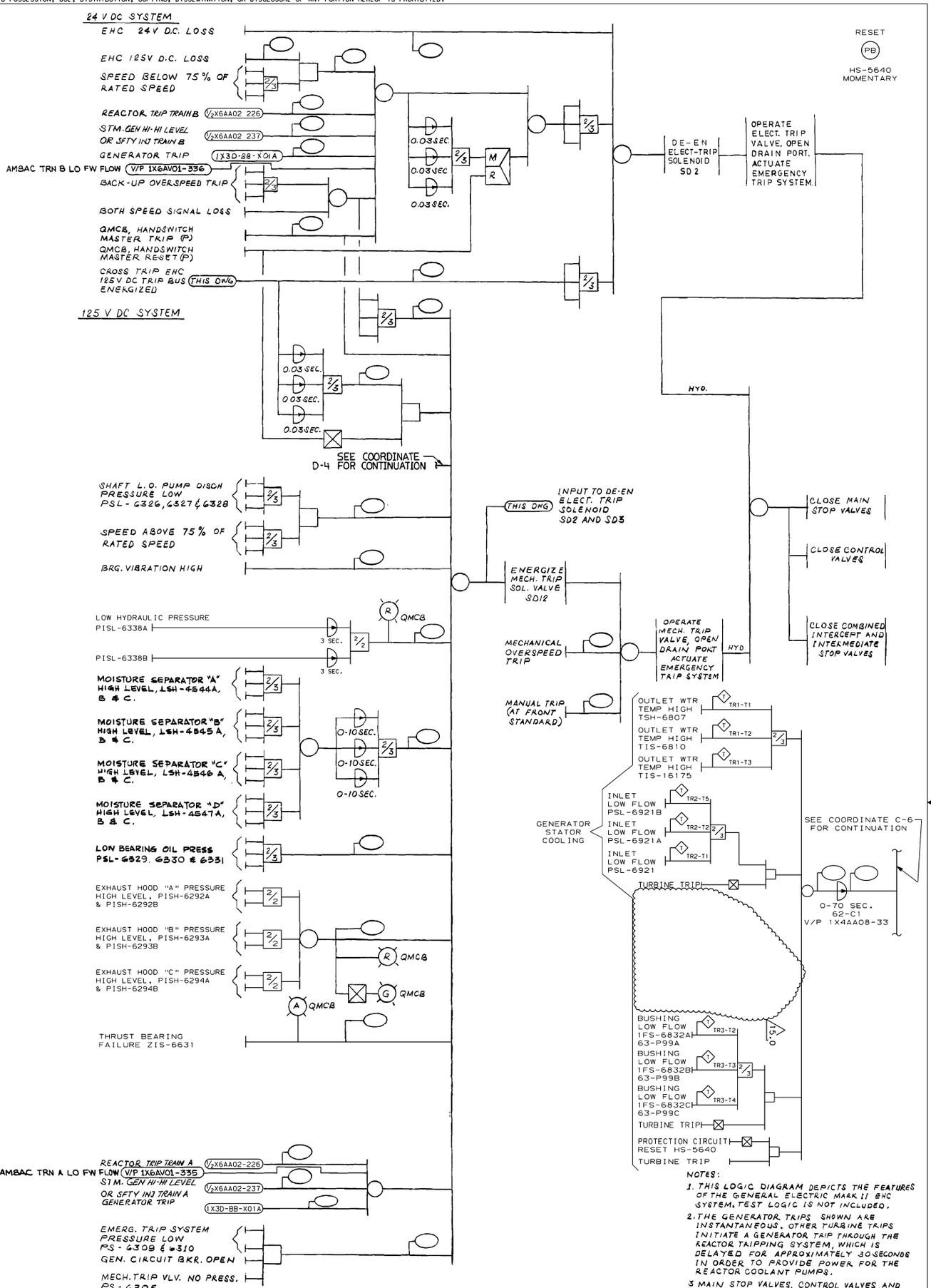


NOTES:

1. THIS LOGIC DIAGRAM DEPICTS THE FEATURES OF THE GENERAL ELECTRIC MARK II EHC SYSTEM. TEST LOGIC IS NOT INCLUDED.
2. THE GENERATOR TRIPS SHOWN ARE INSTANTANEOUS. OTHER TURBINE TRIPS INITIATE A GENERATOR TRIP THROUGH THE REACTOR TRIPPING SYSTEM, WHICH IS DELAYED FOR APPROXIMATELY 30 SECONDS IN ORDER TO PROVIDE POWER FOR THE REACTOR COOLANT PUMPS.
3. MAIN STOP VALVES, CONTROL VALVES, AND COMBINED AND INTERMEDIATE STOP VALVES POSITION ARE INDICATED BY PERCENTAGE INDICATORS.
4. PROVIDE ANNUNCIATOR INPUT FOR CONTROL POWER LOSS ON CUSTOMER TRIP ISOLATION LOGIC (TRAIN A & B)
5. THE SYMBOL ◇ REPRESENTS INTERPOSING TARGET RELAYS WHICH ARE LOCATED ON THE HYDROGEN AND STATOR WATER COOLING SYSTEM CONTROL PANEL (HSC).

REF. DWGS
V/P 1X4AA11-445 UP TO 454
INST LOOP DIAG 1X5DV094.95.96

NO.	DESCRIPTION	DATE	DR	CHK	APPV	DTL
15.0	REVISED PER ABN 1049001501J011, VER 1.0	8/18/09	RBH	MWD	JMR	

VERSIONS

NUCLEAR SAFETY RELATED Q

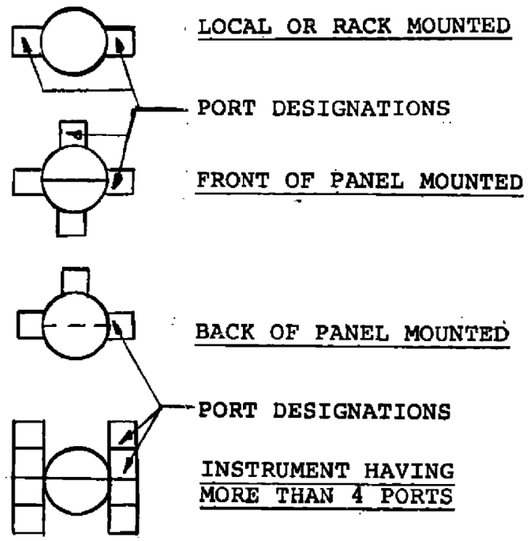
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

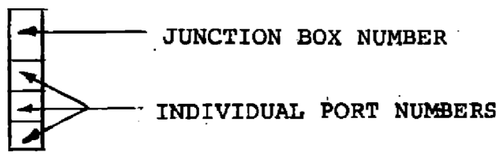
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

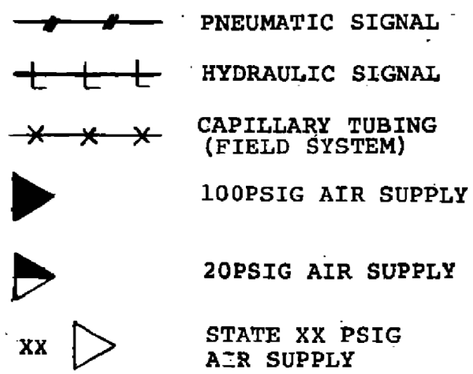
CONTROL LOGIC DIAGRAM
MAIN TURBINE TRIPPING SYSTEM

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X5DN203-1
VER. 15.0
SIZE D 22X34

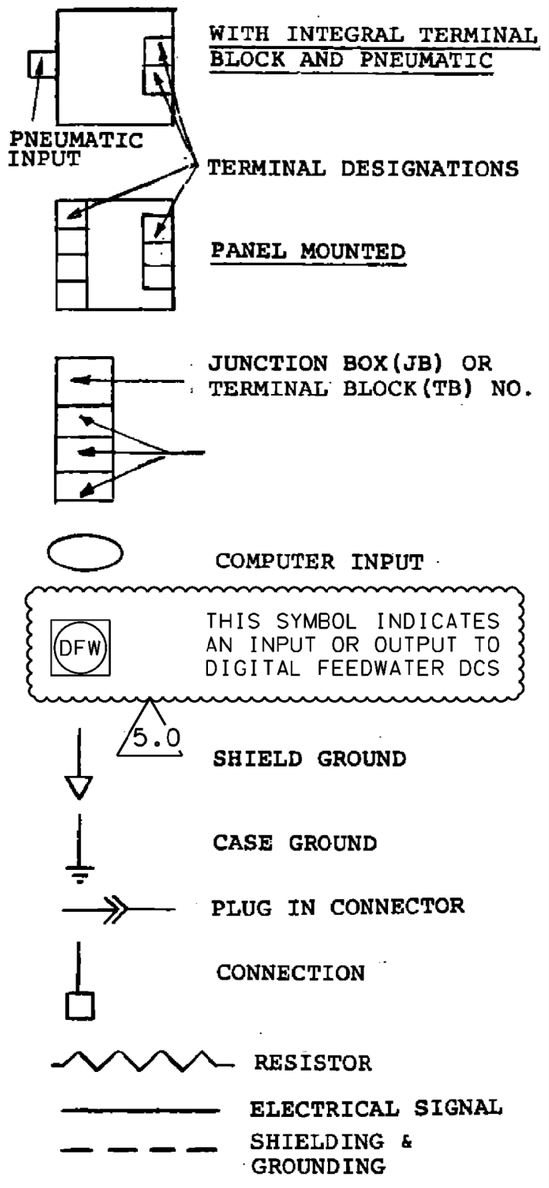
PNEUMATIC INSTRUMENTS







ELECTRONIC INSTRUMENTS



GENERAL NOTES

1. For symbols & abbreviations not covered, refer to:
   a. P&ID Drawing 1X5DV 101 & 1X5DV102
   b. ISA Standard S5.4-1976, Instrument Loop Diagrams.
2. All equipment numbers, line numbers, and instrument tag numbers have a Unit 1 prefix, unless otherwise noted.
3. All computer output signals are 24 VDC nominal, unless otherwise noted.
4. All shield wire to be grounded at the process cabinets, unless otherwise noted.
5. All pneumatic signals are 3-15 PSIG, unless otherwise noted.
6. All resistors are 100 OHM, one watt, one percent tolerance, unless otherwise noted.
7. Information shown on individual loop diagrams includes field instrument, panels/process rack connections, hard-wired indicators/recorders, and transfer switches.
8. Applicable secondary loop information (alarms, computer points, and control interlocks) i.e., cabling information termination information, location number information, can be found on the cross reference drawing identified on the individual loop diagrams, or in the EE-580 Data Base.
9. All loop instruments are direct acting unless otherwise noted.
10. For all OIM stations manual actuation of "RAISE" button opens the valve to increase the process variable, and actuation of "LOWER" button closes the valve to decrease the process variable, unless otherwise noted.

BISTABLE

Process variable normally less than set point or reference. Comparator energizes on increase.

Process variable normally less than set point or reference. Comparator de-energizes on increase.

Process variable normally greater than set point or reference. Comparator energizes on decrease.

Process variable normally greater than set point or reference. Comparator de-energize on decrease.

NOTES ON OUTPUT RELAY ACTION

Output ralay state (A) or (B)

All cabinet outputs from bistable driven relays shall be unpowered isolated dry contacts. Contact state is identified sa follows:

(A) Contact open with bistable de-energized.
(B) Contact closed with bistable de-energized.

- AC Source

- Substract, difference

- Output equals square root of input

- Algebric summation or totalize

- Cabinet mounted card

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

INSTRUMENT LOOP DIAGRAM LEGEND

NO.	VERSIONS	DATE	DR	CHK	APPV
5.0	REVISED PER ABN 1039002501J173, VER 1.0	07/26/13	DCP	MWD	JMR

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X5DV002	5.0

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SIZE B 11x17

SHEET - 1

7243D07

GEORGIA POWER COMPANY

PLANT: VOGTLE ELECTRIC GENERATING PLANT
UNITS: 1 & 2 SPIN: AA5FEC03
STATUS: APPROVED
CERTIFICATION LTR. NO. GP-14217
AUTHORITY: J. L. TAIN
ENGR. LTR. NO. EP/SA-64496

LOGIC SYMBOLS

SYMBOL	LOGIC FUNCTION	DESCRIPTION
	OR	A DEVICE WHICH PRODUCES AN OUTPUT ONLY WHEN ONE INPUT (OR MORE) EXISTS.
	NOT	A DEVICE WHICH PRODUCES AN OUTPUT ONLY WHEN THE INPUT DOES NOT EXIST.
	AND	A DEVICE WHICH PRODUCES AN OUTPUT ONLY WHEN EVERY INPUT EXISTS
2/3	COINCIDENCE (2 OUT OF 3 SHOWN)	A DEVICE WHICH PRODUCES AN OUTPUT WHEN THE PRESCRIBED NUMBER OF INPUTS EXIST (EXAMPLE 2 INPUTS MUST EXIST FOR AN OUTPUT).
	ADJUSTABLE TIME DELAY (ACTUATING)	A DEVICE WHICH PRODUCES AN OUTPUT FOLLOWING DEFINITE INTENTIONAL TIME DELAY AFTER RECEIVING AN INPUT.
	OFF RETURN MEMORY	A DEVICE WHICH RETAINS THE CONDITION OF OUTPUT CORRESPONDING TO THE LAST ENERGIZED INPUT, EXCEPT UPON INTERRUPTION OF POWER IT RETURNS TO THE OFF CONDITION.
	RETENTIVE MEMORY	A DEVICE WHICH RETAINS THE CONDITION OF OUTPUT CORRESPONDING TO THE LAST ENERGIZED INPUT (ALSO UPON INTERRUPTION OF POWER).
L R	RETENTIVE MEMORY WITH ACTUATION BLOCK	A DEVICE HAVING RETENTIVE MEMORY AND ACTUATION SIGNAL BLOCK LOGIC FUNCTIONS AS INDICATED BY THE DIAGRAM BELOW. ACTUATING SIGNAL / RESET (MOMENTARY) / OUTPUT SIGNAL
ANALOG INPUT / CONTROL LOGIC INPUT / ANALOG OUTPUT	ANALOG GATE	A DEVICE WHICH PERMITS AN ANALOG SIGNAL TO PASS IN AN ISOLATED CIRCUIT IF THE CONTROL LOGIC INPUT EXISTS.

SCOPE OF SUPPLY DEMARCATION

REDUNDANCY DEMARCATION

ADDITIONAL SYMBOLS

- PB / 43A — INSTRUMENT CHANNEL BISTABLE
- OUTPUT INDICATOR
  - BISTABLE OUTPUT IS A LOGIC "1" WHEN THE MEASURED PARAMETER IS GREATER THAN THE SETPOINT VALUE.
  - BISTABLE OUTPUT IS A LOGIC "1" WHEN THE MEASURED PARAMETER IS LESS THAN THE SETPOINT VALUE.
  - BISTABLE OUTPUT IS A LOGIC "1" WHEN THE MEASURED PARAMETER DEVIATES FROM THE NORMAL VALUE BY MORE THAN THE SETPOINT AMOUNTS.
  - SAME AS ABOVE EXCEPT WITH AN AUTOMATICALLY ADJUSTED SETPOINT
  - SAME AS ABOVE EXCEPT WITH REQUIRED HYSTERESIS
- 27 — NON-INSTRUMENT BISTABLE
- OUTPUT INDICATOR (SAME AS EXPLAINED ABOVE)
- A_2 — ALARM ANNUNCIATOR (ALARMS ON THE SAME SHEET WITH THE SAME SUBSCRIPT SHARE A COMMON ANNUNCIATOR WINDOW)
- R — REACTOR TRIP "FIRST OUT" ANNUNCIATOR
- T — TURBINE TRIP "FIRST OUT" ANNUNCIATOR
- I — INDICATOR LAMP
  A - ACTUATION STATUS LIGHTS, T - TRIP STATUS LIGHTS,
  P - PERMISSIVE STATUS LIGHTS, B - BYPASS STATUS LIGHTS
- C — COMPUTER INPUT
- E — EXT. COMPUTER INPUT
- L — LOGIC INFORMATION TRANSMISSION
- ANALOG INFORMATION TRANSMISSION
- ANALOG DISPLAY
  I - ANALOG INDICATOR, R - RECORDER, R2 - RECORDER 2 PEN,
  R3 - RECORDER 3 PEN, R8 - RECORDER 8 POINT
- Σ — ANALOG SUMMER

LOGIC SYMBOLS (CONT.)

SYMBOL	LOGIC FUNCTION	DESCRIPTION
TRACK/STORE	TRACK/STORE UNIT	A DEVICE WHICH PRODUCES AN OUTPUT EQUAL TO THE ANALOG INPUT WHEN THE CONTROL LOGIC INPUT DOES NOT EXIST. WHEN THE CONTROL LOGIC INPUT EXISTS, THE OUTPUT IS EQUAL TO THE VALUE OF THE ANALOG INPUT THAT EXISTED AT THE TIME THE LOGIC INPUT WAS ACTUATED.
	VARIABLE TIME DELAY (ACTUATING)	A DEVICE WHICH PRODUCES AN OUTPUT FOLLOWING A TIME DELAY WHICH IS EQUAL TO THE ANALOG INPUT
	ADJUSTABLE TIME DELAY (DEACTUATING)	A DEVICE WHICH PRODUCES AN OUTPUT COINCIDENT WITH THE INPUT AND MAINTAINS THE OUTPUT FOR A DEFINITE INTENTIONAL TIME DELAY AFTER THE INPUT IS REMOVED.

DEVICE FUNCTION LETTERS AND NUMBERS

FB	FLOW CHANNEL
LB	LEVEL CHANNEL
NC	NUCLEAR CHANNEL
PB	PRESSURE CHANNEL
RC	RADIATION CHANNEL
SB	SPEED CHANNEL
TB	TEMPERATURE CHANNEL
ZB	POSITION CHANNEL
20	ELECTRIC OPERATED VALVE
27	UNDERVOLTAGE RELAY
33	POSITION SWITCH
52	AC CIRCUIT BREAKER
63	PRESSURE SWITCH
71	LEVEL SWITCH
80	FLOW SWITCH
81	UNDERFREQUENCY RELAY

GENERAL NOTES: (FOR ALL SHEETS)

1. IN ALL LOGIC CIRCUITS, THE INDICATED ACTUATION OF A SYSTEM OR DEVICE OCCURS WHEN A LOGIC "1" SIGNAL IS PRESENT. EXCEPT WHERE INDICATED OTHERWISE, ALL BISTABLES ARE "DE-ENERGIZE TO ACTUATE" SUCH THAT A LOGIC 1 SIGNAL IS DEFINED TO BE PRESENT WHEN THE BISTABLE OUTPUT VOLTAGE IS OFF.
2. EXCEPT WHERE INDICATED OTHERWISE, THE FOLLOWING IS TRUE: ALL LOGIC CIRCUITS ARE REDUNDANT THAT IS EVERY LOGIC CIRCUIT SHOWN HAS A DUPLICATE LOCATED IN A SEPARATE CABINET. ALL INSTRUMENT CHANNELS, BISTABLES, ANNUNCIATORS, COMPUTER INPUTS, AND INDICATOR LAMPS ARE NOT REDUNDANT. MANUAL CONTROLS DO NOT HAVE REDUNDANT ACTUATORS, BUT DO HAVE REDUNDANT CONTACTS WHERE LOGIC IS REDUNDANT. ALL INDICATOR LAMPS, ANNUNCIATORS, AND COMPUTER INPUTS ARE CONNECTED TO BOTH TRAINS WHERE LOGIC IS REDUNDANT SO THAT A SIGNAL IN EITHER TRAIN WILL ACTUATE.
3. WHENEVER A PROCESS SIGNAL IS USED FOR CONTROL AND IS DERIVED FROM A PROTECTION CHANNEL, ISOLATION MUST BE PROVIDED.
4. THIS SET OF DRAWINGS ILLUSTRATES THE FUNCTIONAL REQUIREMENTS OF THE REACTOR CONTROL AND PROTECTION SYSTEM, INCLUDING ENGINEERED SAFEGUARDS. THESE DRAWINGS DO NOT REPRESENT ACTUAL HARDWARE IMPLEMENTATION. FOR HARDWARE IMPLEMENTATION, REFER TO THE FOLLOWING LIST.

FUNCTIONAL DIAGRAM	BLOCK OR WIRING DIAGRAM
REACTOR PROTECTION SYSTEM (SHEETS 1 TO 8 AND 15 TO 17)	DRAWING NUMBERS: 1087865, 7244042, 3655049, 3655050
REACTOR CONTROL SYSTEM (SHEETS 9 TO 14 AND 18 TO 20)	DRAWING NUMBERS: 5655051, 1185E15, 271CG83, 875704 / 5655052, 7244042, 271CG83

5. THIS SET OF DRAWINGS IS IDENTICAL FOR UNITS 1 AND 2 EXCEPT FOR THE TAG NUMBERS. FOR UNIT 1 TAG NUMBERS ADD A "1" EXAMPLE: 1PB-455E. FOR UNIT 2 TAG NUMBERS ADD A "2". EXAMPLE: 2PB-455E.

INDEX

TITLE	SH. NO	SUBS
INDEX AND SYMBOLS	1	1 2 3 4
REACTOR TRIP SIGNALS	2	6 6 6 6
NUCLEAR INSTR. AND MANUAL TRIP SIGNALS	3	4 4 4 4
NUCLEAR INSTR. PERMISSIVES AND BLOCKS	4	2 2 2 2
PRIMARY COOLANT SYSTEM TRIP SIGNALS	5	6 6 6 6
PRESSURIZER TRIP SIGNALS	6	6 6 6 6
STEAM GENERATOR TRIP SIGNALS	7	10 10 10 10
SAFEGUARDS ACTUATION SIGNALS	8	10 11 11 10
ROD CONTROLS & ROD BLOCKS	9	6 6 6 6
STEAM DUMP CONTROL	10	4 4 4 4
PRESSURIZER PRESSURE CONTROL	11	5 5 5 5
PRESSURIZER HEATER CONTROL	12	5 5 5 5
FEEDWATER CONTROL & ISOLATION	13	5 5 5 5
FEEDWATER CONTROL & ISOLATION	14	5 5 5 5
AUXILIARY FEEDWATER PUMPS STARTUP	15	6 7 7 7
TURBINE TRIPS, RUNBACKS & OTHER SIGNALS	16	5 6 6 7
PRESSURIZER PRESS. R.P. SYS. (TRAIN A)	18	2 2 2 2
PRESSURIZER PRESS. R.P. SYS. (TRAIN B)	19	2 2 2 2
PRESSURIZER LEVEL CONTROL	20	2 2 2 2
SAFEGUARDS ACTUATION SIGNALS (CONTINUED)	21	1 - - -

SHEET 17 DELETED

NO.	DATE	DESCRIPTION	BY
1	9/30/86	CHANGED PER SDCR 10	
2	10/27/86	REVISED SUB BLOCK	
3	11/19/86	REVISED SUB BLOCK	
4	12/3/86	REVISED SUB BLOCK	
5	1/7/87	REVISED SUB BLOCK	
6	1/16/87	REVISED SUB BLOCK	
7	2/27/87	CHANGES CHECKED	
8	3/4/87	REVISED INDEX	
9	3/6/87	REVISED INDEX	
10	3/11/87	REVISED INDEX	
11	3/19/87	REVISED INDEX	
12	4/16/87	REVISED SUB BLOCK	
13	5/5/87	REVISED SUB BLOCK	
14	1/26/89	REVISED INDEX	

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Westinghouse Electric Corporation — NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA. 15230

GEORGIA POWER CO.
ALVIN B VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
INDEX AND SYMBOLS

7243D07
SHEET - 1

DRN. B WEARN — CHK. — APPD. — DRN. — CHK. — APPD. — SPEC. MGR. — DO NOT SCALE

SUB-STANDARD ORIGINAL — NOT SUITABLE FOR LEGIBLE REPRODUCTION — CONTACT NRC APPROVAL (IF REQ'D)

VOGTLE ELECTRIC GENERATING PLANT — JOB NO. 9510

EQUIPMENT TAG NO. VAT2
STARTUP DESIGNATION NO. VA1'
ACTIVITY NO. N/A
SYSTEM NO. VA1'
CATEGORY NO. N/A

RE-FOR FITTING REQUIRED ☐ YES ☒ NO

DISTRIBUTION TO: FOR: REVIEW
- MECHANICAL
- HVAC
- EQUIPMENT
- PROCESS
- CONTROL SYSTEMS
- ELECTRICAL
- WIRING
- CONDUIT
- HAZAROS
- CIVIL/STRUCTURAL
- NUCLEAR
- STRESS/PLANT DESIGN
- CODES AND STANDARDS
- ARCHITECTURAL
- STARTUP
- CONSTRUCTION
- NOT REQ'D BY ENGR
- CLIENT
- EQUIP. QUALIFICATION
- M & QS
- WESTINGHOUSE

IDENTIFYING TITLE OF THIS DOCUM
FUNCTIONAL DIAGRAM
INDEX AND SYMBOLS

Bechtel Log No. 1XG1-02-225-

IMPORTANT

Permission to proceed does not constitu acceptance or approval of design details calculations, analyses, test methods or materials developed or selected by the supplier and does not relieve supplier f full compliance with contractual obliga

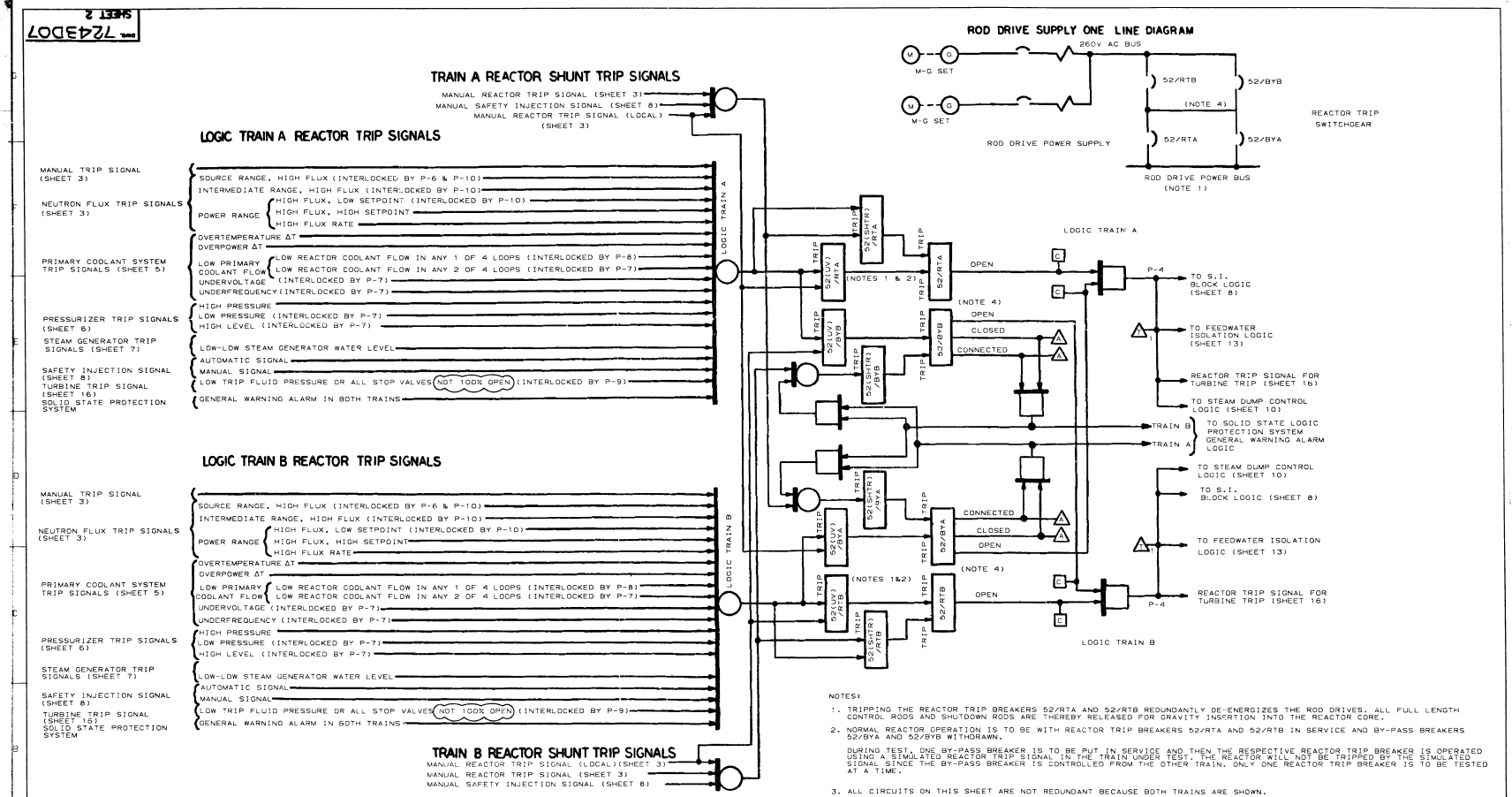
DATE RECEIVED 2-10-89

DOCUMENT STATUS
1. WORK MAY PROCEED
2. REVISE AND RESUBMIT WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED
3. REVISE AND RESUBMIT WORK MAY NOT PROCEED
4. INFORMATION ONLY DISTRIBUTION REQ'D ☐ YES
5. RESUBMIT - NOT ACCEPTABLE FOR MICROFILM, WORK MAY PROCEED

SECTION SUPERVISOR — DATE 5-9-89

CAMERA OPERATOR — DATE 5-9-89

SHEET 2 7243D07



NOTES:

1. TRIPPING THE REACTOR TRIP BREAKERS 52/RTA AND 52/RTB REDUNDANTLY DE-ENERGIZES THE ROD DRIVES. ALL FULL LENGTH CONTROL RODS AND SHUTDOWN RODS ARE THEREBY RELEASED FOR GRAVITY INSERTION INTO THE REACTOR CORE.
2. NORMAL REACTOR OPERATION IS TO BE WITH REACTOR TRIP BREAKERS 52/RTA AND 52/RTB IN SERVICE AND BY-PASS BREAKERS 52/BYA AND 52/BYB WITHDRAWN.

   DURING TEST, ONE BY-PASS BREAKER IS TO BE PUT IN SERVICE AND THEN THE RESPECTIVE REACTOR TRIP BREAKER IS OPERATED USING A SIMULATED REACTOR TRIP SIGNAL IN THE TRAIN UNDER TEST. THE REACTOR WILL NOT BE TRIPPED BY THE SIMULATED SIGNAL SINCE THE BY-PASS BREAKER IS CONTROLLED FROM THE OTHER TRAIN. ONLY ONE REACTOR TRIP BREAKER IS TO BE TESTED AT A TIME.
3. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT BECAUSE BOTH TRAINS ARE SHOWN.
4. OPEN/CLOSED INDICATION FOR EACH TRIP BREAKER AND EACH BYPASS BREAKER IN CONTROL ROOM.

Southern Company Services, Inc.
DATE RECEIVED 12/03/93
DOCUMENT STATUS
1 APPROVED FOR CONST. WORK MAY PROCEED (SUBJECT TO RESOLUTION OF COMMENTS INDICATED)
2 APPROVED AS NOTED (RESUBMITTAL NOT REQUIRED)
3 REVISE AND RESUBMIT (SEE REMARKS)
4 DISAPPROVED (SEE REMARKS)
INFORMATION ONLY
DISTRIBUTION RECD
ACCEPTANCE FOR USE WITH SCP
DATE 12-7-93

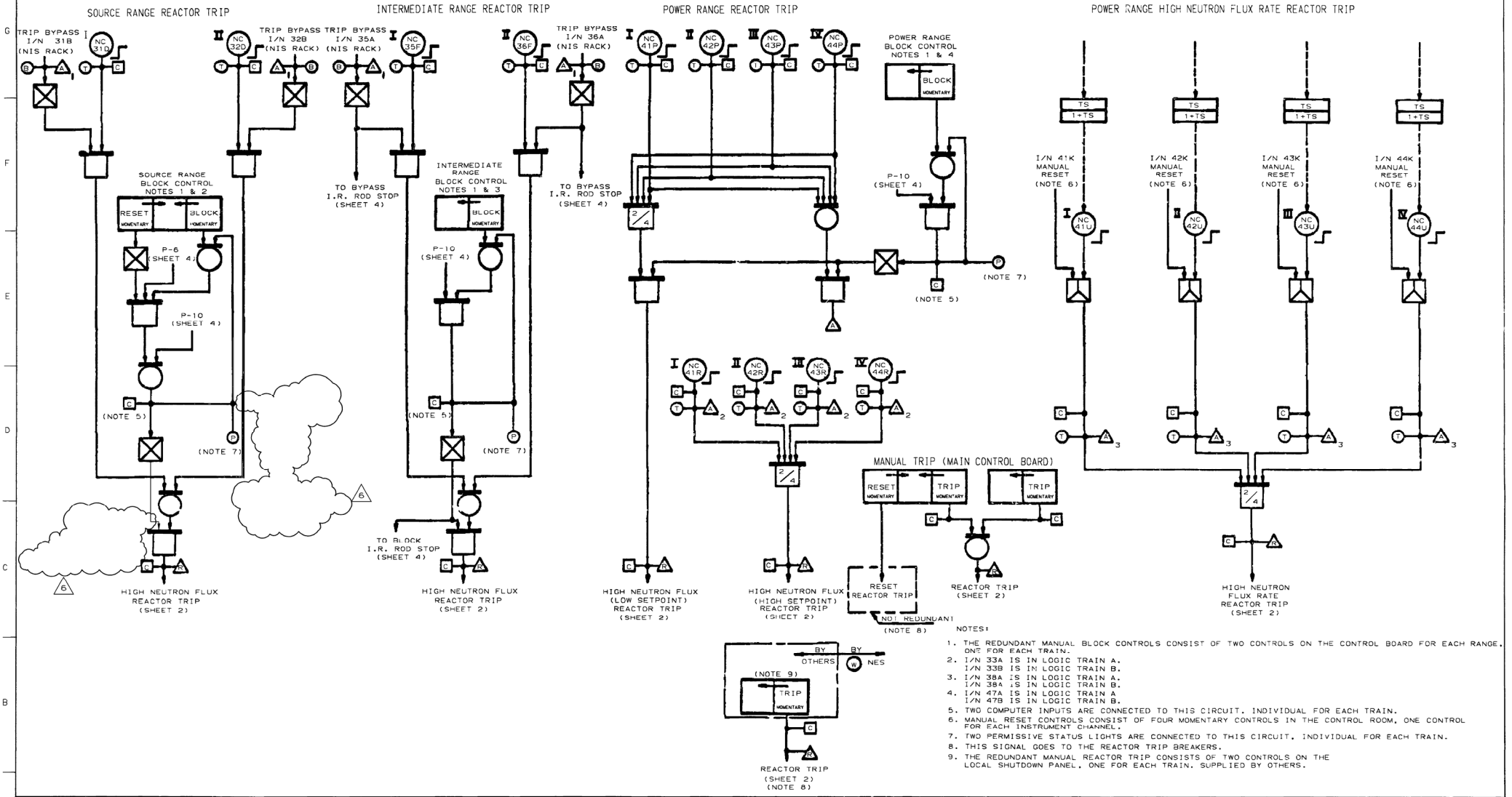
PLANT VOGTLE
1X6AA02-226-9
FUNCTIONAL DIAGRAMS REACTOR TRIP SIGNALS
(REDRAWN AND REVISED BY SCS TO INCORPORATE ABN-05114)
P.O.
VENDOR: WE

ABN-05114 7 REDRAWN BY SCS AND ISSUED TO INCORP. ABN-05114

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS PITTSBURGH PA U.S.A.
TITLE GEORGIA POWER CO ALVIN W. VOGTLE UNITS 1 & 2 FUNCTIONAL DIAGRAMS REACTOR TRIP SIGNALS
SCALE DO NOT SCALE
DIMENSIONS IN INCHES
7243D07 SHEET 2
SUB 1X6AA02-226-9

11 10 9 8 7 6 5 4 3 2 1

7243D07
SHEET - 3



NOTES:

1. THE REDUNDANT MANUAL BLOCK CONTROLS CONSIST OF TWO CONTROLS ON THE CONTROL BOARD FOR EACH RANGE, ONE FOR EACH TRAIN.
2. I/N 33A IS IN LOGIC TRAIN A.
   I/N 33B IS IN LOGIC TRAIN B.
3. I/N 38A IS IN LOGIC TRAIN A.
   I/N 38A IS IN LOGIC TRAIN B.
4. I/N 47A IS IN LOGIC TRAIN A
   I/N 47B IS IN LOGIC TRAIN B.
5. TWO COMPUTER INPUTS ARE CONNECTED TO THIS CIRCUIT. INDIVIDUAL FOR EACH TRAIN.
6. MANUAL RESET CONTROLS CONSIST OF FOUR MOMENTARY CONTROLS IN THE CONTROL ROOM, ONE CONTROL FOR EACH INSTRUMENT CHANNEL.
7. TWO PERMISSIVE STATUS LIGHTS ARE CONNECTED TO THIS CIRCUIT. INDIVIDUAL FOR EACH TRAIN.
8. THIS SIGNAL GOES TO THE REACTOR TRIP BREAKERS.
9. THE REDUNDANT MANUAL REACTOR TRIP CONSISTS OF TWO CONTROLS ON THE LOCAL SHUTDOWN PANEL. ONE FOR EACH TRAIN. SUPPLIED BY OTHERS.

NO.	REVISIONS	DATE	DR	CHK	APPV
6	INCORPORATED PER DCP 96-V1N0044	05/12/99	ELC	EOG	GLB

SCS REVISIONS

DOCUMENT STATUS CODE 1

1X6AA02-00227-9

5	GAE	GBE-300
	ECN-34937	

CHANGES CIRCLED
REDRAWN AND
RE-ISSUED BY SCS
TO INCORP. ABN-11479

DRFTR R HEARN 9/98

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURG, PA. U.S.A.

TITLE: GEORGIA POWER CO.
ALVIN W. VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
NUCLEAR INSTR. & MANUAL TRIP SIGNALS

SCALE DO NOT SCALE

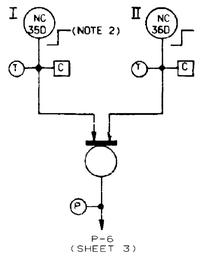
7243D07...
SHEET 3
SIZE 6

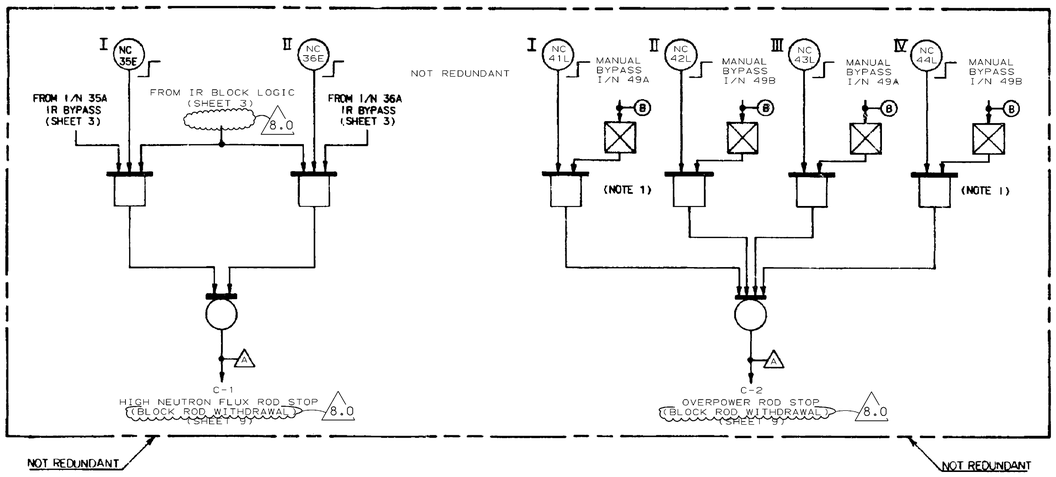
SIZE "D" 1 CAD NAME 16A02227

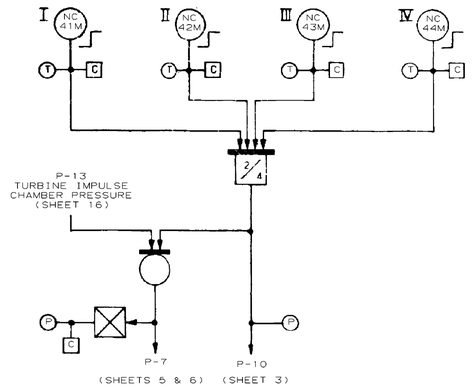
7243D07 SHEET - 4

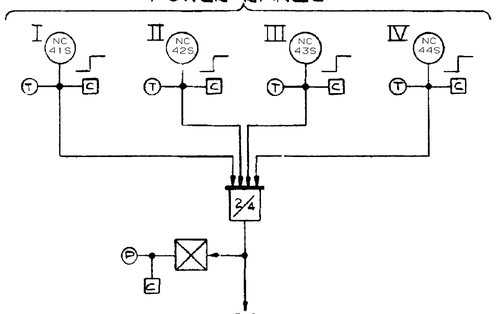
INTERMEDIATE RANGE

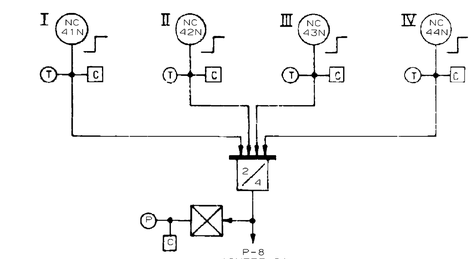
POWER RANGE











NOTES:

1. THE BYPASS SIGNALS ARE MADE UP BY MEANS OF TWO THREE-POSITION SWITCHES ON A NIS RACK. SWITCH I/N 49A BYPASSES EITHER NC-41L OR NC-43L. SWITCH I/N 49B BYPASSES EITHER NC-42L OR NC-44L.

2. THE TWO P-6 BISTABLES NO. NC-35D AND NC-36D ARE "ENERGIZED TO ACTUATE" SUCH THAT A LOGIC 1 SIGNAL IS DEFINED TO BE PRESENT WHEN THE BISTABLE OUTPUT VOLTAGE IS ON.

8.0

REV	CHANGE	
11	GAE/GBE-300	2
10	ECN-31242	
9	REVIS CIRCLED	
8	WEABS 7-13-06	

SO GAE/GBE-300
D 936528/936527

DSTN P. MEARN
CHKD.
DES. ENG.
MFG. ENG.
MTLS. ENG.
APP.
APP.
APP.
DPTG. SUPV.
REF. DWG.
NEXT ASSY.

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

TITLE: GEORGIA POWER CO
ALVIN W. VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
NUCLEAR INSTR. PERMISSIVES & BLOCKS

SCALE
DIMENSIONS IN INCHES
DO NOT SCALE

7243D07
SHEET - 4
SUB 12

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

CAD MICROSTATION CYN 02
1X6AA02-00228.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

VERSION 8.0	DATE 11/04/09
REVISED BY SNC PER ABN-V01192, VER. 2.0.	

SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

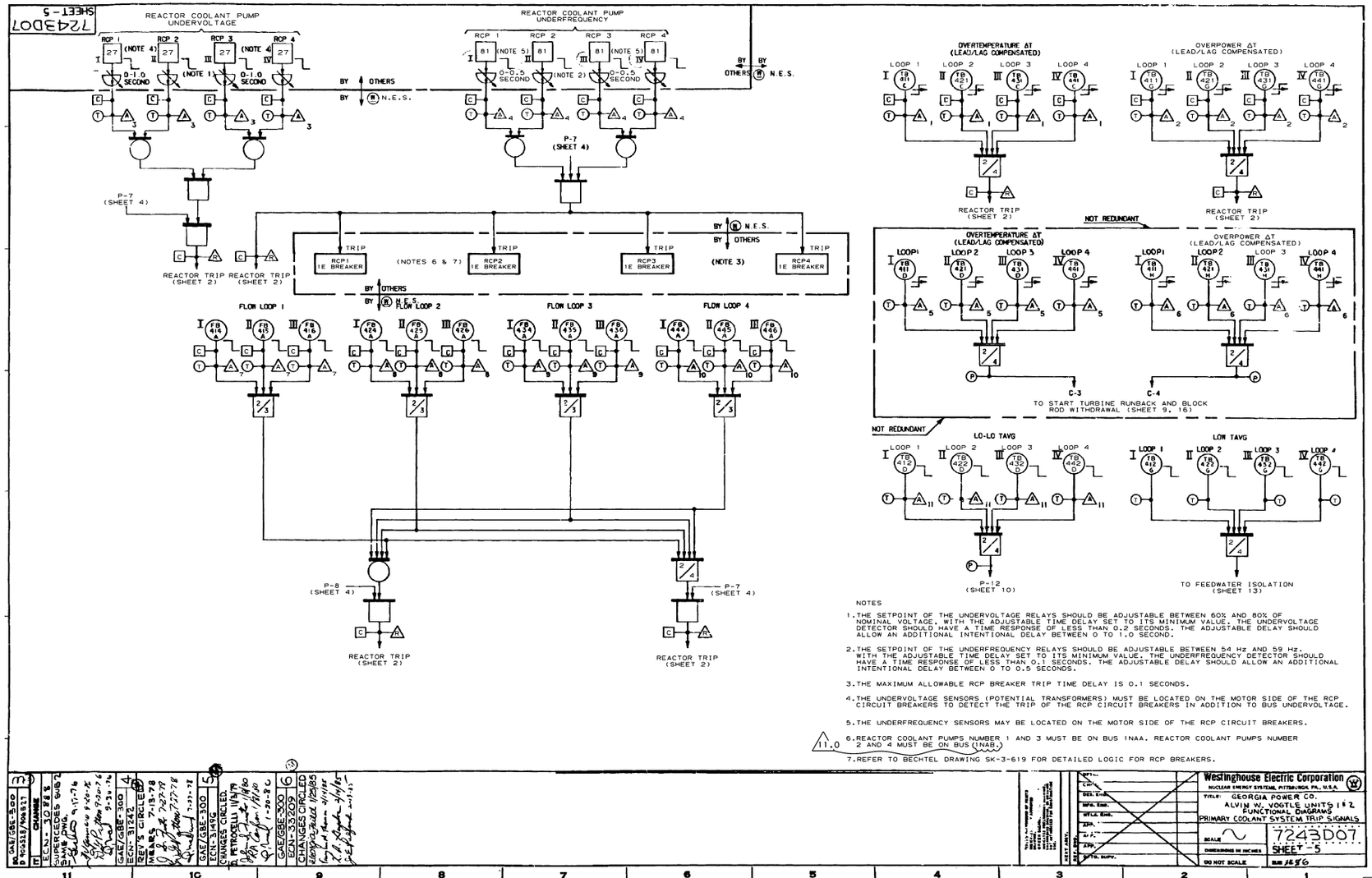
BY	CHK'D	APPR.1	APPR.2
CYN	RKF	PML	

TITLE:
FUNCTIONAL DIAGRAMS
NUCLEAR INSTRUMENT PERMISSIVES
& BLOCKS

VENDOR: WESTINGHOUSE P.O.#:

DRAWING NO. 1X6AA02-00228

SIZE D



NOTES

1. THE SETPOINT OF THE UNDERVOLTAGE RELAYS SHOULD BE ADJUSTABLE BETWEEN 60% AND 80% OF NOMINAL VOLTAGE. WITH THE ADJUSTABLE TIME DELAY SET TO ITS MINIMUM VALUE, THE UNDERVOLTAGE DETECTOR SHOULD HAVE A TIME RESPONSE OF LESS THAN 0.2 SECONDS. THE ADJUSTABLE DELAY SHOULD ALLOW AN ADDITIONAL INTENTIONAL DELAY BETWEEN 0 TO 1.0 SECOND.
2. THE SETPOINT OF THE UNDERFREQUENCY RELAYS SHOULD BE ADJUSTABLE BETWEEN 54 Hz AND 59 Hz. WITH THE ADJUSTABLE TIME DELAY SET TO ITS MINIMUM VALUE, THE UNDERFREQUENCY DETECTOR SHOULD HAVE A TIME RESPONSE OF LESS THAN 0.1 SECONDS. THE ADJUSTABLE DELAY SHOULD ALLOW AN ADDITIONAL INTENTIONAL DELAY BETWEEN 0 TO 0.5 SECONDS.
3. THE MAXIMUM ALLOWABLE RCP BREAKER TRIP TIME DELAY IS 0.1 SECONDS.
4. THE UNDERVOLTAGE SENSORS (POTENTIAL TRANSFORMERS) MUST BE LOCATED ON THE MOTOR SIDE OF THE RCP CIRCUIT BREAKERS TO DETECT THE TRIP OF THE RCP CIRCUIT BREAKERS IN ADDITION TO BUS UNDERVOLTAGE.
5. THE UNDERFREQUENCY SENSORS MAY BE LOCATED ON THE MOTOR SIDE OF THE RCP CIRCUIT BREAKERS.
6. REACTOR COOLANT PUMPS NUMBER 1 AND 3 MUST BE ON BUS 1NAA. REACTOR COOLANT PUMPS NUMBER 2 AND 4 MUST BE ON BUS 1NAB. (11.0)
7. REFER TO BECHTEL DRAWING SK-3-619 FOR DETAILED LOGIC FOR RCP BREAKERS.

BY	CHANGE	DATE
GAE/GBE-300	ECN - 30885 SUPERCEDES SH 2 SAME DWG.	9-17-76
GAE/GBE-300	ECN-31245 REV'S CIRCLED	9-30-76
GAE/GBE-300	ECN-31496 CHANGES CIRCLED	7-18-78
GAE/GBE-300	ECN-33209 CHANGES CIRCLED	11/25/80
GAE/GBE-300	ECN-33209 CHANGES CIRCLED	11/25/80

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.
TITLE: GEORGIA POWER CO. ALVIN W. VOGTLE UNITS 1 & 2 FUNCTIONAL DIAGRAMS PRIMARY COOLANT SYSTEM TRIP SIGNALS
SCALE: NONE | 7243D07 SHEET-5
DIMENSIONS IN INCHES
DO NOT SCALE
SUB 1856

CAD MICROSTATION LPM/03 1X6AA02-00229.DGN

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A	
TAB/SECT.	N/A
PAGE	N/A
FIGURE	N/A

VERSION 11.0	DATE 3-4-11
REVISED BY SNC PER ABN-V02456, VER. 1.0	

SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
LPM	MLH	AAN	

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

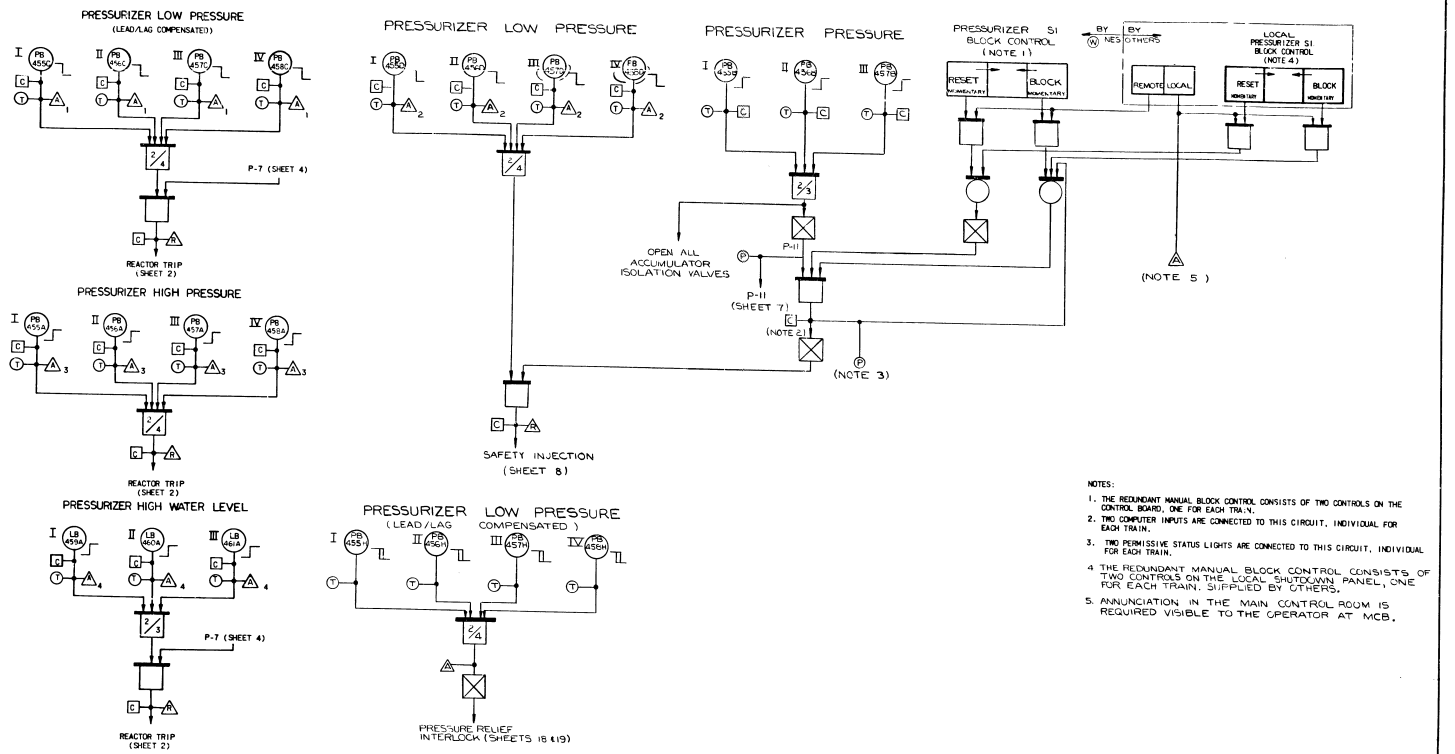
TITLE: FUNCTIONAL DIAGRAMS PRIMARY COOLANT SYSTEM TRIP SIGNALS

VENDOR: WESTINGHOUSE | P.O.#:

DRAWING NO. 1X6AA02-00229

SIZE D

SHEET - 6



NOTES:

1. THE REDUNDANT MANUAL BLOCK CONTROL CONSISTS OF TWO CONTROLS ON THE CONTROL BOARD, ONE FOR EACH TRAIN.
2. TWO COMPUTER INPUTS ARE CONNECTED TO THIS CIRCUIT, INDIVIDUAL FOR EACH TRAIN.
3. TWO PERMISSIVE STATUS LIGHTS ARE CONNECTED TO THIS CIRCUIT, INDIVIDUAL FOR EACH TRAIN.
4. THE REDUNDANT MANUAL BLOCK CONTROL CONSISTS OF TWO CONTROLS ON THE LOCAL SHUTDOWN PANEL, ONE FOR EACH TRAIN, SUPPLIED BY OTHERS.
5. ANNUNCIATION IN THE MAIN CONTROL ROOM IS REQUIRED VISIBLE TO THE OPERATOR AT MCB.

REV.	CHANGE
12	GAE/GBE-300 D 306 E68/304437
11	GAE/GBE-300 CHANGES ARE BACKCIRCLED. P. HEARN 7-17-78
10	GAE/GBE-300 REVISIONS 9-15-78
9	GAE/GBE-300 ECN-31242 REV'S CIRCLED WEARS 7-13-78
8	GAE/GBE-300 ECN 33209 REV. CIRCLED B. STEMMLER 8/1/81
7	GAE/GBE-300 ECN 39569 REV. CIRCLED B. STEMMLER 8/1/81
6	GAE/GBE-300 ECN-33209 CHANGES CIRCLED

GEORGIA POWER COMPANY
PLANT: ALVIN W. VOGTLE NUCLEAR PLANT
UNITS: 1 & 2 STATE: GEORGIA
STATUS: APPROVED
CERTIFICATION LTR. NO. GP-9146
AUTHORITY: J. L. VOTA
ENGR. LTR. NO. EP/SA-47348

DSTR. P. HEARN	5/2	Westinghouse Electric Corporation
CONS.		NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.
DES. ENG.	5/2	TITLE: GEORGIA POWER CO. ALVIN W VOGTLE UNITS 1 & 2 FUNCTIONAL DIAGRAMS PRESSURIZER TRIP SIGNALS
MFG. ENG.		
MTLS. ENG.		
ENG.	5/2	
APP.		SCALE: DO NOT SCALE
APP.		DIMENSIONS IN INCHES
DSTR. SUPV.	5/2	SUB 1238 56

7243D07
SHEET - 6

VOGTLE ELECTRIC GENERATING PLANT — JOB NO. 9510

EQUIPMENT TAG NO. Various

STARTUP DESIGNATION NO. Various
ACTIVITY NO. Various
SYSTEM NO. Various
CATEGORY NO. NA

RETROFITTING REQUIRED ☐ YES ☒ NO

DISTRIBUTION TO: FOR: REVIEW	INFO.
• MECHANICAL	
• HVAC	
NSSS	
BOP	
• CONTROL SYSTEMS	✓
• ELECTRICAL	✓
WIRING	
CONDUIT	
• HAZARDS	
• CIVIL/STRUCTURAL	
• NUCLEAR	
• STRESS/PLANT DESIGN	
CODES AND STANDARDS	
• ARCHITECTURAL	
• STARTUP	
• CONSTRUCTION	
• NOT REQ'D BY ENGRG.	
• CLIENT	
• EQUIP. QUALIFICATION	
• M & OS	
• WESTINGHOUSE	
•	
•	

IDENTIFYING TITLE OF THIS DOCUMENT: Functional Diag.

Bechtel Log No.

1 XGAA07-230-8

IMPORTANT

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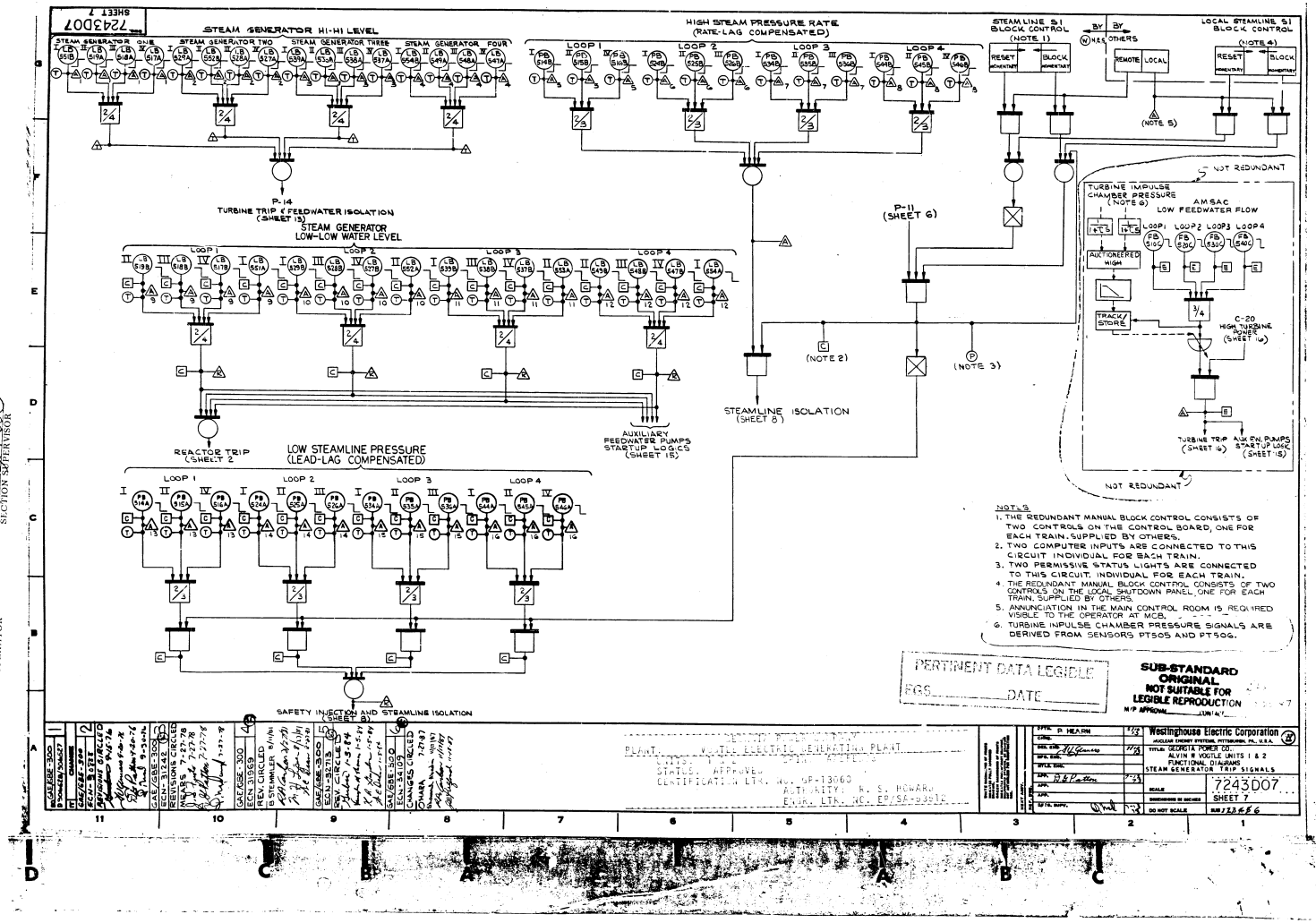
DATE RECEIVED 3-14-85 — SIGNED

DOCUMENT STATUS — DATE 5/4/85

1 ☒ WORK MAY PROCEED
2 ☐ REVISE AND RESUBMIT. WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED.
3 ☐ REVISE AND RESUBMIT. WORK MAY NOT PROCEED.
4 ☐ INFORMATION ONLY. DISTRIBUTION REQ'D? ☐ YES
M ☐ RESUBMIT – NOT ACCEPTABLE FOR MICROFILM. WORK MAY PROCEED.

PF-5987 (9510) 9/84

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL



GPC PLANT VOGTLE CONSTR MICROGRAPHICS

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1/24/02
DATE

CAMERA OPERATOR

SECTION SUPERVISOR

16X

BECHTEL NOTE (APPLICABLE TO UNIT 1 ONLY):
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BECHTEL NO. N/A IS ASSOCIATED WITH THE GPC/WEC NO. UNIT CONSTRUCTION WITH THE SHOULD BE USED. THE WORK IS BEING REFERENCED BY PROJECT DOCUMENT:
IMPLEMENTED BY PROJECT DOCUMENT:
DCP 94V1E012

VOGTLE ELECTRIC GENERATING PLANT — JOB NO. 9510

EQUIPMENT TAG NO. V.U.

STARTUP DESIGNATION NO. V.U.
ACTIVITY NO. V.U.
SYSTEM NO. V.U.
CATEGORY NO. N/A

RETROFITTING REQUIRED ☐ YES ☒ NO

DISTRIBUTION TO: FOR: REVIEW	INFO
• MECHANICAL	
• HVAC	
• EQUIPMENT	
• PROCESS	
• CONTROL SYSTEMS JB JE	V
• ELECTRICAL EE EE	
WIRING	
CONDUIT	
• HAZARDS	
• CIVIL/STRUCTURAL	
• NUCLEAR A.S.M.E	V
• STRESS/PLANT DESIGN	
CODES AND STANDARDS	
• ARCHITECTURAL	
• STARTUP	V
• CONSTRUCTION	
• NOT REQ'D BY ENGRG	
• CLIENT	
• EQUIP. QUALIFICATION	
• M & QS	
• WESTINGHOUSE	
•	
•	

IDENTIFYING TITLE OF THIS DOCUMENT
Functional Diagrams Steam Generator Trip Signals

Bechtel Log No.
1 KGA1202-231-E

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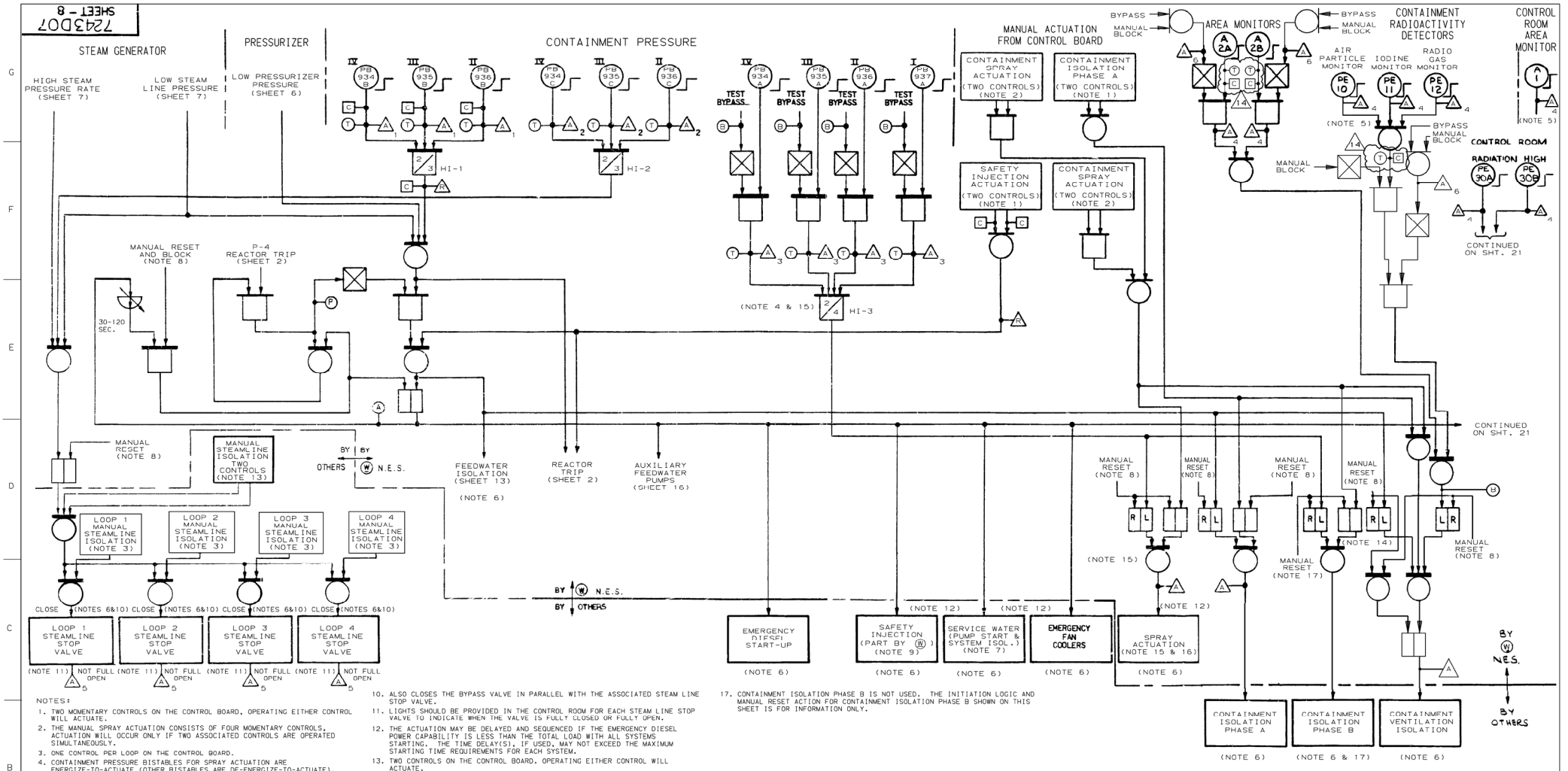
DATE RECEIVED 12-9-97
SIGNED
DATE 12-27-97

DOCUMENT STATUS
1 ☒ WORK MAY PROCEED
2 ☐ REVISE AND RESUBMIT WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED
3 ☐ REVISE AND RESUBMIT WORK MAY NOT PROCEED
4 ☐ INFORMATION ONLY DISTRIBUTION REQ'D ☐ YES
M ☐ RESUBMIT – NOT ACCEPTABLE FOR MICROFILM, WORK MAY PROCEED

PP-3987 (9910) 12/83



NOTES:

1. TWO MOMENTARY CONTROLS ON THE CONTROL BOARD, OPERATING EITHER CONTROL WILL ACTUATE.
2. THE MANUAL SPRAY ACTUATION CONSISTS OF FOUR MOMENTARY CONTROLS, ACTUATION WILL OCCUR ONLY IF TWO ASSOCIATED CONTROLS ARE OPERATED SIMULTANEOUSLY.
3. ONE CONTROL PER LOOP ON THE CONTROL BOARD.
4. CONTAINMENT PRESSURE BISTABLES FOR SPRAY ACTUATION ARE ENERGIZE-TO-ACTUATE (OTHER BISTABLES ARE DE-ENERGIZE-TO-ACTUATE).
5. CIRCUITRY IS NOT PART OF THE SAFEGUARDS SYSTEM, AND IS NOT SAFETY RELATED, HOWEVER, CIRCUITRY IS INPUT TO THE SAFEGUARS SYSTEM THERFORE ISOLATION MUST BE PROVIDED.
6. COMPONENTS ARE ALL INDIVIDUALLY SEALED IN (LATCHED), SO THAT LOSS OF THE ACTUATION SIGNAL WILL NOT CAUSE THESE COMPONENTS TO RETURN TO THE CONDITION HELD PRIOR TO THE ADVENT OF THE ACTUATION SIGNAL.
7. SERVICE WATER SYSTEM ISOLATION IS USED ONLY IF REQUIRED.
8. THE REDUNDANT MANUAL RESET CONSISTS OF TWO MOMENTARY CONTROLS ON THE CONTROL BOARD, ONE FOR EACH TRAIN.
9. SAFETY INJECTION SEQUENCE REQUIREMENTS (IF SEQUENCING IS NECESSARY) ARE SPECIFIED BY (W) NUCLEAR ENERGY SYSTEMS.
10. ALSO CLOSES THE BYPASS VALVE IN PARALLEL WITH THE ASSOCIATED STEAM LINE STOP VALVE.
11. LIGHTS SHOULD BE PROVIDED IN THE CONTROL ROOM FOR EACH STEAM LINE STOP VALVE TO INDICATE WHEN THE VALVE IS FULLY CLOSED OR FULLY OPEN.
12. THE ACTUATION MAY BE DELAYED AND SEQUENCED IF THE EMERGENCY DIESEL POWER CAPABILITY IS LESS THAN THE TOTAL LOAD WITH ALL SYSTEMS STARTING. THE TIME DELAY(S), IF USED, MAY NOT EXCEED THE MAXIMUM STARTING TIME REQUIREMENTS FOR EACH SYSTEM.
13. TWO CONTROLS ON THE CONTROL BOARD, OPERATING EITHER CONTROL WILL ACTUATE.
14. SOME ENGINEERED SAFEGUARDS FUNCTIONS ARE NOT WITHIN THE FUNCTIONAL DESIGN SCOPE OF (W) NUCLEAR ENERGY SYSTEMS BUT ONLY SHOWN ON THIS SHEET AS THE FUNCTIONS ARE BUILT IN THE (W) SUPPLIED EQUIPMENT.
15. THE 2 OUT OF 4 COINCIDENCE, MEMORY, AND "OR" LOGIC ARE DUPLICATED WITHIN EACH TRAIN. SEPARATE OUTPUT RELAYS ARE ALSO PROVIDED IN EACH TRAIN. TO MINIMIZE FALSE CONTAINMENT SPRAY ONE OUTPUT RELAY SHOULD START THE PUMPS WHILE ANOTHER SHOULD OPEN THE SYSTEM VALVES.
16. THE SEQUENCE TO LOAD THE CONTAINMENT SPRAY SYSTEM ON THE EMERGENCY DIESEL POWER SHOULD BE INDEPENDENT OF THE SEQUENCE TO LOAD THE SAFETY INJECTION SYSTEM. IF THE CONDITIONS OF NOTE 12 APPLY THE SEQUENCE INTERLOCK SHOULD BE SUCH THAT SPRAY WILL START WITHIN THE REQUIRED TIME INDEPENDENT OF THE SAFETY INJECTION SIGNAL STATUS.
17. CONTAINMENT ISOLATION PHASE B IS NOT USED. THE INITIATION LOGIC AND MANUAL RESET ACTION FOR CONTAINMENT ISOLATION PHASE B SHOWN ON THIS SHEET IS FOR INFORMATION ONLY.

NO.	REVISIONS	DATE	DR	CHK	APPV
14	INCORPORATED PER DCP 97-V1N0067	11-16-00	ELC	TSL	GLB

SCS REVISIONS

DOCUMENT STATUS CODE 1

1X6AA02-00232-17

WESTINGHOUSE PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF THE WESTINGHOUSE ELECTRIC CORPORATION WATER REACTOR DIVISIONS. IT IS TRANSMITTED TO YOU IN CONFIDENCE AND TRUST, AND IS TO BE RETURNED UPON REQUEST. ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF THE WESTINGHOUSE WATER REACTOR DIVISIONS.

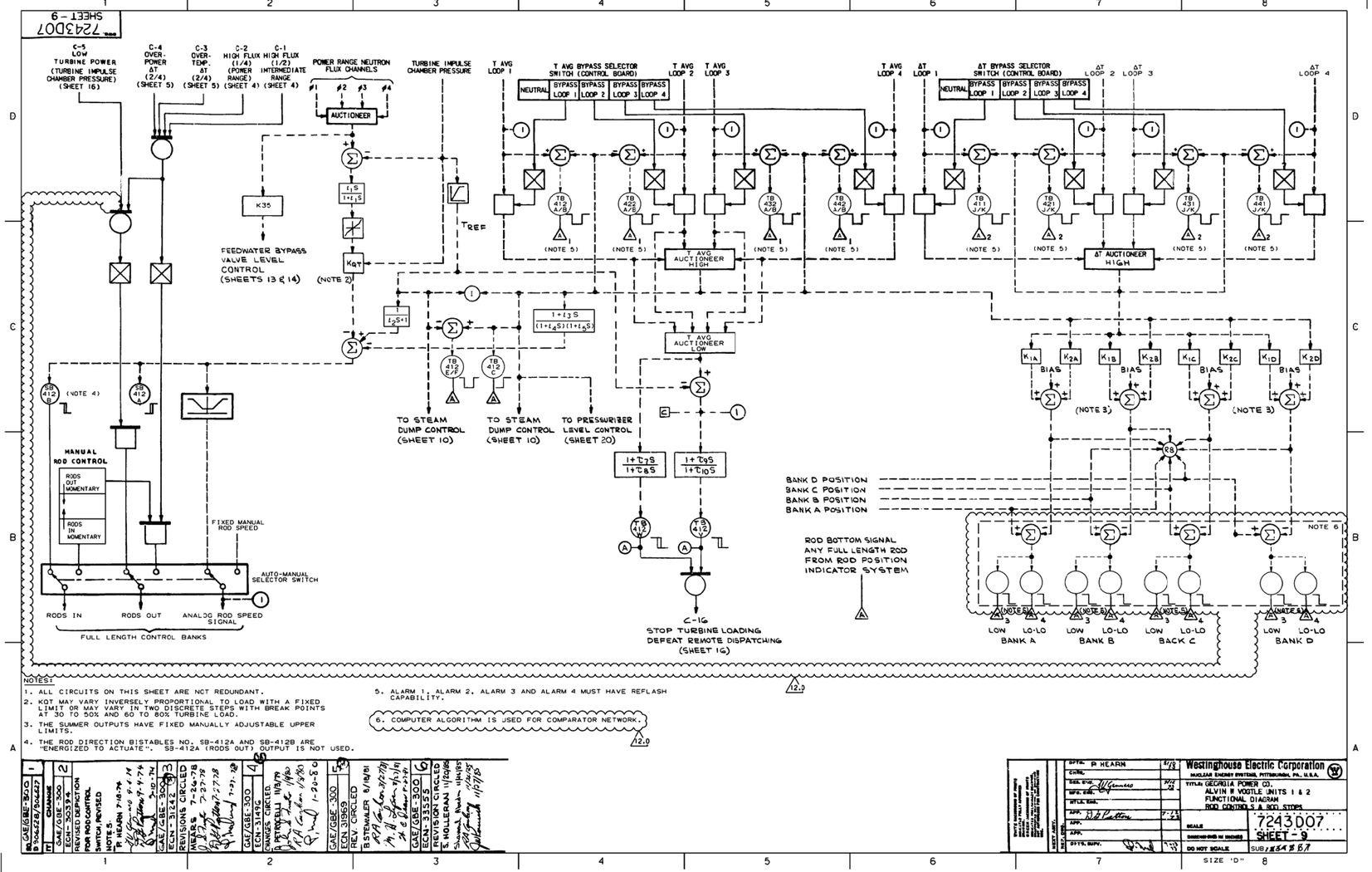
DFTM.	P HEARN	5/10/93
CHKR.		
DES. ENG.		7/12/93
SFTY. ENG.		
APP.		7-12-93
APP.		
APP.		
SFTY. & T.PH.		7/13/93

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

TITLE: GEORGIA POWER CO.
ALVIN W. VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAM
SAFEGUARD ACTUATION SYSTEM

SCALE: DIMENSIONS IN INCHES — DO NOT SCALE

7243D07
SHEET - 8
SUB 13



NOTES:

1. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT.
2. KQT MAY VARY INVERSELY PROPORTIONAL TO LOAD WITH A FIXED LIMIT OR MAY VARY IN TWO DISCRETE STEPS WITH BREAK POINTS AT 30 TO 50% AND 60 TO 80% TURBINE LOAD.
3. THE SUMMER OUTPUTS HAVE FIXED MANUALLY ADJUSTABLE UPPER LIMITS.
4. THE ROD DIRECTION BISTABLES NO. SB-412A AND SB-412B ARE "ENERGIZED TO ACTUATE". SB-412A (RODS OUT) OUTPUT IS NOT USED.
5. ALARM 1, ALARM 2, ALARM 3 AND ALARM 4 MUST HAVE REFLASH CAPABILITY.
6. COMPUTER ALGORITHM IS USED FOR COMPARATOR NETWORK.

REV	CHANGE
1	GAE/GBE-300 ECN-3128 REVISED DESCRIPTION FOR ROD CONTROL SWITCH, REVISED NOTE 5. P. HEARN 7-18-74
2	GAE/GBE-300 ECN-3128 P. HEARN 7-9-74 ECN-3128 3-10-74 ECN-3128 3-22-74
3	REVISIONS CIRCLED WEARS 7-26-78 7-27-78 7-27-78
4	GAE/GBE-300 ECN-31149C CHANGES CIRCLED 1/13/79 1/30/80 1-21-80 1-20-80
5	GAE/GBE-300 ECN 31969 REV. CIRCLED B STEWALER 8/19/81
6	GAE/GBE-300 ECN-33555 REVISION CIRCLED E. HOLLERAN 11/29/85 11/4/85 11/7/85

DSTR.	P HEARN	8/78
CHKR.		
DES. SUP.		3/24/78
MFG. ENG.		
MTLS. ENG.		
APP.		7-27-78
APP.		
DSTR. SUPV.		7-27-78

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

TITLE: GEORGIA POWER CO.
ALVIN B VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAM
ROD CONTROLS & ROD STOPS

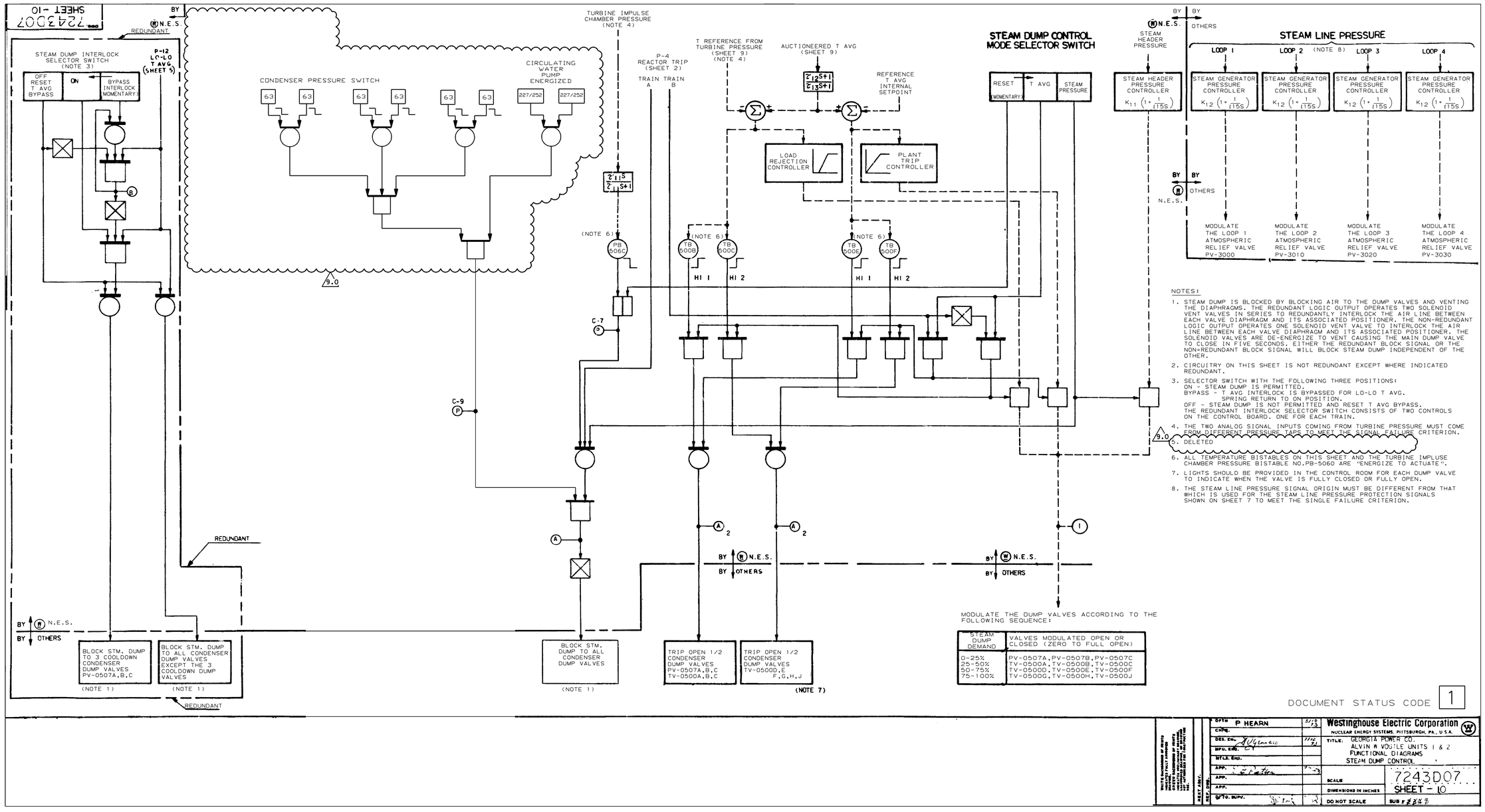
SCALE
DIMENSIONS IN INCHES
DO NOT SCALE

7243D07
SHEET - 9
SUB 1 X 6 A A 0 2

SIZE "D"

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A	CAD MICROSTATION JLO/02 — 1X6AA02-00233.DGN
TAB/SECT.: N/A; PAGE: N/A; FIGURE: N/A	Southern Nuclear Operating Company, Inc. FOR VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1
VERSION 12.0 | DATE 9/18/14	TITLE: FUNCTIONAL DIAGRAM ROD CONTROLS & ROD STOPS
REVISED BY SNC PER ABN 1081846201J240, VER. 1.0	

BY	CHK'D	APPR.1	APPR.2	VENDOR: WESTINGHOUSE ELEC	P.O.#:
JLO	CYN	TSL	X	DRAWING NO.	1X6AA02-00233



NOTES:

1. STEAM DUMP IS BLOCKED BY BLOCKING AIR TO THE DUMP VALVES AND VENTING THE DIAPHRAGMS. THE REDUNDANT LOGIC OUTPUT OPERATES TWO SOLENOID VENT VALVES IN SERIES TO REDUNDANTLY INTERLOCK THE AIR LINE BETWEEN EACH VALVE DIAPHRAGM AND ITS ASSOCIATED POSITIONER. THE NON-REDUNDANT LOGIC OUTPUT OPERATES ONE SOLENOID VENT VALVE TO INTERLOCK THE AIR LINE BETWEEN EACH VALVE DIAPHRAGM AND ITS ASSOCIATED POSITIONER. THE SOLENOID VALVES ARE DE-ENERGIZE TO VENT CAUSING THE MAIN DUMP VALVE TO CLOSE IN FIVE SECONDS. EITHER THE REDUNDANT BLOCK SIGNAL OR THE NON-REDUNDANT BLOCK SIGNAL WILL BLOCK STEAM DUMP INDEPENDENT OF THE OTHER.
2. CIRCUITRY ON THIS SHEET IS NOT REDUNDANT EXCEPT WHERE INDICATED REDUNDANT.
3. SELECTOR SWITCH WITH THE FOLLOWING THREE POSITIONS:
   ON - STEAM DUMP IS PERMITTED.
   BYPASS - T AVG INTERLOCK IS BYPASSED FOR LO-LO T AVG. SPRING RETURN TO ON POSITION.
   OFF - STEAM DUMP IS NOT PERMITTED AND RESET T AVG BYPASS.
   THE REDUNDANT INTERLOCK SELECTOR SWITCH CONSISTS OF TWO CONTROLS ON THE CONTROL BOARD. ONE FOR EACH TRAIN.
4. THE TWO ANALOG SIGNAL INPUTS COMING FROM TURBINE PRESSURE MUST COME FROM DIFFERENT PRESSURE TAPS TO MEET THE SIGNAL FAILURE CRITERION.
5. DELETED
6. ALL TEMPERATURE BISTABLES ON THIS SHEET AND THE TURBINE IMPLUSE CHAMBER PRESSURE BISTABLE NO.PB-5060 ARE "ENERGIZE TO ACTUATE".
7. LIGHTS SHOULD BE PROVIDED IN THE CONTROL ROOM FOR EACH DUMP VALVE TO INDICATE WHEN THE VALVE IS FULLY CLOSED OR FULLY OPEN.
8. THE STEAM LINE PRESSURE SIGNAL ORIGIN MUST BE DIFFERENT FROM THAT WHICH IS USED FOR THE STEAM LINE PRESSURE PROTECTION SIGNALS SHOWN ON SHEET 7 TO MEET THE SINGLE FAILURE CRITERION.

DOCUMENT STATUS CODE 1

DSN P HEARN	5/13	Westinghouse Electric Corporation	
CHKD.		NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.	
DES. ENG.	7/10/71	TITLE: GEORGIA POWER CO.	
APVD. ENG.		ALVIN A VOGTLE UNITS 1 & 2	
APVD. ENG.		FUNCTIONAL DIAGRAMS	
APP.		STEAM DUMP CONTROL	
APP.		SCALE	7243D07...
APP.		DIMENSIONS IN INCHES	SHEET - 10
APVD. SUPV.		DO NOT SCALE	SUB

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

VERSION 9.0 | DATE 10-14-05

REVISED BY SNC PER ABN-80649, VER. 1.0

SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
DFV	WBJ	ASK	x

CAD MICROSTATION DFV/01

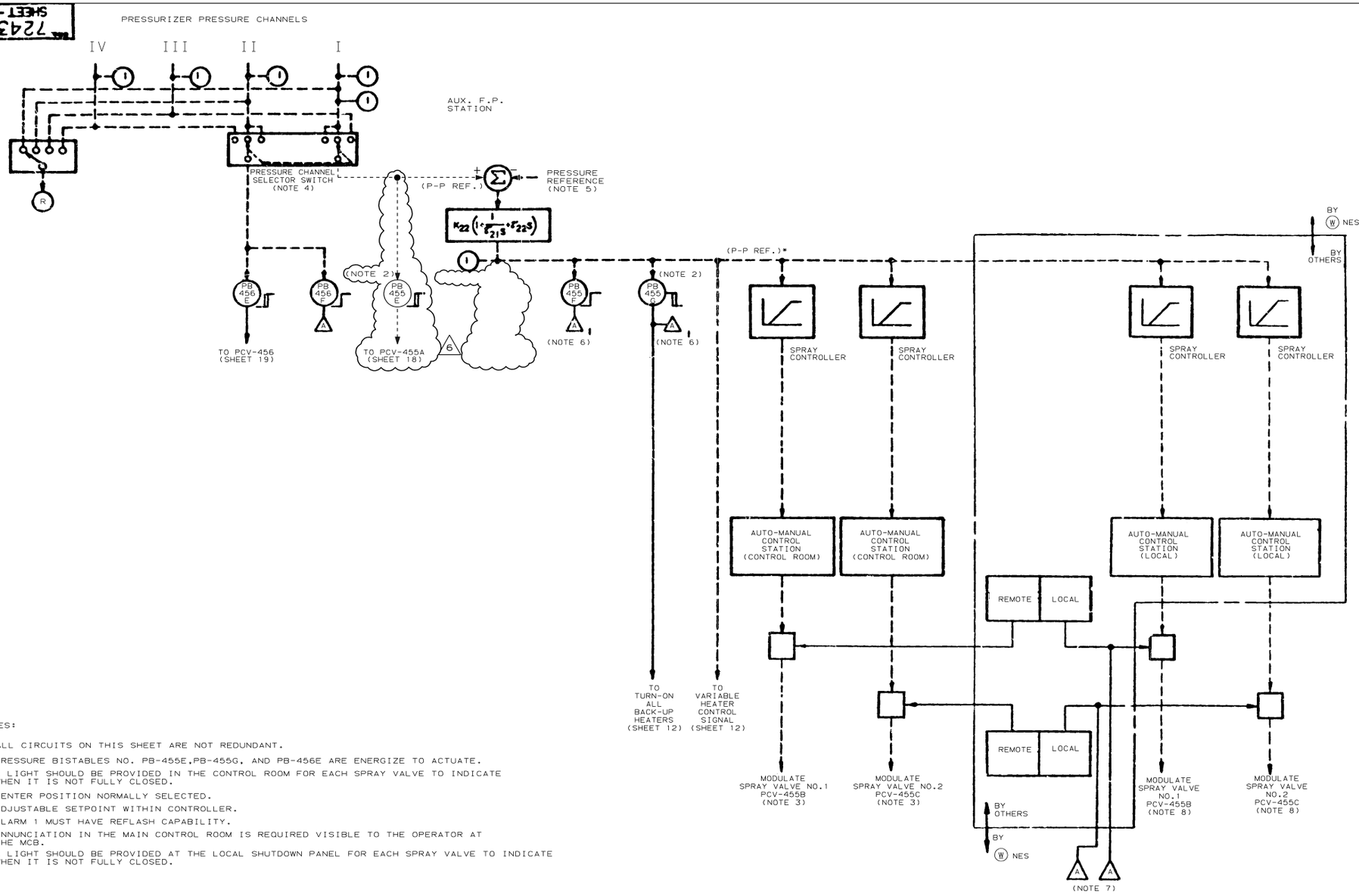
1X6AA02-00234.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
FUNCTIONAL DIAGRAMS
STEAM DUMP CONTROL

VENDOR: WESTINGHOUSE ELEC | P.O.#: –

DRAWING NO. 1X6AA02-00234



NOTES:

1. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT.
2. PRESSURE BISTABLES NO. PB-455E,PB-455G, AND PB-456E ARE ENERGIZE TO ACTUATE.
3. A LIGHT SHOULD BE PROVIDED IN THE CONTROL ROOM FOR EACH SPRAY VALVE TO INDICATE WHEN IT IS NOT FULLY CLOSED.
4. CENTER POSITION NORMALLY SELECTED.
5. ADJUSTABLE SETPOINT WITHIN CONTROLLER.
6. ALARM 1 MUST HAVE REFLASH CAPABILITY.
7. ANNUNCIATION IN THE MAIN CONTROL ROOM IS REQUIRED VISIBLE TO THE OPERATOR AT THE MCB.
8. A LIGHT SHOULD BE PROVIDED AT THE LOCAL SHUTDOWN PANEL FOR EACH SPRAY VALVE TO INDICATE WHEN IT IS NOT FULLY CLOSED.

NO.	REVISIONS	DATE	DR	CHK	APPV
6	INC. PER DCP 98-V1N0061	11-06-00	CD	EOG	MWD
	SCS REVISIONS				

DOCUMENT STATUS CODE 1

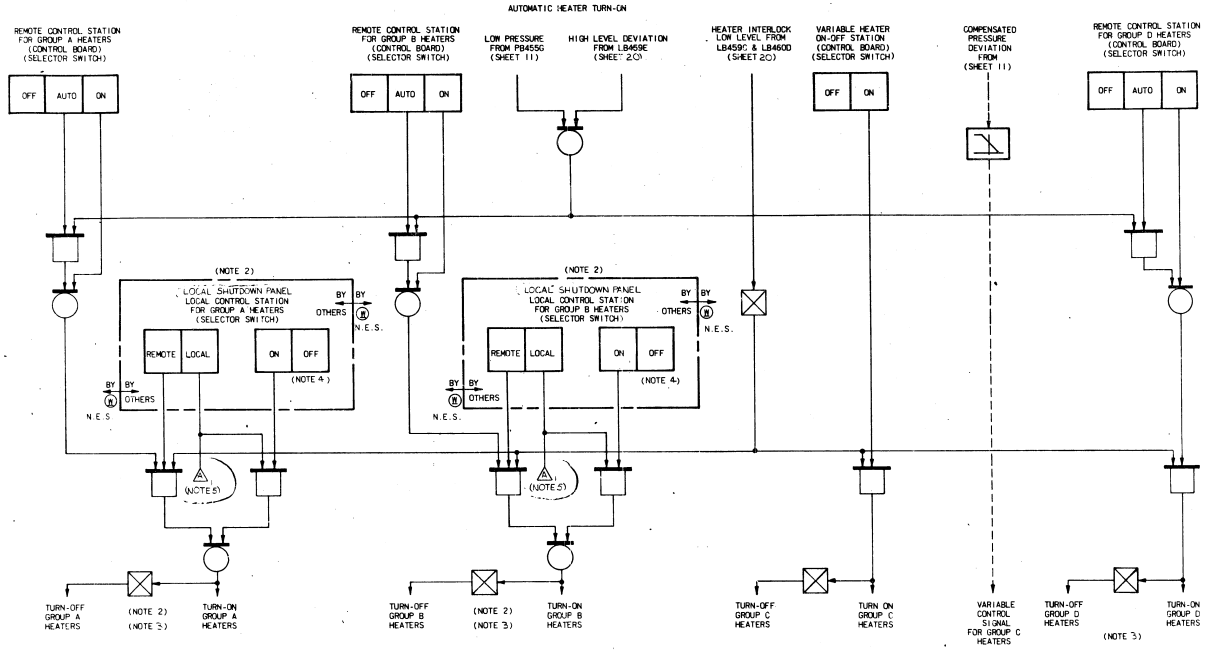
1X6AA02-00235-9

Westinghouse Electric Corporation
GEORGIA POWER CO.
ALVIN W. VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
PRESSURIZER PRESSURE CONTROL
7243D07 SHEET-11
SUB 1 2 3 4 5 6

SHEET -12

7243D07

AUTOMATIC HEATER TURN-ON



NOTES:

1. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT.
2. GROUP A AND GROUP B HEATERS MUST BE ON SEPARATE VITAL POWER SUPPLIES WITH THE LOCAL CONTROL SEPARATED SO THAT ANY SINGLE FAILURE DOES NOT DEFEAT BOTH.
3. BACK UP HEATER STATUS INDICATION IN CONTROL ROOM.
4. PRECAUTIONS SHOULD BE TAKEN TO AVOID MANUAL HEATER OPERATION, WHICH WOULD CAUSE HEATER DAMAGE, IF THE WATER LEVEL UNCOVERS THE HEATERS.
5. ANNUNCIATION IN THE MAIN CONTROL ROOM IS REQUIRED TO BE VISIBLE TO THE OPERATOR AT THE MAIN CONTROL BOARD.

GEORGIA POWER COMPANY
PLANT: ALVIN W. VOGTLE NUCLEAR PLANT
UNITS: 1 & 2
STATUS: APPROVED
CERTIFICATION LTR. NO. GP-3709
AUTHORITY: J. L. VOTA
ENGR. LTR. NO. EP/SA-36110

MICROFILM QUALITY ACCEPTABLE

IT	CHANGE
1	GAE/GBE-300 DS#GBE/SD#627
2	GAE/GBE-300 ECN- 9[illegible] REVISIONS CIRCLED
3	GAE/GBE-300 ECN-31242 REVS CIRCLED
4	GAE/GBE-300 ECN 31969 REV CIRCLED
5	GAE/GBE-300 ECN-32037 CHANGES CIRCLED

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

DFTR.	D. HEARN
CHK.	
DES. ENG.	
MFG. ENG.	
MTLS. ENG.	
APP.	
APP.	
APP.	
DFTG. SUPV.	

TITLE: GEORGIA POWER CO.'S ALVIN W. VOGTLE UNITS 1 & 2 FUNCTIONAL DIAGRAM PRESSURIZER HEATER CONTROL

SCALE: DO NOT SCALE
DIMENSIONS IN INCHES
SUB y Z84 5

7243D07
SHEET - 12

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DATE: CAMERA OPERATOR SECTION SUPERVISOR

DISTRIBUTION TO: FOR: REVIEW | INFO
- MECHANICAL
  - BALANCE OF PLANT
  - BOILER/NSSS
  - PLANT UTILITIES
- PLANT DESIGN
- CONTROL SYSTEMS ✓
- ELECTRICAL ✓
  - WIRING
  - CONDUIT
- MDS
- PAINTING & COATINGS
- CIVIL/STRUCTURAL
- NUCLEAR
- STRESS
- ARCHITECTURAL
- STARTUP
- CONSTRUCTION
- NOT REQ'D BY ENGRG
- CLIENT

IDENTIFYING TITLE OF THIS DOCUMENT:

Resub for micro film quality

Bechtel Log No. 1 KEAAD-236-7

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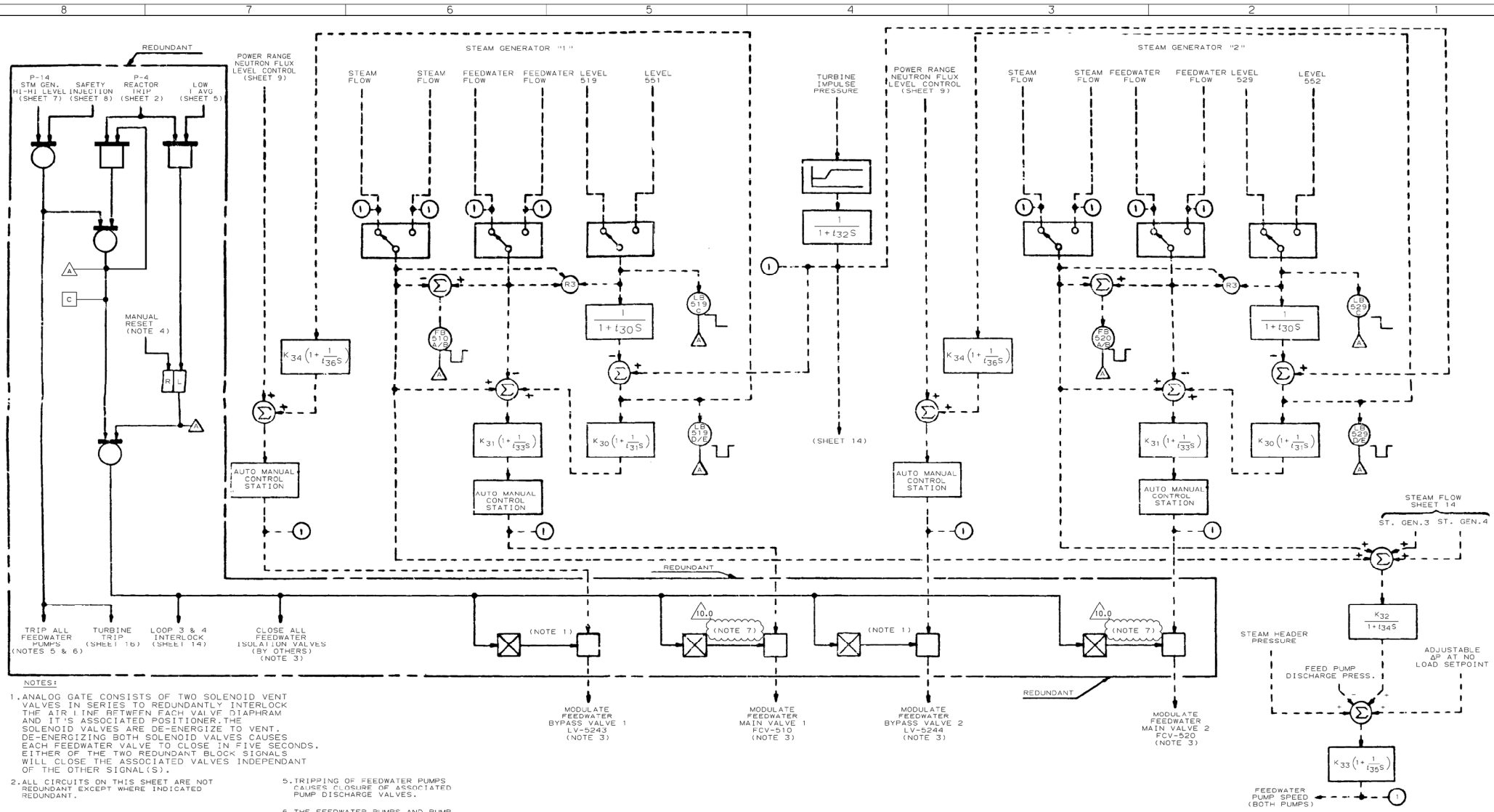
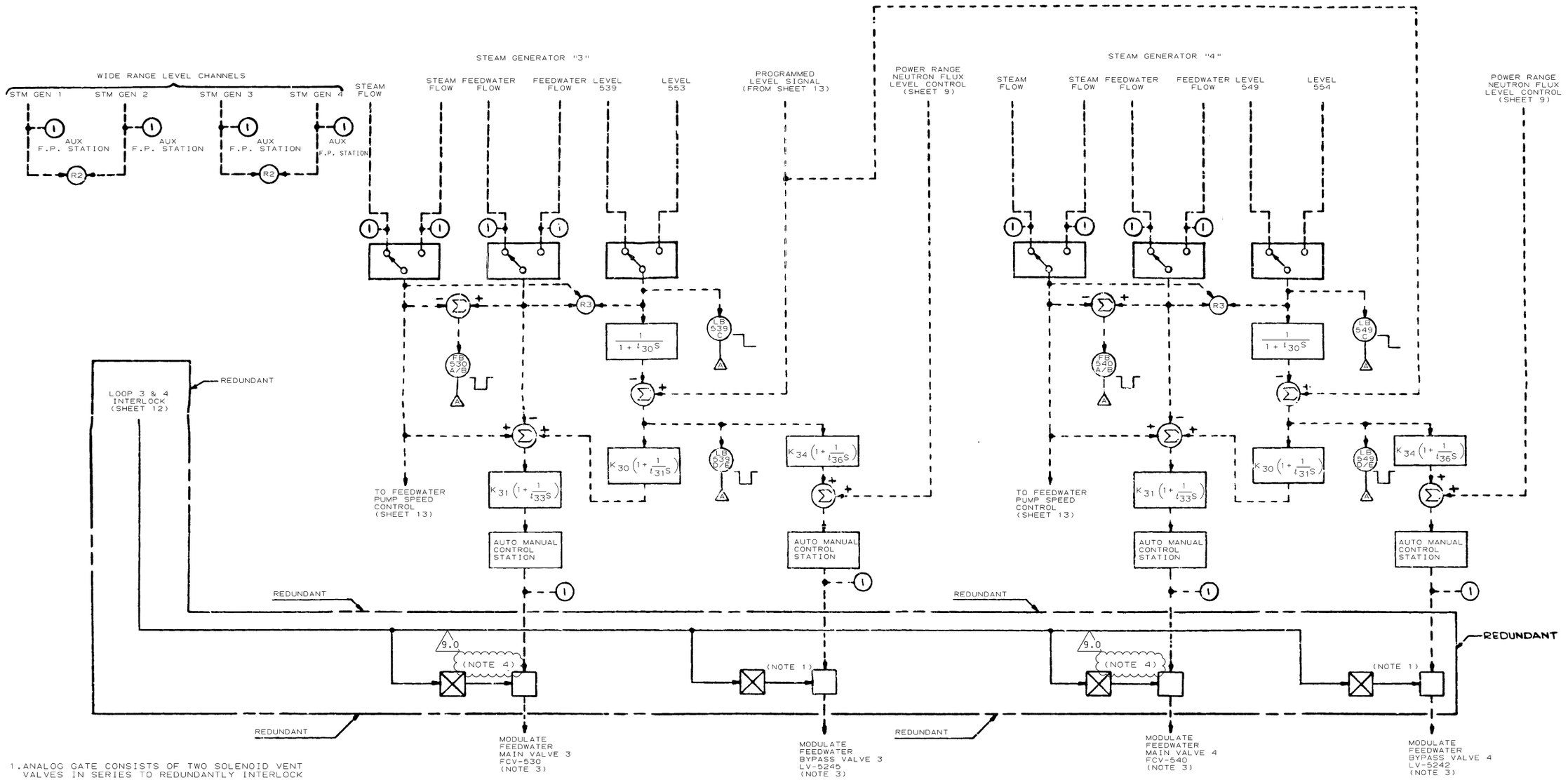
DATE RECEIVED: 4-04-82 | SIGNED: J Mauck
DOCUMENT STATUS
(X) WORK MAY PROCEED
( ) REVISE AND RESUBMIT WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED
( ) REVISE AND RESUBMIT WORK MAY NOT PROCEED
( ) INFORMATION NOT ON DISTRIBUTION REQ'D | DATE: 8-04-82



NOTES:

1. ANALOG GATE CONSISTS OF TWO SOLENOID VENT VALVES IN SERIES TO REDUNDANTLY INTERLOCK THE AIR LINE BETWEEN EACH VALVE DIAPHRAM AND IT'S ASSOCIATED POSITIONER. THE SOLENOID VALVES ARE DE-ENERGIZE TO VENT. DE-ENERGIZING BOTH SOLENOID VALVES CAUSES EACH FEEDWATER VALVE TO CLOSE IN FIVE SECONDS. EITHER OF THE TWO REDUNDANT BLOCK SIGNALS WILL CLOSE THE ASSOCIATED VALVES INDEPENDANT OF THE OTHER SIGNAL(S).
2. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT EXCEPT WHERE INDICATED REDUNDANT.
3. OPEN/SHUT INDICATION FOR EACH FEEDWATER VALVE IN CONTROL ROOM.
4. THE REDUNDANT MANUAL RESET CONSISTS OF TWO MOMENTARY CONTROLS ON THE CONTROL BOARD, ONE FOR EACH TRAIN.
5. TRIPPING OF FEEDWATER PUMPS CAUSES CLOSURE OF ASSOCIATED PUMP DISCHARGE VALVES.
6. THE FEEDWATER PUMPS AND PUMP DISCHARGE VALVES ARE SUPPLIED BY OTHERS.
7. ANALOG GATE CONSISTS OF TWO SOLENOID VALVES. THE SOLENOID VALVES ARE DE-ENERGIZE TO VENT. DE-ENERGIZING BOTH SOLENOID VALVES CAUSES EACH FEEDWATER VALVE TO CLOSE IN FIVE SECONDS. 10.0

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A	CAD MICROSTATION TSL/01 — 1X6AA02-00237.DGN
TAB/SECT. N/A; PAGE N/A; FIGURE N/A	Southern Nuclear Operating Company, Inc. FOR VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1
VERSION 10.0 — DATE 01/11/08	TITLE: FUNCTIONAL DIAGRAM FEEDWATER CONTROL & ISOLATION
REVISED BY SNC PER ABN-V00442, VER.3.0	
SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.	
BY: TSL — CHK'D: RBH — APPR.1: JMR — APPR.2:	VENDOR: WESTINGHOUSE — P.O.#: PAV-00370
	DRAWING NO. 1X6AA02 00237 — SIZE D



1. ANALOG GATE CONSISTS OF TWO SOLENOID VENT VALVES IN SERIES TO REDUNDANTLY INTERLOCK THE AIR LINE BETWEEN EACH VALVE DIAPHRAM AND IT'S ASSOCIATED POSITIONER. THE SOLENOID VALVES ARE DE-ENERGIZE TO VENT. DE-ENERGIZING BOTH SOLENOID VALVES CAUSES EACH FEEDWATER VALVE TO CLOSE IN FIVE SECONDS. EITHER OF THE TWO REDUNDANT BLOCK SIGNALS WILL CLOSE THE ASSOCIATED VALVES INDEPENDANT OF THE OTHER SIGNAL(S).

2. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT, EXCEPT WHERE INDICATED "REDUNDANT"

3. OPEN/SHUT INDICATION FOR EACH FEEDWATER VAVLE IN CONTROL ROOM

4. ANALOG GATE CONSISTS OF TWO SOLENOID VALVES. THE SOLENOID VALVES ARE DE-ENERGIZE TO VENT. DE-ENERGIZING BOTH SOLENOID VALVES CAUSES EACH FEEDWATER VALVE TO CLOSE IN FIVE SECONDS. 9.0

FOR INFORMATION ONLY

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A				CAD MICROSTATION TSL/01 — 1X6AA02-00238.DGN
TAB/SECT. N/A; PAGE N/A; FIGURE N/A				Southern Nuclear Operating Company, Inc. FOR VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1
VERSION 9.0 DATE 01/11/08				TITLE: FUNCTIONAL DIAGRAM FEEDWATER CONTROL & ISOLATION
REVISED BY SNC PER ABN-V00442, VER.3.0				
SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.				
BY	CHK'D	APPR.1	APPR.2	VENDOR: WESTINGHOUSE — P.O.#: PAV-00370
TSL	RBH	JMR		DRAWING NO. 1X6AA02-00238

SIZE D

SAFETY INJECTION SIGNAL (SHEET 8)

STEAM GENERATOR 1 2/4 LOW LOW LEVEL (SHEET 7)

STEAM GENERATOR 2 2/4 LOW LOW LEVEL (SHEET 7)

STEAM GENERATOR 3 2/4 LOW LOW LEVEL (SHEET 7)

STEAM GENERATOR 4 2/4 LOW LOW LEVEL (SHEET 7)

AMSAC ACTUATION SIGNAL (SHEET 7) (NOTE 13)

BY OTHERS / BY (N) N.E.S.

BLACK-OUT SIGNAL (NOTE 12)

TRIP OF MAIN FEED PUMPS FP-1 FP-2

BY OTHERS

NOT REDUNDANT

MANUAL CONTROL CONTROL ROOM (NOTE 7)

MANUAL CONTROL LOCAL (NOTES 2 & 7)

MOTOR DRIVEN AUX. FEED PUMP FEED SYSTEM VALVES (NOTE 4)

MANUAL CONTROL CONTROL ROOM (NOTE 7)

MANUAL CONTROL LOCAL (NOTES 2 & 7)

TURBINE DRIVEN AUX. FEED PUMP FEED SYSTEM VALVES (NOTE 4)

NOT REDUNDANT

MANUAL CONTROL CONTROL ROOM

(NOTE 9) MANUAL CONTROL LOCAL (NOTE 2)

C.R. LOCAL

TURBINE SPEED CONTROL

MANUAL START, CONTROL ROOM (NOTE 8)
MANUAL START, LOCAL (NOTES 2,3 & 8)
MANUAL STOP, CONTROL ROOM (NOTES 4 & 8)
MANUAL STOP, LOCAL (NOTES 2,3 & 8)

MANUAL START, CONTROL ROOM
MANUAL START, LOCAL (NOTES 2 & 3)
MANUAL STOP, CONTROL ROOM (NOTE 4)
MANUAL STOP, LOCAL (NOTE 2 & 3)

START STOP

MOTOR DRIVEN AUX. FEED PUMPS 1 & 2 (NOTES 1 & 8) (NOTES 10 & 11)

CLOSE BLOWDOWN ISOLATION AND SAMPLE LINE VALVES (NOTE 14) FOR ALL STEAM GENERATORS 11.0

(NOTE 11) START STOP

TURBINE DRIVEN AUX. FEED PUMP (NOTE 6)

NOTES:

1. TRAIN A CONTROLS MAFP 1: 1AA-02 BREAKER NUMBER 17
   TRAIN B CONTROLS MAFP 2: 1BA-03 BREAKER NUMBER 21
2. LOCAL CONTROL OVERRIDES ALL OTHER SIGNALS.
3. LOCAL OVERRIDE ACTUATES ALARM IN CONTROL ROOM.
4. MANUAL STOP OVERRIDES THE AUTOMATIC START. MANUAL STOP OVERRIDE ACTUATES ALARM IN CONTROL ROOM.
5. OPEN/SHUT INDICATION IN CONTROL ROOM.
6. PUMP OPERATING LIGHTS IN CONTROL ROOM.
7. INDIVIDUAL FOR EACH VALVE.
8. INDIVIDUAL FOR EACH PUMP.
9. THE TURBINE SPEED CONTROL IS TYPICAL. ACTUAL IMPLEMENTATION MAY NOT INCLUDE SPEED CONTROL.
10. THE PUMP START MAY BE DELAYED AND SEQUENCED IF THE EMERGENCY DIESEL POWER CAPABILITY IS LESS THAN THE TOTAL LOAD WITH ALL SYSTEMS STARTING. THE TIME DELAY, IF USED, MAY NOT EXCEED THE MAXIMUM STARTING TIME REQUIREMENTS FOR THIS SYSTEM.
11. THE PUMP START MUST BE SEALED IN (LATCHED), SO THAT LOSS OF THE ACTUATION SIGNAL WILL NOT CAUSE THE PUMP TO STOP
12. "BLACK-OUT" CORRESPONDS TO A LOSS OF OFFSITE POWER.
13. THE AMSAC IS NOT REDUNDANT. ITS SIGNAL IS DUPLICATED FOR INPUT TO EACH AUXILIARY FEEDWATER ACTUATION CIRCUIT. ISOLATION DEVICES ARE REQUIRED BETWEEN THE NON-1E AMSAC CIRCUITS & THE 1E AUXILIARY FEEDWATER START CIRCUITS.

11.0 14. THE SAMPLE VALVE CIRCUITS ARE TRAIN RELATED.
TRAIN "A" CONTROLS HV-9451 AND HV-9454.
TRAIN "B" CONTROLS HV-9453 AND HV-9452.

10	GAE/GBE-300	5	GAE/GBE-300 ECN-31496 CHANGES CIRCLED. T. PETROCELLI 11/13/79
9	GAE/GBE-300 ECN-30888 REVISIONS CIRCLED	3	GAE/GBE-300 ECN-30394 ADDED NEW NOTES 4 & 11, RESEQUENCE NOTES 5 THRU 10.
8	GAE/GBE-300 ECN-31242 REVS CIRCLED HBEARS 7-17-78	4	GAE/GBE-300 ECN-30394 CHANGES CIRCLED
7	GAE/GBE-300 ECN-34109 CHANGES CIRCLED	7	GAE/GBE-300 ECN-33209 CHANGES CIRCLED
6	GAE/GBE-300 ECN-33209 CHANGES CIRCLED	6	GAE/GBE-300 ECN-30394 CHANGES CIRCLED

DSN. ENG.		
MFG. ENG.		
MTLS. ENG.		
APP.		
APP.		
APP.		
QA REL. ENGR.		

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

TITLE: GEORGIA POWER CO.
ALVIN W VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
AUXILIARY FEEDWATER PUMPS STARTUP

SCALE

DIMENSIONS IN INCHES

DO NOT SCALE

7243D07 SHEET - 15

SUB 1X6AA67

THIS DWG. IS REFERENCED IN VENDOR MANUAL:

TAB/SECT. N/A

PAGE

FIGURE

CAD | MICROSTATION | DFV-01

VERSION 11.0 | DATE 4-27-05

REVISED BY SNC PER ABN-80651, VER. 1.0

SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
DFV	PWG	ASK	X

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
FUNCTIONAL DIAGRAMS
AUXILIARY FEEDWATER PUMPS
STARTUP

VENDOR: WESTINGHOUSE | P.O.#:

DRAWING NO. 1X6AA02-00239

AMSAC ACTUATION LOGIC (SHEET 7) (NOTE 9)

P-4 REACTOR TRIP BY (W) NES (TRAIN A) (SHEET 2)

STEAM GENERATOR HI-HI LEVEL OR SI (TRAIN A) (SHEET 13)

STEAM GENERATOR HI-HI LEVEL OR SI (TRAIN B) (SHEET 13)

P-4 REACTOR TRIP (TRAIN B) (SHEET 2)

BY OTHERS

CONDENSER VACUUM

THRUST BRG. FAILURE

BRG. VIBRATION HIGH
ELECTRIC GENERATOR
PROTECTION RELAYS

LOSS OF STATOR COOLANT

SHAFT LO PUMP
LO PRESS. AT SPEED

LOW EHC HYDRAULIC PRESS.

MOISTURE SEPARATOR
HIGH LEVEL

LOW BRG. OIL PRESS.

TURBINE SPEED
SIGNAL LOSS

BACKUP OVERSPEED

LOSS OF 24V D-C

MANUAL TRIP P.B.

ELECTRIC GENERATOR
PROTECTION RELAYS

LOSS OF EHC
125V D-C AT SPEED

TURBINE SPEED
SIGNAL LOSS

BACK-UP OVERSPEED

LOSS OF 24V D-C

MANUAL TRIP P.B.

125V D-C TRIP

24V D-C TRIP

(NOTE 7)

ENERGIZE

MECHANICAL TRIP SOLENOID VALVE

MECHANICAL OVERSPEED

MANUAL TRIP (AT TURBINE)

OPERATE

MECHANICAL TRIP VALVE

DE-ENERGIZE

ELECT. TRIP SOLENOID VALVE

OPERATE

ELECTRICAL TRIP VALVE

DUMP

DUMP

EMERGENCY TRIP SYSTEM HYDR. FLUID

CLOSE

COMBINED INTERMEDIATE STOP VALVES & INTERCEPT VALVES

CLOSE

CONTROL VALVES

MAIN STOP VALVES (NOTE 3)

I PB 6161

II PB 6162

III PB 6163

(NOTE I)

APPROX. 30 SECONDS (NOTE 5)

P-9 (SHEET 4)

BY OTHERS (W) NES

FAST AUXILIARY BUS TRANSFER (NOTE 6)

TO GENERATOR TRIP

TO REACTOR TRIP (SHEET 2)

REDUNDANT

BY OTHERS (W) NES

TURBINE POWER
(TURBINE IMPULSE CHAMBER PRESSURE)

I PB 5054

II PB 5064

P-13

P-13 TO P-7 (SHEET 4)

REDUNDANT

III PB 5050

C-5

C-5 BLOCK AUTOMATIC [illegible] WITH C-7 ANNUAL

TO [illegible] AT [illegible] ING [illegible] OTHERS)

NOT REDUNDANT

NOT REDUNDANT

TURBINE POWER
(TURBINE IMPULSE CHAMBER PRESSURE)

PB 5059

PB 5060

(NOTE 10)

0-420 SEC.

C-20 PERMIT AMSAC ACTUATION (SHEET 7)

NOT REDUNDANT

TURBINE RUNBACK
VIA LOAD REFERENCE

C-3 OVERTEMPERATURE ΔT (2/4) (SHEET 5)

C-4 OVERPOWER ΔT (2/4) (SHEET 5)

TIME DELAY RELAY LOGIC (CYCLIC)

1-5 SECONDS

ON

OFF

10-60 S

TO DEFEAT REMOTE DISPATCHING (NOTE 4) (BY OTHERS)

REDUCE LOAD REFERENCE AT 133% PER MINUTE (BY OTHERS)

C-16 LOW T AVG. (SHEET 9)

BYPASS FOR TURBINE TESTING/PLANT COOLDOWN (MOMENTARY)

TO DEFEAT REMOTE DISPATCHING (NOTE 4) (BY OTHERS)

STOP TURBINE LOADING (BY OTHERS)

NOTES:

1. THESE SIGNALS INDICATE THE CLOSING OF THE STOP VALVES. POSITION DETECTION IS ACCOMPLISHED BY TWO CONTACTS PER STOP VALVE. ONE FOR EACH TRAIN. THE LOGIC SHOWN IS FOR FOUR STOP VALVES.
2. REDUNDANCY IS INDICATED IN REGARDS TO (W) REQUIREMENTS ONLY.
3. OPEN/SHUT INDICATION IN CONTROL ROOM FOR EACH STOP VALVE.
4. THE REMOTE DISPATCHING IS TYPICAL. ACTUAL IMPLEMENTATION MAY NOT INCLUDE REMOTE DISPATCHING.
5. GENERATOR MOTORING PROTECTION SHOULD NOT DEFEAT THE THIRTY SECOND DELAY.
6. THE REACTOR COOLANT PUMP BUS TRANSFER SHOULD BE COMPLETED WITHIN SIX CYCLES IN ORDER TO INSURE COMPATIBILITY WITH THE UNDERFREQUENCY TRIP OF THE REACTOR COOLANT PUMP CIRCUIT BREAKERS. THE TRANSFER TIME LIMIT MAY BE EXTENDED ANOTHER 4 CYCLES TO A TOTAL OF NOT GREATER THAN 10 CYCLES, IF THE SYSTEM DYNAMICS ARE SUCH THAT A RCP TRIP DOES NOT OCCUR. THE 10 CYCLE LIMIT IS FOR PUMP MOTOR PROTECTION.
7. ISOLATION IS PROVIDED FOR THIS SIGNAL TO AVOID AN INTERCONNECTION BETWEEN TRAIN A AND TRAIN B.
8. THIS FUNCTION IS USED TO BYPASS THE REDUCED TEMPERATURE RETURN TO POWER CONTROL SYSTEM CIRCUITRY TO ALLOW TURBINE TESTING AND/OR TO DEFEAT THE C-16 ANNUNCIATOR SUCH AS WOULD BE REQUIRED DURING A PLANT COOLDOWN.
9. THE AMSAC SIGNAL IS NOT REDUNDANT. ISOLATION DEVICES ARE REQUIRED BETWEEN THE NON-1E AMSAC CIRCUITS & THE 1E TURBINE TRIP CIRCUITS.
10. TURBINE IMPULSE CHAMBER PRESSURE SIGNAL DERIVED FROM SENSORS PT505 AND PT506.

CHANGES												
GA-1/GBE-300	ECN-30394	REV. REL. TURBINE TRIP-PING (ONE)	P. HEARN 7-22-78	GAE/GBE-300	ECN-31242	REV'S CIRCLED	MEARS 7-17-78	GAE/GBE-300	ECN-31496	CHANGES CIRCLED	D. PETROFELLI 11/8/79	GAE/GBE-300	ECN-33002	CHANGES CIRCLED	MR. DERRY 7-31-84
GAE/GBE-300	ECN-34109	CHANGES CIRCLED	O'HARA 7-28-87	GAE/GBE-300	ECN-34476	CHANGES CIRCLED	O'HARA 1-26-89								

GEORGIA POWER COMPANY
PLANT: VOGTLE ELECTRIC GENERATING PLANT
UNITS: 1 & 2 SPIN: AASFED03
STATUS: APPROVED
CERTIFICATION LTR. NO. GP-13060
AUTHORITY: R.S. HOWARD
ENGR. LTR. NO. EP/SA-53512

WESTINGHOUSE PROPRIETARY DATA

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NO.	DESCRIPTION	DATE	DR.	CHK.	OM
1	REV. TO CLOSE ABN 00370		WVG.		
2	REV. TO CLOSE ABN 87-VIF-57A009, ABN 87-VIN0447A002 AND EFCR 42N. NO DRAFTING		C.I.B.		
3	CHANGE REQUIRED FROM WESTINGHOUSE REV. 6				

SOUTHERN COMPANY SERVICES
BIRMINGHAM, ALABAMA

DRAWING NUMBER: IX6AA02-240 REV. 9

DSN.	P HEARN	5/8/78
CHKR.		
DES. ENG.	L U Gemis	11/6/78
RFR. ENG.		
RFLD. ENG.		
APP.	S E Patton	7-11-78
APP.		
APP.		
OPTG. SUPV.		

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS, PITTSBURGH, PA., U.S.A.

TITLE: GEORGIA POWER CO.
ALVIN W VOGTLE UNITS 1 & 2 FUNCT. DIAG. TURBINE TRIP, RUNBACKS, AND OTHER SIGNALS (W REQUIREMENTS)

SCALE: NONE
DIMENSIONS IN INCHES
DO NOT SCALE

7243D07
SHEET - 16
SUB 1238367

Southern Company Services, Inc.
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3) DISAPPROVED — RESUBMITTAL REQUIRED
4) INFORMATION ONLY — DISTRIBUTION REQ'D

PLANT: VOGTLE
1)5AA02-240-13
FUNCT DIAG TURBINE TRIP, RUN BACKS, AND OTHER SIGNALS
VENDOR: WE
P.O.: PAY-2

GPC PLANT VOGTLE CONSTR MICROGRAPHICS
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SECTION SUPERVISOR
CAMERA OPERATOR
DATE
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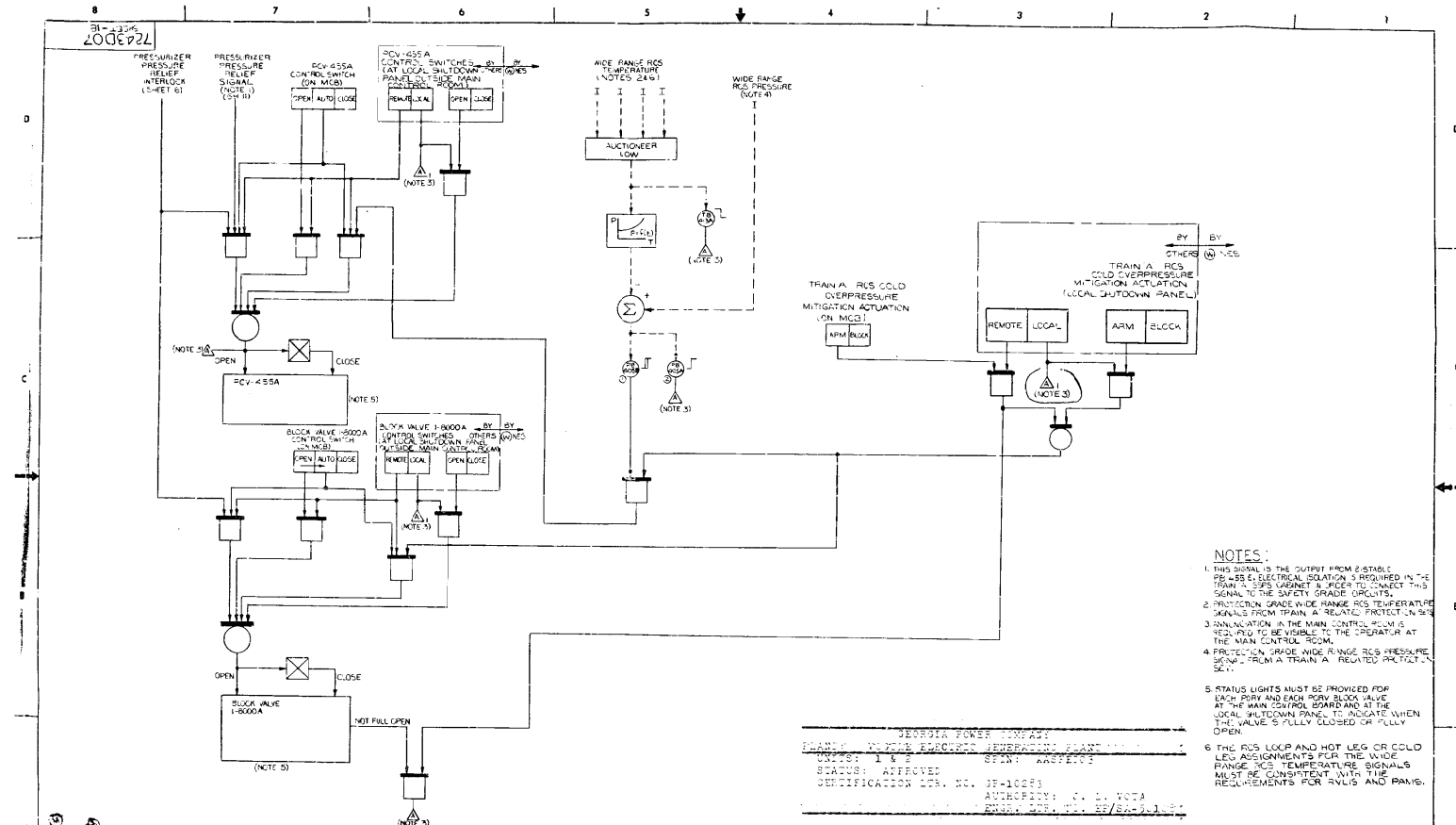

VOGTLE ELECTRIC GENERATING PLANT — JOB NO. 9510

EQUIPMENT TAG NO. VAR

STARTUP DESIGNATION NO. VAR
ACTIVITY NO. VAR
SYSTEM NO. VAR
CATEGORY NO. A

RETROFITTING REQUIRED ☐ YES ☒ NO

DISTRIBUTION TO: FOR: REVIEW	INFO
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• HVAC	
NSSS	
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CONDUIT	
• HAZARDS	
• CIVIL/STRUCTURAL	
• NUCLEAR	
• STRESS/PLANT DESIGN	
CODES AND STANDARDS	
• ARCHITECTURAL	
• STARTUP	
• CONSTRUCTION	
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IDENTIFYING TITLE OF THIS DOCUMENT
Functional Diag

Bechtel Log No 10-22-87

1 X6AA02-494-42

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4 ☐ INFORMATION ONLY DISTRIBUTION REQ'D ☐ YES
5 ☐ RESUBMIT — NOT ACCEPTABLE FOR MICROFILM WORK MAY PROCEED

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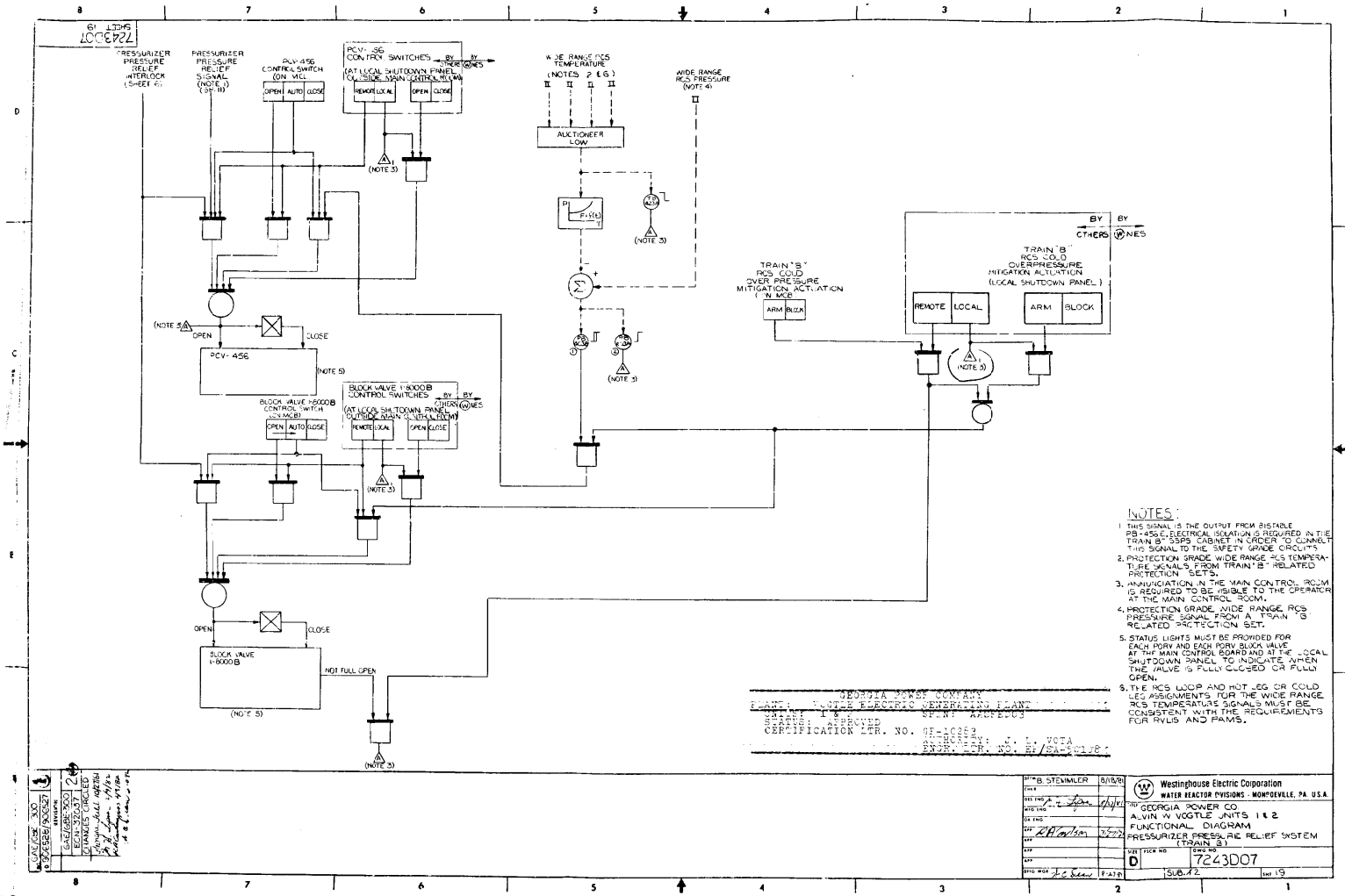
PP-5937 (9510) 9-84



GPC PLANT VOGTLE CONSTR MICROGRAPHICS

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11/16/89
DATE

Nicholas Petrilli
CAMERA OPERATOR

Nicholas Petrilli
SECTION SUPERVISOR

VOGTLE ELECTRIC GENERATING PLANT — JOB NO. 9510

EQUIPMENT TAG NO. VAR

STARTUP DESIGNATION NO. VAR
ACTIVITY NO. VAR
SYSTEM NO. VAR
CATEGORY NO. M4

RETROFITTING REQUIRED ☐ YES ☒ NO

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• HVAC	
NSSS	
BOP	
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• ELECTRICAL	✓
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CONDUIT	
• HAZARDS	
• CIVIL/STRUCTURAL	
• NUCLEAR	
• STRESS/PLANT DESIGN	
CODES AND STANDARDS	
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• STARTUP	
• CONSTRUCTION	
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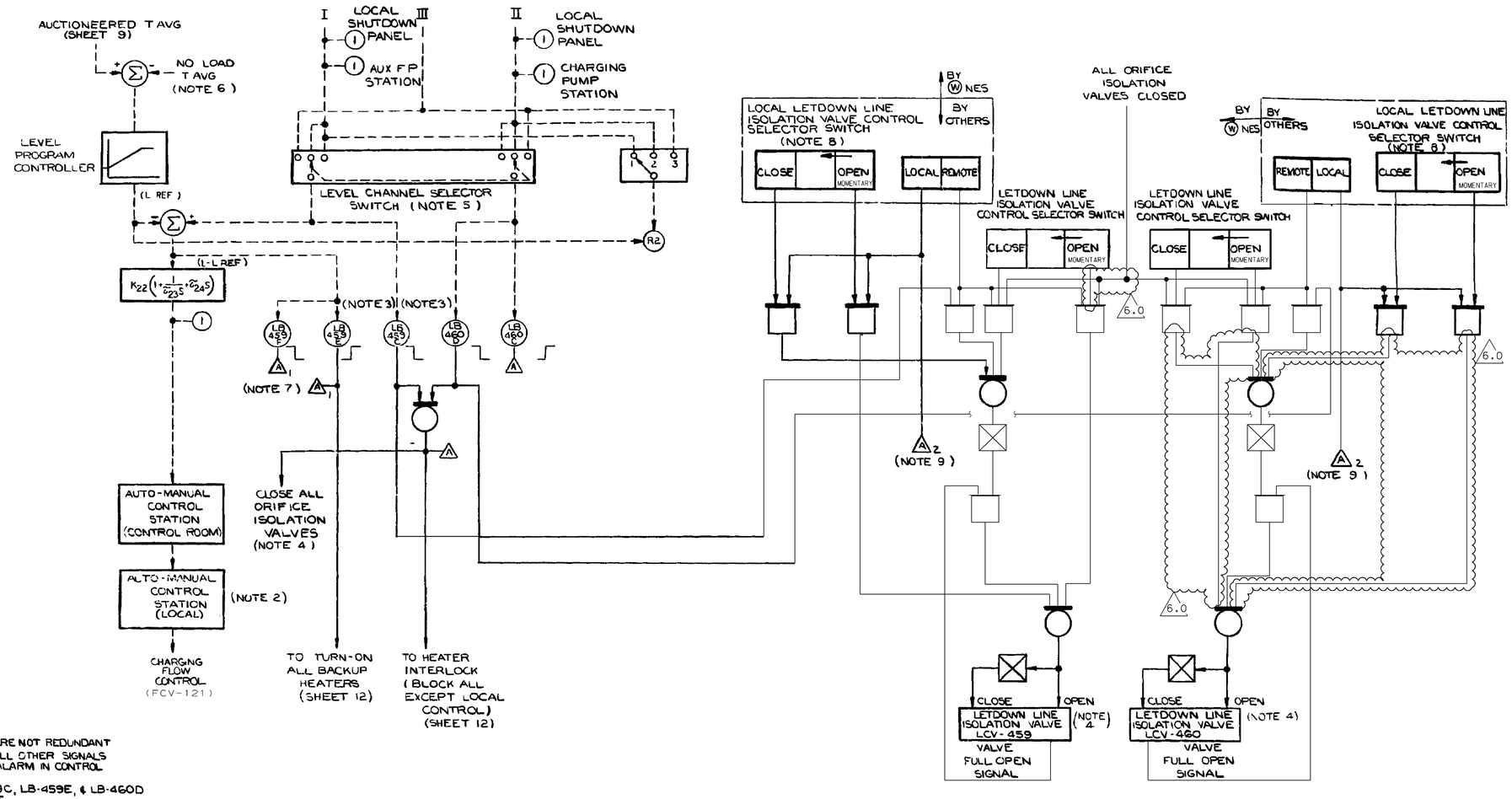
3 ☐ REVISE AND RESUBMIT WORK MAY NOT PROCEED
4 ☐ INFORMATION ONLY DISTRIBUTION REQ'D ☐ YES
M ☐ RESUBMIT – NOT ACCEPTABLE FOR MICROFILM, WORK MAY PROCEED

PF-5987
195101 9/84

7243D07 SHEET -20

PRESSURIZER LEVEL CHANNELS



NOTES

1. ALL CIRCUITS ON THIS SHEET ARE NOT REDUNDANT
2. LOCAL CONTROL OVERRIDES ALL OTHER SIGNALS LOCAL OVERRIDE ACTUATES ALARM IN CONTROL ROOM
3. LEVEL BISTABLES NO LB-459C, LB-459E, & LB-460D ARE ENERGIZE TO ACTUATE.
4. OPEN / SHUT INDICATION IN CONTROL ROOM.
5. CENTER POSITION NORMALLY SELECTED.
6. ADJUSTABLE SETPOINT WITHIN CONTROLLER.
7. ALARM 1 MUST HAVE RESET CAPABILITY.
8. MANUAL CONTROL OF THESE VALVES CONSISTS OF TWO SELECTOR SWITCHES ON THE LOCAL SHUTDOWN PANEL, ONE SWITCH PER VALVE. SUPPLIED BY OTHERS.
9. ANNUNCIATION IN THE MAIN CONTROL ROOM IS REQUIRED VISIBLE TO THE OPERATOR AT THE MCB

1	SO SCALE / GBE-300 NCR-28/82-2527	2/8/82	
	CHANGE		
	GAE/GBE-300 ECN-32037	2/8/82	
	CHANGES CIRCLED	3/1/82	

DRTN B STEMMLER 8/82

CHKR

DES ENG 8/1/82

MFG ENG

WT.S ENG

APD 3/17/82

APD

APD

DRTG SUPV 8/27/82

Westinghouse Electric Corporation
NUCLEAR ENERGY SYSTEMS PITTSBURGH PA U S A

TYPE GEORGIA POWER CO
ALVIN W VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAMS
PRESSURIZER LEVEL CONTROL

SCALE

DIMENSIONS IN INCHES

DO NOT SCALE

7243D07
SHEET 20

SUB # 2348

SIZE D 34X22

CAD MICROSTATION BgT/01

1X6AA02-00496.DGN

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A

TAB/SECT. N/A
PAGE N/A
FIGURE N/A

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

VERSION 6.0	DATE 12-21-05
REVISED BY SNC PER ABN-81187, VER.1.0	

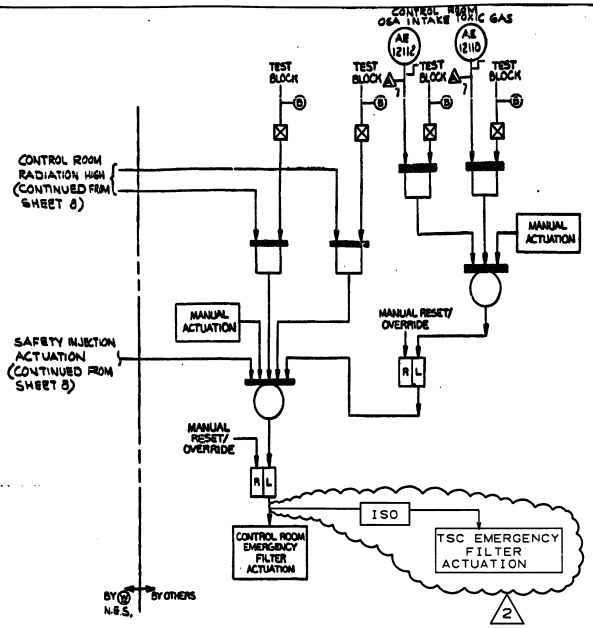
SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

TITLE: FUNCTIONAL DIAGRAM

BY	CHK'D	APPR.1	APPR.2
BgT	PWG	ASK	

VENDOR: WESTINGHOUSE P.O.#:

DRAWING NO. 1X6AA02-00496



NO.	REVISIONS -	DATE	DR	CHK
2	INCORPORATED ABN 9968	9-24-96	GS	

SCS REVISIONS

DOCUMENT STATUS CODE 1

1X6AA02-519-2

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CHK	
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ENGR	
APP	[illegible] 1/24/97
APP	
APP	
DRG APP	[illegible] 1-24-97
SCALE	

TOLERANCE & MACHINE NOTES

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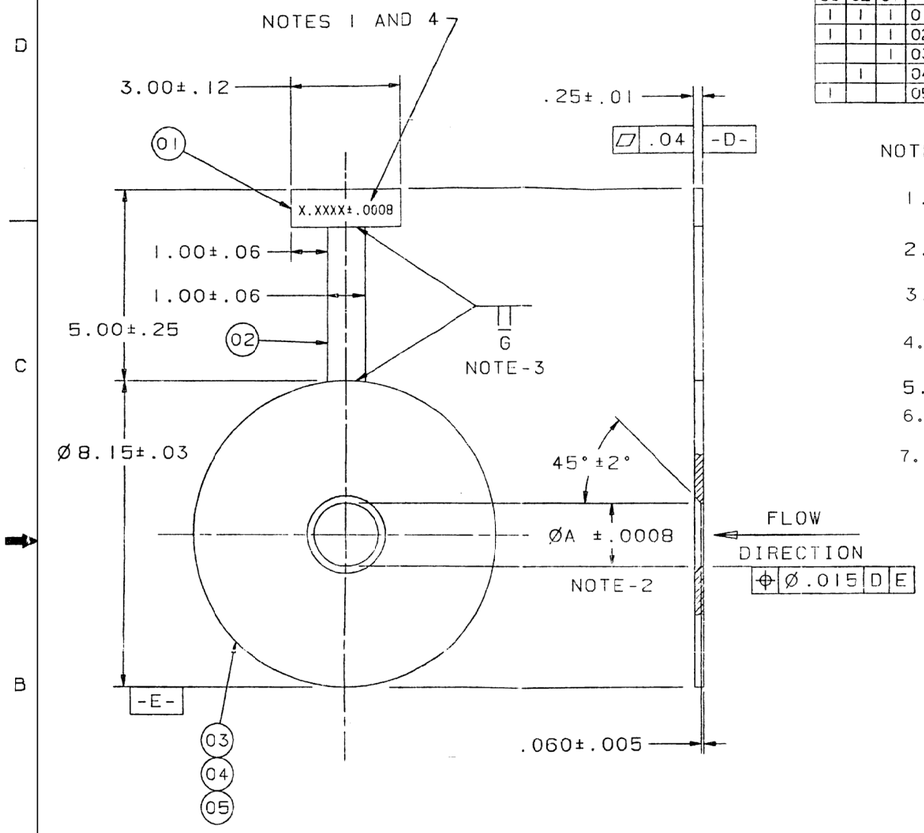
Westinghouse Electric Corporation
WATER REACTOR DIVISIONS - MONROEVILLE, PA. U.S.A.

GEORGIA POWER COMPANY
ALVIN W. VOGTLE UNITS 1 & 2
FUNCTIONAL DIAGRAM (CONT. FROM SHT. 8)
SAFEGUARD ACTUATION SYSTEM

SIZE D | FSCM NO. | DWG NO. 7243D07 | REV. 1

SCALE | WEIGHT | SHT 21

CAD NAME
16A02519



IDENT CLASS							
GROUP NOTE							
REQD			ITEM	NOTE	IDENT	PART NAME	(SIZE) REFERENCE INFORMATION	DEF	MATL SIZE CODE PART NUMBER OR REF DWG
03	02	01					
1	1	1	01			NAME PLATE	ASME SA 240 304-316 SST.		
1	1	1	02			HANDLE	ASME SA 240 304-316 SST.		
		1	03			DISK	ASME SA 240 304-316 SST.		
	1		04			DISK	ASME SA 240 304-316 SST.		
1			05			DISK	ASME SA 240 304-316 SST.		

BILL OF MATERIAL

7.0

NOTES:

1. IDENTIFICATION CHARACTERS MAY BE STAMPED OR ETCHED APPROX. .25 HIGH.
2. INLET EDGE AT DIAMETER "A" MUST BE SQUARE, SHARP, AND FREE FROM DURRS, NICKS OR ROUNDING.
3. OTHER MEANS OF ATTACHMENT MAY BE USED PROVIDED PRIOR W APPROVAL IS GIVEN.
4. NAMEPLATE SHALL IDENTIFY ORIFICE SIZE, ØA, INCLUDING TOLERANCE, GRADE, SERIAL NUMBER, AND HEAT NUMBER. 7.0
5. WESTINGHOUSE SPIN NO. - CSORCP (GAE).
6. FOR GROUP B1,B2 AND B3 USE ITEMS 1,2 AND 5. THESE GROUPS MANUFACTURED ON SITE.
7. ORIFICE PLATE BORE DIAMETERS MUST BE CONSIDERED IN ORDER TO MEET PUMP PERFORMANCE REQUIREMENTS AND SYSTEM RESISTANCE CRITERIA PER TRM 13.5.1.

GROUP	A
01	1.5834
02	1.4995
03	1.7048
04	1.1456
05	1.1206
06	1.1738
B1	1.4136
B2	1.4435
B3	1.4770
B4	3.4380 W/O TAPER

INSTRUMENT TAG NUMBER	GROUP
FO 10118	B4
FO 10122	B1

SO. GAE-205
D. 3D5090-01
IT.
REVISION
—

TOLERANCE & MACHINE NOTES
(UNLESS OTHERWISE SPECIFIED)

DRAWING PRACTICES, GEOMETRIC SYMBOLS, DIMENSIONING, CLEARANCING & INTERPRETATION BASED ON ANSI Y14 SERIES STANDARD & PS 80400HA.

DIMENSIONS IN INCHES BASED ON 68°F(20°C)
RADIUS OR CHAMFER ALL EDGES-----.005-.030
FILLET RADII-----.031-.01
ANGULARITY: CHAMFERS-----±2°

MAXIMUM SURFACE ROUGHNESS-250µIN. (√)AA

SPIN: DWG. REF. | THIRD ANGLE PROJECTION
NEXT ASSY.

WESTINGHOUSE PROPRIETARY DATA

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DFTM	MILLSLAGLE	10.29.91
CHKR		
DES ENG		
MFG ENG		
QA ENG		
APP		
APP		
APP		
APP		
DP10 W-1		10-29-91

INTERACTIVE GRAPHICS
A-MILL-8251C26-01-001-102991

Westinghouse Electric Corporation
WATER REACTOR DIVISIONS - MONROEVILLE, PA. U.S.A.

TITLE
GAE CHARGING PUMP
FLAT PLATE ORIFICES
ORIFICE ASSEMBLY

DIMENSIONS IN INCHES
SIZE C | DWG NO. 8251C26 | REV. 5
SCALE | WEIGHT LBS. | SHT 1 OF 1 SHTS

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

VERSION 7.0 | DATE 01/13/14
REVISED BY SNC PER ABN-V03188, VER 1.0
SEE MICROFILM FOR PREVIOUS REV. SIGNATURES.

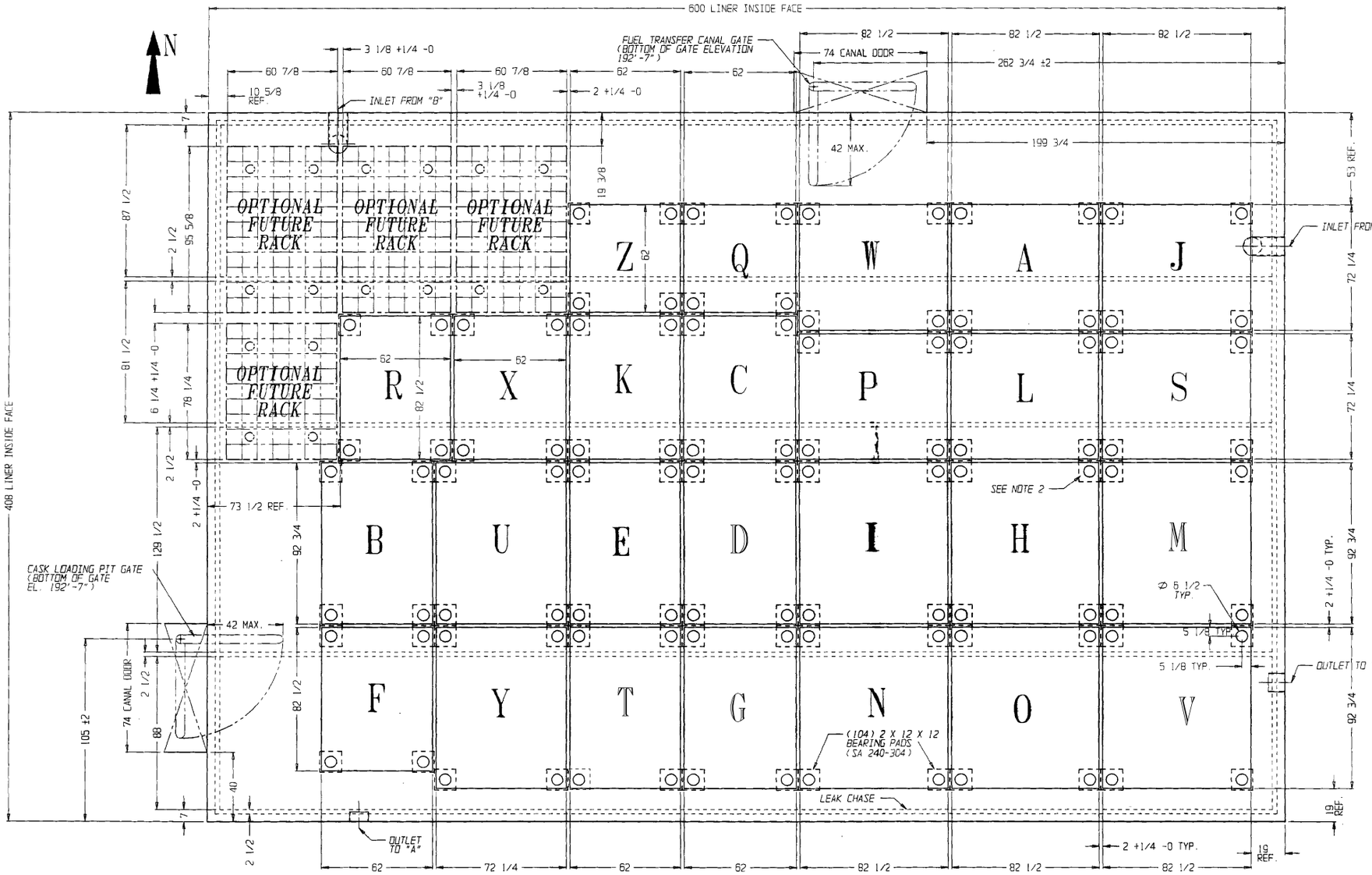
BY	CHK'D	APPR.1	APPR.2
DCP	ELC	JMR	

CAD MICROSTATION DCP/02
1X6AH02-30000.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
GAE CHARGING PUMP
FLAT PLATE ORIFICES
ORIFICE ASSEMBLY

VENDOR: WESTINGHOUSE | P.O.#: PAV-2
DRAWING NO. 1X6AH02-30000
SIZE D



CONFIGURATION CHART

MODULE LETTER	MODULE SERIAL NUMBER	WEIGHT LBS. DRY	SIZE
Z	AD-32129-D-01	12,700	6 X 6
Q	AD-32129-D-02	12,700	6 X 6
W	AD-32132-D-01	17,550	7 X 8
A	AD-32132-D-02	17,550	7 X 8
J	AD-32132-D-03	17,550	7 X 8
R	AD-32130-D-01	16,900	6 X 8
X	AD-32130-D-02	16,900	6 X 8
K	AD-32130-D-03	16,900	6 X 8
C	AD-32130-D-04	16,900	6 X 8
P	AD-32132-D-04	17,550	7 X 8
L	AD-32132-D-05	19,600	7 X 8
S	AD-32132-D-06	19,600	7 X 8
B	AD-32131-D-01	17,100	6 X 9
U	AD-32133-D-01	19,750	7 X 9
E	AD-32131-D-02	17,100	6 X 9
T	AD-32131-D-03	19,050	6 X 9
I	AD-32134-D-01	22,400	8 X 9
H	AD-32134-D-02	22,400	8 X 9
V	AD-32134-D-03	22,400	8 X 9
F	AD-32130-D-05	16,900	6 X 8
Y	AD-32133-D-02	22,050	7 X 9
G	AD-32131-D-04	19,050	6 X 9
D	AD-32131-D-05	19,050	6 X 9
N	AD-32134-D-04	22,400	8 X 9
O	AD-32134-D-05	22,400	8 X 9
M	AD-32134-D-06	25,075	8 X 9

NOTES:
1) ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.
2) INSTALLER MAY USE SINGLE LARGE BEARING PAD AT A TWO, THREE OR FOUR RACK PEDESTAL INTERSECTION AS OPPOSED TO SEPARATE 12" SQUARE BEARING PADS AS LONG AS THE DISTANCE FROM THE EDGE OF A RACK PEDESTAL TO THE EDGE OF THE BEARING PAD IS A MINIMUM OF 2 1/2".
3) RACK MODULE LOCATION IS SIZE DEPENDENT. RACKS OF THE SAME SIZE CAN BE SUBSTITUTED EVEN THOUGH THEY HAVE DIFFERENT WEIGHTS AS LONG AS THE WEIGHT DIFFERENCE DOES NOT EXCEED 3% OF THE WEIGHT OF THE LIGHTEST RACK WITH ALL LOCATIONS FILLED WITH FUEL. FOR EXAMPLE, ALTHOUGH RACK "M" IS SHOWN ON THE SOUTHEAST CORNER, ANY OF THE 8X9 RACKS (T,H,V,N,O, OR M) MAY BE PLACED IN THIS CORNER POSITION.

REVISIONS

REV	DESCRIPTION	PREP	CHECKED	ENGINEERING	D.A.
0	ISSUED FOR APPROVAL	K.SHAW 5-10-97	K.TAYLOR 5-10-97	S.PALLARD 5-10-97	V.DUFFY 5-10-97
1	REV'D AS SHOWN	K.SHAW 5-10-97	K.TAYLOR 5-10-97	S.PALLARD 5-10-97	K.MOORE 5-10-97
2	ADD TOLERANCE & CHANGE RACK I.D.	K.SHAW 7-8-97	K.TAYLOR 7/8/97 S.PALLARD 7/8/97		10-8-97

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HOLTEC INTERNATIONAL

EQUIPMENT DESIGN
ANALYSIS
CONSULTING

DESCRIPTION:
UNIT #1 SPENT FUEL POOL- SPENT FUEL RACK LAYOUT

CLIENT: SOUTHERN NUCLEAR
ALVIN W. VOGTLE NUCLEAR PLANT

COMPANION DRAWINGS: NONE

SHEET NO. SHEET 1 OF 1

PROJECT No.:	SCALE:	DRAWING No.:	REV.:
70241	1/32 = 1	1945	6

P.O. No.: 70290170000

NO.	REVISIONS	DATE	DR	CHK	APPV
6	INCORPORATED PER DCP 97-V1N0001 (CHANGE STATUS FROM P TO 1)	2-16-99	ELC	EOG	GLB

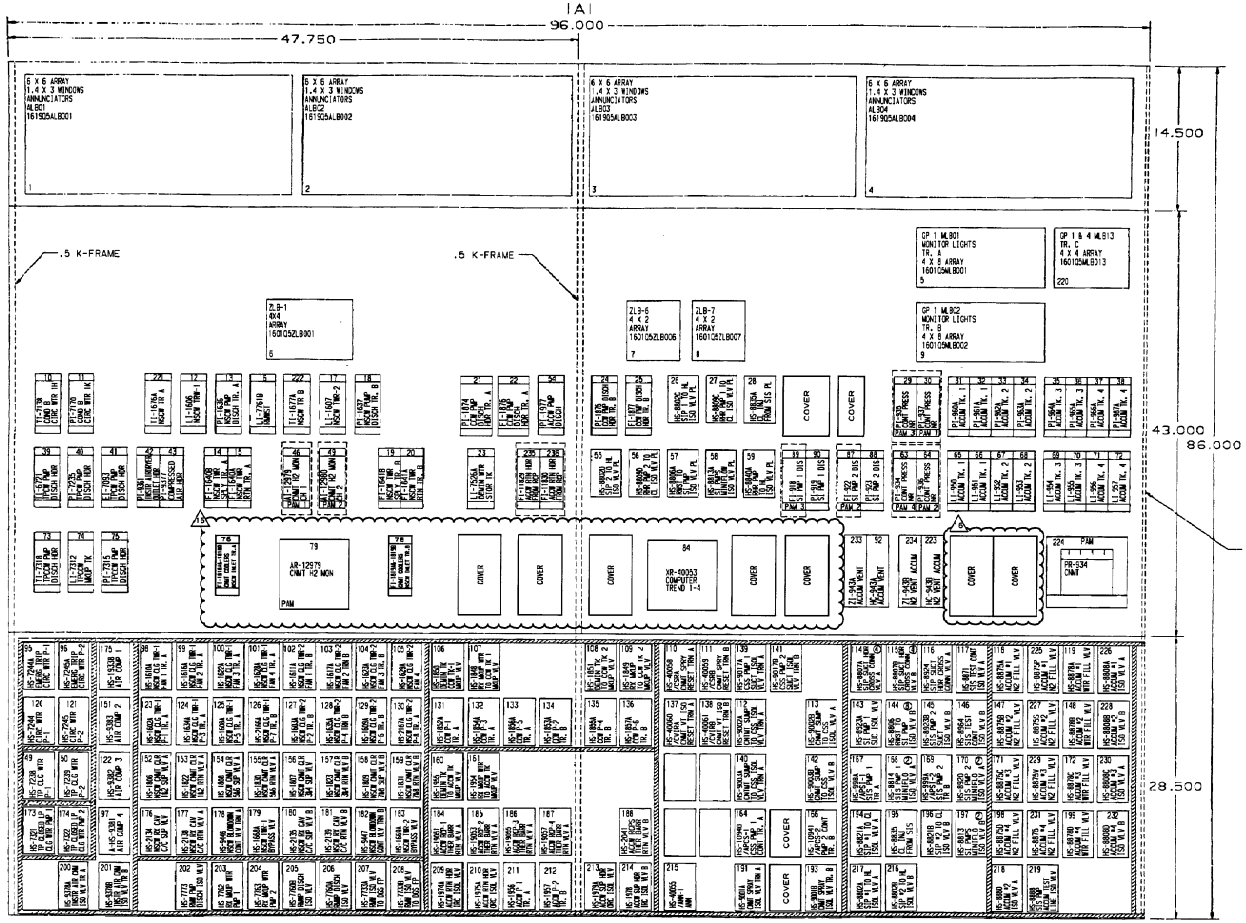
SCS REVISIONS

1X6AN10B-00066-3

DOCUMENT STATUS CODE 1

SIZE D

CAD NAME 16N10B66



NOTE:

FOR LOCATION OF CUTOUTS AND PANEL STIFFENERS SEE W DWG 1548F98 SH.1

PANEL SECTION LOCATION PLAN

INTERACTIVE GRAPHICS
D:ELC PALM:1451E70:XXX:15:001:030989

INDEX

SH	DESCRIPTION	REVISION
1	FRONT VIEW SECTION 1A1	1 2 3 4 5 6 7 8 9 10 11 12
2	FRONT VIEW SECTION 1A2	1 2 3 4 5 6 7 8 8 8 9 10
3	FRONT VIEW SECTION 1C1	1 2 3 4 5 6 7 8 9 10 10 11
4	FRONT VIEW SECTION 1B1	1 2 3 4 5 6 7 8 8 9 10 11
5	FRONT VIEW SECTION 1B2	1 2 3 4 5 6 7 7 7 7 8 9
6	FRONT VIEW SECTION 1D1	X X 1 1 2 3 3 4 5 5 5 5

NO.	REVISIONS	DATE	DR	CHK
16	INCORPORATED PER DCP 94-V1NOC13	6/15/96	SRM	

SCS REVISIONS

1X6AV02-46-18

DOCUMENT STATUS CODE 1

ABN-06119 14

ACUSIM SOFTWARE INCORP. ABN-06119

WESTINGHOUSE PROPRIETARY DATA

DRN H. BELLAS 3-5-79
CHK 3-6-79
APP 3-6-79
ENG 3-6-79
WELDING
MTLS ENG
QA
SAFETY

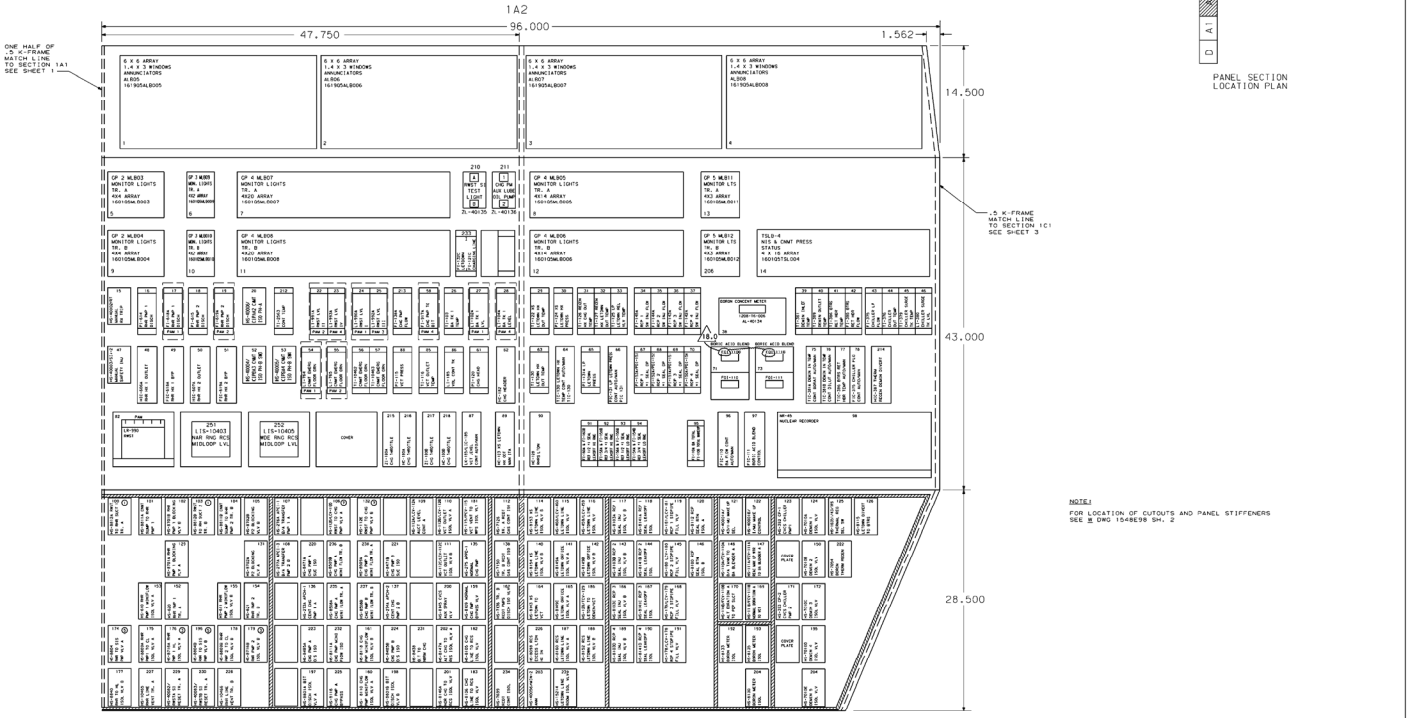
WESTINGHOUSE ELECTRIC CORPORATION NUCLEAR ENERGY SYSTEMS, P.O.BOX 355, PITTSBURGH, PA. 15230

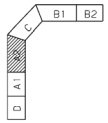
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
MAIN CONTROL BOARD FRONT VIEW
SECTION 1A1

SCALE
DIMENSIONS IN INCHES
DO NOT SCALE

1451E70
SHEET 1

CAD NO. 16AV02





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WESTINGHOUSE ELECTRIC CORPORATION	
NUCLEAR ENERGY SYSTEMS, PGH, PA., U.S.A.	
TITLE: DESCRIPTIVE "POWER COMPANY"	
ALVIN VOGTLE UNIT 1	
MAIN CONTROL BOARD FRONT VIEW	
SECTION 1A2	
SCALE	1451E70
DRAWING IS (NOT) SCALE	SHEET 2
DO NOT SCALE	SIZE E

THIS DWG. IS REFERENCED IN VENDOR MANUAL:	N/A
TAB/SECT.	N/A
PAGE	N/A
FIGURE	N/A
VERSION 18.0	DATE 02/16/99
REVISED BY SNC PER ABV 1062047701E003, VER. 1.0	
SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.	

BY	CHK'D	APPR. 1	APPR. 2
TSL	RBH	JMR	

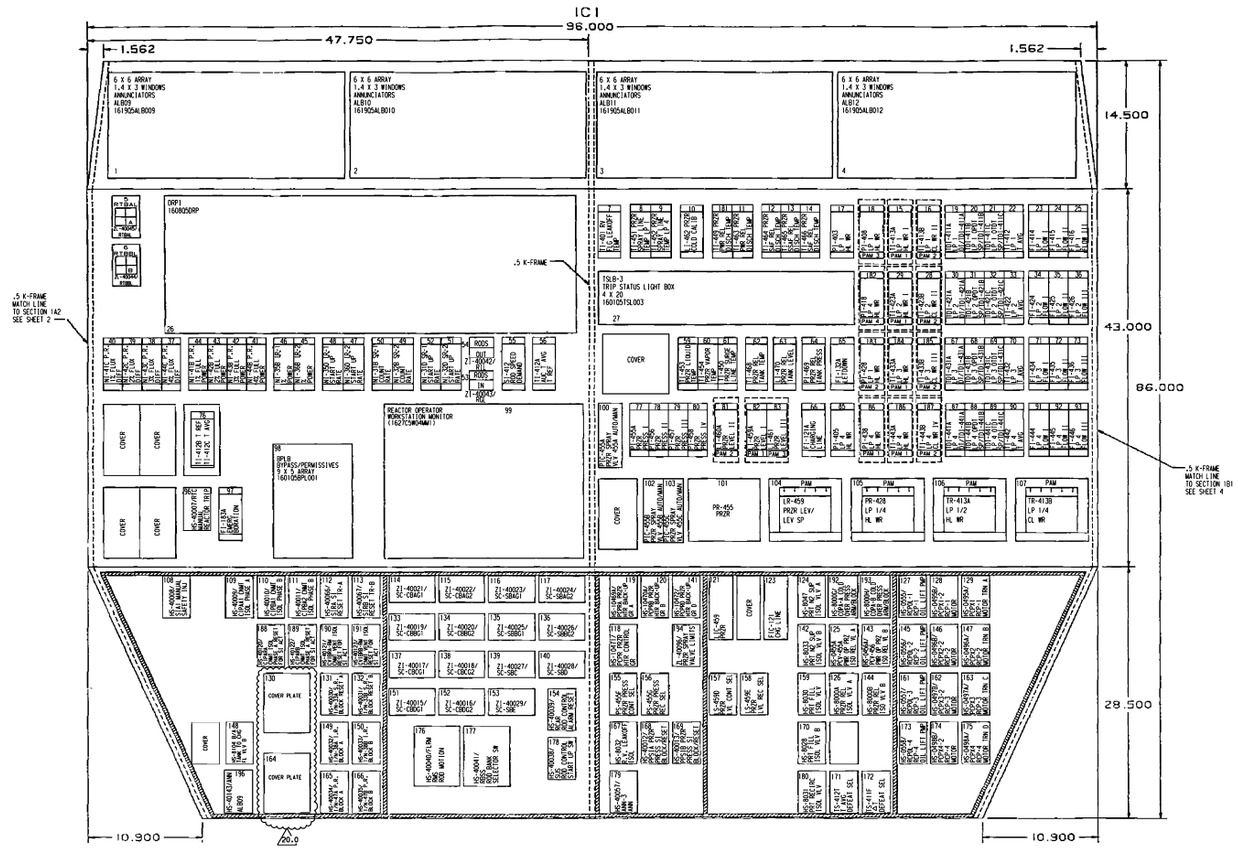
C4D MICROSTATION TSL/01 1X6AV02-00376.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
MAIN CONTROL BOARD
FRONT VIEW
SECTION 1A2

DRAWN BY: WESTINGHOUSE ELEC | P. O. #:

DRAWING NO. 1X6AV02-00047



NOTE:

FOR LOCATION OF CUTOUTS AND PANEL STIFFENERS SEE W DWG 1548E98 SH.3

WESTINGHOUSE PROPRIETARY DATA

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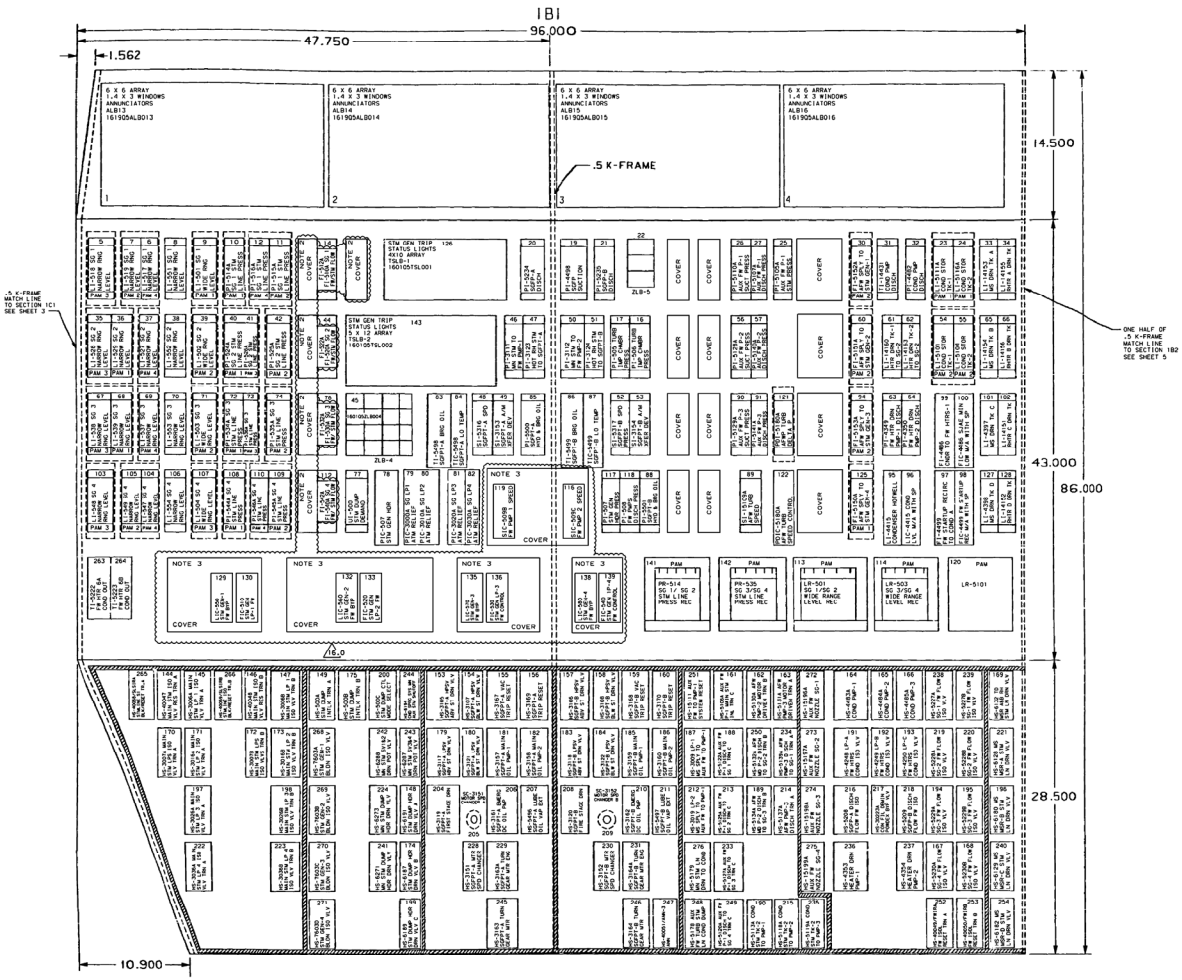
DRN M. DELLAS	6-1-79	EQUIPMENT TAG NO.: 1-1601-05-MCB	
CHK	6-1-79	WESTINGHOUSE ELECTRIC CORPORATION NUCLEAR ENERGY SYSTEMS, PGH.PA.,U.S.A.	
DES ENG	6-1-79	TITLE GEORGIA POWER COMPANY ALVIN W. VOGTLE UNIT 1 MAIN CONTROL BOARD FRONT VIEW SECTION 1C1	
SYS ENG		SCALE	1451E70
NUC ENG	6-1-79	DIMENSIONS IN INCHES	SHEET 3
APP	6-1-79	DO NOT SCALE	SUB 1/ 18
APP			
APP	6-1-79		

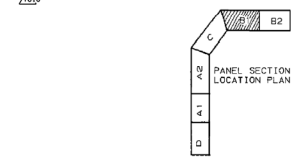
THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A			CAD MICROSTATION TSL01	1X6AV02-00048.DGN
TAB/SECT. N/A			Southern Nuclear Operating Company, Inc. FOR VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1	
PAGE N/A				
FIGURE N/A				
VERSION 20.0	DATE 08/21/09		TITLE: MAIN CONTROL BOARD 1-1601-05-MCB FRONT VIEW - SECTION 1C1	
REVISED BY SNC PER ABN 1052592101J034, VER.1.0 SEE MICROFILM FOR PREVIOS VER. SIGNATURES.				
BY	CHK'D	APPR.1	APPR.2	VENDOR: WESTINGHOUSE P.O.#: PAV-2
TSL	ELC	JMR	X	DRAWING NO. 1X6AV02-00048



NOTE:

1. FOR LOCATION OF CUTOUTS AND PANEL STIFFENERS SEE W DWG 1046E36 Sh. A.
2. SEE AX5ABD1-00001 FOR COVER PLATE DETAILS.
3. SEE AX5ABD1-20008 FOR COVER PLATE DETAILS.

16.0



SECTION 1B1

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D.ELC.SMOO.1451E70.xxx.11.004.011987

SIZE	FSCM. NO.	DWG. NO.	REV.
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SCALE		DFTM. -	SHT. 4 OF SHTS.

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10	4	GAE-3370	ECN-334370	
9	3	GAE-3359	ECN-333359	
8	2	GAE-3348	ECN-332348	
7	1	GAE-3337	ECN-331337	
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		GAE-3194	ECN-318328	
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REVISED LOCATIONS PER YFRB-3388	B. SMOODY	1-13-97
REVISED DEMARCATION PER YFRB-3382	B. SMOODY	12-27-96
REVISED LOCATIONS PER YFRB-3359	B. SMOODY	7-24-96
REVISED LOCATIONS PER YFRB-3317	B. SMOODY	5-20-96
REVISED LOCATIONS PER YFRB-3304	B. SMOODY	1-13-97

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A			
TAB/SECT.: N/A			
PAGE: N/A			
FIGURE: N/A			
VERSION 16.0	DATE 7/18/13		
REVISED BY SNC PER ABN 1039002501J059, VER. 4.0 SEE WORK RAK FOR PREVIOUS REV. SIGNATURES.			
BY	CHK'D	APPR.1	APPR.2
JLO	GDM	JMR	X

CAD MICROSTATION JLO/01 1X6AV02-00049.DGN

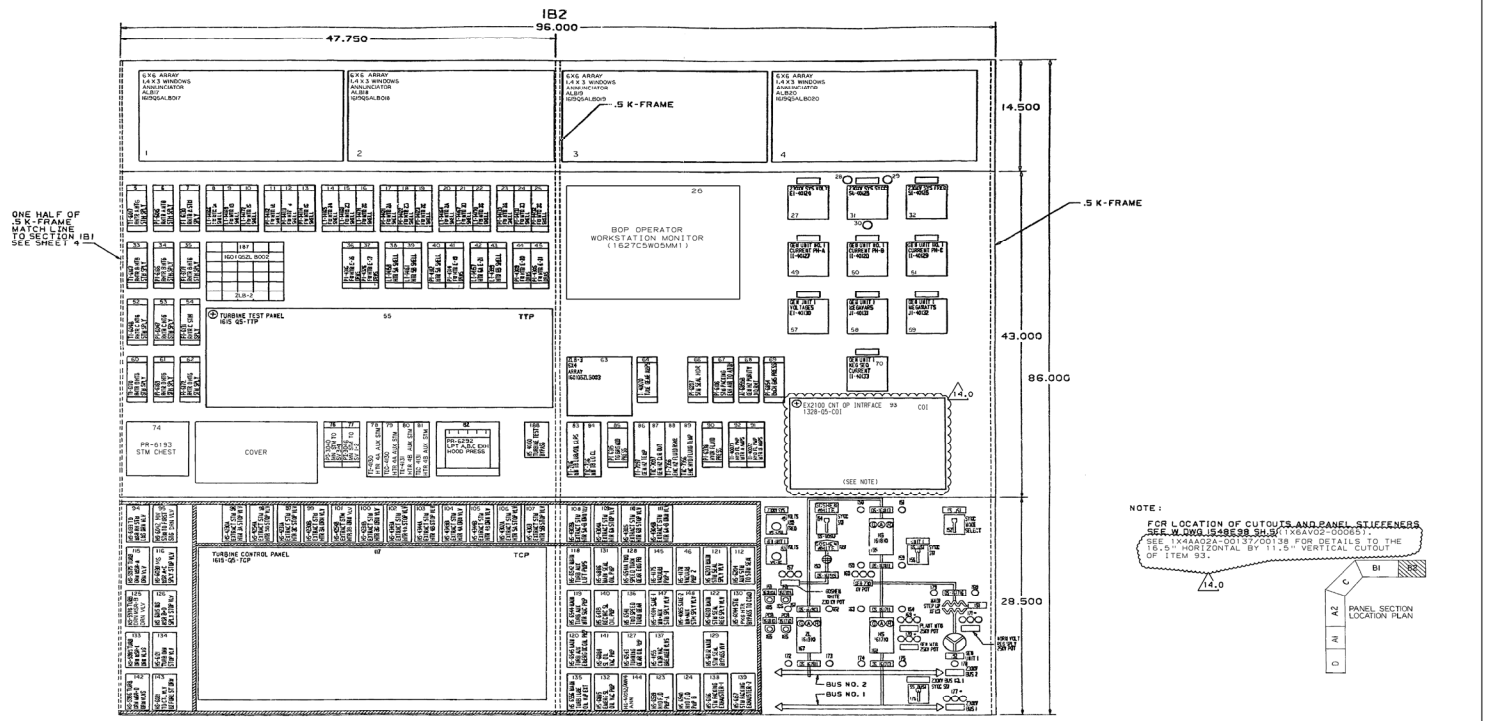
Southern Nuclear Operating Company, Inc.

VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

TITLE: MAIN CONTROL BOARD - FRONT VIEW SECTION 1B1

VENDOR: WESTINGHOUSE P.O.#:

DRAWING NO. 1X6AV02-00049



NOTE:
FOR LOCATION OF CUTOUTS AND PANEL STIFFENERS SEE W DWG 1548E98 SH 3 (1X6AV02-00065).
SEE 1X4AA02A-00137/00138 FOR DETAILS TO THE 16.5" HORIZONTAL BY 11.6" VERTICAL CUTOUT OF ITEM 93.

NO.	DESCRIPTION	DATE
3	ADD CHANGE TEC 49337 REVISED FOR FSAR Update	8/31/09

GEORGIA POWER COMPANY
PLANT: VOGTLE ELECTRIC GENERATING PLANT
UNIT: 1 ITEM=19
STATUS: APPROVED FIELD MODIFICATION
CERTIFICATION LTR. NO. GP-14412
AUTHORITY: J. L. TAIN
ENGR. LTR. NO. ERSA-54758

INTERACTIVE GRAPHICS
D.ELC.JOHN..145E70.XXX.09.000.032389

SOUTHERN COMPANY SERVICES
BIRMINGHAM, ALABAMA

NO.	SOUTHERN COMPANY SERVICES REVISIONS	DATE	DR.	CHK	DM
	REV. TO CLOSE DCN 1 (IDCH-S #5, CCP NO. F-0005-W), NO DWG CHANGE REQD.	08/09	CJB	JM	JJ
	SEE MICROFILM FOR PREVIOUS REVS.				

DRAWING NUMBER 1X6AV02-50 REV. 11

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WESTINGHOUSE ELECTRIC CORPORATION
NUCLEAR ENERGY SYSTEMS, P.O. BOX 355, PITTSBURGH, PA. 15230-0355 U.S.A.
GEORGIA POWER COMPANY
ALVIN VOGTLE UNIT 1
MAIN CONTROL BOARD FRONT VIEW SECTION 1B2
1451E70
SHEET 9

CAD NAME 16AV0250

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

VERSION 14.0 DATE 8/31/09

REVISED BY SNC PER ABN 1049001501J018, V. 1.0
SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.

BY	CHK'D	APPR.1	APPR.2
RBH	MWD	JMR	

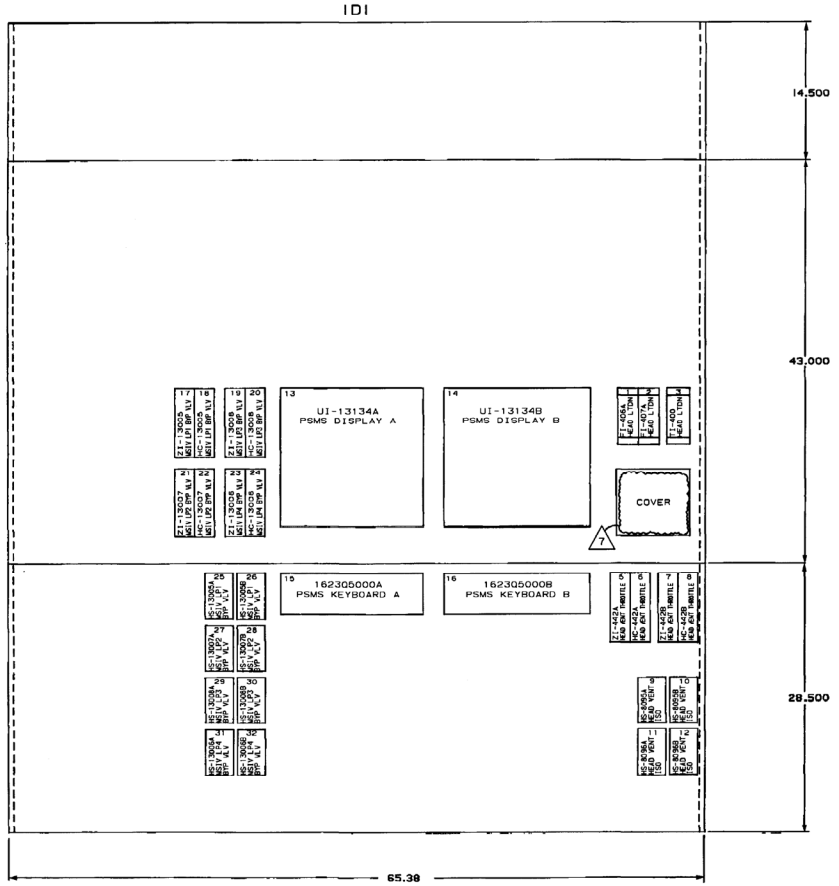
CAD MICROSTATION

Southern Nuclear Operating Company, Inc.
for
VOGTLE ELECTRIC GENERATING PLANT UNIT NO.1

TITLE: MAIN CONTROL BOARD FRONT VIEW SECTION 1B2

VENDOR: WESTINGHOUSE P.O.#:

DRAWING NO. 1X6AV02-00050



NOTE:

FOR LOCATION OF CUTOUTS AND PANEL STIFFENERS SEE W DWG 1548E98 SH.8

NO.	REVISIONS	DATE	DR	CHK
7	INCORPORATED PER DCP 94-V1N0013	8/15/96	JGM	
	SCS REVISIONS			

DOCUMENT STATUS CODE 1

1X6AV02-60-8

PANEL SECTION LOCATION PLAN

INTERACTIVE GRAPHICS
ELC.LEVH.145E70.XXX-05.006.062894

SOUTHERN COMPANY SERVICES
BIRMINGHAM, ALABAMA

DRAWING NUMBER 1X6AV02-60 REV. 7

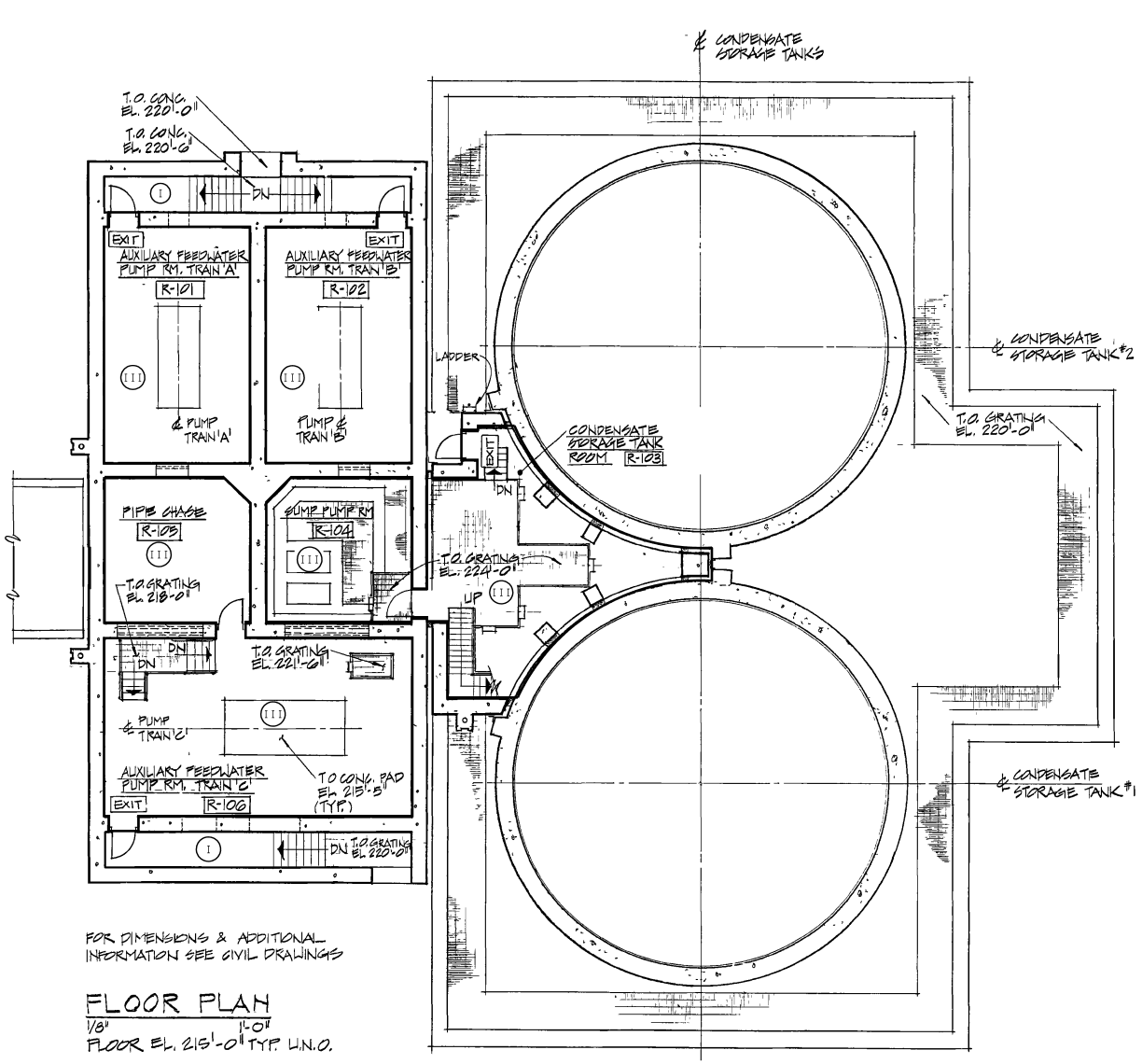
NO.	SOUTHERN COMPANY SERVICES REVISIONS	DATE	DR.	CHK.	QA
4	REV. TO CLOSE DCN'S (EDN-8453, DCN 2 (EDN-8493. NO ENGR. CHANGE REQD. SEE MICROFILM FOR PREVIOUS REVS.	8-9-91	C.L.A.	S.W.	JJ

J. GERBER 2-8-91

WESTINGHOUSE ELECTRIC CORPORATION
NUCLEAR ENERGY SYSTEMS, PGH. PA. U.S.A.
GEORGIA POWER COMPANY
ALVIN W. VOGTLE
MAIN CONTROL BOARD FRONT VIEW
SECTION 1D

1451E70
SHEET 6

CAD NAME
16AV0260

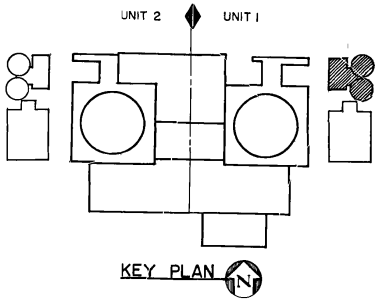


FOR DIMENSIONS & ADDITIONAL INFORMATION SEE CIVIL DRAWINGS

FLOOR PLAN

1/8" = 1'-0"

FLOOR EL. 215'-0" TYP. U.N.O.



RAD ZONE MAP POST TMI 24 HOURS	
ZONE	REM/H
I	< 0.0025
II	0.0025-0.1
III	0.1-1.0
IV	1.0-10.0
V	10.0-100
VI	100-1000
VII	1000-10000
VIII	>10000
———	ZONE BOUNDARY

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

RAD ZONE MAP POST TMI 24 HOURS
AUXILIARY FEEDWATER
PUMP HOUSE FLOOR PLAN

SCALE: NONE | DRAWING NO. 1X6DD101 | REV. 0

JOB NO. 10604

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
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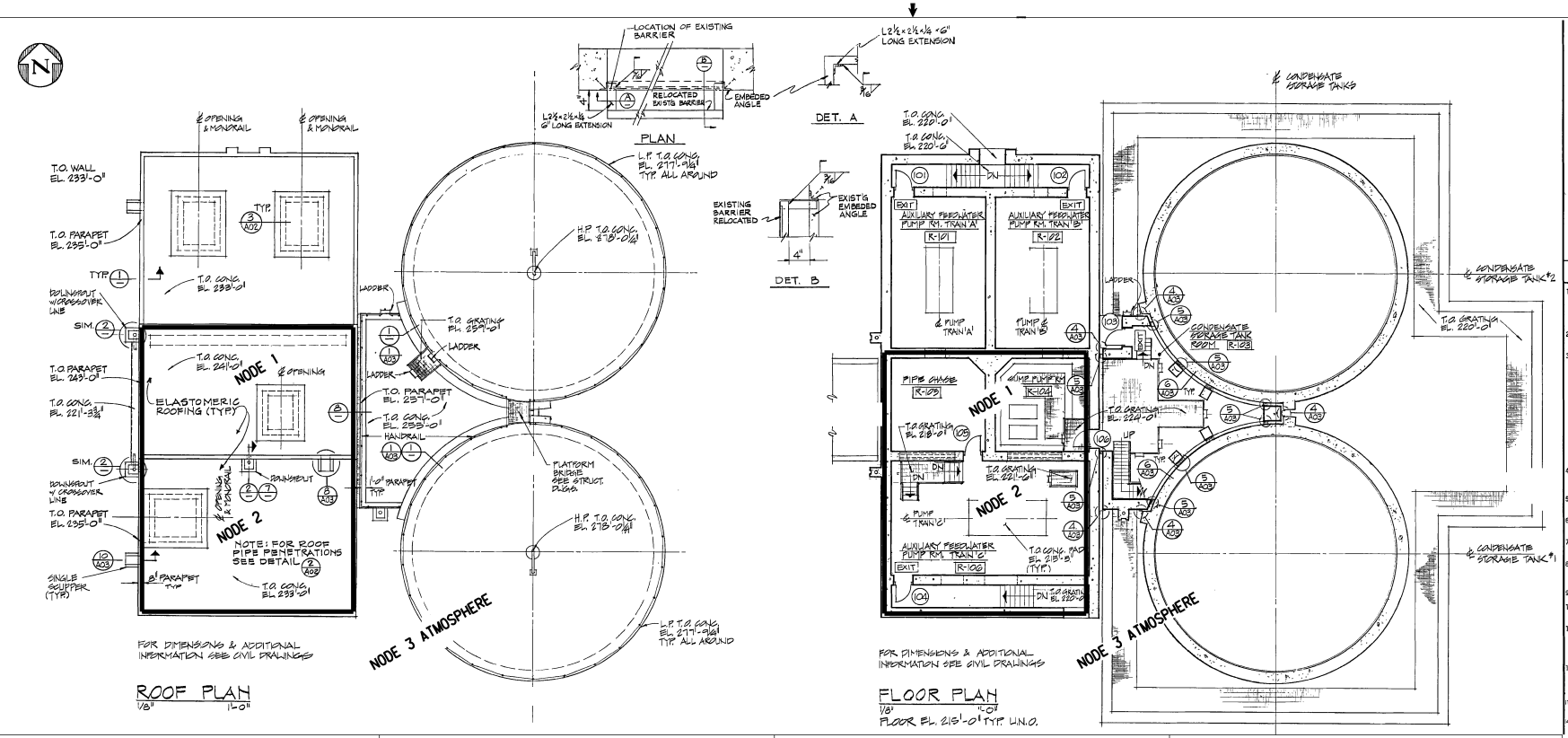
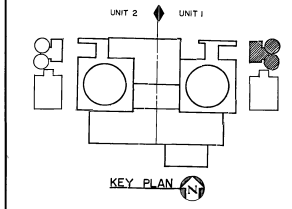
1	ISSUED PER REA 96-VAA054	7-28-98	LVL	RBH	CBH	WFP

SIZE E 34X44

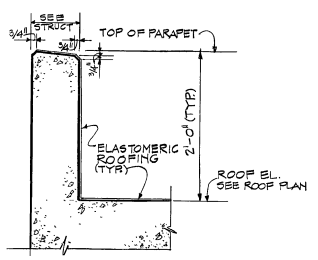
CAD NAME 1X6DD101



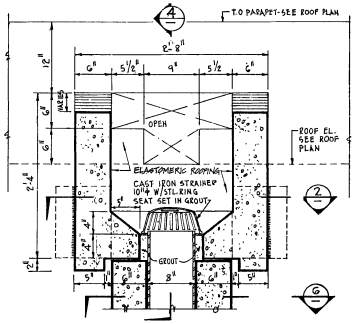


GENERAL NOTES

1. DIMENSIONS:
   - REFER TO STRUCTURAL DRAWINGS FOR DIMENSIONS NOT SHOWN IN ARCHITECTURAL DRAWINGS.
   - ALL DIMENSIONS SHALL HAVE PREFERENCE OVER SCALE.
   - ALL DIMENSIONS SHALL BE VERIFIED IN THE FIELD BEFORE PROCEEDING WITH THE WORK.
2. PROVIDE ALL NECESSARY BLOCKING, BACKING FRAMING FOR HOSE CABINETS, WALL MOUNTED AND RECESSED CABINETS, FIXTURES, ELECTRICAL CABINETS AND ALL OTHER ITEMS REQUIRING SAME.
3. ALL CONSTRUCTION SHALL COMPLY WITH THE "NATIONAL FIRE CODES", 1977 EDITION.
   - BUILDING STRUCTURE FIRE PROTECTION.
   A. ALL BEARING WALLS OR BEARING PORTIONS OF WALLS SHALL HAVE 3 HOUR FIRE RESISTIVE CONSTRUCTION.
   B. ALL PRINCIPAL SUPPORTING MEMBERS INCLUDING COLUMNS, TRUSSES, GIRDERS AND BEAMS SHALL HAVE FIRE-RESISTANCE RATING AS FOLLOWS:
      - 2 HOUR FOR ONE FLOOR OR ROOF ONLY
      - 3 HOUR FOR MORE THAN ONE FLOOR OR ROOF.
   C. ALL SECONDARY FLOOR SUPPORTING MEMBERS - BEAMS, SLABS, JOIST, NOT AFFECTING THE STABILITY OF THE BUILDING SHALL HAVE 2 HOUR FIRE-RESISTANCE RATING.
   D. ALL SECONDARY ROOF SUPPORTING MEMBERS - BEAMS, PURLINS, SLABS NOT AFFECTING THE STABILITY OF THE BUILDING SHALL HAVE 1-1/2 HOUR FIRE-RESISTANCE RATING.
   E. FOR TYPE AND MATERIAL REFER TO DETAIL DRAWINGS AND SPECIFICATION.
4. - FIRE RESISTIVE CONSTRUCTION:
   A. INSTALL FIRE DAMPERS ON DUCTS PASSING THROUGH FIRE RATED WALLS.
   B. ALL FIRE RATED WALLS SHALL EXTEND TO THE UNDERSIDE OF THE STRUCTURE ABOVE, UNLESS NOTED OTHERWISE.
5. ALL EGRESS EXITS SHALL BE OPERABLE FROM THE INSIDE WITHOUT THE USE OF KEY, SPECIAL EFFORT OR KNOWLEDGE. EXIT SHALL BE PROVIDED WITH ILLUMINATED EXIT SIGNS - SEE FLOOR PLANS FOR LOCATION.
6. ALL CABINETS, ELECTRIC PANELS, ETC. RECESSED INTO FIRE RATED WALLS SHALL BE BACKED WITH FIRE RATED CONSTRUCTION TO MATCH THE WALL FIRE RATING.
7. CROSS CHECK ALL DETAILS AND DIMENSIONS SHOWN IN THE ARCHITECTURAL DRAWINGS WITH RELATED REQUIREMENTS ON THE STRUCTURAL, MECHANICAL ELECTRICAL, CIVIL AND/OR CONTROL DRAWINGS.
8. FLOOR AND WALL OPENINGS, SLEEVES, VARIATIONS IN THE STRUCTURAL SLAB ELEVATIONS, DEPRESSED AREAS, AND ALL OTHER CONTROL, MECHANICAL, ELECTRICAL AND/OR STRUCTURAL REQUIREMENTS, MUST BE COORDINATED BEFORE PROCEEDING WITH CONSTRUCTION.
9. DETAILS MARKED "TYPICAL" SHALL APPLY IN ALL SIMILAR CASES UNLESS SPECIFICALLY INDICATED OTHERWISE.
10. WHERE NO SPECIFIC DETAIL IS SHOWN, THE FRAMING OR CONSTRUCTION SHALL BE IDENTICAL OR SIMILAR TO THAT INDICATED FOR LIKE CASES OF CONSTRUCTION.
11. THE PRECISE DEMENSIONS AND LOCATIONS OF ALL LADDERS, EQUIPMENT AND ACCESS HATCHES, SUMPS, EQUIPMENT AND FLOOR DRAINS SHALL BE DETERMINED FROM THE STRUCTURAL, MECHANICAL, ELECTRICAL, AND CONTROL DRAWINGS, AND BE VERIFIED FROM VENDOR SHOP DRAWINGS, EQUIPMENT DATA ETC. AS REQUIRED.
12. SIZES OF MECHANICAL EQUIPMENT ROOM PADS, AND BASES ARE APPROXIMATE ONLY. CONTRACTOR SHALL VERIFY DIMENSIONS OF ALL CONCRETE PADS AND BASES WITH EQUIPMENT MANUFACTURER'S SHOP DRAWINGS.
13. ALL BEARING WALLS OR BEARING PORTIONS OF WALLS SHALL HAVE 2 OR 3 HOUR FIRE RESISTIVE CONSTRUCTION. SEE FIRE PROTECTION DWGS.
14. ALL CONCRETE ROOF SURFACES SHALL RECEIVE ELASTOMERIC ROOFING MATERIAL.
15. EXPOSED CONCRETE SURFACES OF THE TANK CEILING ARE TO RECEIVE FC-13 COATING SYSTEM.



PARAPET DETAIL 1 — 1½" = 1'-0"



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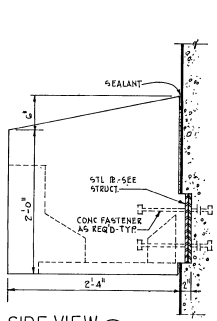
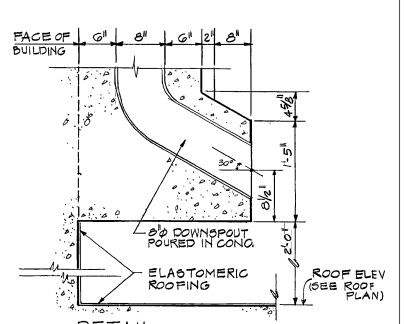
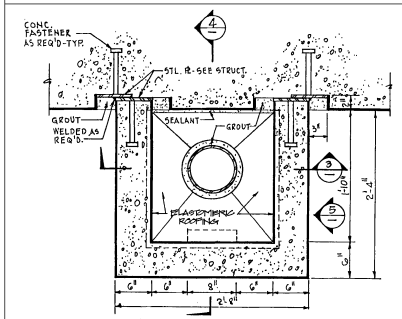


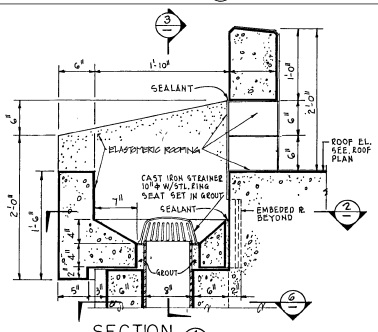
SIDE VIEW 5 — 1 1/2" = 1'-0"



DETAIL 7 — 1½" = 1'-0"



PLAN DETAIL OF DOWNSPOUT 2 — 1 1/2" = 1'-0"



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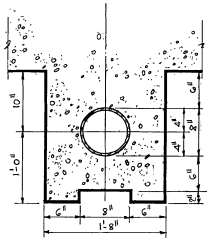
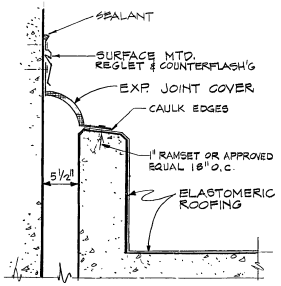
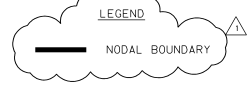


PLAN SECTION 6 — 1 1/2" = 1'-0"



DETAIL 8 — 1½" = 1'-0"



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2						
1	REVISED PER REA 96-VAA054	10-1-98	GS	RBH	CBH	
0	ISSUED PER REA 96-VAA054	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

NODAL BOUNDARY 1
AUXILIARY FEEDWATER PUMP HOUSE AREA

SCALE: 1/8"=1'-0" | JOB NO. 10604 | DRAWING NO. 1X6DD300 | REV. 1

SIZE E 34x44 | CAD NAME 1X6DD300